SYNTHESE LIBRARY

STUDIES IN EPISTEMOLOGY,

LOGIC, METHODOLOGY, AND PHILOSOPHY OF SCIENCE

VOLUME 329

SIGNS OF LOGIC

PEIRCEAN THEMES ON THE PHILOSOPHY OF LANGUAGE, GAMES, AND COMMUNICATION

by

AHTI-VEIKKO PIETARINEN

University of Helsinki, Finland

Springer

A C.I.P. Catalogue record for this book is available from the Library of Congress.

ISBN-10 1-4020-3728-7 (HB)
ISBN-13 978-1-4020-3728-3 (HB)
ISBN-10 1-4020-3729-5 (e-book)
ISBN-13 978-1-4020-3729-0 (e-book)

Published by Springer,
P.O. Box 17, 3300 AA Dordrecht, The Netherlands.

www.springer.com

Printed on acid-free paper

Printed in the Netherlands.

Contents

Preface

Charles Sanders Peirce (1839–1914), the principal subject of this book, was one of the most profound and prolific thinkers and scientists to have come out of the United States. His pragmatic logic and scientific methodology largely represent the application of interactive and intercommunicative triadic processes, best viewed as strategic and dialogic conceptualisations of logical aspects of thought, reasoning and action. These viewpoints also involve pragmatic issues in communicating linguistic signs, and are unified in his diagrammatic logic of existential graphs. The various game-theoretic approaches to the semantics and pragmatics of signs and language, to the theory of communication, and to the evolutionary emergence of signs, provide a contemporary toolkit the relevance of which Peirce envisioned to a wondrous extent.

These are some of the perspectives on Peirce's philosophy that are uncovered in the present work. Many of his most significant writings in this context reflect his later thinking, covering roughly the last 15–20 years of his life. Unfortunately, the bulk of his material from that period is still unpublished. In preparing the essays that comprise this book, I have greatly benefitted from using Peirce's manuscripts available in the microfilm at the University of Helsinki.

The conferences and meetings at which parts of the present work have featured between 2002 and 2004 include *The Charles S. Peirce Society* meeting in Washington D.C., *The Prague International Colloquium* in Prague, *The Italian Analytic Philosophy Society* meeting in Genoa, *The Evolutionary Epistemology in Language and Communication Conference* in Brussels, *The Diagrams IV* in Cambridge, *The Compositionality, Concepts and Cognition Conference* in Düsseldorf, *The International Wittgenstein Symposium* in Kirchberg am Weschel, *The International Conference on Historical Linguistics* in Copenhagen, *The International Cognitive Linguistics Conference* in La Rioja, *The World Congress of Philosophy* in Istanbul, *The Communications on the 21st Century: A New Research Agenda for Philosophy Conference* at the Hungarian Academy of Sciences, *The Philosophical Insights into Logic and Mathematics*

Conference in Nancy, *The International Workshop on Visual Representations and Interpretations* in Liverpool, as well as the *Finnish Philosophical Society*, *The Metaphysical Club*, *The Perspectives on the Philosophy of Charles S. Peirce Symposium* and *The UCLA–Helsinki Logic* meetings at the University of Helsinki.

Communication with the participants and organisers of these events as well as comments by Harry Alanen, Mats Bergman, Michael von Boguslawski, Leila Haaparanta, Risto Hilpinen, Jaakko Hintikka, Nathan Houser, Lauri Järvilehto, Erkki Kilpinen, Ilkka Niiniluoto, Kristof Nyíri, Panu Raatikainen, Shahid Rahman, Veikko Rantala, Henrik Rydenfelt, Gabriel Sandu, Peter Schulman, Lauri Snellman, André De Tienne, Isabelle Verschraegen and Fernando Zalamea have resulted in many improvements to the material. Naturally, they are all absolved from any responsibility for the final product. I must also express my thanks to Joan Nordlund for her supportive revision of the English, and to Malin Grahn for her assistance in preparing the index.

I am grateful to the publishers and editors for permission to include the material based upon previously published papers. Details of the original publications are as follows.

"Peirce's game-theoretic ideas in logic", *Semiotica* 144, 33–47. Chapter 3 is a revised, rewritten and expanded version of this paper. Copyright © 2003 *Walter de Gruyter*.

"Diagrammatic logic and game-playing", in Grant Malcolm (ed.), *Multidisciplinary Approaches to Visual Representations and Interpretations* (Studies in Multidisciplinarity), Oxford: Elsevier, 2004. Revised and partially included in Chapter 4. Copyright © 2004 *Elsevier Science*.

"Peirce's diagrammatic logic in IF perspective", in Alan Blackwell, Kim Marriott and Atsushi Shimojima (eds), *Diagrammatic Representation and Inference: Third International Conference, Diagrams 2004, Cambridge, UK, Lecture Notes in Artificial Intelligence* 2980, Berlin: Springer, 97–111. Revised and partially included in Chapter 5. Copyright © 2004 *Springer-Verlag*.

"Semantic games in logic and epistemology", in Shahid Rahman, Dov Gabbay, Jean Paul Van Bendegem and John Symons (eds), *Logic, Epistemology and the Unity of Science* (Logic, Epistemology and the Unity of Science Series), Dordrecht: Kluwer, 57–103. Rewritten for inclusion in the present volume as Chapter 7. Copyright © 2004 *Kluwer Academic Publishers*.

"Logic, language games and ludics", *Acta Analytica* 18, 89–123. Rewritten and expanded for inclusion in the present volume as Chapter 8. Copyright © 2003 *Society for Analytic Philosophy and Philosophy of Science of Slovenia*.

"Games as formal tools versus games as explanations in logic and science", *Foundations of Science* 8, 317–364. Rewritten and expanded for inclusion in the present volume as Chapter 10. Copyright © 2003 *Kluwer Academic Publishers*.

During the last four years, I have received financial support from the Finnish Cultural Foundation, the Helsingin Sanomat Centenary Foundation, the Osk. Öflund Foundation, the Jenny and Antti Wihuri Foundation, the Osk. Huttunen Foundation, the University of Helsinki, the Ella and Georg Ehrnrooth Foundation, and the Academy of Finland (project numbers 1178561, 1101687, 1104262 and 1103130).

Last but not least, I give my thanks to my parents Pirkko and Juhani, and my Laura, without whom this book would not be dedicated to them.

Ahti-Veikko J. Pietarinen, HELSINKI, JANUARY 2004

Bibliographical abbreviations

References are to the following Peirce collections (in chronological order):

1889–1909. Contributions to *The Century Dictionary and Cyclopedia* and to *The Century Dictionary Supplement*. Edited by William Dwight Whitney (1889–1891) and Benjamin Eli Smith (1894–1909). Century Dictionary Online: www.global-language.com/CENTURY/ (accessed 31 December 2004).

Abbreviated by CD (Century Dictionary and Cyclopedia volumes I–X, 1889–1891), CDS (Century Dictionary Supplement volumes XI–XII, 1909), followed by volume and page number.

1923. *Chance, Love, and Logic: Philosophical Essays*. Edited by R. C. Morris, with an essay by John Dewey, London: Kegan Paul, Trench, Trubner & Co.

Abbreviated by CLL, followed by page number.

1931–1958. *Collected Papers of Charles Sanders Peirce*, 8 volumes, vols. 1–6 edited by Charles Hartshorne and Paul Weiss, vols. 7–8, edited by Arthur W. Burks. Cambridge, Mass.: Harvard University Press.

Abbreviated by volume and paragraph number.

1967. *Manuscripts in the Houghton Library of Harvard University*, as identified by Richard Robin, *Annotated Catalogue of the Papers of Charles S. Peirce* (Amherst: University of Massachusettes Press, 1967), and in The Peirce Papers: A supplementary catalogue, *Transactions of the C. S. Peirce Society* 7 (1971), 37–57.

Abbreviated by MS, followed by page number or alternative identifier (a.p.: assorted draft pages).

1975–1987. Contributions to *The Nation.* Four volumes. Edited by Kenneth Ketner and James Cook. Lubbock: Texas Tech University Press. (Available as cd-rom with a biography of Peirce's published writings via InteLex).

Abbreviated by CN, followed by item and paragraph number.

1976. *The New Elements of Mathematics*. Four volumes in five books. Edited by Carolyn Eisele. Berlin: Mouton Publishers; Atlantic Highlands, NJ: Humanities Press.

Abbreviated by NEM, followed by volume and page number.

1977. *Semiotic and Significs: The Correspondence Between Charles S. Peirce and Victoria Lady Welby*. Edited by Charles S. Hardwick and James Cook. Bloomington: Indiana University Press.

Abbreviated by SS, followed by page number.

1982–. *Writings of Charles S. Peirce: A Chronological Edition*, 6 volumes to 2004. Vol. 1 (1857–1866), edited by Max Fisch et al., 1982; vol. 2 (1867–1871), edited by Edward C. Moore et al., 1984; vols. 3–5 (1872–1878, 1879–1884, 1884–1886), edited by Christian J. W. Kloesel et al., 1986, 1986, 1993; vol. 6 (1887–1890), edited by the Peirce Edition Project, 1999. Bloomington and Indianapolis: Indiana University Press.

Abbreviated by W, followed by volume and page number.

1983/1883. *Studies in Logic, By Members of the Johns Hopkins University (1883)*. Edited by Charles S. Peirce. Amsterdam: John Benjamins. With an introduction by Max H. Fisch and a preface by Achim Eschbach. Originally published by Little, Brown, and Company, Boston, 1883.

Abbreviated by SIL, followed by page number.

1985. *Historical Perspectives on Peirce's Logic of Science. A History of Science*. Two volumes. Edited by Carolyn Eisele. Berlin: Mouton Publishers.

Abbreviated by HP, followed by volume and page number.

1992. *Reasoning and the Logic of Things*. Edited by Kenneth Laine Ketner, with an introduction by Kenneth Laine Ketner and Hilary Putnam. Cambridge, Mass.: Harvard University Press. Abbreviated by RLT, followed by page number.

1992–. *The Essential Peirce. Selected Philosophical Writings*. Vol. 1 (1867–1893), edited by Nathan Houser and Christian J. W. Kloesel, 1992; vol. 2 (1893–1913), edited by the Peirce Edition Project, 1998. Bloomington and Indianapolis: Indiana University Press.

Abbreviated by EP, followed by volume and page number.

1997. *Pragmatism as a Principle and Method of Right Thinking. The 1903 Harvard Lectures on Pragmatism by Charles S. Peirce*. Edited by Patricia Ann Turrisi. Albany: State University of New York Press.

Abbreviated by PP, followed by page number.

1998/1901–1902. Contributions to *The Dictionary of Philosophy and Psychology*. Two volumes. Edited by James Mark Baldwin. Bristol: Thoemmes Press, 1998. Appeared in 1901 (vol. 1) and in 1902 (vol. 2). New York: The Macmillan Company & London: Macmillan and Co., Limited. Partially online: psychclassics.yorku.ca/Baldwin/Dictionary/ (accessed 31 December 2004).

Abbreviated by DPP, followed by page number.

Index

Bibliography

Abramsky, Samson, Gay, S. J. and Nagarajan, Rajagopal, 1996. Interaction categories and the foundations of typed concurrent programming, in M. Broy (ed.), *Proceedings of the 1994 Marktoberdorf Summer School on Deductive Program Design*, Berlin: Springer-Verlag, 35–113.

Abramsky, Samson and Jagadeesan, Radha, 1994. Games and full completeness for multiplicative linear logic, *Journal of Symbolic Logic* 59, 543–574.

Abramsky, Samson and Melliès, Paul-Andre, 1999. Concurrent games and full completeness, in *Proceedings of the 14th Annual IEEE Symposium on Logic in Computer Science*, IEEE Computer Society Press, 431–442.

Albertazzi, Liliana, 2000. Which semantics?, in L. Albertazzi (ed.), *Meaning and Cognition: A Multidisciplinary Approach*, Amsteram: John Benjamins, 1–24.

āl-Fārābī, 1961. *The Fuṣūl al-Madanī of āl-Fārābī (Aphorisms of the Statesman)*, edited with an English translation, introduction and notes by D. M. Dunlop, Cambridge: Cambridge University Press.

de Alfaro, Luca, 2003. Game models for open systems, N. Dershowitz (ed.), *Verification: Theory and Practice, Lecture Notes in Computer Science* 2772, Springer, 269–289.

de Alfaro, Luca and Henzinger, Tom A., 2000. Concurrent Omega-regular Games, in *Proceedings of the 15th Annual IEEE Symposium on Logic in Computer Science*, IEEE Computer Society Press, 141–154.

Anderson, Douglas, 1998. *Strands of System: The Philosophy of Charles Peirce*, West Lafayette: Purdue University Press.

Arrow, Kenneth J., 1951. *Social Choice and Individual Values*, New York, Wiley & Sons.

Asher, Nicholas, 1993. *Reference to Abstract Objects in Discourse*, Dordrecht: Kluwer.

Ashworth, E. Jennifer, 1976. Agostino Nifo's reinterpretation of medieval logic, *Rivista Critica di Storia Della Filosofia* 31, 354–374.

Aull, C. E. and Lowen, R., (eds), 1998–2001. *Handbook of the History of General Topology*, 3 vols., Dordrecht: Kluwer.

Aumann, Robert J., 1964. Markets with a continuum of traders, *Econometrica* 32, 39–50.

Aumann, Robert J., 1981. Survey of repeated games, in *Essays in Game Theory and Mathematical Economics in Honor of Oskar Morgenstern* Mannheim: Wissenschaftsverlag Bibliographisches Institut, 11–42.

Aumann, Robert J., 1985. What is game theory trying to accomplish?, in R. Arrow and S. Honkapohja (eds), *Frontiers in Economics*, Oxford: Blackwell, 28–76.

Aumann, Robert J., 1987a. Game theory, in J. Eatwell, M. Milgate and P. Newman (eds), *The New Palgrave, A Dictionary of Economics* 2, London & Basingstoke: Macmillan, 460–482.

Aumann, Robert J., 1987b. Correlated equilibrium as an expression of Bayesian rationality, *Econometrica* 55, 1–18.

Austin, John L., 1960. *Philosophical Papers*, Oxford: Clarendon Press.

Badesa, Calixto, 2004. *The Birth of Model Theory: Löwenheim's Theorem in the Frame of the Theory of Relatives*, Princeton: Princeton University Press.

Bacharach, Michael, 1985. Some extensions of a claim of Aumann in an axiomatic model of knowledge, *Journal of Economic Theory* 37, 167–190.

Bacharach, M. O. L., Gérard-Varet, L.-A., Mongin, P. and Shin, H. S., (eds), 1997. *Epistemic Logic and the Theory of Games and Decisions*, Dordrecht: Kluwer.

Baliga, Sandeep and Serrano, Roberto, 1995. Multilateral bargaining with imperfect information, *Journal of Economic Theory* 67, 578–589.

Barwise, Jon and Etchemendy, John, 1992. *Tarski's World*, Stanford: CSLI.

Basar, Tamer and Haurie, Alain, (eds), 1994. *Advances in Dynamic Games and Applications*, Boston: Birkhäuser.

Barwise, Jon and Feferman, Solomon, (eds), 1985. *Model-Theoretic Logics*, New York: Springer.

Batali, John, 1998. Computational simulations of the emergence of grammar, in J. R. Hurford, M. Studdert-Kennedy and C. Knight (eds), *Approaches to the Evolution of Language: Social and Cognitive Bases*, Cambridge: Cambridge University Press, 405–426.

Bayart, A., 1958. La correction de la logique de modale du premier et second ordre S5, *Logique et Analyse* 1, 28–44.

Bechtel, William, Mandik, Pete, Mundale, Jennifer and Stufflebeam, Robert S., (eds), 2001. *Philosophy and the Neurosciences: A Reader*, Oxford: Blackwell Publishers.

Bennett, Jonathan, 1955. Iterated modalities, *Philosophical Quarterly* 5, 45–56.

van Benthem, Johan, 1988. Games in logic, in J.Ph. Hoepelman (ed.), *Representation and Reasoning. Proceedings of the Stuttgart Conference Workshop on Discourse Representation, Dialogue Tableaux and Logic Programming*, Tübingen: Niemeyer Verlag, 3–15.

van Benthem, Johan, 1993. Modeling the kinematics of meaning, *Proceedings of the Aristotelian Society* 93, London, 105–122.

van Benthem, Johan, 1999. Wider and still wider... Resetting the bounds of logic', in A. Varzi (ed.), *The European Review of Philosophy* 4, Stanford: CSLI, 21–44.

van Benthem, Johan, 2001. The epistemic logic of IF games, in R. E. Auxier and L. Hahn (eds), *The Philosophy of Jaakko Hintikka*, vol. 30 in the Library of Living Philosophers, Chicago: Open Court.

Bergman, Mats, 2000. Reflections of the role of the communicative sign in semeiotic, *Transactions of the Charles S. Peirce Society: A Quarterly Journal in American Philosophy* 36, 225–254.

Berners-Lee, Tim, 1999. *Weaving the Web*, London: Orion Business Books.

Berners-Lee, Tim and Miller, Eric, 2002. The semantic web lifts off, *ERCIM News* 51, 2002, 9–11. www.ercim.org/publication/Ercim_News/enw51/berners-lee.html (accessed 31 December 2004).

Bertuccelli Papi, Marcella, 2000. Is a diachronic speech act theory possible?, *Journal of Historical Pragmatics* 1, 57–66.

Binmore, Kevin, 1996. A note on imperfect recall, in W. Albers et al., (eds), *Understanding Strategic Interaction: Essays in Honor of Reinhard Selten*, New York: Springer, 51–62.

Blass, Andreas, 1972. Degrees of indeterminacy of games, *Fundamenta Mathematica* 77, 151–166.

Blass, Andreas, 1992. A game semantics for linear logic, *Annals of Pure and Applied Logic* 56, 183–220.

Blass, Andreas, 1994. Is game semantics necessary?, in E. Börger, Y. Gurevich and K. Meinke (eds), *Computer Science Logic, Lecture Notes in Computer Science* 832, Springer, 66–77.

Blass, Andreas, 1997. Some semantical aspects of linear logic, *Logic Journal of the IGPL* 5, 487–502.

Blinov, Arcady, 1994. Semantic games with chance moves, *Synthese* 99, 311–327.

Bloomfield, Maurice, 1895. On assimilation and adaptation in congeneric classes of words, *American Journal of Philology* 16, 409–434.

Bonanno, Giacomo and Battigalli, Pierpaolo, 1997. Synchronic information, knowledge and common knowledge in extensive games, in Bacharach et al. (1997, 235–263).

Boolos, George, 1981. For all A there is a B, *Linguistic Inquiry* 12, 465–467.

Borel, Emil, 1921. La théorie du jeu et les equations intégrales à noyau symétrique, *Comptes Rendus Hebdomadaires des Séances de l'Académie des Sciences* 173, 1304–1308. (Translation by L. J. Savage: 1953. The theory of play and integral equations with skew symmetric kernels, *Econometrica* 21, 97–100.)

van Brakel, Jaap, 2003. Multiculturalising Davidson's triangulation, ms.

Bradfield, Julian C., 2001. Independence: logic and concurrency, in P. G. Clote and H. Schwichtenberg (eds), *Proceedings of the 14th International Workshop on Computer Science Logic, Lecture Notes in Computer Science* 1862, Berlin: Springer, 247–261.

Brady, Geraldine and Trimble, Todd H., 2000. A categorical interpretation of C. S. Peirce's propositional logic Alpha, *Journal of Pure and Applied Algebra* 149, 213–239.

Brady, Geraldine and Trimble, Todd H., 1998. A string diagram calculus for predicate logic, ms.

Bréal, Michel, 1964. *Semantics: Studies in the Science of Meaning*, New York: Dover. (Translated by Mrs. Henry Cust, 1900.)

Brent, Joseph, 1998. *Charles Sanders Peirce: A Life* (2nd edition), Indiana University Press.

Brighton, Henry and Kirby, Simon, 2001. The survival of the smallest: Stability conditions for the cultural evolution of compositional language, in J. Kelemen and P. Sosik, (eds), *Advances in Artificial Life*, Springer, 592–601.

Broca, Paul, 1861. Remarques sur le siège de la faculté du langage articulé; survies d'une observation d'aphemie, *Bulletin de la Société Anatomique* 6, 330–357. Translated by Charles S. Thomas, Remarks on the seat of the faculty of articulate language, followed by an observation of aphemia, in G. von Bonin (ed.), *Some Papers on the Cerebral Cortex*, Springfield: Charles C. Thomas, 1960, 49–72.

Brock, Jarret, 1980. Peirce's anticipation of game-theoretic logic and semantics, in M. Herzfeld and M. D. Lenhart (eds), *Semiotics 1980: Proceedings of 5th Annual Meeting of the Semiotics Society of America*, New York: Plenum, 55–64.

Brouwer, Luitzen Egbertus Jan, 1975. *Collected Works 1: Philosophy and Foundations of Mathematics*, Amsterdam: North-Holland.

Burch, Robert W., 1991. *A Peircean Reduction Thesis*. Austin: Texas Tech University Press.

Burch, Robert W., 1994. Game-theoretical semantics for Peirce's existential graphs, *Synthese* 99, 361–375.

Burch, Robert W., 1997. A Tarski-style semantics for Peirce's beta graphs, in J. Brunning and P. Forster, (eds), *The Rule of Reason: The Philosophy of Charles Sanders Peirce*, Toronto: University of Toronto Press, 81–95.

Burge, Taylor, 1974. Demonstrative constructions, reference and truth, *Journal of Philosophy* 70, 425–439.

Camic, Charles, 1986. The matter of habit, *Americal Journal of Sociology* 91, 1039–1087.

Carlson, Lauri, 1983. *Dialogue Games: An Approach to Discourse Analysis*, Dordrecht: D. Reidel.

Carse, James P., 1986. *Finite and Infinite Games: A Vision of Life as Play and Possibility*, New York: Ballantine Books.

Carston, Robyn, 1988. Implicature, explicature, and truth-theoretic semantics, in R. Kempson (ed.), *Mental Representations: The Interface between Language and Reality*, Cambridge: Cambridge University Press, 155–181.

Changeaux, Jean-Pierre amd Ricoeur, Paul, 2000. *What Makes us Think? A Neuroscientist and a Philosopher Argue about Ethics, Human Nature, and the Brain*, Princeton: Princeton University Press.

Chiasson, Phyllis, 2001. *Peirce's Pragmatism: The Design for Thinking*, Amsterdam: Rodopi.

Chomsky, Noam, 1986. *Knowledge of Language: Its Nature, Origin, and Use*, New York: Praeger Publishers.

Christiansen, Morten H. and Kirby, Simon, 2003. *Language Evolution*, Oxford University Press.

Clark, Herbert H., 1996. *Using Language*, Cambridge: Cambridge University Press.

Clark, Herbert H. and Carlson, T. B., 1982. Speech acts and hearers' beliefs, in N. V. Smith (ed.), *Mutual Knowledge*, London: Academic Press, 1–36.

Combs, James E., 2000. *Play World: The Emergence of the New Ludenic Age*, Westport: London.

Copeland, B. Jack, 2002. The genesis of possible worlds semantics, *Journal of Philosophical Logic* 31, 99–137.

Cressman, Ross, 2003. *Evolutionary Dynamics and Extensive Form Games*, Cambridge, Mass.: MIT Press.

Davidson, Donald, 2001. *Subjective, Intersubjective, Objective*, Oxford: Clarendon Press.

van Damme, Eric, 1984. A relation between perfect equilibria in extensive form games and proper equilibria in normal form games, *International Journal of Game Theory* 13, 1–13.

De Morgan, Augustus, 1849. On the structure of the syllogism, and on the application of the theory of probabilities to questions of argument and authority, *Transactions of the Cambridge Philosophical Society* 8, 379–408.

De Vos, Marine and Vermeir, Dirk, 1999. Choice logic programs and Nash equilibria in strategic games, in J. Flum and M. Rodriguez-Artalejo (eds), *Computer Science Logic, Lecture Notes in Computer Science* 1683, Springer, 266–276.

Deacon, Terrence W., 1997. *The Symbolic Species: The Coevolution of Language and Human Brain*, London: Penguin.

Deane, Paul D., 1996. Neurological evidence for a cognitive theory of syntax: agrammatic aphasia and the spatialization of form hypothesis, in E. H. Casad (ed.), *Cognitive Linguistics in the Redwoods. The Expansion of a New Paradigm in Linguistics*, Berlin: Mouton de Gruyter, 55–115.

Dekker, Paul and van Rooy, Robert, 2000. Bi-directional optimality theory: an application of game theory, *Journal of Semantics* 17, 217–242.

Dewey, John, 1997/1916. Theories of knowledge, in L. Menand (ed.), *Pragmatism: A Reader*, New York: Vintage Books, 205–218. (Originally appeared in *Democracy and Education*, 1916).

Dienes, Zoltan and Perner, Josef, 1999. A theory of implicit and explicit knowledge, *Behavioral and Brain Sciences* 22, 735–755.

Doherty, Patrik, (ed.), 1996. *Partiality, Modality, and Nonmonotonicity*, Stanford: CSLI.

Drake, F. R., 1962. On McKinsey's syntactical characterizations of systems of modal logic, *Journal of Symbolic Logic* 27, 400–406.

Ebbesen, Sten, 1991. Is logic theoretical or practical knowledge?, in J. Biard (ed.), *Itinéraires d'Albert de Saxe Paris–Vienne au* XIV^{e} *Siècle*, Paris: Librairie Philosophique J. Vrin, 267–276.

van Eemeren, Frans H., Grootendorst, Rob and Snoeck Henkemans, Francisca, 1996. *Fundamentals of Argumentation Theory: A Handbook of Historical Backgrounds and Contemporary Developments*, Mahwah: Lawrence Erlbaum.

Fagin, Ronald, Halpern, Joseph Y., Moses, Yoram and Vardi, Moshe Y., 1995. *Reasoning about Knowledge*, Cambridge, Mass.: MIT Press.

Fann, K.T., 1970. *Peirce's Theory of Abduction*, The Hague: Martinus Nijhoff.

Feigenbaum, Joan, Koller, Daphne and Shor, Peter, 1995. A game-theoretic classification of interactive complexity classes, *Proceedings of the 10th Annual IEEE Conference on Structure in Complexity Theory*, IEEE Computer Society Press, 227–237.

Felscher, Walter, 2002. Dialogues as a foundation for intuitionistic logic, in D. Gabbay and F. Guenthner (eds), *Handbook of Philosophical Logic* 5, (2nd ed.), Dordrecht: Kluwer, 115–146.

Fensel, Dieter and Musen, Mark A., 2001. The Semantic Web: a brain for Humankind, *IEEE Intelligent Systems* 16, 24–25.

Fisch, Max and Turquette, Atwell, 1966. Peirce's triadic logic, *Transactions of the Charles S. Peirce Society* 11, 71–85.

Fong, Terry, Nourbakhsh, Illah and Dautenhahn, Kerstin, 2003. A survey of socially interactive robots, *Robotics and Autonomous Systems* 42, 143–166.

Forget, Danielle, Hirschbühler, Martineau France and Rivero, Maria Luisa, (eds), 1998. *Negation and Polarity: Syntax and Semantics*, Amsterdam: John Benjamins.

van Fraassen, Bas C., 1969. Presuppositions, supervaluations, and free logic, in K. Lambert (ed.), *The Logical Way of Doing Things*, New Haven: Yale University Press, 67–91.

Frieden, Roy, 1998. *Physics From Fisher Information*, Cambridge, Mass.: Cambridge University Press.

Gabbay, Dov and Woods, John, 2001. The new logic, *Logic Journal of the Interest Group in Pure and Applied Logic* 9, 141–174.

Gale, David and Stewart, F. M., 1953. Infinite games with perfect information, in H. W. Kuhn, and A. W. Tucker (eds), *Contributions to the Theory of Games*, Vol. 2, Princeton: Princeton University Press, 245–266.

Gärdenfors, Peter, 2002. Cooperation and the evolution of symbolic communication, *Lund University Cognitive Studies* 91.

Gardiner, Alan H., 1932. *The Theory of Speech and Language*, Oxford: Clarendon Press.

Geach, Peter, 1967. Intentional identity, *Journal of Philosophy* 64, 627–632.

Geach, Peter, 1976. Saying and showing in Wittgenstein and Frege, in J. Hintikka (ed.), *Essays in Honour of G. H. von Wright, Acta Philosophica Fennica* 28, Amsterdam: North Holland, 54–70.

Geanakoplos, John, 1989. Game theory without partitions, and applications to speculation and consensus, *Cowles Foundation Discussion Paper* 914, Yale University.

Gellner, Ernest, 1995. *Language and Solitude: Wittgenstein, Malinowski and the Habsburg Dilemma*, Cambridge: Cambridge University Press.

Gentzen, Gerhard, 1969. The consistency of elementary number theory, in M. E. Szabo (ed.), *The collected works of Gerhard Gentzen*, Amsterdam: North-Holland, 132–213.

Giddens, Anthony, 1979. *Central Problems in Social Theory*, London: Macmillan.

Girard, Jean-Yves, 1998. On the meaning of logical rules I: syntax vs. semantics, in Berger and Schwichtenberg (eds), *Computational Logic*, Heidelberg, 215–272. iml.univ-mrs.fr/pub/girard/ (accessed 31 December 2004).

Girard, Jean-Yves, 2001. Locus Solum: from the rules of logic to the logic of rules, *Mathematical Structures in Computer Science* 11, 301–506.

Goldstein, Lawrence, 1989. Wittgenstein and paraconsistency, in G. Priest, F. R. Routley and J. Norman (eds.), *Paraconsistent Logic. Essays on the Inconsistent*, Munich: Philosophia Verlag, 540–562.

Gosser, H. Mark, 1977. *Selected Attempts at Stereoscopic Moving Pictures and their Relationship to the Development of Motion Picture Technology, 1852–1903*, New York: Arno Press.

Guenthner, Franz, 1987. From sentences to discourse: some aspects of the computational treatment of language, *Natural Language at the Computer 1988*, 147–165.

Guillaume, M., 1958. Rapports entre calculs propositionnels modaux et topologie impliqués par certaines extensions de la méthode des tableaux sémantiques. Système de Feys-von Wright, *Comptes Rendus des Séances de l'Académie des Sciences* 246, 1140–1142.

Grice, H. Paul, 1986. Reply to Richards, in R. E. Grandy and R. Warner (eds), *Philosophical Grounds of Rationality*, Oxford: Clarendon Press, 45–106.

Grice, H. Paul, 1989. *Studies in the Way of Words*, Cambridge, Mass.: Harvard University Press.

Gross, Charles G., 1998. *Brain, Vision, Memory: Tales in the History of Neuroscience*, Cambridge, Mass.: MIT Press.

Habermas, Jürgen, 1995. *Peirce and communication*, in K. Laine Ketner (ed.), *Peirce and Contemporary Thought*, New York: Fordham University Press, 243–266.

Habermas, Jürgen, 1998. *On the Pragmatics of Communication*, (ed. by Maeve Cooke), Cambridge, Mass.: MIT Press.

Habermas, Jürgen, 2001. *On the Pragmatics of Social Interaction: Preliminary Studies in the Theory of Communicative Action*, (translated by Barbara Fultner), Cambridge, Mass.: MIT Press.

Hacking, Ian, 1979. What is logic?, *Journal of Philosophy* 76, 285–319.

Hamblin, Charles L., 1971. Mathematical models of dialogue, *Theoria* 37, 130–155.

Hammer, Eric M., 1998. Semantics for existential graphs, *Journal of Philosophical Logic* 27, 489–503.

Hammer, Eric M., 2002. Diagrammatic logic, in D. Gabbay and F. Guenthner (eds), *Handbook of Philosophical Logic* 4 (2nd edition), Dordrecht: Kluwer, 395–422.

Hansen, B., 1970. *A Survey of General Equilibrium Systems*, New York: McGraw-Hill.

Harré, Romano, 1961. *Theories and Things: A Brief Study in Prescriptive Metaphysics*, London: Sheed and Ward.

Harsanyi, John C., 1967. Games with incomplete information played by 'Bayesian' players. Part I: The basic model, *Management Science* 14, 159–182.

van Heijenoort, Jean van, 1967. *From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931*, Cambridge, Mass.: Harvard University Press.

Heim, Irene, 1982. *The Semantics of Definite and Indefinite Noun Phrases*, dissertation, University of Massachusetts at Amherst. (Reproduced and distributed by Graduate Linguistics Students' Association, University of Massachusetts.)

Hendrix, G. G., 1979. Encoding knowledge in partitioned networks, in N. V. Findler (ed.), *Associative Networks: Representation and Use of Knowledge by Computers*, Orlando: Academic Press, 51–92.

Henkin, Leon, 1961. Some remarks on infinitely long formulas, in *Infinistic Methods. Proceedings of the Symposium on Foundations of Mathematics, Warsaw, Panstwowe* (no editor given), Naukowe: Wydawnictwo, New York: Pergamon Press, 167–183.

Hilbert, David, 1926. Über das Unendliche, *Mathematische Annalen* 95, 161–190.

Hilpinen, Risto, 1982. On C. S. Peirce's theory of the proposition: Peirce as a precursor of game-theoretical semantics, *The Monist* 65, 182–188.

Hilpinen, Risto, 1995. Peirce on language and reference, in K. Laine Ketner (ed.), *Peirce and Contemporary Thought*, New York: Fordham University Press, 272–303.

Hilpinen, Risto, 2004. Peirce's logic, in D. M. Gabbay and J. Woods (eds), *Handbook of the History and Philosophy of Logic. Vol. 3: The Rise of Modern Logic From Leibniz to Frege*, Amsterdam: Elsevier, 623–670.

Hintikka, Jaakko, 1957a. Modality as referential multiplicity, *Ajatus* 20, Yearbook for Societas Philosophica Fennica, 49–64.

Hintikka, Jaakko, 1957b. Quantifiers in Deontic Logic, *Societas Scientiarum Fennica, Commentationes Humanarum Litterarum* 23, Helsinki.

Hintikka, Jaakko, 1962. *Knowledge and Belief: An Introduction to the Logic of the Two Notions*, Ithaca: Cornell University Press.

Hintikka, Jaakko, 1969. *Models for Modalities*, Dordrecht: D. Reidel.

Hintikka, Jaakko, 1973a. *Logic, Language-Games and Information*, Oxford: Oxford University Press.

Hintikka, Jaakko, 1973b. Quantifiers vs. quantification theory, *Dialectica* 27, 329–358. (Reprinted in *Linguistic Inquiry* 5, 1974, 153–177.)

Hintikka, Jaakko, 1975. Impossible possible worlds vindicated, *Journal of Philosophical Logic* 4, 475–484.

Hintikka, Jaakko, 1983. Semantic games, the alleged ambiguity of 'is', and Aristotelian categories, *Synthese* 54, 443–467.

Hintikka, Jaakko, 1986. Logic of conversation as a logic of dialogue, in R. E. Grandy and R. Warner (eds), *Philosophical Grounds of Rationality*, Oxford: Clarendon Press, 259–276.

Hintikka, Jaakko, 1987a. Game-theoretical semantics as a synthesis of verificationist and truth-conditional meaning theories, in E. LePore (ed.), *New Directions in Semantics*, Academic Press, 235–258.

Hintikka, Jaakko, 1987b. Language understanding and strategic meaning, *Synthese* 73, 497–529.

Hintikka, Jaakko, 1989. *The Logic of Epistemology and the Epistemology of Logic*, Dordrecht: Kluwer.

Hintikka, Jaakko, 1995. The games of logic and the games of inquiry, *Dialectica* 49, 229–249.

Hintikka, Jaakko, 1996a. *The Principles of Mathematics Revisited*, New York: Cambridge University Press.

Hintikka, Jaakko, 1996b. *Ludwig Wittgenstein: Half-Truths and One-and-a-Half-Truths*, Dordrecht: Kluwer.

Hintikka, Jaakko, 1996c. The place of C. S. Peirce in the history of logical theory, in J. Brunning and P. Forster (eds), *The Rule of Reason: The Philosophy of Charles Sanders Peirce*, Toronto: University of Toronto Press, 13–33.

Hintikka, Jaakko, 1997. No scope for scope?, *Linguistics and Philosophy* 20, 515–544.

Hintikka, Jaakko, 1998. What is abduction? The fundamental problem of contemporary epistemology, *Transactions of the Charles S. Peirce Society* 34, 503–533. (Reprinted in Hintikka 1999.)

Hintikka, Jaakko, 1999. *Inquiry as Inquiry: A Logic of Scientific Discovery*, Dordrecht: Kluwer.

Hintikka, Jaakko, 2002a. Hyperclassical logic (a.k.a. IF logic) and its implications for logical theory, *The Bulleting of Symbolic Logic* 8, 404–423.

Hintikka, Jaakko, 2002b. Negation in logic and in natural language, *Linguistics and Philosophy* 25, 585–600.

Hintikka, Jaakko, Ilpo Halonen and Mutanen, Arto, 2003. Interrogative logic, in D. M. Gabbay, R. H. Johnson, H. J. Ohlbach and J. Woods (eds), *Handbook of the Logic of Argument and Inference. The Turn Towards the Practical*, Dordrecht: Kluwer, 295–337.

Hintikka, Jaakko and Hintikka, Anna-Maija, 2002. Wittgenstein the bewitched writer, in R. Haller and K. Puhl (eds), *Wittgenstein and the Future of Philosophy: A Reassessment after 50 Years*, Vienna: Wittgenstein-Gesellschaft, 131–150.

Hintikka, Jaakko and Hintikka, Merrill B., 1986. *Investigating Wittgenstein*, Oxford: Blackwell.

Hintikka, Jaakko and Kulas, Jack, 1983. *The Game of Language: Studies in Game-Theoretical Semantics and its Applications*, Dordrecht: D. Reidel.

Hintikka, Jaakko and Kulas, Jack, 1985. *Anaphora and Definite Descriptions*, Dordrecht: D. Reidel.

Hintikka, Jaakko and Rantala, Veikko, 1976. A new approach to infinitary languages, *Annals of Mathematical Logic* 10, 95–115.

Hintikka, Jaakko and Sandu, Gabriel, 1989. Informational independence as a semantical phenomenon, in J. E. Fenstad et al., (eds), *Logic, Methodology and Philosophy of Science* 8, Amsterdam: North-Holland, 571–589.

Hintikka, Jaakko and Sandu, Gabriel, 1991. *On the Methodology of Linguistics*, Oxford: Blackwell.

Hintikka, Jaakko and Sandu, Gabriel, 1995. What is the logic of parallel processing?, *International Journal of the Foundations of Computer Science* 6, 27–49.

Hintikka, Jaakko and Sandu, Gabriel, 1997. Game-theoretical semantics, in J. van Benthem and A. ter Meulen (eds), *Handbook of Logic and Language*, Amsterdam: Elsevier, 361–410.

Ho, Y. C., and Sun, F. K., 1974. Value of information in two-team zero-sum problems, *Journal of Optimization Theory and Applications* 14, 557–571.

Hodges, Wilfrid, 1977. *Logic: An Introduction to Elementary Logic*, London: Penguin Books.

Hodges, Wilfrid, 1986. Truth in a structure, *Proceedings of the Aristotelian Society* 86, 135–151.

Hodges, Wilfrid, 1993. The logical content of theories of deduction, *Behavioral and Brain Sciences* 16, 353–354.

Hodges, Wilfrid, 1997a. Compositional semantics for a language of imperfect information, *Logic Journal of the IGPL* 5, 539–563.

Hodges, Wilfrid, 1997b. Games in logic, in P. Dekker, M. Stokhof and Y. Venema (eds), *Proceedings of the 11th Amsterdam Colloquium*, Amsterdam: University of Amsterdam, 13–18.

Hodges, Wilfrid, 1998. An editor recalls some hopeless papers, *The Bulletin of Symbolic Logic* 4, 1–16.

Hodges, Wilfrid, 1999. A formality, *Festschrift for Johan von Benthem*, University of Amsterdam, 1–10.

Hodges, Wilfrid, 2001a. Model theory, in *The Stanford Encyclopedia of Philosophy*, E. N. Zalta (ed.), plato.stanford.edu/entries/model-theory/ (accessed 31 December 2004).

Hodges, Wilfrid, 2001b. Logic and games, in *The Stanford Encyclopedia of Philosophy*, E. N. Zalta (ed.), plato.stanford.edu/entries/logic-and-games/ (accessed 31 December 2004).

Hodges, Wilfrid, 2001c. Dialogue foundations I: a sceptical look, *Proceedings of the Aristotelian Society, supplementary volume* 75, 17–32.

Hodgson, Geoffrey M., 1999. *Evolution and Institutions: On Evolutionary Economics and the Evolution of Economics*, Cheltenham: Edward Elgar.

Hookway, Christopher, 2000. *Truth, Rationality, and Pragmatism: Themes from Peirce*, Oxford: Oxford University Press.

Houser, Nathan, 1992. The fortunes and misfortunes of the Peirce papers, in M. Balat and J. Deledalle-Rhodes (eds), *Signs of Humanity*, Berlin: Mouton de Gruyter, 1259–1268.

Houser, Nathan, Roberts, Don D. and Van Evra, James, (eds), 1997. *Studies in the Logic of Charles Sanders Peirce*. Bloomington: Indiana University Press.

Hulstijn, Joris, 2000. *Dialogue models for inquiry and transaction*, dissertation, University of Twente.

Hurford, James R., 1989. Biological evolution of the saussurean sign as a component of the language acquisition device, *Lingua* 77, 187–222.

Hyland, J. M. E., and Ong, C.-H. L., 2000. On full abstraction for PCF. I. Models, observables and the full abstraction problem, II. Dialogue games and innocent strategies, III. A fully abstract and universal game model, *Information and Computation* 163, 285–408.

Isbell, John, 1957. Finitary games, in D. Dresher, A. W. Tucker and P. Wolfe (eds), *Contributions to the Theory of Games* 3, Princeton: Princeton University Press, 79–96.

Jackendoff, Ray, 2002. *Foundations of Language: Brain, Meaning, Grammar, Evolution*, Oxford: Oxford University Press.

Jackson, Peter, 1987. *A representation language based on a game-theoretic interpretation of logic*, dissertation, University of Leeds.

James, William, 1997/1890. Habit, in Menand, Louis (ed.) *Pragmatism: A Reader*, New York: Vintage Books, 60–68. (Originally appeared in *The Principles of Psychology*, 1890).

Janasik, Tapio and Sandu, Gabriel, 2003. Dynamic game semantics, in J. Peregrin, (ed.), *Meaning: The Dynamic Turn*, Elsevier, 215–240.

Janasik, Tapio, Ahti-Veikko Pietarinen and Sandu, Gabriel, 2002. Anaphora and extensive games, in M. Andronis, E. Debenport, A. Pycha and K. Yoshimura (eds), *Chicago Linguistic Society 38: The Main Session*, Chicago: Chicago Linguistic Society, 285–295.

Janssen, Theo M. V., 2002. On the interpretation of IF logic, *Journal of Logic, Language and Information* 11, 367–387.

Jervell, Hermann R., 1985. Gentzen Games, *Zeitschrift für Mathematische Logic und Grundlagen der Mathematik* 31, 431–439.

Johansen, Jørgen Dines, 1993. *Dialogic Semiosis: An Essay on Signs and Meaning*, Bloomington, IN: Indiana University Press.

Johnson, Alexander Bryan, 1959/1836. *A Treatise on Language*, D. Rynin (ed.), Berkeley: University of California Press. (Contains the works *The Philosophy of Human Knowledge, or a Treatise on Language* (New York, 1828), *A Treatise on Language: or The Relation Which Words Bear to Things* (New York, 1836), *The Meaning of Words: Analyzed into Words and Unverbal Things* (New York, 1854).)

Johnson, Paul E., 1998. *Social Choice: Theory and Research*, Thousand Oaks, CA: Sage University Press.

Johnson-Laird, Peter, 1981. Mutual ignorance: comments on Clark and Carlson's paper, in N. V. Smith (ed.), *Mutual Knowledge*, London: Academic Press, 40–45.

Johnson-Laird, Peter, 1983, *Mental Models: Towards a Cognitive Science of Language, Inference, and Consciousness*, Cambridge: Cambridge University Press.

Jónsson, Bjarni and Tarski, Alfred, 1951. Boolean algebras with operators, *American Journal of Mathematics* 73, 891–939.

Kadmon, Nirit, 2001. *Formal Pragmatics: Semantics, Pragmatics, Presupposition, and Focus*, Oxford: Blackwell.

Kalmár, László, 1928–9. Zur Theorie der abstrakten Spiele, *Acta Scientiarum Mathematicarum (Szeged)* 4, 65–85. (Translation: On the theory of abstract games, in M. A. Dimand, and R. W. Dimand, (eds), 1997. *The Foundations of Game Theory* 1, Cheltenham: Edward Elgar, 247–262.)

Kamp, Hans, 1981. A theory of truth and semantic representation, in J. Groenendijk et al., (eds), *Formal Methods in the Study of Language*, Amsterdam: Mathematical Centre, 475–484.

Kamp, Hans and Reyle, Uwe, 1993. *From Discourse to Logic. Introduction to Modeltheoretic Semantics of Natural Language, Formal Logic and Discourse Representation Theory*, Dordrecht: Kluwer.

Kanger, Stig, 1957a. The morning star paradox, *Theoria* 23, 1–11.

Kanger, Stig, 1957b. *Provability in Logic, Stockholm Studies in Philosophy* 1.

Kant, Immanuel, 1988a. *Prolegomena to Any Future Metaphysics* (Originally published in 1783, a revision of the Carus translation with an introduction by Lewis White Beck), New York: Macmillan.

Kant, Immanuel, 1988b. *Logic*, (Originally published by Friedrich Nicolovius, 1800, under the title *Immanuel Kants Logik: Ein Handbuch zu Vorlesungen*, Köningsberg. Translated, with an introduction, by Robert S. Hartman and Wolfgang Schwarz), New York: Dover.

Kelly, George A., 1955. *The Psychology of Personal Constructs*, New York: Norton.

Kent, Beverley, 1987. *Charles S. Peirce: Logic and the Classification of the Sciences*, Kingston & Montreal: McGill-Queen's University Press.

Ketner, Kenneth L., 1998. *His Glassy Essence: An Autobiography of Charles Sanders Peirce*, Nashville: Vanderbilt University Press.

Kiikeri, Mika, 1997. On the logical structure of learning models, in M. Sintonen (ed.), *Knowledge and Inquiry: Essays on Jaakko Hintikka's Epistemology and Philosophy of Science*, Amsterdam: Rodopi, 287–308.

Kilpinen, Erkki, 2000. *The Enormous Fly-wheel of Society: Pragmatism's Habitual Conception of Action and Social Theory*, dissertation, University of Helsinki.

Kim, K. H., and Roush, F. W., 1987. *Team Theory*, New York: Ellis Horwood.

Kirby, Simon, 1999. Syntax out of learning: the cultural evolution of structured communication in a population of induction algorithms, in D. Floreano, J.–D. Nicoud and F. Mondada (eds), *Advances in Artificial Life*, Springer, 694–703.

Kirby, Simon, 2000. Syntax without natural selection: how compositionality emerges from vocabulary in a population of learners, in C. Knight, M. Studdert-Kennedy and J. Hurford (eds), *The Evolutionary Emergence of Language: Social Function and the Origins of Linguistic Form*, Cambridge: Cambridge University Press, 303–323.

Klir, George J., 2001. *Facets of System Science* (2nd edition), Dordrecht: Kluwer.

Knight, Chris, Studdert-Kennedy, Michael and James Hurford (eds), 2000. *The Evolutionary Emergence of Language: Social Function and the Origins of Linguistic Form*, Cambridge: Cambridge University Press.

Knuuttila, Simo, 1993. *Modalities in Medieval Philosophy*, New York: Routledge.

Komarova, Natalia and Niyogi, Partha, 2004. Optimizing the mutual intelligibility of linguistic agents in a shared world, *Artificial Intelligence* 154, 1–42.

König, Dénes, 1927. Über eine Schlußweise aus dem Endlichen ins Unendliche (On a consequence of passing from the finite to the infinite), *Acta Scientiarum Mathematicarum (Szeged)* 3, 121–130.

Koller, Daphne, 1995. A game-theoretic classification of interactive complexity classes, *Proceedings of the 10th Annual IEEE Conference on Structure in Complexity Theory*, 227–237.

Koller, Daphne and Megiddo, Nimrod, 1992. The complexity of two-person zero-sum games in extensive form, *Games and Economic Behavior* 4, 528–552.

Krabbe, Erik C. W., 2001. Dialogue foundations II: dialogue logic revisited, *Proceedings of the Aristotelian Society, supplementary volume* 75, 33–49.

Kraus, Sarit, 2001. *Strategic Negotiation in Multiagent Environments*, Cambridge, Mass.: MIT Press.

Kratzer, Angelica, 1998. Scope or pseudoscope? Are there wide-scope indefinites?, in S. Rothstein (ed.), *Events in Grammar*, Dordrecht: Kluwer, 163–196.

Kreps, David M. and Wilson, Robert B., 1982. Sequential equilibria, *Econometrica* 50, 863–894.

Kretzmann, Norman, 1967. History of semantics, in P. Edwards (ed.), *The Encyclopedia of Philosophy*, London: MacMillan, 48–406.

Kretzmann, Norman and Stump, Eleonore, 1988. *The Cambridge Translations of Medieval Philosophical Texts* 1, Melbourne: Cambridge University Press.

Kripke, Saul, 1963. Semantical considerations on modal logic, *Acta Philosophica Fennica* 16, Societas Philosophica Fennica, 83–94.

Kripke, Saul, 1976. Is there a problem about substitutional quantification?, in G. Evans and J. McDowell (eds), *Truth and Meaning*, Oxford: Clarendon Press, 325–419.

Krynicki, Michail, 1993. Hierarchies of partially ordered connectives and quantifiers, *Mathematical Logic Quarterly* 39, 287–294.

Krynicki, Michail and Mostowski, Marcin, 1995. Henkin quantifiers, in M. Krynicki, M. Mostowski and L. W. Szczerba (eds), *Quantifiers: Logics, Models and Computation* 1, Dordrecht: Kluwer, 193–262.

Kuhn, Harold W., 1953. Extensive games and the problem of information, in H. W. Kuhn and A. W. Tucker (eds), *Contributions to the Theory of Games* 2, Princeton: Princeton University Press, 193–216.

Langacker, Ronald W., 2000. *Grammar and Conceptualization*, Berlin: Mouton.

Langholm, Tore, 1996. How different is partial logic?, in P. Doherty (ed.), *Partiality, Modality, and Nonmonotonicity*, Stanford: CSLI, 3–43.

Lehmann, Daniel, 1997. What is qualitative? A framework for quantitative and qualitative decision theory, in J. Doyle and R. H. Thomason (eds), *Working papers of the AAAI Spring Symposium on Qualitative Preferences in Deliberation and Practical Reasoning*, Menlo Park, CA: AAAI Press, 65–70.

Leibniz, Gottfried W., 1981. *New Essays on Human Understanding* (Translated and edited by P. Remnant and J. Bennett), Cambridge: Cambridge University Press.

Leonard, Robert J., 1995, From parlor games to social science: von Neumann, Morgenstern, and the creation of game theory 1928–1944, *Journal of Economic Literature* 23, 730–761.

Levinson, Stephen C., 2002. *Presumptive Meanings: The Theory of Generalized Conversational Implicature*, Cambridge, Mass.: MIT Press.

Lewis, David, 1969. *Convention: A Philosophical Study*, Cambridge, Mass.: Harvard University Press.

Lewis, David, 1979. Scorekeeping in a language game, *Journal of Philosophical Logic* 8, 339–359.

Lin, Fanghzen and Shoham, Yoav, 1989. Argument systems — A uniform basis for nonmonotonic reasoning, in *Proceedings of the First International Conference on Principles of Knowledge Representation and Reasoning*, 245–255.

Lindström, Sten, 2001. Quine's interpretation problem and the early development of possible worlds semantics, in E. Carlson and R. Sliwinski (eds), *Omnium-gatherum. Philosophical Essays Dedicated to Jan Österberg on the Occasion of His Sixtieth Birthday*, Uppsala: Uppsala Philosophical Studies.

Liu, Feng-Hsi, 1997. *Scope and Specificity*, Amsterdam: John Benjamins.

Lorenz, Kuno, 1961. *Arithmetic und Logic als Spiele*, dissertation, Universität Kiel. (Partially reprinted in Lorenzen & Lorenz 1978).

Lorenz, Kuno, 2001. Basic objectives of dialogue logic in historical perspective, *Synthese* 127, 255–263.

Lorenzen, Paul, 1955. Einführung in die operative Logik und Mathematik, *Die Grundlehren der mathematischen wissenschaften* 78, Berlin: Springer.

Lorenzen, Paul and Lorenz, Kuno, 1978. *Dialogische Logic*, Darmstadt: Wissenschaftliche Buchgesellschaft.

Luce, R. Duncan and Raiffa, Howard, 1957. *Games and Decisions*, New York: John Wiley.

MacColl, Hugh, 1886–7. The calculus of equivalent statements (fifth paper), *Proceedings of the London Mathematical Society* 28, 156–183.

Malcolm, Norman, 1958, *Ludwig Wittgenstein: a Memoir*, London: Oxford University Press.

Malinowski, Bronislaw, 1935. *The Language of Magic and Gardening*, London: Allen & Unwin.

Mann, William C., 1988. Dialogue games: conventions of human interaction, *Argumentation* 2, 511–532.

Mannoury, Gerrit, 1947. *Les Fondements Psycho-Linguistiques des Mathématiques*, Paris: Éditions du Griffon Neuchatel.

Marshack, Jacob and Radner, Roy, 1972. *Economic Theory of Teams*, New Haven, Conn.: Yale University Press.

Maudet, Nicolas and Moore, David, 2001. Dialogue games as dialogue models for interacting with, and via, computers, *Informal Logic* 21, 219–243.

Maynard Smith, John, 1979. Hypercycles and the origin of life, *Nature* 280, 445–446.

Maynard Smith, John, 1982. *Evolution and the Theory of Games*, Cambridge: Cambridge University Press.

Maynard Smith, John and Price, G., 1973. The logic of animal conflict, *Nature* 246, 15–18.

McCarthy, Jeremiah, 1990. An account on Peirce's Proof of Pragmatism, *Transactions of the C.S. Peirce Society* 26, 63–113.

McKinsey, J. C. C., 1945. On the syntactical construction of systems of modal logic, *Journal of Symbolic Logic* 10, 83–94.

McKinsey, J. C. C., 1954. *Introduction to the Theory of Games*, New York: McGraw-Hill.

McKinsey, J. and Tarski, Alfred, 1944. The algebra of topology, *Annals of Mathematics* 45, 141–191.

McLarty, Colin, 1995. *Elementary Categories, Elementary Toposes*, Oxford: Clarendon Press.

Menger, Karl, 1937. The new logic, *Philosophy of Science* 4, 299–336. (English translation of the German original of 1933.)

Menger, Karl, 1994. *Reminiscences of the Vienna Circle and the Mathematical Colloquium*, Dordrecht: Kluwer.

Meredith, Carew A., 1956. Interpretations of different modal logics in the 'property calculus', mimeographed manuscript, Canterbury University College (with attribution 'C. A. M., August 1956; recorded and expanded A. N. P.'). Reprinted as Meredith, C. A. and Prior, A. N., Interpretations of Different Modal Logics in the 'Property Calculus', in B. J. Copeland, (ed.), 1996. *Logic and Reality: Essays on the Legacy of Arthur Prior*, Oxford: Clarendon Press, 133–34.

Mesarovic, M.D. and Takahara, Y., 1975. *General Systems Theory: Mathematical Foundations*, New York: Academic Press.

Miller, Paul, 2001. Interoperability: what is it and why should I want it?, *Ariadne* 24, 2001, www.ariadne.ac.uk/issue24/interoperability/intro.html (accessed 31 December 2004).

Milner, David A. and Goodale, Melvyn A., 1995. Visual pathways to perception and action, in T. P. Hicks, S. Molotchnikoff and Y. Ono, (eds), *Progress in Brain Research* 95, 317–337.

Mineau, Guy W., 2002. A first step toward the Knowledge Web: interoperability issues among conceptual graph based software agents, part I, in U. Priss, D. Corbett and G. Angelova (eds), *Conceptual Structures: Integration and Interfaces*, *Lecture Notes in Computer Science* 2393, Springer, 250–260.

Mitchell, Oscar Howard, 1982. On a new algebra of logic, in SIL: 72–106.

Mittelstaedt, Peter, 1978. *Quantum Logic*, Dordrecht: Reidel.

Montague, Richard and Kalish, Donald, 1959. That, *Philosophical Studies* 10, 54–61.

de Moor, Aldo, Keeler, Mary and Richmond, Gary, 2002. Towards a Pragmatic Web, in U. Priss, D. Corbett and G. Angelova (eds), *Conceptual Structures: Integration and Interfaces*, *Lecture Notes in Computer Science* 2393, Springer, 235–249.

Morgan, Charles G., 1979. Modality, analogy, and ideal experiments according to C. S. Peirce, *Synthese* 41, 65–83.

Morris, Charles W., 1946. *Signs, Language and Behavior*, New York: Prentice Hall.

Morris, S., 1994. Revising knowledge: a hierarchial approach, in R. Fagin (ed.), *Proceedings of the 5th Conference on Theoretical Aspects of Reasoning about Knowledge*, Morgan Kaufmann, 160–174.

Muggleton, Stephen, 1992. *Inductive Logic Programming*, London: Academic Press.

Murphey, Murray G., 1961. *The Development of Peirce's Philosophy*, Cambridge, Mass.: Harvard University Press.

Muskens, Reinhard, van Benthem, Johan and Visser, Albert, 1997. Dynamics, in J. van Benthem and A. ter Meulen (eds), *Handbook of Logic and Language*, Amsterdam: Elsevier, 587–648.

von Neumann, John, 1928. Zur Theorie der Gesellschaftsspiele, *Mathematische Annalen* 100, 295–320. (Translation by S. Bargmann: On the theory of games of strategy, in A. W. Tucker, and R. D. Luce (eds), *Contributions to the Theory of Games* 4, Princeton: Princeton University Press, 1959, 13–42.)

von Neumann, John, 1953. Communication on the Borel notes, *Econometrica* 21, 124–125.

von Neumann, John and Morgenstern, Oskar, 1944. *Theory of Games and Economic Behavior*, New York: John Wiley.

de Nivelle, Hans, 1998. Resolution games and non-liftable resolution orderings, in L. Pacholski, and J. Tiuryn (eds), *Computer Science Logic*, Berlin: Springer, 279–293.

Nowak, M. A., Plotkin, J. B. and Krakauer, D. C., 1999. The evolutionary language game, *Journal of Theoretical Biology* 200, 147–162.

Nowak, M. A., Plotkin, J. B. and Jansen, V. A. A., 2000. The evolution of syntactic communication, *Nature* 404, 495–498.

Nowak, M. A., Komarova, N. and Niyogi, P., 2001. Evolution of universal grammar, *Science* 291, 114–118.

Ogden, Charles Kay and Richards, Ivor Amstrong, 1923. *The Meaning of Meaning: A Study of the Influence of Language upon Thought and of the Science of Symbolism* (with an introduction by J. P. Postgate, and supplementary essays by B. Malinowski and F. G. Crookshank), London: Kegan Paul, Trench, Trubner.

Øhrstrøm, Peter, 1997. C. S. Peirce and the quest for gamma graphs, in *Conceptual Structures: Fulfilling Peirce's Dream*, *Lecture Notes in Artificial Intelligence* 1257, Berlin: Springer, 357–370.

Oliphant, Michael, 1994. The dilemma of saussurean communication, *Biosystems* 37, 31–38.

Oliphant, Michael and Batali, John, 1996. Learning and the emergence of coordinated communication, *Center for Research on Language Newsletter* 11(1).

Otteson, James R., 2002. Adam Smith's first market: the development of language, *History of Philosophy Quarterly* 19, 65–86.

Pape, Helmut, 1999. Abduction and the topology of human cognition, *Transactions of the Charles S. Peirce Society* 35, 248–269.

Parikh, Prashant, 2001. *The Use of Language*, Stanford: CSLI.

Parker, Kelly A., 1998. *The Continuity of Peirce's Thought*, Nashville: Vanderbilt University Press.

Parmentier, Richard J., 1994. *Signs in Society: Studies in Semiotic Anthropology*, Bloomington: Indiana University Press.

Pauly, Mark, 2001. *Logic for social software*, dissertation, University of Amsterdam.

Piccione, Michael and Rubinstein, Ariel, 1997. On the interpretation of decision problems with imperfect recall, *Games and Economic Behavior* 20, 3–24.

Pietarinen, Ahti-Veikko, 2000. Logic and coherence in the light of competitive games, *Logique et Analyse* 171–172, 371–391.

Pietarinen, Ahti-Veikko, 2001a. Intentional identity revisited, *Nordic Journal of Philosophical Logic* 6, 144–188.

Pietarinen, Ahti-Veikko, 2001b. Most even budged yet: some cases for game-theoretic semantics in natural language, *Theoretical Linguistics* 27, 20–54.

Pietarinen, Ahti-Veikko, 2001c. Propositional logic of imperfect information: foundations and applications, *Notre Dame Journal of Formal Logic* 42, 193–210.

Pietarinen, Ahti-Veikko, 2001d. Varieties of IFing, in M. Pauly and G. Sandu (eds), *Proceedings of the ESSLLI 2001 Workshop on Logic and Games*, University of Helsinki.

Pietarinen, Ahti-Veikko, 2001e. What is a negative polarity item?, *Linguistic Analysis* 31, 235–269.

Pietarinen, Ahti-Veikko, 2002a. Quantum logic and quantum theory in a game-theoretic perspective, *Open Systems & Information Dynamics* 9, 273–290.

Pietarinen, Ahti-Veikko, 2002b. Games and logics of knowledge for multi-agent systems, in C. A. Coello Coello, A. de Albornoz, L. E. Sucar and O. C. Battistutti (eds), *Advances in Artificial Intelligence, Lecture Notes in Artificial Intelligence* 2313, Springer, 214–223.

Pietarinen, Ahti-Veikko, 2002c. Knowledge constructions for artificial intelligence, in M.-S. Hacid, Z. W. Ras, D. A. Zighed and Y. Kodratoff (eds), *Foundations of Intelligent Systems, Lecture Notes in Artificial Intelligence* 2366, Springer, 303–311.

Pietarinen, Ahti-Veikko, 2002d. Negotiation games and conflict resolution in logical semantics, in P. Bouquet (ed.), *Meaning Negotiation: Papers from the AAAI Workshop (MeaN-02)*, Technical Report WS-02-09, AAAI Press, 25–31.

Pietarinen, Ahti-Veikko, 2002e. Some useful social metaphors in logic, in H. R. Arabnia and Y. Mun (eds), *Proceedings of the International Conference on AI* 3, CSREA Press, 1278–1282.

Pietarinen, Ahti-Veikko, 2003a. Peirce's game-theoretic ideas in logic, *Semiotica* 144, 33–47.

Pietarinen, Ahti-Veikko, 2003b. A note on the structural notion of information in extensive games, *Quality & Quantity* 37, 91–98.

Pietarinen, Ahti-Veikko, 2003c. Peirce's theory of communication and its contemporary relevance, in K. Nyíri (ed.), *Mobile Learning: Essays on Philosophy, Psychology and Education*, Vienna: Passagen Verlag, 81–98.

Pietarinen, Ahti-Veikko, 2003d. What do epistemic logic and cognitive science have to do with each other?, *Cognitive Systems Research* 4, 169–190.

Pietarinen, Ahti-Veikko, 2003e. Games as formal tools versus games as explanations in logic and science, *Foundations of Science* 8, 317–364.

Pietarinen, Ahti-Veikko, 2003f. The semantic + pragmatic web = the semiotic web, *Proceedings of the International IADIS/WWW Conference*, IADIS Press, 981–984.

Pietarinen, Ahti-Veikko, 2003g. Wienin piiri, kieli ja signifistiikka (Vienna Circle, language, and significs), *Ajatus* 60, Yearbook for Societas Philosophica Fennica, 327–353.

Pietarinen, Ahti-Veikko, 2003h. Logic, language games and ludics, *Acta Analytica* 18, 89–123.

Pietarinen, Ahti-Veikko, 2004a. Towards cognitive informatics: Awareness in the brain, logic and cognitive neuroscience, in Y. Wang (ed.), *Cognitive Informatics: Exploring the Natural Intelligence*, World Scientific.

Pietarinen, Ahti-Veikko, 2004b. Diagrammatic logic and game-playing, in G. Malcolm (ed.), *Multidisciplinary Approaches to Visual Representations and Interpretations*, Amsterdam: Elsevier, 115–133.

Pietarinen, Ahti-Veikko, 2004c. Semantic games in logic and epistemology, in S. Rahman, D. Gabbay, J. P. Van Bendegem and J. Symons (eds), *Logic, Epistemology and the Unity of Science*, Dordrecht: Kluwer, 57–103.

Pietarinen, Ahti-Veikko, 2004d. Multi-agent systems and game theory — a Peircean manifesto, *International Journal of General Systems* 33, 294–314.

Pietarinen, Ahti-Veikko, 2004e. Grice in the wake of Peirce, *Pragmatics & Cognition* 12, 395–415.

Pietarinen, Ahti-Veikko, 2004f. Logic, phenomenology and neuroscience in cahoots?, in G. Büchel, B. Klein and T. Roth-Berghofer (eds), *Proceedings of the 1st International Workshop on Philosophy and Informatics, CEUR Workshop Proceedings* 112, Technical University of Aachen (RWTH).

Pietarinen, Ahti-Veikko, 2004g. Peirce's diagrammatic logic in IF perspective, in A. Blackwell, K. Marriott and A. Shimojima (eds), *Diagrammatic Representation and Inference: Third International Conference, Diagrams 2004, Cambridge, UK, Lecture Notes in Artificial Intelligence* 2980, Berlin: Springer-Verlag, 97–111.

Pietarinen, Ahti-Veikko, 2005a. Compositionality, relevance and Peirce's logic of existential graphs, *Axiomathes*.

Pietarinen, Ahti-Veikko, 2005b. Peirce's magic lantern: moving pictures of thought, *Transactions of the Charles S. Peirce Society: A Quarterly Journal in American Philosophy*.

Pietarinen, Ahti-Veikko, 2005c. Early cognitive science: a challenge to analytic philosophy?, in H. J. Koskinen, S. Pihlström and R. Vilkko (eds), *Science – A Challenge to Philosophy*.

Pietarinen, Ahti-Veikko, 2005d. Evolutionary game-theoretic semantics and its foundational status, in N. Gontier, J. P. Van Bendegem and D. Aerts (eds), *Evolutionary Epistemology, Language and Culture: A Nonadaptationist Systems-theoretical Approach*, Dordrecht: Springer.

Pietarinen, Ahti-Veikko, 2005e. The composition of concepts and Peirce's pragmatic logic, in E. Machery, M. Werning and G. Schurz (eds), 2005. *The Compositionality of Concepts and Meanings: Foundational Issues*, Ontos-Verlag.

Pietarinen, Ahti-Veikko, 2005f. Some games logic plays, in D. Vanderveken (ed.), *Logic, Thought and Action*, Dordrecht: Springer, 409–431.

Pietarinen, Ahti-Veikko, 2005g. Independence-friendly logic and incomplete information, in J. van Benthem et al. (eds), *The Age of Alternative Logics: Assessing Philosophy of Logic and Mathematics Today*, Dordrecht: Springer.

Pietarinen, Ahti-Veikko, 2005h. Reversed interactive epistemology, to appear.

Pietarinen, Ahti-Veikko, 2005i. Interdisciplinarity and Peirce's classification of the sciences, to appear.

Pietarinen, Ahti-Veikko and Sandu, Gabriel, 1999. Games in philosophical logic, *Nordic Journal of Philosophical Logic* 4, 143–173.

Pietarinen, Ahti-Veikko and Sandu, Gabriel, 2004. IF logic, game-theoretical semantics and philosophy of science, in S. Rahman, D. Gabbay, J. P. Van Bendegem and J. Symons (eds), *Logic, Epistemology and the Unity of Science*, Dordrecht: Kluwer, 105–138.

Pietarinen, Ahti-Veikko and Snellman, Lauri, 2005. "Preface" (MS 654), "Of Reasoning in General" (MS 849), "Reflexions upon Reasoning" (MS 686) (in Finnish), *niin & näin*, 2/2005, Tampere University Press.

Pietarinen, Ahti-Veikko and Tulenheimo, Tero, 2004. *An Introduction to IF Logic*, ESSLLI 2004 Lecture Notes, The 16th European Summer School for Logic, Language and Information, University of Henri Poincaré, Nancy.

Plotkin, Henry, 1994. *The Nature of Knowledge: Concerning Adaptations, Instinct and the Evolution of Intelligence*, London: The Penguin Press.

Putnam, Hilary, 1995. Peirce's continuum, in K. Laine Ketner (ed.), *Peirce and Contemporary Thought*, New York: Fordham University Press, 1–22.

Quine, Willard Van Orman, 1953. *From a Logical Point of View*, Cambridge, Mass.: Harvard University Press.

Rabin, Michael O., 1957. Effective computability of winning strategies, in M. Dresher, A. W. Tucker and P. Wolfe (eds), *Contributions to the Theory of Games* 3, Princeton: Princeton University Press, 147–157.

Radner, Roy, 1962. Team decision problems, *Annals of Mathematical Statistics* 33, 857–881.

Rahman, Shahid, 2002. *The Logic of Connexive Statements in the Early Work of Hugh MacColl*, to appear.

Rahman, Shahid and Rückert, Helge, 2001. Dialogical connexive logic, *Synthese* 127, 105–139.

Ramsey, Frank P., 1991. *On Truth: Original Manuscript Materials (1927–1929) from the Ramsey Collction at the University of Pittsburgh* (ed. by N. Resher and U. Majer), Dordrecht: Kluwer.

Ransdell, Joseph, 1977. Some leading ideas of Peirce's semiotic, *Semiotica* 19, 157–178.

Rantala, Veikko, 1975. Urn models: a new kind of non-standard model for first-order logic, *Journal of Philosophical Logic* 4, 455–474.

Rantala, Veikko, 1982. Impossible worlds and logical omniscience, *Acta Philosophica Fennica* 35, Societas Philosophica Fennica, 106–115.

Rasmusen, Eric, 1994. *Games and Information* (first edition 1989), Cambridge, Mass.: Blackwell.

Rescher, Nicholas, 1978a. *Scientific Progress: A Philosophical Essay on the Economics of Research in Natural Science*, Oxford: Blackwell.

Rescher, Nicholas, 1978b. *Peirce's Philosophy of Science: Critical Studies in His Theory of Induction and Scientific Method*, Notre Dame: University of Notre Dame Press.

Rescher, Nicholas, 1989. *Cognitive Economy: The Economic Dimension of the Theory of Knowledge*, Pittsburgh: University of Pittsburgh Press.

Reid, Thomas, 1967/1895. *Philosophical Works II, with notes and supplementary dissertations by Sir William Hamilton*, Hildesheim: Georg Olms Verlagsbuchhandlung.

Rhees, Rush, 2002, Five topics in conversation with Wittgenstein (*Numbers; concept-formation; time-reactions; induction; causality*), *Philosophical Investigations* 25, 1–19.

de Rijke, Maarten, 1995. The logic of Peirce algebras, *Journal of Logic, Language and Information* 4, 227–250.

Roberts, Don D., 1973a. *The Existential Graphs of Charles S. Peirce*, The Hague: Mouton.

Roberts, Don D., 1973b. An introduction to Peirce's proof of pragmatism, *Transactions of the Charles S. Peirce Society* 14, 128–134.

Robin, Richard S., 1997. Classical Pragmatism and Pragmatism's Proof, in J. Brunning and P. Forster (eds), *The Rule of Reason: The Philosophy of Charles Sanders Peirce*, Toronto: University of Toronto Press. 13–152.

Rosenschein, J. S. and Zlotkin, G., 1994. *Rules of Encounter: Designing Conventions for Automated Negotiation among Computers*, Cambridge, Mass.: MIT Press.

Rossell, Deac, 1998. *Living Pictures: The Origins of the Movies*, Albany: State University of New York Press.

Rubinstein, Ariel, 1998. *Modeling Bounded Rationality*, Cambridge, Mass.: MIT Press.

Rubinstein, Ariel, 2000. *Economics and Language: Five Essays*, Cambridge: Cambridge University Press.

Russell, Bertrand, 1946. Foreword, in Feibleman, James, 1946. *An Introduction to Peirce's Philosophy Interpreted as a System*, New York: Harper.

Saarinen, Esa, (ed.), 1979. *Game-Theoretical Semantics: Essays on Semantics by Hintikka, Carlson, Peacocke, Rantala, and Saarinen*. Dordrecht: D. Reidel.

Samuelson, Larry, 1997. *Evolutionary Games and Equilibrium Selection*, Cambridge, Mass.: MIT Press.

Sandholm, Tuomas, 1999. Distributed rational decision making, in G. Weiss, G., (ed.) *Multiagent Systems*, Cambridge, Mass., MIT Press, 201–258.

Sandu, Gabriel, 1993. On the logic of informational independence and its applications, *Journal of Philosophical Logic* 22, 29–60.

Sandu, Gabriel, 1997. On the theory of anaphora: dynamic predicate logic vs. game-theoretical semantics, *Linguistics and Philosophy* 20, 147–174.

Sandu, Gabriel and Hintikka, Jaakko, 2001. Aspects of compositionality, *Journal of Logic, Language and Information* 10, 49–61.

Sandu, Gabriel and Pietarinen, Ahti-Veikko, 2001. Partiality and games: propositional logic, *Logic Journal of the Interest Group of Pure and Applied Logic* 9, 107–127.

Sandu, Gabriel and Pietarinen Ahti-Veikko, 2003. Informationally independent connectives, in G. Mints and R. Muskens, (eds), *Games, Logic, and Constructive Sets*, Stanford: CSLI, 23–41.

Savage, Leonard J., 1972. *The Foundations of Statistics* (first edition 1954), New York: Dover.

Schelling, Thomas C., 1960. *The Strategy of Conflict*, Cambridge: Harvard University Press.

Schiffer, Stephen, 1972. *Meaning*, Oxford: Clarendon Press.

Schmitz, H. Walter, 1984. Searle ist in Mode, Mannoury nicht: Sprech- und Hörakt im niederländischen Signifik-Kreis, *Zeitschrift für Semiotik* 6, 445–463.

Schmitz, H. Walter, 1990. *Essays on Significs: Papers Presented on the Occasion of the 150th Anniversary of the Birth of Victoria Lady Welby (1837–1912)*, Amsterdam: John Benjamins.

Schotter, Andrew, (ed.) 1976. *Selected Economic Writings of Oskar Morgenstern*, New York: New York University Press.

Schwalbe, U. and Walker, P., 2001. Zermelo and the early history of game theory, *Games and Economic Behaviour* 34, 123–137.

Scott, Dana, 1993. A game-theoretical interpretation of logical formulae (original manuscript 1968). *Jahrbuch 1991 der Kurt-Goedel-Gesellschaft*, 47–48, Wien: Kurt-Goedel-Gesellschaft.

Selten, Richard, 1983. Evolutionary stability in extensive two-person games, *Mathematical Social Sciences* 5, 269–363.

The Semantic Web Agreement Group, 2001. What is the Semantic Web?, swag.webns.net/whatIsSW (accessed 31 December 2004).

Seuren, Pieter A. M., 2000. Presupposition, negation and trivalence, *Journal of Linguistics* 36, 261–297.

Shapiro, Michael and Haley, Michael C., (eds), 1999. *The Peirce Seminar Papers: Essays in Semiotic Analysis* 4, Berghahn Books.

Shin, Sun-Joo, 2002. *The Iconic Logic of Peirce's Graphs*, Cambridge, Mass.: MIT Press.

Short, T. L., 1999. Teleology and linguistic change, in M. Shapiro (ed.), *The Peirce Seminar Papers: Essays in Semiotic Analysis* 4, New York: Berghahn Books.

Shubik, Martin, 1985. *Game Theory in the Social Sciences*, Cambridge, Mass.: MIT Press.

Shubik, Martin, 1992. Game theory at Princeton, 1949–1955: a personal reminiscence, in E. R. Weintraub, (ed.), *Toward a History of Game Theory*, Durham: Duke University Press, 151–163.

Skolem, Thoralf, 1920. Logico-combinatorial investigations in the satisfiability or provability of mathematical propositions: a simplified proof of a theorem by L. Löwenheim and generalizations of the theorem, in J. van Heijenoort, (ed.), (1967). *From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931*, Cambridge, Mass.: Harvard University Press, 254–263. (Originally published in: Logisch-kombinatorische Untersuchungen über die Erfüllbarkeit und Beweisbarkeit mathematischen Sätze nebst einem Theoreme über dichte Mengen, *Skrifter utgit av Videnskabsselskapet i Kristiania* 1, Matematisk-naturvidenskabelig klasse 4, 1–36. Reprinted in Skolem, Thoralf, 1970. *Selected Works in Logic*, J. E. Fenstad (ed.), Oslo: Universitetsforlaget, 103–136.)

Skyrms, Brian, 1996. *Evolution of the Social Contract*, Cambridge: Cambridge University Press.

Smith, Adam, 1970/1767. *A Dissertation on the Origin of Languages, or Considerations Concerning the First Formation of Languages and the Different Genius of Original and Compounded Languages*, Tübingen: Gunter Narr, 507–538.

Smith, Adam, 1980/1795. *Essays on Philosophical Subjects, with Dugald Stewart's Account of Adam Smith*, A. P. D. Wightman and J. C. Bryce, (eds), Oxford: Clarendon Press.

Smith, Adam, 1991/1776. *The Wealth of Nations*, New York: Prometheus Books.

Smith, Barry, 1992, Characteristica universalis, in K. Mulligan, (ed.) *Language, Truth and Ontology*, Dordrecht: Kluwer, 48–77.

Sowa, John F., 1984. *Conceptual Structures: Information Processing in Mind and Machine*, Reading, Mass.: Addison-Wesley.

Sowa, John F., 2000. *Knowledge Representation: Logical, Philosophical, and Computational Foundations*, New York: Brooks/Cole.

Sowa, John F., 2001. Existential Graphs: MS 514 by Charles Sanders Peirce with commentary by John F. Sowa, users.bestweb.net/˜sowa/peirce/ms514.htm (accessed 31 December 2004).

Sperber, Dan and Wilson, Deidre, 1987. Précis of Relevance: Communication and Cognition, *Behavioural And Brain Sciences* 10, 697–754.

Sperber, Dan and Wilson, Deidre, 1995. *Relevance Theory: Communication and Cognition* (2nd ed.), Oxford: Blackwell.

Stalnaker, Robert, 1972. Pragmatics, in D. Davidson and G. Harman (ed.), *Semantics of Natural Language*, Boston: D. Reidel.

Stalnaker, Robert, 1974. Pragmatic presuppositions, in M. K. Munitz and P. K. Unger (eds), *Semantics and Philosophy*, New York: New York University Press.

Stalnaker, Robert, 1978. Assertion, *Syntax and Semantics* 9, 315–332.

Stalnaker, Robert, 1999. Extensive and strategic forms: games and models for games, *Research in Economics* 53, 293–319.

Stalnaker, Robert, 2002. Common ground, *Linguistics and Philosophy* 25, 701–921.

Steels, Luc, 1998. Synthesizing the origins of language and meaning using coevolution, self-organization and level formation, in J. R. Hurford, M. Studdert-Kennedy and C. Knight (eds), *Approached to the Evolution of Language: Social and Cognitive Bases*, Cambridge: Cambridge University Press, 384–404.

Steels, Luc and McIntyre, Angus, 1999. Spatially distributed naming games, in *Advances in Complex Systems* 1, 301–323.

Steinhaus, Hugo, 1960/1925. Definitions for a theory of games and pursuits, *Naval Research Logistic Quarterly* 7, 105–108. (Originally in *Mysl Akademicka* 1, 13–14, 1925, University of Lvov.)

Stenning, Keith, 2000. Distinctions with differences: comparing criteria for distinguishing diagrammatic from sentential systems, M. Anderson, et al., (eds), *Diagrams 2000, Lecture Notes in Artificial Intelligence* 1889, Springer, 132–148.

Street, Warren R., 1994. *A Chronology of Noteworthy Events in American Psychology*, Washington DC, American Psychological Association.

Strotz, Robert H. 1956. Myopia and inconsistency in dynamic utility maximization, *Review of Economic Studies* 23, 165–180.

Subrahmanian, V. S., Bonatti, Piero, Dix, Jürgen, Eiter, Thomas, Kraus, Sarit, Ozcan, Fatma and Ross, Robert, 2000. *Heterogeneous Agent Systems*, Cambridge, Mass.: MIT Press.

Sudgen, Robert, 1995. A theory of focal points, *The Economics Journal* 105, 533–550.

Suppe, Frederick, 1989. *The Semantic Conception of Theories and Scientific Realism*, Urbana, IL: University of Illinois Press.

Tennant, Neil, 1986. The withering away of formal semantics?, *Mind and Language* 1, 302–318.

Tennant, Neil, 1998. Games some people would have all of us play, Critical study/Book review of Hintikka (1996a), *Philosophia Mathematica* 6, 90–115.

Todd, Charles L. and Blackwood, Russell T., 1967. *Language and Value: Proceedings of the Centennial Conference on the Life and Works of Alexander Bryan Johnson*, New York: Greenwood.

Tuomela, Raimo, 2000. *Cooperation: A Philosophical Study*, Dordrecht: Kluwer.

Turner, Ken, 1999. *The Semantics/Pragmatics Interface from Different Points of View*, Oxford: Elsevier.

Ulam, Stanislaw M., 1958. John von Neumann, 1903–1957, *Bulletin of the American Mathematical Society* 64, 1–49.

Ulam, Stanislaw M., 1960. *A Collection of Mathematical Problems*, Groningen: Interscience Publishers.

Underwood, Geoffrey, 1996. *Implicit Cognition*, Oxford: Oxford University Press.

Ungerleider, L. and Mishkin, M., 1982. Two cortical visual systems, in D. J. Ingle, M. A. Goodale and R. J. W. Mansfield (eds), *Analysis of Motor Behavior*, Cambridge, Mass.: MIT Press, 549–586.

Vaina, Lucia, 1987. *Matters of Intelligence*, Dordrecht: D. Reidel.

Valiant, Leslie G., 1984. A theory of the learnable, *Communications of the ACM* 27, 1134–1142.

Van Essen, D. C. and Gallant, J. L., 1984. Neural mechanisms of form and motion processing in the primate visual system, *Neuron* 13, 1–10.

Väänänen, Jouko, 2001. Second-order logic and the foundations of mathematics, *Bulletin of Symbolic Logic* 7, 504–520.

Watts, Richard J., 2000. Introduction, *Journal of Historical Pragmatics* 1, 165–173.

Wegener, Philipp, 1885. *Unterschungen über die Grundfragen des Sprachlebens*, Halle: Max Niemeyer.

Weibull, Jörgen W., 1995. *Evolutionary Game Theory*, Cambridge, Mass.: MIT Press.

Weiss, Paul, 1934. Peirce, Charles Saunders, *Dictionary of American Biography* 14, 398–403.

Weiss, Paul and Burks, Arthur, 1945. Peirce's sixty-six division of signs, *The Journal of Philosophy* 42, 383–388.

Westerståhl, Dag, 2002. On the compositionality of idioms, in D. Barker-Plummer et al., (eds), *Words, Proofs and Diagrams*, Stanford: CSLI, 241–272.

Whitney, William D., 1971. *Whitney on Language: Selected Writings of William Dwight Whitney* (ed. by M. Silverstein with an introductory essay by R. Jakobson), Cambridge, Mass.: MIT Press.

Wible, James R., 1998. Charles Sanders Peirce, in *The Handbook of Economic Methodology*, J. B. Davis, D. W. Hands and U. Mäki (eds), 357–364, Cheltenham: Edward Elgar.

Wible, James R., 2000. Prediction, complexity, and pluralism: more on Rescher's post-postmodern economic philosophy of science, *Journal of Economic Methodology* 7, 294–301.

Wilson, Deirdre and Sperber, Dan, 1979. Ordered entailments: an alternative to presuppositional theories, in C. K. Oh and D. Dineen (eds.), *Syntax and Semantics* 2, New York: Academic Press.

Wittgenstein, Ludwig, 1953. *Philosophical Investigations*, (3rd ed. 1967), Oxford: Blackwell.

Wittgenstein, Ludwig, 1956. *Remarks on the Foundations of Mathematics*, Oxford: Blackwell.

Wittgenstein, Ludwig, 1969. *On Certainty*, New York: Harper Torchbooks.

Wittgenstein, Ludwig, 1970. *Zettel*, Berkeley and Los Angeles: University of California Press.

Wittgenstein, Ludwig, 1978. *Philosophical Grammar*, Columbia: University of California Press.

Wittgenstein, Ludwig, 1984. *Bemerkungen über die Grundlagen der Mathematik*, Frankfurt am Main: Suhrkamp.

Wittgenstein, Ludwig, 2000–. *Wittgenstein's Nachlass, The Bergen Electronic Edition*, The Wittgenstein Trustees, The University of Bergen, Oxford University Press.

von Wright, Georg Henrik, 1982, The Wittgenstein Papers, in G. H. von Wright, *Wittgenstein*, Oxford: Blackwell, 36–62.

Wooldridge, Michael, 2000. *Reasoning about Rational Agents*, Cambridge, Mass.: MIT Press.

Wray, Alice, (ed.), 2002. *The Transition to Language*, Oxford: Oxford University Press.

Yates, C. E. M., 1976, Banach-Mazur games, comeager sets, and degrees of unsolvability, *Mathematical Proceedings of Cambridge Philosophical Society* 79, 195–220.

Yrjönsuuri, Mikko, (ed.), 2001. *Medieval Formal Logic: Obligations, Insolubles and Consequences*, Dordrecht, Kluwer.

Zalamea, Fernando, 2003. Peirce's logic of continuity: existential graphs and non-cantorian continuum, *The Review of Modern Logic* 9, 115–162.

Zeman, Jay J., 1964. *The Graphical Logic of C.S. Peirce*, dissertation, University of Chicago. Online edition, 2002, web.clas.ufl.edu/users/jzeman/ (accessed 31 December 2004).

Zeman, Jay J., 1974. Peirce's logical graphs. *Semiotica* 12, 239–256.

Zeman, Jay J., 1986. Peirce's philosophy of logic, *Transactions of the Charles S. Peirce Society* 22, 1–22.

Zermelo, Ernst, 1913. Über eine Anwendung der Mengenlehre auf die Theorie des Schachspiels, 501–504 in E. W. Hobson and A. E. H. Love, (eds), *Proceedings of the Fifth International Congress of Mathematicians* 2, Cambridge: Cambridge University Press. (Translation: On an application of set theory to the theory of the game of chess, in Appendix to Schwalbe & Walker 2001.)

Chapter 1

AN INTRODUCTION TO PEIRCE'S LOGIC AND SEMEIOTICS

> CHARLES SANTIAGO SANDERS PEIRCE (b. 1839), son of the mathematician Benjamin P., brought up in a circle of physicists and naturalists, and specially educated as a chemist, derived his first introduction to philosophy from the K.d.R.V. [Immanuel Kant's Critique of Pure Reason, published in 1781] and other celebrated German works, and only later made acquaintance with English, Greek, and Scholastic philosophy. Accepting unreservedly Kant's opinion that the metaphysical conceptions are merely the logical conceptions differently applied, he inferred that logic ought to be studied in the spirit of the exact sciences, and regarded Kant's table of functions of judgment as culpably superficial. (MS L 107: 1, 26 October 1904, *Auto-Biography for Matthew Mattoon Curtis*, Draft C, marked "final" by Peirce).

1. Kant's influence and the logical roots of pragmatism

The last great effort In the autumn 1904, Peirce was 65 years old, in a state of ferment and excitement as he was drafting and re-drafting his newly-emerged logical ideas. He had started his "last great effort", despite the fact that he had just failed to raise funds for that purpose. He had applied for money for a major project of "Logic" two years earlier from the Carnegie Institute, but the application was turned down.[1]

These constantly flowing new ideas promised new ways of approaching the structure and nature of thought in logical reasoning and representation. He thought he was not far from having accomplished a comprehensive theory of logic through his detailed investigation of his recently discovered iconic, diagrammatic theory of existential graphs (EGs). "Since I am now sixty three years old and since all this is matter calculated to make a difference in man's future intellectual development", wrote Peirce in 1903, "I can only say that if the *genus homo* is so foolish as not to set me at the task, I shall lean back in my chair and take my ease. I have done a great work wholly without any kind of aid, and now I am willing to undergo the last great effort which must finish me up in order to give men the benefit of that which I have done. . . . I have reached the age when I think of my home as being on the other side rather than on this uninteresting planet".[2] EGs were among the endless number of other logical, philosophical and scientific discoveries that Peirce had already made and published — and perhaps more importantly, had left unpublished or were habitually rejected by the publishers and editors of various journals.

Leaving aside the details of this diagrammatic part of his work on logic and semeiotics for the moment (see Chapters 3–6), I will approach Peirce's philosophical ideas here by comparing them with some of the characteristic features of Kant's thought. It is undisputable that Peirce was inspired and affected, albeit by no means singularly persuaded, by Kant's thinking.

> He also thought that Kant's reply to the question how are synthetical judgments *a posteriori* possible was altogether insufficient, and that an exact inquiry into it would probably throw some light upon judgments *apriori*. (MS L 107: 1).

Peirce's own assessment of the similarities and differences between him and Kant was that Kant "set out from the obvious truth that cognition is the result of the interaction of two independent agents, the mind and the real object" (MS 280: 7).[3] He nevertheless thought that Kant was led astray by the further assumption that cognition, or "cognition through concepts", as Kant considered thinking (*cognition discursiva*)[4] to be, has two parts, matter determined by the object, and form determined by the mind.

The way in which Peirce used the word "mind" was different from Kantian usage. Peirce took it to refer to that entity which determines the forms of cognition. Kant's choice of words was that of "generality" (Kant, 1988b, p. 96). The difference is negligible. The two-part division of cognition follows only if we assume the criteria for it to be in Kant's categories of universality and necessity. Peirce asserts that Kant learned these from Leibniz' *Nouveaux Essais* (MS L 107). According to Leibniz, necessity is the capability of observing the determinability of man's actions by a perfect mind having complete knowledge:

> If by 'necessity' we understood a man's being inevitably determined, as could be foreseen by a perfect Mind provided with a complete knowledge of everything going on outside and inside that man, then, since thoughts are as determined as the movements which they represent, it is certain that every free act would be necessary; but we must distinguish what is necessary from what is contingent though determined. (Leibniz, 1981, Book II, *xxi*).

The "perfect generality" is capable of seeing what is necessary, and if universality in any proposition is, in turn, that which employs a general term for its subject, then generality becomes a variety of necessity. In this sense the two uses of the terms "generality" and the "mind" are indeed interchangeable.

The susceptibility (or "crudeness", see MS 280: 15) of Kant's assumption lies, according to Peirce, in the inseparability of the two factors of cognition that Kant does not take into account. For such inseparability is in conflict with the assumption that an element of cognition, be it matter or form, may result from one of the factors alone.

The salvation of these "blunders of those great minds" (MS 280: 18) that Peirce has in the offing lies in logical analysis, albeit one that is quite different from that which Leibniz or Kant envisioned. A proper analysis is to be carried out by means of a graphical, diagrammatic system of EGs. Peirce had begun developing these systems towards the end of 1896, although he claimed to have already been in possession of the essential ideas some fourteen years earlier. By 1906, the system was at a fairly advanced level, even though it had some imperfections, especially in relation to the gamma part, which among other things was set out to deal with modalities (Chapter 4).

Peirce was utterly exited about these inventions, and when finally a publication channel was opened to him in The Monist for the *Prolegomena* series, he started very actively and very strenuously to draft manuscripts that would not only encompass a thorough exposition of EGs, but would also tie them in with these grander philosophical aims.

The phrase "logical analysis", which Kant had used to clarify the nature of thought, was in Peirce's view just tolerably apt, not because of the difficulties associated with the notion of analysis, but because of the difficulties associated with logic, its unsettled and ever-expanding nature, and its dependence, on the whole, on the broad spectrum of underlying semiotic deliberations concerning scientific inquiry.

Peirce suggests analysing propositions in terms of their parts that are not propositions themselves, but blank forms of incomplete expression waiting to be filled in with proper names. Only after the process of such saturation the intended expressions acquire the status of propositions. He calls such blank forms rhemas or rhemata, the term in full swing in the system of EGs. According to him, natural language does not lend itself well to the analysis of the nature of rhemas, and so a special logical inquiry is called for.

What is a rhema? Peirce explained that "a rhema is an indispensable part of speech in every language. Every verb is a rhema".[5] A verb itself "is a fragment of a possible proposition having blanks which being filled with proper names make the verb a proposition".[6] In modern logic, it corresponds to an uninterpreted predicate term. Like predicates, functions and relations, rhemas have a valency, that is, a fixed number of argument places, ranging from a zero ("medad") valency to higher n-place valencies. A medad, which correlates with an individual constant, can, according to him, only be a mental proposition, and for that reason may be a judgement. It does not need to be asserted or mentally assented to, it suffices just to be comprehended. Monadic, dyadic and triadic rhemas are not judgements but proper constituents of assertions.

Rhemas are free forms of expression adjoined by blank lines into which names are plugged: "Each rhema is equivalent to a blank form such that if all its blanks are filled with proper names, it becomes a proposition, or symbol capable of assertion".[7] Peirce admitted that "logically their [rhema's and term's] meaning is the same", the only difference being that the rhema "contains no explicit recognition of its own fragmentary nature" (NEM 4:246). Being uninterpreted, yet potentially interpretable by their proper names, rhemas are neither true nor false, whereas the propositions that they comprise when filled with suitable proper names are typically either true or false.[8]

Proof of pragmaticism Rhemas were put into full use in the context of EGs. By the time Peirce had made several attempts to prove (by sound philosophical argumentation rather than strictly mathematical means) the truthfulness of pragmaticism, beginning in 1905, he decided to employ the logic of EGs as the chief ingredient in that proof. This was a considerable step forward towards the proof.

Many expositions concerning Peirce's attempted proof have underrated the role of diagrammatisation. Hookaway's discussion of the motivations for Peirce's desire to achieve that proof mentions EGs only as an ephemeral curiosity (Hookway, 2000, p. 285). In Robin (1997), the role of graphs is nonetheless recognised in terms of presenting the proof itself. Nathan Houser's introduction to Essential Peirce, volume 2, refers to EGs as those "which Peirce would later choose as the preferred medium for the presentation of his proof of pragmatism" (p. xxvii). Robin (1997, p. 142) notes that in the multi-layered draft *The First Part of an Apology for Pragmaticism*, Peirce recorded that "it will make no essential difference in my argument for the truth of Pragmaticism" (MS 296: 4, 1907–8). However, Peirce used the term "essential difference" not with reference to the question of whether to employ EGs or some other logical system in the exposition of the argument, but rather to that of whether to incorporate a new "feature" (MS 296: 4) into the theory of EGs. By this novel feature he meant study through which different varieties of modality could be exposed, something that he had begun to work with during the eighteen months that had elapsed since the *Prolegomena* was written.[9] Peirce lamented that, while he had applied EGs to a range of logical inquiries, as he admitted in the Prolegomena, they in fact lacked "the treatment of modality" (MS 296: 15 a.p.).

A moment's reflection on the role of EGs in argument aimed at the proof reveals that such graphs were not only employed as a convenient tool or medium within which to set out the proof. EGs are the exact and indispensable core of what is necessarily analytic in any logical tackle on pragmatic principles. Houser continues his introduction by noting that, by 1906, "Peirce had decided that it was by means of the Existential Graphs that he could most convincingly set out his proof, which was to follow in subsequent articles (although it is significantly previewed in this one). Peirce had decided to use his system of graphs for his proof for three principal reasons: it employed the fewest possible arbitrary conventions for representing propositions, its syntax was iconic, and it facilitated the most complete analysis" (pp. xxviii–xxix). It is certainly true that, by that date, Peirce thought he had finally settled the question of what the conventions of EGs were to be (the conventions for the beta part of the system, the one that extends the propositional alpha part with quantification and identities, Chapter 4), and so the number of conventions that permit one to soundly transform one EG into another was reduced to a minimum. It is also true that the iconicity of the system was of central importance for representing propositions; the term Peirce coined in his last years was "diagrammatic syntax".[10] The system of diagrammatic syntax, in fact, came to possess several features that are nowadays recognised as semantic. The key question is: How do EGs facilitate the "most complete" analysis of pragmaticism?[11]

The importance of rhemas is related to the idea of their role as the unanalysed, incomplete component of logic. In diagrammatic contexts, Peirce termed the their representations spots. More precisely, a spot is

> a graph any replica of which occupies a simple bounded portion of a surface, which portion has qualities distinguishing it from the replica of any other spot; and upon the boundary of this surface occupied by the spot are certain points, called the *hooks* of the spot, to each of which, if permitted, one extremity of one line of identity can be attached. Two lines of identity cannot be attached to the same hook; nor can both ends of the same line.[12]

Boundaries, hooks, the lines of identity and their extremities, are technical notions of EGs we can safely close our eyes to for the time being, as I will explain the special terminology in more detail in Chapter 4. Important to note is that spots are diagrammatic marks that are one of two kinds, rhemata or onomata. In contrast to rhemas, whose blanks are assumed to be filled with proper names, onomas admit of indefinites:

> Each onoma is an arbitrary index of an indefinite individual. A connecting line may abut upon it, and this has the effect of attaching the onoma, as a designation, to the individual which that line denotes. I usually write capital letters for onomata. . . . A spot has a definite place upon its periphery, called a *hook*, corresponding to each blank; and to each hook an extremity of a line of connection may be attached with the effect of filling the blank with a designation of the individual denoted by the line. (MS 491: 3–4).

The common aspect in both rhemas and onomas is that they are incomplete predicates, acquiring their values and thus growing into proper constituents of propositions through a semeiotic process in which either proper or indefinite names are selected.

This was to mark a great improvement in the ambivalent field of logical analysis. The profit in logical investigations of the nature of signs and their relation to minds, as well as of the status of pragmaticism, is unmistakable. Rhemas and onomas illustrate Peirce's "would-bes", preliminary, primordial concepts of what may or will happen when we enquire the world, or the inner thought, or the parts of the worlds or thoughts connected with our mutually agreed universes of discourse. Their role is, in that sense, similar to the transcendental arguments put forward by Kant, without being transcendental in the least.[13]

The entities to be stored in the blank spaces of rhemas and onomas are names of things, suggested by those who undertake the task of uttering and interpreting propositions. Chapters 2, 4 and 13 concern these communicative aspects of Peirce's logic. Let us note that the propositions submitted to interpretation are taken to have assertive force. Rhemas and onomas await to be saturated so as to give rise to proper concepts, that is, to give rise to interpreted predicates that make up propositions that can then be employed in assertions. Consequently, the propositions so formed are not simply true or false, but possess a potentiality of being of one of these truth-values. No wonder that Peirce was drawn into an agonising investigation of the connections between probabilities and modalities in the last couple of years of his life (Chapter 6). The mere possibility of an action, say, raising an arm, struck him as a very real thing. The fact that there exists a possibility warrants that there must be a reasonable sense in which possibilities are just as real as actual events.

The incomplete status of the proof of pragmaticism — or pragmatism as it was again termed, secured from the kidnappers by that time — is one of the characteristic results of the incomplete and unfinished business Peirce had with the concepts of probabilities and possibilities, especially with reference to their relation to the rhema–onoma interpretations in EGs.[14]

Which one is really more far-reaching and salient, the concept of rhemas that call for proper names in order to take part in the composition of an assertion and fit for use in arguments, or the indexical concept of onomas, the actual content of which is produced by introducing values for indefinite expressions? The above quotations suggest that, in fact, onomas were thought by Peirce later on to have succeeded in achieving a more prominent and consequential role in logical analysis, despite the fact that they were discovered and singled out as separate parts of the diagrammatic analysis of the logical content of thought at a much later date than rhemas.

This is an unmistakable mark of Peirce's increased recognition of the centrality of not only the indexical character of signs, but also the communicational core of logic. Names as instances or values assigned to indefinites is bound to remain one-sided as such. It is just one party (the utterer, the speaker, the assertor) who is handed the responsibility of interpreting indefinite expressions, and who is free to choose such names that the proposition in question is taken to assert within the limits of discourse. The value of that choice is, ceteris paribus and within the limits understood, not disclosed to the other party (the interpreter, the hearer, the critic).[15]

That only the utterer has names for onomas in mind is all the more salient when sentences with a silent or overt indefinite, *a certain*, are interpreted. Direct textual evidence shows that Peirce recognised both the centrality of the reciprocal nature of indefinites and the meaning of *a certain* as distinct from both universal and particular quantifications: "'A certain' followed by a noun has a force very different from 'any' and equally different from 'a definite'. It implies that the writer has in mind a *single individual* object of the character implied by the noun, and limits his predicates to that single one, although he does not tell the reader what one it is".[16]

The distinction between the two kinds of spots, diagrammatically speaking, was one of Peirce's remarkable late discoveries in logical analysis. Their central role, not only in his attempted proof of pragmatism, but also in philosophy and reasoning, has not been acknowledged in the literature before. Unlike Kant, whose processual approach to the construction of concepts was limited to the production of representatives in the mind that correspond to individuals, in other words the construction that was confined to representatives operating within the fixed context of proper names, Peirce broke off from this tradition and thought of the construction of concepts (analogous to Kant's intuition, *Anschauung*) as a selection of not only the representatives of proper names, but also of the representative instances of indefinite expressions. Even further, as he already had established a comprehensive logical, mathematical and philosophical machinery to deal with different kinds of indefinites, he never saw any compelling reason to set artificial limits on these profoundly analytic facets of EGs.

Analysis through diagrams Do these facets in toto constitute a perspective from which Peirce's system of logic facilitates the most complete analysis of propositions? Two points may be noted. His graphs certainly superseded by far any non-symbolic system of reasoning that was known at that time, including the early proposals of Raymundus Lullus' (Ramón Llull, 1235–1315) conceptual system of Ars Magna of the 14th century, followed by Giordano Bruno's (1548–1600) suggestions for artificial memory systems in the 16th century, Leonhard Euler's (1707–1783) graphical diagrams for syllogisms, or the geometric diagrams of the 19th century produced by Jean-Victor Poncelet (1788–1867), John Venn (1834–1923) and others. Gottlob Frege's (1848–1925) two-dimensional Begriffsschrift (ideography) was developed for syntactic purposes and is not to be counted in. According to Peirce's own testimony, it was Juan Luis Vives (1492–1540) who first came to propose diagrammatic methods to mind mapping (Chapter 4). In a different shape, diagrammatic methods came later to define the foundations of cognitive approaches to language, especially cognitive semantics and the semantic field theory.

Second, the analytic dimension of Peirce's graphs was profound. It was carried to the extreme, leaving little unanalysed. The analyticity of diagrammatic forms of representation demonstrates the thoughtlessness of what was suggested a hundred years later in Tennant (1986), who expresses hasty reservations on any privileged status of visual geometric devices in logic:

> [The graphical diagram] is only an heuristic to prompt certain trains of inference; that it is dispensable as a proof-theoretic device; indeed, that it has no proper place in the proof as such. For the proof is a syntactic object consisting only of sentences arranged in a finite and inspectable array. (Tennant, 1986, p. 304).

This viewpoint derives from a misguided analogy between the use of diagrams for deductive reasoning and in the semantic modelling of language.

Moreover, such views suggest that their author has not taken note of Peirce's distinction between theorematic and corollarial reasoning, the nature of synthetic reasoning in geometrical constructions, or the role of iconic signs in reasoning and representation. All these contributed to one of his the most spectacular discoveries in logic. Despite all this, slightly later we learn that

> If one draws the appropriate diagram to help one appreciate the validity of the argument from 'all F's are G's' and 'there is an F' to 'there is a G', one may take oneself to be constructing the 'general model' in which the premisses are true, and checking that in this model the conclusion is indeed true also. But the general model is as much a chimera as the general triangle. It does no more than recapitulate information already available in the obvious natural deduction of the conclusion from the premisses. . . . The situation in semantics generally is analogous to that in the case of general models and general triangles. I intend to show that models have no peculiar and irreducible role to play in our understanding of language. (Tennant, 1986, pp. 304–305).

Not only is this view flawed on model-theoretic accounts of logic, but denounces all empirical tests in support of the view that there is an element of 'mental models' in human reasoning (Chapter 2).

It is not the idea of uninterpreted terms, with any process of filling in their blanks, that alone suffices to smooth out the "crudeness" of Kant's thinking. What is essential is not only the interpretation of graphs, but also their construction. Two interrelated processes are involved.

First, one sets out to express or present thoughts on a sheet of assertion associated with the universe of discourse, by scribing (i.e. partly writing and partly drawing) certain elements on it.[17] To scribe something is to propose modifications to the content of thought. This is done by one of the parties engaged in the *cognition discursiva*. The Graphist is the utterer who asserts propositions that come to be scribed on the sheet. The other half of the discourse evokes the Grapheus, who creates the universe and either accepts or rejects the modifications put forward by the Graphist.

Second, the graph, thus considered a complete and accurate presentation of the thoughts on the sheet of assertion, is interpreted by considering one of its components at a time, proceeding in a well-defined order starting from the outmost part (correlated with the selectives, i.e. certain proto-forms of quantifiers and their interpretation, see Chapters 6 and 9) and proceeding until the process terminates in the atomic, unanalysable graph. This is the endoporeutic interpretation of graphs, scrutinised in Chapters 4–6.

The discourse-taking between two is prevalent throughout Peirce's corpus. The intended proof of pragmatism would not be comprehended without it. Its elements were already present in his algebraic logic of the early 1870s. It formed a cunning theme in his semeiotics and became more and more prominent as his explorations progressed. Its cumulative thrust, unlike almost anything else in his philosophy, has been invariably missed by later commentators. In EGs, we have the privilege of enjoying one of its most mature descriptions.

Any further exposition of these fundamental questions would necessitate a spacious exploration of a number of other aspects of Peirce's logic, including proper names and indefinites as subjects of the proposition, as well as the distinction between theorematic and corollarial reasoning in logic. To quote:

> There are two kinds of Deduction; and it is truly significant that it should have been left for me to discover this. I first found, and subsequently *proved*, that every Deduction involves the observation of a Diagram (whether Optical, Tactical, or Acoustic) and having drawn the diagram (for I myself always work with Optical Diagrams) one finds the conclusion to be represented by it. Of course, a diagram is required to comprehend any assertion. My two genera of Deductions are first those in which any Diagram of a state of things in which the premisses are true represents the conclusion to be true and such reasoning I call *Corollarial* because all the corollaries that different editors have added to Euclid's *Elements* are of this nature. Second kind. To the Diagram of the truth of the Premisses something else has to be added, which is usually a mere May-be, and then the conclusion appears. I call this *Theorematic* reasoning because all the most important theorems are of this nature. (EP 2:502, 1909, *Letter to William James*).

Considerable further light is thrown on these aspects of Peirce in Hilpinen (1995) and Hintikka (1996b).[18]

Let us next provide an overview of a couple of key aspects of Peirce's logic, signs and semeiotics. But first, let me give a thumbnail sketch of Peirce's life, and my views on the state and availability of the publications of Peirceania.

2. On this uninteresting planet: a biographical sketch

Charles Sanders Santiago Peirce was born in Cambridge, Massachusetts, on 10 September 1839. He was the second son of Benjamin Peirce, noted Harvard University mathematics and astronomy professor. He died in Milford, Pennsylvania on 19 April 1914. He graduated from Harvard College and received a degree in chemistry from the Lawrence Scientific School at Harvard at the age of 20. In 1861 he began working for the United States Coastal and Geodetic Survey as a scientific assistant and consultant, which he continued until 1891. Among the subjects he worked on were the measurement and theory of gravity, spectroscopy, mathematical methods of measurement, the economics of research, metrology and geodesy. From 1879 to 1884 he was employed by the Johns Hopkins University in Baltimore as a lecturer in logic. In 1889 he retired with his second wife to Arisbe, his home in Milford, Pennsylvania, never to return to academia other than to deliver the occasional lecture. From 1891 onwards he worked as a private and independent researcher, employed by among others, William Dwight Whitney, the editor-in-chief of the Century Dictionary for which he wrote over 5.000 definitions and edited some 11.000 more on all aspects of science, and by the editor-in-chief James Mark Baldwin to contribute to the Dictionary of Philosophy and Psychology, the first volume of which appeared in 1901. His publications add up to 800 pieces, including almost 400 reviews, but his unpublished and unfinished manuscripts are much more voluminous and catholic.

At the tender age of twelve, Peirce wrote that he disagreed with his father about the role of logic in mathematics. For Benjamin, logic was an idle activity compared, for instance, to the force of geometric demonstration in mathematics. For Charlie, it was the fountain of all inquiry, an indispensable component in developing instinctive capacities for reasoning, not a static and immutable faculty as Benjamin may have thought, but the dynamic, educable source on which all reasoning ultimately reposes. There was no lack of respect, however: "My father was a dynamical and astronomical mathematician of great distinction, who broadened as he grew old, and was at all ages remarkable for his esthetical discrimination" (MS 296: 4 a.p.).

"He loved and hated, and quarreled with almost everyone he came in contact with, wives, relatives, and associates", wrote Benjamin Peirce Ellis, Charles' sister Helen Peirce's son, in his private memoirs as a reaction to the obituary by Juliette, Peirce's second wife, which appeared in the New York Herald on 21 April 1914.[19] Even if we could question the accuracy of the tone of this remark, Peirce's personal qualities and his failure to apply quietly the "business logic" of ordinary life were two of the main reasons why his prospects of a conventional institutional career diminished. This shows in his audacious and defiant attempts at getting his major works on philosophy published and printed, especially in book form. And only one book written solely by him was published, The Photometric Researches (1878), in which he defined the length of the meter based on the wavelength of light.[20] One other book, edited by Peirce, was the Studies in Logic by the Members of Johns Hopkins University, which appeared in 1883. Among numerous other books that were planned was one that was to be entitled — in the Spencerian fashion — The Principles of Philosophy (12 vols.) announced by Henry Holt Co. in 1894, which, judging by its scope and the breadth and quality of the already extant material that Peirce had for it, would no doubt have become his *magnum opus* on philosophy.[21]

These attempts were repeatedly frustrated. He never found a publisher, despite agonising efforts. The account of these events constitutes one more chapter in the thick book of resistance to original contributors in philosophy, work that failed to get proper recognition by the majority of contemporary scholars and editors, even though the sheer volume of his output far exceeded any average measures, and even though his masterly devotion was well acknowledged by his fellows and correspondents. Moreover, his grant applications were invariably rejected.[22] Gradually becoming unable to earn enough money, he lived his last years in poverty and destitute, given occasional support by his old friends from academia, most notably William James (1842–1910) and Josiah Royce (1855–1916). The upshot was that Peirce remained in the shadows of most of 20th century philosophy.

For the dismissal to be doubly ironic, the route by which Peirce found its most faithful way to the 21st-century logic was not associated with that of analytic philosophy, the members of which were one of the most brutal in their misuse and disquotation of Peirce's ideas, but that of the continuous passage from the early studies of information science, starting from the Significs Movement in Amsterdam, repeatedly ignored by the Unity of Science Movement and the Anglo-Saxon ordinary language philosophers, via the growing interest in semiotics in arts and literature and via the mid-century resurgence of the inquiry dubbed artificial intelligence, whence it gradually found its way to the computational sciences and informatics of the late 20th century.

Only in 1923, a selection of his essays were edited by M. R. Cohen and published as Chance, Love and Logic. This title got into the hands of Wittgenstein, shaping his later thought.

According to Paul Weiss, the editor of the Collected Papers, in later years Peirce was

> a frustrated, isolated man, still working on his logic, without a publisher, with scarcely a disciple, unknown to the public at large. (Weiss, 1934, p. 403).

By the end of the 20th century, his ordeal had come to an end. He was transferred to the front line of philosophy. What had been a grave injustice is currently being set right at an increasing pace. However, one should emphatically remember that positive discrimination can be a double-edged sword in philosophy. (Just ask yourself what has happened to the Wittgenstein of the late 20th century.)

The project of publishing, in chronological order, a critical edition of a selection of all that survives from Peirce is underway (Writings of Charles S. Peirce: A Chronological Edition, 1982–). The Peirce Edition Project began in the 1970s, and the sixth volume of that edition appeared in print in 1999.[23] By that date, his 50th productive year had been reached, covering the period from January 1887 to April 1890. About thirty volumes are projected. It will take several decades for a good number of his manuscripts to appear in print. The effort is remarkable, but it has downsides. The decision by the editors to accommodate all of his writings implies page after page of material of little significance, including calculations, tables and other marginalia. A considerable amount of outmoded material from Peirce the scientist is also included, much of which has little present-day interest. Yet the critical edition means that a large amount of material will not be published at all. The residue includes alternative draft documents and even longer incomplete manuscripts, notes and references that Peirce produced during the course of writing his intended papers. Many of these drafts are not just earlier versions of more complete, or more accurate, or otherwise preferred variants of something that is to be included, and for that reason out of date and irrelevant. They often contain significant explanations and substitute formulations of the more complex principal drafts. The existence of different versions is valuable, as it reveals the evolution of Peirce's thought in the course of his actual process of writing, as so often happened. The excluded sheets were not invariably marked by Peirce as being definitely discarded. Often, they were labelled "keep for reference". Sometimes the strikeout marks on the draft sheets are very small, typically appearing in the marginal rather than on top of text, indicating that he by no means wanted to disregard their contents. Such papers are not to be included in the editions. One hopes that future electronic editions of the Writings, which will supplement the volumes in a hardcopy form by including digitised images of the manuscripts, will include a substantial number of pages not transcribed to the body of the printed text. Nevertheless, roughly one third of Peirce's work is to be edited in the thirty or so volumes of the Writings.

As a result, critical editions not only slow the turnaround of further editions-to-be, but also imply that a complete publication of Peirce's most important and topical work is likely to not be accomplished in the near future, one that would include his later writings on logic, semeiotics and EGs. Furthermore, the transcriptions are not diplomatic, and such editions are unlikely to ever be made. The assumption that the reader will track all the changes and emendations that the editors have made from the notes and indices that will run to hundreds of pages in total is demanding: while the cd-rom version of the Writings may ease the burden it does not do away with the tedious back-and-forth task prompted by the mammoth editorial apparatus and documentation.

What is included in the Writings is always very careful and considered, and the outcomes are of immense value. However, my feeling is that the manuscripts themselves, at least when finally renumbered and reorganised — itself an enormous task: "No human being could ever put together the fragments. I could not myself do so"[24] (MS 302) — will serve as the most accessible and the most accurate resource for Peirce studies, unsurpassed in the end by the critical editions. Peirce's hand reads perfectly, and the occasional rubbings out and additions that he made are not just minor.

Above all, the manuscripts are all there is of the material from the later and the most productive and synthesising period of his life. The complexity of organising these later manuscript sheets increases exponentially when we move on to the material from the turn of the century and later. The eight-volume Collected Papers is still a decent and respectable overview. Errors, obscure and misleading organisation, discontinuous fragments, numerous silent emendations and changes of terms for the sake of consistency, and not properly documented by the editors, will for most part be excused given the earnest attempt at overall relevance and comprehension. It is unlikely that ultimate final critical editions, hampered by the accumulating decision-making task concerning what to include and how to revise the drafts most faithful to Peirce's intentions (and, surpassing him in many respects), and containing all the manuscripts dating from the last fifteen years of his life, will be completed in the decades to come.

3. Signs, logic and semeiotics

Kant's influence shows up in a number of junctures in Peirce's philosophy, but his overall aims rest on quite different concepts and theories. Unlike Kant, he never approved the distinction between things in themselves and the phenomenal world. However, the architectonic nature of his philosophy is eminent in his doctrine of categories. I will begin with those and then move to his interrelated kingdoms of sign theory, logic and semeiotics.

Firstness, Secondness, Thirdness The categories Peirce ended up advocating, termed cenopythagorean categories, are a definitive simplification of Kant's twelve categories of metaphysics into just three. They comprise one of the most fundamental trichotomies Peirce makes, namely that of the division between what pertains to the category of firstness, what pertains to the category of secondness, and what pertains to the category of thirdness.

Very briefly, the division between firstness, secondness and thirdness refers to the three categories of quality, reaction and representation. Alternatively, the characteristic notions are those of possibility, actuality and law, respectively. I will consider each of these categories in turn. They were not taken to be universally applicable to all aspects of inquiry, but are most pertinent in one branch of philosophy, namely that of phenomenology. Later in his life, Peirce renamed phenomenology phaneroscopy. According to Peirce, its purpose is to contemplate universal phenomena and discern ubiquitous elements of these three categories.[25] The second major division of inquiry is "Normative Science, which investigates the universal and necessary laws of the relation of Phenomena to *Ends*, that is, perhaps, to Truth, Right, and Beauty" (5.121). The third division is "Metaphysics, which endeavors to comprehend the Reality of Phenomena. Now Reality is an affair of Thirdness as Thirdness, that is, in its mediation between Secondness and Firstness" (5.121).

> *Firstness* is the mode or element of being by which any subject is such as it is, *positively* and regardless of everything else; or rather, the category is not bound down to this particular conception but is the element which is characteristic and peculiar in this definition and is a prominent ingredient in the ideas of quality, qualitativeness, absoluteness, originality, variety, chance, possibility, form, essence, feeling, etc. (MS L 107: 21).

Firstness contains pure modes of being, objects and things without any intervention by the human mind. Besides the characteristics listed above are the ideas of "freshness, life, freedom" (1.302, c.1894). Anything that falls under firstness has to be unanalysable, immediate and absolutely simple, and must not depend on any further cognition by a mind or an action by an agent. Firstness may be a feeling, but it has to be immediate, placed upon one who experiences and feels suddenly, without deliberation, in a timeless and instantaneous fashion. Examples are "the quality of the emotion upon contemplating a fine mathematical construction, the quality of falling in love" (1.304, c.1904), and "a feeling of stillness" (8.330, 1904). In the Logic Notebook Peirce speaks of a "flavor" of all that is present, the quality of what is as it is, regardless of anything else: "all that is collectively taken in its absolute simplicity is flavor".[26]

On an entirely different plane,

> *secondness* is that mode or element of being by which any subject is such as it is in a second subject regardless of any third; or rather, the category is the leading and characteristic element in this definition, which is prominent in the ideas of dyadic relativity or relation, action, effort, existence, individuality, opposition, negation, dependence, blind force. (MS L 107: 22).

Unlike firstness and its still-life quality of the lack of action, secondness is the dynamic, two-sided encounter and opposition found in pairs and polarities such as action-reaction, effort-resistance, time-like passing from one instant to another, and in the contact of the ego (the mind) with the non-ego (the non-mind). These all involve effort in acting and perceiving, the struggle to achieve and feel something, shocks in the sense of change. Only one of these pairs cannot stand alone, and its contrast, complementary polarity, is needed. Action cannot exist without reaction, and effort cannot exist without resistance. Singulars maintain subsistence only in so far as there is a counterpart resisting or acting against it.

In a letter to Lady Welby, Peirce explained firstness, secondness and the transitional phase to thirdness.[27]

> The type of an idea of Secondness is the experience of effort, prescinded from the idea of a purpose. It may be said that there is no such experience, that a purpose is always in view as long as the effort is cognized. This may be open to doubt; for in sustained effort we soon let the purpose drop out of view. However, I abstain from psychology which has nothing to do with ideoscopy. The existence of the word *effort* is sufficient proof that people think they have such an idea; and that is enough. The experience of effort cannot exist without the experience of resistance. Effort only is effort by virtue of its being opposed; and no third element enters. Note that I speak of the *experience*, not of the *feeling*, of effort. Imagine yourself to be seated alone at night in the basket of a balloon, far above earth, calmly enjoying the absolute calm and stillness. Suddenly the piercing shriek of a steam-whistle breaks upon you, and continues for a good while. The impression of stillness was an idea of Firstness, a quality of feeling. The piercing whistle does not allow you to think or do anything but suffer. So that too is absolutely simple. Another Firstness. But the breaking of the silence by the noise was an experience. The person in his inertness identifies himself with the precedent state of feeling, and the new feeling which comes in spite of him is the non-ego. He has a two-sided consciousness of an ego and a non-ego. That consciousness of the action of a new feeling in destroying the old feeling is what I call an *experience*. Experience generally is what the course of life has *compelled* me to think. Secondness is either *genuine* or *degenerate*. There are many degrees of genuineness. Generally speaking genuine secondness consists in one thing acting upon another, — brute action. I say brute, because so far as the idea of any *law* or *reason* comes in, Thirdness comes in. When a stone falls to the ground, the law of gravitation does not act to make it fall. The law of gravitation is the judge upon the bench who may pronounce the law till doomsday, but unless the strong arm of the law, the brutal sheriff, gives effect to the law, it amounts to nothing. True, the judge can create a sheriff if need be; but he must have one. The stone's actually falling is purely the affair of the stone and the earth at the time. This is a case of *reaction*. So is *existence* which is the mode of being of that which reacts with other things. But there is also action without reaction. *Such is the action of the previous upon the subsequent*. (8.330, 1904, *Letter to Welby*).

There are many parallel issues running in this paragraph, such as the distinction between generate and degenerate forms of secondness, which Peirce was forced to introduce in order to avoid the pitfalls concerning the demarcation between what belongs to secondness and what belongs to thirdness. He also mentions ideoscopy, which "consists in describing and classifying the ideas that belong to ordinary experience or that naturally arise in connection with ordinary life, without regard to their being valid or invalid or to their psychology".[28]

In addition, in this quotation Peirce makes it clear that psychology is not to be introduced in the effort to provide principles for logical and semeiotic investigation. This idea is reminiscent of a similar conviction in Kant, who noted the following.

> Some logicians presuppose *psychological* principles in logic. But to bring such principles into logic is as absurd as taking morality from life. If we took the principles from psychology, i.e. from observations about our understanding, we would merely see *how* thinking occurs and *how* it *is* under manifold hindrances and conditions; this would therefore lead to the cognition of merely *contingent* laws. In logic, however, the question is not one of *contingent* but of *necessary* rules, not how we think, but how we ought to think. (Kant, 1988a, p. 16).

The seeds of another doctrine of Peirce's occur in the last sentence, namely that logic, like ethics, is a normative science. This idea was absorbed by Frank Plumpton Ramsey (1903–1930) soon after Peirce's death, and Ramsey further transferred it to Wittgenstein (Chapter 8). The idea never gained fertile ground in the foundations of the 20th century logic. In linguistics, parallel discussions on the normative basis of language emerged in the wake of the early structuralist-functionalist-conventionalist debate, only to be brushed aside from the mainstream linguistics, too.

The quotation from Peirce's letter continues with a discussion of whether the idea of determination by action that involves considerations of time as the previous upon the subsequent is the "pure idea of secondness", or whether it ought to involve thirdness in the Kantian sense that indeterminacy, belonging to the future, is an idea that does not concern existence, because existence is determinate. Peirce remained indecisive about whether there is a sensible secondness/thirdness interface in the notion of "action without reaction".

Despite these animadversions, a distinct category of thirdness exists, which he describes as follows.

> *Thirdness* is that mode or element of being whereby a subject is such as it is to a second and for a third; or rather, it is the characteristic ingredient of this definition, which is prominent in the ideas of instrument, organon, method, means, mediation, betweenness, representation, communication, community, composition, generality, regularity, continuity, totality, system, understanding, cognition, abstraction, etc. (MS L 107: 22–23).

Peirce was convinced of the utmost importance of the category of thirdness. It stands out in the ideas of our minds, as something that cannot be covered by considerations of secondness. There were many motivations for establishing this category, but the essential one was his proof, which he set up in algebraic form but which shows up in his overall conviction outside purely algebraic and mathematical considerations, that triadic relations are irreducible to monadic and dyadic relations. In any triadic relation, Peirce claimed, there is an element of the mental. One example is the three-place relation of giving, such as in "giving a horse to a trainer". It involves a law that makes the receiver (here the indirect object "trainer") a possessor of something (the absolute term "a horse") by the sender's action. It cannot be reduced to two identities of (i) one agent putting away a thing and (ii) the other agent receiving it later.

Peirce also considered thirdness a naturally leading category simply because so many of his other distinctions came in three parts. One of these is the sign-object-interpretant trichotomy: "In its genuine form, Thirdness is the triadic relation existing between a sign, its object, and the interpreting thought, itself a sign, considered as constituting the mode of being of a sign. A sign mediates between the *interpretant* sign and its object. . . . A *Third* is something which brings a First into relation to a Second" (8.332, 1904). I will address the different notions of interpretants shortly, but essentially, they embrace things such as thoughts, actions, experiences, and qualities of feeling.

Secondness also comes into play in the all-important notion of communication (Chapter 2), although not in the full sense that would fit it in the category of thirdness. Without doubt, understanding the interplay between secondness and thirdness is one of the keys, and also one of the challenges in Peirce's philosophy, notwithstanding the fact that it was recognised to be of major importance only quite recently. I will address some of the applied sides of the sign-theoretic view of communication in Chapters 2 and 13, having found their homes in recent theories of information and computation.

Peirce was particularly eager to avoid the infiltration of psychological notions into his all-pervading concept of the sign. Among the sentiments he was quite sensitive to were "that a Sign brings a Second, its Object, into *cognitive* relation to a Third", "that a Sign brings a Second into the same relation to a first in which it stands itself to that First", and that "if we insist on *consciousness*, we must say what we mean by consciousness of an object. Shall we say we mean Feeling? Shall we say we mean association, or Habit?"[29]

Avoiding such psychological undertones, Peirce emphasised the concept of communicational relations that mark the difference between a sign and the mind. The terminology may be mentalistic, but an immediate lapse into the sea of psychic vernacular is circumvented.

How the categories are related to some other key notions in Peirce's philosophical inquiry is clarified in the following passage. I will take up the notions of signs, thought, interpretants and knowledge as the work progresses.

> The essential function of a sign is to render inefficient relations efficient, — not to set them into action, but to establish a habit or general rule whereby they will act on occasion. . . . a sign is something by knowing which we know something more. . . . all our thought and knowledge is by signs. A sign therefore is an object which is in relation to its object on the one hand and to an interpretant on the other, in such a way as to bring the interpretant into a relation to the object, corresponding to its own relation to the object. I might say 'similar to its own' for a correspondence consists in a similarity; but perhaps correspondence is narrower. (8.332, 1904, *Letter to Lady Welby*).

The communicative dimension implicit here is a major perspective and a pervasive thread in my understanding of Peirce's philosophy.

The division of all there is into three categories may raise the question of the explanatory value of such a triangulation. Some commentators have considered it too coarse to have the explanatory virtues of Kant's metaphysics. Granted, the three categories are applicable throughout the grassland of phenomenology, but it is not clear whether they capture the other two grand divisions of inquiry.

Speculative grammar, critic, methodeutic The second grand division of Peirce's concept of inquiry, theoretical normative science, or the science of what ought to be or what ideals things ought to possess, falls into three parts: esthetics, ethics and logic (1.191, 1903). Moreover, quite apart from what esthetics (fine art) and ethics (the art of the conduct of life) are — Peirce was not as keen to study them as he was to study the third part — the subject of logic, as the art of reasoning, the theory of self-controlled, deliberate thought and hence part of the other two branches of normative science, is equally divided into three separate subjects, each division depending on that which precedes it.

The first he called speculative grammar (pure grammar, stecheotic, stoicheiology). It is about the nature of icons, indices and symbols. It aims to ascertain what is true of signs (also called representamens) so that they can embody any meaning (2.229, c.1897). The second is critic (logic proper, stoechelogy), the science of classifying arguments and their validity and degree of force, aimed at ascertaining what is true of signs so that they may hold good of objects. The third is speculative rhetoric (methodeutic, transuasional logic, methodic, methodology), which is the study of methods in the application of truth, a doctrine about the reference of signs to what they aim at determining, namely the interpretants. It is the study of meaning, and it aims to ascertain the ways in which signs beget other signs, as well as the ways in which thoughts beget other thoughts.

In the Logic Notebook Peirce distributes these so as to encompass the following classification (LN: 145r, 29 September 1898): (i) Speculative grammar explains quality, collection, dyadic relation, and triadic relation of copies, signs and symbols, plus what is necessary for the expression of thought. It also addresses the graphical component of algebra. The index-icon-symbol trichotomy pertains here. (ii) Logic proper deals with the doctrines of terms (rhemas), propositions and inferences. (iii) Speculative rhetoric is about methodology, applications of logic to mathematics and other areas of inquiry, and logic in its objective sense. He devised this categorisation in 1898, but he noted that, in fact, the term speculative rhetoric "is bad". Later on he coined methodeutic.

Peirce never saw any reason to give up what has probably become the broadest conception of logic that has ever been written. When he wrote "logic" he almost invariably meant semeiotics, "the doctrine of the necessary principles of signs".[30] As observed, this doctrine has the three parts of grammar, logic and methodeutic. As such, it suffices to cover vast aspects of human inquiry, and is applicable to virtually any discipline, branch of knowledge, nook, cranny or speciality of scientific inquiry. I will argue in Chapter 12 that this trichotomy is not, however, to be equated with that of syntax/semantics/pragmatics.

Signs, objects, interpretants A sign, that sweeping and by far the most general and at the same time one of the most sublime concepts in Peirce's architectonics, appears to be practically anything. Even so, it is a useful concept onto which to project various restrictions. For its essence is in representation: a sign stands to something for something in some respect or capacity (2.228, c.1897). This is one of the most often referenced of Peirce's explanations of what signs are. By reason of their "standing for something", he habitually calls them representamens. A caveat here is that representamens have a more limited and technical meaning than signs. Signs are there because they create in the minds of their interpreters other signs, called their interpretants. That for which a sign stands is its object, with reference to some idea, impression, perception or essence that were called the grounds of representamens in the earlier expositions of his sign theory. This happened roughly up to 1903. Later on he decided to include the grounds in the notion of interpretants. The reference of the object to an idea is either something that is shared between the multiple minds of the interpreters who conceive it, or a private discourse and contemplation within a single mind.

What a sign is, is nonetheless made slightly Delphic by Peirce's surprising list of what counts as signs. He says that, among other things, a human being, the universe, a thought and our knowledge are all signs. It promptly turns out that only experience and habit, two cornerstones of Peirce's phenomenology and metaphysics, do not fall under the umbrella of being a sign.

Every sign has its representative quality, its meaning. Meaning, in turn, is a habit, and it is derived from experience. Experience is a reaction between two phases of the mind, the ego and the non-ego, and thus exemplifies secondness. Pure quality, without a contemplating mind, is firstness. Habit, connected with the mind, is thirdness. According to Peirce, we have direct knowledge of all these three categories, and they must, for that reason, be beyond doubt.[31] The direct knowledge of signs is the thought, the knowledge of a reaction is experience, and the knowledge of quality is the feeling that it generates. Beyond these, the kind of metaphysics that takes these categories and applies them to the totality of the universe is "inferential", by which Peirce means that it represents the process of coming to have knowledge that is obtained through signs.

In his excellent book on Peirce's semeiotics, Johansen (1993, p. 55) observes that signs are the only medium through which to communicate and bring about knowledge. This is a slightly problematic claim, even though it is based on textual evidence from Peirce. He remarked in several places that "signs are a species of medium of communication".[32] If so, they are the 'third', in between objects and interpretants. This prompts the question of what else there is in communicational situations other than signs that can mediate something, and in particular, presumably mediate something other than knowledge. Peirce claims that all our knowledge and thought is acquired through signs (8.332).

Even more overtly, he remarks in several places that every thought is a sign (or that every thought is in signs, cf. the following quotation). Thus, it must be the case that ways of communicating other than those providing knowledge are not connected with thought. The only exception to the doctrine of the non-existence of knowledge unmediated by signs presented by Peirce is instantaneous knowledge, which exemplifies the contents of consciousness. This is inherent in the claim that we possess direct knowledge of the three cenopythagorean categories. Even if this relation turns out not to be the case, it seems to me that what such "non-signs" mediate constitutes one of the major puzzles in Peirce's philosophy of signs that remains unanswered.

In his early article, *Questions Concerning Certain Faculties Claimed for Man* (5.265, 1868),[33] Peirce addressed the question of whether it is at all possible to think without signs.

> From the proposition that every thought is a sign, it follows that every thought must address itself to some other, must determine some other, since that is the essence of a sign. This, after all, is but another form of the familiar axiom, that in intuition, i.e., in the immediate present, there is no thought, or, that all which is reflected upon has past. *Hinc loquor inde est.* That, since any thought, there must have been a thought, has its analogue in the fact that, since any past time, there must have been an infinite series of times. To say, therefore, that thought cannot happen in an instant, but requires a time, is but another way of saying that every thought must be interpreted in another, or that all thought is in signs.

One uses thought to analyse and understand thought; there is no other way. Since thinking is a concrete, yet temporal activity, when one thing is interpreted and another thing follows it in chronological succession, the latter thought becomes a representation of the former, and thus has to stand as its sign.

Yet signs are not "real" things,[34] because they are representations, capable of being uttered many times and capable of being precisely alike while representing separate things. Propositions, which are true or false, or rather are capable of turning out to be true or false, are perfections of signs in that they distance themselves from objects in order to produce their representations. A sign is intended to determine, in the minds of its interpreter, its interpretant, which is typically different from the sign. As Peirce puts it, even in its most imaginary form, the purpose of the sign is to "communicate ideas" (MS 283: 101) from one state of mind to another, future state of mind. A sign is an "implement of intercommunication" (MS 283: 106), a version of a medium of communication. In some places, he even describes it as a "determination of a quasi-mind" (MS 283: 131), which emphasises its abstract processual and effectual character in addition to communicative and dialogical aspects.

Precisely what is meant by determination here was a considerable source of struggle for Peirce. He despaired of clarifying the notion of determination in the rest of that manuscript and elsewhere. He articulated the nature of thought and thinking as being the determination of something that "corresponds" to a mind or a quasi-mind.[35] This may be too easy a way out, but at least it is perfectly consistent with his view that every thought is a sign.

The following quotation, which is an excerpt from one of Peirce's numerous letters to Welby, endeavours to explain the nature of signs, objects and interpretants, together with their interplay, in a most instructive and lucid manner.

> It seems to me that one of the first useful steps toward a science of *semeiotic* (σημειωτική), or the cenoscopic science of signs, must be the accurate definition, or logical analysis, of the concepts of the science. I define a *Sign* as anything which on the one hand is so determined by an Object and on the other hand so determines an idea in a person's mind, that this latter determination, which I term the *Interpretant* of the sign, is thereby mediately determined by that Object. A sign, therefore, has a triadic relation to its Object and to its Interpretant. But it is necessary to distinguish the *Immediate Object*, or the Object as the Sign represents it, from the *Dynamical Object*, or really efficient but not immediately present Object. It is likewise requisite to distinguish the *Immediate Interpretant*, i.e. the Interpretant represented or signified in the Sign, from the *Dynamic Interpretant*, or effect actually produced on the mind by the Sign; and both of these from the *Normal Interpretant*, or effect that would be produced on the mind by the Sign after sufficient development of thought. On these considerations I base a recognition of ten respects in which Signs may be divided. I do not say that these divisions are enough. But since every one of them turns out to be a triplet, it follows that in order to decide what classes of signs result from them, I have 3^{10} or 59049, difficult questions to carefully consider; and therefore I will not undertake to carry my systematical division of signs any further, but will leave that for future explorers. (8.343, 1908, *Letter to Lady Welby*).

The last sentence echoes my reason for not to attempting any comprehensive presentation of further divisions of signs in this semeiotic system, apart from the three major ones, namely the rhema–proposition–argument, icon–index–symbol and qualisign–sinsign–legisign. For instance, later in his life, Peirce worked out a classification of signs into sixty-six divisions, regenerated from the ten basic genera that is the most well-known classification. There is no consensus on how this regeneration should be conducted. This is yet another reason to avoid any mystical numbers, be they 66 or 59049, as future explorers would all too easily be swamped by the indefinite number of ever-reproducing classifications, and any such enterprise may turn out to be of little significance given Peirce's overall and more important aims.

More pressing is the question of the systematic connections between the theory of signs and logic. First of all, signs have to be abstracted away from any psychological influences and psychological circumstances. Peirce continues in his notable MS 499 to express his views on this by saying that, in order to see the formal relation of signs to their interpreting minds, they have to be studied by disregarding the qualities of consciousness, the effort of attention, and other psychic ingredients of reasoning. Nevertheless, a sign cannot function "unless it be ultimately interpreted by [a] personal mind; so that if we limit ourselves to *concepts*, or the mental interpretations of signs, we shall therein include every sign that is a sign in actual function, while if we consider signs regardless of the relation of each to a mind, we neglect to consider a most essential characteristic of signs, and thus make room for errors of logic" (MS 499). Mental discourse is thus quite admissible in logic, while psychologism is not, in other words the discourse of importing psychological theories and principles into logical lands. In fact, Peirce was also deeply interested in psychology, in that it was a subject that might eventually contribute to logic in some novel ways, but he did not regard logical reasoning itself amenable to psychological theorising. Most importantly, he believed that it was the EGs that provided the key to the door of logical relationship between signs and the interpreting mind. I will return to some of the issues this perspective generates in Chapters 2, 4 and elsewhere.

As far as objects in this troika of signs, objects and interpretants are concerned, their role is derived from principles resembling Kant's *Dinge as sich*, pure modes of being, but unlike Kant, Peirce held it meaningless to think that there are objects beyond the reach of intelligent cognition and comprehension.

In manuscript 499, Peirce uses slightly different terminology in explaining the triangle of signs, objects and interpretants and their interplay, suggesting that "the object of the sign is the sign's *determinant*; the interpretant is the *determinand* of the sign". The determinant is that which determines something, and the determinand is that which something determines. Being determined by and determining something are in no way complete methods of dealing with elements in semeiotic theory, however, as is witnessed by the remark that "a sign has an object and an interpretant. In the interpretant, which is a partially indeterminate thing, the sign determines a Firstness, not absolute but relative to the object. The pairedness it brings about is not absolute or brute, but recognizes the sign as its creator" (LN: 109r, c.1898). Pairedness obtains between the object and its interpretant, and is mediated by the sign. Signs, so to speak, look backward to the object and forward to the interpretant. The interpretant is partially indeterminate because a sign only determines the firstness, or pure qualities, of the object, and the secondness and thirdness must come from there being action of the object with its interpreter, plus the presence of the interpreting mind. The latter two are realised only in the full semeiotic process.

In a letter to Christine Ladd-Franklin (1847–1930), written in order to correct an error in one of his earlier letters, Peirce expounds the sign–object–interpretant relation from a slightly different and unprecedented perspective. He writes, "A sign is an object made by a party we will call the *utterer*, and determined by his idea, which is the sense or *depth* of the sign, in order to create in the mind of the interpreter an *interpretant* idea of the same *object*. The object is itself really of the nature of a sign, too".[36] As will be explained later, he admitted that there were signs that had no utterer, and that often the utterer and the object were assimilated. All this is perfectly consistent with his earlier views on signs, but an interesting twist here is that he admitted more: that signs are objects, and that objects are of the nature of signs.

Another point he made was that signs, in fact, have three relations: they not only look back to the object and forward to the interpretant, they are also recursive in relating to their "senses" or "depths". The sense of the sign is its "better self" (MS L 237: 1). The analogues to the relation between the depth and the sign that he draws are those of "an idea to an ideal" and "memory to vivid hallucination", and to the relation between a sign and its interpretant that of a "seed to the plant that grows from it". The upshot is that signs are typically quite incomplete representations, they have no clear-cut boundaries, and they elude any strict identification unless submitted to rigorous semeiotic scrutiny.

There is a virtually open-ended number of questions concerning the basic characters and nature Peirce assumes signs to have. Some of them are discussed in what follows in connection with the trichotomies of his semeiotics and logic. Many of these questions were originally posed by Peirce, and many of them were left unanswered by him. For instance, in the Logic Notebook he asked what the object of an imperative sign could be (LN: 254r, 9 October 1905). He noted that its interpretant was the desired state of things, while no easy solution was to be found as regards its object. This is but one peculiarity illustrating the unfinished but also intrinsically endless questioning in Peirce's writings. It is a fractal that by its very nature generates endless new questions, proposes new solutions and ways forward, but never finds the equilibrium, the ultimate, final interpretation of all its knotty threads, plots and pieces.

The queen bee of the sciences As the previous remarks testify, Peirce searched in vain for the right forms of expression of his ideas in spoken or written natural language. His inability to finalise his drafting and redrafting, and his endless formulations of interconnected brainwork, are poignant reminders of the proverb 'thought flies, words go on foot'.

He also lamented his singular ineptitude for language, which is perhaps surprising given his extensive studies on classification and history of words, synonyms, dictionaries of choice English words, color names, changes in spelling and pronunciation of English, vowel changes, English grammar and orthography, tenses, commas, punctuation, let alone his typological investigations on at least French, Latin, Ancient Greek, Italian, Spanish, Arabic, Japanese, Tibetan, Dravidian, Inuktitut, Hebrew, Adelaide, Tagalog, Gaelic, Welsh, Hieroglyphs and Cuneiform.

He was anxious not to choose words that may give his readers the wrong idea, but occasionally he had to admit that to be almost impossible, and he was forced to end up deciding between equally misleading verbalisations. Just to give an example, he envisioned the known objects of some agreed universe in EGs as being of one of two kinds, but alas, the pairs of terms "represented as internal" and "represented as external", or "represented as real" and "represented as imaginary", both carried false connotations. He was not entirely happy with the terms "active" and "passive", either. However, he had to bite the bullet and stick to using these latter terms to refer to his conviction that two kinds of objects are present to interpreters. In his opinion, the two kinds of objects could, albeit less favourably, be termed the "actual" and the "possible" (LN: 274r, 9 March 1906). This tedious choice of words reflected the thorny question of modality that Peirce was to struggle with for the rest of his life.

It was precisely these kinds of grapples, plus his general incapacity in thinking in natural language about logical issues, that contributed to the sense of approval and release when he made his discoveries pertaining to iconic, diagrammatical modes of expressing thought. His EGs soon provided an inexhaustible source and inspiration for scores of logical investigations.

Partly because of the deficiencies in the language, Peirce coined and employed an entire armoury of new words to be used for analytical purposes. As a true scientist, he maintained that the discovery of new scientific phenomena warranted the invention of technical words, but that philosophy, being entirely different, should stick to vernacular expressions. However, these vernacular expressions have a similar rank as the technical terms in the special sciences. The chief task of philosophy is to express the meaning of the body of vaguely signifying words adopted by the discipline in order to express the vague ideas of ordinary life that it aims to analyse by means of expressions put into the nomenclature of technical terms (MS 280: 6). This explains the abundance of all kinds of peculiar and special terms of Latin or Greek origin that we find among his philosophical writings, as he strove to school and fertilise "the queen bee of the sciences", his pet term for philosophy.

Rhema, proposition, argument The categorisation of signs into rhemas, propositions and arguments is a trichotomy that comprises one of the major building blocks of Peirce's logical and semeiotic architectonics. These are major components of the received conception of logic of the 20th century, too. They refer to what in our nomenclature are better known as (uninterpreted) predicate terms, sentences with some propositional content, and proofs. In addition to mathematical proofs, also derivations, deductions, sequents, demonstrations, dialogues, disputes and so on are subspecies of arguments or argumentative constructions. Peirce allowed that arguments could also be non-demonstrative and abductive. Other idiosyncrasies he used were the following: for rhemas — signs of qualitative possibility, rhemes, rhemata, terms, seme, sumisign; for propositions — signs of fact, dicisigns, dicent signs, phemes; and for arguments — signs of reason, delomes, suadisigns.

Because it remains uninterpreted, a rhema does not provide information about the objects of the sign, and cannot by itself receive a definite truth value. Nevertheless, Peirce saw its role in the philosophy of logic to be unmistakable in overcoming Kant's deficient programme of logical analysis. Propositions, in contrast, provide information about the objects of the signs. However, both rhemas and propositions are would-bes: the former forbears the object and the interpretant and leaves them undetermined, while the latter forbears the interpretant and leaves it unrealised.

> Symbols, and in some sort other Signs, are either *Terms*, *Propositions*, or *Arguments*. A *Term* is a sign which leaves its Object, and a fortiori its Interpretant, to be what it may. A *Proposition* is a sign which distinctly indicates the Object which it denotes, called its *Subject*, but leaves its Interpretant

> to be what it may. An *Argument* is a sign which distinctly represents the Interpretant, called its *Conclusion*, which it is intended to determine. That which remains of a Proposition after removal of its Subject is a Term (a rhema) called its Predicate. (2.95, c.1902, *General and Historical Survey of Logic: Partial Synopsis of a Proposed work in Logic*).

According to Peirce, these symbolic signs are representamens:

> A representamen [as symbol] is either a *rhema*, a *proposition*, or an *argument*. An *argument* is a representamen which separately shows what interpretant it is intended to determine. A *proposition* is a representamen which is not an argument, but which separately indicates what object it is intended to represent. A *rhema* is a simple representation without such separate parts. (5.139, 1903, *Lectures on Pragmatism: The Three Kinds of Goodness*).

The notion of interpretant in this triplex typically refers to final interpretants, as it concerns the result of a sufficient inquiry that is undertaken by communities of sign users. As we shall see, only such interpretants are worth using in the exact science of logic.

Abduction, deduction, induction Peirce subdivided simple arguments into three kinds: abduction, deduction and induction. I will confine my remarks here to the question of how he sees these types of arguments as correlating with the icon–index–symbol triad.

An abductive argument has a relation of similarity between the facts stated in the premisses and the facts stated in the conclusion, without compelling one to accept the truth of the conclusion when the premisses are true. Peirce goes on to say that the facts in the premisses of an abductive argument constitute an icon of the facts in the conclusion, asserted positively and admitted with suitable inclination. It is in this sense that abduction starts a new idea; in Peirce's words, it is "originary" (2.96).

Deduction is, in Peirce's words, "an argument representing facts in the Premiss, such that when we come to represent them in a Diagram we find ourselves compelled to represent the fact stated in the Conclusion" (2.96). The notion of index arises here, in that "the Conclusion is drawn in acknowledgment that the facts stated in the Premiss constitute an Index of the fact which it is thus compelled to acknowledge" (2.96). It is in this sense that deduction is demonstrative reasoning, "obsistent" and "compulsive" in Peirce's terms.

Induction is an argument starting from a hypothesis that is a result of abduction, interspersed with results of possible experiments deduced from hypotheses and selected independently of any epistemic access to its truth value. Peirce called them "virtual predictions". The hypothesis is concluded "in the measure in which those predictions are verified, this conclusion, however, being held subject to probable modification to suit future experiments" (2.96). The relation between the facts stated in the premisses and the facts stated in the conclusion of inductive arguments is symbolic, as "the significance of the facts stated in the premisses depends upon their predictive character, which they could not have had if the conclusion had not been hypothetically entertained"

(2.96). In Peirce's terminology, inductive arguments are "transuasive" in their assurance of the amplification of positive knowledge.

Icon, index, symbol Inherent in the above distinction is another triad, namely that of index, icon and symbol. Peirce held this to be a central trichotomy. It is explained in his famous article *A Guess at the Riddle*, which is one of his few attempts to actually unify and draw together his philosophy, as follows:[37]

> There may be a mere relation of reason between the sign and the thing signified; in that case the sign is an *icon*. Or there may be a direct physical connection; in that case, the sign is an *index*. Or there may be a relation which consists in the fact that the mind associates the sign with its object; in that case the sign is a *name* [or *symbol*]. Now consider the difference between a logical *term*, a *proposition*, and an *inference*. A term is a mere general description, and as neither *icon* nor *index* possesses generality, it must be a name; and it is nothing more. A proposition is also a general description, but it differs from a term in that it purports to be in a real relation to the fact, to be really determined by it; thus, a proposition can only be formed of the conjunction of a name and an index. An inference, too, contains a general description. (MS 909; 1.372, 1887–88).[38]

A related composition provides an explication of icons, indices and symbols:

> There are three kinds of signs which are all indispensable in all reasoning; the first is the diagrammatic sign or *icon*, which exhibits a similarity or analogy to the subject of discourse; the second is the *index*, which like a pronoun demonstrative or relative, forces the attention to the particular object intended without describing it; the third [or *symbol*] is the general name or description which signifies its object by means of an association of ideas or habitual connection between the name and the character signified. (1.369, c.1885).

Modern examples of partly iconic reasoning include the theory of discourse representation, together with similar diagrammatic systems of logical representation of language and discourse (Chapter 4). In AI, iconic representations flourish in the research on conceptual graphs (Sowa, 1984, 2000), the descendants of Peirce's EGs. The origin of this rapid emergence of automated reasoning systems of the late 20th century was the second industrial revolution of the late 19th century, an era that witnessed the first union between technological innovations and scientific discoveries. Indexical signs such as those exhibited in demonstratives, anaphoric pronouns and related intentional statements are, despite Peirce's assertion of their direct physical connection to objects, still very hard to comprehend in terms of automated reasoning systems. This is because what is meant by a "physical connection" is no means obvious, unique or undisguised. The post-Peirce research on logic has adopted the symbolic aspect of the sign as dominant, because it is the one that can be regimented by general conventions and recursive mathematical definitions.

Genuine and degenerate signs According to Peirce, signs may also be genuine or degenerate, and to varying degrees. He describes these divisions in *Partial Synopsis of a Proposed Work in Logic* (2.92, c.1902), in that a sign that is degenerate to a lesser degree is an "obsistent sign" (index), which has a genuine relation to its object independently of what its interpretant is. Examples of obsistent signs are exclamations as indicative of danger, and a rap on the door as indicative of a visitor.

A sign that is degenerate to a greater degree is an "originalian sign" (icon), which has its significance purely in its quality. An example of this is imagining

> how I would act under certain circumstances, as showing me how another man would be likely to act. We say that the portrait of a person we have not seen is *convincing*. So far as, on the ground merely of what I see in it, I am led to form an idea of the person it represents, it is an Icon. But, in fact, it is not a pure Icon, because I am greatly influenced by knowing that it is an *effect*, through the artist, caused by the original's appearance, and is thus in a genuine Obsistent relation to that original. Besides, I know that portraits have but the slightest resemblance to their originals, except in certain conventional respects, and after a conventional scale of values, etc. (2.92).

Third, a genuine sign is, in Peirce's terms, "transuasional" (symbol), a sign that is only realised by its interpretant and exemplified in any utterance of speech: "The words only stand for the objects they do, and signify the qualities they do, because they will determine, in the mind of the auditor, corresponding signs. The importance of the above divisions, although they are new, has been acknowledged by all logicians who have seriously considered them" (2.92).

Peirce went on to draw the grand triune of firstness, secondness and thirdness into his dichotomy of genuine and degenerate signs:

> An Index or Seme (séma) is a Representamen whose Representative character consists in its being an individual second. If the Secondness is an existential relation, the Index is genuine. If the Secondness is a reference, the Index is degenerate. A genuine Index and its Object must be existent individuals (whether things or facts), and its immediate Interpretant must be of the same character. But since every individual must have characters, it follows that a genuine Index may contain a Firstness, and so an Icon as a constituent part of it. Any individual is a degenerate Index of its own characters. (2.283, 1902–03, *Speculative Grammar: The Icon, Index, and Symbol*).

Existentially bound variables of first-order logic are present-day examples of genuine indexical signs, whereas free variables are degenerate.

Particular, universal, singular propositions General subjects of propositions are either particular or universal. Propositions themselves may also be singular. According to Peirce, a particular proposition is one which transfers the "liberty of choice to the other party, the utterer, and consequently the defender of the proposition".[39] Similarly, a singular proposition is "one which leaves no liberty of choice as to the singular instance, to either party" (MS 515: 20).

These statements derive from more essential considerations. According to Peirce, every proposition is in every aspect either definite or individual. Definite signs or propositions are those to which the principle of contradiction applies, and individual signs or propositions are those to which the principle of excluded middle applies. These two logical principles are, so he claims, themselves derived from more general considerations about logical activities, and are not as such beyond dispute. They pertain to the most general relations between signs, logical reasoning and the notion of communication between the utterers and the interpreters of the signs (Chapter 2). Notwithstanding these considerations, which Peirce was under enormous pressure to settle but never completely succeeded in presenting an overall picture, universal propositions are those that are non-individual and thus definite, whereas particular propositions are indefinite and thus individual. Singular propositions, then, are those that are both definite and individual.[40]

Qualisigns, sinsigns, legisigns According to Peirce, signs are divisible by three major trichotomies. The icon–index–symbol one is the second of them, and it is, as noted above, made according to the sign's characters in its relation to its object or its interpretant. The first division concerns the sign's quality, its actual existence, and its general law, and the third derives from whether the interpretant represents it as a sign of possibility, of fact or of reason.[41] The latter is the rhema–proposition (dicent sign)–argument trichotomy.

The first division differentiates qualisigns, sinsigns and legisigns: "A Qualisign is a quality which is a Sign. It cannot actually act as a sign until it is embodied; but the embodiment has nothing to do with its character as a sign" (2.244). It is by means of signs of qualities that one defines simple signs, or the sinsigns (tokens), signs of an actually extant entity or event. A sinsign can only exist "through its qualities; so that it involves a qualisign, or rather, several qualisigns" (2.245). Thirdly, a legisign (type) is a law that is a sign, "usually established by men" (2.246). An example is a conventional sign of a general type, the signification of which is by its application. As each individual instance of such an application is a sinsign, every legisign requires sinsigns.

This triplet divides signs with respect to firstness, and is thus only one of three respects in which they can be categorised. The icon–index–symbol trivision applies to secondness, and rhemas, propositions and arguments to thirdness.

Ten genera of signs Bringing the previous trichotomies together, we get the three-fold classification of signs:

	Sign	**Object**	**Interpretant**
Firstness	Qualisign	Icon	Rhema
Secondness	Sinsign	Index	Proposition
Thirdness	Legisign	Symbol	Argument

The reductions and interrelations that Peirce discovered later in his life and argued for were the following. (i) Every qualisign is an icon, (ii) every icon is a rhema, (iii) every argument is a symbol, and (iv) every symbol is a legisign (MS L 237: 4–5, 27 July 1904). Consequently, he ended up with ten genera of signs: qualisigns, iconic sinsigns, rhematic indexical sinsigns, propositional (dicisignificant) sinsigns, iconic legisigns, rhematic indexical legisigns, propositional (dicisignificant) indexical legisigns, rhematic symbols, propositional (dicisignificant) symbols and arguments (2.264, c.1903).

It is frequently acknowledged that the main types of signs and their interrelations and contact points may be represented in a triangular form that has ten nodes corresponding to the ten signs that Peirce considered irreducible. Alternatively, they may be depicted in a three-dimensional format with indexical signs in the centre, which allows for most interconnections with other signs.[42] The central place that indexical signs enjoy is by no means coincidental or without consequences: when Peirce discovered this new division of signs and their interrelations, he started to regard indexical notions as having an increasingly prominent role not only in his general sign theory, but also in the emerging logic of quantifiers and the iconic EGs. Indexical signs were not reducible to any other genera. In Peirce's opinion, in these logical systems, that which is called the particular or existential quantifier and that which is called the selective are both inherently indexical and do not need to carry in themselves extraneous existential presuppositions (2.283; see Chapter 6). Whether index-centrality is also preserved in a comparable manner for his more extensive categorisations, such as his sixty-six-part division, remains to be seen.

Informed, essential, and substantial breadth and depth These triads take us back to the early phase of Peirce's architectonics. Logically, the act of predication, namely the joining of the object to its subject, is a way of increasing the logical breadth of a sign without diminishing its logical depth. Breadth and depth refer to more customary denotation and connotation of logical terms.

> Sir William Hamilton has borrowed from certain late Greek writers the terms breadth and depth, for extension and comprehension respectively. . . . "wide" learning is, in ordinary parlance, learning of many things; "deep" learning, much knowledge of some things. I shall, therefore, give the preference to these terms. Extension is also called sphere and circuit; and comprehension, matter and content. (2.394, 1867, *Speculative Grammar: Terms*).

Peirce went on to categorise breadth and depth as informed, essential and substantial. The informed breadth of a term means "all the real things of which it is predicable, with logical truth on the whole in a supposed state of information" (2.407, 1867). The informed depth means "all the real characters which can be predicated of it (with logical truth, on the whole) in a supposed state of information" (2.408).

The ground of the object was taken in Peirce's early philosophy to refer to the connotation of a symbolic sign, the second of the three-way notion of reference. The first is the direct reference of a symbol to its objects, namely the denotation of the symbol. The second, connotation, is the reference of the symbol to the common characters of its object, in other words to its ground through its object. The third is the reference of the symbol to its interpretants through its object, which Peirce termed the information of the symbol.

The symbol's direct reference to its object is an example of informed breadth, and its reference to the ground of the object is an example of informed depth.

Whatever reference there is to its interpretant is the information concerning the symbol (2.418). Later, Peirce subsumed the ground under interpretants.[43]

The informed breadth and depth of a term lie between the two extremes of the states of information of which no fact is known and of which there is perfect knowledge of all there is. So two other states of information correspond to these extremities. First, Peirce distinguishes the essential depth of a term, by which he means "the really conceivable qualities predicated of it in its definition" (2.410). The second is the substantial breadth of the term, "the aggregate of real substances of which alone a term is predicable with absolute truth". The substantial depth, in turn, "is the real concrete form which belongs to everything of which a term is predicable with absolute truth" (2.414).

For completeness, the essential breadth is all the objects the sign refers to by virtue of the definition of the object offered through the interpretant.

The three aspects of breadth and the three aspects of depth were not differentiated from each other by the distinction between supposed vs. imaginary states of information. One might expect this kind of division in view of the idea that matured only much later, namely that of intensions, or modal statements, as involving multiplicities of different 'possible worlds' in which statements are evaluated. The possible-worlds approach to intensions, while anticipated by medieval writers (Knuuttila, 1993), is routinely said to have materialised only with the inception of the relational theory of possible-worlds semantics in the 1950s and the early 1960s.[44] Nor did Peirce's contemporary, Hugh MacColl (1837–1909), who in fact was contemplating versions of modal logic sometime before Peirce, come to conceive of 'possible worlds' in any clearly articulated sense of accessibilities between something like states, different sheets of assertions (Chapter 4) or various scenarios.[45] This idea was also evinced in semantic theories of information, which aimed at spelling out the information of proposition in terms of those states of affairs that the proposition includes plus those that it excludes. Montague grammar and the variants succeeding it also resorted the idea of possible states of affairs.

In line with this idea, Peirce's notion of a "perfect state of information", as it was phrased in the late MS 664, *The Rationale of Reasoning*, written in November 1910, could be taken to correspond to sets of possible worlds according to which all worlds are linked by equivalence relations. The "perfect state" thus explains what is means that the representation of all the characters involved in the uttered word is one that has no ignorance.

Accordingly, whilst Peirce considered breadth and depth in his early writings merely in terms of an insufficient dichotomy, the dependence of signs on the dynamics of concepts, varying states of knowledge and the information contained in them, soon called for more extensive trichotomics of signs, also taking objects and interpretants into account. In particular, Peirce classified interpretants in manifold ways to present us with a truly dynamic theory of signs.

Varieties of interpretants Signs cannot stand alone. The ways in which they refer to objects, in the sense of being representamens of them, is able to vary indefinitely, but that does not support any self-sustaining reality. A sign is always interpreted as being something or as signifying things for someone, who then becomes the interpreter of it. A sign cannot be eradicated from its bearer. Together with its representational function, the other major side to any sign is its interpretation. Interpretation is itself a sign, but a subsequent one that follows the sign that is being interpreted. This subsequent sign, produced by an interpretation, is the interpretant of the previous sign. One way of viewing it is as a functional interpretation of a sign. Indeed, interpretant is a commonplace concept that unfortunately has not achieved any position in ordinary parlance of English or, for that matter, of any other language either.

The relation of every sign to its object and its interpretant is a three-place one, and thus a manifestation of Peirce's thesis that three-place relations cannot be composed of one- and two-place relations. While a sign represents objects, and in virtue of that conveys a meaning, the idea that it gives rise to is its interpretant. The endlessly iterating series of interpretations and objects of representation may have a final, terminating nexus that reveals the reality of the object. Or, the process may be endless. This overall process of sign interpretation is routinely termed semiosis. It was defined by Peirce as "an action, or influence, which is, or involves, a coöperation of *three* subjects, such as a sign, its object, and its interpretant, this tri-relative influence not being in any way resolvable into actions between pairs".[46] It is not as frequent a term in Peirce as its later popularity would have us believe.

The representation of an object by a sign is therefore mediated in its interpretants. Interpretants do not necessarily materialise in the mind, in which case they could be referred to as interpretations of objects that are deferred indefinitely into the future. They do not need to be actual. There is thus an element of potentiality and modality in them.

In the following, a distinction between immediate, imperfect and indirect interpretants is made, corresponding to indexical, symbolic and iconic signs.

> Although the immediate Interpretant of an Index must be an Index, yet since its Object may be the Object of an Individual [Singular] Symbol, the Index may have such a Symbol for its indirect Interpretant. Even a genuine Symbol may be an imperfect Interpretant of it. So an *icon* may have a degenerate Index, or an Abstract Symbol, for an indirect Interpretant, and a genuine Index or Symbol for an imperfect Interpretant. (2.294, c.1902).

This outlines some general features of interpretants. But in reality, they are categorised into much more diverse classes. Peirce himself provides these classifications, but unfortunately he used varying terminology, and the scattered nature of the fragments in which they appear make the classifications uncertain. Johansen (1993, pp. 171–174) divides interpretants into three major classes. He argues that there are nine main concepts, which may be further subsumed under three main headings according to their semeiotic role.

The above quotation suggests that the first refers to immediate interpretants, related to the antecedent, anterior states of information and initial periods of semiosis. They give what is typically called the meaning of a sign. Among them are (i) essential interpretants, which are found in minimal states of information in the sense of the notion of the essential depth of a symbol, (ii) the lowest grades of word meaning, which are the initial states of the information where the semiosis commences, (iii) intentional interpretants, which are those determined in the mind of the utterer of the sign, and (iv) immediate interpretants, which are expressed by the signs themselves without interference from the utterer or the interpreter, and which include grammatical, morphosyntactic, phonological and semantic aspects of linguistic signs.

The second group includes dynamical interpretants, which carry the objective content of the actual ongoing, synchronic processes of interpretation. Because they are related to interpretation, they do not consist exclusively of dynamic interpretants, defined by Peirce as "the actual effect which the Sign, as a Sign, really determines".[47] They also encompass effectual interpretants, which are the interpretants produced in the minds of the interpreters by the signs uttered, and intended by the utterer (Chapter 2).

At the receiving end of semiosis rest, thirdly, final interpretants. They are "that which would finally be decided to be the true interpretation if consideration of the matter were carried so far that an ultimate opinion were reached".[48] Johansen (1993) describes the four main kinds of final interpretants present in Peirce's system as follows. (i) Communicational interpretants are those that are common to and shared by the partners in semiosis, conceived of as a communicational and dialogical act of interpretation. Their successful mediation requires suitable and adequate common ground based on presuppositions and knowledge drawn from common experiences (Chapter 2). (ii) Final, habitual or logical interpretants are those that give rise to habits, that is, modifications to participants' dispositions and beliefs of acting in a certain way in certain circumstances. (iii) Rational, normal or final interpretants are value-bound, normative interpretants that are used in judging other forms of final interpretants. (iv) Eventual, final, normal or ultimate logical interpretants designate maximal states of information that disallow any ignorance, and the sign's object and the interpretant become one. It is given in the substantial depth and breadth of the symbol.

Around 1906 Peirce had to admit, "I confess that my own conception of this third interpretant is not yet quite free from mist" (4.536). The later writings clarify it in no significant degree, but the emphasis on the role of communicational interpretants that grew after 1906 sheds some new light on this class (Chapters 2 and 13). By August 1906, with reference to his attempts to classify interpretants, Peirce also admitted that all his notions are too narrow, and added that "instead of "Sign," ought I not to say *Medium*?" (LN: 283r). This statement represents one of the culminations when Peirce focussed his attention on the possibility of fleshing out the use of signs in communicational contexts.

As this classification, convincing in its extensive coverage of Peirce's corpus, and in my judgement close to what he would have to largely agreed with shows, interpretants are entities linked with the varying and dynamic informational and epistemic states of those participating in the process of semiosis. The accuracy of this classification, given in a synoptic form in the Logic Notebook [288r–289r, 23 October 1906], is vindicated by his admission two days later, according to which he had made this synopsis without recollecting anything about the antedating six divisions of interpretants.

Were we to believe the veracity of this admission or not, one of the chief aspects of semiosis is that the participants are not only real utterers and interpreters of authentic conversational settings, but can also be thought of as quasi-minds of quasi-utterers and quasi-interpreters.

Furthermore, the following passage confirms that objects also need to be categorised according to similar principles. There are two kinds of objects, immediate and dynamic, even though there are three main classes of interpretants.

> I have already noted that a Sign has an Object and an Interpretant, the latter being that which the Sign produces in the Quasi-mind that is the Interpreter by determining the latter to a feeling, to an exertion, or to a Sign, which determination is the Interpretant. But it remains to point out that there are usually two Objects, and more than two Interpretants. Namely, we have to distinguish the Immediate Object, which is the Object as the Sign itself represents it, and whose Being is thus dependent upon the Representation of it in the Sign, from the Dynamical Object, which is the Reality which by some means contrives to determine the Sign to its Representation. (4.536).

This is corroborated in the passage written around the same time: "There are an Immediate Interpretant and a Dynamical Interpretant corresponding closely to the Immediate and Dynamical Objects. But there is, in addition, a Final Interpretant, to which no particular kind of object corresponds".[49]

He puts forward also another three-part division of interpretants in 5.475–476 [1906–07, *A Survey of Pragmaticism*]. The first is the emotional interpretant, which concerns the feeling produced by the sign. Not surprisingly, Peirce did not consider this to be very useful in terms of truth, but it is sometimes the sole interpretant that the sign produces. Second, there is the energetic interpretant, which is a further significant effect of the emotional interpretant, a mental or physical effort or act. Finally, the logical interpretant, as described above, produces an effect so forceful as to give rise to a habit change. Whether the first and the second reduce to one another, and whether the second implies the third, are questions to which he fell short of offering clear solutions. Accordingly, they have provoked much further discussion among commentators.

Another recurrent question in Peirce scholarship concerns the relation between logical and final interpretants. It appears from the following extract that ultimate logical interpretants are final, but that a logical interpretant producing a habit change is not yet sufficiently full-grown or educated to be final.

> The activity takes the form of experimentation in the inner world; and the conclusion (if it comes to a definite conclusion), is that under given conditions, the interpreter will have formed the habit of acting in a given way whenever he may desire a given kind of result. The real and living logical conclusion *is* that habit; the verbal formulation merely expresses it. I do not deny that a concept, proposition, or argument may be a logical interpretant. I only insist that it cannot be the final logical interpretant, for the reason that it is itself a sign of that very kind that has itself a logical interpretant. The habit alone, which though it may be a sign in some other way, is not a sign in that way in which that sign of which it is the logical interpretant is the sign. The habit conjoined with the motive and the conditions has the action for its energetic interpretant; but action cannot be a logical interpretant, because it lacks generality. The concept which is a logical interpretant is only imperfectly so. It somewhat partakes of the nature of a verbal definition, and is as inferior to the habit, and much in the same way, as a verbal definition is inferior to the real definition. The deliberately formed, self-analyzing habit — self-analyzing because formed by the aid of analysis of the exercises that nourished it — is the living definition, the veritable and final logical interpretant. Consequently, the most perfect account of a concept that words can convey will consist in a description of the habit which that concept is calculated to produce. But how otherwise can a habit be described than by a description of the kind of action to which it gives rise, with the specification of the conditions and of the motive? (5.491, 1906–07, *A Survey of Pragmaticism*).

There are alternative readings of this paragraph that refer to the role of the notion of habit in interpretation. This perspective has significant logical repercussions, which are addressed in Chapter 3.

The tentative reductions that can be gleaned from Peirce's records in the Logic Notebook are that emotional, energetic and logical interpretants are found in the notions of immediate and dynamic ones, whereas final ones are always logical (LN: 299r, 23 October 1906). As the above quotation asserts, not all logical interpretants are final, however. An immediate interpretant, in its firstness, may be a quality of feeling and thus emotional, and a dynamic interpretant, in its secondness, encompasses an exertion of force and is thus energetic. Since dynamic interpretants are not final, this effect produced upon the interpreter may be a feeling merely, and thus also emotional. For instance, it is possible for logical interpretants to contain elements of emotional and energetic interpretants.

There are further divisions. The triplet of impressional, factual and habitual interpretants is put forward in LN: 283r [30 August 1906]. Earlier, the "proper interpretant" (MS L 427: 4, 25 July 1904) is meant to be the sign's relation in respect to its aspect of thirdness, over and above the respects of firstness and secondness. It is the "naïve understanding of the sign" (MS L 427: 4), which does not possess the qualities of a "reflective" interpretant. Not all of these interpretants are covered by the classification suggested by Johansen (1993).

Occasionally, Peirce also refers to objective, actual, middle, rational and normal interpretants, but his descriptions of these remained incomplete.

He held that the differences in interpretants that characterised his later writings were central to pragmatism. He never saw any reason to modify the very basic triad of immediate, dynamic and final interpretants, or to add any new ones beyond this triplet, though on April 1909, he made the following confession.

> Let me give a *little* fuller explanation of my distinction between the Immediate, the Dynamical, and the Final Interpretants. . . The Dynamical Interpretant is whatever interpretation any mind actually makes of a sign. This Interpretant derives its character from the Dyadic category, the category of

> Action. This has two aspects, the Active and the Passive, which are not merely opposite aspects but make relative contrasts between different influences of this Category as More Active and More Passive. . . . Thus every actual interpretation is dyadic. . . . [As] pragmaticism says . . . the meaning of any sign for anybody consists in the way he reacts to the sign. . . . In its Active/Passive forms, the Dynamical Interpretant indefinitely approaches the character of the Final/Immediate Interpretant; and yet the distinction is absolute. The Final Interpretant does not consist in the way in which any mind does act but in the way in which every mind would act. That is, it consists in a truth which might be expressed in a conditional proposition of this type: "If so and so were to happen to any mind this sign would determine that mind to such and such *conduct*." By "conduct" I mean *action* under an intention of self-control. No event that occurs to any mind, no action of any mind can constitute the truth of that conditional proposition. The Immediate Interpretant consists in the *Quality* of the Impression that a sign is fit to produce, not to any actual reaction. Thus the Immediate and Final Interpretants seem to me absolutely distinct from the Dynamical Interpretant and from each other. And if there be any fourth kind of Interpretant on the same footing as those three, there must be a dreadful rupture of my mental retina, for I can't see it at all. (8.315, *Letter to William James*).

Collections Collections, plurals, and groups of individuals were important subjects for Peirce, not only in relation to mathematical questions concerning the nature of sets — questions that were rapidly emerging in the late 19th century and at the beginning of the 20th century due to Georg Cantor's (1845–1918) work on the mathematical theory of sets — but also in relation to metaphysics and the logical representation of natural language. Peirce preferred to use the term 'collections' rather than Cantor's suggestion of 'sets'. By the multitude of a collection, he meant, roughly, its cardinality. He held that, although collections behave as logically individual, their individuality comes into being when the vagueness concerning them is cleared. As observed, non-vague propositions are those to which the principles of excluded middle and contradiction apply. The principle of excluded middle applies to indefinite subjects of a proposition, namely to particulars and singulars, while that of contradiction applies to its definite subjects, namely to universals and singulars. When both apply, the subject of the proposition is singular. Formal logic characteristically assumes that propositions are non-vague.

> A *collection* is a hypostatic abstraction which keeps within the bounds of ordinary logic, because its existence, instead of depending upon the truth of a general predicate depends upon the existence of independent concrete objects. Alexander, Caesar, and Hannibal make a collection. Our thinking them together, the nominalist will say, makes the collection. The reason he says so is that, owing to his admitting but one mode of being, — which is the essence of nominalism, — he is forced to say that, or be drawn into absurdity. But we who admit *esse in futuro*, and all that that carries along with it, are not forced into that falsification, and can simply and truly say that the existence of the trio consists in the mere existence of Caesar, Alexander, and Hannibal. Take away Hannibal, whether Napoleon be substituted or not, or add Napoleon, and we have a different collection. A collection, like anything else, may be described in general terms. (MS 690).

As such, a collection is *ens rationis*, an abstraction that is built up, or consists, of individuals. The individuals are confined to those of a certain kind actually extant. Collections differ from generals, which are capable of determination, but which do not have a mode of being different from the modes of being of its individuals, which do not need to exist. An example of a general is the future time, which is just as real as past time (LN: 212r, 20 March 1902). A true continuum is also a general that is not built up from singulars, but is itself singular. A collection stands on its own, in so far as it is itself logically singular. Collections are the subject matter of logical studies. However, a logical approach lends itself badly to the study of generals such as continuants that are not collections of singulars. The universe of the truth, the subject matter of logic, is typically singular.

Peirce attempted to define the notion of abstraction by means of signs rather than sets. Abstraction means that one is allowed to assert that some sign applies to the predicate instead of merely or actually applying it: "Thus, I say Napoleon was a man to whom the term Great could be applied, instead of merely saying Napoleon was a great man" (LN: 129r, 1898).

Any further attempt to characterise generals would require delving far too deeply into Peirce's metaphysics and his synechism, the doctrine of continuity, so I will leave the matter here.

In the light of these considerations, Peirce was led to revise some of his former opinions concerning the existence of collections: "I talk of a collection as being in essence not in existence when, by the very definition of it, it is an individual. How can it be in essence merely without violating its very definition? I think I said in the *Monist*, Vol. 7, that an individual must be known to exist by the utterer and interpreter; and that it must be known to each that the other knows this. This needs some modification" (MS 690; HP: 737). The modification he goes on to discuss relies on the idea that to take something to be extant is not to require that there is a predicate that is universally applicable to it. The idea that existence is a universal predicate is vacuous, because anything can be universally predicated of individuals that are non-existent. Notable in this quotation is also its reference to common, mutual knowledge of the utterer and interpreter about the existence of an individual. Common knowledge is an idea that emerged in the field of pragmatics only much later; I will return to its development and role in the era of post-Peircean pragmatics in Chapter 12.

The dual of the universal proposition, the particular proposition, does not presuppose existence, either. In fact, existence itself consists of the fact that some particulars are true of them. Given a particular statement, *some P*, is true not only when *some P* will be a particular statement, but also when *P* in the statement is replaced by something that is actually extant. Defining existence is an altogether different matter. For Peirce, it involves the category of secondness, examples of which being a confrontation of action and reaction, force and resistance, acts of perception without genuine freedom to choose and interpret, an ongoing duel between the ego of the momentary self and the non-ego of another momentary self. Secondness culminated in his definition of existence as a choice that has no genuine options concerning the application of a predicate: "A proposition which, in like manner, leaves its interpreter no freedom of choice as to what it is to be applied to, namely, a singular or a particular proposition, asserts existence, — i.e. not merely universally predicates existence, but represents that there is, will be, (or *would be*, but this amounts to nothing unless it leads to a 'will be') a perceptive act in which that which is indicated is forced upon said interpreter" (MS 690).[50]

Other kinds of abstractions besides collections should also be considered, from which they differ in that collections have a double mode of being: the "*essence* of a collection depends only on the general", while its *existence* "depends only on the singulars" (LN: 216r, 23 March 1902). A collection exists by virtue of its members, and its identity is given or determined by their identities. This is not merely its being, which is given or determined by the predicate that defines the collection. In several assorted draft pages of MS 283, Peirce often emphasises that a plural may be indefinite, but that it is not the opposite of singularity because it is itself "the singularity of a single collection" (MS 283: 138 a.p.). This is, of course, an instance of the application of hypostatic abstraction when the property or thought of being a singular individual is transposed onto another level that makes assertions about that thought.[51]

These two notions, the essence and the existence, delineate the context in which it is seen that Peirce's theory of collections does not agree with the theory of sets in the mathematical sense. In mathematics (if we look away from proper classes and universes of all sets), all there is are the sets, and no reference to the identities of their members is needed.

However, Peirce wanted to dispel existential presuppositions from the notion of collection, but he needed to ascertain this in such a way as to prevent him from falling back to saying that the mode of being of a collection is in essence. One of the principal issues here is that the utterer and the interpreter must somehow be able to find and choose suitable collections as laid out by the proposition at issue, even if they are non-existent. How can Peirce ensure this? According to the received first-order logic, the instances of individuals are *ens rationis*. In possible-worlds semantics individuals are likewise concrete and available to the interlocutors seeking and selecting them within a reasonable amount of time, space and memory. Are there similar activities for evaluating expressions in the logic of collections?

According to Peirce, existence is, logically speaking, not really anything more than occurrence in the universe of discourse. There are countless existences, just as there are countless occurrences. A singular collection may be picked from the universe by virtue of its being one occurrence among others, some of them exhibiting existence, but not necessarily all having a mode of existence. A similar argument applies to other universes of discourse, or dimensions of the universe as Peirce would have put it, especially to those concerning modalities, possible individuals, eventualities, actions, interrogatives, imperatives, or tenses. The occurrences of singulars are the focal points on which the utterer and the interpreter hook their attention. For example, "when the subject is not a proper name, or other designation of an individual within the experience (proximate or remote) of both speaker and auditor, the place of such designation is taken by a virtual precept stating how the hearer is to proceed in order to find an object to which the proposition is intended to refer".[52] There is no need for any separate existential presupposition for singular existence.

This idea appears highly interesting, and has not been given due attention in the extant literature on Peirce's logic. However, it also prompts new questions. For instance, we appear to need to be able to identify occurrences, or events, in addition to individual objects. This calls for its own philosophical and logical groundwork that addresses the identification of occurrences ('no eventuality without identity'). It may be that the same entities occur repeatedly in the universe of discourse, or that the occurrences, and not just the objects are vague and indefinite. Above all, events and occurrences appear to involve time, another complication for logical investigations.[53]

When a collection is no longer vague, it becomes, logically speaking, an individual. A singular individual is a typical example of a collection consisting of elements of the same kind, such as that consisting exclusively of the inhabitants of Mars. Overall, the account that Peirce provides is a forerunner of many later theories of plurals and collections, which often take the meaning of plural expressions in language to intend complex or compound individuals that dispense with quantification over second-order variables. Thus, these theories avoid falling back on everything to do with the mathematical concept of sets, which they circumvent by defining the denotations of plurals as compound singular objects, or individual sums, of individuals or sets of individuals.

> Just as we speak of whatever inhabitant of Mars there may be, so we can speak of whatever population of Mars there may be, although there may be none, and although for aught we know, it may be identical with the collection Alexander, Caesar, Hannibal, and Napoleon, should it turn out that, at the present writing, they are the only inhabitants of Mars. As an inhabitant of Mars is an individual *in essence*, whose individuality and identity are indeterminate, so the population of Mars is a collection *in essence*, which until it exists, is indeterminate. [...] Mathematicians and logicians speak of O, as a collection. At first sight, this appears to be not only a non-existent collection, but one whose existence is not even logically possible; and I have made the mistake in the text of saying it is only a collection in essence, not in existence. But it is not so. A collection is a singular whose existence *consists* in the existence of its members; that is, it is *sufficient* for its existence that whatever are its members should exist. Consequently, the collection O exists, even if nothing in the concrete universe exists. Hence, there is but one individual O collection; and the collection of no dogs is identical ... with the collection of no trees. Another point: Caesar and the collection of which Caesar is the sole member are not identical. For the existence of Caesar does not properly depend upon, or consist in, the existence of anything. Caesar is not a hypostatic abstraction or singular whose existence consists in the existence of something else, in the same sense in which the collection consisting of Caesar alone is so, by the very definition of a collection. (MS 690; HP: 741–742).

This summarises some of his main views on collections and plurals.

Logica utens and logica docens In relation to the logical reasoning that is most germane to mathematical activities, Peirce made a notable distinction between two logical faculties, the form of reasoning that resorts to the faculty of logica utens (logic in 'use' or in 'action') and reasoning that resorts to the faculty of logica docens ('theoretical', 'scientific' of 'educational' logic). The main difference is that the logica docens is the educable, improving, nurtured and schooled facility for reasoning while the logica utens is not. The latter is a native, stable, acquired, invariable, constitutional, secure, enduring and instinctive form of reasoning — like mathematics, as Peirce says, which "performs its reasonings by a *logica utens* which it develops for itself, and has no need of any appeal to a *logica docens*; for no disputes about reasoning arise in mathematics which need to be submitted to the principles of the philosophy of thought for decision".[54]

Peirce is thought to have adopted these terms from the scholastic philosophers. Indeed, they appear in John Buridan's (c.1295–c.1360) works dating from mid-14th century. Later, according to Ebbesen (1991), the Albertists of the late 15th century (followers of the logician Albert of Saxony, c.1316–c.1390), in their search for reliable knowledge involved in argumentation and disputation, rendered the distinction as one between the principally theoretical and the practically-oriented faculties. These terms did not appear in Albert's own writings, but his commentaries on Aristotle's Posterior Analytics in Perutilis logica ('Very Useful Logic', Venice 1522) follow in the footsteps of Buridan's commentaries, thus endorsing the view that logic is, in essence, practical knowledge.

Even earlier, the distinction was evoked in the writings of the Arabians in their cultivation and reinstatement of Aristotelian logic. If, moreover, the distinction is taken to resemble the general distinction between practical and theoretical reasoning, its origins are even older than this. In antiquity, it was reflected in art (wisdom) vs. techne. The difference transcends the boundaries of logic and science — take instinctive musicality vs. educable musicality, for example.

One might also wish to make a comparable distinction between pure and applied logic, as did the Aristotelian commentator āl-Fārābī (c.870–c.950, āl-Fārābī 1961), perhaps the first to highlight the Aristotelian distinction between theoretical and practical in relation to the rational faculty of human reasoning.[55] In our day and age, however, masking the docens/utens distinction on the pure/applied one may no longer be particularly happy, because what is being applied in applied logic is frequently based on some theories or systems originating and developed within the purview of logica docens.

Nevertheless, Peirce maintained that a proper theory of logic cannot exist without a reasoner having some general idea of what the preferred logical reasoning is.[56] He explained the logica utens as that which is found in the reasoners' instinctive and ingrained, inner life. By drawing attention to it, they become aware of one of their most fundamental native capacities, which thereby constitutes the basis of well-mannered theories of logical reasoning, argumentation and representation of logical concepts. In this sense, the logica utens is both a reasoning instinct and some elementary acquaintance or awareness of a theory of logic of what the ingrained reasoning of a species is.[57]

Logica utens is the bedrock of fundamental mathematical statements and the rules on which the truths of mathematical propositions hang. The validities of statements that this facility produces are beyond any doubt.

It is worth noting that, in his attempt to explain what a logical doctrine based on the utens could be, Peirce arrived at his vigorous criticism of the Cartesian view of philosophy and its prominence in the history of Western thought.

> Nothing is more irrational than false pretence. Yet the Cartesian philosophy, which ruled Europe for so long, is founded upon it. It pretended to doubt what it did not doubt. Let us not fall into that vice. You think that your *logica utens* is more or less unsatisfactory. But you do not doubt that there is *some* truth in it. Nor do I; nor does any man. Why cannot men see that what we do not doubt, we do not doubt; so that it is false pretence to pretend to call it in question? There are certain parts of your *logica utens* which nobody really doubts. Hegel and his [?] have loyally endeavored to cast a doubt upon it. The effort has been praiseworthy; but it has not succeeded. The truth of it is too evident. Mathematical reasoning holds. Why should it not? It relates only to the creations of the mind, concerning which there is no obstacle to our learning whatever is true of them. . . . It is fallible, as everything human is fallible. Twice two may perhaps not be four. But there is no more satisfactory way of assuring ourselves of anything than the mathematical way of assuring ourselves of mathematical theorems. No aid from the science of logic is called for in that field. As a fact, I have not the slightest doubt that twice two is four; nor have you. Then let us not pretend to doubt mathematical demonstrations of mathematical propositions so long as they are not open to mathematical criticism and have been submitted to sufficient examination and revision. The only concern that logic has with this sort of reasoning is to describe it. (2.192, c.1902, *General and Historical Survey of Logic: Why Study Logic?*).

The notion of mathematics that Peirce outlines in this passage refers to creations of the mind. This reminds us of his phenomenology. What is more, Peirce is seen to take a step towards constructivism in mathematics. I will return to Peirce's constructivist leanings in Chapter 6.

Because the logica utens is, according to Peirce, uncontrollable, it is not "subject to any normative laws" (2.204, c.1901–02). It is "neither good nor bad; it neither subserves an end nor fails to do so" (2.204). This implies that all reasoning refers to its general means of classifying arguments, shaping the foundations for any systematic study of the subject. It contrasts with the way in which educated logical reasoning is performed in scientific reasoning, the logica docens, in which reasoning is subservient to theory development. It is the propensity for anyone who practices reasoning. Logical laws such as the excluded middle, the non-contradiction, and the fact that at least some indubitable propositions are true, belong here and not to the realm of utens.

In my opinion, we should interpret the division between these two faculties to mean that any constructivist tendency we might entertain concerning the origins and status of mathematical constructs is not to be based on the validity of purely logical laws such as the law of excluded middle, but that derives at least partly from the specifics of our logica utens. This would be in accordance with Peirce's stout belief that mathematics is primary to logic. Foundations of mathematical systems change not because of changes in logic, but because of changes in what we take there to be in the general and common properties of the human mind that practices mathematics.

This is not to say that logica utens is in some sense pretheoretical, or not subject to its own laws and intrinsic structure of how it comes to be constructed. Neither is utens based on any institutionalised, conventional or impersonal theory concerning the admissible rules of some logical system. It is "instinctive logic" (MS L 75). Remarkably, the masking of utens as a form of instinct is not only consistent with, but also paramount to, Peirce's desire to implant instinctive aspects of reasoning in the notion of habitual reasoning.

Habits of reason According to Peirce, reasoning does not have its beginning in that focal moment at which judgements are found, but precedes them in determining judgements according to the general habits of reason, "which a reasoner may not be able precisely to formulate, but which he approves as conducive to true knowledge" (2.773, c.1901–02). True knowledge, stemming from reliable sources of information, is the locus in which belief beyond any doubt ultimately rests. To have a belief that cannot be doubted is constituted by a process of logical approval, but such a process lacks the essence of reasoning.

Logica utens refers to a doctrine that everyone has to accept. Unconscious in essence, it is not controllable by the active, reflective and self-aware mind. It is therefore not subject to criticism and does not form the subject matter of logic proper, or critic, which accomplishes quite the opposite in classifying arguments according to their value. For that which precedes the control of conscious processes is not subject to considerations of logical goodness or badness and hence truth, but is by its very nature logical reasoning that can be nothing other than good: if no fault is found in such antedating object of cognition, "it must be taken at its own valuation".[58]

Pretheoretical or unexcogitated notions of reasoning also constitute the subject of Peirce's fascinating unpublished manuscript 596, *Reason's Rules* [c.1902–03]. It is here that logica utens finds its home among nine other beliefs Peirce thinks are the presumed initial beliefs of the reasoners concerning their innermost reckoning and thought. The utens is described as a principle that enables reasoners, no matter how completely they have been brainwashed by systems and studies of logic, to distinguish between forms of reasoning that will be approved to lead to the truth and those to be considered dangerous.

The nine other beliefs concerning presumed initial beliefs of the reasoning are mutually understood to hold between active, deliberating agents, and thus forming a part of the common ground of the interlocutors.[59] These commonly and reciprocally recognised principles are the following. (i) The reasoner is in the state of doubt concerning some question and in the state of belief concerning others. (ii) The aim of inquiry is to produce a mental representation which shall be true, in other words accord with the real state of things. (iii) People consider some of their beliefs to be false. (iv) Certain "firm" beliefs beyond reasoner's doubt exist that are unanswerable if questioned by merely "yes" or "no". (v) There are also such firm beliefs that arise directly upon perception. (vi) On reflecting perceptual beliefs they appear infallible, that is, the seeming and the belief that something seems are in reasoner's opinion one and the same. (vii) To certain firm beliefs, both "yes" and "no" may be justifiable answers, provided that the question asked is not definite. (viii) If all acquired notions of logic could be cast aside, each reason would be judged by the agent's own sense of reasonableness. (ix) The agent's own logical judgements are in some degree erroneous and imperfect.[60]

If the faculty that performs reliable reasoning is so central and agglutinated unto our everyday life, what is it, then, that implements and triggers that faculty?

The answer is found in the notion of a habit, which is mentioned at a number of key junctures of Peirce's architectonics related to thought, cognition, interpretation and belief. It is one of the most far-reaching cross-categorial concepts in his philosophy. While its status has more or less been acknowledged in phenomenology and metaphysics, its bearings on logic have not been brought to the limelight. Even so, it is the last resort in his notion of the logica utens, as well as the key notion in the formation of the interpretants that are final in the sense detailed in his theory of signs.[61]

I believe that the return of the habit to our intellectual scenery is underway after its long overdue siesta. It will not only revive the utens/docens distinction but also contribute to the post-symbolic era of logical studies.

Despite there being a range of dimensions that habits may take, the relevant one concerns habits of reasoning that Peirce thought individual agents to possess. However, habits concern not so much the deductive task of logic than the inductive testing of hypotheses. In contrast to induction, the deductive and abductive tasks are related to sensory presentations and volitional reasoning, not to the learned, grown and cultivated processes that individuals have acquired during their existence and interaction with the environment. Habits are linked with evolutionary aspects of Peirce's metaphysics and cosmology such as continuity and law-formation, rather than with the processes of producing hypotheses or deriving conclusions in arguments. This is how the role of habits in Peirce's philosophy has traditionally been observed. But they are congenial also to the third compartment of logic, their role in inductive tasks.

Induction and the very substance of the logica utens are thus entangled. Their cultivation happens not only through presentation and perceptual input, but also through intensive cycles of communication. As Peirce's sign theory teaches us, to understand communication is not to confine it to the intersubjective notion, for instance that which Jürgen Habermas has advocated. He attempted to make assertions a public and linguistic, cognitively testable activities devoid of strategic concerns (Habermas 2001; see Chapter 2). In contrast, Peirce refers to processes that the utterer and the interpreter, conceived through aspects or phases of the mind, or encounters between the ego and the non-ego, are constantly undertaking, the entities who in Peirce's words possess "definite general tendencies of a tolerably stable nature".[62] The tendencies leading to stable outcomes remind us of the ways in which final interpretants of communicators are achieved, how various notions of equilibria arise in theories of actions, decisions, or general systems, or what the propensities are that pilot consistent inquirers to new scientific truths.

This wide notion of strategic communication is what is needed to expose common aspects of reality, in other words to generalise, invent and improve on the basis of information and reasoning, through habituated and pre-programmed responses to what is presented to the mind, its 'run-off'.

It is vital to recall, however, that the concept of a habit was by no means Peirce's own invention (although he tended to use the indefinite article). The term was frequently used for explanatory purposes in many branches of science in the 19th century and earlier. It was thought to be the key to many doors of philosophy, not only in relation to philosophers from David Hume (1711–1776) and Kant to 19th-century pragmatists including John Dewey (1859–1952) and James, but also (and more commonly) for a number of political and social scientists, psychologists, institutional economists, and biologists throughout the 19th century and the early 20th century.

Camic (1986) has traced some of the sociological history of this fascinating concept from the past, with a fleeting look back at its philosophical significance. He makes no mention of Peirce, but on page 1046 notes that Dewey had considered habit to play a considerable role in pragmatics, being the dynamic and projective systematisation of human action.[63] James gave it a psychological twist, which perhaps comes closest to what we nowadays understand by it.[64] The notion even infiltrated the then-quite-antipragmatic atmosphere of logical empiricists, especially through Otto Neurath (1882–1945) and his intrepid use of the habit as custom. At that time, regrettably, habits had already acquired behaviouristic overtones.

The notion was far-reaching in Peirce's systematisation of his pragmatism, and had significant repercussions on the improvements upon logic. A broader use of habits and their evolution across scientific disciplines starting from as far back as Aristotle's *ethos* (ἔθοσ: 'custom, habit') has not yet been assigned its proper place within the history of the development of intellectual ideas.[65] But the story continues in later chapters.

Notes

1 The application is MS L 75 (1902), *Logic, Regarded as Semeiotic (The Carnegie application of 1902)*, available electronically at www.door.net/arisbe/arisbe.htm (accessed 31 December 2004). An editorial introduction to it is available at the Arisbe by Joseph Ransdell, "The significance of Peirce's application to the Carnegie Institution", 1998. See also HP 2:1022–41.

2 MS 302, one of the drafts of the *Lowell Lectures* series for the Lowell Institute in Harvard, Boston, delivered in winter of 1903–1904. For these Lectures, see MSS 454–457.

3 MS 280 is one of the versions in the series of drafts entitled *The Basis of Pragmaticism*, composed mostly in autumn 1905, all of them left unpublished. Their follow-up was the series entitled *Prolegomena to an Apology for Pragmaticism*, three of which Peirce published in The Monist between 1905 and 1906. The fourth paper in this series, in which he promised to provide the "proof of pragmaticism", never appeared. Several scholars have attempted to recover and reconstruct this intended draft, in the hope of rebuilding the alleged proof (Hookway 2000; McCarthy 1990; Roberts 1973b; Robin 1997).

4 See Kant (1988b, p. 96): "All cognitions, that is, all presentations consciously referred to an object, are either *intuitions* or *concepts*. Intuition is a *singular* presentation (*repraesentio singularis*), the concept is a general (*repraesentation per notas communes*) or *reflected* presentation (*repraesentatio discursiva*). Cognition through concepts is called thinking (*cognition discursiva*)".

5 MS 516: 39, *On the Basic Rules of Logical Transformation*.

6 MS 483, c.1901, *On Existential Graphs*.

7 MS 491: 3–4, c.1903, *Logical Tracts. No. 1. On Existential Graphs*.

8 The qualification "typically" refers to the possibility of there being vague propositions, which are those to which the law of contradiction does not apply. I will not go on to discuss Peirce's theory of vagueness and indeterminacy in any great length in this work, apart from a few pointers.

9 The Prolegomena appeared in *The Monist* 16, pp. 492–546, 1906.

10 MS 500, 6–9 December, 1911, *A Diagrammatic Syntax*, Letter to Risteen on Existential Graphs.

11 Around the time Peirce regarded EGs as the key to logical analysis, he came to hold the algebraic logic of relatives secondary to his diagrammatic system. He by no means abandoned algebraic logic, and continued to improve upon this earlier pet topic of his in many of his very latest manuscripts.

12 MS 508, probably 1903, B.4: *Syllabus B*, early draft of 4.414–417.

13 The term "transcendental" in *Critique of Pure Reason* has caused misapprehension and confusion. Kant himself was quick to note the possibility of misconceptions and preferred to term his philosophy "critical idealism": "My idealism concerns not the existence of things (the doubting of which, however, constitutes idealism in the ordinary sense), since it never came into my head to doubt it, but it concerns the sensuous representations of things to which space and time especially belong. Of these [namely, space and time], consequently of all appearances in general, I have only shown that they are neither things (but mere modes of representation) nor determinations belonging to things in themselves. But the word 'transcendental', which with me never means a reference of our knowledge to things, but only to the cognitive faculty, was meant to obviate this misconception. Yet rather than give further occasion to it by this word, I now retract it and desire this idealism of mine to be called 'critical"' (Kant, 1988a, p. 41).

14 Accordingly, I will henceforth use the term pragmatism to refer to Peirce's pragmaticism. His late draft, *The argument for Pragmatism anachazomenally or recessively stated* (MS 330), is quite helpful in sketching how the proof is to be accomplished in thirteen steps. Peirce aims at showing that, via these steps, the premiss "The meaning of an intellectual concept consists in the general manner in which it might modify deliberate conduct" amounts to the conclusion that "the only essence of the concept — its logical interpretant — is the generalized habit of conduct". One of the crucial steps is to demonstrate that, "Any existence that involves continuity cannot be singular but must be a compound of $> n$ triads". Accordingly, it is the continuous nature of existence in diagrammatic logic of graphs that Peirce must have believed to be utterly instructive in showing how this crucial step ought to be made.

15 The idea is related to 'specific indefinites' widely discussed by contemporary semanticists. Unfortunately for them, the discussion is quite corrupt because it fails to take the requisite dialogical perspective into account. Typically, in the specific indefinites idea one hopes to be able to clarify the nature of indefinites merely in terms of differences in the scope relations of quantificational expressions, possibly interspersed with some magical referential intentions assumed of the indefinites.

16 MS 646: 21, 19 January 1910, *Definition, 4th Draught (Studies of Logical Analysis)*.

17 Incidentally, the origins of the verb *to write* in many languages (e.g. in Finnish, cf. *kirjoittaa* [to write] vs. *kirjoa* [to inscribe, embroider]) are to be found in the arts of drawing, presenting pictorially, or putting forth a texture. These activities require the presence of suitable media, and are used for communicative purposes in quite the same way as the writing that is based on the linear structure of language.

18 See also my www.helsinki.fi/˜pietarin/courses/ (accessed 31 December 2004) for an online learning resource concerning Peirce's logic and philosophy.

19 The quote by Benjamin Peirce Ellis is from Ketner (1998, p. 18).

20 *The Photometric Researches: Made in the Years 1872–1875*, *Annals of the Astronomical Observatory of Harvard College* 9 (1878), 181 pp.

21 The material in 1.176–179 (c.1896) is a foreword to the projected first volume of The Principles. In the same year that the project was announced (1894), Herbert Spencer's *Principles of Psychology* appeared, of which Peirce commented extensively.

22 One is reminded here of Kurt Gödel's reply to the question posed by Karl Menger, asking Gödel, who at one time tried to account Leibniz' literary remains, "Who had an *interest* in destroying Leibniz' writings?" Gödel's answer was laconic: "Naturally those people who do not want man to become more intelligent" (Menger, 1994, p. 223). As Houser (1992) makes it clear, Harvard University to which Peirce's literary remains were eventually deposited, had in its possession a monster easier to lock up than harness.

23 The Peirce Edition Project: www.iupui.edu/˜peirce/web/ (accessed 31 December 2004).

24 Murphey (1961) pointed out that Peirce's scepticism refers to the task he set to himself in terms of his unsuccessful Carnegie Institute grant application, not his whole oeuvre.

25 5.121, 1903, *Lectures on Pragmaticism: The Three Kinds of Goodness*.

26 MS 339: 222r, 22 July 1902. I abbreviate the Logic Notebook by LN, followed, if provided, by page, date and title.

27 Victoria Lady Welby (1837–1912) was Peirce's active correspondent in England. The founder and conceptualiser of significs, her theory of signs on which the analysis of meaning was based provided an influential cultural locus of an early Victorian era. She was a prominent writer and critique on broad spectrum of communicational, epistemological, cultural and ethical aspects of science (Schmitz, 1990). The term significs was well received and appreciated in early 1900s, but it did not survive and was later replaced by the misplaced semantics/pragmatics division suggested by Charles W. Morris (1901–1979) and a year later by Rudolf Carnap (1891–1970), together with many others who followed. The term 'semantics' is due to Michel Bréal (1832–1915), prominent linguist in France with whom Lady Welby corresponded. In his work, the term lost philosophical and logical connotations and was bound to remain detached from Welby's theory of significs. Among the notable offspring of her work was the Signific Movement in the Netherlands, the activities of which continued until 1950s, and in the sessions such luminaries as Luitzen Egbertus Jan Brouwer (1881–1966) took spry part. The influential early work The Meaning of Meaning by Ogden & Richards (1923) was also the product of this era of thinking. The book was a crucial link in transmitting parts of the still-today-significant Peirce–Welby correspondence on the theory of signs and meaning, and making their affine ideas available for a wider audience for the first time. What is the relevance for significs in today's science? Its ideas prefigured those of speech act theories, but aimed at setting them into a wider context of communicative acts. It is likely, though unfortunate, that the emergence of cognitive science, psycholinguistics and various communicational sciences including artificial intelligence pre-empted the natural habitat for any future science of significs. But this points to a longer story that is being told elsewhere (Pietarinen, 2003g).

28 8.328, 1904, *Letter to Lady Welby*.

29 MS 499, 1906, *On the System of Existential Graphs Considered as an Instrument for the Investigation of Logic*. MS 499 is a follow-up of MS 498 (*On Existential Graphs as an Instrument of Logical Research*). Both drafts were prepared as preliminary addresses to one of the 1906 meetings of the National Academy of Sciences. MS 490, 1906, *[Introduction to Existential Graphs and an Improvement on the Gamma Graphs]*, is the version Peirce ended up presenting at April 1906 meeting in Washington.

30 MS L 427: 2, *Letter to Charles Augustus Strong*, 25 July 1904, cf. MS L 75.

31 MS 8: 5–6, c.1903?, *On the Foundations of Mathematics*.

32 MS 283: 106, 1905, *The Basis of Pragmaticism*; cf. MS 793, probably 1906, [*On Signs*].

33 The paper appeared in the *Journal of Speculative Philosophy* 2, (1868), pp. 103–114.

34 MS 9: 1, c.1903, *Foundations of Mathematics*.

35 MS 292: 23, c.1906, *Prolegomena*.
36 MS L 237: 1, 12 June 1902, *Letter to Ladd-Franklin*.
37 *A Guess at the Riddle*: MS 909; 1.354–416; EP 1:245–279; W 6:166–210.
38 The term–proposition–inference triad is the same as the rhema–proposition–argument one.
39 MS 515: 20, *On the First Principles of Logical Algebra (First Print)*.
40 See MS 515: 20; MS 690, c.1901, *On the Logic of Drawing History from Ancient Documents, Especially from Testimonies*, reprinted partially in EP 2:75–114 and in 7.164–231 and, most comprehensively, in W6.
41 2.243, c.1903, *Speculative Grammar: Division of Signs*.
42 A marvellous applet by Priscial Farias that shows the interactions between signs in 3D is available at www.digitalpeirce.fee.unicamp.br/p-intfar.htm (accessed 31 December 2004).
43 Even so, Peirce did not maintain an account of information comparable to the emergence of the plethora of different varieties of interpretants. There is little evidence that Peirce showed similar interest in the notion of information in his later works, so one is well-advised to check the early account for consistency with the later classification of interpretants.
44 A. Bayart, Marcel Guillaume, Jaakko Hintikka, Stig Kanger, George Kelly, Saul Kripke, Carew A. Meredith and Arthur Prior, Richard Montague and D. Kalish were among those who came up with ideas of an accessibility or an alternativeness relation between possible worlds. The references here are to Kelly (1955); Meredith (1956); Kanger (1957a,b); Hintikka (1957a), Hintikka (1957b) (especially pp. 61–62), Bayart (1958); Guillaume (1958); Montague & Kalish (1959), and Kripke (1963). The role of Kelly's work on the geometry of psychology in this cluster of early works is unclear, and yet to be documented. He assumes 'contingent schemata', which begins with beliefs about the way the world is, and projecting onto the set of beliefs a conceptual framework for dividing up a cognitive space. His 'construct' may perhaps be viewed as the actual world and 'dichotomy' and 'poles' as valuations. He then introduced the 'range of convenience', not a relation in the mathematical sense but a version of accessibility in the modal sense. Jónsson & Tarski (1951) used a binary relation, but for a different purpose. See Lindström (2001); Copeland (2002) for studies on the development of possible-worlds semantics, certainly not the last words on the subject.
45 MacColl's views on modalities are expressed in MacColl (1886–7). He divided propositions into three classes: certainties (propositions that are necessarily and always true), impossibilities (propositions that are necessarily and always false) and variables (propositions that are neither of above). The law of excluded fourth holds, however, in that every proposition belongs to one of these three classes.
46 5.484, c.1907, *A Survey of Pragmaticism*.
47 4.536, 1905–06, *Prolegomena*.
48 8.184, *Review of Lady Welby, What is Meaning?*
49 MS 295: 28, c.1906, Rejected pages for the *Monist* article of 1906.
50 From the missing page 2 of the original HP: 737, reprinted in *Companion to EP 2*, Selection 8, Page 98, Line 12, Note 31, Peirce Edition Project, Electronic Supplement.
51 For instance, according to the discourse representation theory (DRT) of Kamp & Reyle (1993), abstraction is defined in order to treat plurals (see Chapter 5). Their terminology is somewhat inept, because what they mean by abstraction is quite the converse of what it means in Peirce's jargon, namely it refers to the process of introducing a 'discourse referent' which is a union of previously-introduced individual discourse referents and hence a set, but formed so as to enable one to refer to its components by plural anaphoric pronouns. Moreover, the use of the term 'sets' is misleading in DRT, since such 'sets' do not relate to any set theory in a mathematical sense, the reason being that the latter have no need to recognise the identities of individuals.
52 2.357, 1901, *Speculative Grammar: Propositions*.
53 Of course, time itself may be the substance of logic. Peirce conducted several studies not only on Kant's conception of time but also on the possibility of having a real logic of time. He did not take time in any way to be an extra-logical matter. In LN: 340r [7 January 1909] he considered propositions that "are true sometimes" and those that are true "under all circumstances". Earlier in 20 September 1905 he hinted that probability calculus depends on an essential property of time of "the future being like the past", which ought to afford the key to the nature of time (LN: 249r). Accordingly, he did not land very far from the possible-worlds conception of much later versions of temporal logic. Øhrstrøm (1997) studies elements of temporal logic in Peirce's system of gamma graphs; cf. Chapter 4.

54 1.417, c.1896, *Phenomenology: The Three Categories*.

55 "The rational faculty . . . is partly practical and partly theoretical. The practical is partly a matter of skill and partly reflective. The theoretical is that by which man knows the existents which are not such that we can make them and alter them from one condition to another, e.g. three is an odd and four an even number. . . . The practical is that by which are distinguished the things which are such that *we* can make them and alter them from one condition to another" (al-Fārābī, 1961, pp. 30–31). "Practical" may be interpreted also as pertaining to "calculative", "speculative" or, as I will be suggesting, the "strategic" compartments of reasoning.

56 2.186, c.1902, *General and Historical Survey of Logic. Why Study Logic? Logica Utens*.

57 And so Fann (1970)'s view according to which Peirce held only the utens to be about a theory of what constitutes a good reasoning, is wrong. Chiasson (2001) seems to hold — admittedly in a nebulous dialogical prose that is hard to pin down — the narrow view that attempts to equate the docens with formal logic, which according to Peirce's classification would pertain to mathematics. But not all theories of logic need to be formal in the least.

58 5.114, 1903, *Lectures on Pragmatism: The Reality of Thirdness. Normative Judgments*. Taking something "at its own valuation" does not make the utens normative, since there is no *decision* that could be made between good and bad.

59 The common ground, central for Peirce, will recur in several contexts in later chapters.

60 A little later on, Peirce continues by commenting on these opinions. Only the commentary part of the manuscript was published in the CP, as 5.538, *Belief and Judgement*.

61 Such final interpretants may be thought as equilibrium or saddle points in the general sense of equilibrium systems in systems theory, or as local optima in various optimisation tasks, stable sets in game theory, and so on.

62 MS 280: 30, c.1905, *The Basis of Pragmaticism*.

63 The textual context reads as follows: "The function of knowledge is to make one experience freely available in other experiences. The word "freely" marks the difference between the principle of knowledge and that of habit. Habit means that an individual undergoes a modification through an experience, which modification forms a predisposition to easier and more effective action in a like direction in the future. Thus it also has the function of making one experience available in subsequent experiences" (Dewey, 1997/1916, p. 212).

64 According to James, "When we look at living creatures from an outward point of view, one of the things that strike us is that they are bundles of habits. In wild animals, the usual round of daily behavior seems a necessity implanted at birth; in animals domesticated, and especially in man, it seems, to a great extent, to be the result of education. The habits to which there is an innate tendency are called instincts; some of those due to education would by most persons be called acts of reason. It thus appears that habit covers a very large part of life, and that one engaged in studying the objective manifestations of mind is bound at the very outset to define clearly what its limits are" (James, 1997/1890, p. 60).

65 Kilpinen (2000) provides a comprehensive socio-philosophical study of the history and the development of the notion of a habit in and about the pragmatist genre.

Chapter 2

FROM PRAGMATISM TO THE PRAGMATICS OF COMMUNICATION

> SOOTHSAYER: If thou dost play with him at any game,
> Thou art sure to lose; and, of that natural luck,
> He beats thee 'gainst the odds: thy lustre thickens,
> When he shines by: I say again, thy spirit
> Is all afraid to govern thee near him;
> But, he away, 'tis noble.
> — Antony and Cleopatra, *Act II, scene iii*.

From the introductory atmosphere of the previous chapter I will switch to the systematic consideration of a couple of themes that are central to the forthcoming chapters. Two such key issues are the notion of communication and its multifarious use in Peirce's philosophy, and the early ideas anticipating the science of pragmatics. A thus far slighted aspect of Peirce's work concerns the relation of his sign theory to neuroscience, a topic treated in the appendix.

1. Peirce, communication and formal pragmatics

The significance of the communicational perspective in Peirce's philosophy pervades a number of logical and semeiotic topics. His conviction was that there is no fundamental separation between logical and semeiotic fundamentals of philosophy. He broadened the scope of what later 20th-century logicians came to consider to fall within the purview of logic to include, not only speculative grammar and logic proper (critic) but also methodeutic (speculative rhetoric). These three are routinely taken to correspond to the study of syntax, semantics and pragmatics, respectively. This view is itself certainly limited and inaccurate. However, the latest advancements in formal pragmatics are already a step towards the Peircean ideal of logic as the unlimited, unrestrained and ever-broadening study of all signs and their associated cognitive actions.[1]

Critique of Habermas Space permits no review of the advancements made in formal pragmatics, but one perspective, less formal and more philosophical, is adopted in Jürgen Habermas' writings on the pragmatics of communication.[2] In particular, the role of communicative elements in Peirce's philosophy was one of the subjects of discussion and criticism in Habermas' later works (Habermas, 1995). This is significant, because it is only quite recently that the substance of Peirce's theory of communicative signs has contributed to a better understanding of the overall theory and nature of signs and logic.

Habermas tends to read Peirce rather dismissively. According to him, Peirce's characterisation of communication in semeiotic composition of thought is so abstract, and so detached from what communication ordinarily should be about, namely pertaining to linguistic or at least symbolic signs, that he ran the risk of losing the interpersonal notion of agenthood and patienthood from the processual notion of semiosis altogether. And this is what Habermas thinks did happen: according to him, Peirce shifted focus to an attempt to characterise abstract communication in a non-interpersonal way, but this characterisation fell short of accounting for logical aspects of sign processes.

The abstract characterisation of communication that Peirce describes admittedly distances itself from the mere interpretation of symbolic signs, and is meant to refer to all possible relations between signs and their interpreting minds. These interpreting minds were referred to by what he sometimes termed quasi-minds, sometimes theatres of consciousness, and sometimes still something else (Chapter 3). What they all have in common is that they mean entities that can be thought of as bearing a relation to a communicative situation. They play different roles in different contexts, theories and applications. Communication, in turn, is closely related to the representational character of signs. Just as all signs are representations, or stand for something with regard to some respect or capacity, communication is the abstract relation that obtains between the sign and its interpreter. If the interpreter, the recipient or the presence of the sign, is the sign itself, and there is no separate entity that functions as the interpreter, it may be any one period in the sequence of moments of successive sign appearances that becomes the interpreter of the original sign. The ensuing interpretation becomes the next moment of appearance of the sign in the potentially endless cycle of interpretations.

There is not necessarily any person originating the sign, or interpreting it, although, as Peirce notes, "it will contribute to perspicuity to use language as if such were the case, and to speak of the *utterer* and the *interpreter*".[3] Moreover, "It is convenient to conceive the comparison of an assertion with the real universe to be conducted by a sufficiently informed defender of the assertion and a sufficiently informed opponent of it" (MS 491). He went even further, acknowledging that the *necessity* of imagining two persons was not obvious (MS 280: 30–31 a.p.). He then sketched the following argument. It starts from the three assumptions that signs exist, that they convey ideas, and that they spring from the factory of ideas, conventionally termed the "minds". These assumptions lead to the conclusion that signs are subordinate to interpretation by something or someone who understands this factory, by virtue of the fact that these "understanders" must arrive at the agreement or convention concerning signs. These agents, typically of the opposing polarities of the defenders (the utterers) and the opponents (the interpreters) of a given assertion or some other sign must be of some general nature, because otherwise they cannot understand the mind as a sign producer or generator (cf. the assimilation between Peircean "mind" and Kantian "generality" argued for in Chapter 1). In addition to the agreement or understanding, these agents and patients of some general nature must recognise when an agreement is reached in order to terminate the process of interpretation. Although the impact of the latter point was not explained very cogently by Peirce in the context of this argument, it appears to tie in with his evolutionary views on signs, which take chance and mutation, necessity and spontaneity, as prerequisite in the growth of law and order of Nature. The decidability of signs, or the termination of interpretation, is thus needed in order for these sign factories to survive and regenerate.

The personalised view of dialogues, in which effects of objects and interpretants are determined upon a person, was "a sop to Cerberus", as Peirce was struggling to make his conception of signs and logic understood (EP 2:478). There are various stances that can be taken with respect to this sop; if the identification is taken literally, the dialogues will take place between actual speakers and hearers of the language, and thus act according to the principles of linguistic communication, such as conversational maxims, conventions governing acceptable use, and speaker and hearer's linguistic competence. If the relation is contingent, the references to utterers and interpreters, where convenient, may function as merely a technical device in theories of logic and language, with no compelling need to assimilate actors within a theory with actual persons. The latter stance is typically taken in game-theoretic semantics (GTS; Chapter 7) while the former is commonplace in systems of natural language dialogues and in inquiries concerning social contexts of language. Neither semantic nor pragmatic theories of language claim any distinction between the two per se, and nothing would be gained by trying to lump Peirce's theory of communication within the artificial boundaries drawn by these two points of view.

Peirce allowed for both possibilities. Assertions may be taken to be produced by real persons in real communicative situations in linguistic communities, in which case they relate to those pragmatic theories that aim at accounting theoretical underpinnings of communicational action in societies of language users. However, in his sign theory, there is considerable leeway for one to take utterers and interpreters as abstract notions congenial to objects and interpretants of the signs, in which case the personality is more or less illusory (Chapter 13).

Despite this freedom, the nature of utterers and interpreters remained a puzzle to which Peirce did not find a satisfactory solution. In his manuscripts 498 and 499, he sets out to consider the question of what value existential graphs (EGs) might have in illustrating the relation of a sign to the mind of its utterer and to the mind of its interpreter, and the question of the mode of composition of concepts such as the structure of a judgement.[4]

What these writings eventually convey is a set of desperate attempts to tackle these questions without ever finding a comprehensive explanation. What he wanted to show was that "all deliberative mediation, or thinking, takes the form of a dialogue. The person divides himself into two parties which endeavour to persuade each other. From this and sundry other strong reasons, it appears that all cognitive thought is of the nature of a sign or communication from an uttering mind to an interpreting mind" (MS 498). Moreover, in another manuscript we find, "all genuine reasoning is carried on in the form of a dialogue, the self of one moment appealing to the self of the next moment. Now dialogue can only be carried on in signs" (MS 296: 11).

What he succeeded in doing was to rework the puzzle into a format in which it boils down to the question of the possibility of two communicating minds becoming fused or welded into one mind. This comes close to his idea that in communication, a common overlapping mind emerges, the *commens*, occupied by the utterers and the interpreters of signs and constituted by the parts of their minds that is presented with their common ground (Chapter 13). Nevertheless, the purported answer, the only answer that Peirce thought was available, is provided by EGs. As in so many other instances too, he presents the system in a quite detailed fashion (MS 498), but never gets into the business of sketching the answer to the initial question.[5]

Nevertheless, it is not true that, faced with the impenetrability of these puzzles, Peirce wanted to move away from the communicational issues concerning the interpretation of signs. Such issues simply could not have been suppressed. They mushroomed even in his most formal and detailed studies in logic, including his theory of EGs, his 1885 work on quantifiers in logic, and his overall communicational and dialogical outlook on the logical structure of natural language expressions. The mere 'as if' character of the personalised conception of agents and patients was not a sufficient reason to repudiate them. Accordingly, he decided to pursue precisely this communicational line and thought it to add to the perspicuity of logical studies. He also urged others to continue pursuing it in many of his very latest manuscripts dating from the last years and months of his life. He was quite convinced that this was the way ahead in logic, especially in developing further the system of EGs to cover new areas that would take into account non-declarative assertions and non-propositionality.

The Symbolic Turn However, the logic of the 20th century took a different turn. It betrayed Peirce. It embarked on quite different principles, not on those he brought into the open, but on those that were suggested by symbolically-inclined philosophical and mathematical logicians, many of whom were acquainted, to some extent at least, with what Peirce was doing, but who all the same decided to turn a blind eye to it. The betrayal was thoroughgoing. Not only was Peirce's agenda ignored *en masse*, but also the logic of the 20th century considered the time to be right for some major refurbishment and deep-mouthed discoveries in the area what of what Peirce had pigeonholed to "symbolic logic".[6] These investigations, already anticipated by Peirce to "not be rated as much higher than puerile" (MS 499), found the fundamental value of symbolic logic to be in its mathematical apparatus as a calculus.[7]

These logicians and philosophers did not take the 'as if' character of there being persons, or corresponding theoretical entities, in any way helpful in logic. The reason for this is not altogether clear. Most remained ignorant of Peirce, and did not invent anything comparable to his architectonics themselves. A practical reason was the long-term unavailability of his published work in a decent book form during his lifetime and after his death, although this lack of access cannot explain everything. The communicational aspect of logic, being the species of thirdness, was long thought of by the commentators on, and the interpreters of, Peirce as excess curiosity, or of minor importance in logic in its own right, even an object of ridicule and abhorrence. This lasted until the end of the 20th century. Only the very latest advancements, many of them due to computational sciences, have finally vindicated this interactional and dialogical character of signs and representations (Chapter 8).

Apart from the turn to the informational and computational sciences that philosophical logic took during the late 20th century, this conceptual monster (if its existence was ever genuine in the sense of being a conceptually plausible undertaking) has morphed into the transdisciplinary sciences of logic, language and information. The cataclysmal increase in the existence of concrete physical communication systems and devices in our society has served as an eye-opener for many neglected aspects of Peirce's philosophy. This should be no surprise, if one recalls the effect that the second industrial revolution and the accompanying post-Civil War innovations of the late 19th century had on the scientific world-view of that era. Then, too, technological inventions and mass-communicative hype were all-pervasive. One thing of note is that the invention of moving images and the emerging film industry of the1890s served as a dramatic catalyst on the directions Peirce's logic was taking, namely towards visual representation of all assertions through his EGs (Chapter 4).

Why, then, did the earlier logicians and philosophers of the 20th century tend to ignore the possibility of the immanent dialogical or communicational character of logic, especially one involving the category of thirdness? Several educated scholars studied Peirce's manuscripts at Harvard soon after his death. But why did Clarence Irving Lewis not appropriate this aspect of Peirce's logic while he was virtually living with Peirce's manuscripts at Harvard for two years (Houser, 1992), to complement his modal logical approach to hypotheticals?[8] Why did Bertrand Russell not acknowledge Peirce's influence half-way adequately?[9]

Given the mediation of at least some aspects of Peirce's philosophy by Frank Ramsey to Ludwig Wittgenstein, why did Wittgenstein, too, continue to practice his custom of not acknowledging the obvious sources, even though he decided in the preface to the Philosophical Investigations to acknowledge Ramsey, who is still believed to be his chief informant on pragmatic philosophy?[10] Above all, what were the true reasons that contributed to the unanimous overshadowing of Peirce's work not only by the practitioners of symbolic logic, but also by the philosophical and scientific community at large?

I have no intention of even trying to suggest any definite answers to these weighty questions within the confines of this study. Lewis, Russell and Wittgenstein worked out their influential systems on their own, but did not lack predecessors given the order in the history of intellectual ideas. Nevertheless, these figures served as both the prisoners and the guardians of a cultural era that no longer encouraged an entirely free flow of thought in logic, or in science more generally. Apart from this sociopolitical romantisation, the overall view of language at the beginning of the 20th century was radically different form Peirce's. Hintikka has reminded in several places that Russell, Wittgenstein and many others, including Frege, Neurath and Quine, did not consider language or its supposed logical form to be an indefinitely re-interpretable calculus (Hintikka, 1996c). Instead, they took language to be a universal, symbolic medium for communicating all thought that cannot be studied from a perspective outside of the system that was being studied, that is, outside of language itself. Peirce did not endorse this view, as his frequent remarks on his ineptitude in expressing his views using linguistic means readily testify. As I hope to show in later chapters, Peirce, in fact, promoted a model-theoretic outlook on logic that was even more advanced than currently believed.

Communication, competition, cooperation What is also worth addressing here is another point that Habermas makes, concerning the wedge he wishes to drive between what is communicative and what is strategic in the sign-theoretic framework of linguistic action. He considers these two modifiers as fundamentally different in his own account of communicational practices, and proceeds to suggest that a similar division ought to apply to Peirce's views on communication (Habermas, 1995). His motivation in drawing this division is not to show that the attitudes of the participants may differ in the two, communicative and strategic, settings, but because he sees some transparent structural differences in them. In communication, the structure of how language is used is "superimposed" on the goal-driven action structure (Habermas, 1998, p. 205). Communication is replete with notions such as presuppositions, performatives, and other less objective constraints than strategic action. Its essence is in the idea of interaction, but in Habermas' view not in one that ought to take in strategic considerations. On the other side, strategic interaction is parasitic on communicative interaction.

I do not seek to disagree with the view that there is a tangible difference between how various types of actions involving pragmatic constraints on language ought to be characterised. However, I believe it is misguided to try to draw these distinctions in terms of what is strategic and what is non-strategic, because strategic action can be performed either cooperatively or non-cooperatively. The kind of communication designated to the understanding and interpretation of utterances rather than to serving some further goals beyond comprehension lies still largely within the purview of strategic situations. Formally, we may well have variable-sum games in which it is the outcomes assigned to total strategies that mark the varying levels of understanding and interpretation with respect to the communicating participants.

What Habermas seeks to explain is that, if the crux of the strategic interaction falls within the principle of utility maximisation and hence in self-interest, then it is incompatible with the principles of language use aimed at mutual, interchanged and reciprocal common understanding. However, the principle of utility maximisation, as operationalised within the interactive setting of game theory by means of the notion of strategies (or perhaps by 'plans of actions', a slightly more condensed notion), may well be one that satisfies all of the desiderata of the kind of communication that Habermas wants to have, namely reaching understanding and agreement, having coordination, and having cooperation. One just needs to shift the focus to more permissible classes of strategic interaction, such as cooperation, negotiation, bargaining, variable sums, and so on. This does not diminish the scope of the study of communication in the least; on the contrary, one inherits more precise tools for tackling questions concerning the structure of communication and discourse for the field of inquiry that Habermas entitles formal pragmatics.

Habermas addresses little of the truly formal side of pragmatics, however. His goal is rather to reconstruct the meaning of linguistic competence and awareness of the rules of language. He assumes that language has an in-built notion of validity of which assertions make use. The force that certain acts, such as illocutionary ones have, is based on the assumption that such assertions can be checked for validity. For one thing, this has obvious commonalties with Peirce's view of assertions as acts where the utterers are accepting responsibilities of their truth. Second, because these acts are obliging they are able to convince the hearers. This latter fact is expressed by Habermas as follows:

> With their illocutionary acts, speaker and hearer raise validity claims and demand that they be recognized. But this recognition need not follow irrationally, since the validity claims have a cognitive character and can be tested. I would like, therefore, to defend the following thesis: *In the final analysis, the speaker can illocutionarily influence the hearer, and vice versa, because speech-act-typical obligations are connected with cognitively testable validity claims* — that is, because

> the reciprocal binding and bonding relationship has a rational basis. The speaker who commits herself normally connects the specific sense in which she would like to take up an interpersonal relationship with a thematically stressed validity claim and thereby chooses a specific mode of communication. (Habermas, 2001, p. 85).[11]

It should thus be understood that the term "formal" is to be taken in the sense of "rational reconstruction" rather than in its logico-mathematical sense. This does not provide any excuse for Habermas to insist on the illusory difference between what is cooperative and what is strategic, however. Had his investigation moved on a somewhat more detailed level of rational reconstruction, facets of this illusion might have been exposed much earlier.

To put the point in simple terms, what I believe Habermas is after is that communicative action aims at reaching understanding, whereas strategic action exerts influence on other members of linguistic community. Contrary to what he sees as implications of this division, the former is not devoid of purpose. In explaining what people do, its formal accounts need goal-driven action structures just as they are needed in the latter. This is not to say that the application of strategic games to cooperative behaviour is straightforward. But its potential in dealing with communicative aspects of behaviour and reaching understanding has been recognised since the early phases of the formation of the theory of games:

> Even if the theory of noncooperative games were in a completely satisfactory state, there appear to be difficulties in connection with the reduction of cooperative games to noncooperative games. It is extremely difficult in practice to introduce into the cooperative games the moves corresponding to negotiations in a way which will reflect all the infinite variety permissible in the cooperative game, and to do this without giving one player an artificial advantage (because of his having the first chance to make an offer, let us say). (McKinsey, 1954, p. 359).

The infinite variety permissible in cooperative games is precisely the kind of problem we encounter in studies of formal pragmatics, which not only deal with conventional signs but, among other things, also cover interpretants of indexical and iconic signs. They are very difficult to be reduced to some formal framework of contingent but observable behaviour.

Even more candidly, thirty years later Shubik (1985, p. 293) reiterates this point, suggesting that

> in much of actual bargaining and negotiation, communication about contingent behaviour is in words or gestures, sometimes with and sometimes without contracts and binding agreements. A major difficulty in applying game theory to the study of bargaining or negotiation is that the theory is not designed to deal with words and gestures — especially when they are deliberately ambiguous — as moves. Verbal sallies pose two unresolved problems in game-theoretic modelling: (1) how to code words, (2) how to describe the degree of commitment.

While my study, confined as it is to a few aspects of Peirce's views on language, logic, dialogue and communication, is not indented to directly respond to the challenges of these two passages, it is in order to note that a merger of not only the formal theory of games and GTS, but also of Peirce's pragmatic theory of signs and the dialogical study of linguistic assertions is a step towards a better understanding of the issue of noncooperation versus cooperation. The game-theoretic framework presented in Chapter 7 takes into account not only lexical items, but also complete linguistic expressions, and can be generalised to the study of discourse (Janasik et al., 2002). The strategic component copes with the pragmatic and even non-linguistic communication that is pervasive among language users. It would thus be misleading to label the approach strictly noncooperative, because an element of mutual agreement and reciprocal purpose is prevalent in sign action. The passage quoted above is nonetheless instructive in that game-theorists have started to recognise the indispensability of language to their argumentation. For example, Rubinstein (2000) presents an application of game theory as a possible systematisation of Grice's pragmatic programme. There have been many other similar recent suggestions, putting the somewhat rigid equilibrium-plagued analysis in games into the much more colourful out-and-about multiple-equilibria solutions and complement-equilibria perspectives. Related to these turns are some commonalties between Peirce's pragmatist ideas and much more recent issues taken during the early phases of the pragmatics of interaction (Chapter 12).

When communicative issues are viewed in a strategic setting, what is the degree of rationality in interaction targeted at common understanding? I do not wish to take up this subject at any considerable length here; the ongoing programme that studies non-hyperrational forms of rationality in strategic action is meant to address the issue. It concerns the foundations of game theory that is not merely aimed at dispensing with utilitarian lore, but also seeks to relax the somewhat restraining structural forms of interactant information in many of the game-theoretic approaches to communicative and dialogical situations.[12]

The upshot is that one is encouraged to seek new forms of communicative interaction, be it through the general theory of pragmatic constraints in linguistic and symbolic understanding and interpretation, in dialogical and game-theoretic theories, or anything in between. For instance, an increasingly important endeavour in computational sciences involves the characterisation of computational processes and ultimately of what is meant by computation, in terms of the structure and properties of the interaction between the computational system and its environment rather than in terms of the end products of those processes. (I address some related issues in Chapters 8 and 10.) The perspectives taken here in relation to Peirce's philosophy and his theory of communication are meant both to put these attempts into historical perspective and to give them some additional foundational thrust.

2. Common ground and natural language

Preamble For any sign mediation to take place, there has to be a shared, common view of representations and basic assumptions concerning language. This is possible, according to Peirce, if common traces exist in the utterer's and the interpreter's history of experiences and their collateral observations. This reciprocity is also the reason why it is possible to use a symbolic system such as language for communicational purposes in the first place.

It rings a platitude to say that language users have to have a sufficient supply of background knowledge and mutual beliefs concerning one another's knowledge and belief in order for utterances to be understood. Nevertheless, what is common between discussants can, in fact, be approached in a converse manner. It could be assumed that the possibility of building shared representations of the environment is one of the key justifications for there being a symbolic code of language at all. Under this view, the possibility of constructing shared representations is a precondition for the emergence of language. That is, to build a common, shared model of the utterances taking place between two (possibly imaginary) participants engaged either in an intrapersonal soliloquising or *tête-à-tête* communicative situation is derivative of the presupposition of the existence of the common ground.

As I hope will become clear during the course of this study, the notion of a common ground was of utmost importance to Peirce. He wrote a series of manuscript drafts for his projected book entitled simply Logic concerning the notion in communicative situations that are abundant with different kinds of signs.[13] He came to be the precursor of the pragmatic theories of meaning that have been using the surprisingly similarly conceived concept of the common ground and common knowledge that much later penetrated the pragmatically-oriented linguistic research, mostly attributed to the influential work of Stalnaker (1972, 2002) and Lewis (1969). For instance, according to Stalnaker (1978), the general concept of the common ground refers to the set of propositions whose truth is agreed by the discussants prior to conversation. Game-theoretically, it refers to the mutual knowledge of the structure of the game, especially its payoff structure, along with some key attributes concerning one's partners. I have more to say on the eventful development of these ideas and their impact in the field of pragmatics in Chapter 12.

For Peirce, common ground is the evolving resource of all that is given in the continuous situatedness and mutual experience of communicating agents. It is subordinate to laws of evolution just as Nature is subordinate to them. It evolves and is enriched and nurtured because during any discourse, there is a full body of assertions which all parties subscribe to whenever the assertions have not turned out to be false. This need not rule out that the information conveyed by false assertions can be added to it.

Moreover, it is reasonable to take the nine initial beliefs concerning reasoners' capacities that I discussed in connection with logica utens vs. logica docens in the previous chapter to be congenial to it. The common ground and ways of its built-up is also consistent with Peirce's evolutionary metaphysics (Chapter 3, sect. 3). The value of the common ground is given in the principle of *summum* *bonum*, that rational goal of inquiry towards which any reasonableness of linguistic use, pragmatic communication and cooperation, much in the same way as any body of scientific propositions, is destined to tend.

Contemporary reflections Long after Peirce, the general idea of communication, not principally as a method of transmitting signs, but as a method of constructing shared models or interpretations of sign representations and linguistic utterances, has prevailed in various guises. In the mental models idea of Johnson-Laird (1983) and Guenthner (1987), communicating participants build and update a 'mental data base' from which to draw inferences that work for communicative purposes. The participants have the mutual purpose of making these models overlap, blend or mirror one another. Like Peirce's approach, the mental-models theory is epistemic in that both parties presuppose mutual or common knowledge of the relevant parts of the universe of discourse, its elements, and relations between them. However, the approach endorses a modicum of the kind of psychologism that Peirce came to abhor after some initial hesitation, namely that accounts of how humans reason, choose hypotheses, generalise, formulate conjectures, and so on, are matters that belong in psychology. That anti-psychologism leads to the province of logic was, of course, Peirce's conviction, but his sentiments could also be formulated as a weaker doctrine that merely refers to the relation between logic and psychology, namely that they are incompatible in that their laws do not contradict each other.

The mental-models account does not necessitate commitment to full-blown psychologism. The idea that interpretation proceeds by way of such models does not itself mean that how humans reason is or ought to be a matter of psychology. To my opinion, one of its main contributions is that it only brings aspects of thirdness, the slighted chapter in the history of 20th century-logic, to bear on logical issues. It brings the notion of the mind in true contact with logic. The resulting theories may be mentalistic, and Peirce's theory of logic and semeiotics is certainly mentalistic, but they do not have to be psychologistic. They may purport to take into account, as Kant did, the general aspects of what goes on in the mind when humans think, deliberate, make decisions, cogitate, contemplate hypotheses, and derive conclusions. Human reasoning may have, and this Johnson-Laird (1983) wishes to demonstrate by empirical tests, properties that do not instantiate formal accounts of deductive reasoning, but are better depicted in terms of mental models of various sorts.

What is dubious in mental-models theory is not that what it has been after, namely an appropriate cognitive understanding of the notion of models, would not be the right arena in which logical inquiry may be performed, but that the medium in which these models are tried to be captured does not conform well to logicians' requirement of being objective aspects of figments of reality, with contextually interpreted non-logical symbols. Maybe the most fruitful way of resolving such a dilemma is to revisit logicians' understanding of the concept of a model in Peirce's vein, namely as a Phaneron quite as objective as a mathematical structure, rather than to hold mental-models approach as altogether alien to the inherited image of model theory (cf. Chapter 9).

It would thus be entirely wrong for the purposes of logic to dismiss every aspect of what goes on in the mind (or in the quasi-mind) as irrelevant in model-theoretic studies. That would amount to a considerably diminished input into logical inquiry and an even more inert and protagonist field than the mere puerile, post-Peircean symbolic logic as a mere calculus is able to provide.[14]

But there are others. According to the discourse-representation theory (DRT) (Kamp, 1981; Kamp & Reyle, 1993), discourse-representation structures encode text-level information as the interpreted discourse proceeds. It builds interpretations incrementally, taking the context of prior utterances into account. Its interrelated goals are to provide a general theory of indefinite and definite noun-phrase expressions, and to provide a theory that would predict the range of anaphoric relations of coreference in discourse. To do this, it uses a graphical representation of nested discourse structures, and assumes an intermediate level of representation in terms of special discourse referent variables.[15] Such referents assert existence, and thus the theory dispenses with at least some of the quantificational scope-related problems in representing indefinite and definite expressions of natural language. In this manner the discourse referents contribute to the building of the common ground, in tandem with the set of propositions encountered in the prior discourse.

According to Irene Heim's (1982) file-change semantics, the early ally to DRT, the totality of 'files' provides the generic notion of the common ground. A common ground of a particular context is then a subset of such files, or perhaps a single file. Any individual 'file card' represents a salient feature or a guise of the object taken up along discourse, which work as values for anaphoric pronouns and which come to be updated as conversation progresses.

As will be seen in later chapters, in Peirce's diagrammatic theory of EGs, the blank sheet of assertion represents the parts of the universe that are mutually agreed to exist and to be true by the two imaginary parties associated with the construction and interpretation of the graphs, the Graphist (the utterer) and the Grapheus (the interpreter). Second, juxtaposition of graphs denotes conjunction. Third, cuts around graphs express negation. Fourth, continuous lines of identities together with their dialogical interpretation express the connections between pronouns and their intended values.

Kadmon (2001, p. 102) proposes that the discourse-representation structures may be extended to encompass the content of all that is implicated in a particular discourse, thus contributing to the truly pragmatic force of the theory. Accordingly, similar suggestions may be voiced with regard to EGs, which to date have been extended to cover virtually nothing of all that phenomena that has been tried to be accommodated into DRT or one of its extensions.

However, DRT is ill suited to the fluid representation of a broad notion of communication simply because there is no distinction between the different roles of the speaker and the interpreter. Therefore, the basic approach is, after all, not epistemic, as there is no role for common knowledge and belief.[16] Consequently, the discourse structures do not naturally lend themselves to the possibility of there being a shared representation of the environment, one of the presuppositions for the emergence of linguistic communication.

Furthermore, DRT has not been suggested as showing commitment to psychologism, and indeed, it is not even mentalistic in the sense of overtly expressing any need for separate communicators connected with mind like processors that generate and interpret the discourse-representation structures.

According to the relevance theory of Dan Sperber and Deidre Wilson (Sperber & Wilson 1987, 1995), the inferential model of communication involves attempts to share, distribute and recognise acts of intentions, emotions and other modalities delivered in the communicative information. These attempts are what contribute to the relevance of the utterance that communicates some intended piece of information. What agents recognise as relevant is largely due to common traces in their experience. The notion of context is thus central in this theory. What is relevant is, shortly put, that which produces a tangible contextual effect. Relevant factors are those that intrude into the context of discourse (Sperber & Wilson, 1987, p. 586). According to them, the goal of communication is to maximise the relevance of the phenomena available to humans who use language, while minimising the amount of mental processing that is needed to do so. This hints at an element of utilitarianism in the definition of communicative goals in terms of the maximisation of something (here: the relevance). The goal of Sperber and Wilson is to provide a non-Gricean (or maybe a limited-Gricean) theory of communication that would function as a reference for a host of pragmatic phenomena running wild in linguistic populations. Grice himself never completed the systematisation of his desideratum concerning the question of what it actually (and formally) means, for example, to ‘be relevant’, as his conversational maxim of relation tells us to be.

The basic idea of relevance theory is thus neither entirely psychologistic nor epistemic, while admitting a fair modicum of both. It aims at providing a psychologically and psycholinguistically realistic theory, but not overly so. It is an attempt to make sense of linguistic pragmatics on the cognitive level which, according to Carston (1988, p. 713), is “the first account of pragmatics which is grounded in psychology”. Maybe this is not a major compliment after all. Most have regarded Grice’s program of analysing literal meaning in a public language by conversational maxims as psychological, because it involves notions of the speaker’s and the hearer’s intentions and beliefs. I find this assimilation a gross oversimplification already on Grice's own account, a point to be returned to later in this study.

That the aim of relevance is in psychological explanation has some drawbacks. Even though announced as one of the aims of the theory, the idea of relevance is not yet tame enough to suit the needs of a rigorous logical modelling of discourse, as such an enterprise would hinge on effective ways of representing contextual information and its change. As will be seen in Chapter 4, EGs, alongside game-theoretic approaches to the study of logic and language, tackle contextual phenomena by virtue of their strategic, non-compositional and holistic method of interpretation, extracting contextual input from the early stages of the totality of interpretation. The later stages are then open for the processing of further utterances with this readily extracted content.

Apart from ways of dealing with aspects of relevance in terms of the contextual effects it produces, I will propose in Chapter 12 a complimentary way of capturing relevance in terms of its reliance on Peirce's pragmatic principles.

By way of contrast to these three major approaches to pragmatics, the mental models, the discourse-representational, and the relevance-theoretic, Peirce's semeiotics has remarkable virtues of both the communicational dimension and the logically rigorous approach to representing discourse. It shows the character of shared signs. It also endorses an epistemic and logical approach to the common knowledge and common ground that is assumed of any sensible and rational interlocution. This richness can be summarised in the following five paragraphs. I will not go on to discuss these in more detail here, but they represent common themes that run through many of the forthcoming chapters.

(i) In Peirce's early semeiotics, there is a three-way notion of reference: the direct reference of a symbol to its objects (denotation), the reference of the symbol to its ground through its object (reference to the common characters of its objects, i.e. connotation), and its reference to its interpretants through its object (reference to all of the synthetical propositions in which its objects in common are subject or predicate, i.e. information). The second kind of reference, that of a symbol to the ground, or common characters of the object, gives the informed depth of the symbol (Chapter 1). It is related to the notion of comprehension, typically the amount of the set of attributes describing objects, or the intension of the given concept or a word.

(ii) The copula, in its function of relating terms to the elements of the universe of discourse, presupposes that "the universe must be well known and mutually known to be known and agreed to exist, in some sense, between speaker and hearer, between the mind as appealing to its own further consideration and the mind as so appealed to, or there can be no communication, or 'common ground,' at all".[17] The notion of the common ground as an existence of experiences without which communication does not succeed prevails in both formal (algebraic and diagrammatic) logic and in what Peirce believed to be the proof of demonstrating the validity of the philosophical doctrine of pragmatism. The reverberation of these notions is sensed, a hundred years later, in what became the increasingly topical field of linguistic pragmatics. Caution should be performed when putting forward findings in the loosely-defined field of pragmatics, however. In Chapter 12 I will present just a tiny handful of such ideas concerning language use and its philosophy that were foreshadowed by Peirce's contemporaries and linguists of late 19th and early 20th century.

(iii) The unifying sign in EGs, the line of identity, is capable of representing not only identity between concepts (predicate terms, rhemas), but also the notions of particularity, existence, choice or selection, predication, class inclusion, and anaphoric coreference. This gives a compelling argument for there being no separate underlying logical signs for *being*, a result to which I will return in Chapter 6. All of these notions can be analytically represented in a unifying way. Furthermore, as noted above, EGs presuppose a reasonable amount of common ground between the Graphist and the Grapheus of the graphs, which is the *sine qua non* for the truth to be exposed.

(iv) The two interpretants emerging in communicative situations, the intentional (determined in the mind of the speaker) and the effectual (determined in the mind of the interpreter), give rise to the communicational interpretant, which is the determination of that notion of the mind into which the minds of the speaker and the interpreted are welded in sign-theoretic communication. There are varying degrees to which such communication depicts person-to-person mediation, and to which such mediation is quasi-personal. This topic is taken up in Chapter 13.

(v) According to the paradigmatic version of the pragmatic maxim:

> The rule for attaining the third grade of clearness of apprehension is as follows: Consider what effects, that might conceivably have practical bearings, we conceive the object of our conception to have. Then, our conception of these effects is the whole of our conception of the object. (5.402, 1878, *How to Make our Ideas Clear*).[18]

We may think of practical bearings as contextual bearings that an item of information, or a belief, has to the context within which it is located. The implications that the infiltration of such items into the context may have is thus reminiscent of Sperber and Wilson's relevance theory, which balances the inferences that are made in choosing between possible rival interpretations with the notion of the cost of making such inferences. In the light of the pragmatic maxim, their theory, mutatis mutandis, may turn out to be a truly pragmatic theory of pragmatics.[19]

Supplementing point (v), the effect of such belief-strengthening has its correlate in Peirce's notion of a habit change, an update on the beliefs of the communicators, giving rise to logical interpretants of representations. Whenever that happens, one enters into the domain of logic and can therefore hope to be able to make the broad concept of what is relevant in discourse better understood, on a wider footing than Grice displayed. What is also notable is the appeal to the concept of interaction between relevant information and pre-existing assumptions of the speaker and hearer concerning the world in the relevance theory. This innocent-looking notion implicit in the theory may be unravelled by reflecting it against Peirce's interactional interpretation between the producer of the information (the utterer) and the receiver who has assumptions about the world that are contested by the information produced by the utterer. Since the degree of effort required in changing the background assumptions measures the degree of relevance, the induced minimax reasonings may be fitted into the strategic framework of game theory, making it explicit that the context update in discourse is a rational matter of (optimal) strategy selection.

Whether or not the effort of bringing forth relevant information is recommendable depends on the outcomes (payoffs) of the relevant strategies of the associated game of discourse interpretation. Likewise, the costs incurred by inferences to the best (in the sense of the most relevant) interpretation need to be deducted from the payoff values assigned to such strategies.

In all of these cases, then, the strategies are chosen according to the general principle of the rationality of actions, but as they encode information about the context in which discourse is performed, and are not confined to isolated utterances, they make the relevance-theoretic notion of the context-change potential of information introduced in communication amenable to rationalistic, but not necessarily hyperrationalistic, game-theoretic analysis.[20]

There are many relations between these five points. Let me bring out three of them.

(a) Although left unspecified by Peirce, the notion of the ground of objects laid out in *On the New List of Categories* (EP 1:1–10) provides the early basis onto which the later notion of the common ground shared by communities of sign and language users may be mapped. To see this, observe that Peirce defined the ground as all the common characters of the object to which a symbol (either a concept or a word) refers. It is the connotation of the symbol, which must be recognised to be such via its public nature of being a shared sign. The notion of information that this understanding of the reference produces is the informed depth of the symbol. The fact that Peirce later on subsumed the notion of the ground of the object under the notion of the interpretant only serves to vindicate my argument here, because it emphasises the fact that there are always two parties to sort out the meaning of signs, and brings out the role of the interpreter in recognising these common characters of symbolic signs.

Furthermore, the informed depth of a symbol referring to the ground of its object is related not only to the common ground of language users but also to salience derived from the work of Lewis (1969).[21] Both ground and salience pertain to the category of firstness. A piercing shriek of a steam-whistle, a rabbit running across a visual scenery where there is nothing else in sight, the Central Station of a major city where two persons are to meet, and numbers 1 and 100 can all be taken to be salient attributes of objects or events that reach consciousness while requiring little interpretation. Salience, however, requires some reference to an interpretant, in which case it can be seen as an imputed quality of an object or an event, in the same sense as Peirce holds that a reference to a ground, in so far as it cannot be prescinded from a reference to an interpretant, is such a quality (EP 1:7). The grounds or imputed qualities are determined by symbols, which are conventional terms occupying a central place in any proposition. Not all interpretations of the object or the event occupy the same place and rank in the minds of the interpreters, however. If the interpreters are persons, their consciousness will play a crucial role in measuring the informed depth of the utterances.

The notion of salience has become ubiquitous in pragmatics, from anaphora resolution to questions concerning the evolution and emergence of language and conventionalised systems in human societies. I will return to these topics in Chapter 11.

(b) Unlike Lewis (1979), the notion of the common ground is not related to prescriptions of what are legitimate moves in the ensuing dialogue game, but involves strategic (habitual) considerations. Communicators have to construct it, and to decide what to include in it. This decision concerns not only shared presuppositions but also the relevance or the degree of relevance of propositions admitted to the common ground.

The questions that such a process involves are similar to questions concerning the reduction of cooperative games to noncooperative games, but the scope of the former is wider (Aumann, 1985). They pertain not only to person-to-person pre-play negotiations about some main principles concerning the actual game, involving both linguistic and non-linguistic elements of communication, but also to the aspects of reaching a mutual assent and compliance that ought to be agreed on without having to fall back on any specific round of negotiations.

(c) The emergence of the communicational interpretant is perhaps best to be understood with reference to the common experience of the interlocutors, plus the *summum bonum*, the "continual increase of the embodiment of the idea-potentiality" (MS 283), of their communicative goals. The ensuing picture, which will be the main subject of Chapter 13, is, in the sense to be discussed, a counterpart of a generalisation of the triangulation scheme of Davidson (2001) into multiple worldview schemas. What Donald Davidson has argued is that putting together more than one mind that is capable of contemplating beliefs produces a public locus that comprises perceptual beliefs, accumulated through ostension, learning from common experience, and other traces in the series of perceptual observations. Triangulation happens when more than one participant (the utterer and the 'radical' interpreter) share reactions to common and sufficiently salient stimuli. This presupposes three interdependent varieties of knowledge, namely knowledge of oneself, of the others who communicate, and of the world. These versions of knowledge were taken by Peirce to constitute, among other things, the common ground of the subjects presupposed in communication. The idea of triangulation is therefore not far removed from the goals of communicative situations set out by Peirce in his trichotomy of the utterer's and the interpreter's interpretants, aimed at a fusion into one common, or at least sufficiently overlapping space.

More generally, mutual points of contact also exist between Davidson and Habermas, as they both endorse a communicative dimension to the objective truth of propositions. They both take communication in a realistic, interpersonal sense. To complete the picture, bringing Peirce's triadic concept of a sign to bear on the issue results in a fuller and wider notion of communication that still looms large as the source of propositional truth. There are also stark contrasts in this trio: the degree to which truth so conceived is to be taken to be socially constructed varies considerably. For Habermas, the focus is on the competence of those whose goal is to reach consensual judgement concerning what is true. For Davidson, any non-singular situation brings in new aspects of knowledge and belief about each other's beliefs, cohering into agreed conceptions of propositional truth. In addition to these, Peirce would have brought in the ideal limit of an inquiry, confident that it is within such limits that questions arising in consensual and triangulative situations will be resolved.

A natural caveat is that the communicative dimensions of Peirce's sign theory are by no means exhausted by these novel insights. One aspect that is avoided in Peirce's theory of communication is the untoward tendency in current theories to reduce the variabilities in what linguistic meaning is into the one-sided problem of speaker's meaning and the recognition of his or her intentions. This castration is performed in relevance theory, for instance. Peirce would not have approved any one-shot interpretation, for whom the reciprocal, open-ended and triadic nature of sign meaning is irreducible. I suppose that the reason for the reductionism advocated by Sperber and Wilson lies in the somewhat unpremeditated domination of Grice's original proposal, in which he laid emphasis on the role of speaker-meaning in linguistic comprehension. The followers of Grice took his suggestions too literally: Grice never claimed that by focussing on what is different and what is similar in speaker-meaning vs. literal meaning one would get an exhaustive account of linguistic and logical meaning.

The soi-disant followers were misled by what they took to be Grice's key suggestion: that the proper exposition of speaker-meaning ought to be conducted, first and foremost, by psychological means. In reality, this suggestion was, at best, an afterthought for Grice. He de-emphasised using psychological notions in explaining speaker-meaning, the fact which comes out very clearly and forcefully in his writings once it is realised that (i) Grice's main occupation was the meaning of logical particles (most notably of conditionals and other logical connectives) rather than linguistic utterances, and that (ii) his remark that psychological concepts, needed for the formulation of an adequate theory of language, refers to intensional concepts of believing and intending, which can be tackled by logical means. After all, in Grice's writings, references to psychological terminology are few and far between. His theory of meaning is no more psychological than, say, game theory or epistemic logic are theories of human psychology.

Universes of discourse The importance of the common ground also shows in Peirce's concept of the universe of discourse.[22] It is not only an interpretable resource to which dialogue partners draw their attention and from which, by adjusting their beliefs and attitudes, they choose the elements and values intended by the statements. It is also a readily interpreted pool of mutually known and known to be mutually known and mutually observed facts concerning aspects of language and grammar, linguistic competence, coinciding traces of experience, self-awareness, and common knowledge. These elements are, of course, parts of the folklore from sociolinguistics to linguistic pragmatics, but as noted above, they are all found in Peirce's multiple-draft essays about the notion of common ground (MSS 611–614; see Chapters 12 and 13). The universe of discourse is not limited to the logicians' conception as a prearranged structure of elements, either, let alone a perfectly interpreted, closed totality of such elements, but is indefinitely extendible and calculated to offer considerable latitude for the interpreters in the process of adjusting and modifying their acquaintance of it.

The role of the copula is indispensable in semiosis in that it relates objects of signs to elements of the universe of discourse. This goes beyond what is expressible merely in language or in logic by using a valuation function. For the copula to perform its job, experience has to be presupposed, and the common traces of experience provide the basis without which no coding system (such as language) can instigate sign communication. (There would be no need ever to document a single line of software if computer programs were not executed in different systems and in different environments.) Universes of experience, even if they were to occur in relative isolation from one another and limited to single interpreters, are the pragmatic constitutions of what Edmund Husserl's termed "lebenswelts" of human experiencers. When referring to the community of language users, they become the ground on which a multiplicity of different forms of communication is permitted to evolve.

Universes of discourse are manifold. They are not confined to particular or singular individuals, but may also refer to modalities, qualities, ideas, and other subjects involving extra logical dimensions. While Peirce never ventured very far into detail concerning the formal treatment of various dimensions, his remarks demonstrate the flexibility he assigned to the overall concept of logic. This he did by emphasising the central place that universes of discourse play in logic, thereby vindicating the view that the universes the discourse is to be about are not closed totalities of what there is, but that they come together with reinterpretation and extendibility. For instance, he notes that a universe may contain, among other things, qualities and collections: "I propose to use the term 'universe' to denote that class of individuals about which alone the whole discourse is understood to run. The universe, therefore, in this sense, as in Mr. De Morgan's, is different on different occasions. In this sense, moreover, discourse may run upon something which is not a subjective part of the universe; for instance, upon the qualities or collections of the individuals it contains".[23]

The early proponent for the separation of the first- and the higher-order components of the logic of quantification was David Hilbert. Although Peirce did not consider quantification in algebraic logic of relatives in the primary sense of 'ranging over' objects of a certain type, such as over individuals in case of first-order logic, Peirce's distinction, later bolstered by Ernst Schröder, between first-intentional (referring to concepts that have real individuals as their objects, i.e., as values of indices) and second-intentional signs (referring to concepts that have first-intentional things as their objects) was a definite signal of the need to qualify such a distinction. How Peirce came to understood quantification was, in essence, as an interactive process that aims at capturing the intended meaning of expressions, together with the order and informational attributes of such processual choices as vitally material (Chapters 4 and 6).

3. Conclusions

Peirce did not consider the concepts of utterers and interpreters in the literal sense of actually existent beings. There are signs, such as signs in nature, symptoms of disease, and signs of the weather, that do not have utterers, and it is conceivable that some signs, such as encrypted data, do not have interpreters — hence the prefix 'quasi'. Given no clear-cut demarcation between objects and their interpretants, which rather constitute a continuum, the utterer, if there is one, is found toward the object end, whereas the interpreter, again if one exists, is found toward the interpretant end. In this sense, the utterer is in the object of the sign, and the interpreter is in its interpretant. It is no longer meaningful to ask whether utterers exist without objects or whether interpreters exist without interpretants. On the other hand, it is useful to attempt to measure the 'distance' between objects and interpretants in some informative scale more fine-grained than what would be produced by a mere assortment of different kinds of interpretants. For only in the ideal case of ultimate logical interpretants, in which epistemological and ontological viewpoints converge, is the measure of distance something determinate.

The elimination of utterers and interpreters from the domain of sign action could also be suggested by the forceful tendency in Peirce to repudiate all psychological influences from the provinces of logic and semeiotics. Echoing Kant, he wrote that psychology has as much to do with logic as "if I were to discourse of the ingredients of the ink I use".[24] (Nowadays the reference would be to the properties of the word-processing software one is using.) He classifies his viewpoint by saying that the utterers and the interpretants, conceived as the objects and interpretants, are welded in one sign.[25] Hence, even though communication, prima facie, takes place just between two, these entities enter the irreducibly triadic sign-play in the larger picture.

Nevertheless, in the special case of the sign being symbolic, and more particularly linguistic, it appears that there must be separate entities that utter and interpret signs of natural language, even if they were not to be strictly assimilated with those of actual populations of human language users. Moreover, according to Peirce, if the signs have propositional content, and by virtue of that function as assertions, these entities that are separated from objects and interpretants incur a liability for being responsible for the utterances. This shows up in the dialogical actions that discourse participants take, and in their deliberations concerning continuations of utterances in discourse. The interpreter likewise incurs a responsibility for his interpretations, and runs the risk of punishment if the utterances are not interpreted in the most charitable or humane way. Besides reminding us of some of the principles of communication associated with Grice's and Davidson's philosophies, these remarks may be interpreted in the game-theoretic sense of the participants of discourse being agents acting according to rationality postulates. The possibility of there being systematic connections between game-theoretic rationality principles and Grice's pragmatic program of linguistic communication have not gone unnoticed (Hintikka, 1986; Parikh, 2001), but their cognate origins in Peirce's theory of signs has not been similarly recorded.

Given Peirce's communal principle according to which "a person is not absolutely an individual",[26] any stark contrast between the intrapersonal, multiple-minds perspective on conversational dialogues and the correlated evolution of thought, on the one hand, and interpersonal, social communication, on the other hand, assumes much less importance. In both cases, it is the continuous community of partakers who decide upon the truth and the propositional content of the asserted signs, even if they are merely the quasi-players of semeiotic roles, positions in semiosis, or actually personalised and anthropomorphised entities.

Notes

1 Note that Peirce used the term pragmatics in his classification of the sciences, e.g. in MS 1345, *On the Classification of the Sciences*. There he proposed a threefold primary division of the whole of inquiry into "I. *Mathematics*, the study of ideal constructions without reference to their real existence, II. *Empirics*, the study of phenomena with the purpose of identifying their forms with those mathematics has studied, III. *Pragmatics*, the study of how we ought to behave in the light of the truths of empirics. [. . .] Pragmatics studies the processes by which the outer world is to be brought into accordance with our wishes."

2 Kadmon (2001) is a comprehensive study of a wealth of model-theoretic and generative phenomena that are located somewhere within and across the borderline between semantics and pragmatics.

3 MS 10, 1903, [*Foundations of Mathematics*].

4 Pietarinen (2005d) discusses the composition of concepts in EGs.

5 What is worse, manuscript 499 ends abruptly before even roughing out the wherewithals of EGs in this context.

6 See Peirce's entry on Symbolic Logic in Mark Baldwin's *Dictionary of Philosophy and Psychology*.

7 In contrast, logic is for Peirce neither a universal system of expression nor a calculus in its limited sense: "This system [of logical algebras and graphs] is not intended to serve as a universal language for mathematicians or other reasoners. . . . This system is not intended as a calculus, or apparatus by which conclusions can be reached and problems solved with greater facility than by more familiar systems of expression" (4.424).

8 Lewis was also influenced by the modal logical systems of the Scottish philosopher Hugh MacColl . MacColl and Peirce knew about each other's work and were at one time in regular correspondence. See the special issue "Hugh MacColl and the Tradition of Logic" published in *Nordic Journal of Philosophical Logic* 3, 1999.

9 Much later, in 1946, Russell admitted: "I am — I confess to my shame — an illustration of the undue neglect from which Peirce has suffered in Europe. I heard of him first from William James when I stayed with that eminent man in Harvard in 1896. But I read nothing of him until 1900" (Russell, 1946, p. xv).

10 See Chapter 8 for some indications on how pragmatism arrived at the early 1930s Vienna.

11 This is perhaps something reminiscent of Peirce's opinion on "the German professors' habit of using a hundred words [here: in 117, A-V.P.] to disguise an idea that might have been precisely expressed in ten" (CN: 2.248, 17 May 1900, *History of Ancient Philosophy*).

12 I have discussed some senses in which information structures are restraining in extensive-form games in Pietarinen (2003b). The notion of hyperrationality has been previously tried to be relaxed, for instance, by changing the underlying epistemic logic of games into those that dispenses with logical omniscience, that is, the untoward fact that in normal epistemic logics one knows all logical truths. One way of doing this is by differentiating between implicit and explicit aspects of knowledge. If the idea is generalised to cover other kinds of epistemic attitudes, too, one is advised to look into what the neuroscientific findings that have supported the related distinction actually are. I have discussed themes in the interplay between epistemic logic and cognitive neuroscience in Pietarinen (2003d).

13 MSS 611–615, especially MSS 613–615, *Logic, Book I, Analysis of Thought, Chapter 1: Common Ground*, from 16 November to 1 December 1908; Chapter 13.

14 Aspects of thirdness are increasingly visible in the programme of 'New Logic' as announced by Gabbay & Woods (2001). It sets logical inquiry in the larger context of intellectual inquiry and purports to define it as resonant with 'pragmatic theory of cognitive agency'. A 'New Logic' was already propounded by Karl Menger after his encounter with Paul Weiss and Peirce's philosophy at Harvard in early 1930s (Menger, 1937, 1994).

15 The coinage of 'discourse referents' goes back to Lauri Karttunen's work in the late 1960s.

16 Some epistemic extensions have since been proposed, see Asher (1993).

17 3.621, c.1901–02, *Notes on Symbolic Logic and Mathematics*, cf. MSS 611–615.

18 This formulation of the pragmatic maxim first appeared in the January issue of the *Popular Science Monthly*, 1878. Several versions exist in Peirce's corpus. A very succinct one is: "The maxim of logic that the meaning of a word lies in the use that is to be made of it" (CN: 2.184, 2 February 1899, *Matter, Energy, Force and Work*, vol. 68). This is a striking foreshadowing of Wittgenstein's mantra that "the word has meaning by the particular use we make of it" (Wittgenstein 2000–, item 147, p. 39v, *Grosses Notizbuch*; cf. Chapter 8). Peirce recurrently reflected on his pragmatic maxim in his late writings. In one of the drafts of a letter intended for *The Nation*, MS 324 (c.1907), Peirce strongly emphasised the modal interpretation: "Of all literary forms, the one that, from my point of view, seems most appropriate to clothe a pragmatistic interpretation, is that of the maxim; and accordingly, it was as a maxim that I first enunciated the method of pragmatism, as follows: Consider what effects that *might conceivably* have practical bearings, — especially in modifying habits or as implying capacities, — you conceive the object of your conception to have. Then your (interpretational) conception of these effects is the whole (meaning of) your conception of the object. (The words between dashes and in parentheses have been added in transcribing the maxim, in hopes of rendering it clearer, without substantially modifying it.)" (pp. 11–12; cf. MS 290: 33, 1905, *Issues of Pragmaticism*; MS 313: 28, 1903, *The Harvard Lectures VI*). Another reflection is in MS 319: 9, 1907, *Pragmatism*, which appears to slightly antedate the writing of the previous quotation: "Of all literary form the one that, from my point of view, seems most appropriate to a pragmatistic interpretation is that of the maxim; and accordingly it was as a maxim that I ~~first~~ originally defined the method of pragmatism, as follows: Consider what effects that *might conceivably* have practical bearings, especially in modifying ~~one's~~ your habits, — you conceive the object of your conception to have. Then your ~~general~~ comprehensive conception of these effects is the whole of your conception of the object. (In order to make the rule plainer and more explicit, the words between the dashes, and the adjective 'comprehensive' have been inserted. The intention of the maxim remains just what it originally was.)"

19 Not all theories of pragmatics are pragmatic in the same sense. For instance, Montague's theory of grammar claimed the resolution of contextual matters within a hybrid of higher-order and possible-worlds concepts.

20 The relevance-theoretic notion of context-change potential should not be confused with dynamic theories of meaning, although one might sense some similarities.

21 As such, the idea of salience is much older, and can be found in the works of Karl Bühler, Philipp Wegener and others; see Chapter 12.

22 According to Peirce, exact logic dates from 6 November 1846, the day when Augustus De Morgan introduced the term "universe of discourse" (MS 450: 7, 1903, *Lowell Lectures 1903 [Lecture I]*). We may christen that 'the domain-day'. According to De Morgan (1849): "Writers on logic, it is true, do not find elbow-room enough in anything less than the whole universe of possible conceptions; but the universe of a particular assertion or argument may be limited in any matter expressed or understood. . . . By not dwelling on this power of making what we may properly . . . call the universe of a proposition, or of a name, matter of expressive definition, all rules remaining the same, writers on logic deprive themselves of much useful illustration".

23 3.65, 1870, *Description of a Notation for the Logic of Relatives*.

24 MS L 463: 26, 1904–09, *Letter to Lady Welby*.

25 4.551, c.1905–06, *Prolegomena to an Apology for Pragmaticism*.

26 5.421, 1905, *What Pragmatism Is*.

Appendix 2.A: The early dawn of neuroscience

The relevance of Peirce's semeiotics to the rapidly increasing investment in neuroscientific research has remained by and large unexplored. Several facts speak in favour of there being mutual points of interest between the two, some of which I will put forward in this supplementary section. First, some historical remarks.

In 1861 an important discovery was made that marked the beginnings of sciences with the epithet 'neuro', most pertinently, neuropsychology.[1] The French surgeon and anthropologist Paul Pierre Broca (1824–1880), the founder of the Société d'Anthropologie of Paris, announced that he had shown that there was a connection between lesions in the middle posterior parts of the frontal lobe of the left hemisphere of the human brain and the functional incapacity in humans, here concerning the loss of speech, aphasia.[2] Peirce became aware of this finding, and mentioned it in relation to some religious questions that he was thinking about in his *Answers to Questions Concerning My Belief in God* from c.1906. He did not deny the dependence of mental action upon the brain, which he took to have been demonstrated by these newly-discovered facts, in particular by the finding that lesions in Broca's area greatly affect the use of language (6.520). He nonetheless embedded these findings of a neuropsychological nature within the larger question of immortality, that is, the question of whether memory or consciousness perseveres after the brain and the body are deceased. This was, incidentally, yet another symptom of the two fires he was caught between, compelled as he was to place the results of research on logic and mathematics, as well as experimental findings in science, in religious and theological perspectives.

According to the minutes of the American Psychological Association founded in 1892, Peirce was elected the first psychologist to the National Academy of Sciences on 18 April 1877 (Street, 1994). He was actually honoured for his work on the logic of science, but the title was deemed appropriate by the Association given that he was first to hold empirical psychological experiments on colour vision and difference tresholds in America. His conviction that psychologising would do little good when practiced in connection with logic and semeiotics did not affect the decision to call Peirce the first American psychologist. That was to emerge in the early 1900s, although one may sense some anticipation from 1893 onwards (4.85).

In charting the relevance neurological research may have for sign-theoretic systems and vice versa, it is useful to bring in the concept of pre-representations, and try to see whether it relates to some notion of an interpretant in Peirce's classification. Changeaux & Ricoeur (2000) describes pre-representations as working models the brain produces in the mind, not yet finished, finalised thoughts that could be subsumed under critical self-consciousness. They are fluctuating, non-stable hypotheses and anticipations of them; schemes or frames according to which subjects try to make do in their surroundings in a habitual, non-programmed manner. They are the neuronal counterparts to the hypothesis generation (functional anticipation) and hypothesis testing (verification) performed in science. From this perspective, pre-representations relate to two things in Peirce's thinking: his characterisation of the logic of scientific method, and his idea of the importance of projection, that there are repeating cycles of the ego and the non-ego acting upon one another in creating consciousness of what exists and what the subject's experiences of such existence is.

I would like to suggest that the idea of such pre-representations, is not altogether different from Peirce's idea of the dynamical interpretant. They are both constituted of the floating or changing material evolving in interpretations of signs. They are both the actual effects which the sign, *qua* a sign, determines, while doing this in a not-yet settled process, open to modifications that new information and new input may suggest. That the dynamical interpretant is taken as the actual effect of the sign, being located in the brain (if we limit ourselves to non-quasi minds), does not yet distinguish it from other interpretants such as the communicative ones also determined 'in the minds' of communicating subjects.

Nevertheless, dynamical interpretants are the ones most open to neuroscientific scrutiny. They are represented in instinctive thought, as what determines the immediate interpretants.[3] Still, what such interpretants are is not something that falls within the purview of psychology, as it describes the "logic of mental operations" (4.539), although they differ from other interpretants in being effective, and in this sense are concrete representations of actual mental operations.

Evidence for this can be gleaned from Peirce's later writings. By 1909, he had reached the stage of being able to present a somewhat fuller explanation of the distinction between the immediate, the dynamical, and the final interpretants. As far as the dynamical interpretant is concerned, it is "whatever interpretation any mind actually makes of a sign. This Interpretant derives its character from the Dyadic category the category of Action. This has two aspects, the

Active and the Passive, which are not merely opposite aspects but make relative contrasts between different influences of this Category as More Active and More Passive" (8.315, 1 April 1909). This was to be one of Peirce's final attempts to characterise the main trichotomy of interpretants. We can also detect its compatibility with the previously-mentioned two-fold division of objects into immediate and dynamic counterparts.

Peirce then went on to claim that there was a parallel to the active and passive parts of secondness found in psychology, namely in the category of molition, "in its active aspect of a force and its passive aspect as a resistance" (8.315). This is, of course, quite surprising, given Peirce's drive to avoid the intrusion of psychologism into semeiotics. However, molition remains merely an analogy in this context. It is what volitional attitudes of a person exhibit when stripped of their characters of desire and purpose. Peirce attempts to explain this somewhat esoteric psychological attitude by presenting the following example:

> Hold a dumbbell out at arm's length and tell yourself at the outset not to do anything whatever with that arm, — neither pulling it up nor putting it down, — until you give yourself the word. After a while gravity will catch the arm in such a state that it will take a little step down, which is not surprising, since gravity is pulling at it unceasingly, while the state of the arm is not one of ceaseless inertia. But what is a little surprising is that after each of these little descents the arm springs up a little, although you did not tell it to. For you, I am supposing, have given the arm no orders of any kind since you told it not to move until you gave the word. It comes up with what appears like an elastic rebound. All this time, you have made no exertion whatever. You have been perfectly quiet, but you have felt a certain pain. Now that you are about to give the mental word for your arm to come down, be on alert to see whether you have any Feelings of that giving of the word. Now, actually give the word, and the arm comes down so instantly that you cannot tell which reached your brain first, the report that your order has been received, or the report that the arm was falling. You even suspect the latter report came first. But the significant circumstance is that there was nothing like a *Feeling* connected with giving the mental word. (MS 645, 22 December 1909–12 January 1910, *How to Define (Definition: 3rd Draught)*).

We have all had experiences like this. The point is the momentous sense of awareness without the sense of the quality — the resisting hand moving autonomously, without the subject being able to capture the feeling that the kinematics of the hand were orchestrated by the subject. Research on curious neurological disorders has subsequently brought to light numerous, sometimes quite extreme, dysfunctions ranging from uncontrolled bodily movements and loss of feeling of the spatial arrangement of one's body to alienation from one's bodily parts altogether. These are evidence of 'unaware awareness', implicit aspects of consciousness and propositional attitudes, states of mind in which there may either be a loss of some capacity without the subject being conscious of it, or else an implicit mechanism of performing something without explicit awareness of what one is capable of doing.[4] This is connected not only with psychological sensations, but also with the logical idea of communicative agents in the sense that the necessity for intercommunication on the basis of which parties assent to and recognise each other's assent to propositions involves "voluntary, deliberate molition" (MS 296: 20) by all participants. It is a voluntary act in which the conversing parties take part in building, resolving and interpreting logical representations, just as they build up representations into their cognisant minds.

Cumulative evidence thus exists for trying to locate the neuronal correlates of dynamical interpretants, the 'floating signified', as other semioticians succeeding Peirce have sometimes put it. This refers to material that is detached and distinct from the intentional interpretant subjective to the mind of the utterer, and likewise, it refers to material that is detached and distinct from other, more determinate types of interpretants, such as the effectual interpretant created in the mind of the interpreter. Dynamical interpretants, do not possess the capability of being true, which is the property Peirce deferred to final interpretants. In this sense, parts of what dynamical interpretants are pertain to subjective components of experience.

There is, in fact, direct textual evidence of the possibility of tracking down some neuronal correlates in semiosis. Peirce refers in MS 293[5] to the fundamentals of understanding and interpreting EGs by experimenting upon such visual diagrammatic representations: "The Diagram sufficiently partakes of the percussivity of a Percept to determine, as its Dynamic, or Middle, Interpretant, a state activity in the Interpreter, mingled with curiosity. As usual, this mixture leads to Experimentation. It is the normal logical effect; that is to say, it not only happens in the cortex of the human brain, but must plainly happen in every Quasi-mind in which Signs of all kinds have a vitality of their own" (MS 293: 14–15).

Further aspects related to this train of thought are to be found in the Logic Notebook, in which Peirce wrote that "the *Dynamical Interpretant*, is the actual effect produced upon a given interpreter on a given occasion in a given stage of his consideration of the sign. This again may be 1st a feeling merely, or 2nd an action, or 3rd a habit" (LN 288r, 23 October 1906). This is a subdivision of interpretants that accords with the division that Peirce made between the emotional, the energetic and the logical interpretants (Chapter 1).

Not all of what dynamic interpretants are can be revealed by empirical techniques of neuroimaging. There are areas in the brain that correlate with certain excitements and responses, and so we come to study action that is induced in the subject. We know that the subject learns many things as a result of a habit change, but we still do not get into subjective experiences. Then again, we have to remember that Peirce's notions are not restricted to actual conscious subjects with real brains, but apply to any entity, human or otherwise, with the general character of a sign-carrier or a sign-vehicle.

As to the final interpretant, it is described in the LN as "the ultimate effect of the sign, so far as it is intended or destined, from the character of the sign, being more or less of a habitual and formal nature". This habitual, ultimate character of final interpretants means that they evade any objective, empirical methods so much in vogue in present-day neuroscience.

Even so, there is a great deal of relevance in the details of Peirce's work to the issues that are much debated in neuroscience. Just one, highly interesting example, is the possibility of there being certain determinate neuronal counterparts to the mechanisms that are evoked by what is going on in predication. Predication seems to presuppose sound coordination between what are called the ventral and the dorsal visual pathways in the frontal cortex of the brain. The former is responsible for the public object identification, whereas the latter is responsible for the relative identification of the object in one's visual scenery. Of relevance in this context are Peirce's remarks on how logical constants such as rhemas are interpreted in a two-stage process. First, one of the functionaries chooses an object, and second, he, she or it uses this selection so that the rhema will apply to it. The former activity corresponds to the relative perceptual identification of the object according to its spatial coordinates, whereas the latter corresponds to the capability of definitely using a name for the object in order to apply it in the context of the predicate and the proposition within which it appears.

In neuroscientific studies on vision, this distinction is customarily made between two functional systems, one being responsible for object perception and the other for perception between the relations of objects (Milner & Goodale, 1995; Ungerleider & Mishkin, 1982; Vaina, 1987). A lively discussion continues whether this distinction is to be made on the basis of different functional roles of the two cortical pathways or on the basis of their neurological structure. The neurological structure seems to be more complicated in the light of recent evidence (Milner & Goodale, 1995). Already Van Essen & Gallant (1984) presented data on visual processing by identifying the visual cortex as a complex hierarchy of brain areas that have their own functional specialisation while retaining complex patterns of connectivity between them. This reflects the diversity of visual tasks, including the role of motion in visual perception. At the very least, there seems to be a good deal of interaction between the two systems. Bechtel et al. (2001) has emphasised that the idea of two visual streams plays an important integratory role in theories of visual processing.

As has been aptly put by Dienes & Perner (1999), these two visual information processing systems identified with different neurophysiological pathways (dorsal and ventral) can be characterized by saying that the information in the dorsal system (the 'what' system) is unconscious, not used for locutions about external things, is reliable in the sense of not producing illusions, is used for action, and is of limited duration. On the other hand, the information in the ventral path (the 'where' system) is conscious, illusory, about statements concerning perception, and used for action after some delay. However, if the distinction is put in these terms, it follows that the distinction between implicit information (as represented by the 'what' system) and explicit information (as represented by the 'where' system) is applicable.[6]

A more general perspective is to be drawn here, already hinted at in the preceding paragraphs. The illustrious distinction between two functional extrastriate visual areas in the brain, the ventral pathway responsible for object perception and the dorsal pathway responsible for spatial perception between the relations of objects, is seen to correlate logically with the processes of choosing suitable actions (which correspond to the ventral 'what' systems, i.e., identity), while the processes of instantiating the elements so chosen as suitable attributes of predicates require coordination (which corresponds to the dorsal 'where' system, i.e., localisation). Two main caveats concerning this assimilation of logic and neuroscience that have to be borne in mind are that (i) these two systems in the visual system are interconnected just are objects are to their spatial locations, and that (ii) they both subdivide into several 'subroutines', including the spatial vs. temporal and parallel vs. serial identifications (cf. Chapter 11).

It is of vital importance to recognise that the two chief logical actions that correlate with the two key neural processes have existed as long as logic itself. In general, the question here is about the nature of predicates, their arguments, and their conjugation. Aristotle drew the distinction between predicates (said or spoken phrases or things) and subjects. This distinction prevailed in various guises. For instance, in Kant's logical analysis it was the question of the formation of the interpretation of monadic predicates being related to the power of the mind of synthesising experiences in consciousness, and eventually in Peirce's philosophy being about many-place rhemas as blank forms of expressions being filled with new information derived for them by interactive processes between the utterer and the interpreter. In linguistics, the related division has likewise recurred, comprising the dichotomies such as thema vs. rhema, topic vs. comment, new vs. old information, and so on.

Neuroscience and neuropsychology were only just emerging as scientific discipline in Peirce's times, and massive advances have been made during the decades that followed.[7] However, as shown by the aforementioned points, the logical and sign-theoretic relevance to aspects of what goes on in the brain has not changed much since then.

* * *

It is definitely one of the central questions, not only in Peirce but also in the philosophy of mind in general, whether the system of signs succeeds in bringing two discourses, those of the mental and of the physical, into a conceivable relation, that is, to establish the real distinction between them. This was Descartes' main concern. In his sixth Meditation, he went on to advocate the employment of a third discourse to account the inexplicable union.

> The intellect is distinguished from the imagination; the criteria for this distinction are explained; the mind is proved to be really distinct from the body, but is shown, notwithstanding, to be so closely joined to it that the mind and the body make up a kind of unit. (Rene Descartes, *Mediations on First Philosophy, Synopsis of the Following Six Meditations*).

Peirce was not in the least sympathetic to the Cartesian nominalist approach to philosophy, marking it as "hodge-podge"[8] and "false pretence",[9] the severest and the most illegitimate form of pretense.[10] It is not entirely clear, however, what significance Peirce eventually gave to Descartes' solemn attempts to seek the third discourse. All the same, in seeking such discourse in the interaction of any two agents or notions, the mind and the non-mind, or the non-body and the body, I believe that triadic divisions of signs are a good starting point. If we consider one of the interpretants — and I believe the final interpretant is the most developed and prolific one for that purpose — as an end-product of the synthesis of the two discourses (a kind of hybrid concept of 'mental object', or a version of objective phenomenology), Cartesian interactionism dissolves into dyadic interactionism between the ego and the non-ego, or as Peirce would put it, into the tenet that will need to be classified within the category of secondness. It lacks thirdness, the mediation of the two, not allowing representation and purpose. Cartesian interactionism is merely able to provide the framework within which the puzzle of transmission between the two polarities of the soul and the body engaged in communication can be expressed, but it is devoid of any further purpose and genuine communicative action.

In contrast, in the triadic sign-theoretic philosophy, the ensuing functional notions of agents, or objects and subjects of a meaningful discourse, are not simply to be assimilated into the notions of the subject and the object, the mental and physical, or something like consciousness and non-consciousness, because of the omnipresent and evolving interpretation of signs with the third, the interpretant, linking the two. This is precisely why we find in Peirce no traditional philosophical arrangement that creates a mind–body problem, the archetype of false pretence with its overwrought terminology that is so familiar but so barren and sterile in present-day texts on the philosophy of mind.

Notes

1 A century later, neuropsychology was defined in the editorial to *Neuropsychologia* 1(1), January 1963, as "a particular area of neurology of common interest to neurologists, psychiatrists, psychologists and neurophysiologists. This interest is focused mainly, though not of course exclusively, on the cerebral cortex. Among topics of particular concern to us are disorders of language, perception and action".

2 The publication that followed was Broca (1861).

3 4.539, 1905, *Prolegomena to an Apology for Pragmaticism*.

4 See Pietarinen (2004a) for the exposition of some hitherto unacknowledged relations between logic and neuroscience in terms of these explicit vs. implicit aspects of knowledge, belief, memory, perception and other propositional attitudes.

5 A draft for the Monist *Prolegomena* series, c.1906.

6 A consequence is that some key neurological distinctions are reflected already at the level of propositional epistemic logic dispensing with quantification (Pietarinen, 2004a).

7 For a concise account of the history of neuroscience, see e.g. Gross (1998).

8 5.81, 1903, *The Categories Continued*.

9 2.192, 1902, *General and Historical Survey of Logic: Why Study Logic?*

10 See also *Questions Concerning Certain Faculties Claimed for Man*, 5.265, published in *Journal of Speculative Philosophy* 2, (1868), 103–114, in which extensive criticism is made of the usefulness and explanatory value of the notion of intuition and introspection in philosophical argument.

Chapter 3

PEIRCE'S GAME-THEORETIC IDEAS IN LOGIC

> THINKING ALWAYS PROCEEDS IN THE FORM OF A DIALOGUE, — a dialogue between different phases of the *ego*, — so that, being dialogical, it is essentially composed of signs, as its Matter, in the sense in which a game of chess has the chessmen for its matter. (MS 298: 6, 1905, *Phaneroscopy*).

1. Introduction

My title of this chapter is, and is intended to be, openly anachronistic in the sense that game theory as we know it today had, of course, not yet been developed during Peirce's lifetime. However, my purpose is to demonstrate that, on many occasions, Peirce did conceive his logical and semeiotic ideas in ways that allow faithful translation into game-theoretic terminology. One particularly illustrative example of this has been noted in earlier literature: For example, Brock (1980) and Hilpinen (1982, 2004) have shown that in Peirce's logical system — the system that through its later developments came to be known as first-order logic — existential and universal quantifiers, as indeed connectives and negation, are quite explicitly understood as integral parts of an interpretation game or a dialogue between two parties or "functionaries" (MS 500: 13), most often termed the Utterer and the Interpreter.

He alluded to countless terms to describe his dialogical approach to logic and cognition, including

the utterer – the interpreter
the proponent – the opponent
the defender – the attacker
the speaker – the hearer
the addressor – the addressee
the assertor – the critic
the graphist – the grapheus

the Artifex of Nature[1] – the Interpreter of Nature
the symboliser – the thinker
the scribe – the user
the affirmer – the denier
the ego – the non-ego
the quasi-utterer – the quasi-interpreter
the quasi-proponent – the quasi-opponent
the delineator – the interpreter
the concurrent – the antagonist
the compeller – the resister
the agent – the patient
the putter forth – the auditor
the writer – the reader
the Me – the Against-Me
the interlocutor – the receiver.

Also in use were the descriptions "an omniscient being desirous of supporting his assertion" versus "the most knowing antagonist of the assertion".[2] Both members of these pairs are the "repositories of thought" or "theatres of consciousness" (MS 318: 18). Adjectives prefixed to these subjects include "enlightened" (MS 539: 25, c.1903), "right", "rational" and "virtual".[3] The term "virtual" refers to intrapersonal conversation as opposed to person-to-person speaker-addressee relation, and is in this sense as real as the latter.[4]

The roles exemplified by these terms are essential for any genuine semiosis or sign interpretation to take place. One of their main purposes is to establish the meaning of logical expressions asserted in propositions. In many instances, we may go further and study the idea of dialogues or semeiotic interactions themselves from the viewpoint of the mathematical theory of games, which of course was not available in Peirce's time. These games have subsequently found wide use in formal and natural language semantics, computation, and in the philosophy of logic and language.

My aim in this chapter is to explore a couple of aspects concerning the relation between the side of Peirce's thinking from the perspective of game-theoretic conceptualisation, and his logical studies dating especially from the later period of his life. For one thing, such affinity is evident in his overall diagrammatic approach to logic and reasoning, and in his theory of existential graphs (EGs). Despite these findings, his logic does not share many of the modern features of the game-theoretic interpretation, such as winning strategies, crucial in connecting the truth or falsity of formulas to the processes involved in their interpretation. I will argue, however, that early anticipation of the game-theoretic notion of strategy is buried in his concept of a habit.

The modern era of game-theoretic semantics started with the contributions of Henkin (1961), Hintikka (1973a), and Scott (1993). The explicit connection between truth-values and the existence of winning strategies was noted in Hintikka (1973a). Some further philosophical and linguistic applications of game-theoretic semantics are investigated in Hintikka & Kulas (1983); Hintikka & Sandu (1991, 1997). The otherwise extensive and well-represented study of Houser et al. (1997) does not recognise this important dialogical or game-theoretic side of Peirce's logic.

That language and other symbolic systems can be compared with games, interactions or contests of different kinds is, of course, an ancient idea. The generalisation of the idea of the polar opposition of two into metaphysical portions is found in the *chôra* and *kosmos* of Plato's philosophy, in the feminine and the masculine, the distracted and the ordered, the changing and the permanent, the diachronic and the synchronic, and the inseparable and infolded parts of the fluctuating universe's primeval constituents. At times, one of the oppositions dominates, but soon the other will take over. Apart from the theologico-cosmological overtones, which will not be discussed here, the pattern has recurred in philosophical attempts to understand ideas ranging from the Newtonian concept of force to Friedrich Nietzsche's (1844–1650) will to power, and from Cartesian motion to all that rationalists would explain by action versus passion. It permeated Kant's philosophy and the Hegelian Ego, much of the latter only to be kidnapped and put into psychoanalytic services.

As a metaphor for argumentation, Aristotle's Topics and its later incarnations such as *ars obligatoria* are set up as dialogical duels. In the 20th century, dialogue logics resorted to strategisation in their goal of clearing up the concept of proofs. The idea retained its topicality in many of the theories of computation. Attempts have been made to recast the notion of computation itself in terms of interaction between the Computing System and its Environment (Nature). Such attempts might reflect the embodied notions of cognition, but may also be traced to Peirce's understanding of the mind as the "*sign-creatory in connection with a reaction machine*" (MS 318: 18, original emphasis).[5] I will provide a more extensive review of the range of fields in which the notion of games has found its intellectual home in exact and natural sciences in Chapter 10.

Reflecting some genuine plays and recreations such as chess, language has been the object of interest not only in Wittgenstein's philosophy of language, but also in the work of the semioticians of the early 20th century. Ferdinand de Saussure, a notable pioneer of structural linguistics, considered the game of chess to be the man-made counterpart of what language provides in a form that has emerged by natural processes. He was soon accompanied by Louis Hjelmslev, Roman Jakobson and many others in the semiological tradition. The closer comparison that de Saussure went on to make is essentially the presentation of a list of some analogies between language and chess, such as their dynamics, conventionality of rules, and positionality (history-freeness). The difference he suggests is in the notion of deliberation: while in chess the player intends various moves, in language moves are spontaneous and fortuitous. If we interpret this as the difference between what is strategic and what is non-strategic, de Saussure ends up advocating a difference that runs the risk of erasing practically all he wants to be chess-like in language from it.

The main reason why de Saussure advocated synchronic linguistics was that he took games to be secondary to rules. The converse, according to which games precede the rule-governed system of constitutive rules, is what diachronic linguistics professes. All game-theoretic actions contribute to the meaning, including the history of how the positions are arrived at. I argue in Chapter 8 that this was Wittgenstein's mature position.

Many others then followed de Saussure and Wittgenstein, inspired by the potential of comparing language and thinking to the game of chess to a varying degree and with varying success. Peirce was ahead of the others, however, and he put the comparison into a profound logical perspective.

Language games are not only an illustration of Vaihingerial "as if" philosophy, or something that was termed by Richard Harré "the weakest of all forms of theory — the use of metaphors", namely the last resort of theoreticians of science engaged in the process of trying to select the "candidate for reality" from among the multiplicity of models (Harre, 1961, p. 26) (Harré was attacking the idea of mathematical and formal models as such candidates). They should also be put to the test by harnessing them to methods that pertain to the mathematical theory of games. This theory was largely developed by John von Neumann and his successors, but there were others, too (Chapter 7). It soon started to gain credence in neighbouring disciplines, most notably in economics.

Few have acknowledged Peirce's significant role in the development of economic theories (Hodgson, 1999; Wible, 1998, 2000). Peirce and James influenced Thorstein Veblen's (1857–1929) economic thinking. Veblen boasted economics to be post-Darwinian evolutionary science. The concepts of equilibria and utility were then used during the early phases of economic modelling, but were precisely formalised in game-theoretic terms much later, although this was at least faintly envisaged by some early economists of the 19th century. For instance, Peirce studied and appreciated Francis Ysidro Edgeworth's (1845–1926) Mathematical Psychics, an early work on econometrics and the application of mathematics and science to psychological phenomena, and a precursor of equilibria analysis of social phenomena, even game theory, published in 1881. The extent of this influence does not appear particularly broad, however, and Peirce was quick to dismiss Edgeworth's views on probability as quite untenable, albeit unorthodox and atypical.[6]

Despite this dismissal, the ideas of these two men were not altogether different. Edgeworth realised that if one abstracts away from institutionalised economics, one may treat the remaining material (the core in gamespeak) as applicable to a much broader range of cases. Likewise, Peirce's notion of a habit (see below) was not evolving on individuals' actions, but rather starting out from the motivational and dispositional attitudes of those individuals, the "habit-taking" tendencies of all beings.[7]

As an independent prospect, games returned to the philosophical scene as a logical theory of the interpretation of formal languages, vindicating Peirce's vision in several respects. The rest of this chapter is devoted to the question of what this vision was.

2. The emergence of the notion of strategy

At a very early stage, Peirce appears to have advanced the connections between dialogical or game-like conceptualisations and logical notions. It is mostly his unpublished writings that contain explanations of the meaning of an algebraically notated quantifier as an interaction between the Utterer and the Interpreter. Many descriptions of this idea also surface in his published work (the special terminology is explained in later chapters):

> Begin by saying: "Take any things you please, namely," and name the letters representing bonds not encircled; then add, "Then suitably select objects, namely," and name the letters representing bonds each once encircled; then add, "Then take any things you please, namely," and name the letters representing bonds each twice encircled. Proceed in this way until all the letters representing bonds have been named, no letter being named until all those encircled fewer times have been named; and each hecceity [proper name, A.-V.P.] corresponding to a letter encircled odd times is to be suitably chosen according to the intent of the assertor of the medad proposition, while each hecceity corresponding to a bond encircled even times is to be taken as the interpreter or the opponent of the proposition pleases. (3.479, c.1896–07, *The Logic of Relatives*).

To the same effect, consider also: "In the sentence 'Every man dies,' 'Every man' implies that the interpreter is at liberty to pick out a man and consider the proposition as applying to him".[8]

It also turns out that Peirce envisioned an elementary game-theoretic interpretation of modal notions, as shown in the passage in which he considered the question of identities of individuals in terms of whether it is possible that similar principles to do with general, essential or determinate characters that distinguish actual objects from one another carry over to general characters that pertain to possible objects:

> How can a character be *general* which cannot possibly belong to more than one possible object? Let us devote a minute to the examination of the nature of the impossibility of two objects precisely alike in all respects being two, in the case of necessities, actualities, and possibilities. A necessary proposition is one which makes its predication of whatever case the interpreter of it may imagine, as contradistinguished from a universal proposition which allows the interpreter a choice only among existent cases. Two forms of proposition predicating the same determinations of any object chosen in either way are one and the same proposition. So two general terms applicable to whatever the interpreter may choose under the same limitations are the same. (NEM 2:516; cf. Hilpinen 1995).

An interesting open question here is whether Peirce thought of something like the accessibility relation familiar in modal logic as a prerequisite for conceiving alternative states of affairs. This passage tentatively suggests that such relations may not be essential for possible scenarios that participants can "imagine". I will return to this question later on in this and the next two chapters.

The motivation for using the terminology of interactive game situations was, for Peirce as indeed for many of those who succeeded him, to spell out the meaning of quantified statements, rather than to explore in detail how games can actually be played. Such prioritising of the activities of contestants over actually playable situations may provide a clue as to why he did not come to use the term 'game' in relation to his logic. Only later has it become clear how Peirce's logic may be neatly pigeonholed.

Peirce did conduct a number of studies on other kinds of games, such as chess, Tit-Tat-Too, various card games, betting, games of chance (in relation to probability calculus), and many others.[9]

While he did not make any genuine use of the actual term 'game' in his logical investigations, he frequently admitted to having been deeply impressed by Friedrich C.S. Schiller's conception of play in Aesthetische Briefe (MS L 463: 25), a work in which Schiller discusses 'World-spirit's *Spiel-trieb*' (dubbed by Peirce "mere amusement" or "the play of musement", Ger. 'play-drive'). He even judged 'game' as one of those entries in the Century Dictionary that deserve a fairly extensive definition.[10]

Recreational plays and games are quite different, but they also have similarities. No one would deign to play a game with those who do not observe the rules, and are thus do not take it seriously, but alas, if they react to the outcomes, other participants are prone to accuse them of taking it too seriously. There is a trade-off between the amount of musement that can be garnered from a single game and the complexity of its constitutive and regulative rules.

As far as Peirce's use of, or intention to use, game-related concepts is concerned, the reason why he did not come to develop the connection between logic (semeiotics) and games (dialogues) further was the lack of one of the most important game-theoretic conceptualisations, namely the notion of strategy. Hintikka (1998, p. 515) has remarked that "the concept of habit was one of the notions [Peirce] used to serve some of the same purposes as the notion of strategy has been introduced by later thinkers to serve". Indeed, the theory of games deals entirely with games of strategy, and other recreational, non-strategic games are destined to fall outside.

However, there are reasons to believe that Peirce's game-theoretic characterisation of logical notions — which he devised not only for quantifiers and modalities, but also for logical connectives[11] and negation[12] — are far more advanced than has been acknowledged in the literature so far. He did appear to perceive, however remotely, the need for a concept of strategy, albeit in the disguise of his sweeping notion of a habit.

Some preliminary evidence for this suspicion is to be found in places in which Peirce discusses habits in relation to interpretation: "The interpreter will have formed the habit of acting in a given way whenever he may desire a given kind of result" (5.491). This statement is interesting, because here he addresses one participant of the game of language, the interpreter, and emphasises his or her decisions based on the concept of desire. In addition, he writes elsewhere: "A habit arises, when, having had the sensation of performing a certain act, $\mathfrak{m}$, on several occasions $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$, we come to do it upon every occurrence of the general event, $\mathfrak{l}$, of which $\mathfrak{a}, \mathfrak{b}$ and $\mathfrak{c}$ are special cases".[13] One possible interpretation of this is to take the character of a strategy to be an abstraction of a regulative rule that looks away from any single position in which the player may be located in the relevant semantic game. A motivation is that the outcome of the game — perhaps in terms of payoffs given in the normal-form matrix or in the extensive-form terminal nodes (Chapter 7) — is typically assigned only to total strategies, not to any isolated and individual action or a set of actions, since more relevant than mere observable actions are how actions are constituted. In other words, game theory is concerned with seeking solution concepts that associate payoff profiles with outcomes so that some indication of the right or good courses of action may be perceived.

There is further evidence in Peirce's writings to support the view that habit contains, implicitly, aspects of strategic behaviour and action: "Action cannot be a logical interpretant, because it lacks generality ... But how otherwise can a habit be described than by a description of the kind of action to which it gives rise, with the specification of the conditions and of the motive?" (5.491).[14] In the terminology of semantic games (Chapter 7), the motives Peirce refers to are the purposes of the two players, the verifier and the falsifier, the former aiming to verify a sentence or an expression and the latter aiming to falsify it.

Further, he writes: "It would be necessary, in order to define a man's habit, to describe how it would lead him to behave and upon what sort of occasion — albeit this statement would by no means imply that the habit *consists* in that action".[15] If we take it that a logical interpretant of a sign is its meaning, what Peirce is in effect saying is that no single action or sequence of actions, that is, no choice or sequence of choices as consecutive moves in a game, can spell out the meaning of the signs in question, because it does not put in the picture how one arrives at such choices. There has to be abstraction and generalisation. In order to do that one needs to effectively employ a strategy that leads to those actions. But strategic content is often difficult to identify, and one needs to content with analysing some extraneous output that becomes visible after the strategies have been applied.

For example, logical interpretants are formed by habits upon which an agent can exercise some real effect. A habit, then, organises interpretants into a general concept that only in certain cases gives rise to sentence meanings.

According to Peirce, this is not to say that there is no way we can know the inner structure of habit, however: "Even from the human mind we only collect external information about habit. Our knowledge of its inner nature must come to us from logic" (NEM 4:142).

Habits are the operational heart of pragmatism and the general rule of its logic. Relevant for Peirce here are "voluntary habits", which are subject to "some measure of self-control", under the necessary condition of "circumstances [having] a triadic influence [in] strengthening or weakening the disposition to do the like on a new occasion".[16] According to Peirce, this amounts to voluntary habit being "conscious habit". What is more, "Meaning of a general physical predicate consists in the conception of the habit of its subject that it implies. And such must be the meaning of a physical predicate".[17] Quite similar "conception of the habit" is involved in the game-theoretic process of interpreting sentences of language or logic.

Moreover, "the habits must be known by experience which however exhibits singulars only".[18] Likewise, strategies are exemplified by individual actions. Since "Our mind must generalize these [singulars]", and that generalisation is achieved by the a "triadic consciousness" or "purpose to act in certain ways (including motive) on certain conditions",[19] we may reinterpret the pragmatist rule of logic in the setting of purposeful, strategic action guided by the triadic relationship between modalities, actions and payoffs.[20]

Admittedly, many other things may be distilled from this all-purpose notion of a habit, but seeing it as a normative rule of action indeed comes close to what nowadays is meant by a strategy. As general rules of action, habits are non-deterministic functions from possible situations to actions. In the game-theoretic setting they acquire purpose specified in the payoff structure. To be noted well is that the relevance of the expectations concerning future courses of events to the constitution of meaning does not lead to semantic holism, since the players do not have perfect foresight and thus cannot compute the habitual relevance of all actions.

Further support for these conclusions comes from studies that address the history and evolution of the concept of habit, such as Camic (1986)'s sociological study. These studies have not quite noted the affinity between habits and strategic actions, but indirectly vindicate it by discussing the across-the-board explanatory character of habits that was in use in early studies of psychological and biological behaviour and adaptation, as well as in socio-economic considerations of Western social thinkers. What happened to that notion was that it was soon "intentionally expunged from the vocabulary of sociology as American sociologists attempted to establish the autonomy of their discipline by severing its ties with the field of psychology, where (esp. in connection with the growth of behaviourism) a restricted notion of habit had come into very widespread usage" (Camic, 1986, p. 1077). Indeed, the original meaning of 'behaviour' came from the Latin *habere* ('to have'), but this was distorted in behaviourism to harmonise with action and doing, not with being in possession of some general capacity or respect.

What, then, are the measurable quanta that are won or lost in the plays that produce the actions Peirce describes in his passages just quoted? If we look at his notion of assertion, it is fascinating how close to the idea of strategic interaction it comes. Assertions bear a responsibility to its utterer: if they are not true, the utterer will be penalised. Similar normative and disciplinary aspects also apply to the receiving end of the communication, the interpreter.

In order to set these into a game-theoretic scheme, one simply needs to quantify the value of the strategy a player follows, taking the amount of punishment that the actions may bear into account in describing such payoffs. What Peirce endorses is the normativity of language, in other words taking language as a system of norms prescribing its correct use and semantic meaning relations. From the use and meaning normativity the normative component may then escalate to the prosodic, phonetic, morphological, grammatical and syntactic structures of language. The pronouncement of logic as a normative science is, of course, much better known and better documented in Peirce's writings than the normativity of language, but it is only natural that the latter is no different.

In addition to the quantificational part, there need to be rules that define the legal choices in the game, plus some generator that acts accordingly as principles of rationality. The most obvious generator for Peirce was the mind, which, as noted, he referred to as "sign-creatory", something resonant with machine-like systems that put forward and reproduce endless cycles of further signs. For him, rationality was a term perhaps best seen as a principle welded into the notion of a habit, in accordance with the principle of the *summum bonum*, the ultimate good that looms on the horizon. The regulative rules that define which moves are permissible in a game and which are not presents no great difficulty; if the context is natural language then it contains a variety of rules appropriate for the interpretation of natural-language expressions. Some such rules have been described in the literature on game-theoretic semantics (Hintikka & Kulas 1983; Pietarinen 2001b). As a secondary component, the set of defining rules also comprises principles that govern conversation and communication between different parties in typical situations of language use in general. If the context is logic, the rules will be those given by the semantic game for the logical language in question (Chapter 7). The sense in which evolutionary considerations enter the picture when the semiosis tends to infinity is covered in Chapter 11.

Furthermore, epistemic dimensions permeate Peirce's habit. He introduced them to facilitate its use as an organisational principle that could bring an organism's modes of response, or its interaction with an environment, into a unified and strategic rule of action. These organisms need not be conscious, though the control of habits is more or less voluntary. Such habitual responses are becoming increasingly relevant in evolutionary approaches to language and in game-theoretic approaches to biology, where principles of reflection and rationality are being mitigated.

Such activities, and the frequent references Peirce makes to epistemic, temporal, dynamic and interactive notions, suggest that one ought to place aspects of a habit into a category that roughly corresponds to the contemporary paradigm of 'computation as interaction' (Chapter 8). Meaning of a program, sentence or logical formula may be viewed in computational theories endorsing this classification as a relation between a sign (program) and the interpreter (environment).

It would be productive to explore these connections further. For example, one could think of interactive computing (or game semantics in the so-called programme of 'geometry of interaction') as an epistemic game that computes strategy functions (habits), and of game-theoretic semantics as giving meaning to constituents as a system or environment. There is a ring of the pragmatic maxim here. Accordingly, I will examine some interrelations further in Chapter 8, bringing Wittgenstein into the picture. Further, Chapter 13 concerns other recently emerged forms of communication that can be positioned in the pragmatic tradition. However, I will not discuss Peirce's speculations about the possibility of logical reasoning machines, or whether it can be shown that such machines are impossible (which Peirce did not believe). What the contemporary street value of his suggestions about logical machines is has to be left for further deliberation.

Peirce's notion of a habit surfaces most frequently in his semeiotic and metaphysical work, not in his logical studies. It shows up in psychological (roughly, habit as a disposition), attitudinal (habit as a belief), and physical (habit as regularity in Nature) contexts. However, he did not connect this notion with his otherwise rather advanced game-theoretic outlook on logic in any unequivocal manner. The connection is bound to remain incomplete. For example, the relation between winning strategies and the truth of a proposition is absent. Such a connection is of paramount importance to the modern theory of semantic games (Chapter 7). Admittedly, some fragmentary indications exist: "The duality of the *ego* and *non-ego* is the chief constituent of the idea of the Truth" (MS 515: 24), but this is as far as the connection extends.

I think that part of the reason for this failure is Peirce's mixing of what were later on characterised as syntactic, semantic and pragmatic elements in logic. This cocktail is particularly evident in his groundbreaking theory of EGs that provided a remarkably advanced view of later modal-theoretic approaches to different logics, including first-order and modal logics. Peirce's logic has a clear tendency toward notions that, from the later perspective, are recognised as semantic, but the demarcation between syntax, semantics and pragmatics is not a marked one, and in all likelihoods was not even intended to be something definite.

Despite its rarity in logical contexts, a habit provoked interesting remarks in 4.572 [1905, *Prolegomena*]: "The logical relation of the Conclusion to the Premisses might be asserted; but that would not be an Argument, which is essentially intended to be understood as representing what it represents only in virtue of the logical habit which would bring any logical Interpreter to assent to it". If instead of reading "logical habit" in this passage, we were to read "strategy", we would gain a lucid explication of the meaning of signs (here: arguments) as interpreted by strategic actions undertaken by one of the participants engaged in the process of dialogically or game-theoretically depicted interpretation.

* * *

According to Wittgenstein, we recall, a game, like language and a rule, is an institution (Wittgenstein, 1978, VI, 32). Peirce writes in similar tones. However, for both of them, the notion of a game may impose either too much or too little structure. As far as Peirce was concerned, the term 'game', if we are permitted to use it anachronistically in the sense of the theory of games, is too rigid in much the same way as the real line of numbers is too rigid (or too small in multitude) to uphold the true notion of continua, but too loose in the sense that it does not bring into the picture what the reality of expectations, the would-bes of future actions, might consist of and how they would affect one's decisions. For Wittgenstein, the whole affair of games runs the risk of arbitrariness, that is, they appear to employ arbitrary rules for any given particular situation. For instance, non-cooperative games would need to be repeated in order for one to even begin to gain glimpses of how individuals follow a rule. This latter drawback is much the same as expressed by Peirce: the epistemic considerations pertaining to future deliberations are still missing in the major classes of games.

Indeed, even after some seventy-five-odd years of the existence of game theory, very little is known about all that out-of-equilibria that dispense with hyper-rationality, perfect communication, information, memory, and competition, and that include dynamic or non-stationary payoffs, very large sets of players, syndicates, negotiation with incomplete information, cheating, lying, threat, sympathy and so on.

Even if such notions could one day be accommodated into game theory, with the obvious risk of rendering it technically abstruse and something the critics would no longer be well acquainted with, the theory is not likely to account for all kinds of language games, since there are always those that have no regard for many of these notions. Not all interactions involve judgements to be made about percepts, as many function by way of habitual and spontaneous responses. Some such activities, the critics may well maintain, just are not amenable to mathematical scrutiny.

I will not profess to any full response to such criticism, but some of the issues concerning Peirce's and Wittgenstein's views on language, logic and games are exposed in Chapter 8. The idea of institutions, or societies of agents and inquirers, prevails in game-theoretic arguments concerning the evolution of semantics and pragmatics of language (Chapters 11 and 12) and the role and use of communities of agents in computation (Chapter 14).

It is worth noting that in the original account of Darwin's theory of evolution, habits might have been playing the role of adaptation. A passage from his autobiography is illustrious:

> I soon perceived that Selection was the key-stone of man's success in making useful races of animals and plants. But how selection could be applied to organisms living in a state of nature remained for some time a mystery to me. In October 1838 . . . I happened to read for my amusement 'Malthus on Population', and being well prepared to appreciate the struggle for existence which everywhere goes on from long-continued observation of *the habits of animals and plants*, it at once struck me that under these circumstances favourable variations would tend to be preserved and unfavourable ones to be destroyed. The result of this would be the formation of new species. (Quoted in Plotkin 1994, pp. 28–29, emphasis added).

Evolution that Peirce endorsed was nevertheless quite different and of a broader range and signifigance than it was under Darwin's conception.

3. The economics of research and evolutionary metaphysics

Even though Peirce moved away from any particular 'solution concepts' for a game of interpretation, his theory of ideal inquiry, which, seen from the institutional perspective, might suggest a candidate for such a solution. The scientific inquiry is performed by a community of agents that in the long run tends to agree on the final opinions concerning the truth of their scientific goals. Likewise, the institutional theory of economics may be viewed as a solution to a game of inquiry, a game in which one puts questions to Nature and holds it to be a relatively reliable source of information. Experimentation on diagrams, the moving pictures of thought as will be observed in Chapters 4 and 5, is one such form of communal activity, albeit that it may well take place within the single mind of one inquirer, consisting of a series of phases of dialogue and deliberation rather than any actual cooperation in societal settings.

Without such an institutionalised and communal approach to experimentation, the limiting goals of inquiry may not succeed. No individual is capable of approaching this limit to anything other than an extremely insignificant degree. In order to come to terms with this brute fact of life, Peirce frequently spoke about the economy of research. The intention was to outline a methodology that can be best upheld within the group of investigators seeking such tests that delineate good hypotheses among a class of possible ones. He introduced the notions of caution, breadth and incomplexity as the principles according to which inquiry is, economically speaking, best conducted.

The economics of inquiry was Peirce's brainchild. He perceived that science is of human making, not free from considerations of what self-interested and self-desired agents would do in social and economic contexts. He argued that as any commodity in the market, knowledge and information could be priced, managed, marketed and sold: "Knowledge, even of a purely scientific kind, has a money value".[21] Similar views were propagated later by Freidrich Hayek, Ludwig von Mises, and other predominantly Viennese economists, soon to be joined by von Neumann and other ludents. Knowledge beseech a habit-change to take place in the *homo œconomicus honorabilis*, but it also marks a measurable process that we try to harness and turn into a useful good for commercialisation.

What Peirce suggested, and what Nicholas Rescher and others have advanced, is the cost-benefit analysis of what researchers do, what they should do, how would they develop the most reliable methods, and how they should communicate in the most efficient way, given the expenditure of finite amount of time effort, energy and money. A typical domain for the cost-benefit analysis is the minimisation of costs of testing hypotheses while maximising the information that comes out from the tests.[22] The general question that it addresses is that, given a set of prima facie plausible hypotheses produced in a given research project, given investible capita (either fiscal or cognitive), which of the hypotheses disqualify? The point is that this analysis is conducted with the benefit of all science, or the totality of the community of inquirers, in mind, not with reference to some specific group of individuals and their welfare. In this sense, it is the answer to the Gospel of Greed that depressed Peirce in societies in which progress no longer comes from individuals merging their individualities in sympathy with their neighbours.[23]

Peirce distinguished three components in his argument for research economy in relation to hypotheses. First, there is the quality of caution, according to which a hypothesis is broken up into its smallest logical components. For instance, big questions are better to be divided into series of small questions, and why-questions are better to be divided into series of yes-no questions.

According to the component of breadth, a hypothesis is evaluated by its applicability for the same underlying phenomena occurring in other related subjects across contexts and circumstances. Several explanations of the same phenomena should be evaluated according to their consequences that can meaningfully be sorted and classified.

Finally, the quality of incomplexity (absence of complexity, simplicity, artlessness) says, among other things, that because hardly any hypothesis is optimal, such not altogether successful and complete hypotheses ought at least to "give a good leave" (EP 2:110). That is to say, in case one wishes to refute a hypothesis, it could be viewed as setting an example of a good conduct to be followed, by attempting as large a 'break' as possible from it, for example, or by necessitating the opponent the 'use of sides', thus referring forward to new hypotheses. Underdetermination, being a result of accident rather than of design, is the property of billiard balls that Nature it likely to bring off.[24]

On the other side of the metaphysical fence there stands the evolutionary philosophy. Peirce's approach to metaphysics was in essence that the universe consists of continuous processes, not things, and that these processes are continuously connected. Thoughts and intellectual ideas alike are likewise instances of such synechist processes. The universe is governed by three interrelated principles: absolute chance, mechanical necessity, and their synergetic evolutionary love, agapism, of which chance and necessity are degenerate forms. Chance begets order, and order begets laws. Chance, the disorderly property of the chôra of the universe, cannot and ought not to be explained by any theories that we possess. But laws, including natural laws, are not immutable and eternal. They are evolutionary. They are habits of cosmos just as logical truth of a proposition is a habit of its assertor. Habits grow. Universes are progenies or experiments in which laws of the most fundamental sciences are contested, challenged and litigated. We happen to live in one of them. Others, where physical or biological constants may have had different values, had different regularities and irregularities, and were not so lucky and did not survive.

What follows is that no law of physics or invention of experimental science is needed to understand natural laws. In recent cosmological theories of quantum gravity, similar lines of thought are present as in Peirce's metaphysics. They allow room for physical laws that may undergo change, and that the universe is a network of relatives rather than things. They attempt to demonstrate that not by experiments, but by finding a consistent mathematical description in terms of some constructive logic such as topos theory. They allow for observer-related perspective to systems, like non-commutative geometry, which allows for uncertainties in geometric structures. These technical developments are unlikely to answer to the philosophical questions of what natural laws are, how they originated, and what their explanatory value is, but they are a "condiment to excite" the "own proper observation" prompted by these questions (1.241).

The prospects of agapistic evolutionary philosophy were not rosy. Like habits, it was soon superseded by utilitarianism and neoclassical economics promoting free market, bolstered — not entirely justifiably, because of the omnipresence of chance and trial-and-error — by the Darwinian theory of evolution. Why, then, did Peirce accept the economic approach to research in terms of the cost-benefit analysis, apparently so in some discrepancy with agapism?

The answer is that the project Peirce dubbed the economics of research actually contains more than just maximisation of expected utilities given an adequate span of time, energy, intellectual capital and labour. It has to do with the methodology of inquiry. It is thereby linked with methodeutic (speculative rhetoric, formal rhetoric) of normative logic and semeiotics (Chapter 1).

To begin with the methodology, the project states that only plausible hypotheses come to be considered by abduction, much in the same way as good chess players only care to consider a tiny fragment of possible scenarios. The means of doing so have been implanted into humans during the millennia of their existence, and in this historical development their habits have been nurtured and methods arisen that give instructions of how to block the generation of any (that is, any conceivable but implausible) hypothesis. This historical part of economics is governed by chance (tychism), necessity and their synthesis, and is related to the habitual metaphysical evolution of natural laws.

The relation between habits and pragmatic maxim is that the habits, which may be viewed as propositional attitudes of beliefs of agents, refer to those circumstances upon which agents are prepared to act. There is thus an element of possibility to them. Just as possible scenarios or figments of worlds represent states of affairs for which we must be prepared, so are habits, grown and unravelled as we explore the contingencies, the hypothetical states of affairs that constitute the practical bearings or conceivable effects of the concept that we strive to understand and evaluate.

Alongside with the habits of research that mainly contribute to abduction, the economy of research recommends explicit strategies for conducting actual, institutionalised research. These include methods of taking the likelihood of plausible hypotheses into account in tasks related to induction, performing experiments on complex visual representations of scientific data, or in setting up costly machinery to test infinite divisibility of matter. Further, Peirce explains, “Among hypotheses choose one whose elements are well understood, so that unknown complications, and consequent expense of energy cannot arise. Prefer general hypotheses to special ones, provided the more general are so by being simpler; if they are so by being complex, it is necessary to consider the economics of testing them more particularly” (MS L 75). It is in these optimisation and decision-making tasks that the cost-benefit analysis is best applicable.

The bifurcation of economics into the evolutionary and utility-expenditure compartments implies that Peirce’s project of the economy of research is not conflicting with his agapism. The *summum bonum* of open-systems inquiry is in most respects increased by heeding the principles of economy (Chapter 13).

Furthermore, according to him, methodeutic, alongside with critic, is cheap to employ, and by virtue of the principle of economics of research should hence be used throughout. As it studies the power of signs as they appeal to the mind, and as it tries to uncover how the interpretants are linked with symbols, indices and icons, this component of inquiry thus leads us straight to the heart of pragmatism. Methodeutic is also related to albeit more general than pragmatics, which has predominantly been a linguistic discipline. Hence pragmatics of language ought to be studied in connection with methodological questions (Chapter 12). Like methodeutic, its main purpose is to find out how the energies contained in assertions are transmitted into the minds of their interpreters, and what their consequences are. In this sense, speaking of the assertoric force of statements is apt, but its area of application broader than what the received speech act theories suggest. For instance, questions should be studies in tandem with assertions, also having a degree of energy — potential if not kinetic — in their capacity of requests for information. This is the route that paves the way in which economic considerations are pertinent to logic and the methodeutic of interrogation in inquiry. After all, according to Peirce, "It is a question whether [economics] is not a branch of logic" (MS L 75).

4. Graphs, semeiotics and language

The theory of existential graphs To return to the notion of a habit in logical contexts, it is virtually non-existent in Peirce's *chef d'œuvre*, or in the main component of his logical analysis, his EGs (4.347–584, 1901–02; Roberts 1973a; Zeman 1964), despite the fact that EGs are not at all devoid of game-theoretic ideas. He remarks himself on the role of such activities in this context. For instance, in relation to his tinctured (multi-modal) EGs, game-theoretic concepts rear their heads in his interpretation of the whole theory as collaboration between two parties. These parties or functionaries are to be understood in a broader sense, not necessarily only as persons or animate players but also as mental attitudes or states of mind in a single individual, such as the 'Ego' and the 'Non-Ego' (MS 515: 22, 24).[25]

In fact in most contexts, Peirce laid out the meaning of EGs in terms of two such functionaries: the Graphist who scribes a graph instance on a sheet, and the Grapheus, the interpreter of the graph, who is allowed to manipulate the given graphs according to the rules of the system.[26] He calls the actual starting point of the investigation "the Phemic sheet" (the universe of discourse) designated by one of the participants. Other sheets are placed upon the Phemic sheet as the need arises. Negation, for example, is interpreted as a game-theoretic role switch: "Should the Graphist desire to negative a Graph, he must scribe it on the *verso*, and then, before delivery to the Interpreter, must make an incision" (4.556, 1905). He adds in a footnote: "I am tempted to say that it is the reversal alone that effects the denial" (these statements return with vengeance in Chapter 4). In fact, we can go as far as near assimilation of all the five conventions given for tinctured EGs in 4.552–563 [1905] with general definitions of a game and its intended structure. Furthermore, the four methods of interpretation, or permissions (4.565–569, 1905) are remarkably similar to the constitutive rules of such a game, that is, to the rules prescribing which moves are legitimate and which are not. Similar comments may be voiced on the conventions and permissions crafted for other systems of EGs besides the beta part.

These remarks should suffice to substantiate the finding that Peirce, to some extent at least, did conceive of modalities as related to game-theoretic activities, and that the game approach in general is useful in theorising about modalities. This comes close to the semantic evaluation of modal possible-worlds models as a piecemeal exploration of accessible and conceivable states. Even further, a moment's reflection on Peirce's tinctured EGs gives us a decent idea about how they could be interpreted in terms of possible-worlds semantics: by varying the tinctures instead of the books of sheets we would get other types of alternatives accessible from the actual sheet, namely those of some other agents whose modalities or propositional attitudes also need to be represented. Taken in totality, then, Peirce's theory of EGs marked the beginning of the development of modern possible-worlds semantics, while at the same time incorporating several game-theoretically recognisable ideas into an overall logical and semeiotic system of diagrammatisation.

The gamma part of EGs (4.510–584; Chapter 4) played a key role in anticipating, and to some extent even in contributing to, the later development of the semantics of modal notions. However, the primary reason why gamma graphs did not actually succeed in forming a semantically articulated modal system was the lack of a clearly defined accessibility relation. Yet, some scattered remarks suggest that something like an accessibility relation between "states of information" (Peirce's term, 4.517, 1903) was what Peirce had in mind, for he occasionally referred to such relations as "selectives", drawn as lines between the states. Sometimes these relations were even crossed to distinguish a particular state.[27]

Furthermore, this idea bears a resemblance to indexical notions of time and location. Nowadays, such notions are customarily couched in two- or multidimensional modal semantics.

The use of selectives also serves to assign meanings to proper names, something that Peirce took to be produced by two or more instances of a name attached to the graphs. The selectives are meant to denote the identities of individuals, such as by their proper names. At times, Peirce expressed some doubts about the status of selectives in EGs, and thought of them as redundant given the machinery of subject names linked with rhemas. It has also been claimed that selectives function like bound variables in first-order logic. It would be more accurate to say that they represent instantiations or substitutions of the dots of the rhemas that play the role of bound variables.[28]

Although mainly developed in the beta part of EGs, identity in the gamma part as a continuous connection between spots is not far removed from the contemporary concept of identification in semantics for predicate modal logic (Chapter 4). For Peirce, the notion of identification meant that the interpreter has to meet with a proper name many times in several contexts, or else he fails to be fully acquainted with it. The first cycle of interpretation connected with a name starts with a selective (that is, the outermost occurrence of the name), which then has to be presented to the interpreter repeatedly, on different occasions and in different contexts (4.568, 1905). Note also that, "By a Proper Name, I mean a Sign whose Object a name of anything considered as a single thing; and this thing which the Proper Name denominates must have been one which the Interpreter was already acquainted by direct or indirect experience".[29] "A single thing" does not need to be a denotation of a proper name as a single existent individual, but may also represent an idea or a character.

A logical approach to epistemic notions was not alien territory to Peirce, for he took it that a modal proposition is "about the universe of facts that one is in a state of information sufficient to know" (4.520, 1903). He also recognised that propositions have to be evaluated against *conceivable states of information*: "Suppose, however, we wish to assert that there is a conceivable state of information of which it would not be true that, in that state, the knower would not be in condition to know that [the graph] g is true" (4.520). He used a special cross mark sign in his gamma graphs to distinguish a particular state of information from the one to which it refers. The selectives to which these marks refer have, as he remarked, "the additional peculiarity of having a definite order of succession", and thus "are of great use in cleaning up the confused doctrine of *modal propositions* as well as the subject of logical breath and depth".[30]

These remarks may be deciphered by setting them in a modern context. What he was thinking about was logical depth in terms of modal depth as nested occurrences of modalities. Its counterpart in existential-graph semantics is the succession of the states of information by means of the special cross marking. The nesting of knowledge, or a version of the KK-thesis (that is, 'knowing entails knowing that one knows', an issue extensively discussed in the first semantic treatment of the logic of knowledge and belief in Hintikka (1962) and in the subsequent literature, and with its precursors in the later middle ages, see Knuuttila 1993), is traceable in the form of "peculiar and interesting little rules, owing to the fact that what one knows, one has the means of knowing that one knows" (4.521, 1903). However, Peirce refuted the straightforward rule that "whatever one knows, one knows that one knows, which is manifestly false" (4.521). It is immediately after these remarks — probably the only place in his writings — that he introduced the arrow-like notation signifying the fact that one state of information follows another.

In the light of these findings, it is justified to conclude that, in addition to tinctured EGs, the gamma part contains elements of possible-worlds semantics, although this did not come to have a distinctly defined accessibility relation in the sense in which it is used today (Zeman, 1986).

It should be noted, however, that Peirce referred to the projected delta part of his EGs a few times. It is not known what the delta part was to be about, but in all probability it was meant to deal with modal predicate logic in a more explicit manner than the gamma part, and to repair many of its shortcomings, while the gamma part, with which he struggled in his later years, revising it several times, was converging towards a system that could represent and reason about abstractions, collections and higher-order notions. There is textual evidence for this, for one of the few places in which he refers to the delta part is in the context in which he thought this was what one still had to "add . . . in order to deal with modals" (MS 500: 3) No known document discloses what kind of system Peirce had in mind with his delta graphs. In places he mentions the possibility of extending the graphical system beyond assertions, lamenting that he came to confine his sheets to assertions, and suggesting that other sheets would do as well — including platforms for non-declarative moods such as interrogatives and imperatives, and even those concerning feelings that abound in the arts of music and painting (MS 500). These do not belong to the province of necessary reasoning, such as the anatomy of deductive reasoning in mathematical arguments, however, and for that reason were excluded from diagrammatic studies by Peirce as late as 1911.

The incompleteness of his theory of modalities and his failure to fully recognise the need for a special relation of accessibility between various states of information (bearing in mind that we do not know what was planned in the delta part of EGs) serve to further illustrate the fact that he did not draw together all the detached pieces and ideas he had developed — presumably largely independently of whether he realised some connection between his ideas or not, and presumably largely independently of whether he had developed such ideas in his semeiotic programme of signs or in his more calculi-oriented, although by no means exclusively calculi-like, diagrammatic logical corpus.

A fuller discussion of these points, tracing the exact timing of Peirce's remarks on his dissatisfaction with gamma graphs and on his new projected work on EGs in relation to his other plans and the technological and other innovations of that era, is found in Chapters 4–6.

Logical semeiotics in perspective It was not only the obvious reasons involving the disorderly state of Peirce's writings, his constant lack of time and misplaced orientation, together with the regrettable failure of his colleagues to urge him to put his vast observations into a more coherent presentation, that were behind his failure to collate his rich ideas. There were deeper reasons, illustrated perhaps by a few words about his general outlook on logic.

Logic, as he came to conceive of the notion, formed only part of a much larger project of understanding thought and reality, including things that can be established by means of rules of inference, and things that can be established by means of concepts that we nowadays recognise as semantic, but is not exhausted by either of these methods. This larger semeiotic project, which he attempted to bring to light in the incomplete *Logic viewed as Semeiotics* (1.286–287, c.1904), promised to reveal several logics and calculi that he (and to some extent his collaborators and contemporaries) had developed, many of which clearly involved a semantic, or more strikingly, a model-theoretic component.[31]

Nevertheless, what we usually recognise as semantic in logic may not have sufficed for Peirce when the key problem is not only one of understanding the language–world or language–model relationships, but also of saying something meaningful about the model–world associations. As to the latter, the project that has been dubbed logical semantics, at least since Tarski made his contributions, would have had little to add to Peirce's general project in which the foundational questions address the role of human thought and action in mediating these links, in tandem with the pragmatic overtones that such roles will echo. In view of this, it is likely that Peirce would have remained singularly unaffected by the course research in logic took at the beginning of the twentieth century and later, rapidly becoming bogged down in a sanitised discipline in which pragmatic issues were isolated from the semantic domain, and in which there was a tendency to avoid any contamination that Peirce could have caused.

One distinguishing feature of the received game-theoretic interpretation of logic is that it evaluates formulas by starting with the outermost component and then proceeding from the outside in, ending when an atomic formula is reached (Hintikka, 1973a). Interestingly, we can trace this approach back to Peirce's treatment of EGs. He coined the method "endoporeutic" (*endon* 'within'; *poros* 'passage, pore', see Chapter 6). For example: "The rule of interpretation which necessarily follows from the diagrammatization is that the interpretation is 'endoporeutic' (or proceeds inwardly)" (MS 514: 16). This method was shown to be at work in Peirce's account of the evaluation of proper names, for instance. More precisely, the first occurrence of a proper name to the interpreter (the selective) has to be the outermost one, proceeding towards further, contextually-constrained occurrences and their interpretations. I have found very few passages in which Peirce elaborated this term in so many words, however. He even mentions the "Endoporeutic Principle" (MS 293: 53), but just a handful of scattered references to it exists elsewhere. Nevertheless, in many places in which the term is not explicitly mentioned, it is clearly being assumed as the principle behind the right direction of the flow of information in logical and linguistic interpretation processes. It is also a principle that would not endorse the so-called Frege principle of compositionality.[32]

There is thus room for conjecture. Had the endoporeutic method become more popular, we might have witnessed the game-theoretic development of logic in full, instead of the more prevalent Tarski semantics.[33] It is of some interest that it was only much later that the usefulness of game-theoretic methods was demonstrated in corners of logic in which the more prevalent methods failed. In retrospect, such developments have vindicated Peirce in that one of the most prominent methods in logical semantics in the early part of the last century only merits an isolated chapter in the study of logic in general, and a fortiori represents only a special case in Peirce's general semeiotic and endoporeutic programme of logic.

What kind of characteristics did Peirce assume, then, for the activities involved in uncovering the meaning of logical statements? There is not much evidence to suggest that he took games to form any fixed system with a predetermined structure and rules of operation. Yet in his (tinctured) EGs, for example, there are definitions of the game structure and the rules that players must obey. In addition, the various drafts of MS 295: 55–59 show the evolution of the transformation rules for EGs into rules that have a game-like character. But where did he stand on other general characteristics of such games? Is it meaningful to ask whether he thought that they should be ones of perfect or imperfect information? Should they be competitive or non-competitive? What about cooperation versus non-cooperation?

Yes, these are meaningful questions. The question about players' information has, in fact, already been answered (Hilpinen, 1982). For in MS 9: 4, Peirce remarked, "Whichever of the two makes his choice of the object he is to choose, after the other has made his choice, is supposed to know what that choice was. This is an advantage to the defence or attack, as the case may be". Hilpinen comments that, in modern terminology, this means that such games are ones of perfect information, and so is the logic. This is consonant with the fact that Peirce took the law of excluded middle to hold in the non-vague part of logic: "A Proposition is either True or False".[34]

On the question of the competitiveness of the players, Peirce remarked: "The utterer is essentially a defender of his own proposition, and wishes to interpret it so that it will be defensible. The interpreter, not being so interested, is *relatively* in a hostile attitude, or looks for the interpretation least defensible" (MS 9: 3–4). Hilpinen (1982) notes that Peirce meant his system to be what we currently recognise as congenial to zero-sum or strictly competitive games, in which players have competing and conflicting aims that they try to achieve (Chapter 7). Indeed, Peirce typically does not assume cooperation between the Interpreter and the Utterer, and so this conclusion is justified. However, it needs to be added that, at times, he characterised the game between the Graphist and the Interpreter, as in his tinctured EGs, as "collaborative" (4.552), and thus no similar competitive setting was being assumed therein.

Natural language Peirce was using a great deal of natural-language examples to draw motivation for as well as to give content to his logical investigations. Yet, natural language is one of the least-analysed aspects of his semeiotics, although his work provides a rich source for uncovering the meaning of sentence in terms of confrontation of two contestants:

> Instead of the selection of the instance being left — as it is, when we say "any man is not good" — to the opponent of the proposition, when we say "some man is not good," this selection is transferred to the opponent's opponent, that is to the defender of the proposition. Repeat the some, and the selection goes to the opponent's opponent's opponent, that is, to the opponent again, and it becomes equivalent to *any*. (3.481).

Here, the role switch is applied in the interpretation process to the phrases *some* versus *any*. Peirce frequently considered these quantifiers, especially when adducing examples of expressions that prompt the choice of an individual by one of the parties in the game.

> In my paper on the Classification of arguments, I endeavored to make out that *Some* could be so conceived that its iteration abolished it ('~~An~~ double particularization makes universal'). This I did, I believe, by conceiving ~~the select~~ *Some* to mean that a selection was to be thrown upon the interlocutor.
>
> All A is B. Take any A *you* please and it is B.
> Some A is B. Transfer to your interlocutor (me) the choice of an A and it is B.
> Some Some A is B. Retransfer to you the choice of A and it will be B.
>
> There are but few languages in which two negatives make an affirmative. If *not* means 'less than one' or 'fewer than one', fewer than fewer than one is simply fewer than one. The new signs I propose make *Some Some*, *All*. (MS L 237, 12 November 1900, *Letter to Christine Ladd-Franklin*).

The quantifier *any* was moreover held by him to be a universal quantifier just as *All*, although he recognised *Any*'s commonplace free-choice use.

There is a great deal of further evidence of the kinship between Peirce's logic and the more recent game-theoretic conceptualisations of natural language. According to Peirce, the sentence

> "Any man will die," allows the interpreter, after collateral observation has disclosed what single universe is meant, to take any individual of that universe as the Object of the proposition, giving, in the above example, the equivalent "If you take any individual you please of the universe of existent things, and if that individual is a man, it will die". (EP 2:408).

This is similar, both in spirit and in content, to the interpretation game-theoretic semantics assigns to sentences containing the universal *any* (Saarinen 1979):

> If the game has reached the sentence
>
> X – any Y who Z – W,
>
> then Nature may choose an individual and give it a proper name (if it did not have one already), say 'b'. The game is continued with respect to
>
> X – b – W, b is a(n) Y, and (if) b Z.

It is not only simple individual nominals, but also generalised quantifications that fall within the purview of game-theoretic interpretations. The prehistory of generalised quantifiers is indeed vital, yet still uncharted. As is well known, Frege considered quantifiers as variable-binding operators denoting second-order relations. What seems not to have been noted before is that the need to have generalised quantifiers in logic was already shown by Peirce more than a hundred years ago. For one thing, he refers on many occasions to "hemilogical quantifiers" in addition to universal and existential ones. They were taken to mean phrases such as *all but one*, *all but two* and so on. For example, the algebraic quantifiers $\Pi', \Pi'' \ldots$ were taken to mean products of all individuals except one, except two, and so on (SIL: 203).[35] He even attempted to characterise sentences containing phrases such as "there are at least three things in the universe that are lovers of themselves" according to such hemilogical quantifiers (SIL: 203).

Further evidence of the importance he attached to the generalised notion of quantifiers is to be found in the following:

> Two varieties of [selective pronouns] are particularly important in logic, the *universal selectives,* . . . such as *any, every, all, no, none, whatever, whoever, everybody, anybody, nobody.* These mean that the hearer is at liberty to select any instance he likes within limits expressed or understood, and the assertion is intended to apply to that one. The other logically important variety consists of the *particular selectives, . . . some, something, somebody, a, a certain, some or other, a suitable, one.*
>
> Allied to the above pronouns are such expressions as *all but one, one or two, a few, nearly all, every other one,* etc. Along with pronouns are to be classed adverbs of place and time, etc.
>
> Not very unlike these are, *the first, the last, the seventh, two-thirds of, thousands of,* etc. (2.289, 1893, *Speculative Grammar: The Icon, Index, and Symbol*).

Peirce did not interpret these quantifiers in relational terms as ones that quantify sets and express relations that would hold between things and predicates. Rather, it is more likely that, had he continued his development of generalised quantifiers, we would have witnessed the development of game-theoretic semantics for them on a par with games for the usual existential and universal quantifiers during his lifetime.[36] The projected course of such developments can be gleaned from the following passage.

> A subject should be so described as to be neither Universal nor Particular; as in *exceptives (Summulae)* as "Every man but one is a sinner." The same may be said of all kinds of numerical propositions, as "Any insect has an even number of legs." But these may be regarded as Particular Collective Subjects. An example of a Universal Collective subject would be "Any two persons shut up together will quarrel." A collection is logically an individual. (2.324, c.1902–03, *Speculative Grammar: Propositions*).

Accordingly, Peirce came to stress the importance of having generalised quantifiers in logical approaches to language and meaning. In his terminology, the logic concerns particular collective subjects. While some issues related to his view on collections were discussed in Chapter 1, there is ample room for more research on the largely unexplored terrain of these directions into which, as I have argued, he continued to push the study of the logic of collective subjects.

5. Conclusions

If we take logic to constitute a major part of human decision making and cognitive ability, then the game-theoretic approach is utterly natural. In the nomenclature of one of the major parts of game theory, the theory of extensive forms of games (Chapter 7), a strategy has to specify an action for each history, even for those that are on the off-equilibrium paths and never lead to a win, and the strategy has to be specified for every possible choice in a game. Interestingly, this can be emphatically compared with similar sentiments in 5.400 [1893, *How to Make Our Ideas Clear*]: "Now, the identity of a habit depends on how it might lead us to act, not merely under such circumstances as are likely to arise, but under such as might possibly occur, no matter how improbable they may be".

That one needs to take the would-bes, the possible future courses of events, into account is preserved in extensions of the notion of equilibrium that applies to games in which there is imperfect information, such as sequential equilibrium. Moreover, one needs to act as rationally and as optimally in those as in histories that lie on the equilibrium paths. Here, strategies codify interim beliefs plus an assessment of the beliefs concerning other players' types and past and future actions. What else is this extension, then, but a broadened conception of a strategy that just makes more use of habits than non-sequential strategies, and includes the aspects of habits that relate to beliefs, while resting peacefully within the purview of this broad notion?

One of the themes that permeate Peirce's philosophy is the notion of continuity, his synechism, both in the mathematical sense represented by a pseudo-continuum, and in the metaphysical sense, which he considers the real continuum. This suggests that continuity ought to be respected in any of the game-theoretic schemes in which Peirce's ideas may be reflected. One candidate here would be the class of differential games. These are games in which time is continuous rather than discrete, and decisions have to be made at each instant or point along its continuous measure.

Alternatively, a class of continuous games exists in use in which continuity relates to the idea that pure strategies form a continuum. Whether any of these would constitute a viable continuation of Peirce's outcroppings is yet to be seen, but these points should be investigated with an eye on Peirce's notion of the continuum, far exceeding that of the real line, that "pseudo-continuum" of multitude c that did not suffice for him (and so one needs to bring infinitesimals into play, cf. NEM 2:169, Putnam 1995). What such 'non-standard' differential or continuous theories of games would amount to have been little investigated.[37]

The upshot of these observations is that there are many instances in which strategic plan of action and the habit have overlapping significations. Some repercussions of this convergence are imminent in current theories of computation, however blind these theories may be to the complex history and development of this intellectual idea, in which the connection between a thought — say a symbolic term — and its meaning is found habit-changes that such a thought provokes, mostly in observational outcomes of a computational process.

However, this is only one of the many points at which elements of the theory of strategic decisions and Peirce's thoughts on logic and semeiotics meet, and I hope to have uncovered a few more. Such a kinship remains mostly logical in nature, and does not show up particularly often in his economic studies, even though I have here indicated a few such links, and even though "It is a question whether [economics] is not a branch of logic". Soon after his death, the theory of games started to emerge in the works of Ernst Zermelo, Émile Borel, John von Neumann, and many others. The first explicit technical connection between games and logic was that between Skolem functions and winning strategies, discovered by Henkin (1961). I will return to these discoveries in Chapter 7.

Notes

1 Referring to his theory of existential graphs (see sect. 4 below and later chapters), Peirce wrote: "The Graphist is really Plastic Nature, or the Artifex of Nature; and the special permissions are the experiences given to the interpreter of Nature, to the man, to which he is at liberty to attend, or not to attend at all, or to attend and immediately put out of sight, as he will" (MS 280: 23). Peirce means hylozoism of Richard (Ralph) Cudworth (1617–1688) and the idea of an inferior soul or substance by which purposeful behaviour may be attributed to the activities of physical nature. Among the assorted pages of MS 280 we find that "the Graphist *must* be regarded as corresponding to the 'Plastic Nature' of Cudworth, or else to the Artifex of Nature" (30 a.p., emphasis added). Peirce had marked down the neo-Platonist philosophy of hylozoism slightly earlier, in August 1904, in a draft of The Monist article *What Pragmatism Is* (EP 2:331–345) entitled *Nichol's Cosmology and Pragmaticism*, within the context of a dialogue between the Questioner and the Pragmaticist: "*Questioner*: The narrowness of your view of reality only appears more and more strikingly as you go on. You ~~simply~~ are, as you ~~say~~ [yourself well phrased it], simply color-blind to ~~the being~~ [the idea of existence] in itself. *Pragmaticist*: Hylozoism, the doctrine that all matter feels, is an idle ~~theory~~ [and senseless apology for a theory] as long as there is not [a] way of bringing it to the test of experiment; but as soon as such a way shall be found it will ~~be~~ [become] a working hypothesis particularly well worth trying" (MS 329: 22, Copy C).

2 MS 430: 62, 1902, *Minute Logic. Chapter III. The Simplest Mathematics*.

3 MS 25, 1897, *Multitude and Number*.

4 "In all discourse, or reasoning, there are *virtually* two parties. Either there are actually two parties, as when one speaker addresses an audience of one or more persons; or else one person reasons out something with himself, and even then, the difference between his conceptions and opinions before and after a given operation of thought results in his influencing himself much as one person influences another; so that we may say that even in this case there are *virtually* two parties" (MS 25: 2).

5 This and several other related pages, which contain material not found elsewhere in other versions and drafts under the title of *Pragmatism*, were not printed in the transcription of MS 318 in EP 2: 398–433. The claim in the headnote, written by the editors of the Peirce Edition Project, that Peirce's proof of pragmatism is complete in this transcription is thus a little doubtful.

6 Edgeworth was among the key players in the neoclassicist 'utility' movement that grew out of the 'energetics' of physical sciences, popular since about the mid-1800s. Its idea was that energy and its fluctuations are central not only in mathematical and physical investigations, but that they also nurture the human soul, aiming at the maximisation of that nurture. This suggests some tempting analogies with the 'informatics' movement in our late-20th-century computational sciences.

7 6.262, 1891, *Man's Glassy Essence*.

8 5.542, c.1902, *Belief and Judgment*.

9 See, for instance, MSS 1135 (c.1903) and 1525–1537. They had little or no bearing on his logical and semeiotic studies, however.

10 See CD III:2447, CDS XI:0509. Apart from Whatley's *Logick*, Schiller's book was the first philosophical book Peirce is said to have read.

11 MS 290, 1905, *Issues of Pragmaticism*.

12 MS 1147: 3, subentry *Negation*, cf. DPP; 3.480–3.482, c.1896–97, *The Logic of Relatives*.

13 5.297, 1868, *Some Consequences of Four Incapacities*.

14 This statement has been interpreted by the Peirce Edition Project to express the conclusion of Peirce's proof of pragmatism.

15 2.665, 1910, *Critical Logic: The Doctrine of Chances*.

16 MS 318: 48, Note at Pragmatism; cf. EP 2:431.

17 MS 318: 44, Note to Pragmatism continued.

18 *ibid.*: 44.

19 *ibid.*: 44.

20 "No number of existential meanings can be adequate to the meaning of an intellectual concept, since the latter is general; and no collection of individuals, nowever multitudinous, can be adequate to a general" (MS 318: 20).

21 1.122, 1896, *The Scientific Attitude*.

22 See e.g. Rescher (1978a,b, 1989).

23 6.294, 1891, *Evolutionary Love*.

24 Similar qualities will be applied to the methodology of historical pragmatics in Chapter 12.

25 See also: "Though these two functionaries [the utterer and the interpreter] may live in one brain, they are nevertheless two" (MS 500: 13), and "Whenever a person thinks over any question in his own mind, he carries on a sort of conversation. His mind of one minute appeals to his mind of the next minute to agree with it and say whether so-and-so is not reasonable; and then the mind of the next minute ~~says~~ [may say] either 'Certainly by all means, and I wish all future minutes of my mind to take note that this is my decided opinion, after close examination' or else he may say 'Well, that seems so, at first glance, but I don't feel quite so sure of it as that mind of the last minute wished me to be' or he may even think 'Well, look you, my mind of the future, before whom my last minute mind and I, *this* minute's mind are arguing (for we both submit to you as knowing more than either of us do) — it appears to me that that last minute's mind was a goose and entirely failed to perceive the real state of the case', etc etc." (MS 514: 45–46, 1909, [*Fragments on Existential Graphs*]).

26 4.431, 1903; MS 293: 50, c.1906; MS 295: 41, 45, c.1906, Rejected pages for the Monist article of 1906; cf. Roberts 1973a, p. 92.

27 4.518, 1903, *The Gamma Part of Existential Graphs*; MS 467, 1903, *Lowell Lectures. Lecture IV*. Roberts (1973a, p. 86) and Zeman (1964, p. 253) were the first to note the use of a special sign Peirce had in connecting states of information with an arrow-like pointer (a special spot that has been abstracted) attached to them. Peirce's intended meaning was that one state of information follows another. As Zeman notes, one interpretation of this precedence notation is certainly as an accessibility relation. Peirce did not seem to have exploited this device further.

28 This terminology is explained in Chapter 4.

29 MS 612: 33–34, *Common Ground*, 12 November 1908.

30 5.518, 1903, *Consequences of Critical Common-Sensism*.

31 For example, the work of Peirce's gifted student Oscar Howard Mitchell was particularly notable in the development of the notion of the quantifier, and of the logic of multiple dimensions, see Mitchell (1982). His premature death at the age of 38, gone scarcely noticed, surely affected the pace in which the first-orderisation of elementary logic was arrived at by Hilbert and others.

32 The notion of compositionality indicates just one way in which language is built up and understood, not as a necessity but as a property that has become highly popular (Chapters 4, 6).

33 Following Peirce, we could dub Tarski semantics 'ectoporeutic'.

34 4.547, 1905, *Prolegomena*.

35 According to Peirce–Mitchell logic, the sign Σ, denoting the algebraic sum of relative products of terms, corresponds to the existential quantifier, and the sign Π, denoting the product of relative sums of terms, is the universal quantifier. One of the first instances of the name 'Quantifier' is in 3.396 [1885, *On the Algebra of Logic*].

36 Pietarinen (2001b) defines, among other things, some game rules for generalised quantifiers.

37 Aumann (1964) originated the study of infinitesimal notions of players.

Chapter 4

MOVING PICTURES OF THOUGHT I

SERIOUSLY, it is quite the luckiest find that has been gained in exact logic since Boole. (MS 280: 22, 1905, *The Basis of Pragmaticism*).

Peirce was a visual interpreter of language. This led him to adopt the idea of the diagrammatisation of logic and so to the theory of existential graphs (EGs), which he claimed put before us "moving pictures of thought" (4.8, c.1905).[1] My purpose in this and the follow-up chapter is to show that the 'movement' of these 'pictures of thought' may conveniently be viewed as actual moves in correlated games in terms of the theory of extensive games. However, these games lend themselves to a diagrammatic and semeiotic approach to the meaning of propositions or complex concepts in a natural way. Accordingly, I will identify and investigate some of the relations between the resulting dialogical or game-theoretic interpretation and Peirce's theory of communicative semiosis. I will consider the method of endoporeutic, participants' information in the process of diagrammatic interpretation, topology, the notion of the common ground, and the more recent genus of EGs, the heterogeneous systems of logic.

1. Introduction

Peirce interpreted language visually. While such capacity may, to some extent, be indispensable to us all in understanding language, in his case it led him to search for visual methods that could rigorously capture some of the most vibrant aspects of thought and reasoning. This will to visualise and animate the essential content of thought was something he was not hesitant to emphasise again and again, and he frequently complained that he had a singular incapacity to think purely within the confines of the verbal or written, linear structure of language. "I do not think I ever *reflect* in words: I employ visual diagrams, firstly, because this way of thinking is my natural language of self-communion, and secondly, because I am convinced that it is the best system for the purpose".[2]

He attributed this incapacity to his left-handedness. Aside from the peculiarity of thinking outside of language, he did not show any particular interest in artistic endeavours. Probably this did not fit in with his teenage conviction that his life was built upon a theory. As he was one of the greatest of all the logicians, it is not surprising that his earlier investigations into algebraic logic helped him on his way to one of the most spectacular discoveries ever made in logic, the system of EGs.

The struggle Peirce had in understanding the logical structure of natural language led him to develop a quite unprecedented diagrammatic system. This shows that logical thinking and reasoning are linked with linguistic considerations. The discovery is all the more intriguing given that the parts of his system of EGs, namely the alpha and the beta parts, formed (with minor modifications) sound and complete systems, as shown much later by Zeman (1964) and Roberts (1973a). The concepts of soundness and completeness were rigorously delineated only after his death, and these later definitions and findings were not originally inspired by considerations of natural language at all.

Aside from EGs, two closely-related systems of diagrammatic representation of language that have recently been scrutinised and applied, especially to problems of knowledge representation and natural language phenomena, are the system of Conceptual Graphs (CG), similar to EGs but invented independently by John Sowa (1984), and the discourse-representation theory (DRT) of Hans Kamp (1981). Both of these developed after the late 1970s. These theories have not lent themselves to dialogical interpretation, and thus fail to show themselves as genuinely dynamic, interactive theories of logic and language. They rather throw light on what goes on in the one-sided case of a single hearer of a sentence, or in the monologic comprehension of discourse. I will return to these theories in the next chapter.

Peirce claimed that the visual representation of sentences by means of his EGs put before us "moving pictures of thought", "a portraiture of Thought" (4.11), "a moving picture of the action of the mind in thought",[3] and "a moving picture of the action of thought" (MS 296: 6).[4] Concerning the last claim, he said that he would not stop to defend it, because it would be too tedious. Unfortunately no detailed defence was forthcoming elsewhere in his work, either. He deflects the issue by saying, "It is so elaborate and so unfamiliar in substance, that any tolerable clear exposition of it would occupy more pages than it would be decent to ask our good and admirable editor to allow . . . that theory, even if it were developed, would probably seem still more dubious to you than does the proposition that, to my mind, it sufficiently justifies" (MS 298: 17). Regrettably so, because these phrases are significant in that they comprise one of the starting points in the argument in which Peirce promised to show that pragmatism is true (MS 298: 4). As I argued in Chapter 1, the full status and impact of the role of EGs in that suggested argument has not been sufficiently well recognised, and one obstacle is, I submit, that he offers no obvious explanation as to how these graphs lend us truly 'moving pictures' — be it of thought, action of thought, or action of the mind in thought. In addition to the game-theoretic elaboration, I will recommend a topological interpretation of these graphs in the sequel to this chapter, which I hope will contribute to a better understanding of these motivations.

Elsewhere, Peirce wrote, "Every logical evolution of thought should be dialogic" (4.551). This goes with his assertion that, presuming EGs "furnish a moving picture of the intellect" (MS 298: 10 a.p.; cf. "system for diagrammatizing intellectual cognition", MS 292: 41), this should not be taken to imply that it is human thinking that is in operation here. One needs to take into consideration the fact that "all thought is dialogical, and is embodied in signs". This is the "essence of the thought", the realisation of dialogical performance in the mind. There needs to be "self-development and growth" in thought, "without which a 'moving picture could mean nothing"' (MS 298: 11 a.p.). A consequence of this is the fact that all thought must be embodied in signs. This is admirably in accordance with the overall communicative character of his theory of signs.

While a diagram is a precise and non-vague snapshot of any particular thought, seen as a representation of the mind, it gives a "rough and generalised" picture of what the mind is (MS 490). Depicting the mind requires the use of logic that captures general and indefinite propositions, while depicting thought hinges upon a definite and determinate diagram. A mind itself could be understood along the lines of being a "*sign-creatory in connection with a reaction-machine*" (MS 318: 18).

Equally, the epithet "moving" does not appear in manuscript 296, in which Peirce maintains that EGs play the "*rôle* of being a picture of thought" (MS 296: 19). He continues by claiming that if he were to regard only what he considered reasonable concerning such a role, he would say that it consisted of "some mutual intercommunication of assent", but he wished to avoid the murky paths that this might lead to and reduced his explanation to the assumption that there is "some voluntary act, some mental molition of some kind, in which both parties take part" (MS 296: 20). What is noticeable here is the psychologically predisposed concept of molition, which was shown in Chapter 1 to be one of the open paths in his explorations pointing at the heart of the logic of EGs. Overall, the impact of these psychic concepts in the logical setting of diagrams is, I am afraid, bound to remain a mystery that neither Peirce, nor his scholars, have revisited.

Yet another reason why Peirce did not put before us his thoughts concerning the 'moving-picture' idea in full is that, at around the time the phrase was mobilised, roughly by 1905, the representation of modality lacked the "pictorial, or Iconic, character which is so striking in the representation in the same system of every feature of propositions *de inesse*" (MS 298: 18). Among the assorted pages of the same manuscript we find that Peirce is not yet satisfied with the representation of modality, and that the argument against moving pictures of thought by virtue of modality not being diagrammatisable is well taken.[5] Partially by way of an antidote to this, he soon started to think of modality as an ingredient of the conception of negation.[6] This led to the development of the gamma part with the new sign of a broken cut, which weakens the negation to the representation of an expression that something is "possibly not" (see below).

There are any number of further textual samples advertising both the dialogical and picture-like view of thought, the formation of its content, and the explanation of its action in the course of the inner mind in conversation with itself. Diagrams are thus something like, silent, rough and incomplete photographic pictures, still images of these actions captured by the process of diagrammatisation.

However, as happens so often when we attempt to understand Peirce's overall aims, we are left wondering whether his writings form a coherent whole. In the present context, we would like to ask whether his earlier semeiotic considerations, which emerged together with his investigations on algebraic logic, are smoothly carried over to his diagrammatic system of EGs. Are the sign theories applied in both realms not only coherent with each other, but also compatible enough to produce mutually beneficial, if not identical, theories of signs?

Even though the precise nature of Peirce's theory of signs and its evolution in the course of his logical writings are beyond the scope of this book, the initial response, which is also at the heart of this chapter, is a positive one. Peirce did manage to incorporate essential semeiotic ideas concerning algebraic developments of logic into his theory of diagrammatic logic and reasoning. He was able to do this because what was perhaps the most vital idea that he had used in algebraic logic, the idea of dialogue, is the defining character of his category of secondness. As he preserved the triadic categorisation of his overall philosophy into firstness, secondness and thirdness throughout his entire career, it was only natural that whatever the diagrammatic approach to logic and language accomplished in the end, it was able to reinforce the idea of the relation between two (imaginary or real) interlocutors, interspersed with the thirdness of the sign mediation. Even so, in certain ways this reconciliation necessitates going beyond Peirce's own implementations and suggestions, simply because many of the requisite logical innovations were made only sometime later.

In the light of the infiltration of dialogical concepts into his theory of signs and into his logical investigations, the issue I will be mostly dealing with here is the relationship between his diagrammatic theory of EGs and his idea of dialogue. To be more accurate, instead of merely endorsing the view of dialogue and the related concept of communication, I will venture further and examine his diagrammatic system of graphs by means of game-theoretic conceptualisations.[7] Dialogues have their roots in his early logic, but were seriously studied only much later by philosophers and logicians of the late 20th century, most likely to have been unlettered of this prehistory.

One of the disappointing aspects in the interplay of EGs and the intercommunicative approach to thought is that Peirce failed to lay a strong enough emphasis on their intimate connection. This is the main reason why he eventually fell short of articulating what moving pictures of thought are all about. By taking the kinematic idea of ‘moving’ seriously, new light is shed on dialogical thinking by putting these proposed pictures ‘on the move’ in the sense of the theory of games. Much of the discussion about how to read off these graphs, especially with reference to the implicit notion of quantification employed by Peirce, would have been avoided if they had been considered amenable to dynamic and game-theoretic (endoporeutic) interpretations in the subsequent studies.

Commentators have nonetheless almost unilaterally missed this perspective. Shin (2002), who discusses at length the interpretation that Zeman (1964) gives to beta graphs, proposes a new approach that fails to take into consideration the dialogical and dynamic aspects of interpretation. It is essentially an algorithmic translation of any beta graph into a predicate logic formula (a formula with identity signs, being put in its negation normal form). Among other things, this glosses over all the iconic characters of signs that Peirce thought the pragmatic maxim obliged one to preserve in representing the essential content of concepts.

Before proceeding, a couple of conceptual clarifications are in order. Systems of diagrammatic logic and reasoning are in vogue, especially in the applied fields of knowledge representation, AI and cognitive science. It is popular to see such systems in terms of visual representations of something. However, we should understand a graphical system of diagrams not only in terms of what is provoked by visual excitation, but also in terms of auditory or even tactile and haptic signals. It is not even obvious that the possibility of diagrammatisation hinges on data given by the senses. Diagrams may be partly imaginary. They may be percepts at the same time as representations prior to perceptual judgements. By representation, I thus mean the general method of the diagrammatisation of logical propositions. Likewise, the interpretation of such systems may also be, in some sense, ‘visual’. I take this to be in accordance with Peirce’s intentions. Visual is thus to be understood broadly, meaning any diagrammatic logic. By interpretation, I mean the semantics of diagrammatic systems, a method that assigns semantic attributes to the constituents of the system. Here, the terminology differs from that of Peirce, for whom semantics, in the sense pertaining to the interpretation of signs, was a theory of translation. From the perspective of this study, interpretation rather assigns meanings to constituents of propositions in accordance with their diagrammatic representation.

According to Peirce, diagrammatic logic refers to any iconic system of logic that is scribed (i.e. partly written and partly drawn) on a paper or a blackboard — or nowadays perhaps programmed on computer. That on which diagrams are scribed is the sheet of assertion. What is scribed consists not only of symbols and letters of logic, but also of lines (graphs) spread in two dimensions. Being iconic roughly means that some semblance obtains between the signs of the system and aspects of its objects.

It is customary nowadays to say that such diagrammatic logical systems of representation are heterogeneous. However, heterogeneous logics intermingle symbolic with iconic signs. This was not Peirce's goal.[8] If interspersed with predicate symbols, EGs may perhaps illustrate fairly prototypical such heterogeneous systems, but their importance is by no means fully assessed by saying only that, and in all other aspects happily correlating the system with what is traditionally understood by symbolic logic. As noted, additional interest in these graphs arises from Peirce's claims that they project true moving pictures of thought, thereby connecting the method with the philosophical and semeiotic parts of his work, including continuity, iconicity and tendencies to take habits.

2. Existential graphs in a historical context

Let us have a brief historical interlude. These claims were made shortly after the first real moving pictures hit the cinema screen. This was not coincidental. The years 1895 and 1896 were perhaps the most important and electrifying years in the history of the film industry. Although there were many precedents, November 1895 was when moving pictures began their career in cinemas around the world, targeted on the public at large. Interest increased rapidly, so that within six months commercial shows had swept across many countries, and dozens of cinemas were showing short films. It was not yet known what use these moving images would have, and science, entertainment, communication and news broadcasting competed for them. Demonstrations of Kinetoscope, constructed under the supervision of Thomas Alva Edison in 1891, had gone public two years earlier, in 1893. The discoveries were announced in many issues of Scientific American and elsewhere (Rossell, 1998, p. 161).

In view of this, it is not very likely that Peirce remained ignorant of, let alone unaffected by, these significant events. Indeed, he was acquainted with magic lanterns (W 4:48), one of the predecessors of projector technology. The first appearance of the epithet "moving-pictures" (with a hyphen) in his text that I have been able to determine is from 1905, but in 1893 he already spoke of "the living influence upon us of a *diagram*, or *icon*".[9] Even though he had lived in Milford, the town near his Arisbe home for several years by then, he was still very much at the heart of scientific circles. Starting in 1897–98, films were shown in Philadelphia, 75 miles from Milford. In 1896, one of the first films, borrowed from Edison, was demonstrated at the Franklin Institute in Philadelphia by C. Francis Jenkins, who used the newly-developed projector technology called a Phantoscope. (Edison had to buy that device from Jenkins to achieve projection himself, and lawsuits abounded.) From 1897 onwards, the successful Lubin Film Company established regular shows in Philadelphia. However, Peirce was on a prolonged sojourn in New York from August 1895 until the end of 1897 (Brent, 1998), but returned around the time of Lubin's exploits in Philadelphia. In fact, in April 1896, films were premièred in New York under Edison's name using the Vitascope. At that time, Peirce was in competition with Edison on another subject concerning the development of cheap lightning for houses. He also wrote for The Monist and submitted many contributions to The Nation during this productive year. Most importantly, he catalogised the term Vitascope in his MS 1135, *An Attempted Classification of Ends* (c.1903), under the subcategory of Amusements. The likely scenario thus is that Peirce was on the aisle during his New York episodes.

Also in 1896, Peirce announced his diagrammatic EGs. It has to be kept in mind that his investigations into logical algebra were their precursors. We should not attribute the merit of originating diagrams per se to any of these fanciful kinemato-phanto-muto-vitascopic devices. Peirce himself said that the essential ideas had occurred to him some fourteen years earlier, which would have been around 1882, the reference probably being to his letter to his student and colleague at Johns Hopkins Oscar H. Mitchell (MS L 294, 21 December 1882). In that letter he noted that his "notation of the logic of relatives can be somewhat simplified by spreading the formulae over two dimensions".[10] In fact, he offered improvements to Alfred B. Kempe's (1849–1922) method of using diagrams as early as 1880, having learned of his work prior to its publication via the Scientific Association at the Johns Hopkins University. This urgency arose from Kempe's announced proof of the four-colour theorem, a stimulus that is explained in Roberts (1973a, pp. 20–25).

Still earlier, William K. Clifford and James J. Sylvester had suggested in 1878 that there were analogies between diagrams representing bonds and valencies in chemical formulas and algebraic invariants (Murphey, 1961, pp. 196–197). This provoked Peirce's "valental graphs",[11] special cases of logical graphs in which the arities of relations correspond to the valencies of chemical formulas. This, in turn, goes all the way back to Peirce's 1870 theory of relatives, in which ingredients of the graphical and diagrammatic systems were thus present.

Since Peirce probably wrote down the slogan 'moving-picture of thought' for the first time a couple of years after these cinematoscopical events, chances are that it was bound to be an afterthought calculated to serve as an advertisement that would ride on the back of commercialised moving-pictures, rather than actually providing any novel logical insight into his own systems. Alas, not a single EG was printed until 1906 in The Monist.[12]

Let us also remind ourselves that by December 1901, Peirce had translated from French, interspersed with many comments, the French physiologist Étienne-Jules Marey's (1830–1904) *Exhibition of Instruments and Photographs appertaining to the History of Chronophotography,* in which motion was studied through representations by various photographic devices (MS 1514, 44 pages).[13] Marey differed with Peirce in believing graphical representations to provide a universal language, but his inventions were certainly a rich source for Peirce regarding visual representations of thought and movement. In 1885, Marey had presented movement in terms of photographic sequences of human beings and animals, suggesting the relevant devices as early as in 1857. Peirce also translated Marey's 1878 paper on the analysis on motion (MS 1515).

Even if the title 'moving-picture' was merely an *hors* to his *chef*, as a guiding metaphor it was a productive stimulus. The exact context of some of these ideas remains a mystery, however. For instance, in 22 June 1911 he remarked, "At great pains, I learned to think in diagrams, which is a much superior method [to algebraic symbols]. I am convinced that there is a far better one, capable of wonders; but the great cost of the apparatus forbids my learning it. It consists in thinking in stereoscopic moving pictures".[14] A stereoscopic camera, known since 1838, is a double camera giving two pictures on the same place, so that changes in the altitudes of the target become discernible (CD VII:5935).[15] However, what this far better diagrammatic method was supposed to be was revealed nowhere. The Century Dictionary (CD II:1589), within the entry referring to Diagram, has a subentry Stereoscopic diagrams, describing "A pair of diagrams, perspective representations of a solid diagrammatic figure, intended to be optically combined by means of a stereoscope". Among the draft sheets for proposed entries in this dictionary are both "diagram" and "diagrammatic", and so it is likely that Peirce wrote all these entries.[16]

Zeman (1964) has noted that "Hopes for the stereoscopic gamma went a-glimmering".[17] Gamma was the unfinished systems of modal and higher-order notions (abstractions, collections), including reasoning about and representing the graphs themselves using graphs. However, it is more likely that stereoscopic diagrams were something to be postponed to the delta part, which was calculated to "deal with modals" (MS 500: 3). This statement was made in December of the same year, 1911. Peirce documented no further reflection upon stereoscopic or delta continuations of diagrammatic explorations. I propose in the next chapter an extension of EGs as an elaboration of the stereoscopic idea.

Peirce attributed the fundamental idea of there being diagrammatic content in the actions of the mind to the Spanish renaissance humanist Ludovico Vives (alias Ludovicus Vives, Juan Luis Vives, 1492–1540) — interestingly bypassing the Swiss Leonhard Euler (1707–1783) on this matter — and appealed to the remarks Friedrich Albert Lange (1828–1875) made in his *Logische Studien* (1877).[18] This path from Vives to Lange has not been studied since, and more often than not, the origins of the logical status of diagrams have somewhat erroneously been credited, ever since John Venn's adulation, to the Helvetic mathematician.

3. The magic lantern lit up

Some initial considerations What are Peirce's EGs? In principle, he divided them into three parts. Roughly and customarily, the alpha part corresponds to propositional logic, the beta part to predicate logic with identity, and the gamma part, which was left incomplete and whose status is still somewhat uncertain, is capable of encompassing modalities, higher-order notions, abstraction, and reasoning about graphs themselves. There was also a projected delta part, which never materialised. Because of their iconic character, it is not quite right to state that these theories equal those of propositional, predicate and modal logic. Because they were originally designed for different purposes, embarking on quite different principles concerning signs other than symbolic logic, one cannot exhaustively be analysed by means of the other. Moreover, later sections will reveal that subtle differences emerge when representing logic by EGs that have no equal in symbolic expressions.

Even if alpha and beta are, as theories, isomorphic to propositional and predicate logic, respectively, the two sides are motivated by essentially different ideas. As far as quantification is concerned, it is not sufficient just to plug in values for variables and observe, via satisfaction, whether the formulas are true in a model. As in game-theoretic semantics (GTS), Peirce's dialogical interpretation implies that suitable individuals, or names for individuals otherwise indesignate, are found in the universe of discourse to function as selectives next to the hooked extremities of the lines of identities (read on for the explanation of this terminology). In addition to that, however, in order to ensure identity, this selective has to be continuously connected with another selective at another hooked extremity of the line or ligature. Symbolic formulas have no way of expressing such things in an equally unitary fashion.

What, then, is a diagram? According to Peirce, "A Diagram is an Icon of a set of rationally related objects. By rationally related, I mean that there is between them, not merely one of those relations which we know by experience, but know how to comprehend, but one of those relations which anybody who reasons at all must have an inward acquaintance with" (MS 293: 11). Elsewhere, he states, "A *Diagram* is a representamen which is predominantly an icon of relations and is aided to be so by conventions. Indices are also more or less used. It should be carried out upon a perfectly consistent system of representation, one founded upon a simple and easily intelligible basic idea".[19] A diagram is thus a sign that is an iconic representamen. It should be "as iconic as possible" in order to represent relations by "visible relations analogous" (MS 492: 22) to the relations intended. This profound iconicity is again related to the idea of diagram construction and propositional assertions as utterances employing "any method of graphic communication" (MS 492: 24), operationalised by the dialogue between the utterer and the interpreter, an idea familiar from Peirce's algebra of relatives, but now set within the context of EGs.

Apart from being essentially iconic, a diagram is also both definite and determinate. By definite, Peirce means that there is no vagueness in it. This suggests that diagrams do not lend themselves well to the study of the logic of vagueness, at one point suspiciously acclaimed by him to have been investigated to an extent something like completeness. By determinate, he means that diagrams are not general. They represent one assertion in the iconised form of a proposition at one time. Diagrams render the content of thoughts precise and rigorous, and the universe that they speak of is determinate at any given time at which they are being interpreted. Thought that is not regimented in a like manner — which probably is the case most of the time — will be both vague and general.[20]

The starting point for investigating diagrams is the concept of a sheet, or the phemic sheet, which, according to Peirce, is "a surface upon which the utterer and interpreter will, by force of a voluntarily and actually contracted habit, recognize that whatever is scribed upon it and is interpretable as an assertion is to be recognized as an assertion, although it may refer to a mere idea as its subject".[21] This already shows the dialogical heart of Peirce's diagrams, later on to be employed for a variety of purposes.

The phemic sheet is a sheet of assertion:

> [A] sheet on which the graphs are written (called the *sheet of assertion*, as well as each position of it, is a graph asserting that a recognized universe is definite (so that assertion can be both true and false of it), individual (so that any assertion is either true or false of it), and real (so that what is true and what is false of it is independent of any judgement of man or men, unless it be that of the creator of the universe; in case this is fictive); any graph written upon this sheet is thereby asserted of that universe; and any multitude of graphs written disconnectedly upon the sheet are all asserted of the universe. (MS 491: 3).

Definite, individual and real are the characteristics of the universe of discourse that the idea of the sheet on which the graphs are scribed is intended to capture. There is thus no vagueness and no generality, and whatever the diagrams represent by their truth-values are internally real or non-relative.

Given the fact that diagrams are iconic representamens, it follows that Peirce did not take their role in logic to be in any way bound to achieve the status of a universal sign or a universal medium of representation and communication. Despite all its diversity, diagrammatic logic is subordinate to the overall theory of semeiotics, confined by the limits each part of the system of EGs is taken to assume. The emphasis is on the words 'part' and 'system', for they are apt to illustrate what is going on in diagrammatic logic, in much the same way as the term 'system' illustrates what is going on in other sciences of inquiry such as physics, namely by denoting the fractions, nooks and corners of the universe to which the applicability of a theory, or a part or system of EGs, is confined at the time. Peirce wrote, as the closing sentence of his *The Bed-Rock Beneath Pragmaticism*, that the "fundamental idea" of graphs is that the "Phemic sheet itself represents the Universe, or primal subject of all the Discourse" (MS 300: 48). The phrase 'all' is to be understood as pertaining to any particular discourse at the time, quantifying over the objects that are within the "field of attention" of the discourse participants, not to the universe writ large. The "general objects" of the field of attention belong to the "Universe of Discourse" and are represented in the Phemic Sheet (MS 300: 39.5).

What is worth bringing out is the fact that Kempe had published a system for expressing relations in 1886, but the virtue of Peirce's system (in its entitative incarnation) was, by way of his own comparison, that it "depends upon considering how the diagram is to be connected with nature".[22] This is a remarkable observation, because one of the side effects is that, by connecting diagrams with certain isolated, small and regimented parts of nature, its own 'systems' with its own fixed interpretation for primitive non-logical concepts, we can understand what the model theory that developed much later is, in Peirce's mind, about (Chapter 6). It concerns the conditions in which propositions hold in such fixed and isolated models. EGs are perhaps the first instance in logic in which the prospects of what was much later called model theory were realised in depth.

In fact, Peirce went even further, continuing his previous comment thus: "It is not surprising that the idea of thirdness, or mediation, should be scarcely discernible when the representative character of the diagram is left out of account". This means that the iconic and representative nature of graphs makes the linking of logic to the world possible via the mediating signs connected with the minds or quasi-minds of the utterers and the interpreters, or in EGs, with the Graphist and the Grapheus. By experimenting upon diagrams, one 'puts questions to Nature'. This implies not only the study of the conditions that obtain in mathematically constructed models, but also the desire to make use of information in the common ground of the collaborating parties who are building up these diagrams. From this perspective, the common ground is the natural history of logic, concerning how one fixes the interpretation of non-logical constants, among other things. This Peirce aimed at by sharing the actions between the Graphist and the Grapheus, whose functioning is habituated through the common ground. The common ground typically refers to shared presuppositions by language users, including background information concerning the beliefs that the others have concerning one's own presuppositions. What is salient here is the public and, in a sense that is subject to qualifications in later chapters, the social character of the common ground.[23]

Considerations such as these, venturing beyond the purely mathematical task of fixing the boundaries of models, were rarely considered in the post-1950s mainstream model theory. But this is precisely what Peirce strove to do. The common ground is, in effect, a closer friend to natural language pragmatists. However, neither Robert Stalnaker, David Lewis, nor the followers of Grice's programme have been aware that Peirce crafted multiple writings on the common ground, or that his use of it involved the iterative notion of common knowledge in the formation of truth in conversational contexts (Chapter 12).

Peirce took the graph to be a type (legisign), and the graph actually scribed on the paper as a token, a graph-replica. He writes: “A *Graph* is a diagram consisting of no more than, first, the sheet upon which it is written, secondly, spots (or their equivalents) having various visible qualities (as colors, etc.), third, lines of connection (commonly of only two kinds, those that are drawn and those that are left undrawn), and fourth enclosing ovals” (MS 491). An *existential graph* is then “a logical graph constructed upon a perfectly consistent system of representation such that any unenclosed partial graph shall assert something asserted by the entire graph”. He defined a *logical graph* as one that “asserts something, or represents an assertion, concerning a recognized universe, real or fictive” (MS 491), and an *entire graph* as “all that is at any time scribed upon the Phemic sheet”. A *partial graph* is “every part thereof which, as it is in the entire graph, is itself a graph” (MS 295: 70). A *graph-replica* is any individual instance of a graph, including the sheet of assertion.

Peirce drafted dozens of slightly varied definitions and expositions of the essentials of the system of EGs during the period of 1896–1911, sometimes with long introductions to how the system should be conceived of prior to its formal definition. No edited volume has yet appeared that reproduced even a small fragment of these drafts and assorted pages, which would have not only made these systems available to a wider audience, but would also have precisely tracked the changes and developments that took place during his most heated period of logical investigation. For reasons noted in Chapter 1, the chronological edition may not be the right place for such an undertaking.

Among the notable changes is the renaming of the sheet of assertion as the sheet of truth in MS 514, reflecting the emphasis he laid on assertions that are binding, namely those who utter them are responsible for their utterances. That on the sheet only true assertions can be scribed upon is then merely an illustration of this jurisdiction that no assertion is to be taken lightly, in other words they convey force that makes them binding. To represent that a proposition is false, one needs to strike such propositions out from the sheet by making an incision. Peirce even considered the possibility that any sheet of truth is an instance of the sheet of “All Truth” (MS 514: 7). A blank sheet, or a blank graph on a sheet, is tautology by virtue of it being “too obvious to take the trouble to say” (MS 514: 21), just as the principle of excluded middle is tautology. He suggested that it might have been better if he had called the sheet one of affirmation rather than assertion because “whatever state of things you represent on this page, you will be understood to affirm as existing somewhere, or, at least, consistently to make believe to affirm”.[24]

Even the way in which Peirce represented the cut underwent several phases of development. It changed from a simple closed line first to a ‘monochromatic’ mode of scribing spots and negations, and again to a representation of negation by means of shaded areas of graphs.[25]

One of the reasons why the course of logic that followed Peirce was predominantly symbolic was that he did not provide inductive definitions of well-formed graphs. His definitions were explicit rather than recursive. There is no 'syntax' in EGs in the sense of what Frege, Russell, Carnap, Tarski and others came to mean by it. Peirce did not make a clear-cut division between syntax, semantics and pragmatics — indeed a somewhat forced trichotomy that was later argued for by many of the semioticians, linguistics and logicians who succeeded. During the last years of his life, he even coined the term 'diagrammatic syntax' to refer to many of the conceptions that we would nowadays recognise as semantic.

True, there is a tendency in his last writings to extract syntactic elements from semantic and pragmatic ones. Such divisions were far from unambiguous, however, since he did not wish to downgrade the value and expediency of a proper model-theoretic conception of logic and semeiotics.

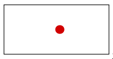
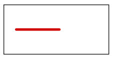
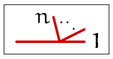
Essentials of the existential graphs explained Peirce termed iconic versions of rhemas spots. We denote them by predicates $S_i, i = 1 \ldots k$ surrounded by a finite number n of hooks at their periphery: S_1 The number of hooks corresponds to the arity of a predicate.

Spots may be linguistic assertions. For instance, the spot of the rhema ___ *is good* has one hook (the blank line of expression), the spot of the rhema ___ *loves*___ has two hooks (two blank lines of expression), and the spot of the rhema ___ *gives*___ *to*___ has three hooks (three blank lines of expression).

The set of beta EGs is the smallest set G_β satisfying:

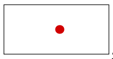
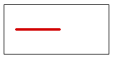
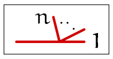
1 A sheet of assertion (SA)  $\in G_\beta$.

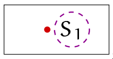
We may think of an SA as an interpreted structure. If nothing is scribed on an SA, it represents tautology ($\top$, *verum*).

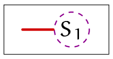
2 A dot , line of identity (LI)  and a finitly branching LI  $\in G_\beta$.

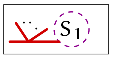
LIs are thick continuous lines composed of sets of contiguous dots.

3 Closure under spots:

If ,  and  $\in G_\beta$, then



, 

and 

$\in G_\beta$.

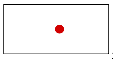
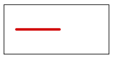
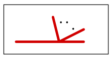
An attached LI is an iconic analogue to existential quantification, identity and predication of first-order logic.

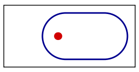
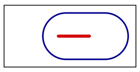
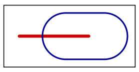
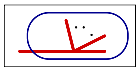
4 Closure under juxtaposition:

If $\varphi_1 \in G_\beta \ldots \varphi_n \in G_\beta$, then $\boxed{\varphi_1 \ldots \varphi_n} \in G_\beta$.

Juxtaposition is an iconic analogue to Boolean conjunction.

5 Closure under cuts, which are thin, simple, non-overlapping closed curves enclosing $G_i \in G_\beta$:

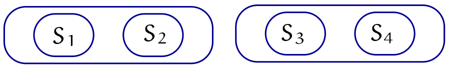
If ,  and  $\in G_\beta$, then

, ,  and  $\in G_\beta$.

A cut is an iconic analogue to Boolean negation.

In what follows, we omit drawing out the SAs and the hooks of spots.

Dropping the second and third formation rules and having closure under cuts with respect to atomic graphs amounts to the alpha part of EGs. For example, on the left below is an alpha graph that diagrammatises the propositional formula on the right:



$(S_1 \vee S_2) \wedge (S_3 \vee S_4)$

Taking juxtaposition to define isotopy-equivalence classes, it is seen that the orientation of juxtaposed graphs on an SA does not matter for truth or falsity, and so the operation of conjunction it defines is commutative and associative.

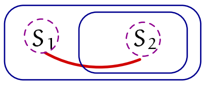
A dot on an SA singles out an individual subsisting in the universe of discourse, and thus denotes existence. An LI connects spots such that any hook at the periphery of a spot may be connected by an LI. At most one LI or a dot may occupy any one hook. Any unconnected extremity of an LI is a loose end.

As with juxtaposition, those LIs whose outermost loose ends are enclosed within the same area of a cut give rise to an isotopy-equivalence class. Therefore, their order of interpretation is irrelevant to the truth-value of a graph.

LIs may be connected to each other. The totality of connected LIs gives rise to a ligature. Ligatures are not graph-instances but "composites" of several graphs-instances.[26] Any line that crosses a cut is, actually, a ligature composed out of two lines. Like graph-instances on an SA, LIs in ligatures are compositions read as juxtaposed signs ('there exists b *and* this b is not S').

Dispensing with any correlate to free variables, the number of occupied hooks at the periphery of a spot S_i^n of the beta system may be taken to correspond to the arity of a predicate as well as to the number of bound variables.

For example, the following beta graph is a diagrammatic counterpart of the first-order sentence on the right:

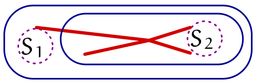


$$\forall x\,(S_1 x \to S_2 x)$$

This EG has two nested cuts, two spots S_1 and S_2, and one ligature abutting a cut, composed of two LIs that meet at the boundary of the inner cut. The LI asserts at its outermost extremity that any individual exists in the universe of discourse of the SA. The EG asserts that, given any individual object of the universe of discourse of SA of which S_1 is true, S_2 is true. The two nested cuts denote implication (Peirce termed it the scroll). The LI asserts at its outermost extremity that there is an individual that we are speaking of in the universe of discourse. Its continuation all the way to the inner extremity, abutting the inner cut and ending at S_2, asserts that this individual is identical ("numerically identical", MS 513: 54) to the one of which S_2 is true. The something at one end of the line is "the very same individual" (MS 514: 50) as the something at the other end.

LIs are thus iconic ways to represent identity, not to be achieved just by producing multiple instances of the same sign (e.g. a variable) and then asserting their identity by means of a conventional symbolic sign to denote a two-place identity relation. Moreover, they assume not only the role of identity but also quantification. The outermost free extremity of an LI or ligature determines whether the quantification it represents is universal (the free extremity lies within an odd number of cuts) or existential (the free extremity lies within an even number of cuts).

The beta graph below contains one ligature composed of five LIs two of which have outermost free ends. They correspond to the universal ($\forall x$) and existential ($\exists y$) quantifiers of first-order logic, in that order. The LIs abut one another within the innermost positive area, thus expressing identity, and connect to two hooks of the spot S_2. The graph may be correlated with the sentence of first-order logic on the right.



$$\forall x\,(S_1 x \to \exists y\,((x = y) \land S_2 xy))$$

Because of the identity, the sentence is logically equivalent to $\forall x\,(S_1 x \to S_2 xx)$. Parallelly, the branch in the graph that is correlated with the existential quantification may be removed by deiteration as well as introduced by iteration.

Beta graphs are different from the alpha part in that the sheet of assertion comprises signs not found in the alpha fragment. Retaining all the signs pertaining to the alpha part, beta is a conservative extension of alpha with spots and ligatures as the additional signs.

Like alpha graphs, the beta part has the signs of cuts, simple non-crossing and self-returning lines.[27] They are not replicas, but sever the area that lies within them from the sheet of assertion. Together with the area that is thus severed from the sheet, the cut is called the enclosure. In symbolic logic, it corresponds to the one-place operation of negation and all that occurs in its scope.

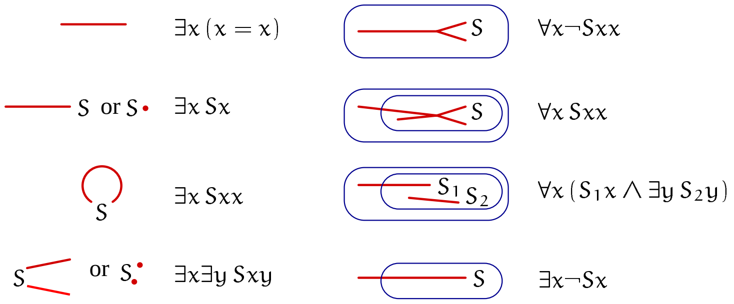
The placement of replicas upon the sheet has a special logical significance of juxtaposition. It corresponds to the two-place logical connective of commutative and associative conjunction.

To recapitulate, the new beta signs are as follows. LIs are graph-replicas of a heavy line with two ends. They are not in contact with any other sign except at their extremities. The extremities of LIs are connected to invisible or imaginary hooks on the boundaries or peripheries of the surface of the sheet that are occupied by the spots. Spots are subgraphs with distinct properties or qualities that distinguish them from other spots. No two LIs may be attached to the same hook. A loose end is "an extremity of a Line of Identity not abutting upon any Spot. Such is the end of a Line of Identity on the Area of a Cut which abuts upon the Cut, itself" (MS 293: 31). The LI is itself a graph, but the totality of all lines connected to one another is not a graph, since it may lie partially within the area of a cut and partially outside. Such a collection of lines is a ligature. It is thus a sign that may cross a cut, and is a composition of lines that may branch in several directions.[28]

Essentially, then, the characteristic features of EGs in their relation to modern symbolic logic are the following.

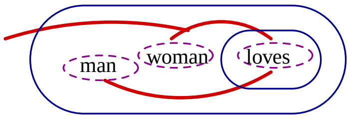
- *Conjunction* is represented by juxtaposing the graph-replicas on the sheet of assertion, in any order.
- *Quantified variables and identity* are represented by LIs attached to predicate terms. Multiple occurrences of the same variable are represented by extending LIs so as to connect different predicates.
- *Negations* are scribed as simple closed lines around graphs.

Let us observe how the eight EGs below diagrammatise the correlated symbolic formulas:

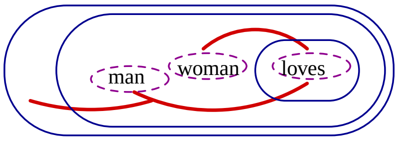


The place of attachments of the LI is a place for the subject, and the symbols so denoted are the verbs. It the line does not hook up with a verb, it denotes an indefinite, utterer's decision of "*some* suitably chosen individual object of the universe of discourse".[29] Actually, to preserve the indefinite reading is to hook the LI up to a spot, the outermost extremity of that LI being enclosed within an even number of cuts, therefore read as 'something suitably chosen by the utterer of the graph'. Dually, spots hooked with oddly enclosed extremities of the LIs are read as referring to anything the interpreter intends to produce, in other words universal statements.

Accordingly, what has been termed the 'scope' of a quantifier in the received conception of quantification corresponds to the extent of the enclosures in Peirce's system. The more iterations of quantifiers, the more enclosures there are. For instance, a so-called 'wide scope' reading of the sentence "Every man loves a woman" is diagrammatised as a graph in which the outermost part is the extremity of an LI not enclosed within any cuts and connected with the spot "being a woman", and thus the assertion that the graph intends is that "There is a woman (possibly the same) who every man loves":[30]



In contrast, the following depicts a 'narrow-scope' reading of "Every man loves a woman":



There are natural reasons why, in Peirce's time, the notion of scope was not in use in graphical or algebraic logic. Such reasons had nothing to do with expressive limitations or the inadequacy of the signs of logic. The notion of scope as later formulated by Frege, Russell and others would simply have been deemed an unnecessary restriction on what diagrammatisation is intended to be, namely a comprehensive representation of the series of images of what the movements of iconic signs of thought and their occurrences in the context of other signs are determined to be. While nestings of cuts bring out the priorities of the order of interpretation of subgraphs, what the identities and bindings of different tokens of selectives on the sheet are need not be limited to any particular level of nestings. For example, LIs may function as coreference markers, in other words denoting connectivity between subgraphs that are separated in not being nested within the other, that is, not belonging to the priority scope of the other.[31] In the next chapter I extend diagrams such that the priority aspect of scoping may in fact be distinguished from the binding aspect of ligatures.

The upshot is that it is better to refer to contexts rather than to scopes in EGs. Contexts have, indeed, been adopted in conceptual graphs in AI and computer science, and are increasingly being used, say, in computational tasks related to speech recognition in dialogue systems.

The emphasis on the contextual viewpoint was well recognised by Peirce. The interpretation of EGs that he came to advocate was, in his terminology, endoporeutic. This peculiar word has not been preserved in the history of logic, and is unlikely to be found in logic textbooks.[32] It means that the flow of information is from the outside in: the outermost occurrence of a graph is examined first and the examination proceeds stepwise towards terminal graphs (a terminal graph is either a spot, an empty SA of tautology, or a "pseudo-graph" of contradiction, which is an empty cut on the SA).

For example, the outermost existential quantifier in a formula of first-order logic is denoted by an LI in the outermost zone of the graph. According to these definitions, conditionals are symbolised by two nested circles (the scroll), the outer one denoting the antecedent and the inner one denoting the consequent.

Context and compositionality The context dependence of logic has not been invariably applauded, however. Customarily, it has been downplayed because it hampers what many regard as the essential ingredient of not only any feasible logical system, but also of the 'guilty secret' logicians and linguists have concerning natural language, compositionality.[33] If proper subgraphs or subformulas are context dependent, they do not necessarily have a self-supporting meaning. For this reason, they are not proper constituents of a larger unit, typically a formula or a sentence, the meaning of which ought to be morphically imaged on those constituents, advocates of the compositional approach acclaim.

I do not plan to dwell on various shapes and shades of what in the end is meant by compositionality, or what the typically quoted results — that at least algebraically, any system of language that has finitely generated syntax will endorse a compositional interpretation — will show about it. I will rather arguer that the claim put forward by Shin (2002), namely that Peirce's obsession with endoporeutic interpretation has in fact foiled a comprehensive understanding of his system and its setting in a wider perspective, is mistaken. Is this claim a plea for compositionality? Shin does not use such a term, but she laments that no challenge has been made to the endoporeutic method of reading graphs. It does not "reflect visually clear facts in the system", and in fact "forces us to read a graph in only one way" (Shin, 2002, p. 63). What are these allegedly visually clear facts? Shin refers to the impossibility of reading graphs so as to determine which are oddly enclosed and which are evenly enclosed by cuts. While this endoporeutic reading may give correct truth conditions for graphs, nestings of cuts are often needlessly forced to be substituted by corresponding implications. Besides, there is no mention in such a reading of a disjunction, namely the juxtapositions of encircled subgraphs within an even number of cuts. Likewise, Shin claims, no vocal difference obtains between existential and universal quantification in the endoporeutic outside-in reading of graphs.

To rectify these defects, Shin strives to reproduce the alpha and beta parts in their respective equivalent negation normal forms, namely those in which cuts are pushed in as deeply as possible.

Shin's remarks reflect more far-reaching issues implicit in her discussion than ones merely related to the direction of reading off these graphs. She terms the preferred reading a "direct reading" or a "multiple reading", while the endoporeutic reading is "indirect". Keith Stenning (2000) has employed similarly confusing terminology in his discussion on the iconicity of diagrammatic methods. There is no mention of the dialogical, communicational interpretation of graphs in these works, however. I strongly doubt that a comprehensive overview of EGs can be obtained without such an interpretation. For instance, the difference between existential and universal quantification is precisely that which determines which one of the parties, the Graphist or the Grapheus is to choose an instance from the universe of discourse as intended by the proposition depicted by the graph. The switching between them is then facilitated by the system of cuts. Likewise, the distinction between reading the graphs as representing conjunctions and reading them as representing disjunctions amounts to this role-playing view of dialogical or game-theoretic interpretation.

Although she does not spell it out directly, the absence of these two interlocutors in Shin's discussion seems to be a symptom of a more general presupposition concerning logic and language, namely that of compositionality. It is this presupposition of compositionality that lurks behind her intention to read those iconic features of graphs in a way that she claims is not brought out by endoporeutic reading. This is an assumption that has preoccupied a number of logicians and logically minded linguists in their search for suitable expressive representations of their everyday jobs. It should therefore be recognised that Shin's theory maintains, however inadvertently and tacitly, this presupposition.

The dialogical interpretation of EGs thus shows what is ill conceived in such preoccupations. Given a full endoporeutic interpretation of graphs, which Shin does not present, there is nothing missing in the way these graphs are read, as the distinction between different quantifiers and different logical connectives is exposed by the system of choices performed by the utterers and the interpreters associated with the graph, the meaning of which is to be disclosed.

Another way of putting a similar point across is to note the equivocation in Shin's use of the term endoporeutic and Peirce's use of it. Shin speaks about the "endoporeutic reading" of the graphs, while for Peirce, the endoporeutic principle was not a matter of reading the graphs, but a necessary follow-on from the diagrammatisation that dictates how to interpret them. This principle provides us with a method for expressing the truth of the graphs, and only perhaps as a consequence, a method for reading them. Full understanding involves the two interlocutors, who choose and assign semantic values to each component in a certain order that respects the passage from the outer instances to the inner, contextually constrained instances. This is unambiguously spelled out in the game-theoretic interpretation, and it comes out in Peirce's notion of "nests" of graphs (4.472, c.1903; 4.494, c.1903; 4.617, 1908; cf. Chapter 6).[34]

In contrast to the animadversions professed in Shin's account of EGs, Peirce did not possess comparable predispositions concerning the preferred interpretation of natural language. The reason was historical rather than systematic. His sin was not that he would have overlooked the distinction between the compositional and non-compositional systems of logic, simply because no such distinction was forthcoming in EGs. His sin was not that he remained preoccupied with only one, the endoporeutic, reading of his graphs either, because it was the only context-abiding game of interpretation in town. His sin was (if it was a sin) simply to fix his ideas by presenting a system of graphs that far exceeded what was comprehensible and printable at that time.

Peirce lucidly and distinctly noted the importance of the contextual issues encountered by those who interpret EGs: "The rule that the interpretation of a graph must be endoporeutic, that is, that the graph of the *place* of a cut must be understood to be the subject or condition of the graph of its *area*, is clearly a necessary consequence of the fundamental idea that the Phemic Sheet itself represents the Universe, or primal subject of all the discourse" (MS 500: 48). The usefulness of this interpretation is beyond dispute in the interpretation and understanding of anaphoric expressions (not only nominal ones but also those involving temporal coreferences). For how otherwise can the values of anaphoric pronouns be brought into being in discourse than by looking up what has happened in previous rounds of interpretation, interspersed with contextual and environmental matters that the phemic sheet, all that is well understood between the utterer and the interpreter, is presupposed to provide?

According to Peirce, "Endoporeusis, or inward-going" (MS 650: 18), is like a global clock that synchronises interpretation and arranges it in a definite inside-out order. Or, it is like

> a march to a band of music, where every other step only is regulated by the arsis or beat of the music, while the alternate steps go on of themselves. For it is only the iteration into an evenly-enclosed area that depends upon the outer occurrence of the iterated graph, the iteration into an oddly enclosed area being justified by your right to insert whatever graph you please into such an area, without being strengthened or confirmed in the least by the previous occurrence of the graph on an evenly-enclosed area. So the analogy to a march is pretty close. (MS 650: 18–19).

Thus, the question for Peirce was not how to read off what the graphs are intended to convey. There may be more than one way of doing that. The question was what is the most appropriate "system for expressing truth endoporeutically" (MS 650: 19). This is what the pragmatist has in view, "a definite purpose in investigating logical questions", as "he wishes to ascertain the general conditions of truth" (2.379, 1901). He predicted that "if anybody were to find [a] fault with the system . . . I should be disposed to admit that it is a poetical fault" (MS 650: 19). In pure aesthetic terms, there may be alternatives, but such alternatives ought to accomplish at least as simple a set of rules as endoporeusis does, and in addition predict the same merits of its essential features.

The alternatives to the endoporeutic method of interpretation have not fulfilled these predictions. Shin's proposal to reduce alpha graphs to their negation normal forms involves a recursive definition from atomic graphs outwards. Replacing Peirce's five transformation rules of erasure, insertion, iteration, deiteration and double cut, Shin's reformulated inference rules come in seven parts, four of which are partial decompositions of the rules of iteration and deiteration into the rules that have been in use in natural deduction systems. Albeit more complex in beta graphs, there are still just five rules for their transformation, which Shin reformulates into nine rules divided into twenty-six subcases.

In general, compositional methods of interpretation are far more complex in that the associated proof system is likely to require a considerably larger number of transformation rules than Peirce proposed. What is more, extending EGs such as adding imperfect information (Chapter 5) is bound to increase the complexity of any compositional alternative to the endoporeutic method to an unnatural extent.

Several scholars have attempted to understand EGs from a modernised perspective. Burch (1997) suggests a Tarskian interpretation of beta graphs. This means variables, infinite sequences of members of the domain of the interpretation, the structure with the universe of discourse of individuals, and, above all, some notion of satisfaction with well-formed formulas by the infinite sequences of individuals derived from the universe. Since beta graphs do not have variables, Burch assumes that their role is replaced by hooks, that is, by a place that, according to Peirce, "shall be appropriated to each blank of the rheme; and such a place shall be called a *hook* of the spot" (4.403, 1903). According to Burch, this is in harmony with Peirce's intentions, shown by the fact that he wanted to make up propositions by filling in the blanks of the rhemas (spots) by individuals (or designations of individuals). Peirce's manuscripts suggest that he did not quite have variables in mind when he resorted to the terminology of blanks, hooks, dots and lines, but their iconic analogues. For instance, blanks are correlated with hooks onto which the extremities of LIs are to be attached:

> A spot has a definite place upon its periphery, called a *hook*, corresponding to each blank; and to each hook an extremity of a line of connection may be attached, with the effect of filling the blank with a designation of the individual denoted by the line. When all the hooks have received such attachments, the spot with these attachments becomes a graph signifying a proposition. (MS 491: 4).

The places upon the periphery of a spot onto which several things that a spot connects are the hooks, and they will have to have a definite order to distinguish between the transitive, intransitive and ditransitive verbs that the spot represents. A verb itself does not morph into a spot before such orders have been agreed and fixed.

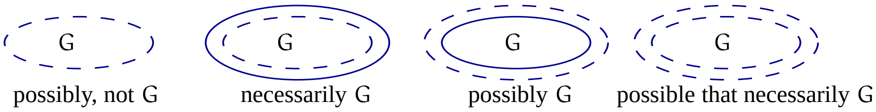
In a closely similar spirit, Hammer (1998) introduced variables into the beta part of graphs, thus giving them a Tarski semantics. To accomplish this, Hammer argues, one needs a "bookkeeping device" (Hammer, 1998, p. 492) that keeps track of the LIs that have already been interpreted as one propagates from context to inner occurrences of cuts. This he does by the system of variables introduced in the interpretation. As will be shown in later sections, there is a more natural way of doing a similar thing in terms of the game-theoretic interpretation, which in fact comes very close to Peirce's own intentions. The interpretation can be written out in game-tree form, and we automatically have all the necessary tools for such a bookkeeping device in the form of derivational game histories showing which LIs have already been met and interpreted.

Gamma systems Peirce held that the essential feature of the beta system consists of the notions of individual identity and individuality.[35] This distinguishes it from the alpha part, which does not possess the ligatures, the networks of LIs that denote relations that hold between different instances of individuals. It is clear from this formulation that with his beta graphs, Peirce's tendency was to create a first-order ('first-intentional') level of representation that makes references only to individuals, not to possibilities, collections or any other higher-order entities.

However, he soon found that there were concepts that cannot be expressed by the beta system. For instance, "Aristotle has all the virtues of a philosopher" cannot be expressed by means of LIs, cuts and spots representing propositions, and is thus "beta-impossible", and so is the following: "A certain institution will pay every dollar it has borrowed or shall borrow with a borrowed dollar; and the payment of a dollar cannot balance debts of more than one dollar. Nonetheless, there will be some dollars borrowed that never will be repaid".[36] The possibility that this statement refers to via the insertion of the modal words *will* and *shall* "consists in a predicted endless future that never can become a positive fact" (MS 462: 6).

With beta graphs one is thus "unable to reason about abstractions. It cannot reason for example about qualities nor about relations as subjects to be reasoned about. It cannot reason about ideas" (MS 467: 4,6). What Peirce was after was the logic that could repair these imperfections.

One way in which he tried to account for this was by suggesting that the third, gamma part of EGs should be enriched by a new form of one of the earlier signs, the broken cut. Any graph, say one that asserts "it is raining", scribed within its enclosure, would have a special meaning. According to Peirce, such a graph "does not assert that it does not rain. It only asserts that the alpha and beta rules do not compel me to admit that it rains, or what comes to the same thing, a person altogether ignorant, except that he was well versed in logic so far as it embodied in the alpha and beta parts of existential graphs, would not know that it rained” (MS 467: 42; 4.515). If a reading is marked so that a broken cut does not compel the interpreter to admit that the graph G within its enclosure is true, its vernacular translation is “it is possible that not G”.



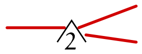
These correspond to the modal-logical formulas $\Diamond\neg p$, $\neg\Diamond\neg p$ $(:= \Box p)$, $\Diamond\neg\neg p$ $(:= \Diamond p)$ and $\Diamond\neg\Diamond\neg p$ $(:= \Diamond\Box p)$, respectively.

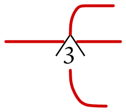
According to Peirce, another special sign required in gamma graphs is one of the eight spots in the class of potentials: “The selectives placed at the left of potentials are proper nouns of a strange kind, since they do not denote ordinary individuals but qualitative possibilities which, in themselves, have no individuality”.[37] This is the innovation that delineated gamma from the beta of actual individuality. The potentials comprise those spots that represent possibilities, and are thus on an entirely different footing than the other main class of signs, the signs of graphs and graph elements.

Spots and rhemas are thus abstracted, for instance, to the potentials scribed on below right:

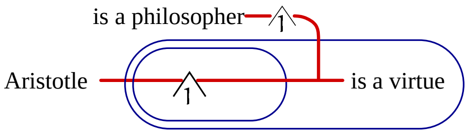
___ *is a proposition/graph* : 

___ *has a quality* ___ : 

___ *stands in a dyadic relation* ___ *to* ___ : 

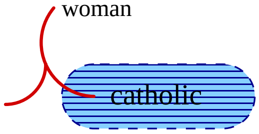
___ *stands in a triadic relation* ___ *to* ___ *for* ___ : 

The following is the abstracted graph for “Aristotle has all the virtues of a philosopher”:



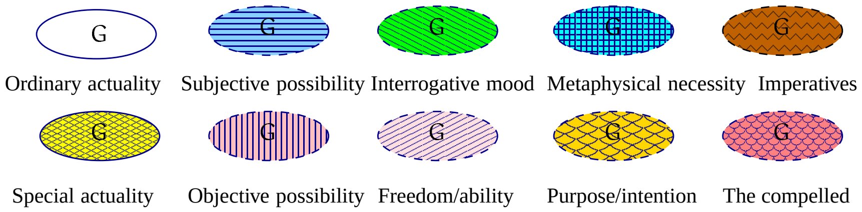
Potentials do not behave like beta spots, and hence different transformation rules are needed. One complication is that the notion of identity is more complicated, since the continuity should link the individual’s actual and possible manifestations.

The notion of possibility that Peirce suggested in MS 490 for expressing things not expressible by beta is that of subjective possibility. The discourse between the Graphist and the Grapheus (the interpreter of the graphs) pertains to a universe that is not one of actually existent individuals, but some other. For instance, to use Peirce's example, this refers to the conditions that prevail in Hell. In that case, to say that there exists a woman and a Catholic is to assert an impossibility, which is diagrammatised by a shaded cut denoting the area in which the assertion that a Catholic exists is denied, while an existent woman is scribed outside the shaded area. What he found was that the *verso* of the assertion sheet on the area of the cut represents a kind of possibility. The reversal of the sheet's *verso*, that is, the *recto*, represents actuality. But in such cases, a ligature (which is a composite of several graph-instances) may connect actually-extant beings scribed on the *recto* of the sheet and mere possibilities scribed on the *verso*, because this connection presupposes composition of "something is a woman" and "something is other than any possible catholic". Below is a gamma EG of "There is a woman who is not and could not be identical with any possible catholic":

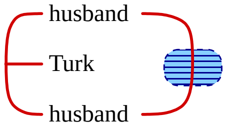


These considerations soon led to tinctured EGs, the main innovation of which was to represent different kinds of universes with different tinctures. This was done not only in order to mark different kinds of possibilities, but also to mark different kinds of actualities, different kinds of possibilities consisting of ignorance, of variety, of power and of futurity, and to mark different kinds of intention.[38]

Tinctured graphs encompass different kinds of modalities by virtue of the fact that the *verso* of the SA on the area of the broken cut represents possibility, while the *recto* represents actuality. Among the possible tinctures are:



Peirce's own example was "There is a Turk who is the husband of two different persons":



The tincture captures that it would be contrary to what is known by the one who scribes the graphs that the two individuals are identical. In other words, "As far as is known, a Turk exists who is the husband of two different persons".

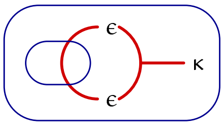
With tinctures, Peirce anticipated not only alethic modal logic, but also epistemic logic, erotetic logic, deontic logic, belief-desire-intention logic, and the logic of imperatives.

Peirce also considered reasoning about graphs by using graphs. For example, Some gamma expressions of alpha graphs would be:

____ *is an enclosure* : ——— κ

____ *is the area of the enclosure* ____ : ——— ϵ———

Example of such a graph-of-graphs gamma expression is:



This says that "No enclosure has more than one area". A similar strategy works for the beta and gamma graphs.

I will not go on to expound further the different compartments of the gamma system, they have been well explained in Roberts (1973a). By way of summary, there are three. The first deals with different modality-types in terms of broken cuts and tinctures. The second comprises abstractions, where one turns a whole thought or a theory into an object of another thought or of another theory. This happens, for instance, in how not only potentials but also collections, logically taken to be individual, are formed, and in assertions that are made about properties of objects and their identities. The third compartment comprises a selection of different spots representing and imbedding the alpha and beta expressions as objects of graphs, the system dealing with the "graphs of graphs". Note here the commonality of the latter with Gödel numbering and the category-theoretic approaches to mathematical theories.

The question of whether it is possible to devise "archegetic" rules for these compartments, namely the saturated kernels or cores that do not logically follow from other rules of the system, was left open by Peirce. He did formulate such rules for the alpha and beta parts, and MS 478 contains an extensive presentation of the rules for the the potentials.[39] Peirce did not claim that these rules comprise a complete system of inference rules, but even to pose the question of such a possibility has to be regarded as a major achievement. The open-ended nature of the gamma system was realised in his note: "*The Gamma Part* supposes the reasoner to invent for himself such additional kinds of signs as he may find desirable".[40] Accordingly, the answer to the archegeticity of the gamma part dealing with potentials is not known.

An interesting codicil to EGs, not clearly either beta or gamma, is to be found in the Logic Notebook (LN: 340r, 7 January 1909). Here, Peirce reads various diagrams that look like beta graphs as "p is true under some circumstances", "p is true under all circumstances", "p is true some times and q is true some times", "p and q are sometimes true", "p is true under some circumstances and q under others" and so on. Thus the line of identity is taken to quantify various circumstances or points of time, not just individuals as before. This is yet another anticipation of possible-worlds semantics for expressing various kinds of modalities, including temporal ones (Chapters 3 and 6). Despite its curious title, in one two-page (341r) and two one-page sketches (342r and 343r, 16–17 February 1909) *Studies of Modal, Temporal, and Other Logical Forms which relate to Special Universes*, no further reflection on modal or temporal amendments is to be found. These drafts rather served to anticipate his invention of the triadic system of logic.

4. Existential graphs on the move

In logical terms, the idea of moving pictures of thought may be operationalised by using game-theoretic techniques — a 'game' in this case loosely meaning any dialogic or discourse-theoretic process of interaction between two contestants or collaborators.[41] Indeed, diagrammatic and dialogic approaches to logic have a lot in common. In Peirce's philosophy, they were intimately connected. Unfortunately, this insight was lost during the bulk of the 20th century, the only exceptions being its first and last decades. Peirce perceived such exceptions in the method of EGs and their interpretation. Their relation to game-theoretic concepts was anticipated in Hilpinen (1982) who revived issue that was further studied in Burch (1994). However, there are several questions to do with this relation that have not been addressed thus far, including several aspects of the nature and characteristics of these game-theoretic explications. The turn logic took in the interim was dominated by what was called symbolic logic, in which, especially in the first half of the 20th century, no additional apparatus of interaction was deemed to be necessary.

There is slight irony here, since according to the entry on Symbolic Logic that Peirce contributed to Baldwin's Dictionary, the term was defined as "logic — treated by means of a special system of symbols [for which] it will be convenient not to confine the symbols used to algebraic symbols, but to include some graphical symbols as well" (DPP: 645). In fact, the actual material in the definition was almost entirely related to graphical or diagrammatic aspects of logic. This definition also included one of the earliest occurrences of the term symbolic logic, other early users being John Venn (1881) and Ernst Schröder (1891). Frege, Russell and many of their followers subsequently employed the term in a much narrower context.

An attempt to restore the relation between diagrammatic representations of logic and games may be made in terms of GTS (Hintikka 1973a; for details, the reader is invited to visit Chapter 7 first). This theory, with its stress on the interactive relations between two functionaries, has been instrumental in unfolding the dynamic concept of truth, but has not quite brought out the actual geometry of the interaction involved, nor its contribution to the idea of diagrammatic, iconic and interactive reasoning in any detail.

What, then, are the kinds of games that lend themselves to diagrammatic systems of logic? Roughly speaking, they are interactions in any sentence of logic between two players, playing two roles on a given model, depending on the type of logical constant encountered in the formula. If the constant is a conjunction or an existential quantifier, then the player will play the role of the verifier of that sentence and choose one of the conjuncts or an individual from the domain of the structure, respectively. If the constant is a disjunction or universal quantifier, then the player will play the role of the falsifier of that sentence and choose one of the disjuncts or an individual from the domain. The player playing the former type of role is often termed Nature, and the player playing the latter type of role is often termed Myself. The game continues with respect to the subformula that was chosen, plus an instantiation of the chosen individual in the case of quantifiers so that the choice fits the intended statement. If a player encounters negation, these roles will be changed throughout the game, and the winning conventions will also change. According to the winning conventions, if an atomic sentence that is true is reached by Myself, she will win that play of the game. Likewise, if an atomic sentence that is false is reached by Nature, he will win that play of the game.[42]

Because this is truly a game, players proceed by way of strategies. This is important, since truth is now defined so that the whole sentence with which the game started will be true if and only if there exists a winning strategy for the player who began it as Myself. Likewise, the whole sentence with which the game started will be false if and only if there exists a winning strategy for the player who began it as Nature.

It is clear from this informal exposition that the reason for the suppression of any visual elements in the dialogue in the sense of GTS is that the games that were taken to form the subject matter of logic were typically similar to the normal (or strategic) forms of games familiar from the mathematical theory. Consequently, only the choice of a strategy profile plus the assigned payoffs for each player are brought out in the description of the game. Thus, one is able to precisely capture which propositions are logically true and which propositions are logically false (plus, possibly, which propositions are altogether undetermined), precisely because truth is defined as the existence of a winning strategy for one of the players. However, one is not necessarily able to see how this truth making proceeds, step-by-step, by means other than putting the strategy functions that exist (in logic, the Skolem functions) in their right places in the formula. This I seek to remedy. In order to arrive at the actual graphics of truth making, one needs to depict the game in its extensive form, which entails writing out the interactive and dynamic process of game playing in a graphical and diagrammatic fashion. It is thus itself an icon of the meaning of an iconic representation of a proposition or a complex concept.

Informally expressed, the idea of the extensive representation of the contest is to depict any game as a finite tree (see the formal definition in Chapter 7). The whole proposition lies at the root of the tree, and the atomic formulas are the leaves. These leaves, or terminal histories, are mapped on payoffs denoting whether the play leading to them is a win or a loss. Any non-terminal history may be extended by an action as given by the game rules informally described above. The game is strictly competitive. What is produced is the diagram of actions and reactions, in other words an information structure showing all the positions plus what has already happened when any position is reached.

The importance of these extensive representations lies, among other things, in the fact that there may be information in the game-playing situation that cannot be, or at least that is not easily, captured by the normal form of the game alone. These include the notions of forgetting and recalling past knowledge or past actions during the plays, or incomplete information about the structure of the game itself, usually implemented by chance moves by Nature (assuming that there are common priors, see Chapter 7), or any combination of the two. Indeed, an exponential amount of information usually needs to be encoded into the strategies when moving from extensive to normal forms of the game.

In a wider perspective, the revival of diagrammatic systems of logic that were witnessed in the late 20th century, even though they resulted in rich heterogeneous theories of logic, they neglected the dialogical aspects of such representations. This neglect shows nowhere more pointedly than in the shortage of strategic resources. Mention of strategies, in the sense of the normal form of the game, is not just cursory. And even if the extensive-form games do not explicitly encode strategies on the representational level, they exist as possibly non-deterministic functions from possible histories to available actions. Moreover, strategies are needed not only in the abstract, truth-conditional sense of delivering the truth-values of the heterogeneous systems in question, but also in order to see how natural language meaning is established. Many natural-language phenomena that are all too easily dubbed pragmatic may be approached from the dialogical and game-theoretic point of view by considering the question of what the strategic meaning of such expressions is, in other words by appreciating the question of what there is in the strategies the players use that makes the meaning of the sentences what it is.[43]

An example of such constructions is natural-language anaphora with its coreference (Janasik et al., 2002). Here, syntactic and lexical clues, contextual and environmental parameters, presuppositions and sensitivity to accommodation, implicatures and presumptive meaning, for example, play an important role. It needs to be acknowledged that we are still far from any good comprehensive theory of strategic meaning for fragments of natural language. I believe the importance of this cannot be underestimated, especially in view of the increasingly prominent role of cross-speaker and cross-cultural dialogue systems aiming at the understanding, interpretation, comprehension and evolution of dialogue, typically interspersed with some computational (but not necessarily recursive) tendencies.

Strategic dimensions are also connected with the idea of multi-modal communication, reasoning and representation, which takes humans and artificial systems alike not only as symbolic and verbal species, but as species encompassing a plethora of indexical and iconic mechanisms in communication.

There are two main aspects to the interconnectedness of EGs and dialogues. First, the graphs are constructed (that is, inscribed) by the interactive process between the Grapheus or the Interpreter of the assertion, who creates the universe, and the Graphist, or the Utterer of the assertion, who proposes modifications to the initially blank sheet on which the graphs are scribed. Any one graph represents one possible state of the universe. The Grapheus determines the characters of the universe as he pleases. This may be deciphered so that the interpretations of the underlying language, that is, the atomic expressions of it, are completely determined by the Grapheus. This is the process of building models. Moreover, there are no partial interpretations, shown by the presupposition that "the blank of the blank sheet may be considered as expressing that the universe, in [a] process of creation by the grapheus, is perfectly definite and entirely determinate" (4.431, c.1903).[44]

Second, the graphs are interpreted through mutual examination and inspection between the Graphist who proposes the assertion that any graph thus created represents and the Grapheus who has created the universe. We could think of this experimentation along the lines of the time-honoured conceptualisation in the history of the philosophy of science of 'putting questions to Nature'.[45] The examination concerns one state of the universe at a time, and it is performed by a communication between the Graphist who puts forward assertions authorised by the Grapheus. The Grapheus will not change his mind about the authorisation because, as noted, Peirce held that the universe that is being communicated upon has to be perfectly determinate.

The following five points elaborate the interconnection further.

(i) Peirce resorts to the notion that continuity or continuous processes concern the assumed creation of the universe, the creation that the Grapheus accomplished "by the continuous development of his idea of it, every interval of time during the process adding some *fact* to the universe, that is, affording justification for some assertion" (MS 492: 18; 4.431). Peirce's synechism, was a vital part of his metaphysics, and its value in EGs should be well acknowledged.[46] In fact, as he proposed to prove the doctrine of pragmatism by using EGs, the crux of the matter was in how continuity was to be incorporated into it (see Chapter 1, note 14, as well as MS 330). The closely-related issue here is the proximity of graphical logic to topology, the logic of continuity. This is considered in the next chapter.[47]

The observation to be made here is that Peirce held the three elements in one of his central trichotomies among the countless others, namely that of rhema, proposition and argument to be continuous with one another.[48] Such continuity is illustrated in beta graphs in terms of spots being continuously connected by LIs or by composition, both of which, topologically speaking, express connectivity between different parts of the surface, thus forming propositions, and also in terms of propositions being connected, in the equally topological sense, under continuous transformations from one graph to another, thus forming inferential arguments. This is but another facet of the centrality of topological considerations that were central to Peirce, but who all the same fell short of possessing some of the key topological concepts.

(ii) Peirce's remark about some of the requisite ingredients of the parties undertaking the scribing and interpreting of graphical proposition is worthy of note. He assumed that the minds of the Graphist and the Grapheus (which could, broadly speaking, be quasi-minds) should be capable of introspection, in order to be able not only to control these processes, but also to develop and educate the habits contained in them.

> Now nothing can be controlled that cannot be observed while it is in action. It is therefore requisite that both minds but especially the Graphist-mind should have a power of self-observation. Moreover, control supposes a capacity in that which is to be controlled of acting in accordance with *definite general tendencies of a tolerable stable nature*, which implies a reality in this governing principle. But these habits, so to call them, must be capable of being modified according to some ideal in the mind of the controlling agent; and this controlling agent is to be the very same as the agent controlled; the control extending even to the modes of control themselves, since we suppose that the interpreter-mind under the guidance of the Graphist-mind discusses the rationale of logic itself. (MS 280: 30–32, emphasis added).

This is yet another striking instance of how the strategic reasoning assumed of agents permeates Peirce's conception of logical tasks (Chapter 3). In other words, it could be said that the attainment of the notion of the finality of interpretants furnishes the maximalisation of the utilities that are the outcomes of actions suggested by the minds of the communities of self-controlled agents (Chapter 11). The result of self-control that amounts to the development and evolution of behaviour, and ultimately to the increase in the *summum bonum* of inquirers, is a habit change, here realised in the guise of "definite general tendencies of a tolerable stable nature".

(iii) Peirce describes the interaction as "collaborative" (4.552), which is notable given that in the customary theory of semantic games, the players draw their actions in a strictly competitive fashion. Yet, if there is collaboration, it is not inconceivable that there be some 'division of surplus' of the truth values of atomic propositions, which leaves the possibility of atomic contradictions. This collaborative form of games is suitable for discourse interpretation, in which the creation of the common ground of the participants is in a more prominent position than in the interpretation of isolated propositions. Peirce did not develop any extension of his ideas towards genuine graphs of discourse, and was strictly committed to the sentence level of assertions as far as the examples went (though some of his examples were complicated sentences filling up a whole page or so, and should have been divided in several parts for comprehension). The viability of such extensions is vindicated by the success of the discourse-representation theory epitomised in Chapter 5.[49]

(iv) What is also notable is the presumption that, in close agreement with the assumption of collaboration between the Graphist and the Grapheus, these functionaries were taken to be in a communicative relationship with each other.

The relationship is slightly asymmetrical, though, as Peirce assumes that "the *grapheus* communicates to the *graphist* from time to time his determinations in regard to the character of the universe. Each such communication *authorizes* the graphist to express it".[50] This is consonant with the idea of interrogating Nature. Peirce continues this paragraph: "*An authorization once given is irrevocable:* this constitutes the universe to be perfectly *definite*". Being perfectly definite and perfectly determinate are not the same thing, however, for: "Should the graphist risk an assertion without authorization, he must hope to receive an authorization later; for *what never will be authorized is forbidden:* this constitutes the universe to be perfectly *determinate*" (MS 492: 17 a.p.). If it happens that a modification needs to be made to the asserted graph, it has to be made according to sound rules of transformation.

(v) The final point is that the twist to what we perhaps take to be an unproblematic notion of soundness, namely one that does not render the outcome of the transformation false whenever the propositions from which we started are true, is that Peirce allows the Grapheus to make further determinations by adding new characters to his universe.

This remark suggests a much more complicated notion of the universe for logical assertions, namely a changing, dynamic domain in which individuals may change, disappear and come into existence while the deductive transformations are in progress. If so, then Peirce's transformation rules need to be set in a quite different light.[51] For, during the transformation prompted by the Graphist's need to make modifications to her initial assertions, the Grapheus may choose to modify the determination of his universe. This does not affect the determinateness of the universe, but one needs to be prepared for the fact that the Grapheus may sometimes cheat.

Peirce took early steps in model theory with precisely these kinds of considerations in mind. According to him, it is in the nature of necessary reasoning, that whenever a consequent B logically follows from an antecedent A, it is the case that "in every universe where A is true, C is true also".[52]

The alpha theory of EGs and GTS were brought into relation in Burch (1994), by mapping the five conventions for the alpha fragment (4.394–402) on the game-theoretic rules of action. This is well in accordance with Peirce's intentions, although Burch does not consider the five points that I outlined above. In essence, the EGs are constructed by the Grapheus, who can be taken to be the *malin genie* determining the truth of the irreducible, terminal graphs. The Grapheus is then willing to play off against the Graphist, who scribes the molecular graphs on the sheet of assertion and begins their examination by engaging in an interactive examination process with the Grapheus.

The mapping from alpha graphs to semantic games is straightforward. To recap, the basic logical components of alpha graphs are the cuts (negation), juxtaposition (conjunction), and the sheet of assertion (verum, the logically true proposition). All these are scribed on the sheet of assertion. Any two graphs scribed on the sheet represent commutative conjunction, as the order in which these are selected is not material. A continuous circle around the graph represents negation. An empty graph represents a dummy action in which nothing is scribed. Likewise, a cut encircling an empty graph is the falsum (the logically false proposition). The Grapheus' universe determines the truth-values as well as the falsity values of atomic graphs.

In semantic games, the Grapheus and the Graphist are mapped on their roles of Nature and Myself, respectively. The mapping is total, that is, at each non-terminal history of the game a player has one of these roles and the adversary has the other. As noted earlier, the rule of interpretation is endoporeutic, starting from an out-most cut or a graph outside a cut, and proceeding towards an atomic or a blank graph. Since the graphs are finite, atomic components are eventually reached. The winning conditions are such that when an atomic graph is reached, the player playing the verifying role (Myself) wins if that graph is true and the player playing the falsifying role (Nature) wins if that graph is false.

The molecular graph itself is true precisely in the case when the player who made the first move as Myself is able to win no matter how her adversary moves. Symmetrically, the graph is false precisely in the case when the player who made the first move as Nature is able to win no matter how his adversary moves. In the terminology of semantic games, we say that, in these cases, there exists a winning strategy for Myself or for Nature.

In the case of the beta part of the theory of EGs, not covered in Burch (1994), there are, in addition, lines of identities. These correspond to quantified variables, identities between the variables, predication and the relation of coreference. Accordingly, their interpretation is such that suitable individuals have to be picked from a domain of discourse by Myself and their names assigned to the corresponding lines. To do this, we assume that a domain of individuals is arranged so as to form an interpreted structure. An interpretation consists of a structure with a non-logical alphabet attached to the domain and definite valuation given to the terms and predicates.

Similar winning conditions and truth definitions apply to beta as alpha graphs, with the sundry addition that the atomic graphs are also interpreted by the Grapheus in terms of checking whether the sequences of individuals chosen during the endoporeutics are to be included in the extensions of the atomic predicates reached at the spots. If so, the current player who is Myself will win. If not, the player who is Nature will win.

Given the game-theoretic interpretation of Peirce's EGs, what is the structure of these games? It turns out that there is a convenient way of representing them in the game-theoretic format of the extensive-form game briefly described in the previous section and to be elaborated in Chapter 7. In doing so, another diagrammatic and iconic representation of logic emerges. In other words, any EG may be turned into an extensive semantic game, adjoined by the payoff conditions that can be taken to be judged by the Grapheus.

In the case of alpha graphs, the tree will consist of binary choices between a subgraph and the rest of the juxtaposition, together with the labelling of the non-terminal histories by the players. The extensive form will be a tree with two successors. The payoff function will assign the terminal histories the values in $\{1, -1\}$, transforming the extensive form into the game proper. In the case of the beta graphs, the branching factor of the tree is the size of the domain for levels at which the lines of identities are interpreted. The edges are also labelled by the names of the individuals chosen from the domain. In the case of gamma graphs, the modalities have to be taken into account too, and the branching factor has to take the cardinality of the different states of information subsisting in the model into account.

This, in short, provides the bookkeeping system for EGs. Most importantly, it is needed for the beta fraction, in which the values acquired by the lines of identities have to be recorded in a way that keeps track of the previous choices. This answers the question posed in Hammer (1998). He asked about the bookkeeping device that would keep track of the lines already interpreted according to the endoporeutic method of interpretation. Extensive semantic games have such a device.

Summarising, a non-cooperative, perfect information, zero-sum semantic game is played between the Graphist and the Grapheus on a given graph-instance G_i and a model $\mathcal{M}$. The game rules are:

1 Juxtaposition of graph-instances on positive (negative) areas: the Grapheus (the Graphist) chooses between two graphs. The evaluation proceeds with respect to that choice. The winning conventions change with respect to choices on negative areas.

2 Polarity of the area of the outermost extremity of a ligature determines whether the Graphist (if positive area) or the Grapheus (if negative area) is to make a choice from the domain of $\mathcal{M}$ to be the interpretation of the ligature. Evaluation proceeds with respect to that choice and with a graph-

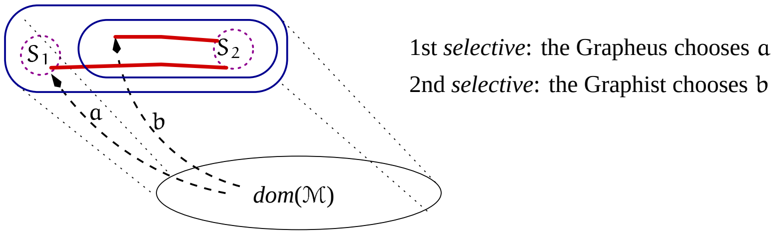


Figure 4.1. EG beta graph, first two steps of the evaluation

instance that has the ligature removed up to its connexions with spots. Dots will be scribed at the hooks of spots.[53]

3 When a spot is reached, its truth-value determines the winner: if a spot is true, then the Graphist wins a particular play of the game; if false, then the Grapheus wins.

4 The existence of a winning strategy for the Graphist agrees with G_i being true in $\mathcal{M}$. Likewise, the existence of a winning strategy for the Grapheus agrees with G_i being false in $\mathcal{M}$. The winning strategy is defined such that the player can arrive at the winning terminal position no matter what the choices of the opponent are.

Graph-instances are interpreted according to the endoporeutics: what is scribed on outermost areas are evaluated on $\mathcal{M}$ before proceeding to nested, contextually constrained graphs-instances, until spots or their cuts are eventually reached.

The evaluation of LIs involves an instantiation of a value to the outermost end of a line, and this value then propagates continuously along the LI towards the interiors of the inner cuts and to the spots to which the lines are attached. Peirce termed the process of instantiation together with the type of the identity line the selective.[54]

As regards the EG in Figure 4.1, the evaluation proceeds, after the first two steps of choosing individuals and naming them a and b, by the Graphist choosing between the spots S_1 and cut-S_2, each of which having one occupied hook. If she chooses S_1 and S_1 is satisfied in $\mathcal{M}$, the Grapheus wins, because the choice was made on the negative area. If the Graphist chooses S_1 and S_1 is not satisfied in $\mathcal{M}$, she wins. The EG is true precisely in the case a winning strategy exists for the Graphist, and false precisely in the case a winning strategy exists for the Grapheus.

As noted, the evaluation takes place between two parties "in our make believe" (MS 280), the Graphist who scribes the graphs and proposes modifications to them, and the Grapheus who authorises the modifications. The graphs scribed by the Graphist are true, because "the truth of the true consists in his being satisfied with it" (MS 280: 29).[55] Hence, the Graphist is the verifier of the graphs. To end with a true atomic graph amounts to a win for the Graphist, and to end with a false one amounts to a win for the Grapheus. The truth of the whole graph agrees with the notion of the existence of a winning strategy for the Graphist, in Peirce's terms a habit that is of "a tolerable stable nature". Likewise, falsity is the existence of such habits for the Grapheus.

The existence of a winning strategy presents an issue that goes somewhat beyond Peirce's own suggestions. It was not until after the 1920s that a game theory that espoused the concept of a winning strategy and gave it a mathematical precision emerged, to be used to provide the conditions for any molecular graph being true under a given interpretation. Peirce did not arrive at the rule that defines when the whole graph is true via any unequivocal game-theoretic route, and the connection is bound to remain informal in the sense that nowhere in his writings do we find a mathematical function that plays the role of any such strategy.

In order to arrive at extensive games, these conventions are applied so that the root is labelled with the entire graph on the SA, and each choice prompted by the conventions extends this root (see Chapter 7 for more detail). A terminal history cannot be extended. Each terminal history is mapped to its payoff from $\{1, -1\}$, determined by the Grapheus during the model-building phase. These values show the wins and losses of plays. Any strategy that invariably leads the falsifying (resp. the verifying) player to the terminal history with the payoff -1 (resp. 1) will be a winning strategy for that player. The set of such strategies agrees with the entire graph being true or false according to the endoporeutic principle of interpretation.

Figure 4.2 reconstructs part of the meaning of the beta graph for "Every man loves (and is loved by) a woman" via an extensive game on the model $\mathcal{M}$ with the universe of discourse containing individuals named by the elements from the set $\{a, b, \ldots\}$. The actual payoff distribution depends on the interpretation of spots given at the outset. After erasing a ligature, any outermost double cuts may be discarded as that would not change the polarity of nested areas.

The phrase 'according to the endoporeutic principle of interpretation' is actually vital, because it almost translates into 'being true or being false in-a-model'. Almost, because the closest Peirce ever gets to this term is his idea of cognition as a "working model" (MS 298: 5). He intended EGs to be true models of dynamic images of thought. Behind this idea, the intention to interpret graphs in-a-model in the sense of the much-later-matured mathematical theory of models is nonetheless unmistakable. Just as an interpretation of a sign can be another sign, the role of the model in question is played by another EG (i.e. an interpreted structure), which is the homomorphic image of the object graph. The choices made by the players from the universe of discourse of this model graph are converses of the assignments of the values to the components of the object graph. I will return to these issues in Chapter 6.

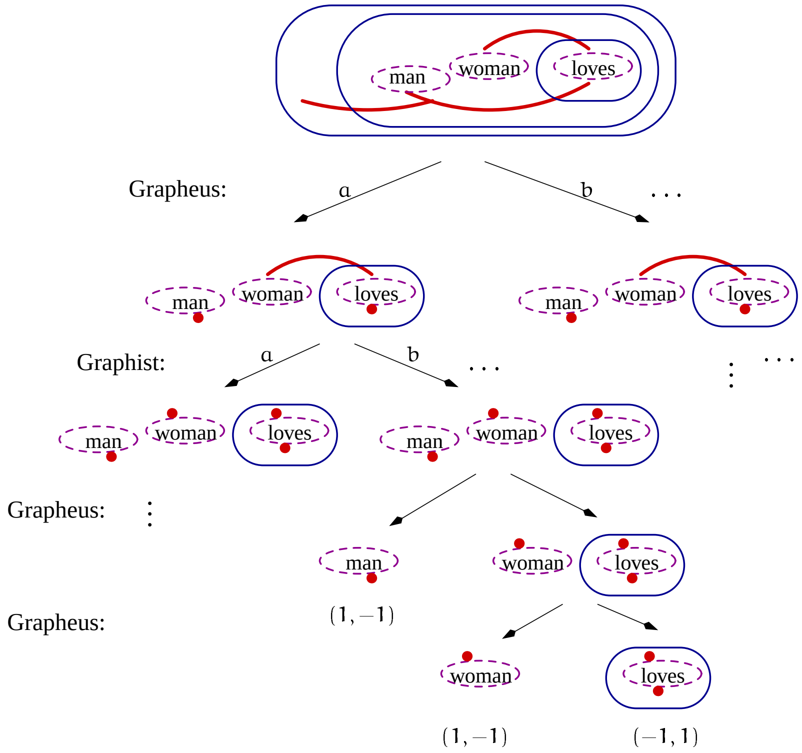


Figure 4.2. An extensive game for the beta EG "Every man loves and is loved by a woman".

In contrast to models of model theory that extend to infinitely deep, those conforming to the endoporeutic interpretation are finite in nature, being confined to the "field of distinct vision" (MS 280: 29) of interpreter's phaneron. No principal mathematical reason exists why they could not be held extendible to approximate standard, infinite, non-extendible models.

That truth-conditions established by the endoporeutic method were one of Peirce's central concerns in his logic is shown, for one thing, in the remark, "Logic proper is the formal science of the conditions of the truth of representations" (2.229, c.1897). He expressed this in MS 593 [1893, *A Search for a Method. Essay VI*] by elaborating on what the conditional proposition 'if A is true, B is true' means: "Every possible state of things in which A should be true would be a state of things in which B would be true. There is a universality and modality involved in it, as the necessity of the conditional is an instance of the universality of the categorical proposition". He refers to the notion of the possible state of things as a universe that consists of a range of possibilities. The invoked labyrinth of modalities was to haunt his logical investigations for the better part of his days, as will be attested in later chapters.

Notes

1 4.8, c.1905, *The Simplest Mathematics: Logic and Mathematics, Preface*.

2 MS 619: 8, 27 March 1909, *Studies in Meaning. The Import of Thought: An Essay in Two Chapters*.

3 MS 298: 1, 1905, *Phaneroscopy*.

4 Cf. Peirce's thought that, "In portraiture, photographs mediate between the original and the likeness" (1.367, c.1890, *A Guess at the Riddle*).

5 MS 289: 4 a.p., c.1905, *Consequences of Pragmaticism*.

6 This interconnectedness of modality and negation is especially evident in MS 300 [1905, *The Bed-Rock Beneath Pragmaticism*] in which the relation is characterised as modality being "an ingredient of the conception of Negation" (MS 300: 40 a.p.).

7 Peirce's semeiotics and its relation to the concept of communication has been aptly investigated in e.g. Bergman (2000) and Johansen (1993), but without game-theoretic twists.

8 At least, he tried to avoid such a mixture. For instance, in repudiating the use of selectives (see sects. 3, 4) in addition to ligatures the main reason was the violation that the graphs were represented in as analytic and iconic a manner as possible, as selectives were bound to be symbolic, not iconic signs. The bulk of MS 300 virtually deals with matters related to non-iconicity of selectives.

9 7.467, *Association and the Law of Mind*.

10 Together with Mitchell, Peirce had already developed an algebraically-motivated logic of quantifiers, with scope conventions and all, at roughly the same time as Frege. Their language and notation were deployed alongside those of EGs until Russell and Whitehead took over. Peirce anticipated the inevitable course of events: "Peano's system is no calculus; it is nothing but a pasigraphy; and while it is undoubtedly useful, if the user of it exercises a discrete freedom in introducing additional signs, few systems of any kind have been so wildly overrated, as I intend to show when the second volume of Russell and Whitehead's Principles of Mathematics appears" (MS 499).

11 MS 484, 1898, *On Existential Graphs*.

12 Note that I wrote "existential". One of its precursors, the theory of entitative graphs, did appear in *The Monist* 3, 1892–93. Peirce was dissatisfied with it, and suggested its existential improvement, but the editors turned down the paper. There is a duality between entitative and EGs: the initially blank sheet of assertion in entitative graphs represents the falsity, whereas in EGs it represents the truth. Consequently, any true graph inside an even number of cuts will, in the former theory, be false, whereas in the latter it will be true, and vice versa for an odd number of cuts. It follows that, in the former, the juxtaposition as a primitive represents a disjunctive assertion, while in the latter it represents a conjunctive assertion. Later on, he thought that asserting denials of propositions, in which one is presented with the true state of affairs where nothing is yet scribed and modifications follow, is "logically" the simpler activity of the two, while its dual, in which one grasps the initial falsum and then goes on to propose modifications to it by asserting affirmative propositions, was "psychologically" simpler (MS 500).

13 Sampling the margin notes we find that, "His [Marey's] inaccurate language is intolerable in mathematics. He does not mean a displacement but a *motion*" (p. 28). In referring to Marey's account of mechanics Peirce noted: "This is inspeakable *rot*! A disgrace to an educated man. As if there ever could be an analogous situation to that of Galileo! Is the camera going to supply intelligence & genius? Does not this mark degeneracy of France? This man is membre de l'Institut. This is enough for me. I want to know no more of this charlatan" (p. 30).

14 MS L 231, 22 June 1911, *Letter to J. H. Kehler*; NEM 3:191.

15 Apart from logic, Peirce was interested in the device for scientific reasons related to map projections, see CD VI:4763. See Gosser (1977) for a comprehensive study of stereoscopic moving pictures.

16 MS 1170, [*Notes for Contributions to the Century Dictionary*].

17 The reference is to the electronic version of the document, without page numbers.

18 MS 856: 7, probably 1911, *A Logical Criticism of the Articles of Religious Faith*. In the published entry 'Modality' in Baldwin's *Dictionary* Peirce says that, "Lange [in *Logische Studien*] thinks the matter is put in the clearest light by the logical diagrams usually attributed to Euler, but really going back to Vives. 'We, therefore, here again see,' he says, 'how spatial intuition, just as in geometry, verifies (begründet) a priority and necessity"' (reprinted in 3.390, cf. Appendix in Chapter 6).

19 MS 492: 1, 1903, *Logical Tracts. No.2. On EGs, Euler's Diagrams, and Logical Algebra*.

20 MS 633: 1:8, 6 September 1909, *Studies in Meaning*.

21 MS 500: 13–14, 8 December 1911.

22 3.423, 1892, *The Critic of Arguments*. Kempe's work appeared as 'A memoir on the theory of mathematical form', *Philosophical Transactions of the Royal Society* 177, 1886, 1–70.

23 See Chapter 12. Contrary to Stalnaker (2002), the common ground (or the "common-ground status") was not a term that H. Paul Grice invented by reading Arthur Prior, but a term that Prior was attracted to by reading Peirce. Grice's own affection for Peirce is unmistakable, although he does not cite him. Grice (1989, p. 36) even speaks of "interpretants" in his William James lectures. The distinction Grice draws is that between a "straightforward interpretant", roughly corresponding to the most salient reading of an utterance, and a "non-straightforward interpretant", roughly corresponding to the subordinate but intended reading or interpretation of the same (or phonemically the same) utterance. I am grateful to Mats Bergman, who pointed out the relevant passage in Grice's work. The related Gricean notions "speaker's meaning" and "literal meaning" have their correlates in Peirce's theory of signs in the form of different interpretants (Chapter 13), to which Grice added new epithets. There are also further similarities, such as the overall role and the nature of the common ground in conversational situations.

24 MS 650: 9, 22 July 1910, *Diversions of Definitions (Essays Definitions)*.

25 See, in particular, MS 670, 12 June 1911, *Assurance through Reasoning*.

26 MS 669, 1911, *Assurance through Reasoning*.

27 Alternative terms were "girdling-edges" (MS 670: 16) and "seps", from the Latin *saepes*, ('a fence').

28 The explanation why this crossing sign is not graphically expressed, according to Peirce, is because "a heavy line traversed by a line signifies 'not', and therefore the existential identity of an object represented by a dot outside and of an object represented by a dot inside the boundary cannot be asserted in the graphical form, just as it cannot be asserted in fact. But non-identity may be asserted, or it may be left uncertain" (MS 483).

29 MS 504: 1, *Peripatetic Talks, No. 6*.

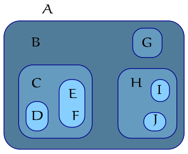
30 The difference between transitive and intransitive verbs is not being made here: "Every man loves a woman" simply means "Every man loves and is loved by a woman". To iconise other relations such as transitive and ditransitive ones would need a stupulation concerning the ordered rotation of hooks on the periphery of the spots.

31 The term 'priority scope' is from Hintikka (1997), and it means the order in which quantifiers enter the interpretation. A different aspect of scope is 'binding', which refers to the extent to which a quantifier maintains the values of the variables across different tokens (Chapters 5 and 7).

32 Roberts (1973a) may have been the first to spot this term in Peirce's work, cf. Chapter 6.

33 See e.g. Sandu & Hintikka (2001).

34 "If there be a collection [. . .] of cuts, of which one is placed in the sheet of assertion, and another encloses no cut at all, while every other cut of the series has the area of another cut of this collection for its place, and has its area for the place of still a third cut of this collection, then I call that collection a *nest*, and the areas of its different cuts *its successive areas*, and I number them ordinally from the sheet of assertion as *origin*, or zero, with an increase of unity for each passage across a cut of the nest inwards that one can imagine some insect to make if it never passes out of an area that it has once entered" (MS 650). For example, in the following alpha graph there are five nests as follows:



1. One of 5 areas, or 4 cuts: A-B-C-E-F

Three of 4 areas or 3 cuts each:

2. A-B-C-D
3. A-B-H-I
4. A-B-H-J
5. One of 3 areas, or 2 cuts: A-B-G.

35 MS 464: 6, 1903, *Lowell Lectures of 1903. Part 1 of 3rd draught of 3rd Lecture*.

36 MS 462: 6, 1903, *Lowell Lectures of 1903. 2nd draught of 3rd Lecture*.

37 MS 508: B.6, *Existential Graphs. Rules of Transformation. Pure Mathematical Definition of Existential Graphs, regardless of their Interpretation*.

38 See MS 670: 19, cf. MS 669, 1911, the antedating draft with the same title, *Assurance through Reasoning*. Roberts (1973a) provides a comprehensive study of both gamma and tinctured graphs. Roberts asserts that, contrary to Peirce's own later dismissal, his heraldic tinctures were not non-sensical.

39 MS 478, 1903, *Syllabus of a course of Lectures at the Lowell Institute beginning 1903, Nov 23. On Some Topics of Logic*.

40 MS 693: 282, *Reason's Conscience: A Practical Treatise on the Theory of Discovery; Wherein logic is conceived as Semeiotic, Note-book IV*.

41 It is apposite that one of the first publicly shown live pictures was a boxing match between 'Gentleman Jim' Corbett and Bob Fitzsimmons, the 'Fisticuffographic Process' in New York in 1897.

42 I will hold Myself to be female and Nature male, and use the gender accordingly.

43 Hintikka (1987b) argues for the importance of the strategic meaning of natural language, and its place within semantic theories of language, not in any separate natural-language pragmatics. Pietarinen & Sandu (2004) unearths further phenomena.

44 Peirce was not entirely consistent in describing the model-building process, however. In MS 280: 28 he writes, "Two parties are concerned in the scribing of graphs. The one is called the *Graphist*, the other the *Interpreter*. On one way of conceiving the matter, the latter does all the scribing; the former authorizes him to scribe a given graph or graphs, and furthermore the interpreter is permitted to make transformations according to a general code of transformations". This is probably an oversight and "the latter" and "the former" have switched sides by mistake. Slightly later (and elsewhere), Peirce refers to the Graphist as the one who does all the scribing: "In our make believe, two parties are feigned to be concerned in all scribing of graphs; the one called the *Graphist*, the other the *interpreter*. Although the sheet that is actually employed may be quite small, we make believe that the so-called sheet of assertion is only a particular region or area of an immense surface, namely that it is the field of 'distinct vision' of the interpreter. It is only the Graphist who has the power to scribe a graph, and the graphs that he scribes are true, because the truth of the true consists in his being satisfied with it. The interpreter, for his part, has the power, with more or less effort, to move the graph-instances over the sheet, out of his field of distinct vision or into it if they are not quite out of his sight" (MS 280: 29–30).

45 See Chapter 10. Further advancement of the phrase 'putting questions to Nature' has been undertaken in the works of Hintikka et al. (2003) in terms of the interrogative approach to inquiry. The importance of such an approach was indeed recognised by Peirce. In terms of the history of ideas, he was particularly inspired by Francis Bacon's work in which the 'interrogating Nature' idea saw the light of day.

46 Zalamea (2003) has taken groundbreaking steps towards the important direction of continuity in EGs.

47 One of the early studies noting the importance of the concept of continuity to various problems that Peirce discusses is Murphey (1961). See also the introduction to RLT by Kenneth Laine Ketner and Hilary Putnam.

48 See SS: 72, 14 Dec 1908, *Letter to Welby*, where Peirce speaks about "continuous predicates", cf. 4.438, c.1903, *On Existential Graphs, Euler's Diagrams, and Logical Algebra*; Pietarinen (2005c).

49 On the possibility and implications of non-strictly competitive semantic games, see Pietarinen (2000).

50 MS 492: 17 a.p., marked with small x in the margin.

51 One perspective on the kind of logic that has varying domains is provided by Rantala's urn models (Rantala, 1975). In such a first-order logic, one may pick elements from the domain without replacement. Another example of the notion of individuals that may subsist in some worlds but disappear in others is given by quantified modal (and especially epistemic) logics with identification semantics.

52 MS 492: 26 a.p., marked with small x in the margin, cf. Chapter 6.

53 Uninstantiated dots attached to hooks may be regarded as free variables of the extension of the system of EGs whose graph-instances would correspond to formulas.

54 "A symbol for a single individual, which individual is more than once referred to, but is not identified as the object of a proper name, shall be termed a *Selective*. The capital letters may be used as selectives, and may be made to abut upon the hooks of spots. Any ligature may be replaced by replicas of one selective placed at every hook and also in the outermost area that it enters. In the interpretation, it is necessary to refer to the outermost replica of each selective first, and generally to proceed in the interpretation from the outside to the inside of all cuts" (4.408, 1903).

55 We may even see the phrase "being satisfied with" used here through Tarski's later and less concrete idea of satisfaction. Löwenheim used similar term of being "satisfied" ["erfüllt"] in his early model-theoretic explorations well before Tarski (Badesa, 2004, p. 139).

Chapter 5

MOVING PICTURES OF THOUGHT II

NATURAL LANGUAGES ARE NEEDLESSLY serial, rife with ambiguity, complicated by alternations that are relevant only to discourse, and cluttered with devices that make no contribution to reasoning.
— S. Pinker & P. Bloom, *Natural language and natural selection* (1990)

THERE EXISTS HERE DIFFERENT logical structures from the ones we are ordinarily used to in logics and mathematics. Thus logics and mathematics in the central nervous system, when viewed as languages, must structurally be essentially different from those languages to which our common experience refers.
— John von Neumann, *The Computer and the Brain* (1958)

1. Information flow in existential graphs

One general and, in reality, quite remarkable implication of the game-theoretic perspective on EGs is that we are able to record the total derivational histories, that is, the maximal plays of the game, and refer back to any of the non-terminal subhistories along such plays at will. Derivational histories are helpful in understanding natural-language anaphora, for instance.[1] Their worth is also recognised in computational logic as history-preserving bisimulations. The collection of derivational histories, the totality of which presents the frame for the whole game, adds a vital informational dimension to the endoporeutic interpretation of EGs.[2] For example, we may ask the following questions: (i) Do the Graphist (the Utterer) and the Grapheus (the Interpreter) always know what the past choices have been? (ii) Is the information that the Graphist and the Grapheus possess persistent throughout the game?

If the answer to either one of these is negative, we are dealing with games that no longer are of perfect information. In the former case, we come across games of imperfect information, and in the latter, we come across games of imperfect recall. Both classes of games have been extensively studied in game theory. A possibility thus exists of extending diagrammatic logic along an entirely new dimension. The diagrammatisation of propositions in EGs is an alternative method for capturing iconically what one's linear, symbolic assertions expressions are intended to express, and how they are to be interpreted. In a similar vein, extensive games assemble formulas into a tree structure, showing the information transmission from one logically active component to another. However, as soon as diagrams are associated with endoporeutic interpretation, a striking generalisation follows, since the flow of information no longer needs to be uninterrupted.

I argue in Chapter 7 that there are propositions correlated with games that have imperfect flow of semantic information. Adopting Peirce's perspective that communication is mediation of knowledge about signs, it is only natural that some noise exists in the media. The family of logics correlated with games of imperfect information are collectively known as 'independence-friendly' (IF) logics.[3] In a nutshell, they are extensions of traditional formulas, no longer written as natural language is written, namely in linear order, but they spread along two dimensions.[4] In natural-language semantics, a subclass of such formulas are known as branching quantifiers. However, even if there were syntactic techniques of writing all IF formulas in way that mimics the linear, one-dimensional line of language, what is not being linearised is the transmission of semantic information. Accordingly, the games associated with sentences of IF logic have imperfect information.

In IF first-order logics, the quantifiers and connectives use the syntactic device of slashing. Its meaning is to delineate the logically active components that are to be independent from the components onto which such slashes are attached (details are provided as we move on). This is not the only way of extending logics with perfect information into those of imperfect information, but it is nowadays the most widespread. For instance, if the choice for the universally quantified variable x in the first-order sentence $\forall x \exists y\, Sxy$ is not visible at the location in which choices for the existentially quantified variable y are made, the sentence is rewritten by using the slash: $\forall x (\exists y / x)\, Sxy$. Semantically, the player who is making the latter move is not informed about what the previous move of the game was (Chapter 7).

As soon as this much is admitted, however, the iconicity of logic suggests that we go all the way. So far, independence has been restricted to universal-existential types in the literature. In general, formulas themselves may be conceived of as graphs with any dependence and independence between its constituents. More liberal scope relations may be seen as one of the motivations behind diagrammatic logics such as EGs in the first place, although Peirce did not come to conceive of their extensions to independent logical constants. Anyhow, the sentence φ of traditional first-order logic L may be rewritten as a tuple $\langle DG, \varphi \rangle$, where DG is a directed graph and φ is a sentence that carries no presupposition about the ordering of its logical constants. The relation between two nodes in DG means that the information concerning the value of the variable instantiated to it — the starting node of a relation — transmits to the receiving node in that relation. In other words, the latter variable depends on the former.

The extreme cases occur when (i) $\mathbb{DG}$ is closed under equivalence relation, in which case all variables and connectives in φ depend on all the others, and when (ii) $\mathbb{DG}$ is a disjoint graph, in which case no variable and no connective in φ depends on anything else, not even on itself.[5]

The generalisation gives rise to systems that resemble Bayesian belief nets and the allied knowledge-representation schemes of semantic nets, both of which aim at depicting unconstrained transmission of semantic values around the graphs in order to formally represent the inheritance of concepts.

The decisive question is: is the theory of EGs, being diagrammatic, able to reflect these suggestions? For instance, what is the beta graph for the sentence $\forall x(\exists y/x)\,Sxy$?

It turns out that one should not try to build any special notational gimmick into EGs in order to capture independence in the sense conveyed in the example, because that would not sanction the iconicity of diagrams. Instead, since neither Peirce's endoporeutic method, nor its modern cousin of GTS, need any inbuilt assumption of perfect information, as soon as the endoporeutic method is cast into the terminology of the mathematical theory of games, it may be extended to accommodate imperfect information. Imperfect information refers to the imperfect communication between the Graphist and the Grapheus, who may be viewed as the (real or imaginary) players playing the roles of Myself and Nature, according to the given conventions of the semantic game.

In principle, then, there need be no changes in the way the graphs are scribed. But it would contribute to the perspicuity of diagrams to have a sign that denotes which parts of the graph are drawn away from the nests of those within which they occur. Such detachment refers to LIs, which should be interpretationally independent of other LIs the extremities of which reside further out in the context. We could attempt to achieve this by scribing special marks at the extremities to denote the dependency relations between different LIs.

LIs play multiple roles in beta graphs. Not just simply counterparts to existential quantification, they also denote identities. Different LIs (or ligatures) are distinguished from each other by their occurrences in different parts of the graph, and especially by the relative depth of their outermost extremities, in other words by the odd or even number of cuts within which they occur. The problem here is that, since LIs do not differ from each other in terms of the variables coming together with them, we do not have an analogous way of referring back to them from some deeper area in the graph, unlike in the case of first-order formulas. However, this is a minor problem, because what we are interested in is the interpretation of graphs, and we may assume that any independence there may be is manifest on the semantic level, and need not be instantiated on the level of syntax. Any imperfectness there may be in the communication between the two agents may be denoted within the extensive-form representation of the associated semantic game.

However, it might be convenient to signify special interpretational provisions by the means of graphs, just to differentiate them from graphs that do not have any such provisions. I propose two ways of achieving this. The first is to do it 'Hammer-style' (Hammer, 1998), and assume that the extremities of LIs give rise to variables, and write the variable at those extremities, providing different names of variables for each occurrence of an extremity. Whenever an outermost line is interpreted, the interpretation will refer to the variable adjoined with the extremity. Variables may then be slashed, so that the right-hand side of the slash lists those variables associated with the LIs the depth of which is less than (nested in a lower number of cuts than) that of the LIs associated by the variables on the left-hand side. This necessitated an extension of Hammer's suggestion such that the traces of the LIs in ligatures need to be retained in order to keep track of the depths of the lines in graphs.

As noted, Hammer uses variables to obtain a bookkeeping device that records the number of cuts. My intention is to retain Peirce's endoporeutic interpretation. It turns out that my approach will also retain an exhaustive record of all relative depths of LIs in terms of the extensive-form representation. One needs not worry about the number of nests within which variables occur.

Using variables means that the LIs are not retracted after interpretation. Moreover, variables are not nearly as iconic as one could hope for. To suggest an alternative, assume that such LIs exist that, while crossing cuts, they may also jump over some of them.[6] Cuts that are thus ignored correlate with the LIs within whose priority scope the jumping lines do not belong.

In order to make this preliminary idea perspicuous, I will extend graphs from two to three dimensions. Graphs scribed on a sheet of assertion are planes or manifolds arranged on a stack in a three-dimensional space. The ensuing graphs are thoroughly iconic. Multiple planes exist whenever graphs have independencies between the logical constants they contain. The sheet of assertion is extended to the space of assertion, in the first instance the three-dimensional space that represents the universe of assertions as constructed by the Grapheus. As before, an empty space of assertion represents a true proposition. A plane that has a cut scribed on it represents a two-dimensional projection of a cylindrical incision through the space of assertion. A false assertion, a pseudo-graph, is thus scribed on all of the sheets and on the space itself.

Thinking of the assertion sheets as layers in the space according to one of its coordinates, then the dimension of this coordinate correlates with the binding scope of showing what the areas are into which the effect of LIs reaches. The priority scope is the separation of the sheets from one another, and their spatial arrangement in the space of assertion. Let us proceed with the development of the aforementioned ideas.

2. Extending existential graphs

IF logic Ordinary classical first-order logic forces quantifiers in a linear order to be read and interpreted from left-to-right, so that any information concerning the values of quantifiers will propagate down to subformulas. IF logic rejects this (Chapter 7; Hintikka 1996a; Hintikka & Sandu 1997; Pietarinen 2003a; Sandu & Pietarinen 2001, 2003). For instance, instead of writing that $\forall x \exists y\, Sxy$ ('For all x, there exists a y such that Sxy'), we may write $\forall x (\exists y / x)\, Sxy$ ('For all x, there exists a y independently of x such that Sxy'). The phrase 'independently of' is given precise meaning in terms of GTS, which, unlike for sentences of classical logic having games of perfect information, now accommodates imperfect information.

We get IF first-order logic by adding to classical formation rules the following. Let $Qx\psi, Q \in \{\forall, \exists\}$ and $\phi \circ \psi, \circ \in \{\bigwedge, \bigvee\}$ be first-order sentences in the scope of $Q_1 x_1 \ldots Q_n x_n$, in which $A = \{x_1 \ldots x_n\}$. If $B \subseteq A$, then $(Qx/B)\,\psi$ and $\phi\,(\circ/B)\,\psi$ are well-formed sentences (wfss) of IF first-order logic. For instance, $\exists x\,(S_1 x\,(\bigvee/x)\,S_2 x)$ and $\sim\forall x_1 \ldots \forall x_n (\exists y / x_1 \ldots x_n)\, Sx_1 \ldots x_n y$ are wfss of IF logic.

A similar idea is known as Henkin quantification (Henkin, 1961). Henkin quantifiers represent independence by organising quantifiers into two-dimensional arrays:

$$\begin{matrix} \forall x_1 \ldots \forall x_n & \exists y \\ \forall z_1 \ldots \forall z_m & \exists u \end{matrix}\; Sx_1 \ldots x_n z_1 \ldots z_m y u\,. \tag{5.1}$$

IF logics generalise the independence to arbitrary patterns.

Let us consider the IF first-order sentence of the form

$$\forall x_1 \ldots \forall x_n \forall z_1 \ldots \forall z_m (\exists y / z_1 \ldots z_m)(\exists u / x_1 \ldots x_n)\, Sx_1 \ldots x_n z_1 \ldots z_m y u\,. \tag{5.2}$$

In models with pairing functions, it follows from the Krynicki normal form theorem (Krynicki, 1993) for Henkin quantifiers that all IF sentences in prenex and negation normal form can be brought into the form (5.2), which in turn is equivalent to the Henkin quantifier (5.1). I will use this result in the next subsection.

Likewise with classical logic, IF sentences may be brought to Skolem normal forms. The Skolem normal forms of (5.2) and (5.1) are both as follows:

$$\exists f_1 \exists f_2 \forall x_1 \ldots x_n \forall z_1 \ldots z_m\, Sx_1 \ldots x_n z_1 \ldots z_m f_1(x_1 \ldots x_n) f_2(z_1 \ldots z_m)\,. \tag{5.3}$$

The difference of these existential second-order sentences to the Skolem normal forms of classical logic is that the Skolem functions f_1, f_2 in (5.3) have fewer arguments than the classical, slash-free counterparts of (5.2) would (i.e., the arrays $z_1 \ldots z_m$ and $x_1 \ldots x_n$ are omitted, respectively, from f_1 and f_2 in (5.3)). This is due to information hiding. In imperfect-information games, not all semantic information passes from players moving earlier to players moving later.

Taking arrays of Skolem functions as winning strategies for one of the players of the semantic game, the existence of winning strategies for the verifying player (resp. the falsifying player) agrees with the notion of the truth (falsity) of an IF sentence in a given model.

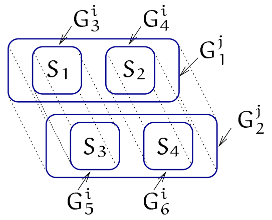
IF extensions of alpha and beta Let us describe an IF extension of EGs for the alpha part first. Consider the sentence (5.4), which is an IF version of $(S_1 \vee S_2) \wedge (S_3 \vee S_4)$ with disjunctions independent of conjunction:

$$(S_1\ (\vee/\wedge)\ S_2) \wedge (S_3\ (\vee/\wedge)\ S_4)\,. \tag{5.4}$$

This is interpreted by parallel moves by the Graphist choosing a disjunct and the Grapheus choosing a conjunct. Since the game-theoretic interpretation is endoporeutic, in the alpha graph in non-IF version of (5.4), a subgraph from one of the two disjoint molecular subgraphs is chosen first, before proceeding to any of the inner, contextually constrained graphs.

I noted in the previous section that one might wish to use some quirk in the syntax and stipulate that the choices between juxtaposed partial graphs are independent of one another. However, this would not preserve the premise of iconicity, according to which symbols are subordinate to visual representation. My solution is that, to capture independence, we move from two to three-dimensional spaces into which different SAs are layered.

Accordingly, we diagrammatise (5.4) by the following IF EG:



(5.5)

Since the interpretation of these graphs is endoporeutic, it does not matter which of the players, the Graphist or the Grapheus, is to choose first, as they may move concurrently. The choice by the Graphist is (after two role-changes prompted by the game rules for cuts) between two indices $i = l, r$ in subgraphs $\{G_3^i, G_4^i\}, \{G_5^i, G_6^i\}$, and the choice by the Grapheus is likewise between the indices $j = l, r$ annotated to graphs G_1^j and G_2^j on the two spatially separated SAs. In other words, the Graphist and the Grapheus are not informed of each others choices. The associated semantic game thus has imperfect information.

The upshot is that the independent moves are not between subgraphs as such but between the indices by which they are annotated. This is simply to prevent players using illicit information in their strategic decisions that they would get by observing what the subgraphs are at any history in which they are to move.

We may stipulate that (5.5) is a new four-place connective $W(\varphi, \psi, \theta, \xi)$. It gives rise to a partial logic and a truth-functionally complete set of connectives when added to the classical set of connectives $\{\wedge, \vee, \sim\}$ (Sandu & Pietarinen, 2001, 2003).

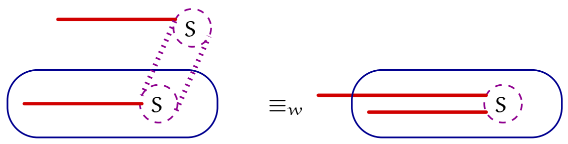
Moving then to the beta part. As in alpha, in which we iconise independence between connectives by a spatial separation along an added dimension, to capture independence between quantifiers (lines of identities) we likewise resort to three dimensions. Thanks to the Krynicki normal form theorem, we may confine ourselves to IF beta graphs that have at most two manifolds (SAs) layered in a space. (In what follows we toggle between IF sentences and sentences with Henkin quantifiers at will. A caveat is that a difference obtains between these two symbolisations that comes to the fore in the next section, points (iii) and (iv), as soon as more complex diagrammatic representations are at issue.)

One peculiarity of such IF graphs as I will call them is that spatially separated sheets on which cuts and LIs are scribed may be connected by spots, because LIs on separate sheets need to be attached to the hooks of spots on both sheets. Such spots are hence scribed as cylinders with a boundary at their peripheries. A finite number of hooks are imagined on the lids of these cylinders, upon which LIs may be attached. Such cylinders make SAs contextually dependent on one other.

Let the relation $\equiv_s$ denote strong equivalence, in other words graphs being true in the same models M and being false in the same models M. Let $\equiv_w$ denote weak equivalence, in other words sentences or graphs being true in the same models M. To start with a simple example, consider two sentences equivalent in M in the following sense:

$$M \models \forall x(\exists y/x)\, Sxy \equiv_s M \models \exists y(\forall x/y)\, Sxy \equiv_w M \models \exists y \forall x\, Sxy\,. \quad (5.6)$$

The diagrammatisation here goes by way of weakly equivalent IF EGs:



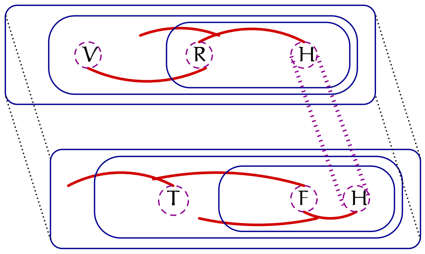
(5.7)

The graph on the left represents the two strongly equivalent sentences of (5.6). Dashed circles depict the lids, the peripheries of the spots S upon which lines are attached. The heavy dotted line on the left graph represents the periphery of the cylinder S. As the order of the quantifiers does not matter for the truth-value, the order in which the two LIs on the left graph in (5.7) are selected is immaterial.[7]

In a similar vein, and to consider a slightly more involved example, let us take the famous sentence used in Hintikka (1973b) to demonstrate the existence of branching (Henkin) quantification in natural language: “Some relative of each villager and some friend of each townsman hate each other”.[8] This sentence may be symbolised (with self-explanatory abbreviations) by the following IF first-order sentence:

$$\forall x \exists y (\forall z / xy)(\exists u / xy)\,[\sim (Vx \wedge Tz) \vee (Rxy \wedge Fzu \wedge Hyu)]\,. \tag{5.8}$$

This sentence is diagrammatically captured by the IF EG with two spatially separated SAs:



(5.9)

The two-place cylinder H is common to both SAs.

Some remarks Let me delineate six key points considering this extension of EGs.

(i) Having exactly three dimensions is not the crucial feature of IF EGs. One could as well scribe unextended beta on three-dimensional Space of Assertion, wherefore the negation would correspond to closed spheres, and quantification, identity, predication and subsumption to Hyperplanes of Identity dissecting the spheres. Its IF extension would then occur in four dimensions, in other words the Spaces of Assertion would be layered in a four-dimensional space in which spots are four-dimensional objects connecting these layers. This process obviously generalises to n dimensions, having its IF extension in $n + 1$ dimensions.[9]

(ii) The spatial separation of LIs shows what the two notions of scope that Hintikka (1997) has distinguished amounts to in the iconic sense. Binding scope refers to the reach of quantification in a formula in the sense of binding different tokens of variables. It corresponds to the prolongation of LIs from end-to-end. Topologically, it expresses connectivity of different areas of cuts. In contrast to binding, priority scope refers to the logical ordering between different quantifiers. It corresponds to the order given by the nesting of cuts. The least number of cuts defines the logically prioritised extremity of an LI.

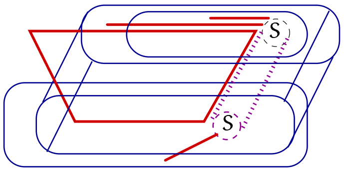
In classical logic, and a fortiori in unextended beta graphs, these two notions are entangled in themselves. In IF logic and in IF EGs, they become separated.

(iii) IF diagrams promote duality in the sense that not only evenly-enclosed LIs need to be functionally dependent on oddly-enclosed lines. Since the polarity of the areas of cuts is the only distinguishing feature between the two kinds of lines, also the oddly-enclosed LIs may be dependent or independent on evenly-enclosed lines.

(iv) One may ask whether we have reached a comprehensive diagrammatic method of representing independence. For instance, how do we represent (5.10), in which all quantifiers not explicitly slashed are assumed to be dependent from those occurring further out?

$$\forall x \exists y \forall z (\exists u/x)\, Sxyzu\,. \tag{5.10}$$

The following is an IF EG of (5.10):

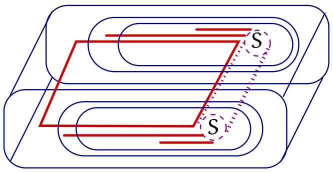


(5.11)

The EG here consists of one plane of identity (PI) that crosses the vertical boundary of a three-dimensional cut. This is permitted as long as the outermost corners of the PI lie either within the positive or within the negative area of a cut. It is also observed here that functional dependency, symbolically captured by priority scope, refers to the iconic property of the nesting of three-dimensional cut enclosures just as of two-dimensional ones.

(v) It is thus commonplace that, if some LI on which lines drawn on spatially separated SAs both depend, the line in question extends to a plane that dissects the cuts. Accordingly, let us call the connected system of planes of identities curtains. A plane that dissects 3D cuts is thus actually a curtain composed of two planes. For instance, the Henkin quantifier sentence (5.12) is represented by the IF EG (5.13) with one curtain (the bold rectangle) spanning the space that has one side connected to the periphery of the spot S and one loose side not connected to any spot.

$$\sim\forall u \begin{array}{cc} \forall x & \exists y \\ \forall z & \exists w \end{array} Suxyzw\,. \tag{5.12}$$



(5.13)

(vi) The operation that the cut defines in IF EGs is no longer the ordinary Boolean negation. If we distinguish from each other the boundary of a cut and its area, as Peirce in effect did (see sect. 4), what we get is the distinction that is analogous to that between closed and open sets, in other words between enclosures of the cuts and areas of the cuts.[10] A closed set contains its boundary, and hence the complement of the closed set is open. Likewise, an open set consists of the interior without the boundary. Its complement is a closed set.

If we think of the structure of graphs in this topological setting, then the cut appears to define complementation, in other words an operation that is the weak, contradictory one. Such a cut would correspond to classical negation. This is fine, as far as it goes. However, what if a graph enclosed by the cut is without a truth value (which may happen in case its spots would be partially interpreted, see Sandu & Pietarinen 2001)? What would it mean to sever an undefined atomic graph from the assertion sheet? It ought to mean nothing. Cutting an undefined graph does not affect the truth-value of its complement. It would do so only if the negation is weak, for which we do not have another iconic notation separate from strong negation. The cut denotes strong negation, which does not tamper with the values of undefined graphs but only changes the role of the player entering the enclosure.

Peirce erroneously thought that this transfer of responsibility between the Graphist and the Grapheus delivers the same concept of negation as the set-theoretic meaning of the cut. It does not. There are two functions that the cut may play. It may obstruct the movement of values and thus the continuity of the ligature that abuts it, so as to distinguish between the interior and the exterior of a cut. It is the interior that is severed from the SA. This means that what is scribed on that area is not asserted. The operation produces the denial of an assertion. It corresponds to classical, weak negation. The other function is that, when the player who is interpreting the graph enters the enclosure, his or her strategic position changes to that of his or her opponent's. This delivers game-theoretic, strong negation. These two different interpretations of a cut give rise to two different notions of negating a statement.[11]

The cuts (as closed simple curves) in IF beta graphs do not denote complementation, and thus do not correspond to classical negation, because complementation applies only in local SAs on which the cuts are scribed. For this reason, the interpretation of cuts should be as game-theoretic role reversals, in other words corresponding to strong negations.

Summary To take a stock, the essential features of IF EGs are such that, from the ordinary first-order sentence

$$\forall x \exists y \forall z \exists u\, Sxyzu, \tag{5.14}$$

we proceed to de-linearise the information flow and rewrite it in the form

$$\forall x \exists y (\forall z/xy)(\exists u/xy)\, Sxyzu\,. \tag{5.15}$$

It cannot be diagrammatically scribed merely by the received beta system. No attempt in two dimensions will do, because the two innermost LIs, corresponding to the variables z and u of (5.15), while pertaining to the binding scopes of the two outer LIs corresponding to x and y in terms of occurring in the same predicate, are nevertheless severed from their priority scopes.

Therefore, the SA needs to be separated into two planars, each with two cuts, layered within the space of assertion so that they match one another in different sheets via the cylinder S. The idea generalises to n dimensions with hyperplanes as cuts in higher dimensions.[12]

Similar extensions are also conceivable for the gamma part. With respect to that compartment of gamma using the broken cuts to represent that something is possibly not the case, the broken cut would need to sever through multiple sheets and hence produce a cylindrical incision so as to expose new possible worlds. Cylinders are also broken, not continuous objects. The interpretation is that the inside region of the cylinder represents the possibility, referring to universes of possibilities in three dimensions. The enclosure of the broken cut is, so to speak, turned around on a space and its recto, representing the universe of possibilities (and not, unlike the verso does, the universe of actually existent things) is exposed to view. Moreover, both lids of the broken cylinder are turned around, so that the universes of possibilities are applied to assertions anywhere within the interior of the cylinder. Various tinctures may be considered as well.

3. The game interpretation fine-tuned

Given the endoporeutic process of interpretation, the IF EG depicted, say, in (5.9) appears to indicate a quarrel between two agents about which of the two lines of identities is resolved first. If this graph is like a "sponge" that is to "absorb water" (MS 650: 18, 24 July 1910), which one of the two lines of identities should be selected first?

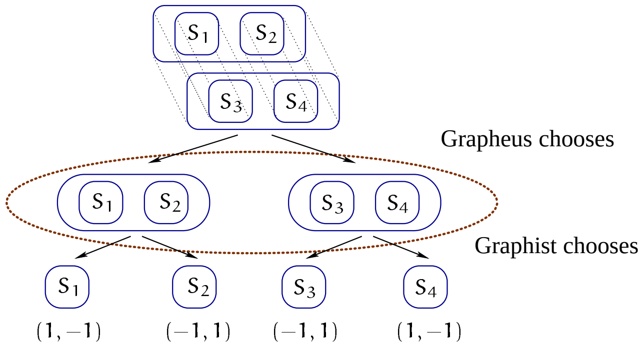
Luckily there is an easy answer. Since concurrent LIs only appear in cases in which they are either all evenly enclosed or all oddly enclosed, they are correlated with the same player. Any discrepancy takes place within the single mind of one of the players, never between them. The tension does not materialise with respect to the truth or falsity of the graphs. However, the concurrent LIs may well have further repercussions concerning the actual implementation of the endoporeutic method, to which I will return anon.

Adding a third dimension to EGs does not confine possible dependencies between LIs to partially-ordered cases. (In the case of linear dependencies, graph reduce to those that can be scribed on two-dimensional sheets of assertion.) For instance, the dependence relations may also be non-transitive, as in (5.11). Any network of dependencies should be equally considered.

The sheets of assertions layered within the space of assertion may refer to their own particular universes of discourse. It would be convenient to take these universes to be partially overlapping, though, to share common vocabulary. This does not need to impinge upon possible-worlds semantics, as no accessibility relation is assumed between the sheets.

The extension is thus not towards any gamma conception of graphs either, in which sheets would, in contrast, represent altogether different universes than one's discourse is predicated to deal with. The sentences so diagrammatised are purely extensional. Indeed, it was customary for Peirce to assume of beta graphs that the universe of discourse is not many-sorted, in other words that it "embraces all the objects of a single category" (MS 292: 26).

By virtue of the game interpretation presented in the previous chapter, the informational loss means partitioning of the game histories of the extensive game into equivalence classes. This provides a simple method of extending imperfect information to connective choices, too, that is, to juxtapositions of different graphs. The sole difference is that these games are of imperfect information in contrast to traditional games of perfect information. The Graphist and the Grapheus may not know the moves made earlier in the game. For example, given the IF alpha graph of (5.5), the following two-stage semantic game in played on it, in which the dotted oval captures that the Grapheus choosing one of the oddly-enclosed graphs is not informed about the choice the Graphist has made for the two evenly-enclosed subgraphs:



The payoffs here reflect the interpretation in which S_1 and S_4 are true and S_2 and S_3 are false. Since the Graphist has to choose in ignorance of the Grapheus' initial choice, which means that she does not know whether her choice takes place on the left history of the information set (the dotted oval) or on the right, there is no winning strategy for her. Nor is there a winning strategy for the Grapheus, and so the graph in question is neither true nor false in the given interpretation.

Consequently, the diagrammatisation of logic by EGs reveals yet another facet of the commonality of the endoporeutic method. Depending on the purpose, we may impose various restrictions on it to produce new classes of games, or relax some of its characteristic attributes.

Peirce did not consider the two questions posed at the beginning of this section seriously from the point of view that would have made him thought of any answer other than a negative one to be viable. For instance, as far as the first question is concerned, namely whether the Graphist and the Grapheus always know what the past choices have been, he did seem to assume his dialogues to be of perfect information. This was noted, for the first time, in Hilpinen (1982). Why he thought so is not so clear, but there is ample evidence of this conclusion in his corpus: "Whichever of the two makes his choice of the object he is to choose, after the other has made his choice, is supposed to know what that choice was. This is an advantage to the defence or attack, as the case may be" (MS 9: 4). In a similar vein:

> The utterer is to determine the meaning of one sign, the interpreter [p. 3] of the other. Whichever of the two has the last choice is supposed to know what the previous determinations were. Consequently, the utterer, who is essentially a defender of his own utterance has an advantage in choosing last; while the interpreter, as not being necessarily a defender of that which he interprets, but rather a critic, and quasi-opponent, is as such, at a relative disadvantage.[13]

A similar idea is implicated in the following passage:

> If the first person is to select one object and a second person another, it will always be an advantage to the former to wait and see what selection the latter makes. Consequently, 3 always follows from 2. For 2, or
>
> (2) Some S has all π's,
>
> asserts that an S can be selected so that after it is selected, and the selection made known, no matter what π be selected, that S will have that π; while 3, or
>
> (3) Every π is had by some S,
>
> asserts that no matter what π be selected, after it is selected, and the selection made known, an S can be so chosen that that S will have that π.[14]

Furthermore, the same conclusion is also confirmed in the manuscript passage from 1902: "When the proponent or opponent has to designate an individual object as a member of the set, he is entitled to know what are the objects so far selected, so that he may shape his choice accordingly".[15] The term "set" refers here to all the values chosen by the speaker (the Graphist) and the auditor (the Grapheus) interpreting a given beta graph. It is in my terms the sequence of actions labelled the nodes of a complete history of the extensive game. To adduce still more textual evidence: "It is the set so made up that the predication is made", and "each individual of the set . . . forms the subject of the assertion" (MS 430: 62). This remark was explicitly made within the context of EGs, not the algebraic logic of relatives. Common in all these is the reference to the epistemic state of the participants, namely the assumption that one of the players, at any point in the interactive process of interpretation, knows the previous selections.

At least as candidly, Peirce also considered the answer to the second question ('Is the information that the Graphist and the Grapheus possess persistent throughout the game?') to be negative. In other words, he explicitly denied that there was information flow in a dialogical interpretation that was not continual for an individual actor:

> Individuals represented by less enclosed lines are to be selected before individuals selected by less [*sic*, ought to be 'more'] enclosed lines. In all cases in which this rule leaves the order of choice indeterminate it makes no difference which order is pursued. The reason this convention is sufficient is that if either proponent or opponent has, at any stage, several individuals to select successively, it is obvious that the order in which he names them will be indifferent, since he will decide upon them in his own mind simultaneously. (MS 430: 62).

This statement refers to the fact that, in the string of existential quantifiers with no intervening negations, values (the "named" individuals) for variables thus quantified may be chosen in any order. For instance, $\exists x \exists y\, Sxy$ and $\exists y \exists x\, Sxy$ are equivalent sentences of first-order logic. What is remarkable in this statement is the observation that the order of naming such variables is indifferent (for the truth-value of the graphs) because of the selections of individuals are made by the same player. If this is so, what is implicit in this statement is the assumption that the player has persistent storage of such values and once chosen, does not forget them.

The possibility of oscillation in the amount of information possessed by a single player arises in EGs scribed within the space of assertion. In game theory the correlated games are known as ones of imperfect recall, are customarily implemented under the assumption that there are multiple decision makers, or 'multiple personalities', within the individual player. I will consider this in Chapter 7.

The previous quotation is also remarkable in the sense that it implies the possibility of the endoporeutic method functioning so that individuals are chosen simultaneously. Concurrency of actions arises in imperfect information and imperfect recall, and both interpretations are valid in terms of choosing simultaneously. This deviates from the idea that Peirce had in that the concurrency in the latter type of simultaneous action concerns the different levels of graphs or lines of identities within different numbers of enclosures, whereas in the former, the depth in which the graphs or the lines of identities subject to simultaneous interpretation is the same.

The problem with an added perspective on and hence a change in the basic characteristics of games is that one falls short of semantic tools other than those provided by the endoporeutic method. The reason is that any move may or may not be taken to depend on another, previous move. If the interpretation is the converse, in another words ectoporeutic, which is formally equivalent to Tarski semantics, such context dependence is glossed over because at any point of interpretation, the inner graphs may be informationally independent of the outer graphs within whose context they reside. However, there is no way of knowing that until the outer instances of the graph are reached. Hence, the ensuing phenomenon of informational independence violates what is known as a version of compositionality of logical systems.

As noted earlier, according to this requirement, the meaning of the composite proposition, or the entire graph, is determined by the meaning of its parts and the way they are put together. The remarkable thing to be noted here is that there is no non-compositionality at the level of diagrammatic or logical syntax. It only arises whenever the intercommunication between the Graphist and the Grapheus is being obstructed in one way or another.

Unlike the received ectoporeutic method, its older brother thus provides a versatile context-dependent system. Furthermore, it may be adjoined to background knowledge by known AI and computational knowledge-representation techniques such as scripts and frames, default reasoning, and many others. On the whole, such information may, by and large, be encoded into the strategy functions of the game, or the initial moves by Nature. From this perspective, the contextual meaning remains semantic in its basic nature.

What also makes the endoporeutic method as a game appealing is that it provides concrete systems in the sense of being potentially playable by real persons or real players. This does not mean that these games are necessarily played by humans, analogously to the socially tedious senses of ‘the games people play’, although the possibility of applying actual human players to them should by no means be ruled out. This means that several crucial aspects should be taken into account in the design of diagrammatic logic, which would enable these games to be played by any non-omniscient theoretical entity or agent with limited or bounded resources. These aspects include the following:

- That the graphs are finite, which is the case as the assertions are finite in length. Thus the interaction will come to a halt in a finite number of steps, and thus also the plays of the associated games are all finite.
- That the universes of discourse may be infinite, which makes the breadth, not the length, of the game tree infinite.
- That the players are not totally acquainted with the universe of discourse on which they pick elements, and are allowed to name the objects when the need arises, namely if their identities are not known. In other words, the universes are not total systems sealed off at the outset.

All these requirements are readily found in Peirce’s logic. For instance, with regard to the first item, let us single out the following remark: “The seps [the cuts] of a nest are restricted to a finite multitude, so far as this rule is concerned. A graph with an endless nest of seps is essentially of doubtful meaning, except in special cases” (4.494, c.1903). It needs to be kept in mind that finiteness is restricted to the endoporeutic method, and not, for example, to semiosis or the scientific method, which Peirce took to be in principle never-ending processes.

As far as the second item is concerned, the universes of discourse may indeed be infinite: “The interpretation [of a quantified formula of a logical algebra] is restricted to cases in which the universe is a collection of individuals, even if not to cases in which it is a finite collection”.[16] Here, Peirce baptises the term “quantifier” as synonymous with the direction “for the order and manner of choosing the individuals of the set to which the second part of the proposition [i.e. the matrix of prenex normal form] is written" (MS 530: 37 a.p.). It is only the length of the histories associated with the nests of cuts that are finitely long. He did not suffer from similarly bad conscience like other up-and-coming mathematicians such as Brouwer in using infinitary constructions in mathematics (Chapter 6).

The status of the third item is perhaps the most controversial. Peirce frequently emphasises there to be suitable and mutually agreed common ground before any communication or genuine dialogue can take place. Specifically, this common ground should encompass that it should be agreed between the utterer and the interpreter that the universe is well known, mutually known to be known, and exist in the case the SA represents actuality (Chapters 1, 6 and 12). Does this, then, not make the universe a closed, determined, or at least a mutually and exhaustively known, totality?

The answer is that assuming the common ground and collateral acquaintance concerning the environment plus background beliefs does not yet mean that proper identification of the objects that are suggested as members of the set on the basis of which a predication is made has already taken place. While a sign can be communicated only if there is suitable familiarity with it (e.g. by acquaintance), plus an understanding that the other party is also appropriately familiar with it, the object in question does not yet need to be identified. Acts of naming objects, or perhaps ostensive definitions of them, are already examples of identification, but they are not examples of genuine moves in the semantic game or dialogue.

It is also worth recalling that, even if such identification fails, a sign can still be put forward, but in that case it is bound to remain indeterminate. Peirce was convinced that to make real progress in logic, vague, indeterminate and indefinite assertions need to be accommodated.

What is also significant in relation to the common ground is that Peirce assumes the universe about which the participants wish to communicate to be common knowledge, too. This is a non-trivial feature of the notion of the common ground that assumes the existence of mutual and infinite knowledge between agents. Furthermore, elsewhere Peirce hints at the fact that common knowledge of rationality is needed for those engaged in communication. This is, of course, an idea that came into being only much later in the theory of games and rational decisions, broadly falling within the contemporary programme of interactive epistemology that lies in the intersection of economics, game theory and logic.

A further feature connected with non-omniscient players is that if they are artificial, say computers, one needs to restrict the strategies to those that are, in some sense, constructive (for example, effective or learnable). As Peirce did not have the precise concept of a strategy at his disposal, this qualification is bound to be obsolete from his point of view. However, there is little doubt that he would not have sympathised with such constructive turns.[17]

4. Topology, graphs and games

Several further points need to be made concerning the 'movement' in EGs. In this section, I will turn to the question of a combination of physical and topological elements in explaining some of Peirce's innovations. Since his EGs are diagrammatic representations within some space (sheet of assertion), it is only natural to consider the question of what such spaces are like, and what properties are preserved under continuous deformations. In this sense, graphs were an early application of topology (or topical geometry or mathematical topics as Peirce preferred to call it) to logic.

However, such an application was never properly accomplished. Peirce's investigations into topology remained on a rather elementary level. According to the entry Topics that he contributed to the Century Dictionary (CD X:1360), topics (topology) is "the most general, fundamental, and naturally elementary branch of geometry, which neither considers lengths, areas, or volumes in their character of being measurable, nor distinguishes straight from curved or crooked lines, nor place from curved or bent surfaces, but studies only the manner in which the parts of places are continuously connected". In the draft of this entry (MS 1170: 539) he had also written that topology "studies only figures which are understood to be deformable in any way, so long as no parts are joined which were at first not continuously connected in the same manner. The characters which topics studies are continuity and betweenness, dimensionality, topical singularities or places within places which differ in their connections from neighboring places within the same places, and Listing numbers". These definitions are quite similar to the one that the almost forgotten mathematician Johann Benedikt Listing (1808–1882), a student of Gauss and a contemporary of Riemann, from whom Peirce learned the fundamental concepts of topology, gave in *Vorstudien zur Topologie*, published in 1847: "The doctrine of the modal features of objects, or of the laws of connection, of relative position and of succession of points, lines, surfaces, bodies and their parts, or aggregates in space, always without regard to matters of measure or quantity".

Peirce was certainly right to urge others to investigate these continuous connections and deformations in mathematical objects, but his means of doing so were somewhat limited (and so were Listing's). There were no equivalence classes (quotient spaces), and thus deformations would not always be reversible (that is, if a continuous mapping exists from a topological space X to its quotient, there is an inverse from an open subset of the quotient space to an open set in X), even though Peirce did attempt to define some anticipatory notion of a "shape-class". For example, a line with a free end, a point and a simple, closed surface are within the same shape-class. These give rise to the property of cyclosis, which was Peirce's term borrowed from Listing, "a number of independent rings" (MS 482: 1), better known as the object's genus.

The forays Peirce made into early topology have been singularly disregarded in subsequent history.[18] The recent three-volume book series on the history of topology (Aull & Lowen, 1998–2001) does not recognise Peirce at all. It was only when Solomon Lefschetz (1884–1972), often referred to as the founding father of topology, unaware of Peirce, once again translated the German word Topologie into Topology in 1931 that the theory caught on among mathematicians. For instance, the census theorem of Listing was only known to Peirce — no one referred to it after him, although everyone knows the related Euler–Poincaré formula. It is also possible that Peirce wanted to defend Listing so sternly that it drove him into an unjust downplay of Arthur Cayley (RLT: 246). At all events, the subsequent developments on topology took an entirely different turn from that predicated by Peirce, who fittingly said, "[Listing's] name must forever be illustrious as that of the father of geometrical topic" (RLT: 254).

Most of Peirce's investigations on topology were carried out roughly from 1895 to 1907, during which he also discovered Cantor's transfinite sets and engaged in intriguing correspondence with him.[19] He did not connect these investigations with his systems of EGs, which were already in full blossom by 1903, in any outspoken manner. According to Zeman (1964), "The existential graphs constitute the 'topology of logic"' — but Zeman did not investigate the graphs by topological means. There was a reason for this, which I will give shortly. In general, mutual points of contact are equal in number to those of the failures to observe the connections between EGs and topological issues.

Three main points of contact between Peirce's logic and topology may be delineated:

- In 1905, Peirce used Listing's terminology to demonstrate the completeness of alpha and beta graphs.[20]
- The function of these graphs may be explained, at least partly, in terms of topological concepts, much in accordance with Peirce's intentions.
- Synechism, or 'together-withness', the nature of continuity in both mathematical ("pseudo-continuum") and metaphysical senses ("true continuum"), influenced, and was influenced by, developments in diagrammatic logic.

As far as the second point is concerned, the way Peirce set up his graphs suggests the following topological interpretation.

Let us start by summarising some of Peirce's terminology. He notes, "From an *ordinary* point on a line a particle can move in two ways" (RLT: 250). A singular point is the topological singularity of a line, and a particle is that object which is movable at any one instance occupying a point. Likewise, a surface has topological singularities of a point and a line. A movement is performed by the material that fills these at any instance, which, in Peirce's terms, is a particle for a point and a filament for a line. The place occupied by a filament given any lapse of time is a surface, and a movable thing at any instant occupying a surface is a film. A solid occupies a space, and so on for more than three dimensions.[21]

These are the basic conceptual tools Peirce used, and they are applied to EGs as follows. As far as the beta part, with a theory isomorphic to the theory of first-order logic with identity is concerned, I only note that isomorphism is a weak condition that does not say anything about the fine details concerning the iconic and representational nature of diagrams.

To recap, beta graphs contain the following signs: the sheet of assertion (phemic sheet), LIs (ligatures), simple closed lines (cuts), juxtapositions, spots (rhemas) and hooks, and selectives. As noted above, the interpretation of LIs is inward and endoporeutic, and the interpreter and the utterer move in one direction by choosing values from the universe of discourse or by choosing one of the juxtaposed partial graphs. The reason why there is only one direction is that the line has an outermost extremity at which one has to start. An interpretation is a dynamic process in which the interpreter or the utterer places a particle, now coming with a selective (typically proper names for rhemas and indefinites for onomas), at the mouth of the outermost extremity of an LI. The extremity is a loose end of the LI not connected with a hook on the periphery of spots. An LI may terminate at a place occupied by a dot, also termed by Peirce a peg, which has been scribed on the periphery and attached to a hook on that periphery. A dot itself suffices to assert existence, but it is customarily lengthened to a short LI.[22] By starting from the place occupied by the dot and moving the particle placed in that position into a position in which another hook has a dot, one makes an assertion of identity between two particles.

Such a movement of a particle carrying the selective leaves a trace or a path that enables one to assert identities between its extremities. To give room for such movement is Peirce's insistence that these LI need to be heavy. This, in turn, suggests that they would, in fact, be better viewed as channels or ruptures of identities transmitting the continuously moving particles rather than as lines created by continuous streams of particles associated with the values given in the selectives. In contrast, the closed lines of cuts are thin because there is no such traffic, they are just boundaries.

The movement may be described as a function $F\colon [0,1] \to \mathbb{R}^2$ that is continuous on the LI described by its extremities $0, 1$ upon a space $\mathbb{R}^2$. At any time, t, $f(t)$ gives the position of the particle in $\mathbb{R}^2$. Identity as continuity is indeed what Peirce had in mind, but a caveat is that this functional explanation is not sufficient for a full description of his intentions. This is evinced in the comment, "'Identity' means a continuity, not necessarily in Place, nor in Date, but in what I may call *aspect*, i.e. a variety of presentation or representation" (MS 300: 44–45). The function from time to place does not itself capture all there is in identity as continuity, since the variety of presentational or representational aspects of individuals asserted to be identical has to be taken into account.

Further consideration of the various aspects that Peirce seems to have hinted at requires bringing in the distinction between percepts and perceptions or perceptual judgements. Representationally, percepts stand for something behind them, be it reality, mental images, or experiences of the duality of secondness. This is one "aspect" under which identity as continuity may hold. Another is the presentational one, according to which a percept could be direct consciousness of reality, losing the intellectual and rational character that is inherent in the experience of secondness. The spatio-temporal modes of identification are alone insufficient to capture the entire variety of possible aspects.

If an LI branches, it gives birth to a point of furcation. For instance, the relation of teridentity consists of three lines of identity, and it asserts identity between three dots. What happens to the particle at the point of furcation is that its degree of freedom is three. However, since the movement is assumed to be irreversible (no two lines of identity may be attached to the same hook), there are just two ways it can move. Given the irreducibility of triadic relations, we take it that the particle splits and continues both paths. In this way the teridentity, not constructible from two identities and a conjunction, is preserved. The dot at the point of furcation, namely at the point not hooked with a spot and not a loose end, thus does not prompt any choice of a value for a selective.

Any cut that a particle may come across obstructs such traffic, breaking down the channels transmitting particles from one LI outside the cut to another inside the cut. Since no LI, being a graph, may cross a cut, any system composed of lines is a ligature, which is not a graph and may so cross cuts. A line of teridentity that crosses a cut before furcating consists of four rather than three lines, and a line of teridentity that crosses a cut before two LIs are welded at the place of the intermediate dot consists of five rather than four lines. Peirce noted that the point of furcation of teridentity ought to be regarded as a dot, too, but it needs to be a special one so as not to prompt a choice of a new selective and solely splits the particle arriving at it into two tokens (MS 293: 32).

I have not yet addressed the interpretation of these graphs under this new light. According to Peirce, "Every heavily marked point [a dot], whether isolated, the extremity of a heavy line, or at a furcation of a heavy line, shall denote a single individual, without in itself indicating what individual it is" (4.405, 1903, *Convention V*). Therefore, individuals enter the picture when the scribing of the graph is finished and the interpretation begins. Note that what Peirce calls a selective is best seen as not just a name (proper or indefinite) for an individual picked from the universe of discourse, but as an instantiation of a name of an individual at the outermost extremity of an LI. Therefore, selectives are not, contrary to what Zeman (1974) suggests, redundant with the LIs and the diagrammatic counterparts of uninstantiated bound variables. Accordingly, contrary to what is suggested in Burch (1997), selectives are not hooks connected with bound variables either. Peirce was, admittedly, hesitant about the proper use and status of selectives in graphs, and at one point wanted to dispense with them. However, by viewing them as shorthand for the whole business of instantiation works out smoothly.

To bring continuity into the picture, according to Peirce, it is "relational generality" (RLT: 258). For instance, if one draws a line on a blackboard, "the boundary between the black and white is neither black, nor white, nor neither, nor both. It is the pairedness of the two. It is for the white the active Secondness of the black; for the black the active Secondness of the white" (RLT: 262). This pairedness, or interaction, is what gives rise to existence. The LIs in beta graphs, for instance, can be read "something exists", but that is secondary; in general they denote particulars.

The boundary is where that which lies on one side of it interacts with that which lies on the other side. In itself it is different from both, although the "black" (let it be a set S) and the "white" (the complement of the set S, S^c) share the same one, that is, by denoting it $\mathrm{b}S$, it equals $\mathrm{b}(S^c)$. In this sense, the previous quotation shows Peirce's anticipation of elementary topology that became known only much later.

What Peirce did not take into consideration was that the boundary between a set S and its complement S^c may have points in common with both. Accordingly, S is called open if and only if it contains none of its boundary points, $\mathrm{d}S \subset S^c$. Likewise, S is closed if and only if it contains all of its boundary points, $\mathrm{d}S \subset S$. In an open set, the entire boundary belongs to its complement. Likewise, in a closed set, the entire boundary belongs to it. A set S may also be neither closed not open ('clopen' sets, i.e. partly open partly closed).[23]

Since Peirce did not consider these different possibilities in which boundaries may enter the surrounding spaces, his investigations on topology came to a dead end. Similarly, he did not succeed connecting topological investigations with logical ones, namely with EGs. If we wish to define graphs as topological we need to embed a disjoint union of graphs into $\mathbb{R}^2$, and define the space of continuous mappings from any graph to $\mathbb{R}^2$, assuming here that the cuts do not belong to its interior, in other words that all graphs are open. This procedure will then define the quotient spaces (set of equivalence classes) of closed circles.

Furthermore, since Peirce evidently did not distinguish between open and closed sets — not because of overlooking the distinction between boundary and the interior, but because he detested the Cantorian concept of a set — he did not have any use for a closure operator, either. Consequently, because he did not have a closure operator, he was blindfolded from the possibility of different interpretations for one of his most important diagrammatic signs, that of cuts.

This brings us back to the game-theoretic interpretation of EGs. This interpretation is itself dynamic and qualifies as an explication of how the step-by-step assignment of semantic attributes to the constituents of graphs comes about. The question is how we should interpret the iconicity of a cut in EGs. Disregarding gamma graphs, Peirce believed that he had only one concept of negation in play in diagrams, simply the "not", the denial of an assertion.[24] This was, of course, the encircling of a graph as a separation.

However, as soon as two different negations exist, they may be taken to form a closure operator that turns the underlying algebra from Boolean to closure. As is well known, closure algebra is a topology (McKinsey & Tarski, 1952).

In fact, Peirce was not too far from this route to topology. In addition to seeing negation as a closed line, he took it as a sign that delivers the responsibility of one of the parties interpreting the graph to its adversary. In other words, the responsibility is overturned from the utterer to the interpreter or from the interpreter to the utterer. This is confirmed, for example, in 4.458 [c.1903]: "Any line of identity whose outermost part is evenly enclosed refers to *something*, and any one whose outermost part is oddly enclosed refers to *anything* there may be. And the interpretation must begin outside of all seps and proceed inward. And spots evenly enclosed are to be taken affirmatively; those oddly enclosed negatively" (italics/romans reversed). On the subject of role-switch he remarked, "This rule makes evident the reversing effect of the encirclements, not only upon the 'quality' of the relatives as affirmative or negative, but also upon the selection of the hecceities [proper names] as performable by advocate or opponent of the proposition, as well as upon the conjunctions of the propositions as disjunctive or conjunctive, or (to avoid this absurd grammatical terminology) as alternative or simultaneous" (3.480, c.1906).

Peirce's sin was to think that this transfer of responsibility between the advocate and the opponent delivers the same negation as the set-theoretic meaning of the cut, namely complementation correlated with contradictory negation. But the cut is janus-faced. It obstructs the movement and thus the continuity of the ligature that abuts it, so that one distinguishes what is the interior and what is the exterior. The interior is severed from the sheet of assertion. What is scribed on that area is not asserted. The operation produces the denial of an assertion, and in this sense corresponds to weak contradictory negation. The other role is that when the player who is interpreting the graph enters the enclosure of the cut, his or her strategic role will change to that of what his or her opponent has. This gives game-theoretic, strong negation.

Peirce comes grippingly close to distinguishing these two senses of negation in his footnote, "I am tempted to say that it is the reversal alone that effects the denial, the Cut merely cutting off the Graph within from assertion concerning the Universe to which the Phemic Sheet refers. But that is not the only possible view, and it would be rash to adopt it definitely, as yet" (4.556, c.1906). By reversal, he meant the responsibility-shift to the opponent of the proposition. It was nevertheless erroneous to assume that the reversal effects the denial.

Rather, it is the cutting that effects it, and the reversal effects something else, namely the strong notion of negation.

Still, in 1911 (MS 500), Peirce held denial and affirmation to be the proper contrasts that are manifested in diagrammatic syntax, although now he noted that "A denial is logically the simpler, because it implies merely that the utterer recognizes, however vaguely, *some* discrepancy between the fact and the speech, while an affirmation implies that he has examined all the implications of the latter and finds no discrepancy with the fact" (MS 500: 11–12, 8 December 1911). On reflection, Peirce notes that, despite this, the affirmation is nonetheless "*psychically* the simpler", and therefore ought to be chosen as the primary mode in which propositions are expressed on the sheet, the denial requiring an additional operation. This was the reason why he preferred EGs to his former, dual system entitative graphs.[25]

Peirce did not suggest that there were no obstacles in relating topics and logic. In one of the variants of manuscript *Topical Geometry* he noted that "How these three doctrines [time, multitude and characters, A.-V. P.] are related to one another and to the doctrine of Logic is not, at present, entirely clear".[26] In the very next paragraph he starts to consider the doctrine of negation: "In the field of logic, it seems desirable to remark that though the principles of contradiction and excluded middle must be considered as defining the relation of *negation*, that does not prevent their embodying two valuable distinctions". The variant trails off. The "two valuable distinctions" probably referred not to negation as a sign in itself, but rather to the distinction between vague, indefinite signs and general, indeterminate signs, the former being those to which the principle of contradiction does not apply, and the latter being ones to which the principle of excluded middle does not apply.

However, despite Peirce's lack of topological appreciation of the distinction between what is open and what is closed, such a distinction may have eventually meant something quite different for him. He did not endorse the conception of a set at all. All he used were hypostatically abstracted collections. Accordingly, he may have wished to treat the concept of the neighbourhood of a set and their boundaries quite differently. In any case, his notion of the area of a cut anticipated openness of a set or a collection. As the cut is not part of the area, the area being disconnected from the sheet, this prefaces the notion of a set whose neighbourhood is not part of it. His notion of the area is thus not unlike what is known in topology as the Jordan region of simple closed curves.

Venturing somewhat further, we could also think of a cut enclosure as a set that has both open and closed segments, in other words as a clopen set. In that case, the location of the ligature that attempts to penetrate the cut will become material. I will leave such an extension to be studied on another occasion.

Summarising, the SA is a representation of "an external continuity" between assertions that are true, and of "a continuity of experiential appearance". It represents "a field of Thought, or of Mental Experience, which is itself directed to the Universe of Discourse". According to Peirce, SA is "the most appropriate Icon possible of the continuity of the Universe of Discourse" (4.561).

Furthermore, LIs assert continuous connections between concepts:

> A heavy line shall be considered as a continuum of contiguous dots; and since contiguous dots denote a single individual, such a line without any point of branching will signify the identity of the individuals denoted by its extremities, and the type of such unbranching line shall be the Graph of Identity, any instance of which (on one area, as every Graph-instance must be) shall be called a *Line of Identity*. (5.561).

LIs in one dimension represent "linear continuity" (4.561), deriving continuity from the SA upon which they are scribed. They are 'channels' or 'ruptures' upon the continuum.

Cuts, on the other hand, are interruptions on continuity, just as if one is to place a point upon a continuous LI in order to break it. "At the cuts we pass into other areas, areas of conceived propositions which are not realized" (4.512).

Given the semantics of EGs according to which values of individuals from the universe of discourse are placed upon the extremities of the line (selectives) and continuously transmitted towards spots, cuts signal a change in the polarity of these semantic values, in other words the change of the roles of the Graphist and the Grapheus. This interruption effects the non-identity as well as negation of assertions.

5. On diagrammatic representations

Emerging diagrammatisation Of late, following the proliferation of graph-theoretic and diagrammatic methods largely due to the expansion of computer science and computational linguistics, cognitive science and artificial intelligence, the time to understand the insights of EGs finally arrived. Sowa (1984) showed that conceptual graphs (CGs) can be mapped on classical predicate calculus or order-sorted logic, and hence are a useful and efficient basis for a graphical logic. Meanwhile, Kamp (1981) developed DRT to facilitate the linguistic representation of natural-language discourse. Its discourse-representation structures are diagrammatic images resulting from the interpretation of linguistic utterances, aimed at providing a precise medium for the information possessed by speakers of language.

Unlike EGs, these structures are heterogeneous rather than iconic, since they use symbolic notions, such as identity relations, to capture concepts that are captures non-symbolically in EGs.

Other widely-used pictorial methods for computational purposes include entity-relationship diagrams, flowcharts, Petri nets, finite-state machines, and semantic nets. Parallel distributed programming, and neural networks in general, are also diagrammatic in nature. In semantic nets, for example, the task of knowledge representation has been to keep the theory close to actual expressions of natural language. In closer contact with logic, Kripke frames and modal models are labelled graphs in which formulas of modal logic are interpreted. Processes of logical reasoning have, in turn, been investigated in projects such as Hyperproof[27] and Tarski's World (Barwise & Etchemendy, 1992). With their capacity for complicated calculations, Feynman diagrams and Penrose tilings have proved advantageous in quantum physics and related fields. All these may benefit from the insights brought into the open by Peirce in his system of EGs.

Many of these approaches were preceded by GTS. This provides an alternative, yet flexible, tool from which formal or linguistic expressions derive their meaning. It accounts for the same underlying linguistic and discourse-related phenomena as DRT, but, as noted, unlike its discourse-representation structures, games additionally draw out the total histories of the evaluation process correlated with linguistic or logical concepts. As discussed in the previous section, this happens as soon as we take the game-theoretic character of the theory seriously and think of games as extensive diagrams of players' actions.

A particularly significant area is the representation and resolution of anaphoric concepts in natural language, which is relatively well studied in DRT, but which nonetheless wavers once one moves from simple sentences to more complex ones, or when sentences involve negative constructions.

The relations between games and CGs have not yet been investigated in the literature in sufficient depth. Yet, they are both foundationally appealing and attractive in terms of applications because they allow one to exploit powerful pattern-matching abilities to a larger extent than classical linear or compositional logical notations. Games and CGs can both be viewed as attempts to build a unified language-modelling tool. However, the main difference that needs to be acknowledged is that CGs aim at incorporating reasoning methods into the theory. On the other hand, game-theoretic methods as conceived here are primarily semantic or, more broadly speaking, semeiotic in nature. If proof-theoretic concepts are to be employed, we can still use games to that effect, but we would need to change the class from semantic to dialogical, or proof-theoretic (Chapters 7 and 10; Lorenzen & Lorenz 1978; Rahman 2002). Such classes are widely researched in computer science as game semantics and linear logic (Chapter 8; Abramsky & Jagadeesan 1994). They often resort to the resources of category theory, itself diagrammatic in nature. Perhaps not unsurprisingly, categorical concepts indeed lurk behind the mathematics of Peirce's EGs (Brady & Trimble, 1998).

Conceptual graphs Sowa (1984) introduced CGs that have grown into a central diagrammatic method and a tool of knowledge representation and reasoning in AI tasks. CGs are iconic just as EGs are in showing the connections between natural-language expressions directly without using variables and variable renaming or typing. The characteristic features of CGs are:

- *Concepts*, denoted by boxes drawn on a sheet. A concept consists of a type, which is a label of the concept, and a *referent*, which is a name or a quantifier. Either classical or generalised (plural) quantifiers may be allowed.
- *Relations* between concepts, denoted by circles.
- *Coreference*, denoted by a dotted line drawn between concepts.

Various labels can be further specified and analysed, for example according to the genus and differentia of the entity they denote. A label is very much like a restrictor limiting the domain of applicability of a quantifier.

A simple CG representing sentence (5.16) is given in Figure 5.A.1 in the Appendix to this chapter.

Every man sees a dragon. (5.16)

Adding nests of CGs produces different kinds of information, especially with regard to the context within which any inner layer of a graph subsides. Take contexts as negations would provide a visualisation of negative information analogous to that of the theory of EGs. For example, the graph in Figure 5.A.2 corresponds to the classical first-order sentence

$$\neg(\exists x(S_1x \wedge S_2x \wedge \neg(S_1x \wedge S_3x))), \tag{5.17}$$

which is equivalent to

$$\forall x((S_1x \wedge S_2x) \rightarrow S_3x)\,. \tag{5.18}$$

Hendrix (1979) shows how CGs may be partitioned and thus used more effectively in visual and diagrammatic expressions of natural language. Even further, non-partitioned structures would be congenial to even more expressive non-compositional diagrams. There are various possibilities of extending the nested system of graphs that were discussed in the previous sections. The game-theoretic interpretation for such non-partitioned nets would then involve imperfect information as do the IF extensions of EGs.

Discourse-representation theory The development of the theory of discourse representation (Kamp 1981; Kamp & Reyle 1993) was motivated by interpreting pronouns in a sequence of natural-language sentences. Its idea is to portray discourse in two steps. The first is to construct a discourse-representation structure involving a set of discourse referents, and the second is to provide a set of conditions (say, predicates) for the discourse referents introduced by the expression in question. The result is the graphical notation of a box, consisting of an upper part with the list of discourse referents and a lower part with the conditions imposed on them.

Figure 5.A.3 illustrates a simple discourse-representation structure for:

A man sees a dragon. (5.19)

Like CGs, these box structures can be nested — a vital move in trying to capture the meaning of anaphoric pronouns. According to the rule of transitivity, the conditions of a box are inherited by those boxes that occur deeper inside it.

Pronominal anaphora Due to its coreferential nature, a test case for the usefulness of diagrammatic representation systems is provided by natural-language anaphora. For example, consider

A man sees a dragon. He escapes. (5.20)

These two sentences form an implication from left-to-right. Diagrammatic representations of this mini-discourse are given in Figures 5.A.4, 5.A.5, 5.A.6 and 5.A.7 for EGs, CGs, DRT discourse structures and extensive forms of GTS, respectively.

Negation in diagrammatic representations In EGs, negation is a cut or separation that severs the enclosed subgraph from the rest of the graph. Alternatively, incision gives rise to the role reversal between graph interpreters. As observed, both topological (geometric) and communicational (game theoretic) methods may be used. As far as the former is concerned, the outer area of the enclosed simple curve is the complement of the space that forms its interior. The interpretation of negation is then classical, namely the negation means complementation, the algebraic counterpart of which is the contradictory forming operation. Alternatively, separation means that the continuity of the sheet on which the graph is scribed is halted in that the participant associated with the what is immediately outside the cut can no longer proceed, and the responsibility for permeating the interior of the cut has to be turned over to the party with the opposite polarity, who will continue the examination of the graph.

Two different conceptions of negation exist even though the difference does not emerge in the received conception of EGs. The received perspective is that, as the theory of alpha graphs is isomorphic to the theory of Boolean algebras (Brady & Trimble, 2000), the notion of negation is unproblematic. Likewise, the system of beta graphs is routinely observed to be isomorphic to the system of first-order calculus of relations (Brady & Trimble, 1998). These results hold if the cut corresponds to weak negation. In actual fact, the negation bifurcates into several senses. In one, classical, sense it relates sets to their complements, and interiors of graphs to their exteriors. In the other, game-theoretic sense, it reverses the roles of the players, but asserts nothing about complements. For example, in IF extension of EGs, these two notions need to be distinguished.

The notion of negation as a cut has interesting applications. Because of its iconicity, it is explained why anaphoric coreference is not always possible. Consider a discourse that is marked (the illicit part is marked by a star):

It is not the case that a man sees a dragon. *He escapes. (5.21)

(We naturally assume that the man does not escape unless he sees the dragon.) The diagrammatic representations of (5.21) are given in Figures 5.A.8, 5.A.9, 5.A.10 and 5.A.11, in which the the cut obstructs the identity, explaining the impossibility of coreference.

Alternatively, it is the role-reversal between the two interlocutors that is the key explanatory concept for the impossibility of coreference in the game graph represented in Figure 5.A.11.

Variables Since CGs do not use variables, they come closer to EGs than to other diagrammatic methods such as DRT. What, then, is the role of GTS here? An alternative non-representational approach to semantic games that can be defined for a number of formal languages is to associate them directly with natural-language expressions, including lexical items. For a game-theoretic interpretation to apply, bound variables are not necessary.[28] It is, of course, possible to use some formalised medium into which linguistic assertions are mapped, but that would be an optional extra that runs the risk of turning the representations less iconic than with direct interpretation.

Modalities One aspect still little developed is the incorporation of modalities into diagrammatic logic. Especially predicate modal extensions are few and far between. Implicit in Peirce's gamma graphs are several precursors of such extensions, together with apt recognitions of the problems that arise with interspersing individuals and modalities. Nonetheless, diagrammatic methods promise a good deal of insight into old problems of quantification in modal logic, including cross-world identities, the *de dicto* versus *de re* distinction, and modal anaphora across attitude contexts. An example of the latter is intentional identity, a special anaphoric coreference in the context of modalities.[29] Diagrammatic theories of modalities, in fact, enable us to tackle the notion of cross-world identity in novel ways, dispensing with the somewhat dubious existence assumptions in the actual world.

At the same time as developing a modal system in the gamma part (and also in tinctured EGs), Peirce was trying to represent the concepts of possibility and necessity using marked relations between the "states of information" of different graphs (4.517; MS 467). This, of course, comes very close to the modern model-theoretic approach to modalities as possible worlds, but apart from an isolated description in his writings, he did not go on exploit this idea much further. The construction of gamma graphs was not comprehensive in the direction of the treatment of modalities, and gamma was used to make sense of higher-order (type-theoretic) logic with abstraction, for example.

Quantified modal logics and gamma graphs are studied in Øhrstrøm (1997). He argues that in his system of transformation rules for gamma, Peirce uses a version of what we recognise as Barcan formulas, namely that from a graph corresponding to $\Box\forall x Sx$ one may infer $\forall x \Box Sx$, and likewise from $\exists x \Diamond Sx$ one may infer $\Diamond \exists x Sx$. These inferences are valid assuming that the domains of universes are constant in all possible worlds (all sheets of assertion). This was nevertheless not assumed by Peirce. He wanted the system of sheets of assertion, tacked together at points, to exemplify multiple and different universes of discourse. Weaker assumptions than the common-domains one hold for these inferences too, but the domains will still need to be related to each other by being ordered in terms of increasingness or in terms of decreasingness.

Further research Not only a knowledge-representation scheme for cognitive and AI purposes, or a pictorial device for writing out discourse structures, the diagrammatic approach also unifies the outlook on logical systems themselves. The motivation for CGs comes from computer science, but their proximity to EGs, as well as to GTS, makes them foundationally rich. However, although structurally similar, CGs and DRT lack the strategic dimensions of game-theoretic systems, and are not as iconic. Many knowledge-based systems aimed at understanding natural language benefit from strategic resources such as world knowledge, collateral information, lexical clues, and various cognitive repertoires. Extensive games are diagrammatic systems tailor-made for such strategic tasks. To mention just one possibility of increasing the contextual dimension of games is to assume an initial move by Nature (the Grapheus) who 'shuffles the deck', thereby designating the contextual or environmental parameters according to which any of the plays of the game are to proceed.

We still need to investigate interactive systems for CGs and for other diagrammatic logics, too, including DRT. One of the key distinctions between GTS and DRT is that DRT does not keep any record of the histories of discourse elements to which we could refer, and from which we could go on to choose the preferred interpretations of, say, anaphoric constructions. Accordingly, the interpretational history is also missing from EGs until we take their dynamic and dialogical character revealed in the apparatus of extensive games.

Nevertheless, Peirce's EGs came with rudimentary forms of strategic meanings of utterances. He often resorted to habits that guide us to the right decision through generalisation. The development of GTS has rewarded his objective, although we are still far from a complete theory of strategic meaning. For example, various forms of bounded rationality that are currently being pursued in game theory and interactive epistemology are pertinent to strategic meaning (Pietarinen & Sandu, 2004).

A rarely noted point concerns the original goals of general systems theory (Klir 2001; Chapter 13). In representing iconic features of what is common between the world and human artefacts, these are but part and parcel of the semeiotic point of view to human cognition. Systems theory aims at investigating the arrangements and relations among the parts that connect them into an organisation. So conceived, systems are independent of the content of their constituents, and hence their methods and principles are applicable across a range of disciplines. However, although some systems are markedly iconic, they lack strategic dimensions and have not been integral to logic.

6. Conclusions

Let me put the findings of the present and the previous chapter into perspective. With his diagrammatic logic, Peirce embarked to improve on the idea of logical analysis not only in the sense of having n-place predicates (spots with n occupations of their hooks) and not only monadic predicates, but also in the sense that the blanks may be filled not only by proper names but also by indefinites, "onomas" (MS 491; Chapter 1). By so doing he prefigured the discourse referent idea of the discourse-representation theory Kamp & Reyle (1993), in which the value of a pronoun or an indefinite is derived from the discourse context and the common ground of the interlocutors. The idea is also related to Peirce's semeiotic concept of the dynamic object constructed and nurtured in the continuous interaction between the utterer and the interpreter of the given assertion. Indeed, assertions scribed on SAs are assertions within some context from which their components derive values. A similar idea is found in GTS in which the values are picked among the derivational histories of the game tree produced by the endoporeutic interpretation of a graph, possibly supplemented by some environmental (deictic) parameters initially assigned to these histories by the Grapheus.

Iconic representation and the dialogical interpretation were the two key elements in Peirce's diagrammatic tactics to unearth the content of thought and cognition. My purpose in these two chapters was to elucidate the key points as to the sense in which he wanted his EGs to put before us true moving pictures of thought. Their intent was to reveal what is essential to all concepts that can be analytically and iconically represented and experienced, and this is their pragmatic value. However, this desideratum was not achieved in full. Peirce's own investigation was conducted on the fairly static level of endoporeutic interpretation. By putting the graphs, so to speak, 'on the move' in the sense of the theory of games, we eventually have accomplished a truly pictorial and dynamic representation of the meaning of all assertions.

Another kind of dynamism is achieved by incorporating the notion of time in reasoning, in the sense of the topical geometry of these graphs, amounting to 'identity-through-time' type of objects. By having to walk further along the path of diagrammatisation than Peirce, we also manage to put his anticipations into a sharper perspective: "A picture is [a] visual representation of the relations between the parts of its objects; a vivid and highly informative representation, rewarding somewhat close examination. Yet ... it cannot directly exhibit all the dimensions of its object, be this physical or psychic. It shows this object only under a certain light, and from a single point of view" (MS 300: 22–23). What is also implicit in this passage is one of the motivations behind Peirce's desire to produce stereoscopic extensions of diagrams.

What he did achieve was a notable break-off from the linear confines of language. He added polyphony to the tenor of language, so to speak. He opened up the path of expressing any consistent assertion. Logic no longer needs to protract in time, as a series of symbolic expressions and their manipulation, but as "diagrams upon a surface",[30] with continuous deformations performed upon them. He continued: "Three dimensions are necessary and sufficient for the expression of all assertions; so that, if man's reason was originally limited to the line of speech (which I do not affirm), it has now outgrown the limitation" (MS 654: 6–7). Peirce assumes here a viable three-dimensional gamma system. What is more, however, is that the prediction holds good for the three-dimensional extension with imperfect information and independence. This avenue can be further extended to the whole gamut of different diagrammatic methods that Peirce introduced under the umbrella of the gamma part.

The reason why Peirce failed to explicate what his moving-picture of thought really meant was that he campaigned on multiple fronts, including taking a dialogical approach to the building and interpretation of EGs, and simultaneously wrestling with issues of topical geometry and the metaphysical and mathematical nature of continuity. He may have wished that all these would merge into one giant diagrammatic system. This was hopeless, and the outcome was a number of significant outcroppings that await integration. Nevertheless, his intent is clear: as a diagram is not a perfect picture of the mind in operation, it aims at preserving its essential structure, in much the same way as in topology one aims at preserving some essential properties of objects, such as a 'nearness of two points on a surface'. I hope that the ideas I mentioned are suggestive of the directions in which further integration is to be found.

Furthermore, given the treatment on anaphora and related phenomena, the next step would be to extend the system of EGs to text. Meaning in text extends sentence boundaries, but ligatures in EGs are tailor-made to accomplish that.

Because of the assumption of perfect information, Peirce thought that the meaning of an EG was compositional in the sense of being determined by its component graph-instances. This is evidenced in MS 280: 35, in which he wrote, "The *meaning* of any graph-instance is the meaning of the composite of all the propositions which that graph-instance would under all circumstances empower the interpreter to scribe".[31] The reasons for holding this assumption are less clear, however, given the contextuality and the endoporeutic interpretation of graphs. As no overlap is allowed, any graph may be represented as a tree structure. Given the reliance on nested systems of contexts, however, it is but a small step to extensions of graphs that transform them into partially-ordered rather than tree structure. These extensions would no longer be compositional.

However, as the graphs are finite, the possibility of a compositional interpretation exists, provided that contextual information is carried along the subgraphs interpreted prior to the larger graphs within which they are embedded.

There is a dazzling juncture at which Peirce comes close to the possibility of non-compositional mapping from models to graphs, however. In MS 490 he discovered a graph that crosses a cut (Chapter 4, pp. 126). In one of his multiple attempts to capture modalities by gamma apparatus, he found that comparisons between actual and possible individuals could not be avoided. A need exists for representing assertions concerning identification (as well as non-identification) between existent things scribed on the blank sheet and possible things scribed on the shaded enclosure of a broken cut, in other words on a graph that refers to possible universes. In that case, an LI, not a ligature, has to signify identity, and a graph that crosses a cut exists.

This, briefly, is the explanation of what Peirce means in 5.583 (a fragment of MS 490) by his assertion that some concepts are not propositional. As noted there, non-propositional signs can only exist as constituents of propositions, while it is not true that a proposition can be built up of non-propositional signs. A non-propositional sign is one that is not assigned a semantic attribute, or a meaning, by either of the interlocutors. In this sense, it violates compositionality, because one version of the principle asserts that the meaning of the proposition has to come from the meaning of its constituents and their proper combinations. According to Peirce, no proposition can be properly assembled from non-propositional signs alone.

A wider perspective opened up pertains to the question of identity between possible individuals and their identities with individuals that actually exist. This is a well-noted quandary in quantified modal logics. It is possible to tackle it by means of semantics that assumes an identification system across the multiplicity of possible worlds, the network of world lines between multiple manifestations of individuals in different possible worlds. Precisely how these lines are drawn depends on what identification modes one is dealing with.[32]

Following the discovery of non-propositional signs in the diagrammatic representation of possibilities, Peirce came to suggest that each LI in fact fans out so as to give rise to teridentity with a loose end. He termed such graphs ones of “indefinitely multiple identity”. The gamma EG on page 126 of “There is a woman who is not and could not be identical with any possible catholic” was Peirce’s example with such a teridentity. What he came to imply by such a suggestion was a strong hint towards the idea that in modal assertions identity involves multiple reference points. In fact, he even used the term “references” in an unexpectedly similar sense:

> In all my attempts to classify relations, I have invariably recognized, as one great class of relations, the class of *references*, as I have called them, where one correlate is an existent, and another is a mere possibility; yet whenever I have undertaken to develop the logic of relations, I have always left these references out of account notwithstanding their manifest importance, simply because the

> algebras or other forms of diagrammatization which I employed did not seem to afford me any means of representing them. I need hardly say that the moment I discovered in the *verso* of the sheet of Existential Graphs a representation of a universe of possibility, I perceived that a *reference* would be represented by a graph which should cross a cut, thus subduing a vast field of thought to the governance and control of exact logic. (MS 490; 4.579).

This was Peirce's extraordinary step towards a wider understanding of modalities. Modal identity is one of the countless issues that is properly addressed only in the unpublished manuscripts. The explanation that it is quite conceivable that a graph crosses another graph, even if to a limited extent, is not forthcoming anywhere in the Collected Papers. What is more, in continuing with these issues that were not published in 5.583 nor elsewhere, he takes up the notion of synechism, the merger of pragmatism and tychism, in relation to the graph of teridentity in the context of possibilities, the graph of indefinitely multiple identity. This was a natural move, however, given that modal identity involves identification across multiple possible worlds (or several tacked sheets of assertions), along with continuity principles concerning questions of how the different aspects or manifestations of individuals transpire in different sheets or different worlds. Unfortunately, that significant insight has remained a well-kept secret in the unpublished papers.[33]

Purely extensionally, the multiple roles played by LIs answers whether one or multiple logical mechanisms do the work for the verb *being*. Beta graphs provide conclusive evidence that different logical senses of *being* are illusory. There is but one *is*. What Hintikka (1983) has called the Frege trichotomy, according to which there is the *is* of existence, the *is* of predication and the *is* of identity, fails according to Peirce's diagrammatic account. The LIs are, at the same time, signs of existence, predication and identity. The only sundry item needed in class-inclusion is the scroll. Identity lines also function as signs for coreference in anaphora.

Of course, the interpretation of this singular sign may well be different according to different uses of *being*, but that does not change the fact that a unitary mechanism works for all. For example, by such a unifying device one may express identities between variables that are parts of the quantifier structure rather than of the matrix of the formula, which is impossible to do with ordinary, symbolic first-order logic.

One remarkable consequence of this is that, even if alpha and beta are, as theories, isomorphic to propositional and predicate logic, respectively, the two sides are motivated by essentially different ideas. As far as quantification is concerned, it is not sufficient just to plug in values for variables and observe, via satisfaction, whether the formulas are true in a model. As in GTS, Peirce's endoporeutic interpretation requires suitable individuals to be found in the universe of discourse to function as the selectives at the hooked extremities of LIs. In order to assure identity, a selective is continuously connected with another selective at the non-hooked extremity of the line or ligature. Symbolic formulas have no way of expressing this.

To make the use of the personalised notion of players in communicative dialogues more precise, it is possible to broaden the concept of the utterer and the interpreter to the theoretical, inanimate entity of a 'quasi-player'. This is what Peirce suggested in so many words. By his own account, the question of who plays these games can be answered by saying "anyone or anything that puts forward a sign". A mind could be understood as being a "*sign-creatory in connection with a reaction-machine*" (MS 318: 18). True, he was at pains to find and explicate the essential ingredients of a "reaction-machine". But even without such explications, in order to properly understand both Peirce's theory of communication, his semeiotics and his logic — let alone their interrelations — it is necessary to reconcile the triadic theory of signs and the communicative theory of dialogues. An outline of a joint understanding of the two was given in Chapters 1 and 2, to be returned to in Chapter 13.

Some may hold that the issue concerning the kinematics of EGs has to do with graph transformations according to the given rules and permissions that Peirce studied in length. This has some truth, and the illative transformations that he defines, for instance, may be viewed as meaning-preserving mappings from one graph-replica to another. Related to this is the sequential view of reasoning according to which "every cognition is determined logically by previous cognitions" (5.265). As signs beget new signs, reasoning protracts from premisses to a conclusion. However, I believe that no proof-theoretic perspective would, by any standards, be an exhaustive description of what Peirce had in mind. The truly dynamical aspects of logic pertain to how interpretations are formed, and the diagrammatisation is but a fraction of the general theory of how one interprets signs and communicates ideas by their means.

Moreover, deductive issues are not devoid of interpretative concerns, for instance in relation to the question between corollarial versus theorematic reasoning in EGs. They involve theorematic elements of reasoning, not only in the sense that one needs to introduce new information by means of new individual elements into graphs, but also in the sense that one needs to point at the right areas on the given sheet of assertion at which any newly-introduced element is to be scribed in the course of the graph transformation. For instance, a rule that says that any graph may be scribed on the oddly-enclosed zone is not deterministic, and does not guide us towards a definite choice of a suitable graph or to the proper location at which the scribing of it ought to be done.

However, there is a further, quite topical line of research related to these remarks, which as far as I know has not yet been pursued. It addresses the question of what insights into the nature of proofs Peirce's EGs are able to produce. In what sense are they congruent with game semantics in computer science, or with its recent spin-offs such as Ludics, both entertaining the idea of interaction at the heart of proofs (Chapter 8)?

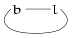
I hope to have shown in this and the previous chapter that the issues raised should make anyone wary of the opinions sometimes voiced to the effect that logical diagrams, our moving pictures of thought, are nothing but alternative means, not different in kind, of expressing the same underlying phenomena. Just to bring out one and by no means an isolated example, it should now be quite palpable what is off beam in statements such as "A diagrammatic logic is simply a logic whose target objects are diagrams rather than sentences. Other than this, diagrammatic logics and logics involving expressions of some language are not different in kind" (Hammer, 2002, p. 421). I believe that these two approaches to logic are not only different in kind but also in the substance the amount of which is gradually beginning to come into sight.

Notes

1 See Janasik & Sandu (2003); Janasik et al. (2002).

2 The frame refers to the fact that, without payoffs, a set of histories is not yet an actual game but only a proto-game, its outline structure that does not tell who wins or loses. This is like frames in modal semantics, distinguishing them from models that come with the valuation.

3 For more details and references, aside from Chapter 7 I refer to Hintikka (1996a); Hintikka & Sandu (1997).

4 One is reminded here of Peirce's 'spread formulae' from 1882: "The notation of the logic of relatives can be somewhat simplified by spreading the formulae over two dimensions. For instance suppose we write $\left(\begin{smallmatrix} b \\ l \end{smallmatrix}\right)$ to express the proposition that something is at once benefactor and lover of something. That is, $\Sigma_x \Sigma_y\, b_{xy} l_{xy} > 0$. We can write  to mean that something is at once benefactor and lover of something, that is, something is benefactor of a lover of itself: $\Sigma_x \Sigma_y\, b_{xy} l_{yx} > 0$" (MS L 294: 1, *Letter to Oscar Howard Mitchell*).

5 Associated semantic games have to be adjusted to reflect these generalisations. For instance, concurrent games are played for those constants that are not in any relation. Games germane to such cases have been developed in the works of Abramsky & Melliès (1999) and de Alfaro & Henzinger (2000), for example. Moreover, reflexive relations are not admitted in disjoint graphs and hence the associated extensive games would not comprise even singleton information sets (i.e. partitions of histories of an extensive game, see Chapter 7).

6 This is not the same as Peirce's "bridges", which are used if two LIs need to cross one another without being connected.

7 Restricted on models $M = \langle I^M, (p^M)_p \rangle$ with I a two element set, this is the diagrammatic counterpart of the four-place connective W in terms of a sentence with restricted quantifiers $\forall j (\exists i/j)\, p_j$.

8 Pietarinen & Tulenheimo (2004) discuss branching quantification in natural language with pointers to the literature.

9 From four dimensions onwards, a caveat is that the topological manifolds may be non-smooth, i.e. non-differentiable. This may affect permissible transformation rules. What the transformation rules are would have to be investigated separately in each dimension. In three dimensions, for instance, it is likely that any set of rules is incomplete.

10 Since Peirce repudiated set theory in its Cantorian sense and wished to replace it with infinitesimals and collections, this terminology of open and closed sets is of course moot. It nevertheless serves to bring out the fundamental difference that these two notions of sets, open and closed, logically give rise to.

11 Similar notion of negation in terms of switching questions and answers is in use in linear logic, which thus is revealed to have had a venerable predecessor in Peirce's system.

12 It is worth asking whether it is also possible to present diagrammatic logic in fewer dimensions, in the one-dimensional case of the real line. Negation would then be iconically represented by a hole on the line. This amounts to a weaker-than-propositional logic. Given that a finite interval would be the one-dimensional projection of an alpha graph, there would be just two possible starting points for the endoporeutic interpretation to commence with. The problem in interpreting such a system is that the matching points at the opposing ends do not necessarily refer to the same token of a cut that appeared on the alpha graph.

13 MS 9: 2, continued in MS 501: 3, *Foundations of Mathematics*. On one of the assorted pages (MS 501: 3 a.p.), the last sentence reads "Whichever chooses last is supposed to know what the choice of the other was, which gives the utterer an advantage in defending the proposition, the interpreter an advantage if he chooses to attack it".

14 2.524, 1883, *Critical Logic. Extension of Aristotelian Syllogistic*.

15 MS 430: 62, 1902, *The Simplest Mathematics (Logic III)*.

16 MS 530: 38 a.p., c.1903, *A Proposed Logical Notation*.

17 One realm in which recursivity and learning raises its head is the semeiotic interpretation of signs, referring as they do to values (interpretants) that previous cycles of interpretations have produced to yield a comprehensive understanding of some task.

18 See e.g. NEM 2:165–191, 477–547.

19 MS L 73, 21–23 December 1900, *Letter to Georg Cantor*.

20 This is claimed in Pape (1999), but there is no documentation.

21 See also PP: 135–136.

22 This would allow us to add free variables into the system as attached, uninstantiated dots.

23 In metric spaces, a set is open if and only if it is closed (i.e. S is both).

24 Though he argued that three notions of negation are in play in sentences with modal expressions: "'An A (whenever there were one) would be a B' has for one negation 'An A would not necessarily be a B' or, otherwise expressed 'An A might not be a B' while for a second negation it has 'An A would be a non-B' and for a third, combining the second and third negations, 'An A might be a B"' (MS 671: 20, c.1911, *First Introduction*).

25 There is also an anticipation here of the semi-decidability of first-order logic, that is, the existence of a complete decision procedure for its true formulas. Moreover, even if some given logic that is different from standard semi-decidable first-order logic did not, unlike this standard first-order logic, have a recursive set of axioms, there may still exist a complete disproof method for it, namely one recursively facilitating a recognition of a discrepancy in the formula (nonvalidity), and which for that reason may still keep the logic manageable, useful, and suitable for foundational purposes.

26 MS 137, 1904, *Variant* B_2; NEM 2:522.

27 www-vil.cs.indiana.edu/Projects/hyperproof.html (accessed 31 December 2004).

28 See e.g. Hintikka & Kulas (1983) for studies of GTS for fragments of natural language. Some of the basics are presented in Chapter 7 and Pietarinen (2001b).

29 On the semantic issues involved in intentional identities, see Pietarinen (2001a).

30 MS 654: 6, 19 August 1910, *Preface to Essays on Meaning*, 1st draft.

31 An alternative formulation states, "The *meaning* of any graph-instance is the meaning of the sum total or aggregate of all the propositions which that graph-instance enables the interpreter to scribe, over and above what he would have been able to scribe" (MS 280: 35 a.p.).

32 On identification semantics for quantified modal logics for knowledge and belief, see Hintikka (1969); Pietarinen (2001a, 2002c, 2003d).

33 For indications towards a continuity interpretation of identification semantics for epistemic notions in terms of differential equations the reader is referred to Hintikka (1989), especially pp. 80–90.

Appendix 5.A: Some diagrammatic representations

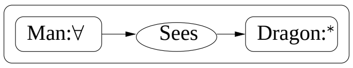


Figure 5.A.1. CG of (5.16).

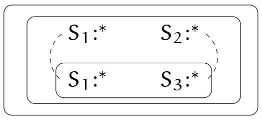


Figure 5.A.2. Nested CG of (5.17).

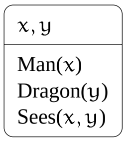


Figure 5.A.3. DRS of (5.19).

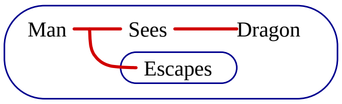


Figure 5.A.4. Beta EG of (5.20).

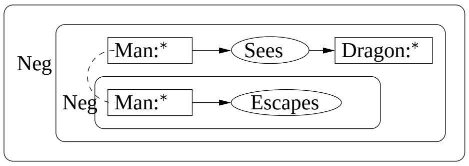


Figure 5.A.5. CG of (5.20).

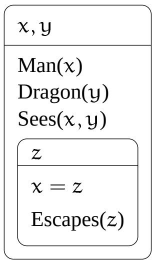


Figure 5.A.6. DRS of (5.20).

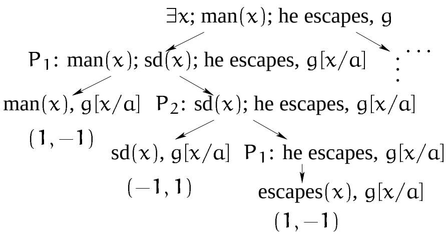


Figure 5.A.7. Extensive semantic game of (5.20).

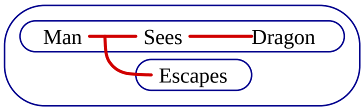


Figure 5.A.8. Beta EG of (5.21), blocking coreference.

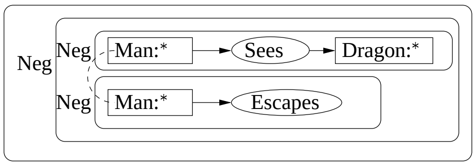


Figure 5.A.9. CG of (5.21), blocking coreference.

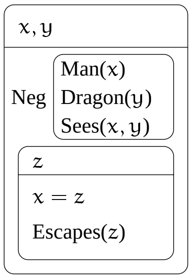


Figure 5.A.10. DRS of (5.21), blocking coreference.

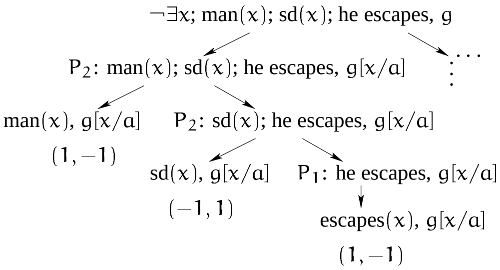


Figure 5.A.11. Extensive semantic game of (5.21), blocking coreference.

Chapter 6

EXISTENCE, CONSTRUCTIVISM, MODELS, MODALITIES

1. Introduction

The purpose of this chapter is to record and expound a couple of perspectives on Peirce's logic that I feel have remained less appreciated than the subject matter they deal with deserves. In short, these are: (i) the peculiar emergence of the assumption of existence with regard to Peirce's quantificational notions, especially the particular selective, (ii) his constructivist leanings in logic, (iii) an apparent discrepancy in dyadism vs. triadism from a communicational point of view, (iv) the early genesis of model theory (the endoporeutic method) in the system of existential graphs (EGs), plus (v) aspects in the relation between his theory of quantification and modality. My intention is to raise some questions concerning the place of these issues in his overall logical system rather than to definitively solve them, although I give some suggestions towards solutions.

Understanding of Peirce's logic is only just evolving. This is mainly due to unavailability of published material from his last and very prolific epoch. I also think that these issues are all interconnected to an extent that is impossible to articulate in a single treatise. Nevertheless, I believe that the connections between, say, the emergence of existential assumptions in quantification, the reduction thesis concerning relational notions, the dialogical approaches to semantics, the tenet of constructivism, and the theory of modalities are all destined to find solid logical home in Peirce's overall semeiotic programme.

The appendix to this chapter provides a diplomatic transcription of the previously unpublished entry on Modality (in MS 1147) that Peirce wrote for Baldwin's Dictionary of Philosophy and Psychology in 1901.[1] Peirce contributed many carefully drafted articles to this dictionary, but the manuscript on Modality among dozens of other draft articles in MS 1147 was considerably modified in the published version. I have annotated this diplomatic transcription with comments, because it contains many lines on Peirce's thought on modalities and logic that have escaped earlier acknowledgement.

2. The emergence of existence in quantificational logic

The origins of logical semantics close to the current understanding emerged when Peirce noted that, in order to render the sentence of the form 'Any A there is, is B' true, one either needs to take A to be empty or else there may exist one or more A's but none that are not B's (MS 593; MS 671). Given that every case, or the state of things, of the truth of A is a case of the truth of B, there are no other exceptions in B.

Peirce was speaking on behalf of the algebraic system that he had been developing since about 1870, building on his earlier work on Boole's algebra. He took this early treatment of the algebra of relations to be insufficient, noting that he did not have "familiarity" with the signs of quantification at that time.[2] Both Peirce and Boole had used the symbols Σ and Π to denote the sum and the product operations, respectively, in their algebraic logic of relations.

By 1903, and probably even earlier, guided in good measure by the studies of Mitchell, Peirce had realised in full the idea of these operators as expressing the notion of scope in terms of the significance of the order of the choices they prompt. Back in the mid-1880s, he preferred to call these symbols quantifiers, attributing the first use of them for quantificational purposes to Mitchell (SIL: 74). To be precise, a "Quantifier", strictly speaking, referred in Peirce's terminology to all of the symbols in front of the "quantified" part, which he occasionally termed "the boolian" or "the Anonymian" part. He also proposed calling the quantifier part *Hopkinsian* (MS 530: 37, 1903).

He also noted the qualification that Σ and Π are, correctly speaking, only similar to quantifiers as sum and product, because the individuals in the universe of discourse may be infinite ("innumerable"), in which case we cannot write quantified propositions in terms of infinite clauses consisting of the Boolean disjunction and conjunction: "In order to render the notation as iconical as possible we may use Σ for *some*, suggesting a sum, and Π for *all*, suggesting a product. . . . It is to be remarked that $\Sigma_i x_i$ and $\Pi_i x_i$ are only *similar* to a sum and a product; they are not strictly of that nature, because the individuals of the universe may be innumerable" (W 5:180, 1885; 3.393).

This observation had repercussions to two related directions. Namely, to do so and to take quantifiers as infinite lists of sums and products was "the biggest mistake" Wittgenstein said to have made in his Tractatus (1921). On the other hand, Ramsey, the convenor of parts of Peirce's philosophy, also opposed the infinitary character of logical products and sums.

In Peirce's treatment, the more prominent of these two quantifiers is Π, which fell from the relative sum in his algebra.[3] We, of course, recognise it today as the universal quantifier $\forall$. This prominence is important in view of the evolution of the meaning of its mirror quantifier Σ, which fell from the relative product operation. While the term "universal" was indeed approved by Peirce, contrary to what is suggested by the received terminology of the better part of the 20th century, the sum operation did not at first refer to existence, however.

Even in EGs, the sum quantifier, as used by him to draw comparisons between symbolic/algebraic and iconic/diagrammatic representations, was associated with the line of identity.[4] Since there is no special notation for quantified variables in the beta part, Peirce took the lines of identity to also depict which individuals, correlated with the predicate names (indices) of the algebraic system, are identical with each other.

The diagrammatic and graphical method of drawing these links has turned out to be effectual, not only because it allows one to intersect cuts by ligatures composed of a collection of lines of identities, but also because one can symbolise any n-ary relations by lines attached to the hooks of spots. The positive part of what Burch (1991) calls Peirce's reduction thesis states that all relations with an arity of four or more are reducible to triadic teridentity relations. The negative part states that teridentities cannot be constructed solely from one or two-place relations. Furthermore, the diagrammatic notation literally forces the interpretation of graphs to be an onion-like process that proceeds from the outermost occurrences towards inner, contextually constrained occurrences. Peirce argued that diagrammatisation implied the "Endoporeutic Principle" (MS 293: 53). The ligatures are not graphs because they may involve some discontinuous, non-compositional parts, not definable by inductive construction. That these lines depict also existence of individuals is merely a by-product of the way in which they cope with the other, more primary topological and continuous notions such as interaction with predicates (predication).

There are two characteristics of lines of identities that are worth pointing out. Unlike what can be said about the existential quantifier of the ordinary language of first-order logic, the line of identity is a "perfect" sign in the sense that it is, at the same time, an icon, an index and a symbol (4.448, c.1903). This semeiotic perspective allows us at once to see what the distinction between the line of identity and the existential quantifier is. The latter is a symbol, because it is not linked to an outside utterer or assertor, and so does not excite any images in consciousness. An index, in contrast, assures us of the existence and presence of objects by one or another mode of identification. This goes beyond the resources of the received idea of quantifiers in first-order logic. The line of identity is iconic, which follows from it being a graph. Stripped of iconicity, its symbolic nature is what remains of the kind of sign that is taken to assert existence by convention.

The consequences of the iconic lines of identity turn out to be far reaching. They put the entire Frege–Russell ambiguity thesis in a new light. What Frege and Russell claimed was that the verb *be* is four-way ambiguous. According to them, it asserts existence ('Socrates exists'), predication ('Socrates is bald'), identity ('Socrates is the main character of Plato's writings') and subsumption ('Plato's writings are on the shelf') (Hintikka, 1983). A moment's reflection on the beta graphs reveals that the line of identity plays a key part in all of these different uses. Aside from predicate terms (rhemas), only in subsumption one needs some additional ingredients, namely two instances of a cut (the scroll) to express the conditional.[5]

Peirce called what was later to become the existential quantifier the particular selective or the particular quantifier. It was meant to inform the utterer of the proposition how she is to pick out a suitable object intended by the given statement. Dually, the universal selective informs the interpreter how any object is to be picked within the limits understood. In both cases, the assertion containing the selective is intended to apply to the chosen object. If the universes are discrete, these choices are individuals, although we must keep in mind the synechist inclination that "in truth a universe may be continuous, so that there is no part of it of which every thing must be either wholly true or wholly false. For example, it is impossible to find a part of a surface which must be all one color. Even a point of that surface may belong indifferently to three or more differently colored parts. But the logic of continuous universes awaits investigation. . . " (2.339, 1895, ellipsis in the original). The crucial thing to note is that whether such a chosen object actually exists is an outcome of an inferential process that starts from the action and the fact that the object applies to the predicate and ends with a conclusion that something exists. The ensuing existence is typically not asserted in the proposition, but is a result of inferential examination. The later logicians, by deciding to call Σ the existential quantifier, sidestepped such inferential practices and glossed its subtle meaning with the non-descriptive epithet 'existential'. In doing so, they decidedly masked the nuances in the early meaning of quantifiers. Over and above that, they masked the pragmatically-lenient difference between being and existence.

The illusion that such nuances were marginal was perpetuated by the presupposed dual nature of quantifiers. Given that the original algebraic meaning of universal selectives was that there is a relation in some universes of discourse so that no object demonstrating some exception to what the universal selective asserts may exist, it was only natural that particulars should then denote existence.

This ought not to be taken to mean that the rudiments of existence in relation to particular quantifiers were deprived or altogether missing in Peirce's logical investigations. The existence of objects or individuals is the end result of a specified act of perception, generated by the utterer of the proposition by representing its intended object (the intentional interpretant) to the interpreter, thus producing the effectual interpretant. There is plenty of evidence for this line of thought in Peirce's manuscripts: "A proposition which . . . leaves its interpreter no freedom of choice as to what it is to be applied to, namely, a singular or a particular proposition, asserts existence, . . . i.e. not merely universally predicates existence, but represents that there is, will be, (or *would be*, but this amounts to nothing unless it leads to 'will be') a perceptive act in which that which is indicated is formed upon said interpreter".[6]

Again, in bringing out the dimension of existence in relation to particular propositions of the algebra of logic, Peirce was influenced by Mitchell's *On a New Algebra of Logic* of 1882. The gradual shift towards particular quantification as existential was a further step in his self-biographical draft (MS L 107: 8), in which Π and Σ were taken to "show whether the individuals are to be selected universally or existentially, that is, by the interpreter or by the utterer".

This shift was finalised in 1913,[7] six months before Peirce's death, when he revisited the fundamentals of his EGs and complemented his general definition with five more conventions (or emendations to the older ones thereof). The first two are of relevance here: that a dot stands for a particularity, a "something" or "a real object", and that a prolonged dot as a line asserts existence, plus identity of the objects denoted by its extremities if extended to dots hooked with spots. (For completeness, the third convention was a branching line of identity asserting co-identity, the more usual meaning of which he took to be a special kind of conjunction such as *and* in "identity *and* identity", i.e. coexistence. The fourth replaces the notion of a cut as a continuous line encircled around any graph by shading an area of the sheet, effecting a denial of what is scribed on that shaded area. The fifth asserts the endoporeutic, "endogenous", flow of interpretation and will be returned to in sect. 5.)

The qualification to the EG conventions in the late writings therefore demonstrate that there is a difference between a dot and a line, the former being static, non-interactive icon for particularity or being, while the latter is the dynamic, interactive sign denoting existence, an encounter of the mind and the non-mind, a rupture to the continuous sheet of assertion. From the pragmatic viewpoint, "to exist" is distinct from "to be". It is reaction against other objects, or object of a percept against a judging perception of a mind, a chain of efforts and resistance exercised by the non-ego of an object and an ego of a mind comprehending it. Any quantification asserting existence per se takes a dishonest and simplifying leap from an active to a passive, non-dynamic interpretation.

In sum, the lines of identity play a quadruple role in EGs (plus the fifth one of coreference if extended to intersentential dialogues): they are used to assert identities between expressions, to depict predication, to account for subsumption of individuals, and to provide instructions concerning the selection of a particular object that fits the intended statement and thus existence.

Moreover, in the gamma part of EGs, the lines of identity assert identities across states of information. In tinctured EGs, which Peirce later abandoned by mistake as "nonsensical", the lines were taken to assert different modes of being of individual objects, according to the indication of the tincture and colour of an area of a sheet (MS 295: 48). This role is different from the others, because in order to accomplish such an identity, the lines need to assert something stronger, namely that the objects drawn as identical also have to be rightly identified in order to make sense of what is meant by an identity when a multiplicity of states of information is in question. This comes close to the infamous problem of cross-identification that has baffled modal logicians since the mid-1900s. I will return to the question of the relationship between modality and quantification later in this chapter.

If the existence was taken as an active process that needs to be accomplished and maintained by an exertion of cognitive force, what are the implications of this to the philosophy of logic and mathematics?

3. The rise of constructivism

In ordinary parlance, one confronts a similarity between the locutions 'what is found/what is chosen' (or 'what can be found/what can be chosen') and 'what exists'. In mathematical prose, it is customary to substitute 'there exists b' by a modalised free-choice version 'one can find b' or the imperative 'choose a suitable b'. As I illustrated above, as far as logic is concerned, Peirce was not especially enthusiastic in laying out the meaning of a particular quantifier — whether taken in the algebraic sense or in the sense of the lines of identity in EGs — in terms of what objects exist in the universe of discourse. Rather, he insisted on taking the meaning of selectives to revolve around what can be chosen by the utterers and the interpreters of the statement in a dialogical setting. This brings his logic of quantifiers very close to game-theoretic semantics (GTS). It also suggests that he was close in spirit to what, in modern terminology, is meant by constructivist foundations of logic, which to some extent overlap the tenet of intuitionism.

However, convincing evidence that Peirce advocated constructivist or intuitionist stances is hard to come by. Such evidence nonetheless exists, although it is not salient. First of all, a constructivist character surfaces in particular statements, which assert existence of nothing until an appropriate collateral examination has been conducted that decides the applicability of the chosen or constructed object.

> Every proposition refers to some index: universal propositions to the universe, through the environment common to speaker and auditor, which is an index of what the speaker is talking about. But the particular proposition asserts that, with sufficient means, in that universe would be found an object to which the subject term would be applicable, and to which further examination would prove that the image called up by the predicate was also applicable. That having been ascertained, it is an *immediate inference*, though not exactly *asserted* in the proposition, that there is some *indicable object* (that is, something *existent*) to which the predicate itself applies; so that the predicate also may be considered as referring to an index. Of course, it is perfectly legitimate, and in some aspects preferable, to formulate the particular proposition thus: "Something is, at once, an inhabitant of Mars and is red haired," and the universal proposition thus: "Everything that exists in the universe

> is, if an inhabitant of Mars, then also red haired." In this case, the universal proposition *asserts* nothing about existence; since it must already be well understood between speaker and auditor that the universe is *there*. The particular proposition in the new form asserts the existence of a vague something to which it pronounces "inhabitant of Mars" and "red haired" to be applicable. (2.369).

The constructivist tone in this passage lies in the idea that the notion of existence is subordinate to the activities of seeking a suitable object for the subject term from the universe of discourse. Existence is a result of inferential processes or cognitive constructs that enable one to assert that one has found an indicable object. It is the indexical relation between the predicate and the object, not what the predicate itself is capable of pronouncing.

What is also noteworthy in the quotation is that the assertion-making, which the speaker and the auditor are responsible for and which carries the risk of punishment if assertions turn out not to be reliable, does not in itself carry existential presuppositions. By asserting something, the interlocutors cannot be accused of making a false (or likewise true) assertion in the form of a proposition merely on the basis of whether the objects adduced by the assertion exist or not.

The normative character of assertions is prevalent in numerous contexts of Peirce's writings. As with the role of communication in his sign theory (Chapters 2, 13), here, too, one should not assume the parties upon which such norms are imposed to be real persons engaged in real conversations. Purely theoretically, an agent can be held responsible for its actions that produce assertions visually, orally, tacticly, or otherwise. When it is a question of the truth of propositions, the responsibilities are not primarily related to socially-constrained factors any more than interpretations within a single quasi-mind can be seen as social. The liability to punishment under social or moral code if a proposition is not true (2.313, 2.315, c.1902), or being subject to the penalties visited on a liar if the proposition asserted is not true (8.337, 1904), or heeding the rule that "if a lie would not endanger the esteem in which the utterer was held, nor otherwise be apt to entail such real effects as he would avoid, the interpreter would have no reason to believe the assertion" (5.546, c.1908), are all examples of strategic content that propositional assertions bear in Peirce's logic. The punishment imposed on one of the parties in the case of failing to assert truthful propositions needs to be brought out in the payoffs of the strategies that the parties engaged in communication will entertain in the course of interpreting propositions.

By calling for such view of assertions, Peirce does not merely demand that the parties to a communicative situation have to act rationally and hence may never intentionally cheat — rationality is one of the presuppositions — he also claims that a genuine chance exists that the assertions will turn out to be of different truth value than was presumed. By tying assertions in with a scale that evaluates them according to value builds truly strategic considerations into dialogical settings.[8]

Although the constructivist and intuitionist approaches to the foundations of logic and mathematics that came into being shortly after Peirce's death have advocated other meanings for the so-called existential quantifier than outright existence, these other meanings alone do not suffice to characterise these approaches. What is needed is an understanding of the negative sign that shows the limited range of applicability of the law of excluded middle, and that a fortiori precludes the use of indirect proofs. Existence of an object may not ensue from the demonstration of the impossibility of its negation.

> The first [general law of thought] is that any unit (or units) whatsoever contemplated in itself without conscious regard to its parts would, were our sense to respond to it, be seen to embody a monad. De Morgan propounded this law, so far as it is pertinent to formal logic, affirming that any collection of objects whatsoever possess universally some character which belongs to no other object at all. For, said he, they at least possess the character of being units of that collection. Considered as a *proof*, this begs the question; but considered as another way of formulating the same phenomenon, and as a way which throws some light upon it, it has its value. This coincides with the principle of excluded middle. Those objects of the universe which do not possess a given character possess another character which, in reference to that universe, is in the relation of negation to the first. Hence, it is impossible to form a single class of dyads; two classes of dyads must be formed at once. Hence, considering all the monads which can appear on the contemplation of sets of units of the universe in their monadic aspect, every single unit is determined to be one subject of a dyad which has any one of those monads as its second subject, namely it is either such a dyad as determines it to have the character of being one of the units which made up the object of the contemplation in which that monad appeared, or it is such a dyad as determines the unit to have the character belonging to all the other units of the universe. (1.450, 1896, *Phenomenology: The Three Categories*).

The need for the system that shows a limited applicability of the law of excluded middle is present in Peirce's triadic logic.[9] The failure of that law also shows up in his logic of vagueness: before one of the functionaries serving as the interpreter of the statement has determined the identity of a given proposition, it remains general and hence may be neither true nor false. Unlike what might be expected from much later-matured partial and fuzzy approaches to vagueness, the proposition to which the law of excluded middle does not apply is not a vague proposition but a general "any proposition you please" (5.448, 1905) and hence an indeterminate, yet a vague, non-general and non-determinate proposition "whose identity I have not yet determined" is one to which the law of contradiction does not apply. A non-determinate proposition is a potential contradiction, but it cannot actualise as long as the functionaries are making rational choices and the dialogue is strictly competitive. In Baldwin's Dictionary, Peirce defined a vague proposition as one in which there are "possible states of things concerning which it is intrinsically uncertain whether, had they been contemplated by the speaker, he would have regarded them as excluded or allowed by the proposition" (DPP 2:748, 1902).

What he meant by "intrinsically uncertain" is worth noting well, for he regarded it in the sequel as something "not uncertain in consequence of any ignorance of the interpreter, but because the speaker's habits of language were indeterminate". As I argued in the previous chapters, strategies in the game-theoretic sense may be regarded as instances of interlocutor habits. What Peirce's explication suggests is that indeterminacy is indeterminacy in the strategies, in other words, the language game that produces vague propositions allows potential contradiction with non-zero-sum payoffs and non-strictly competitive strategies by virtue of such indeterminacy.[10]

Around the same time, but independently of Peirce, the division between different negations and the ensuing critique of the law of excluded middle was proposed by Brouwer's teacher Gerrit Mannoury of the Netherlands (Mannoury, 1947). All the same, limiting the applicability of the law of exclude middle was one of Peirce's major advances in logic, the impact of which is felt, for instance, in the much more recent discussion of first- versus second-order logic in the foundations of mathematics. Unfortunately, he did not link his insights to his numerous other systems of logic such those of multiple dimension or of limited universe of marks.

There is much further evidence of Peirce's constructivist leanings in his affinity with the dialogical and game-theoretic foundations. He sets out the meaning of expressions in terms of collateral inspection and collaboration between the Utterer and the Interpreter of the proposition, or in EGs between the Graphist and the Grapheus. If we take it that such interaction takes place between actual users of natural language or logical expressions, we are typically dealing with the dialogical (proof-theoretic) games along the lines of Lorenzen & Lorenz (1978), but if we take the users to be imaginary, purely theoretical constructs, we are dealing with semantic games. Dialogical games as originally specified account for intuitionist provability, and classical logic is derived only if procedural conventions of the game are adjusted. More precisely, if a player of a dialogical game is not allowed to change her choice of, say, a disjunct, but is required to fix it in advance, it is a question of constructive logic, and if she is allowed to change her mind, it is classical logic.

On the other hand, the road to constructive logic in GTS lies in restricting the set of strategies that the players use to those that are recursive. There may be other strictures within the loose bounds of recursivity such as having only learnable strategies. At all events, constructivism is a live possibility in both dialogical and semantic games. It is only a small step from Peirce's logic to their worlds. I have found no textual evidence that Peirce would have allowed the partakers to change their minds in the course of interpreting a formula, even though he allowed the Grapheus to change his mind concerning the determination of the universe (Chapter 4), and so from the viewpoint of dialogue, his logic is indeed prone to produce a constructive system.

The reason why Peirce did not fully embrace the game approaches to logic was that the concept of strategy that is integral to game theories of rational decision was not available at the time. The first mathematical outcome of games was suggested by Zermelo in 1912 concerning the game of chess. The notion of strategy was properly formalised during the 1920s by Borel, von Neumann and others. However, I argued in Chapter 3 that it is not a considerable step to be taken from the conception of a habit to the conception of strategy.

Furthermore, I noted that assertions bear strategic content. What is more, strategic dimensions of habits are even applicable to evolution theory in that they correlate with stable sets of strategies (Chapter 11). Such evolutionary strategies provide one of the most closely affiliated modern counterparts to Peirce's organic idea of habits. On the more general level of the history of ideas, it remains to be seen precisely how the concept of habit that was widely used in the late 19th century came to mean, as I believe, the institutionalised concept of strategy during the early decades of the 20th century, only to be blasted from mainstream philosophy into oblivion.

Furthermore, a triadic logic may be derived from games by applying the simple fact that from the non-existence of a winning strategy for one of the contestants it does not follow that there exists a winning strategy for his or her adversary. This reflects the understanding of negation in Peirce's three-valued triadic system, as an operation that does not hamper the third truth-value L that according to Peirce refers to the propositions the truth of which are 'not known'.

The reference to the truth-value of 'not known' is in keeping with the idea familiar from GTS that to reveal the truth of a quantificational sentence is to know what sequences of strategies there exist that lead a player invariably to a winning position. If the truth of such a sentence is not known, then it is not known whether such sequences by which a model in which the sentence is evaluated exist. Peirce's motivation in the triadic logic was to come up with this kind of active demonstration of the emergence of the third truth-value, and indeed, he did not speak of any unexplicated truth-value gaps.

Moreover, intuitionists have often, albeit by no means unilaterally, abandoned the reality of infinitary constructions in mathematics. A crucial difference from the common conception of intuitionism is that Peirce's position in mathematics was far from that of the finitists. He by no means thought the mind to not be powerful enough to capture, characterise or complete infinite constructions or processes: "It is the human mind that is infinite" (7.380).[11]

Writings on mathematical approaches to collections abound with concepts requiring infinity. Intuitionism and finitism need not and do not conflate, but many philosophers of mathematics, such as Brouwer, threw finitism into their constructivist shopping cart. The reason for this was the 'demonstrably unfinished' nature of uncountability, in other words the lack of a proper method of construction, which seemed to hinder progress in infinite mathematics.

As an upshot, the claim made in the classical work on the development of Peirce's thought has to be reassessed. According to Murphey (1961, p. 288), "Peirce's theory thus stands between those of the intuitionists and the logistic school. [...] In spirit, however, Peirce has more in common with the logistic school than with intuitionism”. For one thing, Murphey does not take into account Peirce’s EGs, which are geometrical constructions and as such appeal to judgements. Logicists such as Frege would not have approved this. For Peirce, mathematics was autonomous from other sciences, logic included. Nevertheless, it seems correct to say that Peirce was by no means averse to constructivism, while he is likely to have been averse to finitist versions of intuitionism, that is, to the view that eschews any reference to infinite collections.

Whatever intuitionist or constructivist leanings in mathematics Peirce may have eventually advocated would not have been the result of some changes in his thinking of the logica docens, such as the disposal of the law of excluded middle in his triadic logic and in modal logics dealing with would-bes. The point that Peirce realised clearly was that different mathematics ensues not because of different logic, but because of different philosophical views on what essential properties one takes there to be in the minds of those contemplating mathematical questions.

4. Two and three in tension?

Another reason why the meaning of the particular quantifier was laid out in terms of what can be chosen by the utterer of the proposition rather than in terms of existence is Peirce’s overall dialogical approach to the meaning as well as to thinking, consciousness and the evolution of thought. He attempted to illustrate the emergence of experience as dialogical interaction between the inner and the outer, or the potential and the actual.

> Although in all direct experience of reaction, an *ego*, a something within, is one member of the pair, yet we attribute reactions to objects outside of us. When we say that a thing *exists*, what we mean is that it reacts upon other things. That we are transferring to it our direct experience of reaction is shown by our saying that one thing *acts* upon another. (7.534, *Consciousness*).
>
> There are in experience occurrences; and in every experience of an occurrence two things are directly given as opposed, namely, what there was before the occurrence, which now appears as an *ego*, and what the occurrence forces upon the *ego*, a *non-ego*. This is particularly obvious in voluntary acts; but it is equally true of reactions of sense. (7.538).

Peirce also explains phenomena such as surprise and expectation in terms of similar dyadic interaction.[12] In doing so, he is putting himself in an out-and-out Kantian mood. Existence is a capacity for uniting experiences in consciousness. The two interactors, the ego and the non-ego, refer respectively to the inner and the outer worlds of the observer who acts upon Nature.

How, then, does the wide-ranging reliance on dyadic interaction in logic and consciousness fit in with Peirce’s fundamentally triadic constitution of his sign theory? The answer is that, while his semeiotics is permeated by trichotomies, in tackling the nature of thinking and consciousness, he often prefers to move on the level of dyadic duels.

> The idea of second is predominant in the ideas of causation and of statical force. For cause and effect are two; and statical forces always occur between pairs. Constraint is a Secondness. In the flow of

> time in the mind, the past appears to act directly upon the future, its effect being called memory, while the future only acts upon the past through the medium of thirds. Phenomena of this sort in the outward world shall be considered below. In sense and will, there are reactions of Secondness between the *ego* and the *non-ego* (which non-ego may be an object of direct consciousness). In will, the events leading up to the act are internal, and we say that we are agents more than patients. In [a] sense, the antecedent events are not within us; and besides, the object of which we form a perception (though not that which immediately acts upon the nerves) remains unaffected. (1.325, *Phenomenology: The Categories in Detail*).

If secondness is so central, does not a tension arise between dyadism and triadism? Actions and reactions, or dialogues, simply are secondness, aren't they? And thirdness would be dispensable?

I believe that a more accurate description is that secondness, by virtue of standing for reactions, renders both the dyadic and the triadic divisions equally important in logic. It is unfortunate that studies on Peirce have concentrated almost exclusively on some single, isolated triads of his theory of signs, and the general place of interaction and communication has been left unallocated.

One reason for this oversight has been the strong reliance not only on his own arguments for the reducibility thesis, but also on those who have studied the thesis in detail. According to the positive part of this thesis, all relations of arity three or more can be expressed in terms of three-place relations. The negative part states that no non-degenerate three-place relation may be constructed from one or two-place relations. True, the thesis is mathematically uncontestable. Peirce used it to argue against any higher-than-three n-nesses, and the whole architectonic philosophy builds upon it. In one of his last writings he still maintained that "in Logical Analysis no higher valency than 3 is ever called for" (MS L 477, 14 October 1913).

Quinean reduction of all relations to dyadic ones (Quine, 1953) — a mathematical correlate to his barren metaphysics of 'word plus object, third debarred' should thus be mentioned here. Would that have been some interest or value to Peirce? Would such a reduction have encouraged him to put more weight on his dialogical approach to the congregate concepts of thought, consciousness, existence and experience? I contend that any answer to such a misplaced question is bound to be on the negative side. More assumptions are needed in Quinean reduction than would have been allowed by Peirce. The concepts of negation, relative product and the three-place identity relation (teridendity) have very precise meanings in his irreducibility argument. Teridendity is not just identity and identity, or "binidentity" (SS: 199, 9 March 1906), which just prolongs two identity lines, given his understanding of the relative product. For instance, in the realm of EGs, a predicate attached the heavy dots extended to lines of identity does not give rise to teridendity because no two lines may terminate at the place occupied by just one dot. Dots are attached to hooks on the periphery of the rhema, marking the place where the physical correlate to the topological singularity of a line as a particle either starts its movement or has to terminate.

This is the core of the argument for the irreducibility of teridentity. By holding on to this diagrammatic, interactive and topological understanding of the role of the dots and hooks, together with the prolonged continuous lines and particles moving along the lines and occupying the places of the dots, Peirce deliberately refused to release existence from the iconic and continuous clutches of identity.

The picture here would not be complete without reference to game-theoretic approaches in the semantics of logic, natural language, computation and programming languages. In computer science, the iconic label of 'Geometry of Interaction' (Chapter 8; Abramsky et al. 1996) has been in vogue. It has been asserted that it takes precisely two agents to initiate and conduct the rule-governed process of interaction for computational purposes. This is shown in the reduction that enables all n-ary $(n > 2)$ strategies, understood as relations, to be expressed in terms of nested compositions of two-ary ones. In other words, Peircean reduction is augmented by an understanding that teridentity is now composed of two identities plus an ordinary truth-functional conjunction.

This argument, however, is an instance of Quinean reduction. The 0-level of logic with no players merely gives the truth values. As in Tarski semantics, it has no mechanism of telling how the truth-values come about. One player is capable of action, but it takes two to engage in interaction.

But Peirce's logic is congenial to geometries of interaction. The analogue stems from viewing players or agents as sets of strategies, so that no more than two sets are ever called for. One of these is for the proponent of an assertion (the system, the agent, the user, the processor, Myself), and the other, the set of counter-strategies, is for his, her or its adversary, the opponent of the assertion (the environment, the counter-agent, the client, the input, Nature).

The tension that apparently now raises its head is that Peirce is assuming semiosis to take place between two agents and thus between two sets of strategies, while that assumption — with the hindsight of the strategic perspective — bears a different understanding of what the permissible assumptions are from that of the irreducibility of three-place relations. The view of communication that Peirce advances is triadic, not between three agents but being mediated via the third, the sign (Chapter 13).

This extra elements introduced by strategies explains why interaction is not in discrepancy with triadic communication. Strategies add a great deal of organisation and structure to the basic notion of the interactive processes. This added structure enables us to distinguish the basic process-like structure of interaction in logic from the *causa finalis* of purposeful action provided by strategies.

Furthermore, the incorporation of strategies is a step towards a wider appraisal of the category of thirdness in logic and computing, and an acquittal of constructivism via the ensuing computational notion of realisability of strategies (or players' interactive programs). Peirce would have been thrilled to study these later developments.

In the light of these post-Peircean developments, an oversight just as unfortunate as underrating the general impact of communicative aspects on his logic is the hasty exclusion of thirdness from the provinces of semantics and model-theoretic approaches to logic (see the following section). This was a tendency in the writings of his contemporaries, most notably those that he considered to be allotted the epithets of 'symbolic' or 'formal', that worried him. Here, of course, formal refers not to the mathematical component of logic, but to the tendency of manipulating formulas by rules of inference that disregard the interpretation of non-logical constants.[13]

That secondness supersedes firstness and thirdness was an experience rather than a conception for Peirce. It is remarkable that such an understanding is itself an example of a capacity of observing and reflecting on secondness, a kind of 'meta-secondness', where the ego of a philosophising mind and the resisting non-ego meet:

> The practical exigencies of life render Secondness the most prominent of the three. This is not a conception, nor is it a peculiar quality. It is an experience. It comes out most fully in the shock of reaction between ego and non-ego. It is there the double consciousness of effort and resistance. That is something which cannot properly be conceived. For to conceive it is to generalize it; and to generalize it is to miss altogether the *here*ness and *now*ness which is its essence. According to me, the idea of a reaction is not the idea of two *plus* forcefulness. (8.266, 1909, *Letter to William James*).

The significance of dyadic relations in the constitution of the triadic theory of signs, even though not formally needed, is shown by the fact that, in order to be properly interpreted, signs require that there be dyadic relations between the two repositories of a mind, or quasi-minds in the case of inanimate beings. Peirce expresses this in the often-quoted but equally often missed passage thus: "Signs require at least two Quasi-minds; a *Quasi-utterer* and a *Quasi-interpreter*; and although these two are at one (i.e., are one mind) in the sign itself, they must nevertheless be distinct. In the Sign they are, so to say, *welded*. Accordingly, it is not merely a fact of human Psychology, but a necessity of Logic, that every logical evolution of thought should be dialogic" (4.551).

The concept of a quasi-mind is a key to the theory of signs that Peirce despaired of making understandable. In a letter to Lady Welby (SS: 195; MS L 463, 3 March 1906), he draws a parallel between thinking that requires a mind on the one hand, and signs that require, or are determinations of, quasi-minds on the other. There are few indications that this parallelism is strict in terms of the function and purpose of quasi-minds and the minds of real beings. The cut-off point is the intended intension in his concept of commens, which is "that mind into which the minds of utterer and interpreter have to be fused in order that any communication should take place" (SS: 196–197; EP 2:478). This mind determines the dynamic object of cominterpretant in meaningful communication. Such objects are not located outside of the minds, yet they are capable of representing a shared element of communication. Their determination, the commens, is also a central concept in many recent pragmatic theories of language resorting to the notion of the common ground. Moreover, Peirce's concept of the dynamic object, floating in the conversation, bears a relation to the concept of the discourse referent in the discourse-representation theory. Also, it takes in what has much later been termed "accommodation" in the field of pragmatics (Lewis, 1979).

The significance of the commens in the theory of signs is indicated in Peirce's omitted remark that it consists of all that is understood between communicative parties "when the sign in question is just about to be made".[14] This was a phrase that Peirce rejected, and amended to the more secure "in order that the sign in question should fulfil its function".[15]

Moreover, one should not try to explain interaction in terms of some other activity that might appear more comprehensible. Secondness generates experience, and in logic it refers to the idea of evolving thought, reasoning and experimentation. All these aim at veritable experience. In other words, "the duality of the *ego* and *non-ego* is the chief constituent of the idea of the Truth" (MS 515: 24). This duality shows up in modern game-theoretic approaches to meaning in terms of truth-conditional semantics, in which truth is equated with the existence of winning strategies. This explains how dialogic communicative processes tie in with logic, which, as I observed in Chapter 2, is the missing link in Habermas' reading of Peirce, thus believing erreoneously that there is a tangible conundrum between communicational and logical perspectives undermining Peirce's system of semeiotics.

The preceding considerations take us back to our pet passage from earlier chapters, but this time we have a broader context. The fact that diagrammatic logical graphs and dialogical thinking reveal fundamentally the same phenomena is expressed by Peirce as a requisite in order for the reader to

> fully understand the relation of Thought in itself to *thinking*, on the one hand, and to graphs, on the other hand. Those relations being once magisterially grasped, it will be seen that the graphs break to pieces all the really serious barriers, not only to the logical analysis of thought but also to the digestion of a different lesson, by rendering literally visible before one's very eyes the operation of thinking *in actu*. In order that the fact should come to light that the method of Graphs really accomplishes this marvellous result, it is first of all needful, or at least highly desirable, that the reader should have thoroughly assimilated, in all its parts, the truth that thinking always proceeds in the form of a dialogue, — a dialogue between different phases of the *ego* — so that, being dialogical, it is essentially composed of signs, as its Matter, in the sense in which a game of chess has the chessmen for its matter. Not that the particular signs employed *are* themselves the thought! Oh, no; no whit more than the skins of an onion are the onion. (About as much so, however.) One selfsame thought may be carried upon the vehicle of English, German, Greek, or Gaelic; in diagrams, or in equations, or in graphs: all these are but so many skins of the onion, its inessential accidents. Yet that the thought should have some possible expression for some possible interpreter, is the very being of its being. (MS 298: 6; 4.6).

Peirce's motivation for such an allegory was his view of thought as presenting moving images of actions of the mind. He elucidates this MS immediately preceding the previous extract by offering a twofold approach to the structure and working of thought, namely one endowed with the distinctness of geometrical diagrams plus the convincingness of "working models" (MS 298: 5). The editors of the Collected Papers (vol. 4) omitted this important qualification to the distinctness of geometrical diagrams from the publication of the relevant paragraphs of *The Simplest Mathematics*. This is unfortunate, not least because it related to the idea of diagrammatic logic as moving pictures of thought but because the concept of working models turns out to be the culmination point in the run-up to one of Peirce's most spectacular discoveries in logic.

5. The endoporeutic method

It has not been widely recognised that for his logic, and in particular for his system of EGs, Peirce came up with a version of what has subsequently developed under the auspices of model theory. The invention of model theory is habitually credited to people such as Alfred Tarski, Anatolii Mal'tsev and Abraham Robinson, who, in the 1950s, proposed the term for the mathematical discipline that had emerged sometime earlier, most notably during the phases of discoveries in mathematical logic at the beginning of 1930s.

Tarski's purpose was to develop a formal language such that any of its expressions having the same meaning can always be replaced for each other.[16] In model theory, the interpretation A of a sentence S of a language L, by a mapping of non-logical and logical symbols of S to a domain of a structure, is called a model of S, if it makes S true in A. The inverse of this mapping provides an assignment function. Thus the basic predicates, constant symbols and function and relation symbols (if there are any) will acquire an interpretation. Predicate and relation symbols with a fixed arity are interpreted on the product of the relations of the domain, and the constant symbols are interpreted by assigning them an element of that domain of discourse. Furthermore, assignments so defined need to be extended to valuations assigning values also to the variables of the non-logical alphabet.

Here, Peirce's dialogical interpretation enters the picture. In EGs, the work of assignments is done by the players who choose objects from the universe of discourse so that they fit the intended statement. As the interpretation A that delineates a specific class of objects is a structure, and as in EGs a sentence S is scribed in diagrammatic form as a graph G, the structure that interprets G is another graph G' onto which G is homomorphically mapped. It is from the universe of G' that the players choose objects to which the expressions in G refer, and which fit the statements intended by them.

The function of the unextended assignment was understood by Peirce to be all that which is mutually agreed to exist and known to be mutually agreed by the Graphist and the Grapheus (or by the utterer and the interpreter if we step outside the iconic contexts of the diagrams). However, before values (proper names, indefinite names, selectives) may be chosen and attached to the dots or identity lines hooked to the predicates, the domain will have to come with a (non-exhaustive) supply of names for the individuals, and the players would need to have a basic understanding of what the rhemas with blank forms of expression are intended to mean, in order to work as expressions with semantic attributes and thus constituting parts of propositions with definite truth-values. As it happens, the sheet of assertion is such a mutually understood universe. It is not just a domain of individuals, but also an interpreted structure on which complex assertions can be scribed and their status determined according to the truth and falsity. In modern terms, it is the τ-structure of the domain, where τ is the signature (the set of basic symbols of constants, predicates and relations), onto which the scribed graph is mapped by the inverses of the actions taken by the Graphist and authorised by the Grapheus.

How the non-logical constants will receive their interpretations was not for Peirce, unlike for later model theorists, the result of any process of fixing a model. It was part of the 'natural history' of logic. Only via such history interpretations may be determined, through conditions attached to sound thought. That constitutes the starting point of logical inquiry, the point where all study lies open before us. Peirce explains thus, "a logic which is a natural history merely, has done no more than observe that certain conditions have been found attached to sound thought, but has no means of ascertaining whether the attachment be accidental or essential; and quite ignoring the circumstance that the very essence of thought lies open to our study; which study alone it is that men have always called 'logic,' or 'dialectic"' (4.8).

The reason that unextended assignments are no matter of definitions or secondness is due to Peirce's emphasis on logical propositions as assertions put forward in communicative situations. This emphasis on assertoric content and the responsibility of the utterers and the interpreters of their own assertions gets stronger the closer we come to his last writings. In about 1910 he wrote, in shaky hand, "It must be admitted that in all cases an assertion has two parts. One of these [in the simplest cases ?at least is that?] the predicate must refer to a kind of experience and the other, the subject, to a recognizable occasion of an experience of that kind".[17]

This, in brief, is Peirce's model theory, also known as the endoporeutic method in EGs. The term "endoporeutic" (*adj*., cf. "semantic") appears in few places, however.[18] In MS 650: 18, 19 [24 July 1910], Peirce speaks about "endoporeusis" ("inward-going") and the system of graphs expressing the truth "endoporeutically", and in MS L 477 [1913], he uses the adverb "endogenously" as a substitute. Related to these is 4.408 [1903], where he notes that, in general, the interpretation should proceed from the outermost occurrences of selectives towards the insides of the cuts. All this means one and the same thing: the interpretation of graph instances proceeds in an outside-in fashion, starting with the outermost area or outermost extreme of a ligature and proceeding towards the innermost instances. Each step, performed either by the utterer or the interpreter of the graph, peels off one instance, until the atomic instance is left. If there is a cut, the choice of an object from the universe of discourse is transferred to the opponent of the current player.

Statements referring to the endoporeutic method are nevertheless conspicuously rare. Some crucial definitions and passages are likely to have been lost. For instance, among the dozens of technical terms in his index to *Prolegomena* (MS 292: subjacent to page 53), he included "Endoporeutic Interpretation", which was to be found in "531, 23–28".[19]

The term appeared for the first time around 1905, the same year he spoke about "working models", the conception created by the mind's relation to graphs. However, he had already noted in 1903 (MS 491; 4.408) that there was a rule for EGs according to which what is scribed outside an enclosure in a graph is always mentioned before what is inside. The sentences in 514: 16 read: "The rule of interpretation which necessarily follows from the diagrammatization is that the interpretation is 'endoporeutic' (or proceeds inwardly). That is to say a ligature denotes 'something' or 'anything not' according to [whether] its *outermost part* lies on an unshaded or a shaded area, respectively". The reference to shaded and unshaded areas was his alternative notion for negation, which he later changed from a simple closed line to the shading of the area on the sheet of assertion that denotes the negated graph replica.

Endoporeutics was not only semantics — to abuse the more current signification of that term — but it also entitles the rules of iteration and de-iteration: "This right [of iteration and de-iteration] may be expressed by saying that the interpretation of existential graphs is *endoporeutic*, that is proceeds inwardly; so that a nest sucks the meaning from without inwards unto its centre, as a sponge absorbs water" (MS 650: 18, 24 July 1910). The "nest" is a technical term meaning the sequence of areas of cuts from those areas that have the largest number of enclosures to those that have the fewest. It corresponds to the total history (complete path from the root to the terminal history) in the extensive-form representation of EGs (Chapter 4). As the rules of iteration and deiteration are what we nowadays perceive to be proof-theoretic, it is interesting to note how Peirce thought them to be parasitic on model-theoretic considerations.[20] Admittedly, he spoke of diagrammatic "Syntax" being "endoporeutic" (MS 669: 4), but this was a *façon de parler* for not having distinguished syntax and semantics, as by that he meant that lines of identities are understood so that they are interpreted in the outside-in order.

The remark Peirce makes about the "endogenous" interpretation is interesting because there he considers the opposite possibility, that of an "exogenous" interpretation as well.[21] Nevertheless, in his judgement an exogenous interpretation that begins in the inside and proceeds outwards is not to be preferred, because the nested system of cuts (here as in the other examples of shaded graphs) that are read as conditionals (scrolls) — as in the case when there are two nested cuts, A lying on the shaded area and B in the non-shaded area inside the shaded one, read as 'If A is true so is B' (or equivalently, 'It is false that A is false while B is true') — would not assign a proper meaning to implications when read inside out, which in the case of this example amounts to 'B is true and A is false', contradicting the example.

Further textual instances are found among the assorted pages of MS 295. Peirce writes on page 83, "The interpretation of the Entire Graph is to proceed endoporeutically. That is to say, it is the less enclosed parts which must determine to what the more enclosed parts are to be understood as referring, and not the reverse", and on page 87, that the "[?Interpretation?] of all graphs is to be *endoporeutic*; that is to say, the signification of the contents of the Area of any cut is to be taken as depending upon that of the contents of the Place of the cut". This latter statement is particularly significant, since it asserts that the order of interpretation also concerns the areas of the cuts and the places upon which they are asserted but severed from. It is via this ordering that the context-dependency of what is scribed on the areas of cuts is secured. The latter quotation is embellished in MS 300: 48: "The rule that the interpretation of a graph must be endoporeutic, that is, that the graph of the *place* of a cut must be understood to be the subject or condition of the graph of its *area*, is clearly a necessary consequence of the fundamental idea that the Phemic Sheet itself represents the Universe, or primal subject of all the discourse". Again, Peirce is absolutely clear about the requirement that whatever lies in the context must not be lost in the interpretation. Accordingly, the method vindicates the old idea that negation presupposes the affirmation of the proposition.

Besides thus vindicating his model-theoretic and game-theoretic attitude to EGs, he also defined logical equivalence in model-theoretic terms by noting that, "If one rheme [predicate term, represented by letters or words], or verb, would be true in every conceivable case in which the other is true, and conversely, then and only then those two verbs are *logically equivalent*" (SS: 199, 9 March 1906). This is a remarkable statement, especially in light of the fact that conceivable cases are what are nowadays perhaps best viewed as classes of models. Accordingly, what Peirce is anticipating is categoricity (descriptive completeness) of a logic, in other words the uniqueness of models up to isomorphism.

From this definition we also see what Peirce intended logical consequence to be. It is a relation between an antecedent formula and a consequent in which there are no conceivable cases (models) in which the former is true and the latter is false.

Furthermore, in NEM 2:517, Peirce remarks, "In order to demonstrate that any collection A, is greater that any collection B, it is useful to think of the relations r that these collections bear to each other as producing a 'character', which in turn gives rise to a situation which can be thought of as "a sort of diagram, or model, in which we think of the Bs as so may boxes, the As as so many bullets, and the relation r as that of containing". Here Peirce even uses the term 'model' in relation to some conceivable iconic situations, in the context of measuring the multitudes of collections.

Peirce's overall approach to iconicity in EGs may be seen as the proper use of signs as their own models (Hintikka, 1996c). This is not merely an instance of hypostatic abstraction, but also resembles the later semantic techniques of maximally consistent sets in proving completeness for given calculi. The idea was developed by Leon Henkin and many others in the 1940s, using sets of sentences as the models of sentences.

During the last months of his life, Peirce amended the analytic use of diagrammatic methods in deductive logic by noting that every logical term has to be linked or bound to "one real universe" (MS L 477: 11r, 14 October 1913). This is represented in graphs by scribing terms on the sheet describing the universe. The idea of 'being bound to the real universe' comes very close to the model-theoretic idea of being able to speak about interpreted statements by mapping the terms of one's language to the domain of the structure. Peirce claimed to have been struck by this idea in a rejected paper written in January 1897 for the Monist, which is indeed the manuscript in which the theory of EGs was inaugurated.

However, the structures that he had in mind were not set-theoretic, but were applied to any of the more general diagrammatisations intended to capture iconically any thought that the mind produces. In actual cases, these signs are mainly assertions, but also non-declarative moods of commands, questions (in interrogative inquiry), music, imaginations and absurdities were contemplated. Virtually anything that is linked to the mind or the quasi-mind ought to be subsumed to similar iconic and model-theoretic scrutiny. What is more, Peirce was convinced that the study of such structures may be as objective as the study of mathematical and physical structures.

Thus Peirce not only came to provide an early version of model theory and a fresh incentive for its later advancement, he also recognised the vitality of the iconic character of linking scriptures to structures in logical graphs. The importance of the iconic view of logic is well beyond the mere symbolic and in even the received model-theoretic conceptions of logic in mathematics.

A significant question remains open, however. It concerns the fundamental reasons as to why logic after all took the converse 'ectoporeutic' turn, as endorsed by Tarski and many others. Some suggestions for a response to this were put forward in Chapter 4, along the lines that the views that Tarski and others hold on the matter of compositionality occupies the key position in any answer to this question.

Moreover, since Peirce garnered the processual component of dialogical interpretation from the extended notion of an assignment, he would not have run into similar difficulties as one does in cases where the single such assignments no longer work as adequate interpretations for subexpressions. This happens if the underlying language is not compositional. To interpret formulas of such a language in a recursive fashion, one needs to extend the assignment for each subformula or a subgraph into a set of assignments, in order to cope with the would-be interpretations that such non-compositional, endogeneously interpreted subexpressions assume. Such an interpretation increases immensely the complexity of the semantics, however.

Model-theoretic seeds are also discernible in Peirce's more general writings, such as his late *Reflexions upon Reasoning* (MS 686).[22] In that and in much of his work from the last years of his life he tried to tackle the nature of reasoning from several synthesising points of view, making use of the prevailing notions of modalities, assertions, as well as the distinction between logica utens and logica docens (Chapter 1). He wanted to connect reasoning with the notion of "the state of things". The state of things is here a mental concept (logical dynamical interpretant) which, by being conceivable itself, gives rise to other possible and conceivable states of things, again which realise further states, and so on. By a state of things Peirce meant "anything the reality of which might constitute the truth of an assertion" (MS 686: 1). Truths of asserted propositions have to be contemplated in some given or produced state of things at the time, not in the whole sum total of being. Reality itself is not the absolute totality of all there is. By it "is to be understood that part or ingredient of the being of anything which does not depend upon that thing's actually being represented" (MS 686: 1). In this case, states of things possess their real, objective counterparts in some isolated parts or systems of being.

While Peirce's account of the truth of an assertion in states of things may resort to notions of modality and possibility, these states are actualised as soon as the principles of contradiction and the excluded middle are applied. For Peirce, the outcome of having such states was a guarantee that logical reasoning was necessary and certain, no longer open to doubt. In order for reasoning to be reliable, the assertion must hold not of any single state of things, but of all states of things, something of which we have "a trustworthy knowledge of what *would always* be true" (MS 686: 6).

To conclude, many of the aforementioned points bring us to the idea of mental models, familiar from cognitive science (Chapter 1; Johnson-Laird 1983). Hodges (1993) has criticised the theory of mental models on the grounds that it has nothing to do with the kind of model-theoretic reasoning and argumentation in mathematics. I will return to this bone of contention in Chapter 9, but note that both accounts are misguided. Peirce was using a model theory for his EGs fifty years before Tarski suggested the name for what was regarded as a relatively recent mathematical discipline, and thirty years before model-theoretical methods in mathematical and symbolic realms of logic gained credence.

Peirce's model theory would have been quite deficient if not taken to approve the creative link with diagrammatically iconic systems of representation for thought and cognition. The reason for this is the almost total expurgation of what Peirce would have considered the most important part of logical reasoning, the category of thirdness, and the ensuing triadic relation of logical signs to the minds of their interpreters. I will take up a link between Peirce's way of thinking and cognitive linguistics and semantics in Chapter 12.

6. Modality and quantification

Regarding the development of the meaning of quantification, in the first section I noted in passing that the gamma part of EGs would in modern terms amount to several separate ideas. There is no single systems of gamma. One compartment comprises modal logics extended with quantifiers (Øhrstrøm, 1997). It involves the adoption of ideas close to the much later possible-worlds semantics. The other two exhibit the metalogical principle of abstraction and uses graphs corresponding to higher-order type-theoretic logics (Brady & Trimble, 2000) and describing and reasoning about graphs themselves (Roberts, 1973a).

The controversy regarding the gamma status is made all the more captivating by Peirce's references in his unpublished papers to the fourth system in his EGs, namely to the necessity of having yet the delta part: "The better exposition of 1903 divided the system into three parts, distinguished as the Alpha, the Beta, and the Gamma, parts; a division I shall here adhere to, although I shall now have to add a *Delta* part in order to deal with modals" (MS 500: 2–3). He was willing to defer the advancement of modalities by means of gamma graphs when he noticed certain irrevocable inadequacies in them. He was never entirely satisfied with them, but was running short of time to be able to complete the picture himself.

More than enough textual and literary evidence exists of the modal character of gamma graphs. He notes, "The gamma part of the system deals with what can logically be asserted of meanings" (MS 462: 34). (The concept of meaning that he held until 1903 was somewhat different.) Thereafter, he signalled interest in what was taken only much later to reflect meaning: intensions. Elsewhere, he characterised gamma graphs as taking account of abstractions (MS 464: 28), instances of which are hypostatic abstractions. I believe that it is only a minor step from the notion of hypostatic abstraction, which treated relations and predicates themselves as diagrammatic objects of study, to the inception of category theory in mathematics. The germs of the immense conceptual arsenal were already there during Peirce's lifetime.

There is also abundant evidence of higher-order diagrammatic logic in gamma contributions to the logic of collectives.

Aside from the controversy concerning the status of gamma graphs, the logical approach to modalities was one of Peirce's central concerns. He wanted to supersede the mere parallelisms of universality versus necessity and particularity versus possibility, and the idea of modality as involving simultaneous possibilities.[23] He attributed the impetus for the study of gamma graphs to Mitchell's *On a New Algebra of Logic* for rendering the scholastic concepts of modals exact by introducing the technique of the multidimensional logical universe.[24] Peirce perceived the key aspects to do with the affinity of modalities and quantification to be present in Mitchell's contributions to the logic of quantification.

Peirce did not quite succeed in implementing his idea of modality based on Mitchell's concept of a multidimensional logical universe in terms of gamma graphs. He noted that, as far as his conceptual repertoire was concerned, these graphs did not require any radically new kinds of signs beyond those involved in the alpha and beta parts, although such signs may well take some new forms (MS 467: 20). What was required was the replacement of the sheet of assertion by "a book of separate sheets, tacked together at points" (MS 467: 22). He even remarked on the necessity for a relational arrow-like symbol to express relations between the sheets of assertions via abstraction (MS 467: 58, 60).

Peirce's intended meaning with this relation was that one state of information (B) follows another (A). One interpretation of such a precedence notation is certainly as an accessibility relation.

This is the closest he came to the idea that the meaning of modal statements involves relational accessibility between alternative courses of events or alternative states. Of course, this accessibility was the key to the semantics of modal notions, as recognised by a number of inventors of possible-worlds semantics during the 1950s and the early 1960s (Chapter 4). The idea was not properly implemented in gamma graphs and Peirce fares much better verbally. The bottomline was that multiple logical dimensions involve aggregates of hypotheses, the false ones being those that a supposed state of information does not exclude. The whole range of possibility then measures the amount of ignorance in the given state (MS 1147: 3).

There is no evidence that the arrow-like relational symbol introduced in 1903 was linked with remarks about modality such as the one that Peirce provided for Baldwin's dictionary in about 1900. Moreover, as the arrow relation is entirely iconic, his animadversions concerning the lack of iconic character in modalities appear puzzling: "If that be the case, Modality is not, properly speaking, conceivable at all, but the difference, for example, between possibility and actuality is only recognizable much in the same way as we recognize the difference between a dream and waking experience".[25] These animadversions are just the tip of the iceberg in Peirce's at times hasty attempts to navigate the modal deep.

There are numerous junctures at which modal concepts forced Peirce to fall back on the book-of-sheets type of analysis. Mention hypotheticals:

> In a paper which I published in 1880, I gave an imperfect account of the algebra of the copula. I there expressly mentioned the necessity of quantifying the possible case to which a conditional or independential proposition refers. But having at that time no familiarity with the signs of quantification, the algebra of which I developed later, the bulk of the chapter treated of simple consequences *de inesse*. Professor Schröder accepts this first essay as a satisfactory treatment of hypotheticals; and assumes, quite contrary to my doctrine, that the possible cases considered in hypotheticals have no multitudinous universe. This takes away from hypotheticals their most characteristic feature. It is the sole foundation of his section 45, in which he notes various points of contrast, between hypotheticals and categoricals. According to this, hypotheticals are distinguished from categoricals in being more rudimentary and simple assertions; while the usual doctrine of those who maintain that there is a difference between the two forms of assertion is quite the reverse. (2.349, 1895).

Quantifying the possible cases is a most astute reflection of the idea that there is a universe with a set of possible worlds (states of affairs). In Peirce's opinion, Schröder did not advocate this way of viewing hypotheticals, which Peirce thought necessary in order to be able to deal with modalities. He was very clear in this passage in taking the possible cases to refer to the multitudinous universe, the universe consisting of several possible states of affairs or information.

There is plenty of evidence that Peirce supported a fairly advanced outlook on modal semantics, especially in relation to expositions that go beyond the idea that necessity and possibility move on the same plane as universality and particularity. He made frequent use of concepts such as 'states of ignorance', 'range of possibility', 'hypothetical states of universe', 'indistinguishable propositions', and 'conceivable states of ignorance'. All of these contain seeds of modern possible-worlds semantics.

> The propositions analogous to A are all those propositions which in some conceivable state of ignorance would be indistinguishable from A. Error is to be put out of the question; only ignorance is to be considered. This ignorance will consist in its subject being unable to reject certain potentially hypothetical states of the universe, each absolutely determinate in every respect, but all of which are, in fact, false. The aggregate of these unrejected falsities constitute the "range of possibility," or better, "of ignorance." Were there no ignorance, this aggregate would be reduced to zero. The state of knowledge supposed is, in necessary propositions, usually fictitious, in possible propositions more often the actual state of the speaker. The necessary proposition asserts that, in the assumed state of knowledge, there is no case in the whole range of ignorance in which the proposition is false. In this sense it may be said that an impossibility underlies every necessity. The possible proposition asserts that there is a case in which it is true. (2.382, 1901; DPP: 89).

Furthermore, in the same dictionary entry on modality he noted that Lange said (in *Logische Studien*) the nature of modality to be "put in the clearest light by the logical diagrams". This was probably one of the major sources of inspiration for Peirce to embark on the study of gamma in the first place.

Peirce had his reasons for not analysing modalities by means that would lead to relational modal models, in which relations obtain between different states of the model. For what was one of the most important and most difficult questions for Peirce not only in logic and mathematics, but also in how they interface with metaphysics, was the notion of continuity. The synechistic doctrine that ensued from mathematical and logical continuity does not apply to the possible-worlds analysis of modal assertions. But for Peirce, modalities *ens rationis* need to be in a continuous connection with actualities.

Hence, just as he suggested the study of continuous universes as the next big fish to be caught in the science of logic, similar considerations would need to be carried over to bear on possible-worlds conception of modal assertions. The received relational structures are notoriously discrete, however. The prospects of making them to conform with continuity principles are dim, and given the struggles that Peirce had with the notion of continuity across his theories of topology and collections and across his evolutionary metaphysics, he must have foreseen the difficulties that would arise.

7. Conclusions

Even assuming an inbuilt parallel between particularity and possibility, such a parallel offers no direct support for the thesis that particular statements are bearers of existential assumptions. Rather, the connection between particular and possible statements reinforces the validity of the proposal that, as in the theory of quantification, so with the logics of modality, the meaning of possibility is given in terms of what states of information or affairs within the range of possibility the utterer chooses. Peirce's accounts of modality do not support the thesis of particularity as existence. What is possible neither excludes nor includes the fact that the state of things exists. The logical universe of discourse is about extant things, and does not apply to modal assertions. The reasons why the transition from particular to existential quantification was found compelling by the algebraic and mathematical logicians of the 19th and early 20th centuries are to be looked for in other contributions to the logic of quantifiers.

Since existence is rooted in active processes in thought and thinking, the constructive outlook on logic and mathematics was practically forced on Peirce. Logic and mathematics manipulate entities that are dependent on the contents of the mind. The contents may be rendered visible by the distinctness of topological diagrams and the character of the working models in which they are evaluated. Such diagrams are best interpreted endoporeutically, and hence Peirce came up with the semantic and model-theoretic method that has turned out to be akin to the game-theoretic one. Because of the iconicity of diagrams, the strategic dialogues were not in tension with the category of thirdness and the irreducibility of triadic relations.

Notes

1 Despite its age, the dictionary is still a useful reference book for cognitive science, the philosophy of mind, alongside with studies on the history and principles of psychology.

2 2.349, 1895, *Speculative Grammar: Propositions*.

3 Relative sum is a relation between two relations L (*is a lover of*) and B (*is a benefactor of*), in symbols $\mathrm{L} \dagger \mathrm{B}$ (*is a lover of every benefactor of*), just in case there is an ordered pair of individuals (a, b) standing in the relation in which a is the lover of every c that is the benefactor of b. Relative product is a concatenation of two relations L and B, in symbols LB (*is a lover of a benefactor of*), just in case there is an ordered pair of individuals (a, b) standing in the relation in which a is a lover of c and c is a benefactor of b, i.e., the first individual a is a lover of a benefactor of the second individual b.

4 To recap the definitions from Chapter 4, a line of identity "is a Graph any replica of which, also called *a* line of identity is a heavy line with two ends and without other topical singularity (such as a point of branching or a node), not in contact with any other sign except at its extremities. Otherwise, its shape and length are matters of indifference. All lines of identity are replicas of the same graph" (MS 508: B.4).

5 Note that the epithet "existential" in EGs actually means that some matter of fact asserted by an imaginary person ("the Graphist") exists that holds true of the universe at any given time, not that objects exist. EGs represent factual existence independently of the representation of some other fact by another graph that may be written upon some other part of the same sheet of assertion (4.421, c.1903).

6 MS 31: 2, c.1905–07?, *On the Theory of Collections and Multitude*, including *Note on Collections*.

7 In MS L 477: 13–14, 8 November 1913, *Letter to Woods*.

8 One is reminded here of Brouwer's normative views concerning the dark forces of language poisoning our autonomous thought.

9 LN: 344r, 23 February 1909, *Triadic Logic*; cf. Fisch & Turquette (1966). The inception of three-valued logic was a direct consequence of topological considerations concerning the status of the boundary line in blots scribed on sheets, as such lines may belong neither to the interior nor to the exterior (Chapter 5). Thus predicates are determinate as well as not determinate only at the limit. It would be interesting to develop triadic versions EGs along these lines.

10 See Pietarinen (2000) for a study of non-strictly competitive language games within the framework of game-theoretic semantics.

11 Moreover, "There is no thing which is in-itself in the sense of not being relative to the mind, though things which are relative to the mind doubtless are, apart from that relation. The cognitions which thus reach us by this infinite series of inductions and hypotheses (which though infinite *a parte ante logice*, is yet as one continuous process not without a beginning *in time*) are of two kinds, the true and the untrue, or cognitions whose objects are *real* and those whose objects are *unreal*" (5.311). Of course, any finitism would have been impossible by virtue of Peirce's continuum, which far exceeded any multitudes.

12 5.57, 5.58, 1906, *Lectures on Pragmaticism: The Universal Categories*.

13 This neclect endured. Of late, however, this oversight has been partially remedied in logics developed in AI and cognitive science, including, say, the non-monotonic systems based on the idea of preferential models.

14 MS L 463: 29, not preserved in the normalised transcription of SS: 196–197 nor in its reproduction in EP 2:478.

15 A related question appears that I shall leave only as a thought here. May we assume that Peirce took the commens to be extendable to a concept of a quasi-commens, the locus where the determinations of quasi-minds can also meet? (Cf. EP 2:392).

16 There were a number of other requirements, too, which I shall not discuss here. Peirce would have spoken of signification instead of meaning. He also considered a version of the Leibnizian full abstraction in terms of "the principle, that in a system of signs in which no sign is taken in two different senses, two signs which differ only in their manner of representing their object, but which are equivalent in meaning, can always be substituted for one another" (5.323, 1868).

17 MS 1408, c.1910, *Study of Propositions or Possible Assertions*.

18 Probably only in 4.561, 4.568, c.1905–06; MS 293: 51,53, c.1906; MS 514: 16, 1909; MS 669: 2, 25 May 1911, *Assurance through Reasoning*.

19 Peirce's own numbering and pagination.

20 Although proof-theoretic in character, the application of the permissions and transformations that Peirce gave for the valid manipulation of the alpha and beta graphs is nevertheless creative, as it "can only be made by a living intelligence" (MS 669). They do not make theorematic reasoning any easier for a theorem prover.

21 MS L 477: 13–14, 8 November 1913.

22 Written probably sometime in fall 1913. Pietarinen & Snellman (2005) transcribes this and some related unpublished manuscripts on Peirce's late accounts of reasoning (in Finnish, with an introduction).

23 Both of these ideas were already recognised by 12th-century logicians, see Knuuttila (1993).

24 MS 467, paragraphs marked "omit" by the editors of the Collected Papers, vol. 4; cf. MS 1147: 1–2. I will discuss this important manuscript draft on modality in the appendix to this chapter.

25 4.553ff, c.1906, *Phaneroscopy*.

Appendix 6.A: The entry on Modality in MS 1147

Peirce's struggle with modality continued in his attempt to provide a definition of it for Baldwin's Dictionary of Philosophy and Psychology. The final version that appeared in the Dictionary was a modified and shortened version of the draft entries. MS 1147 contains a 12-page draft in the midst of numerous other alphabetised draft entries intended for the same dictionary. The first volume of this dictionary appeared in 1901, with a considerably modified entry on modality. Why Peirce made so many changes and omissions is not known; one possible reason is that the editors preferred survey-type definitions to the somewhat more radical original ideas that Peirce first put forward.

What follows is a transcription of Peirce's draft definition in its original order and composition, interspersed with comments. I will compare this definition not only with Peirce's views on modalities that have appeared elsewhere, but also with some of the logical theories of modality inspired by his account. The entry was written in the period between the inception of the alpha and beta systems of EGs (1896–07), and attempts to inject more expressivity into graphical diagrams through the system of gamma graphs introduced in 1906. The view that Peirce ended up on modality involves not only an examination of his gamma graphs but also the numerous informal accounts of modality scattered around his later unpublished papers, mostly written in 1900–11.

I have decided to provide diplomatic transcription, because some of the minor deletions that Peirce made are instructive, and because such transcriptions are likely never to appear (the Peirce Edition Project aims at providing critical editions of his writings, Chapter 1). The most obvious slips of the pen have been silently corrected.

> **Modality**
>
> [*Ger. Modalität*]
>
> The qualification of a predication on the one hand, or of a truth on the other, in respect to possibility and necessity. Although the assertoriness, with its metaphysical correspondent, actuality, is the absence of modality, yet it has to be considered from the point of view of modality.
>
> The doctrine of modality, even more, if possible, than other topics of logic, remains to this day an arena of dispute. The simplest point of view, from which considerable insight is gained in regard to others, consists in regarding the doctrine as a part of prioristic analytic, that is, in considering what distinctions of this nature are needed for the purposes of deductive inference, without any regard either to the usages of language or to methodeutic principles. Here Professor Mitchell's idea of a multidimensional logical universe, which is one of several fecund conceptions contained in his paper *On a New Algebra of Logic (Studies in Logic by Members of the Johns Hopkins University*. Boston: Little, Brown & Co. 1883. p. 72), serves to render [p. 2] the view of common sense, which was partly developed in the scholastic doctrine of modals, exact.

The opening sentence of this entry sets the theme in a clear-cut manner. Peirce was concerned not only with what possible and necessary predicates mean, but also with the meaning of complete statements involving modalities. Even though sympathetic to what was later to become the field of pragmatics, he thought that such 'semantics' had to be general so that it could be established independently of concerns about the use of language or individual methodologies.

Mitchell was one of the contributors to the book Studies in Logic by Members of the Johns Hopkins University, edited by Peirce. Peirce acknowledges him in several places for making fundamental inventions in logic. Indeed, the development of the logic of quantifiers dates back to Mitchell's exposition in *On a New Algebra of Logic*, although Peirce made several amendments later on, such as to the notion of scope and the order of quantifiers. He also settled the question of what kinds of deductive inferences are allowed in the logic of quantification. The concept of multidimensional logical universes is attributed to Mitchell as one of his major inventions. Regrettably, he did not live long enough to observe their connection with modalities.

Peirce went on to elaborate this connection.

> A logical universe of two or more dimensions must not be confounded with two or more logical universes. When we consider, in addition to the usual limited universe of individual subjects, also a limited universe of marks, we have two logical universes. That which is contained in the one is not contained in the other. But if, in addition to the universe of subjects, we conceive each of these as enduring through more or less time, so that on the one hand, each subject exists through part or all of time, and on the other hand, in each instant of time there exist a part or all of the subjects, we are considering a logical universe of two dimensions, and the same terms have their place in both. The word *dimension* is here applied with perfect propriety; for were we to restrict it to cases in which measurement could be applied, we should be forced to abandon its use in topical geometry, to which no mathematician (and it is a mathematical word) would consent.

One way of looking at the distinction Peirce makes between logical universes in two or more dimensions and two or more logical universes is, in modern terms, that of the distinction between predicate modal logic and many-sorted logic. In predicate modal logic, possible worlds may have world-independent domains, and the question is how the concept of an individual endures in different worlds, maybe through time or merely as objects of knowledge or belief. In many-sorted logic, variables are specified in relation to individuals belonging to some class of individuals, including specifications that delineate between first-order and higher-order objects. The definition of a quantifier does not include all individuals of a totality of a logical universe.

A caveat is that Peirce was at pains to explicate the possible-worlds conception in full. In places other than this, he comes close to the notion of an accessibility relation between the multitudinous "states of information" in relation to the system of gamma graphs he started to develop soon after DPP appeared (4.522, 1903; MS 467: 58, 60). In the end, he hesitates to connect this with the idea of dimensionality of logical universes. The "limited logical universe of marks" is the idea of sorting the universe of discourse into two or more 'sorts', 'types' or 'characters', so that the elements of these sorts do not overlap. According to Peirce (2.453ff, 1893), some logicians hold the idea of the limited universe of marks to be "extra-logical", but this is only because it does not fall within the scope of their own studies.

Another "extra-logical" matter during Peirce's era was time, but as this definition indicates, this is seen as serving as the basic inspiration for introducing the concept of dimension to logical universes in the first place. The term 'mark' was later amended to 'qualities', and when the gamma part was invented, he noted that qualities gave rise to logical possibility. In this sense, many-sortedness and multi-dimensionality converge. It was only much later that steps were taken to show that modal languages could be viewed as many-sorted ones on relational structures (de Rijke, 1995). These steps also facilitated the first representation theorems on Peirce's algebraic logic, although they use methods that go beyond the resources that were available in his time, namely the completeness proof for dynamic modal logic. Neither completeness nor the dynamics of logic were notions articulated in Peirce's writings on relational algebra, although they are implicit in it. Furthermore, the question of whether the notions of quality and modality are of the same nature was a subject of fruitful explorations.

Peirce also makes the observation that the notion of dimension does not imply that the geometry of logical space is metric. If we have dimension, we already have a topological space (topical geometry, topology, topics) that is not subject to measurement. Around 1900 (see e.g. 3.569, *Infinitesimals*), Peirce expressed his displeasure with the state of topical geometry, as no one had yet developed a logical way of reasoning about it.

> But for the purposes of modality, it is not time which is to be considered, [p. 3] but an aggregate of hypotheses, each completely determinate in respect to every question concerning every object in the universe, but each false, which a supposed (and commonly fictitious) state of information does not exclude. This is called the *whole range of possibility*, which measures the ignorance in the supposed state of information. All the scholastic logics rightly teach that a necessary proposition is one which is in a certain respect universal, while a possible proposition is one which is in a certain respect particular; but they fail to define that respect.

Here Peirce dismisses the use of time for modal purposes. He changed his mind, as in a couple of other entries written at a later date he did conceive of time as certainly not "extra-logical" and, in fact, amenable to analysis by the system of gamma graphs (Øhrstrøm, 1997). In its place, however, he introduced two other concepts for the treatment of modality: the (whole) range of possibility and the measure of ignorance in a given state of information. It was by means of these concepts that he attempted to transgress the boundaries of modalities beyond the mere parallelism of the necessary versus the universal and the possible versus the particular advocated by the scholastics. This is the parallelism that was known not only to the scholastic logicians, but also to those of the late 11th century (Knuuttila, 1993). The logicians that Peirce probably had in mind included John Duns Scotus and William Ockham. In the entry that actually appeared, he held that the parallelism account of modality by the scholastics could be considered the simplest account.

> If, disregarding the usages of language, we define a universal proposition as one which ~~possesses the qualifications requisite for~~ [asserts only what it needs to assert in order to serve as] the major premise of a direct syllogism, and a particular proposition as one which precisely denies a universal proposition; then a simple universal proposition will not assert the existence of the subject, but ~~Any~~ 'All A is B' must ~~for~~ [in its] analytic ~~purposes~~ [sense] be equivalent to 'Whatever A there may be is B' or 'An A that is not B does not exist,' while a particular proposition must assert the existence of the subject, and so must assert something which no simple universal [p. 4] proposition asserts.

Here, Peirce digresses to criticise the simple dualism between universal and particular propositions: he notes the peculiarity of asserting existence with regard to a particular proposition as a sheer consequence of duality by which a universal proposition expresses the non-existence of exceptions. The purpose of this digression becomes clearer as we proceed.

> A universal proposition is merely a hypothetical proposition in which the object of an indefinite pronoun in the consequent is identified with that of a similar pronoun in the antecedent. Thus, 'If anything is a man, something is mortal' is hypothetical, but 'If anything is a man, that same thing is mortal' is universal. So a particular proposition is merely a copulative proposition to which such an identification is ~~made~~ [added]. Thus, 'There was a patriarch and there was a translated man' is copulative; but 'There was a patriarch, and *he* was translated' is particular. ~~A universal proposition asserts that a certain description of object (as, an immortal man) *does not exist* in the universe of logical extension; a particular proposition asserts that a certain description of object (as, a translated patriarch) *does exist* in the universe.~~

The difference between hypotheticals and universals is, according to Peirce, that in the latter, there is a connection between the object applied to the pronoun in the consequent of the proposition and in its antecedent. Likewise, the difference between a copulative and a particular proposition is that in the latter, the connection obtains between the two parts of the conjunction.

> In both these respects, unless we are to be guided by the usages of language, the necessary proposition ought, for analytic purposes, to be understood as analogous to the universal, the compossible to the particular proposition. Thus, ~~'A man without sin must be happy'~~, ['If there is a man in the moon, there must have been a woman'], in order to be best adapted for a logical form, should [p. 5] not be understood as asserting that a man ~~without sin is possible~~ [in the moon is possible], but only that in whatever hypothetic state of the universe the contemplated state of information ~~allows~~ [may allow] in which there should be a man in the moon, in that very ~~state~~ [hypothetic] state there would have been a woman; or in other words in the state of information supposed, ~~it would be known that the~~ the existence of a man in the moon without the previous existence of a woman is absolutely excluded.

Peirce now draws the connection between necessary and universal on the one hand, and between possible and particular on the other, in terms of what hypothetical states of the universe are allowed by the contemplated state of information. This is the added element to the treatment of modalities that the scholastic philosophers lacked in the idea of modality as merely analogous to nominal universal and particular quantifiers (propositions). In Peirce's example 'If there is a man in the moon, there must have been a woman', the object of a particular proposition in the antecedent ('a man') exists in the hypothetic state of information, in which case the object of the consequent proposition ('a woman') also exists in that state. (The phrase "previous existence" in the last sentence is slightly mysterious.) The notion of the contemplated state of information coincides with that of the state of information in which the proposition is evaluated, that is, the actual or designated state. We do not yet find any forthright statement that hypothetic states are to be connected with the contemplated state in any relational manner, however.

> Furthermore, ~~'If a man in the moon is possible'~~ [If there be any allowable hypothetic state in which there is a man in the moon], then there is an allowable hypothetic state in which there had been a woman in the moon' is merely hypothetical; but 'If there be any allowable state in which there is a man in the moon, then in that very hypothetic state there had been a woman in the moon' is a necessary proposition. So 'Foreknowledge is an ~~allowable~~ [admissible] hypothesis, and so is freewill,' is merely copulative, but 'There is an ~~allowable~~ [admissible] hypothesis in which there is foreknowledge, that same hypothesis supposing freewill,' is an assertion of compossibility.

If the hypothetic states of information in which the extant objects applying to the indefinites 'a man' and 'a woman' coincide, we are dealing with a necessary proposition, because there is a link between the two indefinites given by the existence within the same hypothetical state of information. A proposition is merely hypothetical when its translation into modal terminology does not assert the existence of such a link. (The fact that the two hypothetical states of information happen together may be purely coincidental.) Similar remarks apply to the distinction of the copulative versus compossible propositions.

> [p. 6] The reason why the range of possibility is confined to *false* hypothetical states of the universe, neglecting the true one, is that in the first place, the hypothetic states excluded, that is, really known not to exist, ~~the true state~~ never can embrace the one true state of the universe; and in the second place, were "excluded" to be taken in the sense of "believed not to exist," we should be taking into account, not merely *ignorance*, but also *error*. Now the doctrine of ignorance and the doctrine of error ought to be kept distinct, and investigated one at the time. The doctrine of ignorance is the simpler; and in prioristic analytic there is no occasion for considering the supposition of erroneous premisses, further than negation covers the ground.

Here Peirce returns to the notion of ignorance, and contrasts it with the notion of error. Apart from in this drafted entry, he perceived substantial differences between ignorance and error in other connections. His general claim is that states of information can accommodate ignorance but not error. For instance, in 2.382 [1901], which consists of the part of the contribution to the Dictionary that actually appeared, he expounded the difference between 'must' and 'may' in the sentences 'A must be true' and 'A may be true' in terms of propositions analogous to A, so that in the former, the meaning of the modality 'must' in the sentence is that all propositions analogous to A are true, and that in the latter, the meaning of 'may' in the sentence is that some proposition analogous to A is true. The notion of being an analogous proposition is, in turn, explicated in terms of the class of propositions that "in some conceivable state of information would be indistinguishable from A" (2.382, 1901; DPP: 89). The range of information or ignorance so defined dictates when necessary and particular propositions are true. The necessary proposition is true "in the assumed state of knowledge" (2.382) if there is no case in the whole range of ignorance in which the proposition is false. The particular proposition is true if there is a case in the range of ignorance in which the proposition is true. Again, Peirce comes very close to the idea of possible-worlds semantics and its indistinguishability relation between states, but once again falls short of going from end to end. Once more, necessity is defined as the non-existence of exceptions, and possibility as its duality, in other words that there is a state of information within the whole range of ignorance.

The idea of a set of propositions analogous to a given proposition did have an impact on the history of logic. This impact was instrumental, even though the idea was soon superseded by the fully-fledged development of the relational possible-worlds semantics developed in the 1950s and at the beginning of the 1960s by a number of philosophers and logicians. J. C. C. McKinsey had already used a similar idea to "the range of propositions being analogous to" in his syntactically-driven treatment of modalities (McKinsey, 1945). In this work, he defines two sentences as having the same form if one can be transformed into another by substituting new constants for the non-logical symbols occurring in it. This is accomplished by assuming a given set of substitutions taking sentences of a language into some other sentences of that language. A possible definition is, then, a sentence whose substitution takes it into a true sentence.

McKinsey was greatly inspired and influenced by Peirce's logical writings. Given the nature of semantics in the Peircean sense, namely as the translation of a proposition into another proposition or possibly another language, then McKinsey's 'syntactic' translation of a modal proposition into its analogues is actually a semantic one.

Peirce's distinction between syntax and semantics is nonetheless not to be assimilated with the distinction that we are nowadays accustomed to draw. He took syntax to refer in most accounts to the grammatical structure and rules of natural languages. Such rules are not capable of adequately bringing out the logical character of natural-language expressions. He restricted the study of semantics, conceived of as a theory of sign meaning, to the process of translation that takes any sign into another system of signs. For instance, dictionary-type definitions and translations into other languages are semantic constructs. The translation process does not pretend to create an identity between the sign and its semantics, it only approximates its meaning.

McKinsey's work was also a source of stimulation to Bennett (1955) and Drake (1962). All these works provide their own links to the great chain of development of the possible-worlds semantics. In the light of Peirce's influence on subsequent philosophers and logicians concerning the meaning of modality, this chain appears much tighter than is currently believed.

Three other key terms that Peirce introduced in the previous passage were false hypothetic states of the universe, the notion of the exclusion of states, and negation. Only sometime later did Peirce observe that the notions of a negation and of possibility are affiliated. This idea can also be found in Wittgenstein.

> In a necessary proposition, the range of possibility, or ignorance, will not practically be assumed to be less than the speaker's actual ignorance. For though it does [p. 7] not logically follow from what is asserted in the proposition 'A person who should know all that I do and much more beside would know that A is true' that 'I know that A is true,' yet the ~~fact~~ [additional premiss] required to make consequence logical is supplied by the mere fact that the former proposition is asserted.

What he takes up here is the notion of the speaker's ignorance. The speaker is the agent who asserts the proposition containing modalities, and becomes liable for such assertions. What is interesting is that, in his study of modal propositions, Peirce compares the meaning of a necessary proposition as given by a range of possibility or ignorance with that of the ignorance of the speaker. This idea of the speaker, although long dormant after Peirce, was revived in Arthur N. Prior's system of modal tense logic in the 1950s and 1960s, in which he took a chosen 'date variable' to represent the date on which the proposition under consideration was uttered. Like McKinsey, Prior was deeply inspired by Peirce's philosophy and logic, although he acknowledged him very sparingly in his published writings.

> This point will be made clear by considering that the copulative proposition 'A is true, and I say so' asserts in addition to 'A is true' only a fact that the utterance of the latter proposition supplies to the hearer. That the two propositions are not logically identical is shown by the difference between their contradictories, which are, 'A is false' and 'If A is true, I do not say that it is so.'

Peirce makes a note about the information that is added to a performatively-asserted proposition. This added information is enough to render the two propositions 'I say that A is true' and 'A is true' logically distinct. Sentences with performatives do not reduce to sentences stripped of them so as to preserve logical identity. There are no modalities in these sentences, but the point Peirce tried to make was that the speaker has to have sufficient background knowledge, drawn from contextual, collateral and common sources of information, in order to assert sentences that are true, and that involves modalities.

> Thus, necessary propositions practically all involve reference to a range of ignorance greater than the speaker's own or than some contemplated state of knowledge; and the kind of ignorance they assume is one of particular cases, ~~special~~ arbitrary conditions, or special experiences, [p. 8] knowledge of laws of one kind or another being supposed to exist. In other words it is an ignorance of *existences*, such as are stated in particular propositions, with a knowledge of *non-existences* such as are stated in universal propositions.

What is worth noting here, as indeed in the two preceding passages, is the introduction and use of epistemic concepts in order to properly tackle modalities. In particular, Peirce refers to the knowledge that the speaker of propositions involving modal expressions has. Such knowledge naturally goes with the notion of necessary ignorance in the meaning of necessary propositions. In modern terminology, this could be paraphrased by saying that there have to be more states accessible from the state in which the proposition is asserted than there are states capturing the speaker's range of ignorance. What is different from this speaker's range is the range of ignorance of particular propositions, which is ignorance of what exists, and also the knowledge that there are no exceptions regarding the meaning of universal propositions. How this ties in with the modal idea of knowledge as the elimination of uncertainty was never completely answered by Peirce, despite the occasional references he made to what could be characterised as an elementary logical investigation of epistemic concepts (4.520, 1903). For example, he claimed that what was later termed the KK-thesis (from knowing the proposition follows the knowing that one knows it) is manifestly false (4.521, 1903). In attempting to explain modalities in terms of a comparison of two different ranges of information (ignorance) — those prompted by the meaning of the modal proposition and those prompted by the speaker's knowledge — Peirce comes close to a logical analysis of epistemic concepts similar to that of modalities.

> A farmer proposes to measure two sides of a triangular field, merely in order to ascertain which is the longer. I tell him that is needless, since the side of adjacent to the smaller angle *must be* the greater. The kind of ~~knowledge that remains~~ [ignorance] determines the kind of necessity. The greater the range of ignorance supposed, the greater is the amount of knowledge the necessary proposition embodies. With the possible proposition it is the other way: the greater the range of ignorance the less the information the possible proposition carries. In practice, therefore, even when a possible proposition relating to a greater range of ignorance than his own is all that is relevant, a speaker will, more frequently [p.9] than not, substitute a statement about what he, in his actual condition, does not know; and thus 'A may be true,' although, strictly admitting the ~~possibility~~ [case] of A being known to be true, is ~~usually~~ [more commonly] used in the copulative

> sense of 'A may be true, and A may not be true, for ~~all~~ [ought] I now know.' The contradictory to this proposition is 'I know whether A is true or not,' which is seldom said; and indeed, there is no form of possible proposition in ordinary use which precisely denies a form of necessary proposition in ordinary use. When possible propositions do not refer to the speaker's actual ~~state~~ [range] of ignorance, to the expression of which the word *may* is commonly appropriated, they usually refer to a state of information superior to his own, often expressed by 'might', 'might even', or 'still might.' ~~As 'the laws of nature might be what they are; yet still, if only one atom body were known to exist, they could not be verified.' For~~ Example: 'If we could prepare pure chemical bodies, it might [p. 10] still be doubtful whether there was any exact relation between the atomic weights,' or 'Even if ~~it were known that~~ the sum of the angles of a triangle were exactly two right angles at one instant of time, it might not be so at another.' But it is not very often that we are able to say, or have any object in saying, that a definite proposition would not be known to be true in a higher state of knowledge than our own; and were ~~all things known~~ [knowledge perfect] there would, of course, be no ignorance, so that possibility and necessity, in reference to such a state coincide, and are equal empty forms.

Apart from illustrating the relation of knowledge of the agent to necessary and possible propositions, what is conspicuous in the preceding passage is that, despite the initial warnings in the opening sentences of his article, Peirce did not entirely avoid propositions in "ordinary use".

The other example presented here illustrates the difference in meaning between the sentences containing the modality *may* and those containing the modality *might*. The difference is spelled out in terms of the increased state of information exceeding the speaker's own state in the latter types of sentence.

> Usually, a possible proposition expresses a leaning toward belief in the *dictum*; so that if one man says 'A may be true,' another who considers the hypothesis gratuitous will reply 'Yes, but may be it is false.' If it be said that this remark is not germane to logic, which has nothing to do with mere inclinations, the reply is that the only logical value which any scientific theory has in its first stage, [p. 11] rests upon the hope that out of any large number of similar guesses some finite proportion would be approximately right. Every scientific doctrine has its germ in a pure guess; and science is an idle dream unless man has an instinct for embodying scientific truth in his hypotheses, as a bee has for embodying it in his cell. All my science is nothing but instinct [???]. It is this which ~~gives~~ [entitles] the possible proposition, or question with an inclination, to a place in logic.
>
> When a particular proposition is asserted apodictically, or a universal proposition problematically, there is a distinction between the composite and the divided sense of the modal. The difference in the case of necessary propositions is between asserting that 'In every admissible hypothetic state of the universe some A *or other* is B,' which is the composite state, and asserting that 'There is some A which *same* A is B in all hypothetic states of the universe,' which is the divided sense. In possible propositions, the difference is between ~~saying~~ [asserting] that 'In an admissible hypothetic state of universe every A there may be is B,' which is the composite sense, [p. 12] and asserting that 'Every A there may be is B in some admissible hypothetic state of the universe,' which is the divided sense. It will be seen that the denial of a composite modal is composite, and of a divided modal is divided. In necessary universal and possible particular propositions the distinction between the two senses disappears. [end of the draft]

The issue that Peirce takes up in the last paragraph is the distinction between composite and divided senses of the necessary particular proposition and possible universal proposition, which does not apply to the more customary necessary universal propositions or possible particular propositions. The distinction appears in DPP in a revised form, supplemented with the conclusion that "the divided sense asserts more than the composite in necessary particular propositions, and less in possible universals" (DPP: 90). What is interesting is the added qualification in DPP that "in most cases the individuals do not remain identifiable throughout the range of possibility" (DPP: 90). This is a hint at a modal logic that also takes individuals into account, and addresses the question of what happens when one needs to consider individuals within a range of states of information or possibility. This was one of the main struggles Peirce had with his system of gamma graphs, which he aimed to sort out after writing his articles for the Dictionary.

The most salient feature missing in the draft entry compared with the one that actually appeared is the added historical review of modality. This review is more than twice as long as Peirce's own explanations, and it contains an account of the theory of modality put forward by Aristotle, Kant's epistemological and metaphysical renderings, and an examination of German thought as it emerged in Peirce's contemporaries Trendelenburg, Lotze and Sigwart. As might be expected, Peirce's review of their thoughts on the philosophy of modality is as lukewarm as ever, and he considers his own logical account of modality to be far superior to the informal and inexact musings of the others.

What is also missing in the supplementary review is any mention of Hugh MacColl's (1837–1909) logic of modality, which appeared in a series of papers entitled *The Calculus of Equivalent Statements* in the Proceedings of the London Mathematical Society (see especially MacColl 1886–7). Peirce did acknowledge MacColl elsewhere, namely in the part of the entry on Symbolic Logic (DPP: 645) that was written by Peirce's close associate Christine Ladd–Franklin and L.C. [sic?], for his contribution to symbolic logic in terms of propositions that are assigned three truth-values (true, false and undefined). The rest of the entry appeared under the initials of both Ladd–Franklin and Peirce.

Because of the shift in focus that the published article finally took due to the addition of historical material, what were perhaps the most interesting aspects of modality around the turn of the 20th century, namely Peirce's own contributions to the meaning of modal statements, did not appear at all (or appeared only in a considerably shortened form so as to render some of the remarks that remained in the published article almost incomprehensible). For example, the idea of taking the speaker's (or what in the article is actually termed the thinker's) own state of knowledge into account is likely to have left any casual reader of the Dictionary bewildered as to why it is needed at all. Admittedly, the idea is not carried through in the draft version either, but at least an earnest attempt is made to relate it to the information-relative account of modalities in terms of ranges of information and ignorance.

Moreover, the idea of a logical universe in multiple dimensions remains without mention in the published version. This is alleviated to some extent in the other entries that Peirce contributed to the same work, especially in the subentry on Dimension that is to be found within the entry on Logic (this entry was exclusively written by Peirce), in which the role of dimension in the study of modality is rightly and visibly underscored (cf. Peirce's 1901 article Syllogism in DPP).

Morgan (1979) argued that what he calls the information-relative account of modalities is presented by Peirce as a special case of the analogue account. Although it is certainly true that Peirce's view evolved and changed over the years, he was quite clear in trying to render the analogue account into a more formal information-relative account in terms of ranges of possibility and ignorance, not taking it as a special case of the analogue account. What is notable in Morgan's presentation of the information-relative account is its exposition in terms of maximally consistent sets, a technique that was to become instrumental in establishing subsequent metamathematical results for both first-order and modal logic after the 1940s.

Seen from a wider perspective, Peirce's struggle with modality was as innovative as it was frustrating. In his *Prolegomena* that appeared in the Monist in 1906, he made an attempt to tackle modalities by EGs. Years ater he admitted that this account had not been satisfactory, and that it was still necessary to add a delta part in order to deal with them. The delta part was never realised, but Peirce envisioned tinctured diagrams that would encompass not only declaratives but also imperatives and questions and answers, or requests for information.

In order to fulfil Peirce's vision, one could reconstruct a delta part by transforming the possible-worlds semantics for predicate modal logic into a diagrammatic system of EGs. It would be useful if such a transformation were to distance itself from the straightforward rendering of traditional Lewis-type modal logics into diagrammatic logic. Apart from the challenges facing what is bound to be a major undertaking, it is certain to give rise to new opportunities. For example, it might be possible to incorporate some of the latest advances of hybrid, dynamic and multi-dimensional modal logics. A reconstruction would enhance our basic understanding of quantification in modal systems.

The following transcribes the assorted pages among the MS 1147.

[p. 1] **Modality**

The qualification of a predication by one of the received modes, *possible*, *necessary*, etc., and the qualification of a fact by analogous metaphysical conceptions.

Prof. O. H. Mitchell in his paper *On a New Algebra of Logic* (*Studies on Logic by Members of the Johns Hopkins University*. Boston. Little, Brown & Co. 1883. p. 72), among other ideas of extraordinary fertility, first put forth that of a logical universe of two or more dimensions. The idea is essentially different from that of two universes, with which Schröder confounds it. When we treat of a limited universe of marks, we have two universes, some objects being in one and ~~some~~ [other] in the other exclusively, but none in both. But in two-dimensional universe, the same object occurs in both. But if we take into account changes in time, any one individual of the dimension of logical extension may extend through all time, and any one instant of time may contain the whole of logical extension. The term *dimension* [p. 2] is applied with the utmost proprierity to such a case; for were dimension to be restricted to measurement, we should be deprived of the word in topical geometry, where no mathematician would consent to dispense with it. This idea is the key of modality. Let one dimension of the logical universe be that of logical extension, embracing the individual things spoken of, let the other dimension embrace all the different states of the first dimension which a state of information, real or fictitious, permits. The two together make up the universe of possibility, in reference to that state of information. 'If any two men are of different heights, then one man must be taller than all the rest,' means that in that state of information in which we are supposed to know that no two men are of equal height, each state of things not excluded is one in which some one man is the tallest.

The merits of this mode of expounding the matter will appear more clearly, after examining the opinions of other modern logicians. Trendelenburg (*Logische Unterschungen*, 3r Aufl. 1879, XIII.4) says: "What then is possibility and necessity? These concepts can only (nur) be defined by means of that of an antecedent [p. 2, sic, 3] (Grund), it has just been set forth. If all the conditions are known so that the fact is understood from its *entire* antecedent, and through quite permeates being (so dass das Denken das Sein völlig durchdringt), the concept of *Necessity* arises. Possibility, which always includes a part of the antecedent, thus prepares the way for necessity." In this last sentence, Trendelenburg recognizes that possibility is, in some sense, prior to necessity. That being the case, since one is merely the other twice negatived, — that is, 'A is possibly B' = 'A is not possibly not B' and 'A is necessarily B' = 'A is not necessarily not B', — why make the definition of the prior depend on that of the posterior? Doubt antecedes reason; and doubt involves the idea of possibility. The concept of necessity and possibility involve something else than the mere idea of consequence. If they involve consequence at all, they also involve the idea that ~~consequence~~ [the antecedent] is *true*. If consequence is to be introduced, [p. 3, sic, 4] at all, into the definition, the ideas of knowledge and ignorance require to be introduced in addition. Now given the ideas of knowledge and ignorance, possibility and necessity are easily defined, without using the idea of consequence at all. The idea of an antecedent (Grund) ~~involves~~ carries with it that of consequence. It is difficult to see how the idea of consequence can be defined without using those of knowledge and ignorance, *plus* the idea of passing from one to the other, which latter does not, in any obvious way, enter into the ideas of possibility and necessity, at all.

Sigwart discuses at extreme length upon modality (*Logic*. Ch. vi). He begins with making the distinction between modals in the composite, or subjective, sense, and those in the divisive, or objective, sense, attributing the former meaning to Kant, the latter to Aristotle. Kant, of course, considered the idea of objective necessity to be merely a metaphysical application of that of subjective necessity. That Aristotle contemplated only divisive modals is the opinion of Scotus, pronounced by Prantl to be very "light". Sigwart then goes on to two other preliminary remarks, the first of which is that the problematic [p. 4, sic, 5] "judgment," A may be B, is no "judgment" at all, since it only expresses the speaker's uncertainty as to whether A is B or not. This involves the singular assumption that the speaker never refers to any state of information except his actual state, and never fails to state all he knows. All logicians have recognized that in the composite, as well as in the divisive, sense, to say 'A may be B' does not necessarily imply that 'A may not be B.' Sigwart then remark that the assertory judgment, including as positive affirmation, does not differ essentially from monkeys; that is to say, the two forms have an essential resemblance. For when the ~~universe~~ [dimension] of possibility I reduced to what it ~~becomes when~~ [would become] were knowledge perfect, the necessary proposition and the proposition *de inesse* become equivalent. A man tells me he has measured two sides of a triangular field and find the one adjacent to the layer angle the shorter. Yes, I reply, it "must be so." That is to say, it not only is so, but, what is more, might be known to be so, in advance of your measurement. Sigwart says the apodictic judgment [p. 5, sic, 6] takes subordinate place as having only derivative certainty. But the point is that it asserts more than the assertory form. It is not necessary to ask in what sense certainty, if absolute, is made subordinate by being derivative. It is the assertion that a predicate belongs to a wider subject than the state of things known to exist.

[3 a.p.] . . . but the aggregate of different ~~false states of~~ [absolutely determinate ~~and~~ [but] false] the universe which a given state of information, generally a fictitious state, does not exclude. That a necessary proposition is a kind of universal proposition and a possible proposition a kind of particular proposi[tion] . . .

[5 a.p.] . . . with obvious inconsistency, suppose a person to be in a state in which he knows has no knowledge at all, except ~~of the~~ [that he has] the most distinct conceptions of meanings of the words he uses, and perfect skill in ~~knowledge~~ [logic]; and I may assert that such a man would know that twice two was four. Such an assertion is another *necessary* proposition in the divisive sense, but expresses mere logical necessity. If, however, I assert that something would be true in every state of the universe which my actual state of knowledge permits . . .

[6 a.p.] The reason why the range of possibility only embraces false hypothetical states of the universe, not the true one, is simply that the one true state of things cannot be really excluded, or *known* not to exist. If the one true state of things is excluded from the range of admitted hypotheses merely, that would constitute, not *ignorance*, but *error*. Now elementary deductive logic, or prioristic analytic, cannot take into account the supposition that the premises of reasoning are erroneous. It must simply assume them to be true. The question of the effects of error is not germane to the doctrine of simple modality.

In a necessary proposition, the ~~state of information~~ [range of possibility, or ignorance] referred to cannot be ~~less than~~ assumed to be less than that of the speaker's actual ignorance; for what he knows another man would know, under any circumstances, he ~~must~~ [will] already know himself and what he knows, everybody who knows all that he knows, and more, will know. Strictly speaking, the assumed range of ignorance might coincide with the speaker's actual ignorance, for that there is a slight difference between asserting that 'A is true,' and that 'I know that A is true' is shown by the considerable difference between the denials of these propositions, namely 'A is [p. 7] false' and 'I do not know that A is true.' Practically, every necessary proposition amounts to the assertion that not only would the *dictum* be known to be true in one state of knowledge but it would be known even were the range of ignorance greater. For example, there is no need of reusing the sides of a triangular field to ascertain which is the longer, since the one that is adjacent to the lesser angle *must be* the longer; that is, to say this can be known without the knowledge that would be gained by measurement. But there is no such limitation in regard to possible propositions. Thus, though if I, knowing the latest discoveries in elasticity, do not know but that the attraction between two particles may depend on the position of a third, ~~a person not~~ I may nevertheless choose to assert less, namely, that for a person ignorant of the latest discoveries it would be possible for the attraction between a pair of particles to depend on the situation of a third. On the other hand, I might have reason to assert that no matter how accurately the properties of elastic bodies were known, it still would be possible [p. 8] that the attraction between two particles should depend on the position of a third. Here we have the extraordinary phenomenon that certain possible propositions widely different have indistinguishable contradictories. An analogous case is that the propositions 'A is false' and 'A is either false of is not said to be true' are quite different; but their denials are 'A is true' and 'A is both true, and is said to be true.' Now when we hear it said that 'A is true,' our ears testify that if ~~is said be true, also~~ [what is said is true, then] it is further true that it both is, and is said to be, true.

[p. 10] . . . for information in regard to the state of ideas on the subject in scholastic times. The Questions on the Physics which make the second volume of the standard edition of Scotus, if by that doctor, at all, must belong to his extreme youth. They are most likely a little earlier. We there find it assumed as a matter of course that a modal is equivalent to some proposition *de inesse* . . .

Chapter 7

SPIEL-TRIEB OPERATIONALISED: SEMANTIC GAMES IN LOGIC AND LANGUAGE

BY THE PLAY-INSTINCT is meant the most energetic part of the cultivation of ideas. The name, which translates Schiller's *Spieltrieb*, has been chosen because the fine arts exemplify the highest action of this instinct. (MS 1343, c.1902, *On the Classification of the Sciences. Second Paper. Of the Practical Sciences*).

SUBDIVISION OF 32312311: *Games and Amusements*, continued. 32312311113 *Games of Logic*. (MS 1135, c.1903, *An Attempted Classification of Ends*).

The purpose of this chapter is to introduce the reader to the principles of game-theoretic semantics (GTS), which was originally developed by Jaakko Hintikka in the 1960s and became one of the main approaches in logical and linguistic semantics, and to chart some of its current directions. The theory has been researched in numerous publications. In the light of Peirce's logic, some of which were expounded in previous chapters, these games are put into a wider historical and systematic context within the overall development of logic.

1. Introduction

Three questions Three major questions are addressed here. (i) What kinds of tools and doctrines GTS provide for the scientific study of logic and language? (ii) What is the structure of such games? (iii) What is the relation between logic, language and games?

The following responses are proposed.

(i) GTS makes available a formal apparatus that can be put to use in logic in new ways, unifying different semantic outlooks on natural language. Its philosophical component is to be found in the analysis of lexical and logical meaning in terms of enriched game-theoretic content.

(ii) Semantic games may be viewed as a special class of extensive forms of games that show the flow of semantic information and the distribution of the strategic actions of the players during the actual playing of a game. Variations in the information structure of the players give rise to different kinds of logics, including the IF (independence-friendly) logics introduced in Hintikka & Sandu (1989) and studied further in Hintikka (1996a); Hintikka & Sandu (1997), for example. Briefly, IF logics are capable of expressing various informational independencies, their formulas being correlated with games of imperfect information. I noted a related idea in the context of EGs in Chapter 5.

(iii) Various logical semantics may be distilled from different classes of games, which also proves to be useful for the study of language. When games are varied, different logics, other than the classical propositional, first-order or modal, are seen to emerge. This again allows us to perceive much more in the structure and semantics of natural language than is currently believed to exist.

Wider vistas In order to make the broader scientific picture easier to discern, it is necessary to outline some of the wider goals and prospects.

Since the early 1980s, theories of discourse representation (DRT), dynamic semantics (or the dynamic theory of meaning), and relational generalised quantifier theory have been the linguistically-driven approaches to semantics that have dominated the main research fields in logic and in the semantics and pragmatics of natural language. These approaches have been complemented more recently by theories referring to the concept of choice functions. While all of these theories have led to many interesting insights into workings of logic and language, their supremacy is unfair. GTS is probably the first dynamic system that was successfully applied to the study of logic and language. Choice functions, in turn, are special cases of the game-theoretic concept of a strategy. This immediately explains what the linguistic role of such functions in the theory ought to be.[1] However, GTS did not take off to the same extent as the other semantic frameworks, despite the fact that there is a vast array of natural-language expressions in the purview of semantic games. This holds even if the expressions let in a modicum of strategic meaning. In addition, a somewhat less-known but widespread phenomenon in language is the cross-categorial notion of informational independence, the treatment of which is typically successful only via game-theoretic apparatus, and which may be put into a unified perspective by such a game-theoretic analysis.

An instance of informational independence is the branched organisation of quantifier phrases. In addition, games are at least as rich and versatile as DRT, dynamic logic and dynamic semantics, or generalised quantifiers. This is witnessed by issues to do with anaphora and functional dependency, tense and aspect, and the logical representation of eventualities. This wider story remains largely untold, and it is not attempted within the confines of this book (see Janasik et al. 2002; Pietarinen 2001b; Pietarinen & Sandu 2004 instead).

One current effort in the study of language involves locating the semantics/pragmatics interface (Turner, 1999) and charting the phenomena within it.

Such crossing points are where games have always naturally operated. Any cast-iron division here may be artificial and uninspiring. While studies of syntax, semantics and pragmatics (the unhappy trichotomy suggested by Morris, who thought it to reflect Peirce's intentions) have been kept quite strictly separate in mainstream linguistics and the philosophy of language, and while there has been an increasing effort to glue these components back together (especially with reference to semantics and pragmatics) and to relocate any possible trichotomies that may ensue as a result, it is the division between studies in grammar, studies in critic (logic proper) and studies in speculative rhetoric suggested by Peirce that provides the only sufficiently broad categorisation to subsume the phenomena that linguistics and linguistically-minded philosophers and logicians are predicted to uncover and typologise (Chapter 12).

The third answer could be supplemented with the remark that, perhaps in its most general sense, the notion of a game could be thought of as regimentation of the idea that whenever two forms contact one another, the befalling mutual action gives rise to content. The forms in question may be a language and its users or a single communicator, patterns of logic, or a computational system and its environment.

One underlying thesis in this chapter is that games provide a first-rate insight into the different aspects of information flow in logical semantics. Within the present context, these streams and their fluctuation will be harnessed for the most part by the theory of extensive games, intermingled with imperfect information and other phenomena that increase their applicability.

Accordingly, the question of what makes games helpful in the study of logic and language that is addressed here by focussing not only on games that are the best known and most thoroughly studied two-player perfect-information games, but also on those involving teams of players and having imperfect information. Accordingly, the following sections concern such imperfect-information team games, the logic they are associated with, plus some applications to the semantics of natural language. A more thorough exposition of the formal theories that resort to game-theoretic concepts from logical, mathematical and computational perspectives is given in Chapter 10.

The rise of modern game theory I will begin my brief resume from what could be called the modern era of game theories, and ignore the interesting proto-history and names such as the 17th-century philosopher and Descartes' contemporary and critic Antoine Arnauld (1612–1694), Leibniz' acquaintance Chevalier de Méré (1607–1684), and the mathematician Daniel Bernoulli (1700–1782) from the 18th century, James Waldegrave (1684–1741) and Nicholas Bernoulli (1687–1759) from the 19th century, and the early 20th-century mathematical economists Joseph Louis François Bertrand (1822–1900), Antoine Augustin Cournot (1801–1877) and Edgeworth, figures in early studies on political economy, econometrics and probability calculus rather than on game theory.

Among the early ludents was Ernst Zermelo (1871–1953), who showed that for a two-player strictly competitive game with finitely many possible positions, a player can avoid losing for only a finite number of moves (if his opponent plays correctly), if and only if the opponent is able to force a win (Zermelo, 1913). The modern received version of the theorem states that every finite, strictly competitive perfect-information two-player game is determined: either player 1 or player 2 has a winning strategy.

Game theory truly kicked off with Émile Borel (1871–1956) and John von Neumann (1903–1957) (Borel 1921, von Neumann 1928), supported by contributions from László Kalmár (1905–1976) and Dénes König (1884–1944) (Kalmár 1928–9, König 1927). One of the driving motivations in König's and Kalmár's papers was to improve upon Zermelo's earlier work. As to von Neumann and Morgenstern's contribution, the game-theoretic concepts put forward in von Neumann (1928) were, according to the author himself (von Neumann, 1953, p. 124), discovered independently of Borel's earlier discovery of pure and mixed strategies: "I developed my ideas on the subject before I read [Borel's] papers". According to Ulam (1958), however, "Early in his work, a paper by Borel on the minimax property led [von Neumann] to develop ... ideas which culminated later in one of his most original creations, the theory of games". (Kalmár acknowledges von Neumann's work in his 1928 paper, though.) All the same, games were doubtless developed into a fully-fledged theory in von Neumann & Morgenstern (1944).

What is notable is how Peirce's views appear in stark contrast to those of one of the early figures in the emergence of game theory in the 1930s and the proponent of axiomatic methods in science, Oskar Morgenstern, who on more than one occasion praised scholars such as Russell, Whitehead, Carnap, Hilbert and Ackermann precisely because they embarked on a project of isolating logic from neighbouring sciences (Schotter, 1976).

After Zermelo, Thoralf Skolem introduced what is known as the Skolem normal form for first-order logic. Although aware of Zermelo's work, Skolem did not explore possible connections between logic and games. The development of the Skolem normal form is nonetheless interesting, and its exact history still needs to be documented. According to Skolem (1920, p. 254), "Löwenheim proves his theorem by means of Schröder's "development" ["Ausführung"] of products and sums, a procedure that takes a Π sign across and to the left of a Σ sign, or vice versa". Schröder used an awkward (sub)subscript notation adopted by Löwenheim. Interpreted as (existentially quantified) functions, they become what are known as Skolem functions. They were termed the "fleeing subscripts" by van Heijenoort (1967, p. 230), in that a subscript $\mathrm{k_i}$ in $\Sigma_{\mathrm{k_i}}$ actually means a function of i, and the twofold sigma symbol that Löwenheim used could be interpreted as standing for a string of sigma symbols (existential quantifiers), expressing the existence of a value of that function for each i coming together with the universal quantifier. Since these symbols may be eliminated by the known quantifier elimination methods, given that post-Tarski logicians have been interested primarily in the satisfiability of the formula, the normal form of Löwenheim is actually slightly different from that of Skolem.

The works of Schröder and his contemporary Peirce are, naturally, related, but their mutual influence is still somewhat unclear. Early versions of the Skolem normal form are nevertheless to be found, for instance in 3.505 [1896, *The Logic of Relatives*], where Peirce his readers "to place Σ's as far to the left and Π's as far to the right as possible" (SIL has more suggestions of this kind). Skolem was aware of Peirce's logic of relatives and referred to him as belonging to a different tradition in the history of mathematical logic from those of Frege, Peano, Russell and Whitehead (van Heijenoort, 1967, p. 512).[2]

The first explicit connection between the Skolem functions games appeared in Henkin (1961). According to him, Skolem normal forms, and infinite quantifier strings in particular, could be conceived of as games. As is well-known, Hilbert used game-related or inspired ideas in his approach to the foundations of mathematics, and, to a degree, so did Gerhard Gentzen.[3]

The modern era of games for logical investigation started with Henkin (1961), Hintikka (1973a) and Scott (1993). Scott presented the earliest game-theoretic elucidation of logic, based on an interpretation of Gödel's Dialectica (functional) translation of first-order logic and arithmetic into a higher-order language. The Dialectica interpretation has resurfaced since in various guises, such as in category theory, delivering abstract notions of games as Chu spaces or Dialectica categories that are used to model linear logic, and in consistency proofs for constructive theories. The connection between the truth-values and the existence of winning strategies was noted in Hintikka (1973a). It is worth recalling that Peirce already gave some fragmentary indications of the connections between interaction and the notion of truth: "The duality of the *ego* and *non-ego* is the chief constituent of the idea of the Truth" (MS 515: 24). This duality and the ensuing dialectic subject of thought have much wider significance in Peirce's general theory of logic and semeiotics. For instance, the experience of an event is conceived of as a duality between consciousness and the object of consciousness, where the new excitement appears as *non-ego*, opposing the old *ego* and instantly passing into it.

Wittgenstein's far-reaching notion of a language game offers a concept that could be compared with semantic games. I will discuss a significant finding from his recently published Nachlass in more detail in Chapter 8, but what it suggests, in effect, is that, in addition to his remarks that at least some language games are ones of verification and falsification, the purposes of players in semantic games can be best accounted for in terms of the activities of showing or telling what one sees. What the players try to achieve is to bring to the fore what they see to be the case in the context of an assertion. They have been prompted to do this by a specific expression, and they aim to show or say what is the case by instantiations of suitable elements: "It is true that the game of "showing or telling what one sees" is one of the most fundamental language games, which means that what we in ordinary life call using language mostly presupposes this game" (Wittgenstein, 2000–, 149: 1).

Contrary to what is claimed in Hodges (1997b), the attributes of winning and losing were made applicable to language games by way of Wittgenstein's own remarks (Chapter 8). Wittgenstein's Nachlass reveals further that game theory was not foreign territory to him, for he remarked that the theory of the game is not arbitrary, although a game itself is (Wittgenstein, 2000–, 161: 15r). He did not show particularly keen interest in such theorising, however.

Dialogues and logic Since the 1950s, dialogues have earned a notable place in the foundations of logic, as well as in numerous applications that involve formal procedures for reasoning and argumentation, such as parliamentary debate and legal cross-examination. The key players have been Paul Lorenzen and Kuno Lorenz. Lorenzen (1955) sparked off these investigation by providing a platform of operative logic that subsequently was tried to be improved upon.[4]

There are two participants in dialogical logic, the Proponent and the Opponent, sometimes also called, quite misleadingly, the Defender and the Attacker. The former proposes a claim while the latter challenges it. The moves are made according to logical and procedural rules. Informally, the logical rules consist of rules for (i) conjunction, prompting a challenge by the Opponent, the chosen conjunct becoming available to be defended by the Proponent; (ii) disjunction, which is not really challenged but just defended by the Proponent by choosing one of the disjuncts, and (iii) negation, which, as in GTS is a signal to change roles. In other words, negated statements are challenged by defending the statement governed by the negation. An existential statement is a request for a witness produced by the Proponent, instantiated as the value of the quantified variable to serve as a claim to be defended in the future. Likewise, a challenge on universal quantification asks for an individual produced by the Opponent, and the result of the instantiation will be the next challenge.

The Proponent is taken to have lost if the claim can no longer be defended, and the Opponent is taken to have lost if the claim can no longer be challenged. As in GTS, the key concept here is the existence of winning strategies, which prescribes when the formulas will be valid. An analogous result to that of the theory of semantic games is that a first-order sentence S can be deduced from the set of first-order sentences Γ ($\Gamma \vdash S$) if and only if $\Gamma \vdash S$ is valid in intuitionistic logic.

Procedural conventions place some restrictions on how dialogical games are played. For example, it is often stipulated that a challenging claim may be answered at most once (or vice versa), or that responses by the Opponent are restricted to the latest challenge not yet defended. There are significant choices to be made between these conventions, as shown by the fact that classical logic can be reproduced by a suitable combination of these rules (see Lorenz 1961; Rahman & Rückert 2001).

I noted in Part I that Peirce had ideas close to applying dialogues to logic, remarking, for instance, that thinking proceeded in the form of a dialogue between different phases of the ego. In many instances he viewed actors as the actual language users. However, as I mentioned, he presented several ideas that could be seen as semantic, concerning what subsequently has became known as the semantic-game approach to logic. Indeed, he intended the ego and its opponency, the non-ego, to transpire within a single mind or a quasi-mind. Thus, Peirce's games were not always games for actual language users. Anyhow, what he seems to have anticipated was not only the semantic but also the dialogical application of games to logical matters.

Evidence of the dialogical nature of Peirce's logic lies in his proposal that, given the existence of diagrammatic representations, what the experimenting upon these diagrams means is that one puts questions to Nature concerning the relations expressed in them. Thus his approach covers interrogative games, too.

Note that diagrams such as EGs are iconic representations of what one takes to be the relation between the representation and the proposition, or an aspect of the content of thought, that is to be represented. Smith (1992) holds that one could improve on diagrams by taking them directly to depict the relations of the ontology concerned, namely as "*diagrams* of reality" (Smith, 1992). This is nominalism. In contrast, Peirce's reference to experimentation ought to be seen as a game in which one player (the Experimenter) puts an initial thesis to the other player (the Experimentee), and vice versa. It is then in the questioning that the players try to elicit additional theses from each other. As soon as we allow the players to use this elicited information as premisses to prove their own thesis, we have Hintikka's interrogative model of inquiry (Hintikka, 1997). The responses the player has made to the adversary also have to be defended by replying to the questions that the adversary poses. What is notable is the nature of a response as an assertion, reflecting Peirce's view of assertions as binding utterances for which the utterer runs the risk of punishment if the assertion turns out not to be the case with reference to the relations concerned in the experimentation. Moreover, the interrogative game for experimentation, while different from semantic games, is intertwined with them in Peirce's diagrammatic logic. The former comes out via 'putting questions to Nature', and in his view that the experimenter's suggested assertions — if playing the role of the Graphist — are authorised by the experimentee, who is viewed as the Grapheus, the one who has undertaken the task of designing the relations. It is against the backdrop of such designs that the diagrams are deemed true or false.

In his comprehensive survey on dialogues, Felscher (2002, p. 125) notes:

> Lorenz [1961] observed that a change of dialogue rules would give rise to a type of dialogue the strategies for which would prove precisely the *classically* provable formulas. For this situation made it perfectly clear that the mathematical arbitrariness of the Theory of Games, being a tool to describe formally such different ways of reasoning as are classical logic and intuitionistic logic, could not possibly produce a philosophical foundation for either of them.

Contrary to this view, however, one aspect in which dialogical games differ from the theory of semantic games is simply that actual game-theoretic concepts have proved instructive in the latter. Such concepts include extensive-form representations of games, uncertainty by information sets, payoffs versus winning, strict versus non-strict competitiveness, and team formation. Consequently, his claim concerning the theory of semantic games is, in the end, obsolete: "Hintikka restricts his attention to the single argumentation forms and nowhere cares to formulate game rules proper (such that the implied reference to mathematical games remains but an incantation)" (Felscher, 2002, p. 126). Another counterexample to this is the computational programme of the 'geometry of interaction' and its game semantics, discussed in Chapters 8 and 10.

2. Game-theoretic semantics

Semantic games What is it that makes games a powerful tool in logic? The basic idea is somewhat simple. You and I confront one another, observing a set of rules telling us which moves are legal, and with the same purpose. We both try to win the game by winning any play of it, and if one of us finds a systematic way of doing so, he or she has a winning strategy. The set of game rules is fixed by the logically active components in language, which in the case of first-order languages comprise the two quantifiers $\exists$ and $\forall$ and sentential connectives.

Rules Let us assume that the structure $\mathfrak{A}$ is a τ-structure with a signature τ of a nonempty domain $|\mathfrak{A}|$ on which the game is being played. A valuation g is a mapping from terms of a language L to the domain of the model, restricted to the free variables of every $\varphi \in$ L. In the game, the formulas are evaluated according to the rules prompted by the logical ingredients encountered in them, starting with the outermost one. Game $\mathcal{G}$ involves player V (the V∃rifier, H∃loïsé, Mys∃lf) and player F (the F∀lsifier, ∀bélard, N∀ture).[5]

The aim of F is to falsify the formula (i.e. to show that it is false in $\mathfrak{A}$), and the aim of V is to verify it (i.e. to show that it is true in $\mathfrak{A}$). For the sake of simplicity, it is assumed, without loss of generality, that the first-order language $\mathcal{L}_{\omega\omega}$ does not contain $\rightarrow$ or $\leftrightarrow$. The symbols $\forall$ and $\bigwedge$ prompt a move by F, and $\exists$ and $\bigvee$ prompt a move by V. When players come across negation, they change roles, and winning conventions will also change. Each move reduces the complexity of the formula, and hence an atomic formula is finally reached. The truth-value of an atomic formula, as established by a given interpretation, determines which player wins the play of a game.

Let $\mathcal{L}_{\omega\omega}$ be a standard first-order language with $\{\vee, \wedge, \sim, \exists, \forall\}$. A strictly competitive non-cooperative game $\mathcal{G}(\varphi, g, \mathfrak{A})$ is defined by induction on the complexity of each $\mathcal{L}_{\omega\omega}$-formula φ between *V* and F:

(G.$\sim$) If $\varphi = \sim\psi$, V and F change roles, and the next choice is in $\mathcal{G}(\psi, g, \mathfrak{A})$.

(G.$\vee$) If $\varphi = \theta \vee \psi$, V chooses either Left or Right, and the next choice is in $\mathcal{G}(\theta, g, \mathfrak{A})$ if Left, and in $\mathcal{G}(\psi, g, \mathfrak{A})$ if Right.

(G.$\wedge$) If $\varphi = \theta \wedge \psi$, F chooses either Left or Right, and the next choice is in $\mathcal{G}(\theta, g, \mathfrak{A})$ if Left, and in $\mathcal{G}(\psi, g, \mathfrak{A})$ if Right.

(G.$\exists$) If $\varphi = \exists x\, \psi$, V chooses an individual (say, a) of the domain of the structure $\mathfrak{A}$, and the next choice is in $\mathcal{G}(\psi, g \cup \{(x, a)\}, \mathfrak{A})$.

(G.$\forall$) If $\varphi = \forall x\, \psi$, F chooses an individual (say, a) of the domain of the structure $\mathfrak{A}$, and the next choice is in $\mathcal{G}(\psi, g \cup \{(x, a)\}, \mathfrak{A})$.

(G.atom) If φ is atomic, the game ends, and V wins if φ is true, and F wins if φ is false.

Strict competitiveness means that if *V* loses then F wins, and if F wins, then *V* loses. Non-cooperation roughly means that players decide the action they take alone. According to the rules for connectives, rather than choosing subformulas, players choose elements from one domain split into two.

Strategies The strategy for each player in $\mathcal{G}(\varphi, g, \mathfrak{A})$ is a complete rule indicating at every contingency in which the player is required to move what his or her choice is. A winning strategy is a sequence of strategies, a strategy profile, by which a player may make operational choices such that every play of the game results in a win for him or her, no matter how the opponent chooses.

Let $\mathcal{G}(\varphi, g, \mathfrak{A})$ be a game for $\mathcal{L}_{\omega\omega}$-formulas φ, and f a strategy profile.

- $(\mathfrak{A}, g) \models \varphi$ if and only if a winning strategy f exists for *V* in $\mathcal{G}(\varphi, g, \mathfrak{A})$;
- $(\mathfrak{A}, g) \not\models \varphi$ if and only if a winning strategy f exists for F in $\mathcal{G}(\varphi, g, \mathfrak{A})$.

The game-theoretic notion of truth invokes the key notion of strategies, which may be viewed as Skolem functions, and a winning strategy as an array of Skolem functions. Moreover, an existential quantifier that is within the scope of a universal quantifier (in the sense of scope expressing the logical priority order of components) is functionally dependent on the universal quantifier. For example, if Pxy is atomic, then $(\mathfrak{A}, g) \models \forall x \exists y\, Pxy$, if and only if there exists a one-place function such that for any individual chosen by F (say, a), Paf(a) is true in $\mathfrak{A}$.

The distinction in the next two sections between 'games that are not played' and 'games that are played' is analogous to the distinction between normal forms and extensive forms of games.

Games that 'are not played' According to the Skolem normal-form theorem, every $\mathcal{L}_{\omega\omega}$-formula φ is equisatisfiable (satisfiable in the same models) with the existential second-order Σ^1_1-formula of the form:

$$\exists f_1 \ldots \exists f_m \forall x_1 \ldots \forall x_n\, \psi, \tag{7.1}$$

where $f_1 \ldots f_m$, $m \in w$ are new function symbols and ψ is a quantifier-free formula. Such normal forms are effectively to be found for every first-order sentence. The resulting Σ^1_1-formula states the existence of a winning strategy for V. Assuming the Axiom of Choice and taking models to be infinite, by the Skolem normal form theorem, it follows that

$$\exists f \forall x\, Pxf(x) \equiv \forall x \exists y\, Pxy.$$

A special type of Skolem normal-form theorem may be used in skolemising connectives. The only difference is that it is possible to conjoin to each disjunct a Skolem function f which has its value in a set of two elements, say $\{\mathsf{Left}, \mathsf{Right}\}$.

For example, let

$$\forall x \exists y \forall z (P_1xyz \vee P_2xy), \tag{7.2}$$

where P_1xyz and P_2xy are atomic. This is then skolemised to

$$\begin{aligned}\exists f \exists g_1 \exists g_2 \forall x \forall z ((P_1xf(x)z \wedge g_1(x,z) = \mathsf{Left}) \vee \\ \vee (P_2xf(x) \wedge g_2(x,z) = \mathsf{Right})).\end{aligned} \tag{7.3}$$

However, one might find the use of Skolem functions as winning strategies somewhat restrictive and not able to capture the true strategic nature of interactive moves. Indeed, these functions can express only functional dependencies, namely the existential quantifiers or disjunctions that are within the scope of universal quantifiers.

Further, if there exists a winning strategy for one of the players, what interest does the other player have in playing the game off against such an invincible opponent? Since all games for first-order logic are determined, that is, there exists a winning strategy for one of the players in a game (and thus the other, given that the games are strictly competitive, loses), the idea of a game as a set of dynamically evolving plays with truly interacting players tends to recede.

Games that 'are played' These qualms are allayed as soon as semantic games are viewed as extensive-form games in the sense of the classical theory of games. In such a framework, one could think of logical games entirely in terms of how information flows in a formula from one component to another, and study various ways in which this flow may be controlled and regulated. This perspective is not confined to functional dependencies, and thus one is able to say something more about the game-theoretic interpretation of logic than would be possible by merely using the existence of winning strategies. This adds credence to the true strategic content of semantic games without suppressing their dynamics.

Extensive games go beyond the normal (strategic) form in the sense that, whereas normal forms conveniently show at a glance, so to speak, which strategies are the winning ones for which player, strategies in extensive games are generated as the game proceeds.[6]

In general, extensive games capture the sequential structure of players' strategic decision problems. They may be represented as (finite) trees with decision nodes (histories) and actions labelling the edges departing from them. The game starts at the root of the tree and ends at the terminal nodes. At each non-terminal node or decision point, the player has to make a decision as to what to choose. The outcome of this decision in a particular play is a choice, while the set of all choices from a node determines a move.

Extensive games were first formulated (set-theoretically) in von Neumann (1928), although the (graph-theoretic) presentation in Kuhn (1953) has become commonplace. von Neumann & Morgenstern (1944) set out the essentials of the graphical conception. Applied to logic, the key definitions are as follows.

Games in extensive forms. Perfect information Let us suppose a family of actions A, in which the finite sequence $\langle a^i\rangle_{i=1}^n$, $n \in w$ represents the consecutive actions of the players in N, $a^i \in A$ (no chance moves as yet, see sect. 4). An extensive game $\mathcal{G}$ with perfect information is a five-tuple

$$\mathcal{G}_A = \langle H, Z, P, N, (u_i)_{i \in N}\rangle.$$

- H is a set of finite sequences of actions $h = \langle a^i\rangle_{i=1}^n$ from A, called histories of the game. It is required that:
  - the empty sequence $\langle\rangle$ is in H;
  - if $h \in H$, then any initial segment of h is in H too, that is, if $h = \langle a^i\rangle_{i=1}^n \in H$ then $pr(h) = \langle a^i\rangle_{i=1}^{n-1} \in H$ for all n, where $pr(h)$ is the immediate predecessor of h ($= \emptyset$ for $h = \emptyset$).
- Z is a set of maximal histories (complete plays) of the game. If a history $h = \langle a^i\rangle_{i=1}^n \in H$ can continue as $h' = \langle a^i\rangle_{i=1}^{n+1} \in H$, h is a non-terminal history and $a^n \in A$ is a non-terminal element. Otherwise they are terminal. Any $h \in Z$ is terminal.

- $P\colon H \setminus Z \to N$ is the player function which assigns to every non-terminal history a player in N whose turn it is to move.
- each u_i, $i \in N$ is the payoff function, that is, a function which specifies for each maximal history the payoff for player i.

For any non-terminal history $h \in H$,

$$A(h) = \{x \in A \mid h \frown x \in H\}.$$

A (pure) strategy for a player i is any function

$$f_i\colon P^{-1}(\{i\}) \to A$$

such that $f_i(h) \in A(h)$, where $P^{-1}(\{i\})$ is the set of all histories in which player i is to move. A strategy also specifies an action for histories that may never be reached.

In a strictly competitive game, $N = \{V, F\}$ and in addition:

- $u_V(h) = -u_F(h)$;
- either $u_V(h) = 1$ or $u_V(h) = -1$ (that is, V either wins or loses);

for all terminal histories $h \in Z$.

Imperfect information Let $\mathcal{G}_A$ be a perfect-information game. To represent imperfect information, let us extend $\mathcal{G}_A$ to a six-tuple

$$\mathcal{G}^*_A = \langle H, Z, P, N, (u_i)_{i \in N}, (\mathfrak{I}_i)_{i \in N} \rangle,$$

where $\mathfrak{I}_i$ is an information partition of $P^{-1}(\{i\})$ (the set of histories in which i moves), such that for all $h, h' \in S^i_j$, $h \frown x \in H$ if and only if $h' \frown x \in H$, $x \in A$, $j = 1 \ldots m$, $i = 1 \ldots k$, $m \leq k$. S^i_j is called an information set.

The games are exactly as before, except that now the players may not have all the information about the past features. This is brought out by an information partition $\mathfrak{I}_i$ of histories into information sets (equivalence classes). The histories that belong to the same information set are indistinguishable to the players, and thus a player takes no notice of what the histories are that have been played.

In imperfect-information games, the strategy function is required to be uniform on indistinguishable histories:

$$\text{If } h, h' \in S^i_j \text{ then } f_i(h) = f_i(h'), \text{ for all } i \in N.$$

The notion of uniformity is routinely disposed of in game theory, because strategies have information sets rather than individual histories as arguments.

Semantic games in extensive forms. Perfect information Let Sub (φ) denote a set of subformulas of φ. An extensive-form semantic game $\mathcal{G}(\varphi, g, \mathfrak{A})$ associated with a first-order $\mathcal{L}_{\omega\omega}$-formula φ is exactly like the game $\mathcal{G}_A$ defined above, except that it has one extra element: a labelling function $L{:}H \rightarrow \mathrm{Sub}(\varphi)$ such that

- $L(\langle\rangle) = \varphi$ (the root);
- for every terminal history $h \in Z$, $L(h)$ is an atomic formula or its negation.

In addition, the components H, L, P, u_V and u_F jointly satisfy the following:

- if $L(h) = {\sim}\varphi$ and $P(h) = V$, then $h \frown \varphi \in H, L(h \frown \varphi) = \varphi, P(h \frown \varphi) = F$;
- if $L(h) = {\sim}\varphi$ and $P(h) = F$, then $h \frown \varphi \in H, L(h \frown \varphi) = \varphi, P(h \frown \varphi) = V$;
- if $L(h) = \psi \vee \theta$ or $L(h) = \psi \wedge \theta$, then $h \frown \mathsf{Left} \in H$, $h \frown \mathsf{Right} \in H$, $L(h \frown \mathsf{Left}) = \psi$, and $L(h \frown \mathsf{Right}) = \theta$;
- if $L(h) = \psi \vee \theta$, then $P(h) = V$;
- if $L(h) = \psi \wedge \theta$, then $P(h) = F$;
- if $L(h) = \exists x\varphi$ or $L(h) = \forall x\varphi$, then $h \frown a \in H$ for every $a \in |\mathfrak{A}|$;
- if $L(h) = \exists x\varphi$, then $P(h) = V$;
- if $L(h) = \forall x\varphi$, then $P(h) = F$;
- for every terminal history $h \in Z$:
  - if $L(h) = Pt_1 \ldots t_m$ and $(\mathfrak{A}, g) \models Pt_1 \ldots t_m$, then $u_V(h) = 1$ and $u_F(h) = -1$;
  - if $L(h) = Pt_1 \ldots t_m$ and $(\mathfrak{A}, g) \not\models Pt_1 \ldots t_m$, then $u_V(h) = -1$ and $u_F(h) = 1$.

The notion of strategy is defined in the same way as before. A winning strategy for $i \in \{V, F\}$ is a set of strategies f_i that leads i to $u_i(h) = 1$ no matter how the player $-i$ (the player other than i) decides to act.

An extensive perfect-information semantic game for an $\mathcal{L}_{\omega\omega}$-formula $\phi = \forall x \exists y\, Pxy$ on a two-element domain $|\mathfrak{A}| = \{a, b\}$ is depicted in Figure 7.1.[7] Since this is a perfect-information game, each non-terminal history forms its own singleton information set. Singleton information sets may be omitted. The choices are marked on the edges of the game tree, and they correspond to the choices made by the player acting at the histories from which these edges depart. The atomic formulas label the terminal histories. Depending on the truth of atomic formulas, either F or V can win particular plays as seen from the payoffs. In this case, V wins the plays amounting to Paa and Pbb and F loses them, and F wins the plays amounting to Pab and Pba while V loses them. A winning strategy exists for V, choosing a when F has chosen a, and b when F has chosen b, while no winning strategy exists for F.

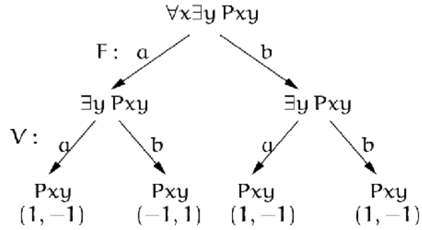


Figure 7.1. Perfect-information semantic game $\mathcal{G}(\varphi, g, \mathfrak{A})$.

Imperfect information If there is imperfect information, the players may not be able to distinguish between some of the game histories. This is indicated by the information partition $(\mathfrak{I}_i)_{i \in N}$, in which the information sets S^i_j denote the information available to the players. When only singleton information sets exist, in other words no two histories belong to the same set, the game is one of perfect information. Otherwise it is one of imperfect information. If needed, semantic games of imperfect information are denoted by $\mathcal{G}^*(\varphi, g, \mathfrak{A})$.

What happens in these semantic games is that the partition may have different properties depending on the language in question and on what syntactic restrictions there might be. I will return to these issues towards the end of the next section. Other qualifications are in order, too.

3. Logic and imperfect information

Independence-friendly logics There are languages in which the assumption of imperfect information fails. An example is a language with Henkin (finite, partially-ordered, branching, parallel) quantifiers (see below). Imperfect-information games also provide semantics for IF logics (Hintikka 1996a; Hintikka & Sandu 1989; Hodges 1997a; Pietarinen 2005f). IF logics use a forward-slash notation that linearises Henkin quantifiers, but makes the information regulations more liberal.

Recapitulating the definition in Chapter 5, let $Qx\psi, Q \in \{\forall, \exists\}$ and $\phi \circ \psi, \circ \in \{\wedge, \vee\}$ be $\mathcal{L}_{\omega\omega}$-formulas in the syntactic scope of $Q_1x_1 \ldots Q_nx_n$, where $A = \{x_1 \ldots x_n\}$. Then the first-order language $\mathcal{L}^*_{\omega\omega}$ with informational independence is formed as follows:

- if $B \subseteq A$, then $(Qx/B)\,\psi$ and $\phi\,(\circ/B)\,\psi$ are wffs of $\mathcal{L}^*_{\omega\omega}$.

Let us call '/' an outscoping device and customarily write $\{x_1 \ldots x_n\}$ as $x_1 \ldots x_n$. $\not\models$ The semantics of an $\mathcal{L}^*_{\omega\omega}$-formula φ is given by the game $\mathcal{G}^*(\varphi, g, \mathfrak{A})$. As before, let us define an $\mathcal{L}_{\omega\omega}$-formula φ as true (resp. false) if and only if there exists a strategy in $\mathcal{G}^*(\varphi, g, \mathfrak{A})$ that is a winning one for V (resp. F) in $\mathcal{G}^*(\varphi, g, \mathfrak{A})$.

Outscoping may be applied to propositional and modal logics. As with propositional fragments, the application of the slash gives rise to formulas in which $\vee$ and $\wedge$ may be replaced by $(\vee/\wedge)$ and $(\wedge/\vee)$. As with quantifiers, in encountering $\varphi\,(\vee/\wedge)\,\psi$, V is not informed about the choice of conjunction made earlier, and in encountering $\varphi\,(\wedge/\vee)\,\psi$, F is not informed about the choice of disjunction. For more complex expressions, we need to distinguish the connective tokens, and the best way is to think of disjunctions and conjunctions as restricted existential and universal quantifiers over a domain with a designated individual. I will not go into detail about these extensions here.[8] Among others, modal extensions have special significance in terms of the semantics of quantified notions in epistemic logic, the problem of intentional identity, and many epistemological questions.

The technique of information hiding may, in principle, be applied to all logics that allow a coherent game-theoretic interpretation, including generalised quantifiers (Pietarinen, 2001b) and non-monotonic logics interpreted via the modal 'only knowing' of inaccessible worlds (Pietarinen, 2002b).

Of particular interest in IF logics is the behaviour of negation. As such, negation $\sim$ denotes strong, game-theoretic negation, prompting a role switch between the two players. If we introduce weak contradictory negation $\neg_w$, then $\neg_w\varphi$ is true if and only if φ is not true. The metalinguistic 'not' is also a contradictory negation.[9] All common laws involving negation, including de Morgan laws and the law of double negation, remain valid, but the law of excluded middle fails. This is because semantic games for IF logic are not determined: if there is no winning strategy for one of the players it does not follow that there is a winning strategy for his, her or its antagonist. An example of such an IF formula in which the law of excluded middle fails is $\forall x(\exists y/x)\, x = y$, interpreted over a two-element domain.

Figure 7.2 is an example of an imperfect-information semantic game for an IF sentence

$$\phi = \forall x(\exists y/x)\, Pxy\,. \tag{7.4}$$

It has one non-trivial information set that includes all the histories in which V is to move. Given the same truth conditions for atomic formulas as in the previous example (Figure 7.1), it is clear that neither F nor V has a winning strategy in this game.

Pietarinen & Sandu (1999) explore some implications of IF logic, and aim to set straight some of the misunderstandings that have occurred in the literature concerning it and its relation to GTS, most notably the misunderstandings and *fauxity* in Tennant (1998). The topics addressed include intuitionism, constructivism, compositionality, truth definitions, mathematical prose, negation in IF logic, and the status of set theory.[10]

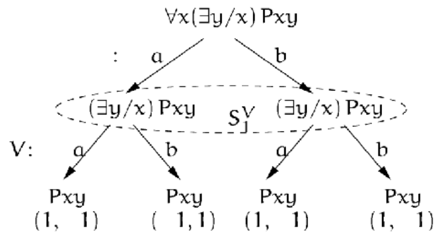


Figure 7.2. Imperfect-information semantic game $\mathcal{G}(\varphi, g, \mathfrak{A})$ with one non-trivial information set S_1^V annotated for V.

Henkin quantifiers Leon Henkin (1961) considered the possibility of extending first-order logic with partially ordered quantifiers:

$$\begin{matrix} \forall x & \exists y \\ \forall z & \exists w \end{matrix} \, Pxyzw \,. \tag{7.5}$$

The meaning of (7.5) is given by the Skolem normal form

$$\exists f \exists g \forall x \forall z \, Pxf(x)zg(z) \,. \tag{7.6}$$

Formula (7.5) is true if for every x there exists y such that for every z there exists w whose choice depends only on z and not on x and y, such that $Pxyzw$. The crucial point is that, whereas in ordinary predicate logic the number of arguments in Skolem functions replacing existential quantifiers corresponds to the number of universal quantifiers within the scope of which the existential quantifiers occur, partially-ordered quantifiers have a reduced number of such arguments.

Henkin had the idea of interpreting quantifiers, especially infinitely alternating and branching quantifiers, through a game played on a structure (Henkin, 1961, p. 179):

> Imagine, for instance, a "game" in which a First Player and a Second Player alternate in choosing an element from a set I; the infinite sequence generated by this alternation of choices then determines the winner. If we let to denote the class of all those sequences for which the First Player is the winner, then the formula $[\exists v_1 \forall v_2 \exists v_3 \forall v_4 \ldots (\pi v_1 v_2 v_3 \ldots)]$ simply expresses the fact that the First Player has a winning strategy.

Henkin quantifiers have been extensively studied, but unlike IF logic, they remain partially ordered and hence do not admit of, say, non-transitive, non-Euclidean, or cyclic quantifier orderings. These further structures may nonetheless be scrutinised in IF logics.

Constraints on information IF logics promote an informational outlook on logic and games. The notion of information may be studied from both logical and game-theoretic perspectives. In particular, I distinguish the following three interrelated notions:

1 *Uniformity* is a property of strategy functions in semantic games of imperfect information. This means that the outcome of an action has to be the same across the indistinguishable histories in such a game.
2 *The assumption of common actions* is a property of imperfect-information games. This means that the set of available actions has to be the same across the indistinguishable histories.
3 *The principle of observed actions* concerns players' knowledge about the game. It means that whenever a player has to make a decision, he or she can observe and identify the totality of available options.

These notions delineate different levels of representing information: The first pertains to the player's strategies, the second concerns the ways in which the structure of extensive games of imperfect information is defined, and the third concerns the player's perception of epistemic features associated with the game.

As noted above, the notion of choosing independently in IF logic may be explicated in terms of the uniformity of the strategy functions. The idea is that nothing in the strategy may signal to the player his or her actual location within an information set. It turns out to be superfluous, because in game theory no separate property is needed for the obvious reason that strategies are defined on information sets, not on individual histories.[11]

The second assumption of common actions, in turn, means that for all $h, h' \in H$: if $h, h' \in S^i_j$ then $A(h) = A(h')$. The idea here is that if a player cannot distinguish between two histories h and h', then the choices available to him or her after h must be the same as those available after h'. For, if $A(h) \neq A(h')$, then by the assumption of the observability of the available options the player could recover the difference between h and h'.

On the other hand, according to the third item, a player observes his or her available options when planning a move. It is of interest that this principle is, in fact, not needed in perfect-information games in which all information sets are singletons. It is thus perfectly legitimate to ask why we suddenly need it in imperfect-information games, which are supposed to solely concern the players' information concerning past actions and not upcoming actions.

By posing such questions one is re-kindling some time-honoured controversies that arose in economics already in the pre-games era. The early economists Léon Walras (1834–1910), the father of general equilibrium theory, and Vilfredo Pareto (1848–1923), the other grand neoclassist, were struggling with questions to do with what an agent can foretell in decision-making situations. After them, perfect foresight was long thought to be a precondition for equilibrium. In economics, such an assumption is all the more dubious the more parameters there are for a *homo œconomicus* to consider, including allocated time, prices, production, income, propositional attitudes involving higher-order beliefs and expectations, and so on. According to Morgenstern, who was unhappy with this general situation, such agents were tantamount to "demi-gods".[12] The term is quite apt in capturing the essence of the hyperrational agenthood in the decision theories of logical empiricists' descent.

Given Morgenstern's axiomatic leanings, he soon sought to fix models of such situations by imposing strict limits to the phenomenon, or system, that is tried to be theoretically captured. The method of fixing the boundaries was at that time heavily influenced by the Frege–Russell conception of logic, which Morgenstern was keen to promote as a conceptual breakthrough in economics. The conception of logic was no longer the Peirce–Peano one, which was dominant until after 1910, when Russell rashly decided to boast about the then-quite-chimerical awareness of Frege's work. There is a need to re-emphasise the influence of the atmosphere around the philosophy of science in Vienna around that time, or the alienation of such promotions from the conceptions of logic that Peirce would have advocated thereof. However, my issue is not the complexity of the situation under attack, but the somewhat inconsistent way that has surfaced in literature in determining whether perfect foresight is assumed in a like manner in perfect-information and imperfect-information games.

Furthermore, in the light of (3), it is not invariably clear that (2) should hold. The usual argument for (2) is that otherwise a player could carry out an infeasible action at some $k \in S^i_j$. If such choices were excluded from the scope of strategies, infeasible yet unattainable actions would ensue, in which case we would have to assume that (3) is thus invalid, too.

Even so, am I able to choose an action if I do not know what it is? Is the identity of actions all the players need to know when planning a move? Do they not need to know the consequences of that action too, enabling them to assess the value or practical bearings of the observable outcomes, making decisions pragmatically feasible, and thus enabling them to make finer distinctions and inferences concerning the actual locations in the game?

Such foresight, concerning not only the identification of immediately available action but also the practical effects of those actions, should not be seen as tantamount to rationality of players. A player may be rational even if he or she has a limited possibility of building a model of the future due to limited information. Foresight is not a precondition for the existence of winning strategies (or more generally equilibrium points), either, because it is something that is built-in into the very notion of strategy, be it a function or a non-deterministic set of relations between the decision point and the available actions.

It is worth observing how close this problem of supposed epistemic states of players concerning available actions is to the problem of cross-identification in the semantics of modal notions in terms of possible-worlds semantics for quantified epistemic logic. Assuming $A(h) = A(h')$ for $h, h' \in S^i_j$ is in modal terms analogous to assuming constant domains whenever questions of the identification of objects in A may arise (or more precisely, whenever $|h| = |h'|$, that is, whenever the 'modal depths' in the game histories coincide). Such a domain restriction amounts to a new type of quantified modal logic, the models of which have 'stratified' layers of individuals.

More generally, these points suggest that game theory is in need of a theory of knowledge that could address these issues. The programme of interactive epistemology that aims at applying logic to equilibrium selection and other game-theoretic problems has so far brought only inadequately expressive propositional logics to bear on such questions. There has been no attempt to analyse different notions of foresight in terms of much stronger quantified epistemic logics. It is interesting to observe that such a need was already expressed in the writings of Morgenstern and others in the 1930s. The emergence of logical notions of knowledge in the late 1950s and early 1960s, in tandem with the possible-worlds semantics for modal logics, did not draw its early motivation from these prevailing problems in economics.

I will refrain from further comment on these questions, and merely observe that the condition met by any IF formula, rather than limiting foresight, constrains the past of indistinguishable histories. To capture the fact that all indistinguishable histories should be composed of indistinguishable pasts, one states that: if $h, h' \in S^i_j$, then $|h| = |h'|$, for all $i \in N$. This condition, sometimes referred to as the von Neumann & Morgenstern condition, also excludes cases of absentmindedness, which are not excluded by (1) or (3): Let $\preccurlyeq$ be a partial order on the tree structure of extensive games $\mathcal{G}$ and $\mathcal{G}^*$, and let the game satisfy the non-absentmindedness condition: $h, h' \in S^i_j$, if $h \preccurlyeq h'$ then $h = h'$. Let depth $d(Q)$ of logical component Q in an $\mathcal{L}^{\mathrm{IF}}_{\omega\omega}$-formula φ be defined inductively in a standard way. Then $\mathcal{G}^*(\varphi, g, \mathfrak{A})$ satisfies non-absentmindedness, because all of the components in Q have a unique depth $d(Q)$, and so every subformula of φ has a unique position in $\mathcal{G}^*$ given by $L(h)$. Thus, for any two subformulas of φ at h, $h' \in H$ within S^i_j, it holds that $h \not\preccurlyeq h'$ and $h' \not\preccurlyeq h$.

4. Directions in game-theoretic semantics

The above description merely places semantic games at the starting point from which to investigate, implement and modify the available game-theoretic apparatus. This kind of variability in the notion of a game has several implications for logic.

Perfect or imperfect information? The first possibility is to drop the assumption that players have perfect information.

- Semantic games are of *perfect information* whenever the flow of information is not constrained. Otherwise they are of *imperfect information*.

In game-theoretic terminology, perfect information means singleton information sets, whereas imperfect information permits non-singleton ones.

Perfect-information games are customarily associated with formulas of ordinary first-order logic, while imperfect-information games are associated with IF formulas and Henkin quantifiers.

Informational independence The assumption of unregulated information flow in logic is one aspect of the informational-dependence assumption, and hence the move to IF logics marks a step from informational dependence to informational independence (Hintikka, 1996a; Sandu, 1993). It is not the rules of the game that one needs to modify for IF logic, but rather the strategic component pertaining to the information available to the players.

For instance, informational independence is needed in the resolution of the problem of intentional identity (Pietarinen, 2001a). It also gives rise to branching quantifiers in natural language (Pietarinen & Tulenheimo, 2004) and partiality in logic.

Partiality and games The field of partial logics has arisen as an independent object of study in logic and linguistics in recent years (Sandu & Pietarinen 2001; Pietarinen 2002b).

- Logic is *partial* whenever it has a third truth-value, Undefined, or has truth-value gaps.

Langholm (1996) has arguments for the case that the truth-value of Undefined and truth-value gaps do not coincide.

Remarks Partial logics are customarily taken to have multiple values. In addition to the two truth-values True and False, there are truth-value gaps, a third truth-value Undetermined. However, partiality should be studied independently of the question of whether logic has two values or more, because what is termed partiality in the literature is a property that naturally emerges from the game-theoretic interpretation. This happens as soon as the transmission of information between participants is not perfect, that is, the players are not perfectly informed of the past features of the game. Partiality is thus a consequence of entirely classical premises concerning the interpretation of language, without any additional postulation of truth-value gaps or third or fourth truth-values. The point is that the lack of an existing winning strategy for one of the players does not presume a winning strategy for the adversary.

Two notions of logical consequence are thus distinguished in partial logics. The notion $\models^+$ means a positive logical consequence (a formula being true in a model), and the notion $\models^-$ means a negative logical consequence (a formula being false in a model). Logical equivalence is thus split in two, weak equivalence (true in the same models) and strong equivalence (true in the same models and false in the same models).

Partiality in propositional logic The following example clarifies the relation between partiality and games. In propositional logic one might apply the slash notation to propositional connectives. Consider the sentence that we already met in Chapter 5, namely

$$(\varphi\ (\wedge/\vee)\ \psi) \vee (\theta\ (\wedge/\vee)\ \chi)\,. \tag{7.7}$$

We think of disjunction as prompting a choice between the two disjuncts (Left or Right), and similarly for conjunction. However, the latter choice is independent of the earlier choice, and hence the second player does not 'know' the earlier choice. We may view (7.7) as a new four-place connective $W(\varphi, \psi, \theta, \chi)$, termed transjunction in Sandu & Pietarinen (2001), and create undefined values by adding this connective to a propositional logic with complete models. Then it can be shown that, together with the usual Boolean connectives, this set of connectives is functionally complete for all partial functions.

Formula (7.7) gives rise to a connective that is motivated from a game-theoretic perspective. It emerges from a two-stage extensive-form semantic game of imperfect information between two players. This kind of game-theoretic approach yields an interpretation according to which partiality is generated as a property of the non-determinacy of games.

With regard to the particular example above, one may ask how the second player is supposed to know that it is his turn to move, without knowing the previous choice. For instance, if the second player is supposed to know that he has to choose between θ and χ, then he can infer that the first player has chosen Right, and if the choice is between φ and ψ, then the inference is that the previous choice was Left. Informational independence brings in some complications involving the notion of the information the players have, their knowledge of the game, and so on. Sandu & Pietarinen (2001) tackle these issues by allowing players to choose elements from a two-sorted domain rather than operating on subformulas. In the light of the informational partition of histories, there is no way a player could recover such information concerning the earlier choices.

Particular forms of imperfect-information games give rise to partial logic through which various forms of informational dependencies and independencies of connectives are studied (Sandu & Pietarinen 2001, 2003; Pietarinen 2001c).

It is possible to apply the analysis to partialised logic in which the law of excluded middle also fails for atomic formulas. In this case, it happens that the payoffs for both players are negative, namely $u_V(h) = -1$ and $u_F(h) = -1, h \in H$. According to IF logic, when the law of excluded middle fails for complex formulas, no such payoff arises. But it is perfectly possible to combine the two, one outcome being that the two notions of negation do not coincide even if the logic contains no slashed expressions, provided that some terminal histories are labelled with payoffs of $(-1, -1)$.

Partially-interpreted logics and games What lies behind the failure of the law of excluded middle in atomic formulas? One answer is that it is not only logically active expressions, but also non-logical constants that may enjoy informational independence. The question that then arises is how to interpret independence in, say, the formula $\forall x P[(c/x), x]$. What does it mean that constant c is independent of the quantified variable x? This is derivative of the question of what the game rules for non-logical constants are. Such rules pertain to the interpretation of language. They state that, when the atomic formula P of an IF first-order formula φ is reached in $\mathcal{G}(\varphi, \mathfrak{A})$, a low-level atomic game $\mathcal{G}^{\mathrm{Atom}}(P, \mathfrak{A})$ is evoked. This low-level game is now simply that whenever a constant is encountered in P, a value is assigned to it.

As soon as we have such rules at our disposal, significant further results may be attained. Instead of the usual winning conventions, according to which

- if P is true then V wins the play of the game $\mathcal{G}(P, \mathfrak{A})$, and
- if P is false then F wins the play of the game $\mathcal{G}(P, \mathfrak{A})$,

we state the converse:

- if V wins the play of $\mathcal{G}^{\mathrm{Atom}}(P, \mathfrak{A})$, then P is associated with the payoff 1 in the parental game $\mathcal{G}(\varphi, g, \mathfrak{A})$, and
- if F wins the play of $\mathcal{G}^{\mathrm{Atom}}(P, \mathfrak{A})$, then P is associated with the payoff -1 in the parental game $\mathcal{G}(\varphi, g, \mathfrak{A})$.

Winning means that the assignment produces the right values as intended by the predicates and other non-logical constants of the assertion, in the sense that they are in accordance with what is the 'natural history' of those formulas together with the use of such assertive propositions. Conversely, losing means the wrong values or failure to produce a value at all.

In semeiotic terms, winning refers to the element of semiosis that the signs, objects and interpretants give rise to, taking place between the interpreter and the utterer of the sign, the conditions of which are determined by the structure of semiosis similarly as they are determined in semantic games.

With these reversed conditions, we are in effect making game-theoretically meaningful distinctions between partially and totally interpreted languages. Partial interpretations arise if neither V nor F wins $\mathcal{G}^{\mathrm{Atom}}(P, \mathfrak{A})$. This is an objective fact of the indefiniteness of the determinations of the universe of discourse. The parental games will in that case exhibit payoff structures of

$u_V(h) = -1, u_F(h) = -1, h \in Z$. Conversely, complete interpretations lack any such indefiniteness. A consequence of these cases is that the law of excluded middle fails either at the level of atomic formulas (partial interpretation) or at the level of complex formulas (complete or partial interpretation).

In the case of partial interpretations, $\mathfrak{A}$ is differentiated in two parts, $\mathfrak{A}^+$ and $\mathfrak{A}^-$. $\mathfrak{A}^+$ is the model in which the formulas of the language are true and $\mathfrak{A}^-$ is the model in which they are false. If the union of $\mathfrak{A}^+$ and $\mathfrak{A}^-$ is not $\mathfrak{A}$, atomic formulas exist that lack an interpretation. They receive the payoffs $(-1, -1)$ in the correlated extensive game.

Remarks Hintikka (2002a) has pointed out that one philosophical consequence of the possibility of defining and actually using game rules for both complex and atomic sentences is that the analytic/synthetic distinction becomes rather feeble. No fundamental distinction obtains between logical and non-logical expressions, as they both fall under similar semantic rules. The reason is that we may well retain the same game-theoretic truth conditions for both, while getting to choose whether to have game rules for non-logical constants separate from those for logical ones.

This point is related to Peirce's concept of interpretants. The essential depth of a symbol is given in an imaginary state of information (Chapter 1). In the terminology used here, this notion could be seen as all the possible interpretations received by an atomic statement irrespective of the position actually reached in the game. It is a collection of labels of terminal histories. According to him, the informed depth of a symbol is, in contrast, all the real characters that can be predicated of it in a particular state of information. In my terminology, this refers to the sequences of assignments reached at terminal histories.

In addition to describing the essential and informed depths of a symbol, Peirce suggested the substantial depth, the hypothetical state of information coinciding with perfect knowledge of all that there is (2.409, 1893; Chapter 1). This could be thought of as the totality of all there is in a game.

The upshot is that essential, informed and substantial variants of the depth of a symbolic expression is given in its interpretants. The choice between such multiplicity in semiosis then depends on the purpose of the sign that gives rise to interpretants and the information conveyed in it.

The interpretants, or in game-theoretic terms anything that makes the interpretations of atomic and non-atomic expressions what they are intended to convey, are themselves cultivated in repeated cycles of semeiotic processes. The nature of that process is explained in Chapter 11. The general conclusion is that, whereas the interpretation of non-logical constants versus that of logical constants is what has typically been taken to be the cut-off point for deciding what is an analytic versus what is a synthetic judgement, similar demarcation no longer carries over to game-theoretic and semeiotic interpretations.

Non-standard partiality Moreover, there are non-standard partialities, both at the levels of winning conventions and at the level of truth conditions. In the latter case, they are given by the existence of Skolem functions (winning strategies). As regards the winning conventions, we define thus:

- If V wins $\mathcal{G}^{\text{Atom}}(P, \mathfrak{A})$ then P is not false. If F wins $\mathcal{G}^{\text{Atom}}(P, \mathfrak{A})$ then P is false.
- If V wins $\mathcal{G}^{\text{Atom}}(P, \mathfrak{A})$ then P is true. If F wins $\mathcal{G}^{\text{Atom}}(P, \mathfrak{A})$ then P is not true.

As regards the truth-conditions, we state that

- φ is not false in $\mathfrak{A}$ if and only if there exists a winning strategy for the player who started the game by playing the role of V.
- φ is not true in $\mathfrak{A}$ if and only if there exists a winning strategy for the player who started the game by playing the role of F.

Instead of verifiers and falsifiers, the players V and F are perhaps best seen as playing the roles of the Non-Falsifier and the Non-Verifier in the clauses defining truth conditions, respectively. These non-standard clauses may be applied to formulas of IF as well as of non-IF logics.[13]

Perfect or imperfect recall? We have seen how semantic games of imperfect information relate to logics. However, the difference between perfect and imperfect information is seen to be subject to further qualification. The class of imperfect-recall games often has to be taken into account too.

- Games are of *perfect recall* whenever the players do not forget their previous information or their previous choices. Otherwise they are of *imperfect recall*.

An illustrative real-life example of the distinction between perfect and imperfect information is the game of chess versus the game of poker. The game of bridge, on the other hand, could be considered a game of both imperfect information and imperfect recall, in that two teams play off against each other.

Imperfect recall is a recurring theme in IF logics and games of imperfect information. The distinction between the player's information and choices may be further characterised with formal precision (Pietarinen, 2001c), but I refrain from doing so here. Imperfect recall is also relevant to the following question.

Two or more players? If the players in semantic games have non-persistent information and hence imperfect recall, we need an effective way of modelling such a phenomenon. A natural way of doing so in game theory is to divide the two principal players into multiple-selves or members of teams.

- A *team* is a (finite) set of non-coordinating players who have identical payoffs but who act individually.

The teams V ('Us') and F ('Them') consist of a finite number of individual members $V_l \in V$ and $F_k \in F$, for the finite positive integers l, k. The members of a team are not allowed to communicate because this destroys the team's ability, when viewed as one player, to forget something about which it has had information. The members of the same team all receive the payoff $u_i(h)$ when the outcome of a play is solved.[14]

Remarks The team approach provides a way to prevent players signalling their choices to other players (Pietarinen, 2001d). The information for the individual team members remains persistent, although the teams, viewed as single players, do not forget it. Hence all the moves made by the individual members are assumed to be member-specific, which means that information sets are assigned to them. However, when each team is viewed as a single player, one could think in terms of an implicit map from the 'information set' $(\mathfrak{I}_{i=V})$ or $(\mathfrak{I}_{i=F})$ of all the information sets of the respective player to the information sets of its members. Thus V and F could coordinate the individual players, and determine who makes the next move or who is to be introduced next, although the decision for the actual choices is made by the individual agents.

The idea of the team (or multi-person or multi-self) approach to games of imperfect recall goes back to the early works of von Neumann & Morgenstern (1944), Strotz (1956) and Isbell (1957). According to von Neumann & Morgenstern (1944, p. 53), "It is worth noting how the necessary "forgetting" of [move$_2$] between [move$_1$] and [move$_3$] was achieved by "splitting the personality" of $[V]$ into $[V_1]$ and $[V_2]$".

The team approach is not, strictly speaking, technically necessary, but it is rather an implementation device to enable us understand the transmission of information in games of imperfect recall. Viewing imperfect recall as a team-theoretic game aims at explaining what happens when information is dispelled from the player's memory. We want information to be persistent for real decision makers, and the team approach provides a way of understanding semantic games for IF logics. What one eventually arrives at is an agent normal form for extensive games, whenever each information set is associated with a separate player in a team. Chapter 14 presents further applications of these games.

There is a concrete twist in semantic games of imperfect recall, however. For example, consider the game correlated with the IF formula $\exists x(\exists y/x)\, x = y$. Is this sentence true over the structure of natural numbers? The answer is affirmative, given the basic notion of semantic games with two players with persistent information, but negative, if the players are considered as teams. In the latter case, team V cannot pool the information (here, two Skolem constants) to yield a winning strategy, in other words it does not have persistent information and thus, not unlike typical approaches to automata, exhibits non-persistent memory storage.

Complete or incomplete information? Despite their generality, imperfect-information games, together with their various refinements, provide just one way of looking at the streams of information between players, or the logical independence and dependence relations thereof. Given any logic that admits of coherent semantic rules, and hence a game-theoretic interpretation, another form of independence, or informational regulation, is seen to emerge. Namely, there may be lack of information about the mathematical structure of the game itself — defined, for example, by its extensive form. This paucity may take many forms. The players may not be fully informed about the other players' payoff functions, about the strategies available to them, about the knowledge the other players have about the game, and so on.

- The game is one of *incomplete information*, if there is a chance move by Nature that is unobserved by at least one player.[15]

The connection between incomplete-information games and logic is in the behaviour of negation. Like quantifiers and connectives, negation, too, may be hidden. What such hiding boils down to is that the information about the role switch may not be available to the players at the later stages. This means that the players are uncertain about which role they should assume, that is, they are not informed about whether they are to act as the verifiers or as the falsifiers of a given sentence.

Logic of payoff independence The types of independence in the literature on IF logics have concerned informational independence, operationalised by games of imperfect information. But also another type exists, not confined to *informational* independence. It need not refer solely to players' knowledge, information or ignorance concerning the choices made in the game. Accordingly, it need not refer solely to possible patterns of dependence and independence between quantifiers or connectives. For other uncertainties may exist in the game that affect players' strategic decisions. Most notably, players may lack information concerning the mathematical structure of the game. In fact, such games are part and parcel of game theory, known as incomplete information, or Bayesian, games.[16]

Lacking an off-the-peg term for such a logic, let us call it logic of payoff independence (PI logic). The idea of referring to payoffs goes back to Harsanyi (1967). What Harsanyi showed was that uncertainties concerning the structure of the game, including players' strategies, may be transformed into uncertainties concerning the values of the players' payoff functions.

What kinds of characteristics of incompleteness can we meaningfully have in logic? Purely game-theoretically, before Harsanyi's pioneering work it was thought that incompleteness concerns players' uncertainty about the rules of the game and thus they cannot act strictly according to the rules. I do not mean rule incompleteness. Players must follow the rules that define the legitimate moves, simply because at each position of the game, the corresponding expression in the sentence completely determines what the legitimate moves of a player are. IF logics do not change this fact. Correlated with games of imperfect information, in IF sentences imperfectness affects players' strategies, not the degree of familiarity about the game rules.

Some examples of incomplete information are:

- Players of a semantic game may lack information about the strategies used in the game. This may concern adversaries' strategies as well as players' own strategies.
- Players may only know the sorts of previous choices made in the game but not the choices themselves.[17]
- Players may be uncertain about various parameters attached to players, including uncertainty about
  - the number of agents there may be in the opponent team;
  - the size of one's own team.
- Players may be uncertain about payoffs.

The uncertainties listed above provide examples of ignorance on the values of payoffs. They are thus instances of the Harsanyi transformation. It asserts that games of incomplete information may be thought of as games of complete but imperfect information with random moves by Nature hidden from subsequent players. This is implemented so that Nature chooses types of players but only reveals to the players their own types. Even the assumption concerning awareness of one's own types may be relaxed.

I will ignore here the uncertainty about the cardinality of teams and about sorts. The former becomes relevant if the games of incomplete information — after the Harsanyi transformation has been performed — exhibit imperfect information and imperfect recall. Imperfect recall is generated by an unrestricted application of slashes, in which case uncertainties concerning the organisation and structure of these associated teams of agents may materialise.

As to the uncertainty concerning strategies, I will limit the discussion to a simple incompleteness in which the (non-empty) type space is at most binary, containing two types corresponding to the two roles that the players may have. The reason for this limitation is that this type of uncertainty is reflected in a natural extension of IF logic to independent negations.

This extension applies slash indicators to strong (dual) negations. Recall that in IF logics, contradictory negation ($\neg_w$) may only be introduced by using the meta-level rule $\mathfrak{A} \models \neg_w\varphi$ iff $\mathfrak{A} \not\models \varphi$. The denial of a proposition asserts truth precisely in those circumstances in which the proposition is false. It lacks the game-theoretic, processual notion of negation (~), which — unlike contradictory negation that reverses the polarities of the *partitions* denoted by the proposition — reverses the polarities of the *processes* correlated with the propositions.

It is assumed that strong negations may occur on either side of the slash.[18] In addition to variables and connectives, a finite sequence of negations may exist on the right-hand side of the slash. These negations, as indeed all occurrences of strong negations, are indexed to distinguish between different tokens in a formula. The case is similar with hidden connective information (Sandu & Pietarinen, 2001, 2003). In contrast to hidden connectives, however, in negation independence we do not have restricted quantification at our disposal that would accomplish informational independence of binary connectives in terms of restricted quantifiers over indices.

More precisely, we take the language $L_{\mathbf{PI}}$ to consist of IF first-order logic plus instances of the following expressions, in which φ is any sentence of IF first-order logic:

- $(\sim_j / W)\,\varphi, W \subseteq A = \{x_1 \dots x_n, \sim_1 \dots \sim_m\}, \sim_j \notin W, j \in \omega.$
- $(Qx/W)\,\varphi, Q \in \{\exists, \forall\}, W \subseteq A = \{x_1 \dots x_n, \sim_1 \dots \sim_m\}, x \notin W.$

Elements in A are those that are already visited in the clauses syntactically superordinate to the slashed expressions.

For example, $\sim_1\sim_2 (\sim_3 /\{\sim_1\})\,\varphi \in L_{\mathbf{PI}}$, $\forall x \sim_1 \forall y(\exists z/\{x, \sim_1\})\,Rxyz \in L_{\mathbf{PI}}$ and $\exists x \sim_1 \bigwedge_{j_1}(\bigvee_{j_1} /\{j_1, \sim_1\})\,(\psi)_{j_1 j_2} \in L_{\mathbf{PI}}$.

The indexing schema may be taken to refer to all the distinct morphological manifestations of negative words (n-words) in natural language.

The semantics for $L_{\mathbf{PI}}$ is given by games of imperfect and incomplete information. However, by the applications of the Harsanyi transformation, the semantic games reduce to those of complete but imperfect information, with hidden chance moves. We let the indices k, l range over a two-element set $\mathfrak{T} = \{\nu, f\}$. At $h, h' \in Z$, the payoffs $u_V^k(h)$ and $u_F^l(h')$ for V^k and F^l will depend not only on strategies s_V^k and s_F^l but also on the types of players.

Also, let R be the role function $R\colon \Phi \times H \to \{\uparrow, \downarrow\}$ from a set of $L_{\mathbf{PI}}$-sentences Φ and a set of histories $H \setminus Z$ of the associated extensive game to the set of two values. The role function is merely a flip-flop register that changes its state every time a negation is encountered in a sentence.

The semantic games $\mathcal{G}(\varphi, \mathfrak{A})$ for $L_{\mathbf{PI}}$-sentences are like those of imperfect information but interspersed with a type space $\mathfrak{T}$ and three players instead of two. The third player, Nature, is, effectively, a probability generator. Let φ be an L_{PI}-sentence. The set of game rules are adjoined with the following three rules:

- If $\varphi = (\neg_j/W)\,\psi$ and $R(\varphi, h) = \top$, then:
$$\begin{cases} \textbf{If} \text{ for any } i \text{ for which } \sim_i \notin W, \text{ then } R(\psi, h \frown \psi) = \bot\,. \\ \text{The game continues as } \mathcal{G}(\psi, \mathfrak{A}). \\ \textbf{Else:} \text{ Nature chooses from } \mathfrak{T} = \{\nu, f\}. \\ \text{The game continues as } \mathcal{G}(\varphi', \mathfrak{A}),\ \varphi' = (\sim_j / W \setminus \{\sim_i\})\,\psi. \end{cases}$$
- If $\varphi = (Qx/W)\,\psi$, $Q \in \{\exists, \forall\}$, then:
$$\begin{cases} \textbf{If} \text{ for any } i \text{ for which } \sim_i \notin W, \text{ then} \\ \text{the game continues as } \mathcal{G}((Qx/W)\,\psi, \mathfrak{A}). \\ \textbf{Else:} \text{ Nature chooses from } \mathfrak{T} = \{\nu, f\}, \text{ and} \\ \text{the game continues as } \mathcal{G}(\varphi', \mathfrak{A}),\ \varphi' = (Qx/W \setminus \{\sim_i\})\,\psi. \\ \text{Later choices for } Qx \text{ are independent of Nature's moves.} \end{cases}$$
- If $\varphi = (Qj_n/W)\,\psi$, $Q \in \{\bigvee, \bigwedge\}$, then:
$$\begin{cases} \textbf{If} \text{ for any } i \text{ for which } \sim_i \notin W, \text{ then} \\ \text{the game continues as } \mathcal{G}((Qj_n/W)(\psi)_{j_n}, \mathfrak{A}). \\ \textbf{Else:} \text{ Nature chooses from } \mathfrak{T} = \{\nu, f\}, \text{ and} \\ \text{the game continues as } \mathcal{G}(\varphi', \mathfrak{A}),\ \varphi' = (Qj_n/W \setminus \{\sim_i\})\,(\psi)_{j_n}. \\ \text{Later choices for } Qj_n \text{ are independent of Nature's moves.} \end{cases}$$

The winning conventions are as before. The truth and the falsity of complex sentences φ of L_{PI} are now defined somewhat differently, however. By f and ν-branches we mean those subgames $\mathcal{G}'$ of $\mathcal{G}$ the roots of which correlate with Nature's choices of the types f and ν, respectively.

- The sentence φ is true in $\mathfrak{A}$ iff in the subgame $\mathcal{G}'$of $\mathcal{G}(\varphi, \mathfrak{A})$ that has no f-branches, there exists a strategy profile S_1 of pure optimal strategies $\{s_1^k, s_2^1 \ldots s_2^m\}$ for the player who started the game $\mathcal{G}(\varphi, \mathfrak{A})$ as playing the role of V, and Nature chooses her type to be ν.[19]
- The sentence φ is false in $\mathfrak{A}$ iff in the subgame $\mathcal{G}'$of $\mathcal{G}(\varphi, \mathfrak{A})$ that has no ν-branches, there exists a strategy profile S_2 of pure optimal strategies $\{s_2^1 \ldots s_2^k, s_1^l\}$ for the player who started the game $\mathcal{G}(\varphi, \mathfrak{A})$ as playing the role of F, and Nature chooses his type to be f.

Nature's choices precede players' assessment of expected values of payoffs and the selection of optimal strategies. Because chance moves are concealed, and unless Nature decides to choose the right types for the players, there is not much a player can do to enforce a win, even if optimal strategies would exist. The aggregate of optimal strategies plus Nature's random choices comprise a winning strategy that agrees with the truth and falsity of L_{PI}-sentences in $\mathfrak{A}$.

It follows that given such chance moves, the law of excluded middle fails even if there were no slashes in an L_{PI}-sentence φ. This holds irrespectively of whether the underlying language is completely or partially interpreted.

By choosing types, Nature's moves affect the strategy sets of players, their information partitions, and payoffs. What Nature thus chooses is a state of the world. The state may denote context, environment, time, mood, collateral information or the common ground according to which the particular plays of the game are to proceed. Such choices are not restricted to initial deals.

The law of double negation is valid for indexed negations in slashes to the right. It suffices to check whether the parity of negations in W is even or odd. Given the probabilistic nature of the system, the double negation $\sim_1 (\sim_2/\{\sim_1\})\, \varphi$ reduces to φ with equal probability of reducing to $\sim_1\sim_2 \varphi$. Likewise, the prima facie contradictory sentences such as $\sim_1\varphi \wedge \sim_2(\sim_3/\{\sim_1\})\, \varphi$ are associated with a conditional probability distribution determined by Nature.

The type, or the state of the world, carries information about the values of the payoff functions. Despite chance moves marking decisions between two options, the notion of a type is not the same thing as the notion of a role of the player. In general, we may think of roles consisting of finitely many types.

Four types of information In extensive-form semantic games, four types of information may be discerned (Rasmusen, 1994).

1 *Certain information*: Nature does not move elsewhere than at the root of the game. Otherwise the game is one of uncertain information.
2 *Complete information*: Nature does not move first, or her initial move is observed by every player. Otherwise the game is one of incomplete information.
3 *Perfect information*: information sets of both players are singletons. Otherwise the game is one of imperfect information.
4 *Symmetric information*: both players get the same information in the game. Otherwise the game is one of asymmetric information.

Certainty thus means initial, unhidden chance moves by Nature.

Examples are provided by the following $\mathrm{L}_{\mathbf{PI}}$-sentences:

$$\phi := \sim_1 (\sim_2/\{\sim_1\})\, \varphi \tag{7.8}$$

$$\phi := \sim_1(\sim_2/\{\sim_1\}) \bigvee_{j_n} (\varphi)_{j_n} \tag{7.9}$$

$$\phi := \sim_1 \forall x(\exists y/\{\sim_1\})\, \varphi. \tag{7.10}$$

The game $\mathcal{G}(\phi, \mathfrak{A})$ correlated with (7.8) is of perfect, complete but uncertain information. $\mathcal{G}(\phi, \mathfrak{A})$ correlated with (7.9) is of imperfect, incomplete and uncertain information. $\mathcal{G}(\phi, \mathfrak{A})$ correlated with (7.10) is likewise of imperfect, incomplete and uncertain information. The third game shows the utility of the role function R: the player in $\{V, F\}$, when choosing for the existentially quantified variable y, may not possess the information that she was the Verifier who in fact chose the value for the universally quantified variable x; however, she might infer that she must have been located within the scope of an odd number of unslashed negations.

On the other hand, all extensive semantic games that are not undetermined and not correlated with formulas with mutual dependencies have asymmetric information.

For instance, the sentence (7.8) reduces to φ with a chance move by Nature. Likewise, (7.9) reduces to $\bigvee_{j_n}(\varphi)_{j_n}$ with a chance move. The sundry effect such reductions have is that they change the game of uncertain information into certain information. Other properties of the information structure are preserved.

Applications and Amplifications Let me suggest a few consequences of having incomplete information in logic.

First, given ignorance concerning payoffs, comparisons of actions under uncertainty vs. actions under risk become possible, since the rival player, although optimal maximiser, can conceal the costs that his or her moves may incur.

In conversational settings, for instance in those provided by game-theoretic formulations of Grice's maxims (Hintikka, 1986; Parikh, 2001), certain moves are costly so as to be charged against the payoffs of the player. But if these costs were concealed, strategic advantages to decision makers are inevitable.

Third, heterogeneous and iconic systems of logic, including Peirce's EGs, are capable for providing analogues to the partially-overlapping scopes of negations of $L_{\mathbf{PI}}$ in terms of non-partitional graphs, if the cuts (the closed lines of separation representing negations) are permitted to overlap. Given two cuts c_1 and c_2, the affirmative area would lie within the double cut area $c_1 \cap c_2$, whereas the negative area is in their union $c_1 \cup c_2$.

Permitting overlaps is not as radical as it may seem. In semantic networks and conceptual graphs akin to EGs, partitioned networks have provided some insight into knowledge representation (Hendrix, 1979). Partitions may then be generalised to non-compositional cases.

Following the recognition of the limitations of standard partitional information structures in extensive games, a growing body of research is emerging on non-partitioned imperfect-information games.[20] The sentences of $L_{\mathbf{PI}}$ are, indeed, more like semantic nets than syntactic constructions, with the dependency relations given by directed graphs.

Linguistically, such nets may depict collections of semantic constraints for possible interpretations of a sentence. They are thus connected with underdeterminacy of scope relations. The perspective of networks as semantic constraints is also applicable to theories of discourse interpretation.

Moreover, PI logic provides the logical counterpart to the distinction between languages that have double negations (such as English) and languages that have negative concord (such as most Romance languages). Negative concord means that multiple occurrences of morphologically negative items compress into a single negative expression.

To come clean on the history of intellectual ideas, Peirce described this phenomenon in 1905 as the distinction between logical languages and quantitative languages:

> There are some languages in which two negatives make an affirmative. Those are the logical languages. [...] In other languages, probably the majority, a double negative remains a negative. These are quantitative languages. We should expect the people who speak them to be more humane and more highly philosophical. The quantitative view of negation ... does not really involve any bad reasoning. (MS 283: 120–1).

In a similar vein, Peirce noted in 1896: "Were ordinary speech of any authority as to the forms of logic, in the overwhelming majority of human tongues two negations intensify one another" (3.481).

The subsequently established idea of drawing this division was in terms of categorising the n-words of such languages. Alternatively, the basic division has been drawn between single and multiple concord (Forget et al., 1998).

In the semantic games for $L_{\mathbf{PI}}$-sentences restricted to negation independence and duplex type spaces, the players are uncertain about which of the two possible subgames are being brought out by Nature. The distribution is furthermore assumed to be fair, and there is an equal chance of playing either of these games. This is reflected in the truth-conditions so that, in addition to the existence of an optimal strategy and the expected values of payoffs, Nature has to side with the player in order for a game to verify (i.e., tagging Truth to the sentence) or falsify (tagging Falsity to it) an $L_{\mathbf{PI}}$-sentence in $\mathfrak{A}$. The sole reason as to why I have restricted the focus to the simplified case of payoff independence is that it falls naturally from the outfit of IF logic via negation independence.

The assumption of common priors that the players have is also simple: the prior probabilities are the same for all elements of the type space, in other words just the probabilities of 0.5. We thus avoid the intricacies of Bayesian reasoning and Bayesian extensions of equilibrium, apart from what was used in the definition of the truth and falsity of formulas. Common priors being common knowledge has for a long been the standard, albeit by no means indispensable or the most realistic assumption in economics, since it looks away from genuine asymmetric higher-order knowledge and belief concerning type distributions.

Moreover, we may even think of there being differently weighed occurrences of dual negations, contributing to logical rules such as the law of double negation with varying degrees.

Also notable is the commonality of common priors as common knowledge to pragmatic theories of language and communication. Following Grice, many have assumed the key element in the creation of the common ground being common knowledge among speakers and hearers (Chapter 12). However, in my case tables can be turned on Grice and his followers, as communicative situations may, alternatively, be modelled applying the Harsanyi doctrine, thus dispensing with any higher-order knowledge. Such communicative situations are far from recondite for the sheer reason that there are strategic advantages in hiding costly payoffs. Assuming common priors, Grice's conversational maxims would then follow from the relatively weak rationality assumptions involved in the Harsanyi doctrine. As far as I know, this possibility has not been pursued by natural-language pragmatists.

Given the probabilistic and context-dependent character of $L_{\mathbf{PI}}$ expressions, a new complication apparently accompanies the advocates of compositional semantics: How to extend the semantics that assigns — instead of sequences of assignments — sets of sequences of assignments to subformulas of a sentence of IF logic, into a compositional semantics for $L_{\mathbf{PI}}$-formulas? This is a genuine complication, since also whether such sets or co-sets go with V or with F (and whether such sets for an $L_{\mathbf{PI}}$-formula φ in $\mathfrak{A}$ are co-sets for an $L_{\mathbf{PI}}$-formula $\sim\varphi$ in $\mathfrak{A}$) fall under imperfect information. This is no mug's game, either, since $L_{\mathbf{PI}}$ is not yet another fabricated language, but a logical reflection of what goes on in commonplace classes of games pursued in game theory.

Finally, the approach should not be confounded with that of Blinov (1994), who also considered the possibility of introducing chance moves into GTS. My approach is very different. Blinov does not use type spaces, and his games are ones of complete information. He also restricts chance moves to initial positions of the game, which may be interpreted as Nature's deal of values for free variables. Blinov considers no incomplete information, as his goal is to find a game semantic correlate to that of supervaluations (van Frassen, 1969).

Much the same ideas that have motivated the passage from traditional first-order logic to its IF extension motivate the move from traditional logics to their payoff-independent extensions, namely the recognition of the ambiguous and restrictive notion of scope (Hintikka, 1997). Negations do have priority orders, too, operationalised via incomplete information.

Strictly or non-strictly competitive games? Yet another variant of semantic games may be created that alters the objectual attitudes they have towards each other — concerning competitiveness, for example. Indeed, strictly competitive games, as the above-mentioned examples all refer to, are rarely considered in game theory and non-strictly competitive games such as variable-sum and mixed-motive games are much more common.

One important feature of IF logics is that negation denotes a strong game-theoretic negation. It is possible to introduce a weak contradictory negation $\neg_w$, but this cannot be captured by any game rules (Hintikka, 1996a, pp. 131–162). The behaviour of classical negation is instead captured by:

(i) $(\mathfrak{A}, g) \models^{+} \neg_{w}\varphi$ if and only if not $(\mathfrak{A}, g) \models^{+} \varphi$
(ii) $(\mathfrak{A}, g) \models^{-} \neg_{w}\varphi$ if and only if not $(\mathfrak{A}, g) \models^{-} \varphi$.

In clause (i), the sentence $\neg_{w}\varphi$ being a truth-consequence of a model $\mathfrak{A}$ says that φ cannot be verified, and in the latter, $\neg_{w}\varphi$, being a falsity-consequence of $\mathfrak{A}$, asserts that φ cannot be falsified. Sentences prefixed with weak negation become assertions about games, indicating when a winning verifying or falsifying strategy does not exist.[21]

Consequently, the occurrence of weak negation introduces the fourth truth-value Over-defined. An alternative approach exists, however, which takes games to be non-strictly competitive.

- For any $\mathcal{L}_{\omega\omega}$ or $\mathcal{L}^{*}_{\omega\omega}$-formula φ, the game $\mathcal{G}(\varphi, g, \mathfrak{A})$ or $\mathcal{G}^{*}(\varphi, g, \mathfrak{A})$ is *strictly competitive*, if (i) a winning strategy f exists for V then a winning strategy g does not exist for F, and (ii) if a winning strategy g exists for F then a winning strategy f does not exist for V.

In non-strictly competitive games, it may happen that both players have a winning strategy. For instance, if there are some terminal nodes that are winning for both V and F, and atomic formulas ψ are interpreted as having the truth-values True and False, that is, they also have the value of Over-defined.

Remarks Non-strictly competitive games are useful in distinguishing between various notions of consistency: although a version of *ex falso sequitur quodlibet* could be tolerable as $\varphi \wedge \sim\varphi$, it is never the case that $\varphi \wedge \neg_{w}\varphi$, for it does not make sense to assert that 'there exists a winning strategy for V in φ, but there does not exist a winning strategy for V in φ', which would denote strong inconsistency.

Given the zero-sum property of strictly competitive games, the partial truth value Undefined means that the attempts of both V and F can be frustrated. Furthermore, in strictly competitive games, players' preferences could become inverses, but if the preferences are not assumed to be strictly oppositional, the presence of a definite truth value in a sentence does not necessarily mean serious deprivation in terms of the purposes and motivation of the other player. For example, the strategy functions in non-strictly competitive games could at times be (partially) revealed to the opponent.

Non-coherence arises as soon as the assumption that the games are strictly competitive is dropped. How viable is this assumption? A number of real-life situations relate to games that are not strictly competitive, such as the prisoner's dilemma, differential games, bargaining and negotiation games or argumentation, which in the context of logics suggests the emergence of yet another game-theoretically motivated logic.[22]

Interestingly, Aristotle observed the game-like character of competitiveness in relation to certain characteristics of an argument:

> If it [what a question says] is partly true and partly false, he [the answerer] must add a remark that it has several meanings and that in one meaning it is false, in the other true; (*Topica VIII*, sect. VII).
>
> He who hinders the common task is a bad partner, and the same is true in argument; for here, too, there is a common purpose, unless the parties are merely competing against one another; for then they cannot both reach the same goal; since more than one cannot be victorious. (*ibid.*: sect. XI).

The point Aristotle makes here is that if argumentative situations are seen as competitions, then only one player can come out as the winner. There are reasons why such a situation is not preferred over mutually beneficial ones, in which participants may concede points made by an opponent. This is commonly viewed as a disputational rule, a rule for dialectics rather than a rule for logic. Nevertheless, Aristotle devoted considerable energy to it in discussing possible exceptions or qualifications to the law of contradiction in the context of logical investigation, too.

Furthermore, one is naturally drawn towards Peirce's remarks concerning the circumstances in which the principle of contradiction does not apply, namely those in which propositions exhibit vagueness (Chapter 6), antithetical analogues of generality unsubject to the principle of excluded middle. According to Peirce, all communication is bound to involve some vagueness. Likewise, potentialities, the signs (spots) of one compartment of gamma graphs, and modal assertions in general, are unsubject to the principle of contradiction. We should remember that these principles are not tantamount to the laws of non-contradiction and excluded middle, but are to be spelled out as ones in which the definite (in the principle of contradiction) and individual (in the principle of excluded middle) non-general subject terms are taken into account in the assertions they incorporate. As it happens, this formulation is closer, in spirit and in letter, to Aristotle than contemporary logic.

5. Semantic games and natural language

As observed in Part I, according to Peirce, the interpretation of phrases such as 'any man is not good' is left to the opponent of the proposition, while when we say 'some man is not good', the respective selection is transferred to the opponent's opponent, the defender of the proposition. The opponent's opponent's opponent is the opponent again, equivalent to *any* (3.481). Thus Peirce came to consider natural-language expressions via dialogic processes.

This basic idea has been developed, and has expanded in several directions, in the game-theoretic works of Hintikka and others. One of the central assumptions in the game-theoretic approach to natural language is that, since there are no variables, proper names are substituted for quantifier phrases such as *every X who Y*. This enables various game rules to be defined for expressions of natural language in a similar way as for formal language.

Moreover, game rules have also been defined for a wide variety of lexical and morphological categories, including modals, intensional verbs, tense operators, pronouns, definite descriptions, possessives, genitives, and prepositional phrases (Saarinen 1979; Hintikka & Kulas 1983, 1985; Hintikka & Sandu 1991).

As in logic, any sentence of English defines a game between two players, the Verifier V and the Falsifier F, the former aiming to show that the sentence is true and the latter aiming to show that the sentence is false. The game rules for familiar quantificational expressions such as *some*, *every*, *a(n)*, and *any* prompt a player to choose an individual from the relevant domain or choice set, giving the individual a name if it does not already have one, the game continuing with respect to an output sentence defined by the game rules. Analogously with GTS for formal languages, the game terminates at the point (roughly corresponding to atomic formulas) at which further applications of the rules are no longer possible. Just as in logic, items that prompt a well-defined game move could be considered either informationally dependent on or independent of the semantic information of earlier items.

There is no obvious limit to how many or what kind of game rules can be defined in order to take in a reasonably wide fragment of natural language. Pietarinen (2001b) introduces and examines a number of new rules, covering a variety of linguistic classes and expressions, including negative and positive polarity items (many of which are lexical), the morpheme *even*, adverbs of quantification, plus both monadic and polyadic generalised quantifiers. There are even prospects of addressing the notion of eventualities in game-theoretic terms, which shows that aspects of informational independence rise above nominal domains. The upshot is that games provide a cross-categorial system of semantics, and also a unifying mechanism across domains of different types.

In general, the use of GTS for natural language is, to a large extent, inspired by the idea that for a specified class of (at least logical) elements there are 'unmarked' cases in which the elements are interpretationally independent, which could then be spelled out in terms of the notion of informational independence.

This idea still needs to be documented in full, but the happy effect is that it would entail scope dependence where found, and an explanation of why we do not find it where it is missing. Among other things, it would explain the behaviour of many of the linguistic expressions perceptively considered in Liu (1997), broached as examples of scopally independent elements in a given sentence. In the light of the totality of these linguistic findings, which will not be considered further in the present work (see Pietarinen 2001b,e instead), it becomes all the more dubious to maintain that informational independence remains a marginal feature of the semantics of natural language.

Among the linguistically relevant issues is the concept of strategic meaning (Hintikka, 1987b). This is different from abstract meaning, which is given by truth-conditional considerations. Strategic meaning refers to a range of aspects related to what the strategies that the players use actually are and what their content is. In addition to anaphoric coreference (Hintikka & Sandu, 1991), it shows up in the meaning of pronouns of laziness, in the nature of aspect in natural language tense constructions, and in the notion of expectations of informativeness that arise in various morphemes, aspectual particles and so on. It is, in fact, quite difficult to find one general term in natural language that does not subsume at least a touch of the strategic side of meaning.

6. Conclusions

Some recent literature The starting point of this chapter was the identification of a class of games that could serve as a unifying framework for different IF logics. In order to set this within the context of current research, I list some the more specific results. (a) Extensive semantic-games for IF logics have a subclass of extensive games of imperfect information satisfying non-repetition, consistency, the von Neumann-Morgenstern condition, and imperfect recall (Pietarinen & Sandu, 1999; Sandu & Pietarinen, 2003). (b) Hodges' uniformity problem arises from violations of game-theoretic consistency in the propositional IF fragment (Sandu & Pietarinen, 2003). (c) A four-place connective ('transjunction') of propositional logic of imperfect information gives rise to a functionally complete set of connectives for all partial functions together with the usual Boolean ones. In addition, compositional semantics has been defined to a propositional IF fragment (Sandu & Pietarinen, 2001). (d) Epistemic (multi-agent, first-order) language of informational independence captures the phenomenon of intentional identity, dispensing with outright pragmatic concerns (Pietarinen, 2001a). The implications for knowledge in multi-agent systems are obvious (Pietarinen, 2002c). (e) GTS may be defined for both monadic and polyadic generalised quantifiers, and for many other cross-categorial linguistic items (such as negative polarity items, adverbs of quantification, the morphemes *even* and *not even*, and eventualities), with consequences for linguistic theorising. Furthermore, generalised quantifiers and eventualities are affected by the phenomenon of informational independence (Pietarinen, 2001b).

Semantic games and logic Once IF formulas are associated with extensive games of imperfect information, the question arises of the kinds of logics giving rise to semantic games that satisfy the consistency, non-repetition and von Neumann–Morgenstern conditions, but which correspond to no known IF formula. For instance, the sheer existence of imperfect information may be conditionalised by a single action, with implications for uniformity and players' knowledge of information sets. There is much more to the phenomenon of informational independence than is currently believed (Pietarinen, 2001d).

Implications for game theory are imminent. How should the assumption of observed options — point (3) above — be understood? In game theory, uniformity means that the number of available actions has to coincide for histories in an information set. Luce & Raiffa (1957, p. 43) express this as follows:

> Each of the moves must have exactly the same number of alternatives. For if one move has r alternatives and another s, where r ≠ s, then the player would need only count the number of alternatives he actually has in order to eliminate the possibility of being at one move or at another.

Even though they later go on to speak of identification, we are not told what kind of presuppositions it bears, or what the identification criteria is.

How, then, can we distinguish between different situations just by counting the number of immediately available actions? It seems that the identities of these actions have to be taken into account as well. In general, is the inspection of alternatives something that can invariably be accomplished? Precisely what are the identity criteria for the alternatives within an information set?

While these are important questions, their status has to be weighed against other presuppositions of semantic games. These need to be examined in another study. Among such presuppositions is that the domain of the game is not a completed totality, but a figment of reality that the players examine and become only gradually aware of.

Moreover, information sets are more dynamic objects than they are usually assumed to be. There just does not seem to be any compelling reason to assimilate a player's decision nodes, that is nodes where he or she has to act, into those where his or her information sets are drawn (Pietarinen, 2003b). One general implication for game theory is the possibility of representing the phenomenon of a player being absentminded about another player's moves, and not only about his or her own moves (Chapter 14).

Further developments In order to make imperfect information more viable, game theory has sought new solution concepts to complement the traditional winning and losing positions. One refined concept is sequential equilibrium (Kreps & Wilson, 1982), according to which players need to form expectations concerning the behaviour and beliefs of other players. Since not all previous moves are known in imperfect information games, players cannot be certain about their opponent's intentions and plans, yet the strategies need to be defined on all decision points, even on out-of-equilibrium ones. Attempts could then be made to capture such uncertain expectations by applying sequential equilibrium.

It is of some interest that formulas with informational independence may give rise to extensive games that are not structurally consistent, which means that there exists a belief system of a game that does not incorporate the fact that there are strategy profiles according to which some information sets are reached with positive probability. In general, these further solution concepts may then be studied in relation to the notion of truth in logic.

Yet another appealing topic is the proximity of evolutionary game theory to logic (Chapter 11). While most research so far has concentrated on cooperative evolutionary games, also a paradigm of non-cooperative games exists, even within the framework of extensive games. It is worthwhile investigating how concepts such as evolutionary stable strategies relate to logic. The foundational value of such an enterprise is in the concept of evolutionary language games, which aim to establish how humans actually acquired their language.

Looking ahead, how far can the idea of controlled information flow be pushed in logic? If we take games themselves as objects of study, could we entertain the notion of independent formulas (or sets of formulas) as well, with concatenated games preserving imperfect information? What if quantifier-free matrices are informationally independent, too?[23] There is nothing sacred about independent atomic formulas as such, and given extensive games it makes perfect sense to have single instantiations independent of some previous choices (such as $P_2 a_1 \ldots (a_i / x_1)$), even if logical constants elsewhere were linearly ordered.

As soon as informational independence exists in logic, why not go all the way? Semantic flows in IF logic are confined to the relaxation of the dependence of existential quantifiers on universal ones. In its most general form, however, independence means that the formulas show all kinds of dependence and non-dependence relations between quantified variables and connectives (and even further, between non-logical constants and subformulas). To attain the most general form of independence, let me recapitulate the suggestion from Chapter 5. Formulas themselves may be represented by directed graphs. The relation between two nodes in such graphs would then mean that the information concerning the value of the variable that is instantiated to the variable of a starting node of the relation is transmitted to the ending node of that relation. The graph closed under equivalence relation represents all of the variables and connectives depending on themselves and on all of the others, and the disjoint graphs representing no variable and no connective depending on anything else, not even on itself, in which case the game even dispenses with singleton information sets.

Notes

1 It has sometimes been claimed that choice functions are somehow 'deictic', viz. prone to pragmatic considerations (Kratzer, 1998). This is not a meaningful claim in GTS, although strategies may of course receive deictic information as input. Deictic information pertains to the theory of the strategic meaning of utterances.

2 It is unfortunate, though, that van Heijenoort's compilation was singularly effective in broadcasting the biased view on the history of modern logic that sidestepped the algebraic tradition, especially Peirce.

3 See Gentzen (1969), cf. Jervell (1985) for a study of what he calls Gentzen games.

4 See e.g. Hamblin (1971); Lorenzen & Lorenz (1978); Lorenz (2001); Mann (1988); Rahman & Rückert (2001) and the extensive literature referenced therein.

5 Allow me to suggest a few other pet names: ∀dam / ∃ve, ∀gape / ∃ros, ∀antony / Cl∃opatra, Mori∀rty / Sh∃rlock, ∀tum / Kh∃pri, M∀sculine / Feminin∃, M∀n / Brut∃, D∀phnis / Chlo∃, ∀gent / ∃nvironment, Edv∀rd / H∃nry, ∀ctual / Pot∃ntial, ∀ll / On∃, ∀chilles / Tortois∃, ∀nalytic / Synth∃tic, He∀ven / H∃ll, ∀ctive / Passiv∃, H∀te / Lov∃, M∀terial / Id∃al, Conc∀ve / Conv∃x, ∀ction / R∃action, P∀in / Pl∃asure, N∀dir / Z∃nith, Terresti∀l / C∃lestial, Sp∀ce / ∃nergy, Inform∀tion / ∃ntropy, ∀scyltos / ∃ncolpius, Thermoplasma-∀cidophilum / Bd∃llavibrio.

6 But see Stalnaker (1999), who suggests that, even from a strategic viewpoint, the distinction between these two forms is somewhat immaterial.

7 We assume here, for simplicity, that the names of the individuals are the same as the individuals of the domain.

8 See Pietarinen (2001a, 2002b, 2003d) for IF modal (epistemic) logic, and Sandu & Pietarinen (2001, 2003) for IF propositional logic.

9 Indeed, according to Wittgenstein, this kind of clause is not a *definition* of negation, because it *constitutes* negation (Chapter 8).

10 Janssen (2002) discusses the game-theoretic interpretation of IF first-order logic, and proposes that there is a difference between the concepts of informational independence and imperfect information, thus putting forward a semantics based on subgames.

11 This property was used in Hodges (1997a) without noting its redundancy.

12 Morgenstern, Oskar, 1935. Volkommene voraussicht und wirtschaftliches gleichgewicht, *Zeitschrift für Nationalökonomie* 6, 337–357. Translated as 'Perfect foresight and economic equilibrium' in Schotter (1976, 169–183).

13 It is worth comparing these clauses to 'no-counterexample' interpretations proposed by Georg Kreisel in the early 1950s.

14 Contrary to what was suggested in van Benthem (2001), coalition games typically assume some coordination and hence do not provide feasible models for understanding informationally independent logics with the kind of imperfect recall in which each member acts individually. Accordingly, they have not usually been considered in relation to imperfect recall in game-theoretic literature.

15 Note the introduction of the third player. This new player should not be confused with the player playing the role of F, often also called Nature.

16 Little hinges on the Bayesian paradigm in the present context, however, since the prior probabilities will be very simple. Moreover, in the case non-partitional information sets (such as in propositional IF logic, see Pietarinen 2005f), Bayesian reasoning will break down.

17 This may happen in many-sorted logics. Take $\forall X(\exists y/X)\, RXx$, for instance, in which V, in choosing an individual for y is not informed about the set chosen for the second-order variable X.

18 A quantifier or an epistemic operator being independent of negation was suggested in Hintikka & Sandu (1989). The meaning that was assigned to such expressions was that the order of the quantifier or the modal operator and the negation is reversed. However, troubles begin as soon as there is more than one negation of which something is independent, or if the negation does not immediately precede the slashed expression and there are some intermediate negations or other constituents on which the slashed expression is dependent. In contrast, we want a general notion of negation independence.

19 Subgames are defined, standardly, as subtrees of the given finite extensive-form tree.

20 See e.g. Bacharach (1985); Geanakoplos (1989); Morris (1994).

21 It would also be possible to study just one direction of previous definitions:

(i) if not $(\mathfrak{A}, g) \models^{+} \varphi$ then $(\mathfrak{A}, g) \models^{+} \neg_{w}\varphi$
(ii) if not $(\mathfrak{A}, g) \models^{-} \varphi$ then $(\mathfrak{A}, g) \models^{-} \neg_{w}\varphi$.

This kind of unidirectional non-truth-functional definition of classical negation have not have been studied in the context of partial and IF logics before.

22 I argue in Pietarinen (2002d) that the kind of non-coherence that may result from non-strict winning strategies that transmit potential contradictions (nonzero-sum payoffs) to the level of complex formulas, may be eliminated by evoking 'negotiation games' of alternating offers. These games, metaphorically speaking, aim at bridging corrupt links between language and reality.

23 E.g., amounting to formulas such as $Qx_1 \ldots Qx_n \begin{pmatrix} P_1 x_1 \ldots x_k \\ P_2 x_{k+1} \ldots x_l \end{pmatrix} P_3 x_{l+1 \ldots n}$ $(k < l < n,\ k, l, n \geq 1)$.

Chapter 8

LOGIC, LANGUAGE GAMES AND LUDICS

THE GAME OF "showing or telling what one sees" is one of the most fundamental language games, which means that what we in ordinary life call using language mostly presupposes this game.
— Ludwig Wittgenstein (1934)

TARSKI'S SEMANTICS represents the most unimaginative expression of Western rationalism.
— Jean-Yves Girard (2001)

1. Introduction

After Peirce, the idea of interactive and dynamic component in the heart of logic raised its head only sporadically and by no means systematically during the 20th century. One of the early proponents was Ludwig Wittgenstein. His views on language games and logic have overlapping significations, and his themes around them are not unlike some of the recently emerged theories of logic and the semantics of computation.

My aim in this chapter is to show how Wittgenstein's language games are put into a wider service by virtue of elements they share with some contemporary opinions concerning logic and the semantics of computation. I will give two examples: manifestations of language games and their possible variations in logical studies, and their role in some of the recent developments in computer science. It turns out that the current paradigm of computation that Girard (2001) termed Ludics bears an eye-catching resemblance to members of the grand family of language games.

Moreover, the kinds of interrelations that are emerging could be scrutinised from the viewpoint of logic that virtually necessitates game-theoretic conceptualisations. These conceptualisations are related to semantic issues to do with language, and demonstrate that the meaning of utterances may, in many situations, be understood as Wittgenstein's language games of 'showing or telling what one sees'. This provides unmistakable motivation for the use of games in relation to logic and formal semantics that some commentators have called for.

The conclusion about Wittgenstein is that the notions of saying and showing converge in his late philosophy.

2. Wittgenstein, language games and logic

The first pages of Wittgenstein's Philosophical Investigations introduce the idea of a language game in order to show that the words of a text or a complete primitive language derive their meaning from the role they have in certain non-linguistic activities that he decided to call games. For Wittgenstein, the first purpose of games was not something that could be found *in* logic or language. He considered them conceptually prior to such symbolic codes, activities from which logic and language derived their meaning. His Nachlass, which long remained unpublished, has recently been published and has turned out to be a very valuable and instructive source on the controversial role of the language game in his philosophy. In what follows, it is one of my major sources.

Wittgenstein (2000–, 149: 18)[1] writes: "For what we call the meaning of the word lies in the game we play with it". In a similar vein (*ibid.*, 148: 36v)[2] he states: "In which case do we say that a sentence has point? That comes to asking in which case do we call something a language game. I can only answer. Look at the family of language games that will show you whatever can be shown about the matter". Indeed, we put conceptual priorities in the wrong order by calling the emerging paradigm one of 'games in logic', because this would largely ignore these foundational aspects. Hodges (1997b) calls the paradigm just that. It would be more fitting to coin catchphrases such as 'logic from games', or 'logic captivated by games'. Later in this chapter I will address some of the misunderstandings that have revolved around the paradigm of 'logical games', which, in this particular case, is the same as the paradigm of 'games in logic'. This story continues in the next chapter.

Although not by way of universal agreement, Wittgenstein's language games have been linked with logical games, in so far as language games server semantic purposes (Hintikka & Hintikka, 1986; Hintikka, 1996b). Such a link, if plausible, would provide some philosophical basis for correlating games, language and logic. For, in that case, one could argue that a relationship exists between Wittgenstein's concept of meaning as a game of language on the one hand, and our understanding of these games as semantic activities linking expressions of language with the world on the other. The ensuing relationship could then be further looked into by finding a suitable application of the mathematical theory of games.

Apart from such suggestions, however, language games have received little acknowledgement from logicians. This oversight is unfortunate, because new opportunities are in sight.

Of late, perhaps the most influential field of study engaged in advancing language games has been the theory of speech acts, a major topic in pragmatics.

Yet, there are reasons to believe that the most interesting forms of these games have little to do with speech acts or correlated interpersonal communicative acts and modes of language use. The most interesting types of games appear to be intra-linguistic, and work in the way they do because language has to function in a certain way, and has to acquire its meaning from several different but interrelated processes. This is connected with aspects of the evolution of semantics and the emergence of meaning in language, a topic that I discuss further in Chapter 11.

In contrast to what the theory of semantic games appears to suggest (see the previous chapter), moves in language games as Wittgenstein came to conceive them were complete sentences, not primitives of language or constituents of logic. This is prima facie evidence for language games as more closely related to speech-act theories than to logical semantics. Wittgenstein seems to have overvalued the role of the game component in this respect, however. If we think of moves as collections of possible individual choices, and plays as sequences of single actions, then the theory of semantic games after all provides a rather fine-grained elucidation of what language games are, namely by switching the perspective from compositions of expressions to sets of possible values for the constituents of full sentences. The way in which the assignment of these values is performed is by starting from the context and proceeding to simpler expressions, and in this manner it is indeed the whole sentence (or even a segment of discourse) that is the target of any single play of the game.

In terms of moves, Wittgenstein could be seen to refer to plays of the game; the processes that constitute parts of the meaning of complete expressions of thought, that terminate, if the game is finite, at the point in which outcomes of actions become observable. Such observational outcomes can only be assigned to such total plays, because they can be identified and associated with particular games. A single move cannot be so identified: "We don't deny that he can make [the] move but we say that the move alone or together with all the sensations, feelings etc. he might have while he is making it does not tell us to what game the move belongs" (Wittgenstein, 2000–, 148: 44v). Of course, subsequent to this, the payoff values can be assigned to total strategies of the game composed of a multiplicity of a system of plays.[3]

Those opposing the attribution of real or formally definable games in understanding Wittgenstein's language games, and the ensuing attribution of them in understanding logical games, have sometimes argued that no one ever wins or loses such games (Hodges, 1997b), and so they should not form the locus in which to attribute the semantic notions of truth and falsity. This view is doubly mistaken. First of all, truth and falsity are not equated with winning and losing in semantic games. Furthermore, Wittgenstein's games form a family, elements of which may be compared by describing them as variations of another, and emphasising their differences and analogies (Wittgenstein, 1978, p. 61). Because of this multiplicity, there is no a priori reason to assume that he would have excluded any kind of game from the scope of the overall notion (possibly with the exception of cases in which the game is unfair, such as when the player who starts it can always force a win and knows how to do this, cf. Wittgenstein 1978, p. 99). More importantly, in many places Wittgenstein openly asserts that some games, at least, are for winning and losing, while others are merely pastimes (Wittgenstein, 1953, p. 66). Even the concept of a winning strategy did not remain foreign to him: "Let us suppose, however, that the game is such that whoever begins can always win by a particular simple trick. But this has not been realised; — so it is a game" (Wittgenstein, 1978, p. 100). Yet even more pointedly: "What constitutes winning and losing in a game (or success in patience)? It isn't of course, just the winning position. A special rule is needed to lay down who is the winner. . . . I would almost like to say: It is true in the game there isn't any 'true' and 'false' but then in arithmetic there isn't any 'winning' and 'losing"' (Wittgenstein, 1978, p. 293).

What all this points to is that games procreate a complicated network of processes that sometimes share mutual characteristics and sometimes do not. By moving from one game to another one can hide some of its features, but also introduce new ones. This is what happens in language, where syntactically equivalent expressions may be used to mean different things in new, unprecedented ways. What is often responsible for such changes in meaning is the fact that the domains of discourse are dynamic and constantly in flux, epitomising one of the most difficult problems for formal theories of linguistic semantics.

If semantic games are instances of language games in Wittgenstein's sense, then the question of a playable semantic game may not even arise in the same sense as it does in other, mathematically defined games for logical problems. Wittgenstein recognised the importance of drawing a fundamental distinction between what is truly a game and what is a non-game. He characterised this distinction as one between 'fair' games with true strategic content and 'unfair' games with only winning strategies that are common knowledge. This by no means implies the semantic plurality that allows us to characterise similar logical structures according to differently defined games. As far as the notion of playability is concerned, language games do not have the kind of rules that precede them, but they nonetheless define a coherent notion of a playable game. All we need to do is to recognise that something is a game, and after that we should try to work out its rules and learn how to play it.

Wittgenstein's most salient reason behind de-emphasising bringing language games closer to logic and formal linguistics is that he never ventured to characterise them. Consequently, he did not provide a solid platform on which to build theories of language games, then to be studied with formal precision. Wittgenstein (1978, p. 75) explains his pessimism about any unitary characterisation: "There is probably no single characteristic which is common to all the things we call games. But it cannot be said either that 'game' just has several independent meanings (rather like the word 'bank'). What we call 'games' are procedures interrelated in various ways with many different transitions between one and another". The key is the contextual shift: "Can't the old game loose [lose] it's [its] point when the circumstances change, so that the expression looses [loses] to have a meaning although of course I can still pronounce it" (Wittgenstein, 2000–, 149: 86).[4] The analogy with games is that one should "remember that a position or a move in a game gets its sense from the game" (Wittgenstein, 2000–, 151: 21).[5]

As a response to the complaints occasionally voiced against using language games for logical purposes it may be noted that Wittgenstein was likely to have no deep interest in any formal theorising about them in the first place, for much the same reason that he would not have had any profound sympathy for any systematic theorising on philosophy in general.

Surprisingly, though, Nachlass has revealed that games as a theoretical discipline was not entirely unbeknownst to him. For example, he remarked, "The theory of the game is *not* arbitrary although a game itself is" (Wittgenstein, 2000–, 161: 15r).[6] These remarks were written after interest in the newly-discovered discipline of the theory of games was on the increase, especially in Vienna in the mid-1930s. Following this passage, Wittgenstein went on to consider the question of whether the theory of a game could be considered pure mathematics and physics. Nevertheless, he was quick to dismiss this possibility by observing that there were some major problems in explaining the notion of decisions assumed in the theory of games.

Apart from these fairly isolated remarks concerning decisions and mathematical proofs, Wittgenstein did not show any particular interest in advancing game-theoretic concepts much further. Nothing indicates that he was aware of the works of von Neumann, Borel, Kalmár, Steinhaus or König, the fathers of modern game theory, who were making seminal contributions already in the 1920s. This is not the end of the story, however. Wittgenstein seldom gives away names. But an isolated reference to economic theorising occurs in Nachlass. This is vital. The reference is analogous to the question of what an appropriate description of a narrowly restricted field of application could be, given a system of communication consisting only of words and commands, a narrowly restricted field in describing what a language could be:

> Augustine describes, we might say, a system of communication; not everything, however, that we call language is this system. (And this one must say in so many cases when the question arises: "is this an appropriate description or not?". The answer is, "Yes, it is appropriate; but only for this narrowly restricted field, not everything that you professed to describe by it." Think of the theories of economists.)

This paragraph, which Wittgenstein modified and of which he omitted the material in parentheses from the published version of Investigations, is immediately followed by:

> It is as though someone explained: "Playing a game consists in moving things about on a surface according to certain rules ...", and we answered him: You seem to be thinking of games on a board; but these aren't all the games there are. You can put your description right by confining it explicitly to those games. (Wittgenstein, 2000–, 226: 2). [7]

The impact of game theory on economics was not well acknowledged at the time the original German version was completed in 1937, and was only gradually emerging. This happened, to a degree, through Oskar Morgenstern's efforts on emphasising this aspects of the Theory of Games and Economic Behavior after its publication. I take Wittgenstein's remark to be an indication of his awareness of the possibility of applying games to economics.

Even the very developers of this approach were sceptical about such applicability. John von Neumann, in a letter to Abraham Flexner (25 May 1934), confessed: "I have the impression that [economics] is not yet ripe... not yet fully enough understood... to be reduced to a small number of fundamental postulates — like geometry or physics" (quoted in Leonard 1995, p. 730).

Therefore, it is likely that the reference to economics was omitted from the published version of Part I of Investigations, completed in 1945, because of the improved state of affairs.

A brief historical interlude. John von Neumann met Morgenstern in 1938 in Princeton. Until the autumn of that year, von Neumann had visited Europe annually since his departure for Princeton in 1931. Morgenstern was Privatdozent in Economics at the University of Vienna from 1928 until the Anschluss of Austria in 1938, when he also moved permanently to Princeton. He met Edgeworth, the one-time inspirer of Peirce in England in 1925, and was a partaker in Moritz Schlick's seminar (which later morphed into the Vienna Circle) and in Karl Menger's Mathematical Colloquium in Vienna in 1928–1936.[8] von Neumann's 1932 presentation at Princeton about economic growth and Brouwer's fixpoints was reproduced in Menger's colloquium in 1936, and his 'expanded economics' model was published in 1937 in German in the proceedings. It then appeared in 1945 (the discovery was preceded by Abraham Wald's existence and uniqueness proof of equilibrium),[9] but it was not until the collaboration began with Morgenstern that his angle on the cogent application of the theory of games to economics was acknowledged. From then on it rapidly gained credence.

I record these events because their order fits in with the picture in which Wittgenstein was aware of what was going on on this front, primarily due to his attachment to the scientific life of Vienna in the 1930s, even though he had left for Cambridge in 1929. This is plausible not only because he is known to have travelled to Vienna several times during the 1930s, for family reasons and otherwise. These events also suggest that he noticed the developments in this particular area some time later, too, when both Vienna meetings had already ended. This is in accordance with the transition that took place in his philosophy in the mid-1930s, alienating him from the lines of thinking to which the former members of the Vienna Circle were accustomed. Once von Neumann, and Morgenstern's collaboration was in full swing, the prospects for economic applications of the theory of games appeared in a much more positive light. The Theory of Games was sent to press in 1943, and it was soon after its publication that Wittgenstein made his final amendments to Investigations, dropping the reference to economics as a "narrowly restricted field" with regard to the question of whether some particular description, such as the theory of games, would be adequate for it. He made very few changes to the material after 1948. Furthermore, most of the cuttings in Zettel are from the immediate post-Theory-of-Games era of 1945–1948. They are largely cohesive with the material in Part II of Investigations, and these clippings indeed contain the most extensive range of references to games and their roles in Wittgenstein's philosophy.

A helpful addition to the who-knew-what-when guesswork is from Karl Menger. He received a fair dosage of Peirce from Paul Weiss, who was editing the Collected Papers in Harvard around the time Menger visited the mathematics department between September 1930 and February 1931. In the 1920's, the philosophical community in Vienna was under the common impression that pragmatism came from William James. On the other hand, Harvard philosophers, who were largely inspired by Peirce's recently discovered treasures, were not well versed in the philosophical atmosphere in Vienna. Nevertheless, Menger seems to have smuggled some of the knowledge back to Europe and to members of the Circle and especially to his Colloquium, after which the groups took a decisive pragmatic and econometric turn. Findings such as Wald's existence and uniqueness for Walras' equations were well-absorbed by Gödel (Menger, 1994, p. 213), and reinvented and extended to cover multi-sectorial balanced growth in von Neumann's contribution to the Colloquium.[10] True, they still reflected Hilbert's axiomatic programme, but at the same time moved away from functional explanations towards causal would-be explanations of economic phenomena. Perhaps somewhat unexpectedly, then, there was even an element of admission to metaphysics in the discussion.[11]

The systematic upshot of these details is that, despite considerable progress in mathematical economics in the 1930s, Wittgenstein thought that theories of games after all had some fundamental impediments in terms of their foundational value, because they were only able to describe various things like actions and decisions, and not really show what these activities truly consisted of. He is in good company here, and by no means ought to be downplayed by game theorists: one of the most deep-seated problems in game theory concerns the prescription of actions that maximise expected utilities.

For Wittgenstein, language games were in the main rule-governed activities. Because of the priority of constitutive rules rather than strategies, one may ask whether game theory can at all contribute to the study of such activities, given the fact that it is presumably concerned only with games that can actually be played. To qualify as playable, a game needs to have sufficiently distinctive and formal characteristics. Wittgenstein's concern was to answer the question: "What now is the relation between a name and the object named, say, the house and its name?" A suggestion he gave was that "we might say that it consists in ... the whole of the usage of the name in the language-game" (Wittgenstein, 2000–, 310: 147).[12] In a similar vein: "'What is the relation between names and the named?' Well, what *is* it? Look at our language game, or at some other language game; that's where you'll see what this relation consists in. A relation may, among various things, consist in the fact that hearing the name calls up an image of the thing in our minds" (Wittgenstein, 2000–, 226: 25). Here, he must have felt game theory itself to be largely irrelevant to the study of such inner meanings, and this happened after he had became aware of its potential and his own reservations from the presentations by economists and mathematicians in and about the Vienna Colloquium.

3. Wittgenstein and Peirce

The aforequoted plus numerous other remarks Wittgenstein made in the Nachlass and elsewhere bear an interesting and, as far as I know, hitherto unexplored relation to Peirce's concept of the interpretant as a sign of another sign. The affinity between the two thinkers is much more than skin deep. Language games are another way of seeing what Peirce's views were on meaning as a dialogue between different phases, between the utterer and the interpreter of an expression. According to Wittgenstein, what is alive in the sentence is what is meant by the thought that is expressed by it (Wittgenstein, 1970, 143). For Peirce, this liveliness is the dialogical relation between successions of signs and their interpretants, the semiosis, grounded in the common and mutually common understanding of the rudiments of what it is to be a sign-carrier, including rudiments of the grammar of language. In the same paragraph 143 of Zettel, Wittgenstein asks what a comparable meaning of a configuration of chess pieces on the game board would be, suggesting that it would be something to be found not only in rules, but also in the experiences that are associated with game positions, and in a common understanding between the players about the usefulness of such a game for some meaningful purpose.

Wittgenstein frequently resorted to the concept of a sign (Zeichen) in trying to clarify elements of language and their function, sometimes meaning indexical, sometimes symbolic, and sometimes iconic things. Signs are always used in the context of a language game. An indexical sign, for instance, points: "Pointing is itself only a sign, and in the language-game it may direct the application of the sentence, and so shew what is meant" (Wittgenstein, 1970, 24).

In conventional signs, on the other hand, "The impression made on us by the signs played no part; [. . .] If the sign is an order, we translate it into action by means of rules, tables. It does not get as far as an impression, like that of a picture; nor are stories written in this language. [. . .] In this case one might say: 'Only in the system has the sign any life"' (Wittgenstein, 1970, 145–146). The significance of such symbolic signs is in convention.

In order to get iconic signs, "we had to use rules, and translate a verbal sentence into a drawing in order to get an *impression* from it" (Wittgenstein, 1970, 147).

What is notable here is iconicity, which may be compared with Wittgenstein's picture theory of language. Such a comparison has, however, provoked strange and unnecessary jargon in earlier literature ranging from homeomorphism of icons to their paramorphic ('iconically metaphoric') character. More often than not, it has been claimed that Wittgenstein abandoned the picture theory. However, aspects of iconicity endured in his writings, which shows that claims to the effect that he discarded the theory are unfounded, or what is worse, that the reasons for his neglect are irrelevant to the question of what the iconicity of some signs is about. A more positive approach would be to ask new questions. What would Wittgenstein's reaction have been had he known of Peirce's diagrammatic logic of EGs, which represented predicate terms, relations and identities as graphs that put forward (incomplete but non-vague) pictures of the entities and their relations as gathered during the interpretation? Would Wittgenstein have maintained the picture theory of language as a more unified account for a longer time than he actually did?

Moreover, it is worth comparing such diagrammatic representations with the kinds of pictures of logical expressions that he envisaged at one time, namely ones that are stripped of all irrelevant details and merely show what the proposition is intended to say. This comparison resonates well with Peirce's theory of diagrammatisation, according to which a diagram is meant to put before us, not a complete thought, but an iconic account of the essential, pragmatically valued content of the action of the mind in thought.

Further probing into Wittgenstein's use of signs shows that he related them to their interpretation in a way that comes close to Peirce's semeiotic conception: "An *interpretation* is something that is given in signs" (Wittgenstein, 1970, 229). An even more striking resemblance to Peirce's theory of signs and his concept of semiosis is found in Wittgenstein (1970, 231): "By 'intention' I mean here what uses a sign in a thought. The intention seems to interpret, to give the final interpretation; which is not a further sign or picture, but something else—the thing that cannot be further interpreted. But what we have reached is a psychological, not a logical terminus". The last sentence gives him away, however. Wittgenstein was not keen to interpret logic in as wide and as liberal a sense as Peirce did. For Peirce, the whole concept of semiosis is a thoroughly logical process, while Wittgenstein was quite sensitive to accommodating its psychological overtones. This is something that shows up on other levels of

Wittgenstein's philosophy, too: according to his earlier thinking, assertions, as distinguished from propositions, are "merely psychological" (Wittgenstein, 2000–, 201a1: A4),[13] and epistemology is "the philosophy of psychology" (Wittgenstein, 2000–, 201a1: B21). There are numerous examples of his psychological inclinations towards the parts of philosophy that, according to Peirce, would have been strictly logical.

Despite stark differences in their attitudes towards the role of psychology in philosophy and logic, there is abundant evidence both of the overall congeniality between Wittgenstein and Peirce's philosophy, and for their overall disparity. In the case of the former, it is worth recalling here Peirce's pragmatic maxim: before applying scientific method to any particular claim to test the truth of it, we must already have conceived of the meaningfulness of the statements or propositions that assert that claim. The way to assert the meaningfulness is to observe the outcomes of the applications of the concepts in question. The paradigmatic version of the pragmatic maxim first appeared in the January issue of Popular Science Monthly, 1878:

> The rule for attaining the third grade of clearness of apprehension is as follows: Consider what effects, that might conceivably have practical bearings, we conceive the object of our conception to have. Then, our conception of these effects is the whole of our conception of the object. (5.402, 1902)

This was paraphrased in 5.439 [1905] via subjunctives:

> The entire intellectual purport of any symbol consists in the total of all general modes of rational conduct which, conditionally upon all the possible different circumstances and desires, would ensue upon the acceptance of the symbol.

There is yet a third formulation in 5.412 [1905]:

> If one can define accurately all the conceivable experimental phenomena which the affirmation of denial of a concept could imply, one will have a complete definition of the concept, and there is *absolutely nothing more* in it.

There are more versions, but I will stop here. This is a profound maxim: even the concept of rationality ought to be tested by it, thus circumventing the sceptical arguments that attempt to show that it is, in truth, a utilitarian maxim falling short of avoiding the pitfalls of the opportunistic ethics of decisions.

Similar ideas are to be found in well-known phrases such as Hilbert's "By their fruits ye shall know them". In proper context he said: "The final test of every new mathematical theory is its success in answering pre-existent questions that the theory was not designed to answer. By their fruits ye shall know them — that applies also to theories".[14] Peirce already expressed this idea, aside from the form of the pragmatic maxim in 1907,[15] within the economy of research. In particular, he considered this in relation to the notion of the "breadth" of the hypothesis, a property that should be taken into account in choosing between rival explanations (EP 2:110; cf. Chapter 6).

In the following passage Wittgenstein apparently sympathises with what could be seen as an implication of the pragmatic maxim, namely that the value of signs is in the observable consequences of their interpretation. Here he considers the possibility that the meaning of a sign or linguistic expression is found in the translation of it into another system of signs:

> "Every sign is capable of interpretation; but the *meaning* mustn't be capable of interpretation. It is the last interpretation." Now I assume that you take the meaning to be a process accompanying the saying, and that it is translatable into, and so far equivalent to, a further sign. You have therefore further to tell me what you take to be the distinguishing mark between *a sign* and *the meaning*. (Wittgenstein, 2000–, 309: 55).[16]

Like Peirce, Wittgenstein maintained that translation provides some 'semantics' of the sign. This, however, is only "so far equivalent to" the translated sign of another system, as further examination that takes in pragmatic, or if you will strategic, considerations, will show in the process of assessing the extent of the similarities between the two signs. Approximate similarities are shown by evaluating the observable outcomes of the sign when it is put to the test, or execution, by the environment (the context, user, system, interpreter).[17]

However, Wittgenstein did not endorse that meaning should be equated with a process. Nevertheless, as Peirce tried to make clear in his sign theory, intrinsically tangled up with the pragmatic maxim are many varieties of the meaning of signs, some of them associated with outcomes of certain processes and some of them independent of such processes. In Peirce's terminology, these varieties are found in the elements within the total spectrum of the sign's interpretants.

The importance of game theory across interdisciplinary fields of inquiry lies in the productivity and changeability of games. The domain of natural-language expressions varies from one sentence or one segment of discourse to another, and so new language games will be constructed that mirror this change. Especially within the framework of extensive games in the sense of the theory of games, there is a virtually unlimited number of meanings that may be assigned to an expression or utterance of a language game. This generative nature of language games is cogently emphasised by Wittgenstein: "But how many kinds of sentences are there? Is it assertions, questions and commands? — There are *innumerable* kinds: innumerable kinds of applications of all that we call 'signs', 'words', 'sentences'. And this variety is nothing that is fixed, given once and for all, but new types of language, new language games — as we may say — come into being and others become obsolete and are forgotten" (Wittgenstein, 2000–, 226: 15). One is reminded here of Peirce's last-ditch desire to transform the sheet (or the universe of truth) on which assertions are scribed into one on which questions, commands, absurdities and other non-declaratives may be scribed as well, and of course in as analytic a manner as the former.

4. Language games in computation

One recent approach in computation, and an integral part of contemporary computational logic, is that of Ludics (Girard, 2001), a deconstruction of game semantics for computation related to the geometry of interaction (Abramsky

& Jagadeesan 1994, dating back to Blass 1972, cf. Blass 1992, 1994, 1997). Given the verbal milieu of this chapter, no comprehensive description of what Ludics is supposed to be is needed.

I will merely give a short explanation. Together with game semantics, both theories aim at interactive and dynamic models of logic. Although targeted at the semantics of computation, they model interaction in a proof-theoretic setting, typically in sequent calculus. The central idea in Ludics is that the objects of derivation in sequent calculus derivation are, unlike ordinary Gentzen-type sequents, the locations of subformulas, not the formulas themselves. Consequently, the object of the proof will also change to what Girard calls 'designs'. Designs are labels of sequential derivations. In locating subformulas, only their relative locations matter, given by an address of the position of the subformulas referred to in the design. This realises a step towards the goal of viewing Ludics as a general logic of space and time. Any interaction gives rise to the set of disputes (plays, i.e., sequences of choices or possible interactions in the game-theoretic sense), which are used in trying to describe what designs are, and how to get the notion of the playing agents from it. These designs then give rise to behaviours, to be roughly equated with what is more commonly viewed as the totality of what there is in an extensive game, with sequences of plays (histories, disputes) giving the interaction and the actions of single players, but without the notion of payoff for all terminal histories.

The ensuing system is proof-theoretic, but unlike the Gentzen-type natural deduction, it makes essential use only of the skeleton of any natural-deduction derivation. The skeleton is a tree structure of the formula under proof, comprising addresses or pointers pointing to the location of the components in the formula. Thus, Ludics assembles formulas into trees by way of storing addresses of their relative locations. This is like extensive game trees in semantic games, which also has their way of denoting the locations of components by histories of the tree.

The idea in Ludics is to present the meaning of a formula (or the meaning of the proof of the formula) by its interaction against observers. Because one operates within a space of proofs, the observers are taken to be those proofs of formulas that appear elsewhere in the system. Proofs, and hence meanings, are created solely by interaction with other proofs.

This is what the essence of operational semantics is in general terms. The meaning of a program, as indeed that of a formula, is given by observations about the results of the evaluation of it as it is being executed in all contexts or situations that are expected to arise.

What is immediately evident is the commonality of this kind of operational meaning with that of Peirce's pragmatic maxim. A consequence of his maxim is that it is possible to give the meaning of a proposition in another proposition. Generalised to the realm of proofs, this means that the meaning of a proof is given by the proofs themselves, and within the realm of rules it means that the meaning of rules is in the rules themselves, and so on, by way of generalising this process into the realm of any intellectual concept whose meaning one tries to capture. Such propositions, proofs, rules and intellectual concepts provide the meaning of the given proposition, proof, rule or concept because the former are general descriptions of all observable phenomena concerning the results of some process, such as interaction or evaluation of the execution of a program in varying environments of suitable type, either predicted or else put forward by the given proposition, proof, rule or concept. The counterparts with whom these concepts interact will mention their observable properties, but only internally, within the contexts of the system. In the terminology of the theory of signs, these counterparts give rise to interpretants of the given concepts, and within these interpretants certain meaning is conveyed.

As a result, some of the latest findings in the semantics of computation have provided a vindication of the pragmatic maxim. This ought to spark considerable interest in its applicability, as I have shown how it can be applied not only within philosophy and the social sciences, but also over exact sciences such as logic and computation.

A couple of remarks are in order here. Ludics features some eccentricities that are not found in the received conception of a sequential proof system. These include the possibility of assuming a proposition without proving it. It may be used if no other rule is applicable. A conclusion may thus be established, or abducted, without presenting any justification for it. The concept of proof is thereby extended to cover paraproofs. As a matter of fact, the meaning of a proposition is thus not actually given by the interaction of a statement with other statements, but rather by an interaction of it with itself, producing an interpretant that is not, in Peirce's terms a logical one, as it is not the result of any sufficient study by the inquirers or the contexts. Paraproofs produce only intended interpretants put forward to proceed with the derivation. Again, this is an instance of the principle of the economy of research, this time of incomplexity. As noted in Chapter 3, it says that hypotheses ought to refer forward to new hypotheses (EP 2:110).

Moreover, in actual interaction, formulas are not manipulated directly, but only by their associated pointers showing their locations in the sequent derivation or in the relevant memory space.

In the spirit of game-theoretic interaction, one then needs to capture properties of the designs that are winning. Unlike the notion of payoff in the theory of games, winning properties do not generally refer to, and are not associated with, the results of an interaction, but belong to the internal structure of the interactions themselves. The internal process of computation is thus taken to be as interesting as the outcome. Nonetheless, the link here is that while the truth of a formula is defined in semantic games as the existence of a winning strategy for Myself, in Ludics the truth of a behaviour is defined as the existence of a winning design.

The overall nature and purpose of this newly invented approach to logic could also be clarified by comparing it to other computational paradigms. For instance, how are the paradigms of game semantics and Ludics located with respect to the division between the denotational and operational understandings of semantics? It soon turns out that denotational tools (e.g., algorithms) are in use in game semantics, while the dynamics of operational methods is incorporated into computational processes. The same holds for Ludics, with certain twists not to be taken up here.

This preliminary observation that game semantics and Ludics do not fall within either of the main computational paradigms is particularly apt because it reminds us of a similar situation concerning the place of semantic games in logical landscape. Semantic games are positioned between truth-conditional (Tarski) semantics and verificationist semantics (Hintikka, 1987a). They are truth-conditional in the sense of delivering truth-values of compound expressions for an interpreted language. However, verificationism comes to the fore in that these truth-values have to be attained by certain processes of verification (and dually of falsification), or at least by knowledge of what counts as the reliable verification or falsification of a sentence in the context of asserting it.

Likewise, game semantics and Ludics make best of both worlds. They derive from denotational semantics the concrete mathematical entities assigned to programs such as functions, relations and logical and arithmetical operations, but draw on computational (operational) mechanisms in computing the actual values of these operations and entities. The meaning of a program is specified by a valuation function that associates an abstract value (a number, a truth-value etc.) with each well-formed syntactic construct. In order to use such an operational mechanism, however, one must also be acquainted with the strategic meaning of expressions. For example, the values of the variable y in the additive operation $y := x + 1$ is interpreted as the results of a floating-point computation, while at the same time the denotational, abstract meaning of the operation following the given rules remains fixed (namely, the rule that states that the store 'y' holds after the operation '$x + 1$'). Operational semantics thus defines an abstract machine and a specification of how the states of the machine are changed, given a suitable set of instructions.

Various game-theoretic conceptualisations such as moves, positions and strategies are also important in Ludics. However, like the related game semantics, it contains very little of the classical theory of games. For instance, payoffs are replaced by internal properties of interaction. Games appear merely a means, or metaphor, of achieving something useful in logic, and to date no attempt has been made to clarify this picture (but see sect. 5). Yet, games for computation consist of a small subclass of all games. In game semantics, moves are typically (strictly) alternating, there are only two players, and one of them — not necessarily the opponent — has to make the first move. None of these restrictions is endorsed by game theory. Furthermore, in game semantics and in Ludics, the strategies are history-free (innocent, positional), and hence do not take the true dynamics of the derivational history of previous choices into account. This latter stricture is imposed in order to make some technical results such as full completeness easier to achieve.

Nonetheless, Ludics is meant to accomplish much more than being just a version of game semantics. It is meant to provide a general logic of space and time. In such a system, one would expect the notion of location, or game history, to assume a central place, for it is by means of such a notion of that one then captures precisely where and when formulas are used as premises in proof derivation. Sequentiality of the calculus and deterministic choices are thus maintained. However, because strategies do not realise the full derivational histories of the plays, the game-theoretic dimension tends to recede. There is thus a risk of seeing computational interaction, after all, as a form of Habermasian communication and cooperation devoid of strategic content.

Besides attesting the reliability of Peirce's pragmatic maxim, Girard's writings on Ludics carry Wittgensteinian undertones. One of the goals is to understand the meaning of logical rules. The slogan Girard has coined is: "The meaning of logical rules is to be found in *the well-hidden geometrical structure of the rules themselves*" (Girard, 1998, p. 1). Again, this may be seen as an instance of the consequence of the pragmatic maxim, but could also be paralleled with Wittgenstein's well-recorded remark, "You can't get behind rules, because there isn't any behind" (Wittgenstein, 1978, p. 244). Likewise, game-theoretic semantics for computation and Ludics take games — or behaviours (sets of designs) and b*i*haviours (sets of 'uniform' designs) as Girard calls them — to be primary elements from which rules follow.[18] Wittgenstein slightly overstates his position here: there are language games that are more fundamental than the material from which the fine-grained structure of the rules gradually evolves, a fact that Wittgenstein did recognise in the later stages of his philosophy.

Furthermore, negation means a polarity switch between the participants in interaction in both the theory of semantic games and Ludics. The true nature of negation resides in the geometrical structure of the actions it is capable of giving rise to, not in the rules that attempt to define it. This is what Wittgenstein endorsed in critisising the rule of double negation: "We would like to say: 'Negation has the property that when it is doubled it yields an affirmation'. But the rule doesn't give a further description of negation, it constitutes negation" (Wittgenstein, 1978, p. 7). The ensuing notion of negation is nonetheless not the classical contradictory one, defined by a meta-rule that asserts about the sentence to which it is prefixed that it is not true. The role-swap between Myself and Nature in semantic games, or between any two interactors (the somewhat high-flying Subject and the Object in Ludics), gives rise to the strong notion of negation about which we could ask further questions related to contradictions.

One of the aspects in the foundations of computational logic developing from these Wittgensteinian ideas is the role played by the notion of consistency. This is far more delicate (and general) phenomenon in Ludics than just a formal contradiction. Given the proliferation of current logical systems, what, in fact, is the reference point of a rule for a logic that creates no inherent contradiction — that is, for a logic that is capable of rivalling those that have earned the epithet 'classical'? According to Girard, consistency "is one of the desirable properties of a logical system, but a rather obscure one" (Girard, 2001, p. 102). Similarly, the principles of excluded middle and contradiction are by no means among the presuppositions of logic, and thus cannot be used in to demarcate between logics that are 'classical' and logics that are 'non-classical'. The non-trivial sense in which consistency is accomplished is attained in Ludics by generalising the notion of proof to the aforementioned paraproof construction, according to which conclusions are justified piecemeal in the sequential derivation, and the termination (termed the 'daemon') assumes the conclusion by simply appealing to its authorisation by the daemon, without the need for any further justification.

Yet, Wittgenstein's concept of consistency is seen in a new light as soon as the games have certain atypical attributes. One such instance is the relaxed character of competitiveness. Given a language game with at least some winning and losing conventions, there is no pre-theoretical reason to presuppose that these conventions are strictly opposed. Such non-strict games, in fact, support Wittgenstein's emphasis on the "civil" nature of strategies in games (Wittgenstein, 1953, p. 125):

> We lay down rules, a technique, for a game, and that then when we follow the rules, things do not turn out as we assumed. That we are therefore as it were entangled in our rules. . . . It throws light on our concept of *meaning* something. For in those cases things turn out otherwise than we had meant, foreseen. That is just what we say when, for example, a contradiction appears: "I didn't mean it like that." The civil status of a contradiction, or its status in civil life: there is the philosophical problem.

In view of this, there does not have to be anything inconsistent in the defining rules of the language game in order for us to end up with non-coherent situations in which both participants may claim success for their own purposes. Yet, a great deal of recent discussion on Wittgenstein's views on contradictories as a result of his way of setting up games presupposes that contradictories should somehow be the end-products of contradictory game *rules* (see e.g., Goldstein 1989). Such a presupposition is not warranted, as shown by the possibility of having semantic games with characteristics that are different from those of ordinary games, which result in inconsistencies simply by changing the strategic rules of the games from strictly to non-strictly competitive (Pietarinen, 2000). Such a move would not interfere with constitutive game rules. Moreover, a steadfast refutation of the assumption of contradictory game rules comes from

Wittgenstein himself: "Why may not the rules contradict each other? Because otherwise they wouldn't be rules" (Wittgenstein, 1978, p. 305).

The primacy of strategic thinking in language games over and above the defining rules is also strongly emphasised by Wittgenstein in the context of mathematics. In his attempt to assess the role of inconsistencies in the foundations of mathematics, there was little point in his arguing that whenever contradictories are faced, different kinds of rules ought to be set up for such unfortunate cases. If certain games lead to contradictory meanings of some mathematical statement, or of some natural-language expression, it is only these games that can reveal any inconsistencies in logical semantics.[19]

Ludics, like game semantics, is a computational theory with its emphasis on the notion of proof. Both were devised in order to further understanding of the sequential proof calculi. In semantic games, on the other hand, proofs do not play any constitutive role, while Ludics is founded upon Gentzen-type sequential calculus, and attempts an analysis of meaning based on the notion of a logical (proof-theoretic) rule.[20] Is this route to the meaning of logical constants a lost cause in logic and computation? According to Ian Hacking (1979), any Gentzen-type meaning analysis is liable to amount to 'do-it-yourself semantics', semantics that attempts to define the meaning of logical constants via sets of syntactic rules. However, unless the concept of a material truth for a language is not first grasped, such attempts are frustrated with no end in sight.

5. On "one of the most fundamental language-games"

In the paradigm of Ludics, as well as in game semantics, the purpose of the players is to achieve a proof, or an algorithm, in an interactive setting between the system and the environment. These theories have been calculated to provide, at long last, a realistic interpretation of the laws of linear logic. Linear logic resorts to nets of proof that in which assumptions are made according to the costs induced by the use of those assumptions within proof sequences.

In contrast, the purpose of players in semantic games could be reworded in a non-technical fashion: it is simply to arrive at true or false atomic formulas, or to demonstrate how to find them. In evolutionary terms, the purpose is the semeiotic process of searching for final interpretants guided by the habits of action (Chapter 11). If the language is completely interpreted, that is, if there are no partially-interpreted models and hence atomic formulas without a truth-value, players arrive at true or false atomic formulas whenever a terminal play of the game is reached.[21] The notion of the existence of a winning strategy then agrees with the notion of a true or false formula or sentence. Does this, then, also answer questions concerning what the fundamental activities that the players perform in a game are, or why we are turning on game-theoretic conceptualisations to capture these activities?

It may be difficult to find a satisfactory answer to the question of players' activities without relating them to a language game. For example, Hintikka (1973a) made an attempt to view semantic games as ones of seeking and finding. According to this proposal, quantifiers prompt an instruction to find a suitable individual from the given universe of discourse, and connectives prompt a bivalent choice marked by a subformula. This is the explication he has since supported, and there have been few alternative suggestions. Hodges (1997b) suggests a different paradigm based on a model of examination, in which semantic games are viewed as representatives of the setting in which Myself is trying to answer the questions posed by Nature. Hodges (2001b), criticising Hintikka's proposal of seeking and finding, states, "Nothing in the logical game corresponds to seeking". There is no support for this rather zany assertion. One could perhaps argue (although Hodges does not) that, in cases in which domains are ordered, or in which there is some algorithmic notion of search associated with the formulas, the concept might gain some initial logical support.

The fundamental trouble with Hodges' contention is that it never gets off the ground. The real question is about the philosophical justification of games in relation to logic. In terms of the purpose of the players, any answer is bound to refer to concepts distinct not only from game-theoretic ones, but also from what is attempted to be expressed through the symbolic conception of logic.[22]

There are further problems for Hodges. Because semantic games encompass neither strictly nor non-strictly alternating sequences of moves, and because they do not have to have an opening move by an opponent, if there is a formula that starts a session with Nature's move, we no longer understand what the examination is supposed to be. Examination is a model for computational game semantics, with polarised games in which the opponent typically moves first, only one player moves in each round, and the players exchange rounds in an alternating manner.

In fact, examinations fall within the Aristotelian model of inquiry in terms of question answering and disputation. Their medieval embodiment was known as *ars obligatoria*. In short, these are dialogisms. Semantic games are not interpersonal games, however. For the most part, they are not games that you and I play in real life, unlike communicative, conversational, dialogic and disputation games, including examination. Other kinds of game-theoretic constructs have been proposed for these activities (Chapter 10).

The explication of semantics as the seek-and-find game is thus the right one after all, for one reason because such activities are much more symmetric than any loose-fitting metaphor of catechism or inquisition.

Here, however, yet another of Wittgenstein's ideas emerges, which immediately puts more weight on language games as the philosophical basis of semantics. For over and above the idea that at least some of the games are those of verification and falsification, and that some of these are games of seeking and finding, the activities and purposes of the players may be made clarified in terms of the activities whose nature Wittgenstein was, in so many words, struggling to spell out in his philosophy. They refer to "showing or telling what one sees":

> "Surely if he knows anything he must know that he sees!" — It is true that the game of "showing [pointing, –A.-V. P.] or telling what one sees" is one of the most fundamental language games, which means that what we in ordinary life call using language [speech-custom, language-utilisation, A.- V.P.] mostly presupposes this game (Wittgenstein, 2000–, 149: 1).[23]

What this means in the context of semantic games is this. The players try to bring to the fore what they see to be the case in the context of an assertion. They have been prompted to do this by the utterance in question, and they aim at showing or saying what is the case, which, logically speaking, happens by instantiating elements of the universe of discourse as suitable values for individual, unsaturated predicate terms. The merit, or the pragmatic value, is assessed by what is understood to be present in the propositional content of those assertions.

What Wittgenstein argues for is that seizing linguistic meaning in ordinary life requires a prior grasp of its use-governed (or application-governed) machinery. Likewise, the semantics of expressions is constructed by agents' ability to apply and utilise expressions according to the habits, customs and practices prescribed by language games.

This explication is so appealing as to warrant a number of explanations and qualifications. First of all, what, if anything, does the language game of showing or telling what one sees have to do with the language-game of seeking and finding, given that the latter also draws its main motivation from some general notion of language games? There is not much difference as to whether we use one or the other of these two notions of conceptualising the practices implicit in, say, interpretation of quantificational expressions and predicate terms. Finding something comes very close to seeing that something is the case, and here we must of course recognise that seeing is by no means confined to visual perception, and also refers to all kinds of ways of coming to understand, realise, recognise, recall and so on. After all, the process of seeing has to begin with something, such as active thinking, and this is what a search or exploration tries to encompass.

In other words, as soon as we think of the process of seeking and finding as a principle of human cognition (such as uniting experiences in consciousness), then the varieties of the notions of seeking and finding seem not to be very different from the processes of seeing that something is the case.[24]

However, in order to show or to say that something is the case is to carry out something more than just the activation of the search process and the eventual finding of suitable individuals. It is something more than just the discovery or production of some such elements from the universe of discourse in question. What it also means is actively communicating those findings. What are these other activities? In some cases they may consist of the naming of objects, but that would not be the whole story. For, to name something is not yet a very complex or effectual activity. It does not, to follow Wittgenstein's remarks, constitute a genuine move in a language game:

> Within naming something we haven't yet made a move in the language game, — any more that you have made a move in chess by putting a piece on the board. We may say: by giving a thing a name *nothing* [has] yet been done. It *hasn't* a name, — except in the game. This is what Frege meant by saying that a word has meaning only in ~~its connection with~~ [the context of] a sentence. (Wittgenstein, 2000–, 226: 36).

Together with seeing that something is the case, naming may also be useful, however. It often suffices to give something a name, and to rest content with that. This nonetheless does not take us very far in the analysis of quantified statements or other logical expressions. And it is not endorsed in game theory, either, since players are typically assumed to be able to effectively observe their available actions, such actions being concerned not only with naming but also actual manipulation of objects or examination of choices between future courses of events.

What is also worth noting is Wittgenstein's reference to word's meaning "in connection with" or "in the context of" a sentence. This came later to be called the Frege Principle (Pietarinen, 2005a). As Wittgenstein notes, naming is not a move. It becomes one when it is actively communicated to other players or other phases of the mind within the context of a play of the game or, analogously, of the interpretation of a sentence. Without communication, there would be no semantic and pragmatic change and variation, even if there were some elementary system of static and immutable protolanguage of some early hominids. Indeed, Wittgenstein was an early advocate of pragmatic and semantic change, or 'diachronic pragmatics', as witnessed by his constantly evolving new language games, being "forms of life" and illustrations of what people simply do. In On Certainty (256, 1969) he remarked, "On the other hand a language game does change with time". In other words, the laws governing language change are laws governing human actions, and from this fact the evolution of pragmatic principles follow.

However, in order to see the true state of affairs in Wittgenstein's "one of the most fundamental language-games", we need to absorb the fact that language games consist of the activities of saying or telling what one sees, and of showing what one sees. Both saying and showing are seen to involve some sense of the notion of communication. Here, two rather fundamental concepts that he tried to keep strictly apart, at least in his earlier philosophy, are made different aspects of one and the same conceptual activity.

Why is it not necessary to distinguish these two notions here? Why is it that the activities of saying and showing both serve as explications of at least one part of these basic language games, the games of communicating what one sees? Let us look more closely at what quantified statements are. Their meaning is established in two stages. First, I (or You) have to find an individual from the domain of discourse, and possibly give it a name. Second, I have to instantiate the name of the individual to the bound variable in question. This latter step relates to saying and showing. Merely seeking and finding an individual does not make information public. In contrast, communicating what this individual is, even if such an act was a private one within a single mind, constitutes an act of publicising and making it accessible to other parties of the relevant language game. This accessibility is important in order for a genuine interaction to emerge. Peirce pointed out this in his algebraic logic of relatives and in his diagrammatic logic.

In any system of logic, some communication is needed to create dependencies between quantified variables and, as a result, to define mathematical relations. Furthermore, identity relations are prototypical in creating channels for continuously transmitting particles, thus asserting identities between individuals selected for the terms (Chapter 5). Similar acts of communication are essential in language to arrive at the meanings of text and to create stable relations between expressions and what they are about.

Yet, it makes no difference, especially from the point of view of the meaning of quantified statements, how this communicative activity is realised in the end. For example, as far as communicative purposes are concerned, it does not seem to matter whether I am able to show that the individuals I have found provide names for indexical expressions such as 'this' or 'that', or whether I simply utter 'this and that are the names of the individuals that I have been looking for'. The oft-noted difference between these activities, as referred to in early Wittgenstein is the difference between saying and *zeigen* ('ostension', Geach 1976). On the whole, however, this untimely contrast is no longer of substantial interest in the context of the most fundamental language games, because both activities subsume different kinds of indexical modes of communicational practices.

The notion of communication in these games may prompt someone to argue that, contrary to what Hintikka argued to be the case in the theory of semantic games, here, in fact, is a clear example of activities that have to be extra-linguistic, ones that resort to something like socially constrained contexts of language use. For, if semantic games presuppose an explicit testimonial to what one sees and shares, they no longer represent activities confined to tasks of establishing the meaning of expressions within a single person or a self.

First of all, Wittgenstein's own remarks add no plausible grist to such mill: "'Surely seeing is one thing, & showing that I see is another thing'. — This certainly is like saying 'skipping is one thing & jumping another'. But there is a supplement to this statement 'skipping is this (showing it) & jumping this (showing it)"' (Wittgenstein, 2000–, 149: 19).

Second, such an attempted argument rests on a sheer fallacy. Utterances, in the same sense as interpretations of the expressions uttered, do not call for social environments in which they may be uttered and in which they are interpreted, in order to be understood and effectively employed according to principles of the correct use of language in ordinary life simply because uttering and interpreting are general, theoretical notions. Not all utterances have an interpreter, not all utterances put forward linguistic signs, and not all utterer-interpreter links and their interactions need to be interpersonal activities bringing together different agents of a linguistic community.

There is another way of putting a related counter-argument. One could try to argue that the notions of saying and showing still differ in late Wittgenstein, because according to him, one cannot describe correct uses of a rule, while it is possible to know with certainty that one acts according to the rule. To what extent does this kind of knowledge, presumably presupposed in any correct use of language, overlap with the kind of showing Wittgenstein argued for earlier, while not overlapping with saying? In other words, does not the notion of knowledge revoke the contrast between saying and showing?

The key lies in the fact that non-verbal aspects of language games, as illustrated in such forms of knowledge as Wittgenstein recognised them, are still to be understood as forms of communication: "And the concept of knowledge is coupled with that of the language-game" (Wittgenstein, 1969, 560). Acts of communicating observations about states of affairs, while presupposing rudiments of language that are present in the common ground of the communicators, thus enabling a correct use of language in ordinary life, need not be interpersonal. The epistemic element of certainty connected with the rule-following pertains to games that do not work by way of appealing to spontaneous or habitual responses to actions. There are games that cannot be trimmed down to rules, typically symbolic instructions (such as ones that, in computational terms, are found in the idea of denotational semantics), and the following of them. The language games of showing what one sees (or perhaps what one experiences) are, as Wittgenstein emphasised, prime examples of the most important of such irreducible games.

Therefore, showing and saying do not portray any fundamental variation in Wittgenstein's later views on language games. Seeing that something is the case with respect to an assertion or and utterance is itself an element of an irreducibility claim for their public character.

It is almost as if Wittgenstein was punning his earlier self.

For these vital reasons, the correct understanding of the principles and precepts of language does not have to be societal or something that is found among the rules that are in some way socially constrained, either, because language games will continue to function without further ado irrespective of any such assumptions. Nothing significant would follow from such excess assumptions. Even if some sense of understanding was, to some extent at least, influenced by rules and principles of language use, the social context or environment provided for various expressions would not affect the most important aspect of language, the grasp and observance of the individuals of what the language speaks about. These individuals are, of course, what quantificational expressions and underlying interpreted languages aim at presenting, contexts of use or interpersonal parameters notwithstanding. What an instantiation of individuals from the logical perspective accomplishes is, after the detection and selection of suitable individuals from the domain of discourse, to make the information about these publicly available. This is not the same thing as actively communicating these individuals in a social context, nor does it entail it (while, of course, social notions of communication entail general notions of communication).

Furthermore, there is the option of not communicating what one sees by not showing it, but this happens in the context of more peculiar and more limited types of games: "'What I show *reveals* what I see'; — in what sense does it do that? The idea is that now you can so to speak look inside me. Whereas I only reveal to you what I see in a game of revealing & hiding which is altogether played with signs of one category 'direct–indirect"' (Wittgenstein, 2000–, 148: 45v).

The category of "direct-indirect" refers to cases in which one has a choice of performing either a linguistic act per se or performing it in terms of other (linguistic or non-linguistic) acts and expressions. It is tempting to speak of indirect speech-acts here, but such language-internal games are not what Wittgenstein was after. He was taking into account the relation between acts and objects ("samples"): "Surely (you wouldn't think that) telling someone what one sees is (could be) a more direct way of communicating than showing him by pointing to a sample!" (*ibid.*, 148: 44r). For instance, the expression of feeling (e.g., of what one sees to be the case) is commonly held to be an indirect way of transmitting the feeling. However, according to Wittgenstein, a direct transmission would "obviate the external medium of communication" (*ibid.*, 310: 167). In the end, Wittgenstein is suspicious whether there is any difference between what is taken to be direct versus indirect communication. He writes: "Where is our idea of direct & indirect communication taken from?" (*ibid.*, 148: 43r). Further, "The one expression is no more direct than the other. The meaning of the expression depends entirely on how we go on using it" (*ibid.*, 147: 42r). So, Wittgenstein is, in fact, suspicious of the relevance of speech acts, too. It is the use that establishes the link between expression and objects (or samples), not merely what the force of the utterance is that gets to be transmitted in a communication. He thus attacks the common idea that "indirect communication is through the medium of speech" (*ibid.*, 148: 43r).

A further consequence of the general point is that the contrast between saying and playing, or that of between saying and forms of life, that is, saying and living (Hintikka, 1996b), is perhaps not entirely satisfactory as an explication of the view at which Wittgenstein arrived in late 1930s or afterwards, the main reason being the appearance of the common denominator of both verbs in the kinds of topmost language games of Wittgenstein's that I tried to decipher and elucidate here. A language game pre-empts any need for such a dichotomy.

6. Wittgenstein and Peirce revisited

In the context of logical calculi, it is important to make the information concerning the individuals in language games freely available. This is one of the key assumptions behind first-order logic. Without it, we would not have the kind of logic pronounced as 'elementary'. This truth of logic was already cogently emphasised already by Peirce, who argued, "When the proponent or opponent has to designate an individual object as a member of the set [subject of the assertion], he is entitled to know what are the objects so far selected, so that he may shape his choice accordingly" (MS 430: 62; cf. Chapter 4). As noted in Chapters 4 and 7, the assumptions of visibility and the public role of choices for quantified expressions can be broken down into some of the more expressive systems exhibiting a restricted interchange of information between the two parties of the semantic game — an interchange that, in Peirce's system, would take place between the utterer and the interpreter.

Peirce held that the parties involved in the game-like activities of logic and language do not have to be persons, but may also be animals or even plants who "make their livings by uttering signs" (MS 318: 17). Like Wittgenstein, he thought that societal dimensions were not pertinent to language games, not even when taken to pertain to phenomena nowadays relegated to the league of cross-culturally influenced pragmatics. In this respect, Peirce's well-recorded declaration to the effect that "logic is rooted in the social principle" [25] has to be taken with a grain of salt as pertaining to the senses in which the concept of rationality and rational action in humans is logical and hence social and normative, in much the same way as the term 'uttering' has to be taken with an even larger grain of salt as a method of how anything or anyone puts forward a sign of any sort.[26]

Even though not the main purpose of this chapter, the material thus far presented supports the view that there is considerable likeness in Wittgenstein's and Peirce's views on logic and language. Many of these views were originated by Peirce, and were partly invented independently by Wittgenstein and partly communicated to him via Frank Ramsey. For instance, in 4.240 [c.1902, *The Simplest Mathematics*] Peirce remarked,

> Formal logic, however, is by no means the whole of logic, or even its principal part. It is hardly to be reckoned as a part of logic proper. Logic has to define its aim; and in doing so is even more dependent upon ethics, or the philosophy of aims, by far, than it is, in the methodeutic branch, upon mathematics.

This quotation is fundamental in pointing out the generality of the science of logic beyond the purview of its purely formal and mathematical use. It is this view that was cogently promoted by Ramsey, who echoed Peirce in calling logic a normative science.

It is not credible that Wittgenstein did not have at least a working knowledge of the essentials of Peirce's philosophy of signs, because he knew well and admired William James' production, and James of course referred to Peirce generously. We may not need Ramsey as the progenitor of some of the pragmatist leanings in Wittgenstein, at least not at this point. In fact, Wittgenstein was about to dismiss Ramsey's significance to his work after their encounter in May 1924, despite later going on record as having had "sehr genußreiche Diskussionen mit Ramsey über Logik etc." (Wittgenstein, 2000–, 105: 4). However, Ramsey soon started to read Peirce, especially on probability, and in his posthumously published On Truth (Ramsey, 1991) he undertakes a thoroughly pragmatic, common-sensical and decision-theoretical analysis of notions of truth and propositional attitudes, repudiating the then-dominating Keynesian notion of probability and replacing it with a Peircean view.

Even so, Wittgenstein's working knowledge of Peirce was probably even much broader than I have dared to suggest here. For instance, in his notes (Rhees, 2002) on conversations with Wittgenstein, Rush Rhees added a citation attributing Wittgenstein's quotation in his statement to Peirce: "To the question 'How is it that a man can observe one fact and straightway pronounce judgment concerning another different fact not involved in the first?' (C.S. Peirce), we might ask instead 'How *do* we?' Otherwise the question seems queer, like 'How can I walk?"'. (Conversations with Wittgenstein, 15 April 1943). The quotation Wittgenstein is alluding to is taken verbatim from CLL: 122, which appeared in 1923,[27] and which Ramsey started to study soon after. Whether or when Wittgenstein got hold of it, and whether this happened via Ramsey, is, as far as I have been able to find out, not known. The conversation between Wittgenstein and Rhees continues with themes familiar from CLL.

When Wittgenstein told Rhees (in 1942 or 1943), "Formalists speak of mathematics as a game" (Rhees, 2002, p. 9), he referred first and foremost to Hilbert and his adherents. Immediately after this, he said, "Frege remarked that the formalists confuse the game and the theory of the game". This must have been a good prediction — there was no such theory yet in the air by 1925, although Borel's *La théorie du jeu et les équations intégrales à noyau symétrique* had appeared in 1921, introducing both 'la théorie du jeu' and the 'méthode de jeu', the latter soon to become the concept of a (pure) strategy. There were also von Neumann's regular visits to Hilbert in 1921–1923 in Göttingen, and his residence there from autumn 1926 until 1932, when he pursued the formalist programmes of the axiomatic foundations of set theory and, later, of quantum mechanics and of economics.

The difference between Peirce and Wittgenstein is often seen in terms of a contrast between Peirce the scientific philosopher and Wittgenstein the anti-scientific philosopher. The evidence reached thus far suggests that Wittgenstein was perhaps more scientifically-minded or at least scientifically-interested than has been admitted in the literature. He appreciated and needed science and scientific discussions and results to reach his perspectives. His *ressentiment* was not towards science, even though it may have been towards mathematicians. Peirce, on the other hand, although an experimental scientist, should not be sloppily categorised as a scientific or naturalist philosopher. His system is extraordinarily resistant to both science and philosophy. It is neither. The struggles he had in finding suitable terminology, as a result of which he coined anything from cenoscopy to idioscopy and back, was no idle part of his system. It was not an attempt to create scientific philosophy or anything anti-scientific, but the establishment of an architecture designed to promote positive knowledge. When the time is ripe, the units are in place and the system is functioning, one has, to so to speak, to discharge the Ministry of Works and kick away the cranes and scaffolding of received notions of science and philosophy.

7. Logical semantics from a game-theoretic perspective

I will conclude this chapter with some remarks that put the concept of the semantic game into a philosophical and historical perspective. As observed, language games share some interesting parallels with Peirce's sign theory, or dialogical semeiotics as it may be called. For both Peirce and Wittgenstein, the concept of interaction, dialogue, or game, regardless of who or what are participating, was fundamental to the understanding logic or, for that matter, the concept of meaning in logic or the language of our natural discourse. Thus, these two philosophers offered some fundamental insights into the relation between these activities and logic, and it is these insights that are needed in order to understand different positions that may be taken up in game-theoretic investigations of the foundations of logic and language.

The idea of a logic game or a language game of Peirce–Wittgenstein origin should first of all be contrasted with an important distinction between two broad kinds of such games. Hintikka & Hintikka (1986) argue that Wittgenstein's language games fall broadly within two categories, the primary and the secondary. Primary games operate by means of spontaneous responses. They do not involve propositional, let alone epistemic attitudes, and they do not seem to have room for any traditional concept of a strategy. Secondary games bank on rationality in the sense of making use of the player's knowledge of his or her own strategies. Since secondary language games do not operate independently of identity criteria for actions, many of the epistemic concepts of our discourse derive their meaning from these games.

In view of this, it is the secondary notion of games that we might attempt to relate to the received notion of games as conceived in game theory. But does this render the theory of games non-viable in the study of logic and language, especially since, in order to make sense of the theoretical notion of a game, surely some kinds of rationality postulates ought to be presupposed?

It quickly becomes evident, however, that there is plenty of room in modern game theory for the concept of a strategy that does not presuppose rationality on the part of the players. The assumption that the strategic evolution of thought is not an exclusive province of the human brain has often proved useful, a case in point being evolutionary game theory (Maynard Smith & Price 1973; Chapter 11). This theory does not advocate winning strategies, but rather typically requires them to be stable, which means that they resist any attempt at invasion by adversary strategies. Stable strategies are associated with non-human actors such as populations, computers, systems and agents. Hence the usage of the term game is not, strictly speaking, a necessity, either. To be sure, the term does not surface in Peirce's writings on logic, although it is rife in his ample writings on recreational matters.

For Wittgenstein, the word sprang to his mind, according to the anecdote in Malcolm (1958, p. 65) — reporting what he once told Freeman Dyson — when he was passing a pitch on which a football game was in progress. My suggestion in the first section of this chapter was that Wittgenstein was aware of the 1930s Vienna taking the economic turn. He was well connected though condescended the Vienna Circle, while a lot more sympathetic with Menger's Colloquium. He took the game idea from his associations with that Viennese environment. And so his comment to Dyson was a hoodwink.

Essential in Wittgenstein is the idea of language as a rule-governed process with variable meaning relations. The language game "is an extension of primitive behaviour. (For our *language-game* is behaviour.) (Instinct)" (Wittgenstein, 1970, 545). What is essential in Peirce is the idea of thought as a dialogue between different phases of a mind, or concerning any agent, entity or role, taking place between the quasi-utterers and the quasi-interpreters of a quasi-mind.

The possibility of applying strategies to situations in which non-hyperrational players take part in the process of interpretation took root in Peirce's evolutionary philosophy of signs, habits and dialogues, and recurred in Wittgenstein's language games as primitive, instinctive behaviour. As I argued in Chapter 3, Peirce's concept of a habit was in no way restricted to rational human agents.

Apart from the differences in the concept of strategy, the division of games into two main categories is strongly reflected in assumptions concerning the structure of the games themselves. This comes to light as soon as we think of semantic games in their extensive form. Thus, primary language games could be seen as those in which the players do not identify the actions available to them across the non-terminal histories in which they move. On the other hand, secondary language games build identification of actions into the game in the sense that strategies cease to be operational if not presented with a range of options. A related distinction was proposed in Chapter 7 to reflect the different notions of information that players may have regarding past moves and also regarding the question of what legitimate future actions are, given their knowledge about them.

As far as identity criteria are concerned, in games of imperfect information, for instance, some actions have to be identified across multiple histories within an information set. It is worth observing well that games in the customary account of extensive games are, in this sense, secondary, as it is assumed that there is perfect foresight in that the set of legitimate actions is available to the players so that they are able to rationally choose their optimal actions from the given set of alternatives.

The upshot is that semantic games call for a re-examination of some of the basic assumptions in game theory. They are not secondary simpliciter. They make public some fundamental hidden assumptions concerning the received notion of a game in the theory of games. Applying the viewpoint of Wittgenstein's On Certainty, in which knowledge depends a great deal on factual information, it is not sufficient for players to simply know or even believe what the available actions are. They should also be able to demonstrate that they are in the position to know or believe them. This makes all the difference in using and applying the concepts such as knowledge, belief and expectation in the search of new solution concepts for games with complex information structures.

The distinction between primary and secondary language games reflects that of games identified through their evolutionary, dynamically stable trajectories, contra players preconceptions regarding the Nash-type rational situations that may come up in the game. The latter have been tried to be conceived through epistemic logic associated with extensive structures, while such a logic is not needed to understand the former, evolutionarily applicable games.

Elements of language and logic ultimately derive their meaning from game-like activities that are already found in Wittgenstein's language games, in geometries of computational interactions such as Ludics, and in the general theory of semantic games. A case in point is the diagrammatic structure of games, or Wittgenstein's show-or-tell game of what you see.[28] In somewhat more picturesque terms, the common element in all these is that whenever two forms — be they individual players, groups or teams in a semantic game, systems and environments in computing, utterers and interpreters of signs — meet, the befallen act gives rise to the content. Varying the ways in which different forms meet casts light on how abstract, triadic communication evolves, and how it contributes to game-theoretic studies on logic, cognition and computation.

Notes

1 *Large Notebook*, called C5, an immediate continuation of 148 (mainly notes for lectures), 1935–36, 96pp. The numbering in the citation is by item and page number. The catalogue numbers and comments are from von Wright (1982).

2 *Logic Notebook*, called C4 (mainly notes for lectures), 1934–35, 96pp.

3 Paraphrasing the so-called Frege's Context Principle a little, one would thus ask, instead of whether a word has a meaning only in the context of a sentence or a complete thought, whether a play can be identified only with a complete game.

4 *Large Notebook*, called C5.

5 *Large Notebook*, called C7 (mainly notes for lectures), 1935–36, 96pp.

6 *Pocket Notebook*, 1939–? Contains drafts for Wittgenstein's lectures on the philosophy of mathematics given in the winter and spring of 1939, 140pp.

7 Translation into English by Rush Rhees, with corrections by Wittgenstein, of the beginning of the pre-war version of Investigations, 1939, 72pp.

8 Menger was a true progenitor and instructor in numerous mathematical, logical and economic fields, including lattice theory and 'fuzzy' set theory (he termed fuzzy sets 'hazy').

9 Published in 1934: "A model of general economic equilibrium", *Review of Economic Studies* 13, 1–9.

10 Walras' — and similarly Marshall's — equations were thought by Peirce to demonstrate the existence of freewill, under the assumption of utilitarianism, in his syllabus for the planned volume VIII of his *Principles of Philosophy*, entitled *Continuity in the Moral and Psychological Sciences*.

11 What is also notable is the influence of Brouwer on the Colloquium. He had worked with Menger for two years in Amsterdam, from 1925, and lectured for the Colloquium in March 1928. The latter part of the first lecture was deemed by Menger to be "obscure remarks on primordial intellectual phenomena and primordial mathematical intuition [and] were not taken seriously by any member of the Circle" (Menger, 1994, pp. 138–139). In that lecture, Brouwer spoke about "das Substrat aller Zweiheiten". Wittgenstein was invited to attend at Menger's suggestion. Wittgenstein moved to Cambridge in January the following year, enjoyed his *annus mirabilis*, and his second coming was already well underway.

12 The so-called Brown Book. English. Dictated to Alice Ambrose and Francis Skinner at Cambridge in the academic year 1934–35.

13 *Notes on Logic*. September 1913, the so-called Russell version. Seven typescript pages were dictated by Wittgenstein and twenty-three manuscript pages are in Russell's hand.

14 In Hilbert (1926), presented in Münster in 4 June 1925.

15 "All pragmatists will further agree that their method of ascertaining the meanings of words and concepts is no other than that experimental method by which all the successful sciences (in which number nobody in his senses would include metaphysics) have reached the degrees of certainty that are severally proper to them today; — this experimental method being itself nothing but a particular application of an older logical rule, 'By their fruits ye shall know them."' (EP 2:400–401, *Pragmatism*, 1907). Cf. this with the passage from MS 324, 1907, *Pragmatism*, in which Peirce sketches an "appeal" for his proof of pragmatism that he never really had presented: "There is a theory that a certain method will perfectly ascertain the meanings of all intellectual concepts. To you, at least, this ~~method~~ [theory] is quite unproved; yet it does not present itself without certain credentials. For ~~it~~ [the method] is but a special application of the sole method by which physical science has achieved its successes during the last three hundred and odd years, having previously to the adoption of this method been universally looked upon as the most uncertain and intractable of all branches of science. It is the experimental or correctly inductive, method; itself ~~a special application of~~ [an amplification or fuller development], of a wider method, exemplified in the maxim, 'By their fruits ye shall know them'. If the reader will only consent actually to take the trouble, — for it will be some trouble — to apply this method to test the truth of the theory of pragmatism itself, — and [p. 15] while he may deem it an imperfect test, yet ~~it~~ he need not, in so using it, assume that it discloses the *entire* meanings of concepts, — what he will surely find is that, though it is no metaphysical doctrine, but only a method of ascertaining implications of words and of concepts, yet a large majority of the hardest knots of metaphysics will fall apart at the lightest touch, once they shall have received their pragmatistic interpretations; while for the few questions that remain unresolved, it will become easy to see that their ~~resolution~~ [unravelment] depends upon our acquiring certain definite experience or information; so that their mystery, their perplexing incomprehensibility, is gone" (pp. 14–15; cf. 325: 11–12).

16 The so-called Blue Book, in English. It was dictated to the class at Cambridge in the academic year 1933–34. 124pp. (There is some variation from the original copies, as Wittgenstein inserted minor corrections.)

17 The term 'semantics' is, of course, quite obsolete in this context. Some of its early beginnings are discussed in Chapter 12. 'Semantics' was proposed also by Gödel to Carnap, who communicated the proposal to Otto Neurath in 1932. Gödel wanted to uphold Alfred Tarski's use of the term, but the history of the term is even older. Neurath's response was that not only was the term ugly, but according to his wife, it resonated with 'semiotics', which made it altogether wrong since the latter was a doctrine about symptoms of disease. The consensus was to stick with the term 'logical syntax'. Like Peirce's diagrammatic syntax (Chapter 4), in Carnap's treatment it was bound to encompass elements that we nowadays see as semantic.

18 It should be noted that correlates to uniform designs, i.e. uniform strategies defined on whole information sets, arise in semantic games for logics that have imperfect information, cf. Chapter 7.

19 There are some interesting remarks concerning some examples of real games resembling linguistic patterns characterised by their winning, losing, or competitiveness conditions in Wittgenstein (2000–, 226: 48).

20 One could also attribute some game-like flavour to Gentzen's sequential calculus. Apart from the structural considerations, this may be seen in the remarks such as: "For the purpose of the *consistency proof* alone, incidentally, the notion of a 'choice' is dispensable, since we are here dealing only with the reduction of a derivation with the endsequent $\rightarrow\ 1 = 2$, and since all reduction steps are *unequivocal* and do not depend on choices" (Gentzen 1969, p. 196, see also p. 197, 198.)

21 For partial interpretations, see Pietarinen (2002b); Sandu & Pietarinen (2001).

22 Evolutionary game theory, at least in its original formulation devised by Maynard Smith & Price 1973, is assumed to be a model of what goes on in nature, not an explanation.

23 "Es stimmt, das Spiel 'Zeigen oder sagen, was man sieht' gehört den fundamentalsten Sprachspielen, was wiederum heißt, daß das, was wir im Alltag als Sprachgebrauch bezeichnen, größtenteils dieses Spiel voraussetzt (Wittgenstein, 1984, p. 56). This passage is not found in the 1956 edition of the *Remarks on the Foundations of Mathematics* the earliest fragments of which date from 1937, nor in the 1967 re-edition.

24 There is ample evidence for the affinity of the two in empirical findings in cognitive neuroscience.

25 2.654, 1893, *The Doctrine of Chances*.

26 Think of performatives, for instance. If someone utters 'I declare that I have never been involved in espionage', who is to judge the truth-value of this statement? It seems clear that it is not solely the utterer that is playing the semantic game, or else we run the risk of deriving truth-values that may result in perjury charges in court. A reliable verification in that case is partly extrinsic; yet it needs to work in a way that does not overlook the performatory preface 'I declare that' in one's declaration.

27 Reprinted in 2.690, 1878, *The Probability of Induction: The Rationale of Synthetic Inference*.

28 It is, of course, also vital to acknowledge the dynamic character of Wittgenstein's thought, and to recognise that his views on language games varied during different phases and episodes in his life. A number of aspects of language games and how they change in Wittgenstein's philosophy are documented in Hintikka (1996a).

Chapter 9

DIALOGUE FOUNDATIONS AND INFORMAL LOGIC

> ALL THE GREAT PHILOSOPHICAL DIALOGUES, those of Plato, the Italian works of the XVI$^{\text{th}}$ century, beginning with those of Pietro Aretino, the most perfect of all but Plato's from the literary point of view, followed by those of Giordano Bruno, Gallileo, and many others, then those of Berkeley and Shaftesbury's Moralists, either narrate actual dialogues, or compress into one a number of such actual conversations. (MS 612: 25, 1908, *Logic. Chapter I: Common Ground*).

Dialogue foundations comprise both informal and formal components. The informal component deals with philosophical assumptions that the use of dialogues for logical purposes takes in. The formal component studies the structure of these dialogues, not the properties of the target language to which they are applied. I will examine the former, informal component, by analysing Wilfrid Hodges' (2002c) recent disordering of 'dialogue foundations', including his mistaken view that dialogue games arise as no more than a 'psychological convenience'.

Appended to this chapter is a conversation between Peirce, Wittgenstein, Hodges and an anonymous referee, who undertake to spend an afternoon discussing these points.

1. Lead-in

In terms of logic, I take the notion of dialogue foundations to comprise both informal and formal components. The informal component deals with a range of philosophical assumptions that any coherent use of dialogues for logical purposes takes in. The formal component concerns the mathematical structure and properties of these dialogues, not the logical properties of the target language to which they are applied (which has nothing to do with foundations).

My purpose here is to examine the former, informal component, by means of setting straight a couple of recurrent misunderstandings that have appeared in the literature, most notably in the paper *Dialogue Foundations: A Scepti-*

cal Look by Hodges that appeared in the supplement to the Proceedings of the Aristotelian Society. Similar misunderstandings recur in his entries on Model Theory and Logic and Games in the Stanford Encyclopedia of Philosophy (Hodges, 2001a,b). In addition to this task, I will set so-called 'dialogue foundations' of informal logic in the perspective of Peirce's pragmatism.

We may broadly understand dialogues as encompassing any interactive process that is effectively strategic in nature. Dialogues occur within may be based on games, decision making, agents and agencies, team formations, *tête-à-tête* conversations, and so on. Let me add that what is strategic and what is non-strategic can itself be understood broadly. For instance, Jürgen Habermas (2001, p. 12) claimed that game-theoretic methods are of little help in understanding natural-language communication, because communication, unlike games, is cooperative and non-competitive. However, this holds only for a limited class of games. It is true that game theory tends to regard cooperation as dispensable and typically relegates it to 'pre-play' situations in which negotiation and bargaining takes place. However, to exclude game theory from the scope of the theory of communication would be too hasty as it overlooks games that are cooperative and non-competitive (Chapter 2).

Another way of putting a closely similar point across is to note that game-theoretic approaches to theories of communicative practice have enjoyed some success. I have in mind pragmatic theories along the lines of that of H. Paul Grice, and the elucidations of such 'logics of conversation' thereof (Chapter 12). Grice's goal was to devise a normative theory of communication that follows from the postulates of rationality and cooperation. However, too much focus has been laid on his maxims of conversation — especially on the maxim of relation currently in vogue in theories of relevance (Sperber & Wilson, 1995), which he took to be imminent outcomes of these postulates. Much less has been said on the overall ethical project to elicit different maxims — the list of which was not claimed to be exhaustive by Grice — from the assumption that dialogue partners are rational and aim to supplement the *summum bonum*, the 'ultimate good' of scholastics, by cooperative practices.

Needless to say, this implementation of the Gricean project suffers from similar drawbacks as the overall theory of games. Just as it is not obvious that agents trying to maximise expected utilities in fact increase the idea-potentiality of the *summum bonum*, it is not clear that language users invariably act according to hyper-rational principles. Nevertheless, this cannot constitute an argument against using game theory in theories of communication and conversation — or in the logic of conversation, as Grice would have us to entitle his project. Rather, it is a preliminary argument to the effect that there is something intrinsically congenial in theories of communication and theories of games.

Attempts have been made to spell this out in a couple of slightly different ways. Hintikka (1986) argued that Grice's programme ought to be operationalised by assigning payoffs to total communicational strategies, not to individual moves that interlocutors make. Only when a certain sequence of communication terminates do we have enough reason to assess the value of the path taken by the speaker or the hearer through a multiplicity of possible conversational situations.

Not referring to Hintikka's proposal, Parikh (2001) investigated similar ways in which game theory could be moulded to yield a framework for a strategic model of communication. Parikh's and Hintikka's ideas have several features in common: for instance, just as Hintikka takes maxims as precepts assigned to conversational strategies, so Parikh assumes that maxims operate on the level of payoff maximisation rather than definition. Payoffs are assigned according to the principle of cooperation, whereas maxims follow from the principle of rationality.

A further argument in favour of actually implementing pragmatic theories of language by strategic and dialogical means is that such implementation would naturally take language use and understanding to be reciprocal, and the responsibilities equally and mutually distributed between the speaker and the hearer. Originally, the maxims pertained only to utterances. Even today, the relevance theory of Sperber and Wilson, which is supposed to be one of the most elaborate theories building on of one of Grice's central maxims — even though it advocates an element of strategic reasoning in terms of agents aiming at the maximisation of linguistic information and the minimisation of the cognitive processing effort required in gleaning relevant information — is rather one-sided in that it ignores the possibility that the interpreter may well get to decide what he or she interprets as relevant in the utterance (Chapter 12). Thus, dialogical models of communicative strategies provide a promising basis for pragmatic theories of language.

Anyhow, the strategic component is vital in both formal and informal dialogues. Formally, strategies refer to the fact that, in relation to the theory of semantic games, the notions of truth and falsity of a proposition in a model agree with the existence of winning strategies. In relation to the theory of dialogue games, the notions of validity and invalidity (formal truth that hinges on the interpretation of non-logical constants) agree with such winning strategies. Informally, strategies are related to other purposes of dialogues, conversation or disputations, such as maintaining consistency and coherence. In argumentation, they pertain to any of the norms that feasible and reliable notions of an argument may suppose. These informal purposes have been prominent since the late scholastic *ars obligatoria* (Yrjönsuuri, 2001).

What Hodges mostly refers to in his ill-conceived critique of 'dialogue foundations' is the theory of dialogue games. This theory was proposed by Paul Lorenzen, and subsequently studied by Kuno Lorenz, Else Barth, Douglas N. Walton, Erik Krabbe, Shahid Rahman, and many others. Its fundamental idea is to resort to two participants, the Proponent and the Opponent. The former proposes a claim while the latter challenges it. The moves are made according to logical and procedural rules. Informally, the logical rules consist of rules for (i) conjunction, prompting a challenge by the Opponent, the chosen conjunct being available for support by the Proponent; (ii) disjunction, which is not really challenged but just defended by the Proponent by choosing one of the disjuncts for the defence, and (iii) negation, which is a signal to change roles. In other words, negated statements are challenged by defending the statement governed by the negation. An existential statement is a request for a witness produced by the Proponent, instantiated as the value of the quantified variable to serve as a claim to be defended in the future. Likewise, a challenge on universal quantification asks for an individual produced by the Opponent, and the result of the instantiation will be the next challenge.

The Proponent is taken to have lost if the claim can no longer be defended, and the Opponent is taken to have lost if the claim can no longer be challenged. The key is the existence of winning strategies, which prescribes when the formulas will be valid. An analogous result to that of the theory of semantic games is that a first-order sentence S can be deduced from the set of first-order sentences T $(T \vdash S)$ if and only if $T \vdash S$ is valid in intuitionistic logic.

Procedural conventions place some restrictions on how dialogical games are played. For example, it is often stipulated that a challenging claim may be answered at most once (or vice versa), or that responses by the Opponent are restricted to the latest challenge not yet defended. There are significant choices to be made between these conventions, as shown by the fact that classical logic can be reproduced by a suitable combination of these rules (Lorenz 1961; Rahman & Rückert 2001).

This short exposé immediately reveals a sore spot in Hodges' argument. He focuses on dialogue games, but at the same time claims that his arguments work for other kinds of logical games, too, including the semantic games of Hintikka and his collaborators. As is well known, these are quite different games meant to serve different purposes. His misunderstandings can thus partly be explained in terms of a sheer confusion between different kinds of dialogues or different kinds of games. One needs to set apart semantic, proof-theoretic, argumentative, rhetoric, discursive, conversational, computational and interrogative games, to mention just a few examples.

Hodges states that he sees no relation between Lorenzen's dialogue games and various real-life argumentative games along the same tradition, but this point is moot, as pointed out by Krabbe (2001).

Among the specific claims that are easily seen to be misguided and false in Hodges' papers are the following. (i) Peirce's dialogical approach to logic and cognition left unspecified what the roles and purposes of the players in a dialogue are. (ii) Wittgenstein's language games bear little resemblance to

Hintikka's semantic games, among other things because no one wins or loses Wittgenstein's language games. (iii) The participants of dialogues do not search for and pick elements because such notions are not well defined in logic. (iv) The metaphor of examination would be fitting for game-theoretic dialogues. (v) Dialogues or games somehow emerge as a 'psychological convenience' (Hodges, 2001c, p. 31) in analysing logical concepts. (vi) Lorenzen's games give no access to logical necessity, and bear no relation to real-life encounters that in some reasonable sense are logical.

My method of addressing these points is to use a jigsaw-type dialogue between Peirce, Wittgenstein, Hodges and a referee (N.N.), presented in the Appendix. I will summarise the arguments in the following section. While this dialogue is primarily aimed at establishing the six contentions listed above and showing the respective individuals' replies to them, as in any dialogue, some emergent implications are exposed. Among them is the question of what is similar and what is dissimilar in Peirce's and Wittgenstein's thinking. (I addressed this question in the previous chapter.) Furthermore, in establishing points (ii) and (iii), this dialogue is seen to take some further and unexpected turns. For instance, it goes on to address the methodology of logic, including what the conversants presumed the distinction between formal and informal accounts or logic is. Likewise, on point (v), the conversation subsumes a discussion about diagrammatic methods and the theory of mental models.

The bulk of the répliques in the dialogue are from Peirce's Collected Papers, from Wittgenstein's recently published Nachlass, referred to by item and page number, and from Hodges (2001c), referred to by page number. Exceptions are indicated. The material from Peirce falls within the period of 1867–1910 (mainly 1902–1905), that from Wittgenstein falls within 1929–1946, and that from Hodges is from 1977–2001. All of the emphases are in the original texts. What Wittgenstein said or dictated is quoted from the Diplomatic Transcription.

The part of Hodges' criticism concerning Lorenzen's approach to dialogue foundations (point (vi) above), is decisively addressed and answered by Krabbe in his reply to Hodges' paper (Krabbe, 2001). I have no additional reason to enter into that particular argument here. Among the main corrections that Krabbe makes is that in dialogue games, one seldom wishes to speak in terms of the opponent attacking the proposition vs. the proponent defending it, since not all dialogue moves count as malicious objections or strict defences. Yet, this is one of Hodges' main points in trying to ridicule game-theoretic methods. It is obvious that a game may be both competitive — which is not the same thing as conflict, contrary to what (Hodges, 2001c, p. 30) believes — and cooperative, which indeed is natural from the viewpoint of the general theory of games. Already by way of Brouwer's remark, if a relation of an utterance to the eventual hearer exists, "it can be one of friendship as well as of hostility" (Brouwer, 1975, p. 467).

As noted above, while fractions of these misunderstandings could be explained away by noting a confusion between different kinds of dialogues, a wider conclusion that I wish to argue for here is that Hodges' discussion implies a more profound fixation to exclude elements of the category of thirdness from the provinces of both formal and informal approaches to logic.

The fundamental conclusion is therefore that the conclusion that Hodges draws to the effect that there is no place for dialogue foundations in logic is based on misinterpretations of and lack of reference to philosophical and semiotic literature on dialogue games and the history of logic. This negative conclusion nonetheless turns out to be instructive, since one obtains, however inadvertently, the much wider and interesting result that there is no sensible philosophical programme of 'dialogues in logic' or 'games in logic'. Instead, the conceptual priority should be along the lines of 'logic captivated by dialogues or games', or even 'logic from games'.

2. Whither dialogue foundations?

Allow me to summarise the five points listed above, and my replies to them. The first and the second claims that Hodges makes are thus:

i Charles S. Peirce's dialogical approach to logic and cognition left unspecified what the roles and purposes of the players in a dialogue are. (Hodges, 2001c).

ii Wittgenstein's language games bear little resemblance to Hintikka's semantic games, among other things because no one wins or loses Wittgenstein's language games. (Hodges, 2001c).

Ad. i, ii: These are definitely rebutted in terms of abundant textual evidence assembled in the Appendix, and thus are not replicated here.

iii The participants of dialogue do not search and pick elements because such notions are not well-defined in logic. (Hodges, 2001b).

Ad. iii: Hodges has claimed on several occasions that Hintikka's original explanation (Hintikka, 1973a) of players' activities, especially in relation to semantic games, namely the seek-and-find game of elements from the domain of discourse, is unsatisfactory. His argument is that nothing in logic corresponds to searching.

Is this a valid objection? The answer hinges on what we take the meaning of quantifiers and logical connectives to be. When the 20th-century conception of elementary logic gradually evolved, such explanations were expelled from the domain of logic, partly due to the uncritical acceptance of Tarski's concept of satisfaction, and partly due to the illusory belief that the concept of a quantifier is unproblematic and its meaning is exhausted by its 'ranging over' some specified domain (as to the latter point, see Hintikka 2002a). If we rest content with these two notions and narrow down the field of logic accordingly, we should never ask how the meaning of a quantifier emerges, or what the fundamental processes behind its interpretation are. Since Hodges operates within this narrow concept of logic, his objection remains an *ignis fatuus*.

Interestingly enough, questions concerning the meaning of quantifiers and connectives were of utmost concern among 19th-century logicians. Peirce frequently spelled out the meaning of quantifiers in terms of two partners, typically termed 'the Utterer' and 'the Interpreter', who are to pick 'names' or 'instances' from the 'universe of discourse', assigned to the 'blank spaces' of 'rhemas', the argument places of predicate terms.

This serves to substantiate the general point that ignorance of the history of logic easily leads to false beliefs concerning the received concepts of logic.

While criticising Hintikka's concept of the seek-and-find game of logic, Hodges wishes to present an alternative explanation:

iv The metaphor of examination would be fitting for game-theoretic dialogues. (Hodges, 2001b).

The idea is that A serves as the examiner who puts questions to E. The elements that E produces are 'answers' to A's questions.

Ad. iv: Five points that speak against this reconstruction:

1 If nothing in logic corresponds to searching, nothing in logic corresponds to question answering, either.

2 In contrast to seeking and finding, question answering is much more asymmetric. There are two different activities that make sense only if they are conducted in a strictly alternating fashion and the opponent has to start. However, there may be no valid *protasis*. For what does it mean to start a session with an answer, which will happen in any proposition first asserting the existence of something?

3 Since values of variables are individuals or names of individuals, it makes little sense to try to identify these with complex interrogative sentences and answers to them. Such activities are linguistic, having to do with presuppositions and conditions for conclusive answers, among other pragmatic and logical phenomena. True, subformulas could be translated into natural language, chosen according to connective instructions, but there is nothing to suggest that they are answers, or that their compounds would miraculously transform into a non-declarative mood of questions.

4 In certain extensions of first-order logic, one could speak of simultaneous, asynchronic activities. It is thus much more plausible to believe that several searches may be conducted independently and concurrently than to believe in the chaotic scenario in which questions and answers are thrown in the air independently of each other.

5 The scheme of examination contains nothing of essence to quantification, but is heir to obligations, the conduct of debates familiar from Aristotle's Sophistical Refutations in which the instructor tries to teach the technique of good dialectics.

An intermediate conclusion is therefore clear: it was a *faux pas* to introduce examinations in this context, and objections to the seek-and-find paradigm back fire.

Therefore, the 'informal' component of logic that may be elicited from attempts to understand the meaning of quantifiers and connectives, if not inherent in the 20th-century conception of symbolic logic, was at least recognised and discussed by its earlier developers, most notably by Peirce. By way of contrast, the existence of the 'informal' component has led Hodges to believe that it has something to do with psychology. This leads to the moot point (v):

v Dialogues or games somehow emerge as a 'psychological convenience' in analysing logical concepts. (Hodges, 2001c, p. 31).

Ad. v: What, if anything, could be psychological in logic? According to one recurrent version of psychologism, the laws of the human psyche may contradict, or be in tension with, the laws of logic. This needs to be distinguished from mentalism, which maintains that references to meaning and interpretation are to be made via an intellect of a broad nature, capable of creating interpretants of signs (and thus interpretation is relative to the mind or the quasi-mind of non-human sign carriers). It does not dispense with propositional attitude constructions such as knowledge, belief and obligation. While mentalism is congenial to logic, no psychologism needs to be admitted. Peirce (and quite different thinkers such as Frege the logicist and Husserl the phenomenologist) opposed any ill-conceived psychology in tasks of reasoning and argumentation. Husserl in particular argued for the irreducible normative character of human language. Psychological concepts just are not essential in understanding the meaning of logical constants or the semantics of our natural language.

In a much similar vein, Grice's pragmatic programme has all too easily been taken to be psychological, thus falling within that broad genre of linguistic theories that attempt to reduce the normative component of language to matters of psycholinguistic theorising. The distinction between mentalism and psychologism goes against any such reduction, the propositional attitudes of Grice's theory notwithstanding. Taking into account how one 'understands' sentences does not change this point, because one still needs to differentiate between understanding something right and understanding something wrong.

The question remains whether there is nevertheless something genuinely psychological (and not just mental) in the new developments that resort to mathematical logic but take into account the human praxis more gingerly than before, including advances in practical logic, non-monotonic reasoning, multi-modal agent systems, decision procedures, or in any other cognitively and computationally driven theories. I do not need to discuss here the extent to which these may have been motivated by considerations of the strange notion of 'psychological convenience'.

Neither do I intend to justify arguments in favour of game-theoretic methods in logic (to that effect, see especially Chapters 7 and 10). In brief, they include the Peirce–Wittgenstein argument according to which interaction, communication and ultimately the system of games are constitutive of creating semantic links between language and the world, that is, are responsible for linguistic and logical meaning and its evolution. However, the philosophies of these thinkers are very different and very similar at the same time (Chapter 8). Quite another vindication is practical in essence: game methods increasingly cover new areas, find new applications, and are used in solving old problems in new ways (Chapter 10; Mauret & Moore 2001).

3. Informal logic from a pragmatist perspective

Implicit in the above discussion, as well as in the appended dialogue, is a provisional agenda for informal logic. The overall argument that emerges implies that informal and formal logic are connected, and that an 'informal' component exists to be distilled from formal logic, closely related to argumentative concerns.

Further work includes the study of the relations between this informal component and the logica utens that Peirce borrowed from the scholastics, namely the inherent, immutable faculty of the mind according to which everyone argues without any theory of argumentation. The immanence is shown by the connection between Peirce's notion of the habit and its partial reincarnation in the form of dialogical strategies in both person-to-person and intrapersonal argumentation (Chapters 1 and 13). Both aim at bringing logic closer to Peircean speculative rhetoric, of which present-day pragmatics is a fragment, by increasing the room for game-theoretic perspectives rather than diminishing it.

The centrality of such speculative rhetoric is also shown by its role in serving as the methodology of individual sciences, which is at the same time one of the primary concerns of informal logic in its task of classifying and evaluating scientific argumentative procedures and protocols.

In fact, such a pragmaticist agenda for informal logic has been suggested before. Arne Næss has been said to have implied that 'The choice of a logical system must be in some relation to the purpose that system and the systems with which it is being compared, are supposed to serve' (in van Eemeren et al. 1996, p. 250). This is a conspicuously pragmatic point that Peirce would have been happy to endorse, that is to say it is an illustration that any logical system, representation technique or piece of reasoning must define its purpose, in other words for what purpose the system is meant or whom or what it is intended to assist.

Logic is always something for somebody, with respect to some capacity or function, the choice between rivals having to be made on methodological grounds. This reflects a central aspect of the category of thirdness. Unfortunately, the systematisation of this important point has occasionally gone somewhat astray. Contrary to what has been suggested by argumentation theorists, the question of purpose is not exhausted by adopting Russell's view that logic is a problem-solving activity. What Peirce started as a promising insight became bogged down in a narrowly focused conception of logic.

To counteract this, programmes such as pragma-dialectics broaden the norm base by introducing other standards besides formal validity and invalidity (or truth and falsity), such as sticking to standpoints, not evading criticism, not imputing anything fictitious to other parties, and relevance, acceptability and sufficiency of premisses (van Eemeren et al., 1996).

I thus believe that logica utens ought to be introduced here as fundamentally congenial to what informal logic and critical thinking have been after. It represents a faculty that is not bound to any particular theory of reasoning and argumentation other than what may be provided by some native, stable, acquired, invariable, constitutional, secure, enduring, instinctive form of reasoning found in the inner life of the reasoner (Chapter 1). This is not unlike mathematics, which, according to Peirce, performs its reasonings by such utens, having no need of appeal to its contrast, the logica docens. Peirce maintained that the docens was the educable, improving, nurtured and schooled faculty for reasoning associated with all evolving theories of logic. Needless to say, it is the docens that has exponentially burgeoned of late, while utens has been lolling in the background. The much-needed move towards the practical logic of cognitive agenthood is a step in an encouraging direction (Gabbay & Woods, 2001).

Unlike theories of logic and argumentation, such innate logic or critical attitude of utens to human thought is not itself to be evaluated as good or bad, but is altogether beyond such normative concerns. It is not to be subsumed under criticism. It is thus an appealing and historically venerable candidate for the locus in which the verification of elementary sentences and the formation of the principles of ortholanguage may take place. This is what the *Erlangen Schule*, most notably in terms of Paul Lorenzen's constructivism and the emergence of dialogue games (Lorenz 1961, 2001), were after — a kind of interpersonal *redepraxis* or ortho-logic for other logical theories. It constitutes a candidate for a unifying reference point for the continuing endeavours to distill out the 'informal' component of logic.

To get the chain of the history of intellectual ideas right, let me note that constructing an ortho-language was not of logical empiricists' origin, despite their famous promotion of logical analysis of language in solving scientific problems. It was a continuation of what the predominantly Dutch group of significians were after since the late 19th century, namely an analysis of language that aims to stratify it in terms of purpose, spelled out in the ever-increasing connection, conventionalisation and interrelationship between words, symbols and terms, starting with what is a 'primordial' language, ending with higher scientific and symbolic (logical) layers, through several intermediate steps.

The term 'primordial' springs from Brouwer, a central significist alongside with Welby, Mannoury, Frederik van Eeden and David van Dantzig. Brouwer's philosophising had an effect, not only on Wittgenstein's return to philosophy but also on Evert W. Beth, Lorenzen and many others who strove to make Brouwer's so-called mystical elements of intuitionism understood by a recourse to game conceptualisations. This was needed, following Brouwer's critique of Hilbert in degenerating mathematics into a meaningless game. Brouwer's comment has to be taken with a grain of salt, however, because at that time Hilbert was, in his attempt to fix models for logical systems, striving to differentiate the interpretation of non-logical symbols — an extraneous consideration by his own words — from the manipulation of inference relations between formulas, a project which actually was closer in spirit to semioticians and significians of his time than the upcoming logical semanticists and symbolic logicians who endorsed a narrower approach. Both Hilbert and the semioticians and significians recognised this difference, although the latter much more dimly. But they all recognised the importance of the pragmatic value of theories not appealing to unsupported intuitions concerning the meaning of primitive symbols.

This Hilbertian stance was implicit in significists' stratification of language, especially in terms of the distinction between primordial language (in which the meaning of all words appeals to profound emotions and strikes one's consciousness much like a child learns and understands words and symbols), and logical languages (in which meaning no longer depends on the content of expressions and has no bearing to the hearer, since the symbols already have their application in the preceding scientific and interactive levels).[1] This resembles Tarski's account of logical consequence, the model-theoretic import of which is that a conclusion follows from the hypotheses precisely when every situation, case or a model in which hypotheses are true is a situation, case or a model in which the conclusion is true. The problem here is the nature of the underlying logical notions that warrant such an explication, and this dialogic constructivism endeavoured to tackle.

Thus, in Brouwer's view, there were two kinds of activities: 'games' or 'recreations' that serve calculations and non-social instruction in mathematics, and those that were embedded in linguistic speech-act practices and other constructions of meaning. The former lack the purpose and life-value of the latter, more contextual and more networked activity aimed at understanding on all levels of language.

Notwithstanding these points Brouwer wishes to impart, in the theory of significs, one of the intermediate manifestations between primordial and symbolic strata of language was as the communicative system of interaction. The point may seem obvious for us, but it is remarkable in terms of setting the history of ideas right in virtue of being yet another observation by the significians that ties visibly in with the Erlangen programme of promoting dialogic analysis of communicative intentions.

Another point is that the significians were also advanced speech-act theorists well ahead of Austin, Searle and Grice. Thus they would not have fallen into the same chaos as Hodges (2001c, pp. 24–25), who overlooked the element of speaker's meaning and the allied perlocutionary force of non-declarative assertions, merely in order to represent them as propositions on a par with declarative ones, and then using his soi-disant construction as a criticism against Lorenzen in not allowing for the relevance of propositions in further cycles of the argument.

One spin-off of the above considerations is that a novel perspective on the relation between informal logic and critical thinking is forthcoming in the analysis of language as a communicative rather than symbolic system. Just to mention one question that arises in this connection is, can thinking be critical without engaging in arguments? Just to give an example, is music an argument? For surely, does not a musical performance, just like any fine piece of art, possess at least some *ethos* and pass on at least some *pathos*? Does not a critical thought or attitude, much like linguistic assertions, extort some degree of energy?

The answer is that the strength of the argument lies in the realm of logic proper, or critic, which is distinct from the methodological concerns of speculative rhetoric: "A proof or genuine argument is a mental process which is open to logical criticism" (2.27). Furthermore, Peirce held that, "Symbols which also independently determine their interpretants, and thus the minds to which they appeal, by premissing a proposition or propositions which such a mind is to admit. These are arguments" (1.559). Therefore, only processes of induction, deduction and abduction are signs that qualify as arguments. Only these three forms represent conclusions, reached from the premisses that determine its interpretant. In view of this, it is seen that critical thinking differs from argumentative processes of logic, however informal. It ought to draw on other genres of Peirce's philosophy, such as his critical common sensism and its shared methodology with the idea of the logica utens, but not on arguments.

4. Conclusions

Peirce's thirdness of logic and the informal approaches to logic overlap, and so do critical thinking and critical common sensism. Likewise notable is that the signific analysis of language antedated German constructivism from which the so-called dialogue foundations evolved. Only a fragment of such mutual points of contact was taken up, but as remarked, the concept of a dialogue is one of the key cohesive factors in establishing these connections. As Hodges' misplaced position for the role of dialogues in logic witnesses, these substantial points of contacts are easily missed from a slender historical perspective.

Moreover, as the advocates of the Erlangen School ought to have shown, the project of dialogue logic is, in line with Wittgenstein's notion of language games, not one of how to apply dialogues, playful dialectics or games to some logical or linguistic tasks. There is no place for such foundations in logic. The activities that such notions have been applied to capture are more fundamental, something from which the rule-governed systems of language gradually evolve.

It is the illicit use of the preposition that has been fateful.

* * *

Finally, before the reader proceeds to the actual conversation that follows, let me mention a caveat, itself from a dialogue written by Peirce:

You I criticize your creation of me, and the whole method of throwing philosophical discussion into the form of dialogue. For a philosopher ought above all things to be sincere and to say just what he means. Now a philosophical dialogue is always a make-believe lower than play-acting. It is just a puppet-show, in which Punch knocks Judy and the policeman and all the rest of the wooden things over the head, and then makes fun of all his lawless doings and of all his victims.

I Well, well, there was plenty of latent heat in the cold steel. (MS 612: 23, 10 November 1908, *Common Ground*).

Notes

1 See Pietarinen (2003g) for a study of significs in relation to some of the central ideas of language that permeated thinkers affiliated with the Vienna Circle.

Appendix 9.A: A dialogue

NN Welcome to all. Will you, Mr. Peirce, begin by describing how you arrived at your dialogic analysis of assertions?

Charles Sanders Peirce When an assertion is made, there really is some speaker, writer, or other signmaker who delivers it; and he supposes there is, or will be, some hearer, reader, or other interpreter who will receive it. It may be a stranger upon a different planet, an æon later; or it may be that very same man as he will be a second after. In any case, the deliverer makes signals to the receiver. Some of these signs (or at least one of them) are supposed to excite in the mind of the receiver familiar images, pictures, or, we might almost say, dreams — that is, reminiscences of sights, sounds, feelings, tastes, smells, or other sensations, now quite detached from the original circumstances of their first occurrence, so that they are free to be attached to new occasions. The deliverer is able to call up these images at will (with more or less effort) in his own mind; and he supposes the receiver can do the same. (3.433)[1]

NN Will you give us a concrete, and if possible, logical example?

CSP Of course. Suppose the assertion is: 'Some woman is adored by all catholics.' The constituent icons are, in the probable understanding of this assertion, three, that of a woman, that of a person, A, adoring another, B, and that of a non-catholic. We combine the two last disjunctively, identifying the noncatholic with A; and then we combine this compound with the first icon conjunctively, identifying the woman with B. The result is the icon expressed by, 'B is a woman, and moreover, either A adores B or else

A is a non-catholic.' The subjects are all the things in the real world past and present. From these the receiver of the assertion is suitably to choose one to occupy the place of B; and then it matters not what one he takes for A. A suitably chosen object is a woman, and any object, no matter what, adores her, unless that object be a noncatholic. This is forced upon the deliverer by experience; and it is by no idiosyncrasy of his; so that it will be forced equally upon the receiver. (3.436)

Wilfrid Hodges At the first sight it seems that the parallel is unconvincing; in the case of ['all'], the interpreter [or the receiver] can pick any [wo]man, but in the case of 'some' the deliverer must pick a suitable [wo]man. But if we view it that way, it's irrelevant who makes the choice, and [you] never had any reason to distinguish the deliverer from the interpreter. (p.17)

CSP The receiver need only ascertain by experiment whether he can distribute any set of indices in the assigned way so as to make the assertion false, in order to put the truth of the assertion to the test. For example, suppose the assertion of logical necessity is the assertion that from the proposition, 'Some woman is adored by all catholics', it logically follows that 'Every catholic adores some woman'. That is as much as to say that, for every imaginable set of subjects, either it is false that some woman is adored by all catholics or it is true that every catholic adores some woman. We try the experiment. In order to avoid making it false that some woman is adored by all catholics, we must choose our set of indices so that there shall be one of them, B, such that, taking any one, A, no matter what, B is a woman, and moreover either A adores B or else A is a non-catholic. But that being the case, no matter what index, A, we may take, either A is a noncatholic or else an index can be found, namely, B, such that B is a woman, and A adores B. We see, then, by this experiment, that it is impossible so to take the set of indices that the proposition of consecution shall be false. The experiment may, it is true, have involved some blunder; but it is so easy to repeat it indefinitely, that we readily acquire any desired degree of certitude for the result. (3.437)

H Then a necessary and sufficient condition for ['Every catholic adores some woman'] to be true is that there is a way for the deliverer to act that will guarantee that she gets her wish if the interpreter chooses the [catholic]. (p.18)

CSP We certainly have habits of reasoning; and our natural judgments as to what is good reasoning accord with those habits. I am willing to grant that it is probable that some of our judgments of rationality of the very simplest kind have at the bottom instincts in the above broad sense. I am inclined to think that even these have been so often furbished up and painted over by reflection upon the nature of things that they are, in mature life, mostly ordinary habits. In more complicated cases, say for example, in that guess about the pair of dice, I believe that our natural judgments as to what is reasonable are due to thinking over, ordinarily in a more or less confused way, what would happen. We imagine cases, place mental diagrams before our mind's eye, and multiply these cases, until a habit is formed of expecting that always to turn out the case, which has been seen to be the result in all the diagrams. (2.170)

Wittgenstein The mental act seems to perform in a miraculous way what could not be performed by any act of manipulating symbols. (309: 67)

CSP To appeal to such a habit is a very different thing from appealing to any immediate instinct of rationality. That the process of forming a habit of reasoning by the use of diagrams is often performed there is no room for doubt. It is perfectly open to consciousness. Why may not all our natural judgments as to what is good reasoning be founded on habits formed in some such ways? (2.171)

W This, of course, doesn't mean that we have shown that peculiar acts of consciousness do not accompany the expressions of our thoughts! (309: 67)

NN Surely not! Now, an emerging view is to consider various strategic processes in logic and in reasoning. [to CSP:] Your notion of a habit appears somewhat broader.

CSP Let us use the word 'habit', throughout this [conversation], not in its narrower, and more proper sense, in which it is opposed to a natural disposition (for the term *acquired habit* will perfectly express that narrower sense), but in its wider and perhaps still more usual sense, in which it denotes such a specialization, original or acquired, of the nature of a man, or an animal, or a vine, or a crystallizable chemical substance, or anything else, that he or it will behave, or always tend to behave, in a way describable in general terms upon every occasion (or upon a considerable proportion of the occasions) that may present itself of a generally describable character. (5.538)

H A strategy for [the deliverer] is a set of rules that tell her how to move when it is her turn, depending on what the previous moves were. The strategy is *winning* if she wins every play in which she follows the strategy. (p.28)

CSP [facing H:] The popular notion is that Reason is far superior to any instinctive way of reaching the truth; and from your desire to study logic, I am perhaps warranted in presuming that such is your opinion. If so, in what respect do you hold reasoning to be superior to instinct? Birds and bees decide rightly hundreds of times for every time that they err. That would suffice to explain their imperfect self-consciousness; for if *error* be not pressed upon the attention of a being, there remains little to mark the distinction between the outer and the inner worlds. A bee or an ant cannot — could not, though he were able to indulge in the pastime of introspection — ever guess that he acted from instinct. Accused of it, he would say, 'Not at all! I am guided entirely by reason'. (2.175)

W [abrubtly:] It is sometimes said: 'animals don't speak, because they lack the necessary intellectual capacities'. And this means: 'they don't think, therefore they don't speak'. But they just don't speak. Or rather [raises voice:] they don't *use language*. [mumbles:] (If we disregard the most primitive forms of language.) Commanding, giving orders, asking questions, recounting — pardon! — describing, prattling, belong to our natural history just as walking, eating, drinking, playing do. It makes no difference here whether speaking is done with the mouth or done with the hand... (226: 17)

CSP [delighted:] ... thought is not necessarily connected with a brain. It appears in the work of bees, of crystals, and throughout the purely physical world; and one can no more deny that it is really there, than that the colors, the shapes, etc., of objects are really there. (4.551)

NN So signs are not connected with brains, either. But isn't language public?

W What does 'now' refer to or 'this' or 'I'. The private object. The naming of the private object. The private language. The game someone plays with himself. When do we call it a game. If it resembles a public game. The diary of Robinson Cr... (166: 2v)

CSP [interrupts:] That footprint that Robinson Crusoe found in the sand, and which has been stamped in the granite of fame, was an Index to him that some creature was on his island, and at the same time, as a Symbol, called up the idea of a man. Each Icon partakes of some more or less overt character of its Object. They, one and all, partake of the most overt character of all lies and deceptions — their Overtness. Yet they have more to do with the living character of truth than have either Symbols or Indices. The Icon does not stand unequivocally for this or that existing thing, as the Index does. (4.531)

W We can indeed imagine a Robinson using a language for himself but then he must *behave* in a certain way or we shouldn't say that he plays language games with himself. (149: 22)

CSP The existence of things consists in their regular behavior (1.411). In order to define a man's habit, [it is necessary] to describe how it would lead him to behave and upon what sort of occasion (2.664). I really know no other way of defining a habit than by describing the kind of behavior in which the habit becomes actualized (2.666).

W We can't pl... The game which we play with the word 't.' entirely depends upon there being a behaviour which we call the expression of t... (149: 29) [pauses] ... One may say that an index *alludes* to something, and such an allusion may be justified in all sorts of ways. (309: 33)

CSP An *index* is a sign which would, at once, lose the character which makes it a sign if its object were removed, but would not lose that character if there were no interpretant. (2.304)

W To get an idea of the enormous variety of language games consider these examples and others: giving and acting according to commands; description of an object by describing what it looks like, or by giving its measurements; producing an object according to a description... (226: 15)

H [interrupts:] ... because people win or lose games, games have the notion of it acting for a purpose built into them, and this is the source of much of their explanatory power. ... [But] nobody wins or loses [your] language game. In fact few ... have tried to make any connection between logical games and [your language] games, and I'm not about to propose a connection. (p.19)

W I want someone to explain to me a certain game and to make it easier I put the question to him: 'tell me what a man does to win the game'.

H From the point of view of game theory, the games that appear in logic are not at all typical. ... We need to understand what is achieved if [the deliverer] does win. (Hodges 2001b, no page numbers in electronic document.)

W Someone might say 'I don't want to know what a man does to win it, this is a question about human beings; I want to know what the game is they play'.

You ask, what does it mean that 'a rod is 6 inches long'. Someone answers one finds out whether it is so by using a measuring rod divided into equal parts. 'What does this mean?' — One divides it into equal parts in such and such a way. I have given a definition in terms of the way of verification. (151: 16)

H If we want [the deliverer's] motivation in a game to have any explanatory value, then we need to understand what is achieved if [the deliverer] does win. [Isn't this...] (Hodges, 2001b)

W 'But isn't this an indirect definition?' Sometimes it is, sometimes it isn't. (151: 16) [losing nerve:] The idea of inconsistency ... is contradiction, and this can only *arise in the true/false game*, i.e. when we are making assertions. (Wittgenstein, 1978, p. 321)

NN I'm sure everybody agrees that it is quite uncharitable to try to refute someone's views by saying that no one has not yet, or that in fact just few have, seen some connection between two things. That not only represents a closed-mind attitude but also makes ones own proposals dubious and is also self-uncharitable. But to clarify ideas, let's get back to that methodological question which didn't quite get going.

CSP My philosophy may be described as the attempt of a physicist to make such conjecture as to the constitution of the universe as the methods of science may permit, with the aid of all that has been done by previous philosophers. I shall support my propositions by such arguments as I can. Demonstrative proof is not to be thought of. The demonstrations of the metaphysicians are all moonshine. The best that can be done is to supply a hypothesis, not devoid of all likelihood, in the general line of growth of scientific ideas, and capable of being verified or refuted by future observers. (1.7)

H Logic can be defined as the study of consistent sets of beliefs; this will be [my] starting point. Some people prefer to define logic as the study of valid arguments. Between them and [I] there is no real disagreement (Hodges, 1977, p. 13)

W Ramsey once, in a discussion with me, stressed the point that logic is a 'normative science'. I can't say, exactly, what idea he had in mind; but it was undoubtedly closely related to that which I only later got hold of: — that in philosophy we often compare the use of words with games, or with calculi having fixed rules, but that we can't say that whoever uses language must play such a game. Ummm... If, however, you say that our languages only approximate to such a calculi, you stand right at the edge of a misunderstanding. For thus it may seem as though in logic we spoke about an ideal language. As though our logic was, so to speak, a logic not taking into account friction and air-resistance...

CSP [suddenly:] Vagueness... is no more to be done away with in the world of logic than friction in mechanics... (5.512)

NN Let's turn to the question of vagueness if there is time!

W ... whereas actually logic doesn't treat of language (or of thought) in the sense in which a natural science treats of a natural phenomenon, and all one might say is that we construct ideal language. (226: 57)

CSP It is pretty generally admitted that logic is a normative science, that is to say, it not only lays down rules which ought to be, but need not be followed; but it is the analysis of the conditions of attainment of something of which purpose is an essential ingredient (1.575). That which renders the word normative needful (and not purely ornamental) is precisely the rather singular fact that, though these sciences do study what ought to be, *i.e.*, ideals, they are the very most purely theoretical of purely theoretical sciences. What was it that Pascal said? 'La vraie morale se moque de la morale'. (1.281)

CSP Normative science has three widely separated divisions: i. Esthetics; ii. Ethics; iii. Logic. ... Logic is the theory of self-controlled, or deliberate, thought; and as such, must appeal to ethics for its principles. It also depends upon phenomenology and upon mathematics. All thought being performed by means of signs, logic may be regarded as the science of the general laws of signs... (1.191)

W [interrupts]: Our craving for generality has another main source: our preoccupation with the method of science. I mean the method of reducing the explanation of natural phenomena to the smallest possible number of primitive natural laws; and, in mathematics, of unifying the treatment of different topics by using a generalization. Philosophers constantly see the method of science before their eyes, and are irresistibly tempted to ask and answer questions in the way science does. This tendency is the real source of metaphysics, and leads the philosopher into complete darkness. I want to say here that it can never be our job to reduce anything to anything, or to explain anything. Philosophy really is 'purely descriptive'. (309: 27) According to the rôle which propositions play in a language-game, we distinguish between orders, questions, explanations, descriptions, and so on. (310: 9) [As I said], giving orders, asking questions, describing, prattling, belong to our natural history just as walking, eating, drinking, playing do.

CSP 'Natural History' is the term applied to the descriptive sciences of nature, that is to say, to sciences which describe different kinds of objects and classify them as well as they can while they still remain ignorant of their essences and of the ultimate agencies of their production, and which seek to explain the properties of those kinds by means of laws which another branch of science called 'Natural Philosophy' has established. Thus a logic which is a natural history merely, has done no more than observe that certain conditions have been found attached to sound thought, but has no means of ascertaining whether the attachment be accidental or essential; and quite ignoring the circumstance that the very essence of thought lies open to our study; which study alone it is that men have always called 'logic', or 'dialectic'. (4.8)

NN How would one characterise what ought to be taken formal and what informal in logic?

H [turning to CSP:] In the course of discussing Kant's Critique [you], in the opening words of the second draft of [your] *On a new list of categories* [used] the phrase [formal logic]. (Hodges, 1999, p. 3).

NN Mr. Peirce has resorted to that phrase hundreds of times, for a variety of purposes. But I think 'informal' is not to be found.

CSP Formal logic classifies arguments by producing forms in which, the letters of the alphabet being replaced by any terms whatever, the result will be a valid, probable, or sophistic argument, as the case may be; material logic is a logic which does not produce such perfectly general forms, but considers a logical universe having peculiar properties.

In most cases material logic is practically a synonym of applied logic. But a system like Hegel's may also properly be termed material logic. The term originated among the English Occamists of the fourteenth century, who declared Aristotle's logic to be material, in that it did not hold good of the doctrine of the Trinity. (2.549)

H [You] used the phrase 'formal logic' to cover both what [you yourself] did in logic and what Kant had discussed under that name. (Hodges, 1999, p. 7)

CSP That there are three elementary forms of categories is the conclusion of Kant, to which Hegel subscribes; and Kant seeks to establish this from the analysis of formal logic. Unfortunately, his study of that subject was so excessively superficial that his argument is destitute of the slightest value. Nevertheless, his conclusion is correct; for the three elements permeate not only the truths of logic, but even to a great extent the very errors of the profounder logicians. (3.422)

NN Is in your view logic part of mathematics?

CSP Logic . . . is categorical in its assertions. True, it is not merely, or even mainly, a mere discovery of what really is, like metaphysics. It is a normative science. It thus has a strongly mathematical character, at least in its methodeutic division; for here it analyzes the problem of how, with given means, a required end is to be pursued. This is, at most, to say that it has to call in the aid of mathematics; that it has a mathematical branch. But so much may be said of every science. There is a mathematical logic, just as there is a mathematical optics and a mathematical economics. Mathematical logic is formal logic. Formal logic, however developed, is mathematics. Formal logic, however, is by no means the whole of logic, or even its principal part. It is hardly to be reckoned as a part of logic proper. Logic has to define its aim; and in doing so is even more dependent upon ethics, or the philosophy of aims, by far, than it is, in the methodeutic branch, upon mathematics. (4.240)

H It is depressing to note that twentieth-century commentators on Kant's Critique regularly refer to 'formal logic', although Kant barely ever used the phrase and his thought never came near what the phrase will suggest to most twentieth-century readers. (Hodges, 1999, p. 3)

CSP Kant gives a half dozen only of his brief pages to the development of the system of logic upon which his whole philosophy rests. (8.41)

— [pause] —

H How [are] logical games . . . helpful for understanding concepts? The notion of competition between two people with contradictory aims seems to be . . . irrelevant. (p.20)

W Let us consider, e.g., the processes we call 'games'. I mean board-games, card games, ball games, athletic contests, etc. What is common to all these? Don't say: 'there must be something common to them all, or they wouldn't be called 'games' — but see whether something is common to them all. For if you look at them, though you won't see anything that's common to all of them, but you will see similarities, connections a whole lot of them. As I said: don't think, but look. Look e.g. at board games and the various connections between them. Now pass to card games; you will find many points of similarity between this and the first class; but many common features disappear and new ones appear. ... Is there always such a thing as winning and losing or rivarly — pardon — a competition between the players? ... I shall say that the 'games' constitute a family. And in the same way the kinds of numbers, for instance, constitute a family. ... And this is how we use the word 'game'. For how — in what way is the concept 'game' circumscribed? What is — when does something begin to be a game, and when does it cease to be one? Can you state where the boundary-lines are? No. You can draw some; for there aren't any drawn yet. [disappointed, sits down:] But this never bothered you, when you used the word 'game'. (226: 47–49)

H But there was no point where... (p.31)

W '... the use of the word is ... regulated, the 'game' which we play with it is... regulated'. — It is not bounded by rules at every point. Hmmm... How would you explain to someone what a game is? And do you know any more youself? Is it just that you can't explain to the other man exactly what a game is?

H This...

W This isn't ignorance, however. You don't know the boundaries because none are drawn. As I said, you may, for some purpose or other, draw a boundary. But is this necessary in order to make it into a useful concept? Not at all — unless you mean, useful for this particular purpose. (226: 50)

CSP The view which pragmatic logic takes of the predicate, in consequence of its assuming that the entire purpose of deductive logic is to ascertain the necessary conditions of the truth of signs, without any regard to the accidents of Indo-European grammar. (2.358)

NN [facing CSP:] Didn't you have an example of a cooperative dialogical approach to logic somewhere?

CSP We are to imagine that two parties collaborate in composing a Pheme — a Sign which is equivalent to a grammatical sentence, whether it be Interrogative, Imperative, or Assertory (4.538) — and in operating upon this so as to develop a Delome... (2.220)

NN Roughly, an argument.

CSP Oh, very well; yes. (1.290) The two collaborating parties shall be called the Graphist and the Interpreter. (4.552)

NN One can think of the Graphist as the deliverer of our initial example.

CSP Certainly. The deliverer makes signals to the receiver. (3.433) The progress of science cannot go far except by collaboration; or, to speak more accurately, no mind can take one step without the aid of other minds. (2.220) An Interpreter ... interprets those propositions and accepts them without dispute. (4.395) A dispute is not a rational consequence of anything. (1.247)

H When the rules distinguish those plays that are wins, we can think of them as describing a *purpose* for [the utterer]. [Raising voice:] In short, these games are a natural setting for talking about situations where *an object has to be constructed in steps so that it meets certain requirements*. (p.26)

W You seem to be thinking of games on a board; but these aren't all the games there are. You can put your description right by confining it explicitly to those games. (226: 2)

H There are plenty of situations of this kind. For example: (1) A tableau proof is constructed from the top downwards, ... (2) A calculation starts from a written expression and goes through the steps of an algorithm to finish up with the value of the expression, ...(3) A handshake between two computer systems is build up in steps, from the time when the systems make contact from time whan they are in full communication, ... (4) We can construct an existentially closed group. (p.26)

W The criteria for [any utterance] being the truth are laid down in language (rules, charts etc.). ...In so far are they are laid down in common language they become just part of the common language game i.e. they don't help me in any particular case. They join the rest of the rules of common language. (149: 37)

CSP Well... Computer's rules... It is quite true that we cannot make a machine that will reason as the human mind reasons until we can make a logical machine — logical machines, of course, exist — which shall not only be automatic, which is a comparatively small matter, but which shall be endowed with a genuine power of self-control; and we have as little hopes of doing that as we have of endowing a machine made of inorganic materials with life. (EP 2:387)

CSP [continuing after a while:] When in the ordinary Boolian algebra we write $\mathfrak{m} \prec \mathfrak{l}$, meaning 'every man is a liar',

NN ... you are using here the sign of conditional...

CSP ... according to me this means 'if $\mathfrak{x}$ (which is any individual object you may choose) is a man, then $\mathfrak{x}$ is a liar', $\mathfrak{m}$ signifying that $\mathfrak{x}$ is a man, and $\mathfrak{l}$, signifying that $\mathfrak{x}$ is a liar. (4.327)

I submit that this is all as clear as your New Hampshire air of October, and as familiar 'comme votre poche', and that I said it plainly enough; only it had to be unintelligible because I said it. (8.292)

But while my forms are perfectly analytic, the need of diagrams to exhibit their meaning to the eye (better than merely giving a separate line to every proposition said to be false) is painfully obtrusive. (4.548)

NN And the exhibition of meaning also uses habits?

CSP [I think I already mentioned that] we imagine cases, place mental diagrams before our mind's eye, and multiply these cases, until a habit is formed of expecting that always to turn out the case, which has been seen to be the result in all the diagrams. To appeal to such a habit is a very different thing from appealing to any immediate instinct of rationality. That the process of forming a habit of reasoning by the use of diagrams is often performed there is no room for doubt. It is perfectly open to consciousness. Why may not all our natural judgments as to what is good reasoning be founded on habits formed in some such ways? (2.170)

H This causes conceptual problems, because it complicates [the utterer's] motivation. She is no longer trying to construct the object required before; so what is she trying to do? (p.29)

NN Yeah, please tell.

H Fortunately there is a paradigm that comes to the rescue here. [The utterer] is sitting an exam on whether she can produce the required object. As in most exams, she knows she will only be asked part of what she is required to know, but she doesn't know which part. [The interpreter] plays the role of the examiner, deciding what questions she needs to answer. To say that she can pass any exam in this subject is to say that she has a winning strategy, and a winning strategy is precisely the object we wanted her to produce. (p.29)

CSP ... a nervous habit, a rule of action... (2.711)

H What?

CSP ... some plants take habits. The stream of water that wears a bed for itself is forming a habit. Every ditcher so thinks of it. (5.492)

NN The exam paradigm dates back to Ancient Greece. [Facing H:] You have offered nothing new here.

CSP In the doctrine of *obligationes*, in logic, 'pertinent' is applied to a proposition whose truth or falsity would necessarily follow from the truth of the proposition to which it was said to be pertinent, and also of a term either necessarily true or necessarily false of another term to which it was said to be pertinent. (2.602)

NN The idea of scholastics. Beyond that, the exam metaphor appears as old as logic itself.

CSP All human research must come to be conducted upon some unitary plan. The pendulum of dispute may swing long; but we must hope it will at last come to rest. To workers for that end this ... is an encouraging signal. For only let exact diagrammatic conceptions, like those of mathematics, once take the place of the vague discourse that has prevailed in modern philosophy since it threw off those wholesome obligations of debate [murmurs:] (which kept the scholastics to precise points and insured their precise arguments' meeting precise criticism). (8.118)

NN If we throw off examinations as unsuitable for semeiotic dialogues, is there something to replace them? We already referred several times to the importance of searching and picking elements.

CSP When the subject is not a proper name, or other designation of an individual within the experience (proximate or remote) of both speaker and auditor, the place of such designation is taken by a virtual precept stating how the hearer is to proceed in order to find an object to which the proposition is intended to refer. If this process does not involve a regular course of experimentation, all cases may be reduced to two with their complications. These are the two cases: first, that in which the auditor is to take any object of a given description, and it is left to him to take any one he likes; and, secondly, the case in which it is stated that a suitable object can be found within a certain range of experience, or among the existent individuals of a certain class. The former gives the *distributed* subject of a *universal* proposition, as, 'Any cockatrice lays eggs'. It is not asserted that any cockatrice exists, but only that, if the hearer can find a cockatrice, to that it is intended that the predicate shall be applicable. The other case gives the *undistributed* subject of a *particular* proposition, as 'Some negro albino is handsome'. This implies that there is at least one negro albino. Among complications of these cases we may reckon such subjects as that of the proposition, 'Every fixed star but one is too distant to show a true disk', and, 'There are at least two points common to all the circles osculating any given curve'. (2.357)

NN We tend to call them generalised quantifiers.

CSP Good! So much the better! (4.78) The subject of a universal proposition may be taken to be, 'Whatever object in the universe be taken'; thus the proposition about the cockatrice might be expressed: 'Any object in the universe having been taken, it will either not be a cockatrice or it will lay eggs'. So understood, the subject is not *asserted* to exist, but it is well known to exist; for the universe must be understood to be familiar to the speaker and hearer, or no communication about it would take place between them; for the universe is only known by experience...

W [interrupts:] ... if you show someone the king in a chess game and say, 'This is the king of chess', you do not thereby explain to him the use of this piece, — unless he already knows the rules of the game except for this last point: the shape of the king piece. We can imagine that he has learned the rules of the game without ever having been shown a real chessman. (226: 20)

CSP Right. The particular proposition may still more naturally be expressed in this way, 'There is something in the universe which is a negro albino that is handsome'. No doubt there are grammatical differences between these ways of stating the fact; but formal logic does not undertake to provide for more than one way of expressing the same fact, unless a second way is requisite for the expression of inferences. (2.357)

W But what is the point in this case of saying that when I describe to myself what I see I describe an object called 'what is seen'? Why talk of a particular object here? Isn't this due to a misunderstanding? (151: 8)

CSP *Index*, which like a pronoun demonstrative or relative, forces the attention to the particular object intended without describing it. (1.369) The index asserts nothing; it only says 'There!' It takes hold of our eyes, as it were, and forcibly directs them to a particular object, and there it stops. (3.361)

W We have thereby given this object a role in our language game, it is now a *means* of description. And the statement: 'If it didn't exist, it could have no *name*', now says as much and as little as: 'If this thing didn't exist, we couldn't use it in our game'. (226: 37)

NN So one searches for objects to fulfil the roles they have in language games.

H Nothing in the logical game corresponds to seeking. (Hodges, 2001b)

W [You] are thinking of a game in which there is an inside in the normal sense. (148: 45v) The expression 'language game' is used here to emphasise that the speaking of the language is part of an activity, part of a way of living of human beings. (226: 15)

H Tarski ... did have a word for a kind of object which satisfies formulas, namely *sequence*. ... We can ask (in English, or in some set-theoretic metalanguage) whether or not some collection of entities (a 'sequence') satisfies a condition expressible in the language. The truth definition in [Tarski's 1935] paper is an inductive definition of this notion of 'satisfying'. (Hodges, 1986, pp. 142–143)

CSP Very well, my obliging opponent, we have now reached an issue. (6.57) True is simply that in cognition which is Satisfactory. As to this doctrine, if it is meant that True and Satisfactory are synonyms, it strikes me that it is not so much a doctrine of philosophy as it is a new contribution to English lexicography. (5.555)

H Suppose we express a law by a formal sentence S, and $\mathbb{A}$ is a structure. Different writers have different ways of saying that the structure $\mathbb{A}$ obeys the law. Some say that $\mathbb{A}$ *satisfies* S, or that $\mathbb{A}$ is a *model* of

S. Many writers say that the sentence S is *true in* the structure A. ... This use of the word *true* seems to be a little over fifty years old. (Hodges, 1986, p. 136) In 1935 there was still no general agreement among logicians about what kind of object a structure or a system is... (Hodges, 1986, p. 138)

CSP ... a 'state of things', ... anything the reality of which might constitute the truth of an assertion. By 'reality' is to be understood that part or ingredient of the being of anything which does not depend upon the thing's actually being represented. (MS 686: 1) Reasoning ... at the very least conceives its inference to be one of a general class of possible inferences on the same model. (MS 845: A7, 1906, *Answers to Questions about my Belief in God*).

H [So, there are] no indexical expressions: there are no demonstrative pronouns or verb tenses or persons. (Hodges, 1986, p. 143)

CSP In the sentence 'Every man dies', 'Every man' implies that the interpreter is at liberty to pick out a man and consider the proposition as applying to him. In the proposition 'Anthony gave a ring to Cleopatra', if the interpreter asks, What ring? the answer is that the indefinite article shows that it is a ring which might have been pointed out to the interpreter if he had been on the spot; and that the proposition is only asserted of the suitably chosen ring. (5.542)

H The non-logical constants of a model-theoretic language are simply indexical expressions. ... There is nothing *sui generis* about truth in a structure. It is the same kind of notion as 'true at the North Pole' or 'true in 1888'. (Hodges, 1986, p. 150)

NN Often the context comes from what is taken to be relevant according to the speaker and the hearer.

CSP If the utterer says 'Fine day!' he does not dream of any possibility of the interpreter's thinking of any mere desire for a fine day that a Finn at the North Cape might have entertained on April 19, 1776. He means, of course, to refer to the actual weather, then and there, where he and the interpreter have it near the surface of their common consciousness. (MS 318: 32–33)

NN In reasoning, to be relevant is to fix the boundaries of models.

CSP Why I did not more simply define reasoning as the passage from judging one state of things to be real to judging another state of things to be real, I should reply that, — to mention only one of reasons, — such simpler forms of statement would overlook an obvious character of all, mathematical reasoning. ... No kind of reasoning concludes that anything ought to be accepted as true in one state of things merely because something is admitted to be true in a quite different state of things. (MS 826: 2–3, c.1880, *The Conception of Infinity*)

NN What about the question about searching?

CSP By directions for finding the Objects, for which I have as yet invented no other word than 'Selectives', I mean such as 'Any' (i.e., any you please), 'Some' (i.e., one properly selected), etc. (8.181)

NN That appears similar to substitution or instantiation, part of what quantification is meant to be.

CSP A precept, which may be called its *quantifier*, prescribes how it [the object] is to be chosen out of a collection, called its universe. (2.339)

NN You seem to be thinking of your concept of a quantifier itself as a rule, or a strategy, for choosing the objects. Perhaps more typically, that is regarded to be a function (which, incidentally, you came pretty close in your 1885 logic...) It is elicited from the concept of quantifiers, and the quantifiers as such are symbols that define only which moves are legitimate and which are not.

CSP Along with such indexical directions of what to do to find the object meant, ought to be classed those pronouns which should be entitled *selective* pronouns — or quantifiers — because they inform the hearer how he is to pick out one of the objects intended, but which grammarians call by the very indefinite designation of *indefinite* pronouns. (2.289)

NN OK. What about the motivation or purpose briefly considered in our very first point of the session?

CSP The pragmaticist has always explicitly stated that the intellectual purport of a concept consists in the truth of certain conditional propositions asserting that if the concept be applicable, and the utterer of the proposition or his fellow have a certain purpose in view, he would act in a certain way. A purpose is essentially general, and so is a way of acting; and a conditional proposition is a proposition about a universe of possibility. At the same time, the conditional proposition refers only to possible individual actions. (5.528)

NN [turning to W:] Aren't you a pragmatist?

W No. For I am not saying that a proposition is true if it is useful.[2] (131: 70)

H Maybe it's a fact of human psychology that we can think about an abstract process more flexibly and reliably if we can imagine that it reflects the actions of a person who has a target and is following some rules for getting to it. (p.27)

CSP Be careful not to confound a proof which needs itself to be experienced with one which requires experience of the object of proof. (3.35) [Here you run the risk of] making logic a question of psychology. But this I deny. Logic does rest on certain facts of experience among which are facts about men, but not upon any theory about the human mind or any theory to explain facts. (5.110) Psychology must depend in its beginnings upon logic, in order to be psychology and to avoid being largely logical analysis. If then logic is to depend upon psychology in its turn, the two sciences, left without any support whatever, are liable to roll in one slough of error and confusion. (2.51) Kant holds that psychology has no influence upon logic. (2.39)

NN What about models here, diagrammatically speaking?

CSP Diagrammatic reasoning is the only really fertile reasoning. If logicians would only embrace this method, we should no longer see attempts to base their science on the fragile foundations of metaphysics or a psychology not based on logical theory; and there would soon be such an advance in logic that every science would feel the benefit of it. (4.571)

H [Many have claimed] that our normal mode of deductive reasoning is proof by cases; that we represent the cases by what [one] calls 'models'... (Hodges, 1998, p. 10)

CSP [As I said], we imagine cases, place mental diagrams before our mind's eye, and multiply these cases, until a habit is formed of expecting that always to turn out the case, which has been seen to be the result in all the diagrams. ...Why may not all our natural judgments as to what is good reasoning be founded on habits formed in some such ways? If it be so, the German doctrine falls to the ground; for to form a notion of right reasoning from diagrams showing what will happen, is to form that notion virtually according to the English doctrine of logic, by reasoning from the nature of things. (2.170) All valid necessary reasoning is in fact thus diagrammatic. (1.54)

H [It has been argued] that we make deductions not by applying rules of inference to representations of the logical forms of our premises but by a process which involves building mental models of the premises and searching among them for counterexamples to the conclusion. (Hodges, 1993, p. 353)

CSP *Deduction* is that mode of reasoning which examines the state of things asserted in the premisses, forms a diagram of that state of things, perceives in the parts of that diagram relations not explicitly mentioned in the premisses, satisfies itself by mental experiments upon the diagram that these relations would always subsist, or at least would do so in a certain proportion of cases, and concludes their necessary, or probable, truth. (1.66)

H The mental-model theory of deduction has a pictorial quality which many people have found appealing and inspiring. (Hodges, 1993, p. 353)

CSP If ... as the English suppose, the feeling of rationality is the product of a sort of subconscious reasoning — by which I mean an operation which would be a reasoning if it were fully conscious and deliberate — the accompanying feeling of evidence may well be due to a dim recollection of the experimentation with diagrams. (2.172)

NN In mental-models theories, I believe it was never claimed that deduction is in any close relation with mathematical model theory, although one may find some commonalities.

H [mumbles:] ...(they are not what model theorists call 'models')... (Hodges, 1998, p. 10)

CSP We form in the imagination some sort of diagrammatic, that is, iconic, representation of the facts, as skeletonized as possible. The impression of the present writer is that with ordinary persons this is always a visual image, or mixed visual and muscular; but this is an opinion not founded on any systematic examination. If visual, it will either be geometrical, that is, such that familiar spatial relations stand for the relations asserted in the premisses, or it will be algebraical, where the relations are expressed by objects which are imagined to be subject to certain rules, whether conventional or experiential. This diagram, which has been constructed to represent intuitively or semi-intuitively the same relations which are abstractly expressed in the premisses, is then observed, and a hypothesis suggests itself that there is a certain relation between some of its parts — or perhaps this hypothesis had already been suggested. In order to test this, various experiments are made upon the diagram, which is changed in various ways. This is a proceeding extremely similar to induction, from which, however, it differs widely, in that it does not deal with a course of experience, but with whether or not a certain state of things can be imagined. Now, since it is part of the hypothesis that only a very limited kind of condition can affect the result, the necessary experimentation can be very quickly completed; and it is seen that the conclusion is compelled to be true by the conditions of the construction of the diagram. This is called 'diagrammatic, or schematic, reasoning'. (2.778)

All deductive reasoning, even simple syllogism, involves an element of observation; namely, deduction consists in constructing an icon or diagram the relations of whose parts shall present a complete analogy with those of the parts of the object of reasoning, of experimenting upon this image in the imagination, and of observing the result so as to discover unnoticed and hidden relations among the parts. (3.363)

NN There was also the pertinent question of the target and following a rule to get at it.

W What we call a rule of a language game can play very different roles in the game. Think of the sort of cases in which we say that a game is played according to a particular rule. The rule may be an aid to instructing people in the game. The pupil is told the rule and is trained to apply it. Or it is an implement of the game itself. Or: a rule is used neither in teaching the game nor in the game itself; nor is it laid down in a book of rules. You learn the game by watching how others play it. (226: 38)

CSP That is an example of following a general rule. (3.341)

W But what does it mean to follow the rule *correctly*? ... How and when is it to be decided which at a particular point is that which is in accordance with the rule as it was *meant*, intended? ... Would you say that there was only one act of meaning, which, however, all these others, or any one of them, followed in turn? But isn't the point just: 'what does follow from the general rule?' You might say, 'Surely I knew when I gave him the rule that I meant him to follow up 100 by 101'. But here you are misled by the grammar of the word 'to know'. ... Or do you mean by knowing some kind of disposition ... (310: 100)

CSP If I may be allowed to use the word 'habit', without any implication as to the time or manner in which it took birth, so as to be equivalent to the corrected phrase 'habit or disposition', that is, as some general principle working in a man's nature to determine how he will act... (1.270)

W — then only experience can teach us what it was a disposition for. (310: 101)

CSP In all kinds of construction the forms must depend, in large part, upon the qualities of the available material. [A man's] catalogue of these kinds of objects might run somewhat as follows: such objects as I may see, hear, or know by touch or pressure; tastes and smells; bodily sensations; emotional feelings; recollections; imaginations; feelings of comparison and, more generally, additional feelings arising upon the assemblage of other feelings; pleasures and impulses of attraction; pains, irritations, and impulses of repulsion; efforts and impulses of effort; resistances and impulse to resist; attentions and efforts to attend; distractions and efforts to dismiss; perceptions of resemblance, contrast, and modes of order, senses of such perceptions and of coming to perceive or understand new modes of order or new elements of resemblance and contrast; recollections or recalls of accompanying circumstances and senses of recollecting; expectations and senses of expecting; senses of having forgotten; surprises, positive and negative; efforts of self-restraint; effort to impress rules upon oneself and senses of being so impressed and of assimilations and assents; efforts to follow rules; sympathetic feelings and senses of sympathizing; senses of puzzle... (MS 611: 21–22)

NN OK, point taken.

CSP [In the argument, 'Men are sinners, and sinners are miserable; ... men are miserable',] 'Sinners are miserable', must be a Rule without exceptions. That is, it says in effect, if you take any sinner, you will find he is miserable. The second person appropriately expresses it, because there is a second premiss which draws attention to certain sinners, and virtually picks them out. If the rule has exceptions, all I can say is, that if you let me pick out the sinner he will turn out miserable. If I guarantee to find a miserable sinner, of course, I guarantee there is a sinner in the world. But if I turn the responsibility of picking out the sinner to you, I do not guarantee you can find one. I only say if you do find one, he will turn out miserable. This is the distinction between Universal and Particular propositions. (2.453)

W We seem to recognise mathematical truth by experience before we can prove them. (161: 9r)

NN To quote one of your own examples, say one finds that $86 \times 91 = 7826$?

W In what cases are we to say this? Only when he's right or also when he is wrong? What about saying: he has arrived at a rule, — which he now acknowledges (as we all do). (161: 9r)

CSP That a habit is a rule active in us, is evident. (2.643)

W We must distinguish between what one might call a 'process' — being *in accordance with* a rule, and, 'a process involving a rule'. (309: 19)

NN Say, rules defining legitimate moves versus habitual ones... We seem to be arriving at some consensus here...

H [irascibly:] There was no point where games emerge as any more than a psychological convenience in analysing other concepts. (p.31)

NN [turning to CSP:] I recall you having written on the idea of 'play of musement' — something related to the aesthetics of logic?

CSP There is no kind of reasoning that I should wish to discourage in Musement; and I should lament to find anybody confining it to a method of such moderate fertility as logical analysis. Only, the Player should bear in mind that the higher weapons in the arsenal of thought are not playthings but edge-tools. In any mere Play they can be used by way of exercise alone; while logical analysis can be put to its full efficiency in Musement. So, continuing the counsels that had been asked of me, I should say, 'Enter your skiff of Musement, push off into the lake of thought, and leave the breath of heaven to swell your sail. With your eyes open, awake to what is about or within you, and open conversation with yourself; for such is all meditation'. It is, however, not a conversation in words alone, but is illustrated, like a lecture, with diagrams and with experiments. (6.461)

NN Is that conception driven by purpose?

CSP [As I said], it is pretty generally admitted that logic is a *normative* science, that is to say, it not only lays down rules which ought to be, but need not be followed; but it is the analysis of the conditions of attainment of something of which purpose is an essential ingredient. It is, therefore, closely related to an art; from which, however, it differs markedly in that its primary interest lies in understanding those conditions, and only secondarily in aiding the accomplishment of the purpose. (1.575)

H Existentially closed groups don't have any particular purpose, they just exist. (p.26)

CSP Secondness is the mode of being of that which is such as it is, with respect to a second but regardless of any third. (8.328) He who wills has a purpose; and that idea of purpose makes the act appear as a *means* to an end. Now the word *means* is almost an exact synonym to the word *third*. It certainly involves Thirdness. Moreover, he who wills is conscious of doing so, in the sense of *representing* to himself that he does so. But representation is precisely genuine Thirdness. You must conceive an instantaneous consciousness that is instantly and totally forgotten and an effort without purpose. (1.532)

The third category — the category of thought, representation, triadic relation, mediation, genuine thirdness, thirdness as such — is an essential ingredient of reality, yet does not by itself constitute reality, since this category (which in that cosmology appears as the element of habit) can have no concrete being without action. (5.436)

The meaning of a word really lies in the way in which it might, in a proper position in a proposition believed, tend to mould the conduct of a person into conformity to that to which it is itself moulded. Not only will meaning always, more or less, in the long run, mould reactions to itself, but it is only in doing so that its own being consists. For this reason I call this element of the phenomenon or object of thought the element of Thirdness. It is that which is what it is by virtue of imparting a quality to reactions in the future. (1.343)

W Let's not be mislead by imagining meaning as an occult process or relation, nay, interaction, between a word a mind and a thing which contains the whole usage of the word as a seed might be said to contain the future tree. (147: 42r)

CSP I fully admit that there is a not uncommon craze for trichotomies. I do not know but the psychiatrists have provided a name for it. If not, they should. 'Trichimania', unfortunately, happens to be preëmpted for a totally different passion; but it might be called triadomany. I am not so afflicted; but I find myself obliged, for truth's sake, to make such a large number of trichotomies that I could not [but] wonder if my readers, especially those of them who are in the way of knowing how common the malady is, should suspect, or even opine, that I am a victim of it. But I am now and here going to convince those who are open to conviction, that it is not so, but that there is a good reason why a thorough student of the subject ... should be led to make trichotomies, that the nature of the science is such that not only is it to be expected that it should involve real trichotomies, but furthermore, that there is a cause that tends to give this form even to faulty divisions, such as a student, thirsting for thoroughness and full of anxiety lest he omit any branch of his subject, will be liable to fall into. Were it not for this cause, the trichotomic form would, as I shall show, be a strong argument in confirmation of the reasoning whose fruit should take this form. (1.568)

— [pause] —

NN Let's turn — briefly, our time's running up — to the question of vagueness in logic.

CSP Logicians have been at fault in giving Vagueness the go-by, so far as not even to analyze it. (5.446) It should never be forgotten that our own thinking is carried on as a dialogue, and though mostly in a lesser degree, is subject to almost every imperfection of language. (5.506)

NN What would a 'vague game' be like?

H It was open to a teacher of obligationes to discover new rules.

W 'It — surely — isn't a game if there is a vagueness in the rules'. — But *isn't* it a game? — 'Well, perhaps you'll call it a game, but anyway it isn't a perfect game'. That's to say it is then impure, and I am interested in the pure article. But what I wish to say is: you're misunderstanding the rôle which the ideal plays in your mode of expression. That is, you too would call it a game; only you're dazzled by the ideal and therefore...

H This open-endedness implies that obligationes are not logical games. (Hodges, 2001b)

W ... you don't see clearly the real application of the word 'game'. (226: 70)

CSP A sign (under which designation I place every kind of thought, and not alone external signs), that is in any respect objectively indeterminate (i.e., whose object is undetermined by the sign itself) is objectively *general* in so far as it extends to the interpreter the privilege of carrying its determination further. ... A sign that is objectively indeterminate in any respect is objectively *vague* in so far as it reserves further determination to be made in some other conceivable sign, or at least does not appoint the interpreter as its deputy in this office. ... Every utterance naturally leaves the right of further exposition in the utterer; and therefore, in so far as a sign is indeterminate, it is vague, unless it is expressly or by a well-understood convention rendered general. Usually, an affirmative predication covers *generally* every essential character of the predicate, while a negative predication *vaguely* denies some essential character. (5.447)

W Refrain to write down any hypothesis and any vague general statement and you have made a philosophical investigation. (155: 40v)

CSP That's right. In another sense, honest people, when not joking, intend to make the meaning of their words determinate, so that there shall be no latitude of interpretation at all. That is to say, the character of their meaning consists in the implications and non-implications of their words; and they intend to fix what is implied and what is not implied. (5.447)

W One may say that the 'object' I am inclined to say I am pointing to in the ostensive definition is not determined by the act of pointing but by the use I make of the word defined. (151: 12)

CSP In so far as the implication is not determinate, it is usually left vague; but there are cases where an unwillingness to dwell on disagreeable subjects causes the utterer to leave the determination of the implication to the interpreter; as if one says, [pointing out:] 'That creature is filthy, in every sense of the term'. (5.447)

W There can't be a vagueness in logic — we wish to say. We live now in the idea: the ideal '*must*' be found in the real world.

CSP The incidental remark [I made] to the effect that words whose meaning should be determinate would leave 'no latitude of interpretation' is more satisfactory, since the context makes it plain that there must be no such latitude either for the interpreter or for the utterer. (5.448ff)

W Then after all you are thinking of games played with others though you left a certain latitude... (149: 52)

CSP The latitude of interpretation which constitutes the indeterminacy of a sign. [It] must be understood as a latitude which might affect the achievement of a purpose. For two signs whose meanings are for all possible purposes equivalent are absolutely equivalent. This, to be sure, is rank pragmaticism; for a purpose is an affection of action. (5.448ff)

NN So we might say that thirdness is even more prominent in logic of vagueness that in necessary reasoning.

CSP *Thirdness* [is] a concept without which there can be no suggestion of such a thing as *logic*, or such a character as *truth*. (4.332)

H Lorenzen's dialogue games do not give us any access to logical necessity, beyond what we already had from the study of formal proofs. But a winning strategy for [the deliverer] is a different form of proof from those in the standard logical calculi, and this is a welcome innovation. (p.30)

W We may say of some philosophizing mathematicians that they are obviously not aware of the difference between the many different usages of the word 'proof'; and that they are not clear about the difference between the uses of the word 'kind', when they talk of kinds of numbers, kinds of proofs, as thought — sorry — as though the word 'kind' here meant the same thing as in the context, 'kinds of apples'. (309: 45)

CSP Agreed. You cannot criticize what you do not doubt; although very many philosophers deceive themselves and others into the belief that they are criticizing what they hardly pretend to doubt, and so 'argue' for foregone conclusions. A proof or genuine argument is a mental process which is open to logical criticism. (2.26)

H Well, there are of course real-life logical contests, where two people try to out-argue each other. One can try to formalize the accepted rules of these contests. Ummm... I see no connection between them and the games we have been considering, and certainly none with the games of Lorenzen. (p.30)

CSP An *incomplete* argumentation is properly called an *enthymeme*, which is often carelessly defined as a syllogism with a suppressed premiss, as if a sorites, or complex argumentation, could not equally give an enthymeme. The ancient definition of an enthymeme was 'a rhetorical argumentation', and this is generally set down as a second meaning of the word. But it comes to the same thing. By a rhetorical argumentation was meant one not depending upon logical necessity, but upon common knowledge as defining a sphere of possibility. Such an argument is rendered logical by adding as a premiss that which it assumes as a leading principle. (2.449)

H It is not open to [Lorenzen] to answer ... that by 'a claim that A' he means a claim to have a proof that A. His whole intention was to build up a logic from the notions involved in a *pre-logical* Redepraxis. Whatever these notions are, they surely don't include that of a proof. (p.25)

W How do I know that I see and that I see red? ...We use the word 'seeing' and 'red' between us in a game we play with one another. (148: 39r)

CSP There is a much more general doctrine to which the name theory of cognition might be applied. Namely, it is that speculative grammar, or analysis of the nature of assertion, which rests upon observations, indeed, but upon observations of the rudest kind, open to the eye of every attentive person who is familiar with the use of language, and which, we may be sure, no rational being, able to converse at all with his fellows, and so to express a doubt of anything, will ever have any doubt...

W [cuts in:] What is the difference between the report, or assertion, 'five slabs.', and the command 'five slabs!'?

NN I suppose at least their context.

W It is the role which the utterance of these words plays in the language game. (226: 13)

CSP Now, proof does not consist in giving superfluous and superpossible certainty to that which nobody ever did or ever will doubt, but in removing doubts which do, or at least might at some time, arise. A man first comes to the study of logic with an immense multitude of opinions upon a vast variety of topics; and they are held with a degree of confidence, upon which, after he has studied logic, he comes to look back with no little amusement. There remains, however, a small minority of opinions that logic never shakes; and among these are certain observations about assertions. The student would never have had a desire to learn logic if he had not paid some little attention to assertion, so as at least to attach a definite signification to assertion. So that, if he has not thought more accurately about assertions, he must at least be conscious, in some out-of-focus fashion, of certain properties of assertion. (3.432)

W Language games are the forms of language with which a child begins to use of words. The study of language-games is the study of primitive forms of language or primitive languages. If we want to study the problems of truth and falsehood, of the agreement and disagreement of propositions with reality, of the nature of assertion, assumption, and question, we shall with great advantage look at primitive forms of language in which these forms of thinking appear without the confusing background of highly complicated processes of thought. (309: 26)

CSP Every assertion has a degree of energy. (MS L 75: 324)

NN So in assertions, not unlike in speech acts but perhaps in a more general form, we find the kind of Redepraxis craved for in logical dialogues. Does that hold in mathematics, too?

CSP Mathematics performs its reasonings by a *logica utens* which it develops for itself, and has no need of any appeal to a *logica docens*; for no disputes about reasoning arise in mathematics which need to be submitted to the principles of the philosophy of thought for decision. (1.417)

NN What is the relation of this to logic?

CSP This constitutes a theory of logic: the scholastics called it the reasoner's *logica utens*. Every reasoner whose attention has been considerably drawn to his inner life must soon become aware of this. (2.186)

NN So logic, or its *utens*, comes from, or is performed via, 'pre-logical', dialogical, or sign-theoretical processes we call games, but we should then acknowledge that the rules that define its legitimate moves are secondary?

CSP The purpose of reasoning is to proceed from the recognition of the truth we already know to the knowledge of novel truth. This we may do by instinct or by a habit of which we are hardly conscious. But the operation is not worthy to be called reasoning unless it be deliberate, critical, self-controlled. In such genuine reasoning we are always conscious of proceeding according to a general rule which we approve. It may not be precisely formulated, but still we do think that all reasoning of that perhaps rather vaguely characterized kind will be safe. This is a doctrine of logic. (4.476) Reasoning does not begin until a judgment has been formed; for the antecedent cognitive operations are not subject to logical approval or disapproval, being subconscious, or not sufficiently near the surface of consciousness, and therefore uncontrollable. (2.773)

W Imagine a person crying out with pain alone in the desert: is he using a language? Could we say that his cry had *meaning*? (151: 29)

CSP If nobody takes the trouble to study the record, there will be no interpreter. So the books of a bank may furnish a complete account of the state of the bank. It remains only to draw up a balance sheet. But if this be not done, while the sign is complete, the human interpreter is wanting. (MS 318; EP 2:404)

W How you say — what we call the meaning of the word lies in the game we play with it. (149: 18)

NN We noted earlier that the rules that one develops in playing such games presuppose experience in order for one to recognise what the rules, or dispositions, are to be rules or dispositions for.

W The word must have a *family* of meanings. Compare: *knowing* and *saying*, how many feet high Mont Blanc is — how the word 'game' is used — what a clarinet sounds like. (226: 55)

CSP Aristotle, ...whose system, like all the greatest systems, was evolutionary, recognized besides an embryonic kind of being, like the being of a tree in its seed, or like the being of a future contingent event, depending on how a man shall decide to act. In a few passages Aristotle seems to have a dim aperçue of a third mode of being in the entelechy. The embryonic being for Aristotle was the being he called matter, which is alike in all things, and which in the course of its development took on form. Form is an element having a different mode of being. The whole philosophy of the scholastic doctors is an attempt to mould this doctrine of Aristotle into harmony with christian truth. This harmony the different doctors attempted to bring about in different ways. But all the realists agree in reversing the order of Aristotle's evolution by making the form come first, and the individuation of that form come later. Thus, they too recognized two modes of being; but they were not the two modes of being of Aristotle. (1.22)

My view is that there are three modes of being. I hold that we can directly observe them in elements of whatever is at any time before the mind in any way. They are the being of positive qualitative possibility, the being of actual fact, and the being of law that will govern facts in the future. (1.23)

— [end] —

Coda

That was our dialogue, establishing, if you dissected it well, the points (i)–(vi) laid out in the introduction. In closing, let me adjoin to the caveat mentioned at the conclusion another fragment from Peirce's dialogue:

You Haven't you written this [dialogue], and was it not you that put the objection into my mouth, and then made believe to be surprised at its force, merely in order to render your triumph over it more striking. What am I, at all, but a puppet of your fabrication, — a puppet cat with whose paw you delight to pull your hot chestnuts out much less for the poor nutriment they afford than for the cruel sport of forcing me to take them.

I You shall be convinced that I was not making fun of you, by any means, although I will not promise not to make fun of your declaration (which you don't believe) that you are only a figment of my creation and at the same time that you are an Idea, both of which are decidedly funny. On the contrary, you are a living man. . . . The plan of setting forth philosophical doctrines in the form of a dialogue involves no such insincerity as you seem to think it does. . . . All the great philosophical dialogues . . . either narrate actual dialogues, or compress into one a number of such actual conversations. The different speakers were intended to represent as many different ways of thinking that were current in the writer's time; but in fact the dialogues were no doubt reminiscences of conversations in which the writer had taken part, filled out, where they must be, by his understanding of what different types of minds would have replied to certain questions. They are, therefore, historical records intended to be veracious. (MS 612: 24–26, 11 November 1908)

Notes

1 For reasons of space, I will suppress the titles to which the quotations refer.

2 "Nein. Denn ich sage nich, der Satz sei wahr, der nützlich ist".

Chapter 10

GAMES AS FORMAL TOOLS VERSUS GAMES AS EXPLANATIONS

WE FACE HERE AND NOW a really conceptual — and not merely technical — difficulty. And it is this problem which the theory of "games of strategy" is mainly devised to meet.
— John von Neumann & Oskar Morgenstern, *Theory of Games and Economic Behavior*, 1944.

This chapter addresses the theoretical notion of a game as it arises across scientific inquiries, exploring its uses as a technical and formal asset versus an explanatory mechanism in logic and science. Games comprise a widely-used method in a broad intellectual realm (including, but not limited to, philosophy, logic, mathematics, cognitive science, artificial intelligence, computation, linguistics, physics and economics). However, each discipline advocates its own methodology and a unified understanding is lacking. The first part of this chapter critically surveys a number of game theories as set out in formal studies. In the second part, the doctrine of games as explanations for logic is assessed, and the relevance to cognitive faculties is discussed. It is suggested that the notion of evolution plays a part in the game-theoretic concept of meaning.

1. Introduction

Throughout the latter part of the 20th century, game-theoretic concepts have prospered in scientific spheres that have embraced formal methods, including economics and econometrics, logic and argumentation theory, linguistics, mathematics, computation, physics and biology. The social sciences and the philosophy of science have also long shared a formal methodology with games at the focal point. Moreover, games lie broadly within the objects of study in cognitive science and artificial intelligence.

Games are one of the oldest paradigms in the study of cognition and reasoning, going back to the art of argumentation in Aristotle's Topics and Socratic *elenchus*. To date, however, this paradigm has not been fully understood. One reason for this is the alleged superficiality of game-related terminology: as such terminology is often dispensable, one is inclined to use less abstract or better-understood notions. The other reason is that games constitute a fairly loose category of formal techniques: applied to divergent problems, they may have remarkably dissimilar characteristics and only some minimal set of common elements, such as a universe, positions, well-defined move rules, winning and losing conventions, and strategies. To borrow a description of category theory in mathematics, games run the risk of becoming 'abstract nonsense' with a frail theoretical status and only a minor importance in their own right.

This chapter is intended to support the conclusion that such a fear is an illusion. An example is provided by logic, in which the notion of a game has found a home both in the foundations and as a technical tool. Game-theoretic concepts have frequently been resorted to when traditional methods have not easily applied, a case in point being infinitary languages in which the truth of formulas with infinite iterations of quantifiers are customarily analysed with the aid of games. Another example comprises the logics that encode informational independencies between logical components in their formulas (Chapter 7). Such formulas are customarily associated with games of imperfect information.

Hence, games provide an interpretational device whenever traditional semantics turn out to be inadequate. Moreover, as is customary in economics, games of imperfect information are needed to understand decisions in concrete situations, buttressed by the pervasive notion of uncertainty across areas such as decision theory, artificial intelligence, cognitive science, economics and psychology.

This reason is supported by the fact that the generality of games is, by and large, due to their Gestalt, non-compositional nature. Since they evaluate logical statements in an outside-in, top-down manner, starting from the outermost ingredient and ending with the atomic components, the contextual information is permitted to reach inner evaluation points. This is something that was recognised by Peirce when he coined the endoporeutic method for his diagrammatic reasoning in EGs (Chapter 4). Although it is known that some sense of compositionality may be restored for any language with a finitely generated syntax by a modicum of algebraic tinkering, games provide more simple and more natural semantics in interpreting complex context-dependent expressions, not to mention discourse that exceeds sentence boundaries.

The other reason for resorting to games is methodological: they guide us towards a deeper understanding of the concepts and activities involved in cognitive reasoning processes. For instance, they may provide explanations for the existence of such processes. This shows an attitude towards games not only as a tool, but also as a scientific viewpoint and methodology. The methodological stance was been spelled out by Hintikka (1973a) in relation to GTS, regarding games as an interactive, dynamic evaluation method assigning meanings to logical expressions, equating the existence of winning strategies with the notion of truth. This contrasts with many other game concepts in logic and computation, in which strategies are usually associated with proofs or state transitions of various kinds. The field of games in computation, which is aimed at investigating issues such as automata and semantics for linear logic and a wide range of programming languages, still lacks a unifying methodology that could be paralleled with that of GTS.

The primary aim in the first part of this chapter is to present a selective survey of games in formal studies, adjoined with some critical remarks. The doctrine of games as explanations is assessed in the second part. To save space, and to set the focus within the confines of the present study, the discussion in the latter part is confined to the theory of semantic games for logic.

2. Game diversity in science and formal studies

Introduction This section provides as an informal survey of game-theoretic methods in the context of formal studies, interspersed with some remarks and suggestions for further research. Because the range of material is vast, only a highly selective inspection is provided here. For alternative expositions, see van Benthem (1988); Hintikka & Sandu (1997).

One possible categorisation of different kinds of games is as follows.

1 *Semantic games* (GTS): Material truth and the language-world relationship (philosophy, linguistics, logic, semeiotics, cognitive science).
2 *Proof games*: dialogue games (mathematics, linguistics, computation, physics); game semantics, interaction games, ludics (computation); obligation games, negotiation games (argumentation). These are for logical truth, validity of propositions, arguments and computational tasks.
3 *Comparison games*: Ehrenfeucht–Fraïssé (back-and-forth) games, bisimulation, game quantifiers, Diophantine games, resolution games, infinitary games, enumeration games (mathematics, computation). These are for model-theoretic purposes.
4 *Division / Building games*: algebraic games, graph games, combinatorial games, the Vapknis–Chervonenkis dimension (machine learning, mathematics, computation); ultimatum games, bargaining (economics).
5 *Linguistic games / Games of actual language users*: optimality-theoretic, conversational and communication games (linguistics, semeiotics, psychology).
6 *Games of scientific inquiry*: interrogative games, experimentation (philosophy of science, physics).

Games are a helpful thing to have since they provide versatile tools for a broad intellectual realm. The approaches are vastly different, however. The common denominator is typically just some minimal set of concepts such as players, universes, positions, rules, actions and strategies.

Before looking at these categories in more detail, I will set the development of game ideas in formal studies in a historical perspective.

On the development of game-theoretic ideas The analytical and formal use of games is certainly not a twentieth-century invention. There are places in Aristotle's Topica VIII in which he discusses dialectical situations and duties that participants must respect.[1] The set up in dialectical situations involves the Answerer, who must defend his or her thesis or *positum*, and the Questioner, who tries to make the Answerer change his positum, that is, to grant the opposite of the thesis.

An anonymous text, *Abbreviatio Montana*, written in the middle of the twelfth century, describes the art of dialectic as follows (Kretzmann & Stump, 1988, p. 40):

> In order to discern the purpose of this art, you have to know that there are two practitioners of the art. Who are they? There is one who acts on the basis of the art, who disputes in accordance with the rules and precepts of the art, and he is called a dialectician — i.e., a disputant. The one who acts in a way that concerns the art is the one who teaches the art and expounds its rules and precepts, and he is named either a master or an expositor (*demonstrator*). And so we ascribe different purposes in association with the different practitioners.

The text goes on to describe the (i) purpose, (ii) function, (iii) subject matter, and (iv) termination of both participants' activities. For the Disputant, they are (i) to prove on the basis of readily believable arguments a thesis that has been proposed, (ii) to dispute properly in keeping with the rules and precepts of the art, (iii) the proof of a thesis, which is the central issue of the dialectical disputation, (iv) to induce belief in the proposed question. For the Expositor, they are (i) to teach the art, (ii) to expound the rules and precepts of the art and to add new ones, (iii) to put forth utterances and to discover the things signified by the utterances, (iv) to discover and judge reasons for the induced beliefs.[2]

Many commentaries on Aristotle's Topics and on the concept of participant roles in dialectical situations were written in medieval and scholastic times. According to these commentaries, Aristotle's ideas were inherited and further developed in a later scholastic period, when there was interest in post-reformation disputations called *ars obligatoria*.[3]

The writings on *obligatio* contain a source of ideas on game-like situations. The Opponens (the Expositor) attacks a thesis defended by the Respondens (the Disputant). The Respondens then has at least two duties: first, he must grant the thesis in the sense that whatever seems to be true of it must be defended. Second, whatever seems to follow from what the Respondens has already granted must also be defended. It might thus happen that the Respondens has to defend a false *positum*. In this case, he would have the new duty of trying to keep his answers consistent. This is an interesting feature of *ars obligatoria*, for in such *position falsa* the Respondens may still survive by keeping the set of answers free of contradictions.[4]

Another way of looking at what goes on here is that, even if the Respondens cannot maintain the *positum*, in other words even if he or she does not have winning strategies for granting the truth of it, the game is not a dead end. For the *positum* can still be defended, since there might not be enough information to judge its status, or it might remain merely a *possibilia* (or *compossibilia*, to follow the terminology of medieval writers).

Agostino Nifo's (Augustinus Niphus) (1470–1538) textbook on logic Dialectica Ludicra ('Playful Dialectics') obligation games, especially the chapter *De obligationibus*, advocated the use of games of obligation in teaching logic, or the art of dialect.[5] Ashworth (1976) has studied the work, but does not focus on the chapter on obligations, which is in the main on dialectic disputations inspired by Topica and instructions for formal debates.

According to Nifo, "*Positio* nevertheless differs from *impositio*, since *position* is the case (*casus*) that is possible in reality (*in re*), and that is called false possibility. *Impositio* is a possible case that is posed in language (*in voce*). *Propositum* is ... something which is true as such or through the preceding *position* or *imposition*, or which is false or is doubtful" (Nifo, Dialectica Ludicra: 71r). In the light of this and other remarks in *De obligationibus*, Nifo's idea may be recast in terms of a combination of semantic games and games of consistency maintenance. In other words, the rule Nifo is advocating is that one should establish material truth for *positio*, but be consistent if *impositio* is presented.

Later, a connection between mathematical reasoning and game-theoretic thinking was discovered by Leibniz, who invented the 'epsilon–delta definition' of continuity and explicated it as a game between two players: the idea that prevailed was that one player uses a function value $f(x)$ to bring home the value of his epsilon-move $\in$, while the other offsets it with delta δ about x.

Leibniz was one of the key contributors to the early dawn of game theory, urging his colleagues to develop "a new kind of logic, concerned with degrees of probability, [...] to pursue the investigation of games of chance" (Leibniz, 1981, p. 467). His wider perspective was that the art of invention (or discovery, *inventer*) would be improved, since the human mind "is more thoroughly displayed in games" ["paraissant mieux dans les jeux"] "than in the most serious pursuits" (*ibid.*: 467). What Leibniz had in mind were games with outcomes independent of players' actions, and thus were not truly strategic games.

In the twentieth century, game interpretations of logic were used, at least occasionally if not systematically, by a number of logicians. Skolem introduced what are known as Skolem functions (Skolem, 1920), and one may view them as winning strategies in the relevant logical game. Hilbert made occasional game-related references in his approach to the foundations of mathematics, which inspired Wittgenstein.[6] A concrete connection was noted by Henkin (1961), who argued that the truth of formulas could be characterised by a game in which Skolem functions are viewed as players' winning strategies.[7]

The work of Zermelo (1913) was forward-looking in that it presents the first formal findings on games. Schwalbe & Walker (2001) argue that, in contrast to what the impression is that one gets from the subsequent literature, Zermelo did not anticipate or invent the method of backward induction in his paper.[8] Nor did he prove the existence of winning strategies (determinacy) of the game of chess. Rather, he was interested in what properties a position in a game must have to enable a player, independently of how the opponent plays, to enforce a win in at most n moves. Zermelo's finding was that it never took more moves than there were positions in the game. His ideas were also considered in the less known but closely related works of the early initiators of game theory, Kalmár (1928–9), König (1927) and Steinhaus (1960/1925).

Some subsequent history hereon was covered in Chapter 7.

Games in mathematics Game conceptualisations in mathematics are not usually purely semantic, but pertain to the categories of comparison and cut-and-choose games. They provide game-theoretic tools that are often resorted to in abstract model theory (Barwise & Feferman, 1985). The idea of games in mathematics goes back to infinite two-player perfect-information analytic games (Borel games) introduced by Gale & Stewart (1953), as well as to similarly infinite, two-player perfect-information Banach–Mazur (forcing) games springing from the 1920s, customarily applied in descriptive set theory (Yates, 1976). The imperfect-information versions of analytic games are known as Blackwell games.

Ehrenfeucht–Fraïssé's (first-order) games are based on comparison, by which one can prove equivalences in finite structures. Two players exist, the Spoiler and the Duplicator, and two structures $\mathfrak{A}$ and $\mathfrak{B}$. The Spoiler begins a game by selecting a point in $\mathfrak{A}$, and the Duplicator follows by selecting a point in $\mathfrak{B}$. This is the first innings, and the game continues up to n rounds, alternating in this manner. The Duplicator wins if the substructure of $\mathfrak{A}$ is isomorphic to $\mathfrak{B}$ (in a certain interpretation). The game is finite, and because finite perfect-information two-player games are determined, so are Ehrenfeucht–Fraïssé games. They may be used to show, for instance, that some property is not expressible in first-order logic. These games are widely used in descriptive complexity theory, according to which a suitable complexity measure is hidden in the process of describing some problem in a logical language.

Game-theoretic principles have been used in algebraic logic to define a game in which two players build a countably infinite sequence of graphs labelled by algebraic elements. One player, the Falsifier (the Spoiler), tries to prove that there is no representation of algebra by picking defects in the graph, whereas the other player, the Verifier (the Duplicator), tries to repair the model and to make it act as the correct representation by repairing the defects the Falsifier has found, or else showing that they were not genuine. If she does not manage this, the Falsifier will win, and if she survives the game without losing she will win. Algebraic graph games are determined as well. They are, in fact, instances of the class of cut-and-choose games, in which the Falsifier cuts by picking some element or a subset of elements, and the Verifier makes choices by accepting (repairing) the cut or refuting it.

There are many other instances of cut-and-choose games. For example, a game on a cardinal number κ is played by two players for w rounds.[9] In round n, the Verifier decomposes κ into the union of countably many disjoint sets, of which the Falsifier chooses one.

In the context of infinite graph games, two-person games are played on a finite graph (since there may be cycles). At any time of a play there is a cursor on a graph that measures the position of the players. The players move this cursor alternately from node to node along the edges of the graph. A play consists of infinitely many moves beginning with a move by the Verifier. The winner of an infinite play is determined by the collected set of nodes that have been visited infinitely often as indicated by the cursor. Like most other games in mathematics, infinite graph games are also determined. They are suitable for computational concurrency in which players move along the graph (often a Petri net) in a distributed, asynchronous fashion.

Related to graph games are enumeration games, which are also infinite two-player perfect-information games, in which players enumerate sets of natural numbers in successive rounds. A formula in the language of the lattice of recursively enumerable sets describes a winning condition: one of the players, say the Enumerator, wins if and only if the enumerated sets satisfy the formula.

Moving towards games in computation, a species of cut-and-choose games from the field of formal learning theory known as PAC (Probably Approximately Correct) learning. This involves the notion of a (finite) Vapknis-Chervonenkis dimension (Valiant, 1984), which means that, in the learning environment, the game setting consists of the Learner who is given random examples according to some probability distribution not known to her. The examples are shattered in a space, and her task is to select a hypothesis that would classify with high probability the new examples in a space under the same probability distribution. There is an Answerer whose task is to give answers to the Learner's questions about the expected values of the distances within some tolerance parameter. The tolerance encodes the noise in a learning environment, and so the Answerer has the power to control the amount of disturbance it creates for the Learner. PAC learning may be viewed as a game in which 'You choose an error $\in$ and I choose a confidence δ' (note the commonality with Leibnizian $\in - \delta$ definitions of continuity in calculus). The connection with learning models is that a typical theorem states that a concept is (potentially) learnable if and only if it has a finite Vapknis-Chervonenkis dimension. This notion has also been applied to statistical problems, computational geometry, and complexity theory.[10]

The game approaches in mathematics mentioned here give just a taste of the totality of games in mathematics, albeit perhaps encompassing the most common ones. The variability of all kinds of games was noted many years ago by Ulam (1960, p. 120), who wrote that it was "rather amusing to consider how one can "gamize" various mathematical situations (or perhaps the verb should be "paizise" from the Greek word $\pi\alpha\iota\zeta\epsilon\iota\nu$, to play".

Are games in mathematics symmetric? One criterion in assessing the role of games in formal inquiry is that the more game-theoretic content a theory has, the less asymmetrical it should be. In certain variants of comparison games, some asymmetries seep in. In what are known as Ajtai–Fagin comparison games, some bias exists towards the Duplicator, because he is given a wider range of options to choose from and hence made likelier to win than in the balanced Ehrenfeucht–Fraïssé game. This is done to make some technical results easier to obtain, but the method is likely to increase asymmetry (sect. 3).[11]

Games in computation The notion of games in computer science is much more recent, but no less significant. Typically, they are needed to account for the dynamic and interactive aspects of computation. For the most part, the paradigm has its intellectual roots in the dialogical school, with the winning strategy playing the role of the proof. Indeed, in computer science proof-theoretic games have been adopted almost exclusively.

Games commonly abound in process algebra, communication in concurrent systems, bisimulation (the cousin of comparison games in a modal setting), the characterisation of complexity classes, and multi-agent systems aimed at capturing knowledge and interaction in distributed environments. A useful notion is to have games interpreted as a category in the sense of category theory, to impose various operations on them, and then study these operations.

These theories are not, in fact, based on purely semantic ideas. An application in which the GTS framework is, however, put into practical use is in the construction of knowledge bases (Jackson, 1987). Recent advances have been made in game semantics and full completeness (Hyland & Ong, 2000), defining game and interaction categories in category theory (Abramsky et al., 1996), game semantics for fragments of linear logic, including affine logic (Blass, 1992), and full completeness for multiplicative (Abramsky & Jagadeesan, 1994) and multiplicative-additive fragments (Abramsky & Melliès, 1999) of linear logic, just to mention a few.[12] One of the goals in these enterprises is to have a precise mathematical analysis of a wide range of programming languages.

These are by no means games that share common characteristics. For example, those of Hyland & Ong (2000) dispense with payoffs and conventions of winning or losing. In general, game semantics springs from the idea of 'computation as interaction', a method that provides an explicit representation of the environment. A game becomes a specification of the interaction between the computing System and its Environment. Strategies denote programs as sets of rules telling the System how it should play the game, and the game itself becomes a method of classifying the behaviours of the System. What is notable is that there are varying views concerning the foundations of such an approach. I noted in Chapter 8 that Girard (1998) describes Ludics in a way that admits comparisons with Wittgenstein's dictum: "You can't get behind rules, because there isn't anything behind" (Wittgenstein, 1978, p. 244).

In game semantics, the usual game specification consists of the moves that the Proponent makes, the moves that the Opponent makes, and a set of available (winning) strategies. It is customary to stipulate that the opponent always makes the first move, and that the game then continues by alternating (either strictly or non-strictly) between the players (although in so-called call-by-value games there are variations in how the game might start, and in Abramsky & Melliès (1999)'s parallel games, any of the players may make the first move, or they may move in parallel). In contrast, GTS is a facility for logical languages that derives the meaning of logical constants and delivers truth-values, and thus similar stipulations do not make much sense. Since the structure of the formulas determines the defining conventions of the game, there are no switching conditions in GTS, whereas they are frequently brought into play in game semantics in order to allow the players to move from one subgame to another, and to interleave the subgames into one big game. There are many differences, and so a slight twist in terminology (semantic games/game semantics) is justified.

Games are rife in complexity theory in characterising complexity classes, in understanding the nature of such classes, and in interpreting the complete problems that arise. For example, the complexity class of polynomial space (PSPACE) is characterised by two-person, perfect-information games in which the length of the game is polynomial in the length of the description of the initial position. These games are usually probabilistic, and they relate to provability in fragments of linear logic. Imperfect-information and imperfect-recall games have also been studied (Koller, 1995): non-deterministically exponential NEXP-complete languages are characterised by games in which the Verifier has imperfect recall (see Chapter 7), and the Falsifier has imperfect information and perfect recall. In a complexity setting, a polynomially definable game system for a language L consists of two arbitrarily powerful players and a polynomial-time Referee R with a common input I. The Verifier claims that $I \in L$, and the Falsifier claims that $I \notin L$; the Referee's job is to decide which of these two claims is true.[13] In games in which there is imperfect information for both players, finding the optimal strategy is NP-hard in the size of the game tree. Related characterisations of complexity have given rise to interactive proof systems.

Automata theory is a renowned field with multiple contacts with games. An automaton is a self-operational system that transmits information by state transformations, with relative notions of memory and a differentiation between deterministic and non-deterministic choices. By rewriting an automaton system in a tree-like form one comes close to the notion of the extensive game, which is formally built up of states and transformations between states (but lacking Peircean thirdness, see Chapter 14). The automaton itself could be viewed as a player. Indeed, together with Dana Scott, Michael Rabin became the initiator of the theory of automata by formulating non-deterministic finite state machines that entertain choices. This invention was preceded by Rabin (1957), who considered how game strategies may be computed by non-human beings. Technically, the theory of automata and games connect via monadic second-order logic, and especially with its subfragments, such as modal μ-calculus that allows one to express fixed points, and modal temporal logics expressing tensed notions. These inventions were not far removed from the contemporaneous discovery of possible-worlds semantics.

Although game semantics in computation has been widely researched, there does not seem to be a coherent unifying view of the role of games as applied to problems of processes, interaction, or game semantics for fragments of linear logic. The semantic models are characterised differently, and some doubts have been voiced concerning whether the game terminology is indispensable, for at least linear logic itself has implicit game-like characteristics (Blass, 1994).[14]

Furthermore, some game models seem to make intrepid assumptions. For example, Abramsky et al. (1996) invokes an assumption of bracketing, and the strategies are taken to be history-free (innocent). The bracketing means that the new strategy should be an extension of the old one, and the history-freeness means that the Proponent's choice should depend only on the Opponent's choice, which is the immediately preceding choice. The reasoning is that because the information about whether the arguments of a strategy are used is not visible farther than the immediately succeeding choice, this assumption, together with the bracketing and the determinacy, is sufficient to characterise the space of programming languages. These assumptions may be reasonable in semantics for fragments of linear logic, but from a more general game-theoretic perspective, they cut back the strategic dimensions. One could think of situations in which strategies are created as the game goes on, so that it becomes necessary to change some of them in the middle, or that the long-distance dependencies affect the outcome so that such properties may no longer be ignored.

A different but related issue is that the meaning of an expression is linked to the history or to part of the history of its interaction with other components, no matter how far or beyond the local context or interaction or discourse limit the other components may occur. Game semantics has imperfect information in a restricted, positional sense: unlike the linear logic on which it is targeted, it does not allow all-round information hiding, let alone selective imperfect recall in terms of an occasional forgetting of actions.

On a different plane, one open line of research is in articulating game-theoretic interpretation for logic programming. De Vos & Vermeir (1999) suggest logic programs with choice. Not unlike these are resolution games, in which the Verifier tries to derive an empty clause from an inconsistent set of (Horn) clauses, while the Falsifier tries to prevent this by changing the indices, that is, by changing the order of each resolvent that has been constructed (de Nivelle, 1998). Does imperfect information make sense in logic programming? Reducing the number of argument sets of Skolem functions amounts to imperfect information (Chapter 7), but does it have applications in computing? Hintikka & Sandu (1995) argue that IF logic is a useful platform for modelling parallel processes. There are some problems in associating functions with processes, because they may have more than one output or not be designed to halt and give an output. Alternatively, the communication between processes may destroy independence and reduce the formulas to traditional first-order ones. Nevertheless, this is a plausible account of the more general theme of 'parallel logics for parallel processing' (Pietarinen & Sandu, 2004).

These suggestions may be applied further. Imperfect information exists in parallel logic programming in the following sense. Usually independent AND-parallelism is more complex than OR-parallelism, since conjuncts are constrained in a special way in the former, and one tries to unificate goals in the body of a rule or a query concurrently. The restriction states that the goals may not share variables, since in operational semantics they are bound run-time. However, no information transmits between clauses processed in parallel, and so may lead to inconsistencies. My suggestion is to use identities, as in the formula $\forall y \forall x \exists z \exists u (Cyz \wedge Sxu \wedge z = u) \rightarrow Uxy$ (where Cyz could mean that 'y is a child of z', Sxu that 'x is a successor of u', and Uxy that 'x is an uncle of y'). To resolve this requires exhaustive bookkeeping in the kind of AND-parallel processing typically assumed in the literature. This may be improved by Skolemising the formula and disregarding the unnecessary information: $\forall y \forall x (Cyf_1(y) \wedge Sxf_2(x) \wedge f_1(y) = f_2(x)) \rightarrow Uxy$ (Chapter 7). This simplifies the bookkeeping, for in logic programming, f_1 and f_2 need not depend on all of the input variables for unification purposes.

The computational idea of meaning as a dialogue between the System and the Environment has far-reaching philosophical repercussions, already reflected in Peirce's semeiotic notion of a habit and the association of meaning with the logical interpretant (Part I). On the other hand, the classical theory of games, which is in the main about strategies, intensify the strategic dimensions in game approaches to computation, and contribute to the abduction and implementation of winning strategies, beyond the assertions concerning their existence.

Games and epistemology The classical theory of games in rational decision-making is a natural arena for logical and epistemological exploration. To mention an example, an approach exists — somewhat misleadingly glossed as interactive epistemology — which aims at bringing game theory and epistemic logic (logic of knowledge and belief) closer to each other (Bacharach et al., 1997). The kind of convergence advocated stems from a host of technical findings rather than from any single unifying insight. The motivation comes from the observation that a decision maker's information structure may be partitioned into elements containing states, and could be said to have knowledge of some property of a total state space within an element if there is an event affecting all the states at which this element is contained in another event that corresponds to the property. In the case that the information structure is partitional, the corresponding epistemic logic is evaluated in possible-worlds structure containing equivalence relations between worlds.

The other observation is that common knowledge of rationality, of which Peirce was able to gain glimpses in outlining the account of the common ground for communication (Chapter 2), is assumed in this paradigm to imply backwards induction in extensive games of perfect information, a result that can be proved using possible-worlds structure of epistemic logic.

The concept of common knowledge in the above connection is omnipresent in game theory and logic. However, a question arises from the literature on common knowledge: Why would one want the logics of common knowledge to be complete, to the extent that completeness appears to be the primary motivation for the work in this area? Why recursive axiom sets? As it happens, game theory usually involves much more than just relating syntax and semantics, as there is very little that is actually syntactic in games. It is not even clear that completeness is useful for the overall purpose of logical modelling, for many important logics are not axiomatisable. What would be more absorbing would be to produce results pertaining to categoricity of common knowledge, meaning a unique intended model modulo isomorphism for a given axiom system. Categoricity is attainable even for (semantically) incomplete theories, and so the deductive completeness of an axiom system (being able to prove φ or $\neg\varphi$ for all sentences of a language) is not necessary.

Moreover, a predominance of propositional epistemic logics exists in interactive epistemology, and a lack of more expressive predicate extensions thereof. The relations between first-order logics and game-theoretic problems just have not yet been studied in full. One reason for this is that one does not need more expressive formalisms as far as the usual solution concepts for game-theoretic problems are concerned. However, finding Nash equilibria for common knowledge appears to require predicate extensions of epistemic logic, for instance.

Furthermore, the convergence of game theory and epistemic logic has been unilateral. Mainstream research has concentrated on analysing game-theoretic concepts by using logical tools derived from the tradition of epistemic logics, while the other direction of the convergence, namely the question of what game theory can do for logic has, by and large, been ignored. Moreover, in the context of interactive epistemology, there is the question of how game theory can help in analysing the logics of knowledge and other epistemic attitudes. There are good prospects in embarking on reverse interactive epistemology (Pietarinen, 2005h), which aims at increasing the epistemological relevance in the marriage.

Nevertheless, GTS is appealing for economic theorising. For instance, in a semantic game on the formula of epistemic logic, two teams of players with distinct purposes play a game on a modal structure, and when a single member of the team encounters epistemic operators, he or she chooses some accessible world in which the game is continued. As is known, possible worlds, like any other logically active component, can be hidden from the players in what amounts to semantic games of imperfect information. Within modal or epistemic logics, one need not move to a first-order level, and it is possible to study information flow between attitude operators and connectives. Since each logically active ingredient may be informationally independent of other components, the information associated with one component need not be included within the information set of the player whose move is associated with the component that is independent of the previous one (Chapter 7).

This kind of game-theoretic take on the epistemic logics of imperfect information induces an alternative definition of knowledge. Since the players might be able to demonstrate the truth of some formulas against any choice of a possible world by their opponent, they could also be said to have knowledge at hand (for suppose the Verifier has a winning strategy in the formula $K_i \exists x\, Sx$, given a possible-worlds model, such that the choice of a possible world made by the Falsifier for the epistemic knowledge operator K_i (where i is the knowing agent) is for some reason hidden from her when picking a value for x in $\exists x$). As it happens, the game model may be recast in a possible-worlds fashion, which is useful in tackling the relation between games and the possible-worlds model of knowledge (Pietarinen, 2002c). All in all, in this arena epistemic logic and game theory are likely to meet increasingly often.

Game theory deals with rationality, probability, preferences, beliefs about the players' own actions, beliefs about other players' beliefs, knowledge, preference, and so on. It is contentious whether one should incorporate these notions into semantic games. Moreover, what if strategies themselves are installed among the semantic primitives of the players? What if imperfect information and imperfect recall are applied to strategies, not only to moves? Losing trace of one's own strategies was witnessed by Wittgenstein on remembering the connection between a sign and a sensation at later decision points (Wittgenstein, 1953). Such notions of remembering could have been prompted by his dyslexia (Hintikka & Hintikka, 2002).

Linguistics and games Wittgenstein's language games have served to stimulate work on semantic games in linguistics and the philosophy of language. The application to fragments of natural language was initiated in Hintikka (1973a). Since there are no variables in natural language the key idea is to substitute proper names for quantifier phrases. In this way, various game rules may be defined for fragments of natural languages as for formal languages. Because the domains of natural language are hard to pin down, the choices of the individuals are restricted, at any move, to the sub-domains of a domain subject to quantification. These subsets provide context to the players, their choice sets, and all plays are relativised to these choice sets. In particular, the treatment of anaphora by means of semantic games has attracted attention (Hintikka & Kulas, 1985; Sandu, 1997). These usually evoke the notion of subgames, which are played on the anaphoric head clause (sentence), and the strategies are then carried over to the next subgame, played on the consequent clause containing the anaphoric pronoun.

Other linguistic theories dressed in a game-theoretic outfit include tasks of optimising interpretation. The idea is that the actual interpretation of an expression, given by the Speaker to the Addressee, should be optimal among the totality of candidates ordered by a preference relation, that is, it should deliver a Nash equilibrium (Chapter 11). A helpful way of thinking about candidate interpretations in optimality theory is as complete game histories from which the optimal interpretations leading to equilibria are selected.

Many other kinds of communication, conversation and signalling games abound. Unlike semantic games, these endorse interpersonality played by actual language users. Like semantic games, they usually carry a modicum of strategic (pragmatic) meaning not captured just by an existence claim concerning the winning strategies.

Pietarinen (2002b) presents some new natural-language items to which GTS is applied. These include negative polarity items, adverbs of quantification, generalised quantifiers and eventualities. In the specific context of anaphora, taking the theory of games seriously turns out to be the key. For extensive games encode their total histories, together with the modalities concerning the alternatives to what has actually been chosen in any play. We can, by and large, dispense with the notion of the choice set in previous treatments of anaphora (Hintikka & Kulas, 1985), and rest content with the explicit representation of previous choices.

Unlike previous game theories, this one makes it possible to take care of complex anaphoric sentences too. In that case, the strategies of the players are also coded into an extensive-form representation. This has to be done because previous expositions have left the notion of the player remembering the verification strategies used by the adversary as somewhat informal and heuristic (Pietarinen & Sandu, 2004).

Thus, there are two stages towards a comprehensive theory of anaphora. The first involves obtaining anaphoric information from the histories of the game: subgames and operations on them are defined so that remembering a strategy amounts to the inheritance of assignments from the top node downwards. Second, anaphoric information may be derived from the players' knowledge. In this case, a strategy is remembered if the player's local state contains information about it, and the relevant information sets in the subgame for the anaphoric antecedent are singletons. No separate choice set is needed, as previous choices for functional anaphora may be recovered from the game history and from what is available at the local states of the players. For details of this approach, see Janasik et al. (2002); Pietarinen & Sandu (2004), in which the use of choice sets is dispensed with, and semantic games are viewed in their extensive form expanding the traces of their derivational histories.

Game theories for the philosophy of science and physics Interactive situations have been utilised in the philosophy of science, most notably in the interrogative approach to inquiry (Hintikka, 1999). In particular, many decision-theoretic approaches to ampliative reasoning, which is subservient to the interrogative model, are sensitive to the contextual aspects required in inductive inference. This applies to inductive logic, and also comes out in the related field of inductive inference that is actively pursued in computer science (Muggleton, 1992). The general induction setting for computational purposes is to find a hypothesis, given a set of observations and consistent background knowledge. As far as the problem of justification is concerned, several confirmation theories, such as the maximum information compression of observations, or upper or lower bounds of background theory, have been proposed to constrain the induction problem, but none of them appears fully satisfactory as yet. To sharpen up this connection, Nature's answers to the Inquirer's questions along the lines of the interrogative model could also be used to provide insights into machine-learning theories and automated induction problems. The question answering would bring some additional non-logical factors into the justification of any one hypothesis in the otherwise fairly under-constrained induction problem.[15]

Game theories have every now and then been considered within the physical sciences. Frieden (1998) supports the process of forging physical laws by acts of extracting information measures in a game between the Observer and Nature. The tactics is in accordance with the venerable philosophical tradition of 'putting questions to Nature', which surfaces in the writings on epistemology and the philosophy of science of scholars such as Francis Bacon, Kant, Peirce, Nicholas Rescher and Hintikka. The sense in which Peirce understood the idea of putting questions to Nature was as experimenting on existential graphs. This is natural, given his conviction that special sciences depend on logic. For instance, the object of investigation for a chemist is the molecular structure, and the reactions and relations of samples to it as well as of the identities of samples with one another are acts of experimentation. Likewise, the object of investigation in diagrammatic approaches to logic is the connectivity between different parts of a graph, most notably the identity relations.[16]

Games also arise in dialogical approaches to quantum theory. According to Mittelstaedt (1978), propositions are defined in the object language of quantum physics by series of challenges and defences by two players in a Lorenzen-style dialogue game, in which connectives are defined by the rules of the dialogue. Thus, one could characterise a quantum logic that does not depend on any specific physical hypotheses. As noted, dialogue games are different from semantic games. Mittelstaedt's dialogue games are proof games for formal validity, aiming to show when a proposition can be formally derived from a theory (a set of propositions), whereas semantic games are truth-value games evaluating sentences in a model, aiming at the truth-values of the output sentence.

In order to find out the usefulness of semantic games in the logical foundations of quantum theory, one could look for several historically and systematically noteworthy relations between the two theoretical constructions, quantum logic and the theory of games. Some of these connections are analysed in Pietarinen (2002a).

Communication games and social metaphors The Huizingian slogan 'the games people play' is without doubt part of our folklore. But very little work seems to have been done on experimenting with the idea of semantic games in cognitive science or psychology, or even in the social sciences in which concepts taken from general game theory are otherwise commonplace.[17] Even so, it is a natural idea to think about games in general as involving human aspects of reasoning, such as debate, discussion, (multi-party) communication, social interaction, behaviour in a group, and other interpersonal relations going beyond the traditional realm of logic. Pietarinen (2002e) studies some social metaphors arising in logic. Other connections include social-choice theory and its links to logical issues (Pauly, 2001). The language of team players in Chapter 7 poses some questions, however, involving the nature of decisions in repeated games, for example. If the game were played again according to some formula under uncertainty, would we get the same team members again, making the same choices, and hence arriving at the same notion of truth as in the previous rounds? For example, Binmore (1996) doubts the feasibility of multiple-self team games since people are liable to repeat their mistakes when making decisions in later information sets.[18]

Nor should the game paradigm be confined to the loose metaphor of 'finite versus infinite games' we get to choose in life (Carse, 1986), or relegated to a teaching and learning device in cognitive concept-acquiring tasks that come to light in psychology and the cognitive science of education. Games based on dialogical concepts give an account of how humans might actually reason, while those that are semantic describe how the logical concepts and models are related to each other.

Semantic games are non-cooperative, but social exchanges may be important in the successful division of labour between the participants, as well as in the strategy-formation tasks that predict the other participants' reasoning abilities in following the rules of successful interaction. In the case of non-coherent formulas, one may even need to 'negotiate contradictions away' by initiating a bargaining process of alternating offers that divides the surplus of truth in non-zero-sum semantic games (Pietarinen, 2002d).

3. Game theories as explanations

The double role of games It is clear from the above exposition that a generous toolkit exists to be applied in indefinite fields of formal inquiry. Even though classical semantic-game method has been found to coincide with classical Tarski semantics for first-order logic, the latter is no longer available once we move to logics with different characteristics, such as those with imperfect information or infinitely long expressions. This adduces some independent technical motivation for resorting to games. Even so, there are some general peculiarities. As indicated, the emphasis on the strategic point of view implies that the whole idea of a game as a collection of dynamically evolving plays loses some of its force. This is all the more relevant given that two general approaches exists: extensive games consisting of players, moves, payoffs and positions, or alternatively one-shot strategic games matrices consisting of players' strategies and their outcomes. As far as logic is concerned, the latter description incorporates the domain of a game, which is the semantic pool of resources consisting of a domain of a structure plus all the conjuncts and disjuncts of a given formula. As the game goes on, the supply is adjoined with the names of individuals picked from the domain (at the beginning of the game, the set of names is empty). The set of strategies, in turn, consists of a set of strategies for the Verifier and a set of counter-strategies for the Falsifier.

These descriptions are loosely related to the distinction between extensive and normal (strategic) forms of games. The normal form describes the totality of all decisions packed into a single one, namely the choice of a strategy profile. The game involves only two moves: the choice of a profile by the Verifier and the choice of a counter-profile by the Falsifier. The extensive form, on the other hand, depicts the choices made by the players, together with the payoffs and the informational partitions of players. Translations between the two are non-trivial because they are calculated to encode different information. Usually, at least an exponential amount of information needs to be decoded in the strategies in going from extensive to normal forms, and thus the translation falls short of being tractable. Stalnaker (1999) advanced the view that, even strategically, these two representations may say the same thing after all. In general, such assimilations depend on the solution concepts employed.

Given that there appears to be two differing descriptions of games at hand, can they be reconciled, or are there two ways of viewing the matter? Perhaps a dual vantage point is needed, so that switching between the two perspectives may be complementary. However, unless some fundamental questions of the game description, such as the possible motivations of the players, are discussed, strict comparison is not possible. Is there some underlying general frame of reference or a model that explains why games are a useful thing to have in the first place? Is such a model explicable without reference to game-theoretic concepts? What motivates the players?

These matters are discussed in the sequel. In general, what is argued is that it is not clear what the unifying purpose of players might be, and that different purposes work for different sets of problems and for different applications. In the interest of saving space, the following discussion is confined to the theory of semantic games.

Failures of the explanatory role The reason for using games in relation to logic could be their explanatory power with respect to logical concepts. Whereas the other methods merely describe how the logical concepts function, games additionally explain why they work in the way they do. In other words, they may spell out what the logical activities involved in concepts such as logical constants, truth and falsehood are. In what sense can semantic games be taken to serve as explanations? A portion of Hodges (1997b) is devoted to this question. We glimpsed at some facets in Chapter 9. Here I will focus on the converse case: When and why might the explanatory force of some classes of semantic games fail?

Possible failures are threefold: (i) games are asymmetric and thus unrealistic; (ii) they lack strategic purpose; (iii) they lack motivational purpose. Failure (i) is a matter of how the game rules are defined, whereas (ii) and (iii) concern the plausibility of the motivations in introducing game-theoretic concepts in logic. On the one hand, understanding the strategies of the players may help us to recognise what they are trying to achieve. On the other hand, a proper grasp of players' motivations gives support in operationalising their aims and actions. Yet it is important to realise that (i) is independent of the reasons for failures (ii) and (iii). Indeed, motivation itself does not entail that a game is asymmetric, since asymmetries pertain to the defining rules of a game, and prescribe which moves are legitimate and which are not. In contrast, (ii) and (iii) are parts of the strategic resource, the characteristics of which are not captured at the definitional level.

According to Hodges (1997b, p. 18), it is not until we spell out the purpose that we can use games as a solid foundation in logic. This shares some similarities with van Benthem (1999, p. 16): "But the jury is still out on what should be the fundamental game model for logical activities, and thus, this intriguing alternative approach still awaits its Turing". Presumably the "fundamental game model" is taken to subsume both strategic and defining rules.

A proposal for a possible purpose suggested in Hodges (1997b) is the paradigm of an examination, according to which the Examiner (taken to correspond to the Falsifier) puts questions to the Examinee (the Verifier). The task of the Examinee is to come up with a correct answer to every question. The Examiner might entertain different motivations. To de-anthropomorphisise this suggestion, one could think of 'client–server' or 'user–data resource' architectures in which the Client or User is perfectly honest and only needs some answers or information. The idea in the examination paradigm is nonetheless that the Examinee's role is central. If she finds a right answer to all the questions she is going to win the game, and the winning strategy is then her habit of arriving at the right answer regardless of the difficulties or complexities of the questions the Examiner puts to her.

Again according to Hodges, the failures (ii) and (iii) may be disposed of by resorting to this paradigm, although it should be noted that this explication does not yet cover all types of games. A case in point is the class of games of imperfect information. Here, the motivational breakdown is due to players' lapses into ignorance about which choices were made previously (the previous choices could have been the Opponent's or the Player's own). In the examination paradigm, however, the Examinee's answers are cumulative in that she is supposed to be able to make use of her earlier responses and is expected to produce as good an answer as possible to the questions posed. If there is imperfect information, this requirement is not always fulfilled, since the Examinee might have been prevented from extracting information from her earlier choices. She may not only fail to know what course her Examiner took, but she may also forget her own answers. Thus one is prompted to ask what kind of examination this is to be. The odd moral Hodges draws is that something might be wrong with semantic games for the logics of imperfect information, although the problem equally undermines the examination paradigm.

Whenever there is imperfect information in logic, it is not only the class of games of imperfect information but also those of imperfect recall (with non-repetition) that are inevitable. This suggests a preliminary remedy. The examination is adjusted such that the Examinee is taken to be, say, a Class, the role of the Examiner being as their Form-teacher. The Class is now being examined on whether it has learned the curriculum and knows, as a whole, how to answer the posed questions. The individual members may fail to know what the other members have previously answered, yet the class can pass the test of knowledge and produce the right answers to all the questions, amounting in logical terms to the existence of winning strategies for the Verifier. Yet this mild modification is bound to remain rather tight-fisted and does not seem to answer the question about the purpose. Aside from this basic question, we still do not have a clear picture of who is playing the role of the coordinator, or to what the dynamic formation of teams corresponds in the examination situation?

Furthermore, in semantic games with imperfect information, the number of members in the group of Verifiers may grow indefinitely large until the truth of a sentence is disentangled. This reveals yet another defect in the metaphor. In terms of the examination paradigm, it means that we need more and more agents to come up with a right answer to every question. Besides, one of the central properties of semantic games of imperfect recall is that the members are not allowed to signal to other members of the same team while the game is being played, since this destroys the team's ability, when viewed as one player, to forget something it once knew (Chapter 7; Pietarinen 2001c). Likewise, there may be no communication between those seeking answers. In an examination, each member would be working in isolation, only on his or her own questions.

What is more, this paradigm is blatantly asymmetric. The Verifier is certainly also allowed to make the first move in the semantic game, although from the perspective of the examination, the Examinee should start the session with an answer. The situation is therefore better suited to proof games in computer science, where it is commonly assumed that the Opponent makes the first move and the play then proceeds by alternating (strictly or non-strictly) between the players. It is also unclear whether an answer is supposed to be an answer only to the most recent question. If so, how does it fit in with the property that the Verifier may derive information from the choices made by the Falsifier that belong to the history, and not only to the most recent moves? The paradigm never explains these questions.

Logical games as cognitive activities If logic is taken to form an inherent part of human decision-making and cognitive capabilities, then GTS could be considered a natural candidate for the meaning of logical constants. In Peircean terms, the potential loss of information or imperfect information in logic resembles the commonplace loss and incompleteness of information in the dialogue between different phases of the Ego and the Non-Ego, but in spite of the losses, we have to be able to grasp the strategies and to find the most suitable actions among the totality of what is available. In extensive games, for instance, this means that a strategy specifies an action for each history in a game tree, even for those that lie on off-equilibrium paths, so that the strategy is specified for every possible sequence of choices, given the game rules and the resource. This has a Peircean niche: "The identity of a habit depends on how it might lead us to act, not merely under such circumstances as are likely to arise, but under such as might possibly occur, no matter how improbable they may be. (No matter if contrary to all previous experience [marginal note, A.-V.P.].)" (5.400).

One could nevertheless venture even further and argue that games are inbuilt in our cognition. The motivation to assess such a hypothesis comes from the observation that sensory input into the brain must first be found, and then processed accordingly. (This holds even at the level of sub-focal or unconscious processing, as rational information processing often takes place implicitly, without interference from conscious thinking.) The answer to the question concerning the purpose of the players is to view semantic games as cognitive activities of seeking and finding. Hintikka (1973a) supported this notion of game processes in logic. According to this view, the basis of logic lies in the human activities of seeking and finding, through which people come to know the propositions of logic. We should perhaps move away from the somewhat misleading titles of the players as the Verifier and the Falsifier, and follow the suggestion in Hintikka (1995): instead of taking the activities of verification and falsification to constitute semantic games, we should understand the processes of searching and eventually finding suitable individuals from the semantic resource as the true fundamental activities that have the potential to establish the wins. Verification and falsification are empirical aspects of logic pertaining to the question of how the winning strategies are found, whereas their plain existence already guarantees the truth or falsity of assertions.

Could the activities of seeking and finding constitute a uniform model for the multiplicity of classes of games for logic? It is indeed attractive to tie in the logical concepts with those of human activities: following Kant, we could state that the existence of an individual is an outcome of the cognitive faculty in the mind uniting experiences in consciousness (it also accounts for information obtained from non-perceptual sources). Why is it, then, that games stand for the connection between logic and cognition, in particular games of seeking and finding? What is the very ground for construing cognition as such a game?

While it is important to maintain awareness of these questions, their delicacy should not be misjudged. Some partial answers are in the offing in the Wittgensteinian notion of a language game, especially in what only quite recently has come to light given the publication of his entire Nachlass. Not only does he claim that at least some language games are ones of verification and falsification, he suggests that the purposes of the players may be explained in terms of the activities of showing or telling what one sees (Chapter 8).

Some implications are worth noting, however. Unlike examinations, the activities of seeking and finding, or showing what one sees, are not asymmetric, although they may well be either cooperative or non-cooperative in nature. Semantics primarily concerns the cognition of a single agent, or a phase within the mind of a single agent, rather than the activities of a group or a collective, and in this sense cooperation turns out to be inessential. We could think of the various faculties in the brain as isolated, with pre-assigned tasks. Suppose that the task of some faculty is to try to demonstrate the existence of an object, say $\mathtt{a}$. There are various ways of achieving this, such as trying out different sensory mechanisms. The received input would be processed, and if identification is carried out successfully, one could say that the subject recognises $\mathtt{a}$. In this case, the search, which actually may consist of diverse input data, is successfully terminated and something was found. Now, and only subsequently to this whole process, the subject is entitled to state that '$\mathtt{a}$ exists'.

Likewise, if the attempt to find objects is frustrated, the perceptual data are not successfully united in consciousness by identification, and the subject cannot assert that 'one can find $\mathtt{a}$'. The important thing is that the same cognitive faculty may account for both cases. The failure to demonstrate the existence is complementary to the act of identification. Strictly speaking, we do not need to distinguish between two separate roles for players, as only a single type of activity suffices.

Towards evolutionary semantics We could push the motivational explorations even further and go on to suggest that, as in evolutionary theory in which the game conceptualisations have frequently been brought into play, comparable motivations may serve for semantic games. I have spoken of players' roles as an instructive thing to have. For instance, in all traditional semantic games, one of the players has to make the first move, and subsequent moves are determined by the roles the players adopt (they may still change roles when encountering negation, but this does not change the point). What happens, then, if the notion of a role is abandoned altogether?

Surely this would not mean abandoning the notion of purpose or the notion of strategy. The players are still trying to achieve something, but they may themselves only have a vague idea of what they are doing. This is a step towards games of evolution (Maynard Smith, 1982). In evolutionary games, some population consisting of actors occupying a type of position they themselves might not be aware of have an access to such strategies that become more and more prevalent as the game goes on. There is no way of distinguishing the players from each other, insofar as they do not play any roles. The winner is the one who goes on longer whatever the task is. Moreover, evolutionary games may be symmetric in the sense that no one is forced to make the first move, and that the best strategy depends on what all the others are doing, albeit in reality symmetry among players is quite rare and there are variations according to their strength. [19]

At least some characteristics of evolutionary games fit the rationale of semantic games. Although most evolutionary games are cooperative and do not assume that players make rational choices (yet intelligence, in one sense or another, is required), some are non-cooperative, usually played many times by boundedly rational players, with little or no information about the game itself. In the context of semantic games, the domain of individuals on which the players play the seek-and-find game could be seen as a supply of resources available to the population. This supply consists of the logically integral objects of one's language, such as individuals in the domain and their names, conjuncts, disjuncts, and in the intensional case, possible courses of action. Players' information sets are then essentially partitions on the sequences of elements picked from the resource. In this light, the paradigm of seeking and finding is again instructive, but the range of the search has to be restricted to the objects that are relevant to evolutionary games.

Furthermore, we could think of the population as the Verifier, and of the logical components of statements as associated with members of the population. Thus imperfect recall can be modelled. As for the opponency, some other population or Nature will serve the purpose.[20] The winning strategies may be reconstructed as evolutionary stable strategies, namely those that cannot be invaded by any other strategy, given the influence of some other evolutionary concepts such as natural selection.[21] In particular, this gives rise to the concept of a semantically stable strategy for finite populations, given the existence of elementary meanings established by a team of players making pure, deterministic choices. Such meanings cannot be invaded by a mutant in the population, such a mutant being an agent playing the role of the Opponent. Meaning should rather be defined as a valuation function that maps atomic propositions to the payoffs of an evolutionary semantic game (Chapter 11).

Among the consequences of this kind of deconstruction of semantic games is the fact that linguistic utterances may be semantically stable among populations, standing up against the game-theoretic approaches to the evolution of language that routinely suggest it is the communicability function that determines the payoffs of expressions of different languages (Kirby 1999; Nowak et al. 1999; Oliphant & Batali 1996). In contrast to such performatory or communicative paradigms, the main component of language evolution should be taken to consist of its semantic attributes.

The evolutionary view of semantics needs to be supplemented. In evolutionary language-games, as in game-theoretic approaches to evolution in general, the outcomes that determine the course of evolution in a game between two members of a population, or between the Agent and the Environment, may be used to guide the adaptation of participants, but will be aimed at an adaptation of the players' linguistically significant structures. As in evolutionary games, inherent in adaptive language games are feedback systems that provide information on how players' linguistic behaviour should be revised or reconsidered in the future. What is interesting in these lines of investigation is that, by means of evolutionary language-games, one can demonstrate how mechanisms of language attain purposive goals in evolutionarily varying contexts.

In this light, the study of imperfect information also becomes increasingly important, especially because countless findings in evolutionary game theory point to the fact that information transmission between biological organisms is hardly perfect, and errors in signalling, recognising phonemes and receiving messages, are liable to occur. Even mutation and crossover could be seen as instances of strategic information hiding. On the other hand, the methodology of evolutionary language-games emphasises the compositional assignment of words with a component of the event in which they occur, whereas the ubiquity of imperfect information debunks this and favours non-compositional variants.[22]

Moreover, the widespread agreement in evolutionary game theory to the effect that information flow in contents between populations may be regulated suggests some further similarities. There is correlation between the choice made by some individual in a population and the choice he, she or it makes later on. Thus, depending on the nature of the correlation, the information is persistent for individual members. There is also a correlation between the choice made by a member and the choice made by one of the adversaries. Likewise, the correlations may not exist. Uncertainty is rampant between players. On the other hand, notions from evolutionary games such as 'increasing aggressiveness' are not so well suited to semantic contexts, in which each choice, at least in ordinary non-probabilistic positions, has to be equally weighted.

Another interesting angle on evolution involves hyper-cycles (Maynard Smith, 1979), and is analogous to symmetric versions of games in which players may start in parallel and make globally simultaneous choices. It may also happen that the information flow is cyclic. Such concurrent games are increasingly common in computation (Abramsky & Melliès, 1999).

Among further potentially useful concepts that have not yet been tried out are repetitive plays of semantic games. It is a common feature of evolutionary game theory that games are played over and over again by agents drawn from large populations, guided by an evolutionary selection process affecting their behaviour. Just as repetition addresses the question of what is common between evolutionary processes and game-theoretic solution concepts, in semantic games, the isolated processes of seeking and finding are not in themselves really the key to knowing the meanings of sentences — they only illustrate the concrete observable outcomes of actions that have practical bearings. For meaning, we need a sufficiently large number of repeated plays, preferably in different contexts and environments that consistently converge in some aggregate behaviour. The next chapter develops upon this perspective.

We also need to have some criteria for knowing when something has been found. In other words, we need a sufficient condition for the termination of the process of searching in semantic games. Sometimes, however, it is difficult to determine the precise circumstances in which termination is likely to happen.

Of course, one could claim to recognise objects in one's visual scenery by declaring, 'I have found a', but a performance of that sort need not add to one's previous knowledge, as the recognition may be based on mere acquaintance with the object, or on the retrieval of relevant information from memory, both trivial end-points of the search. Existence is a more complex process, as Peirce's definition of it as secondness aptly demonstrates (Chapter 6).

Further unpredictability may arise from the actions of the other participants. For example, evolutionary games often entail uncorrelated asymmetries between the Discoverer and the Latecomer that lead further continuation of the search.

Let me offer one final analogy. The law of excluded middle is not invariably valid in logic, because not all games are determined. A winning strategy need not always exist for either player. In the context of evolutionary games, this means that, after a certain number of rounds, it may happen that neither of the populations, nor their individual members, can defeat each other, and both will retreat. The same analogy may be extended to the failure of the law of excluded middle on the level of atomic formulas, giving rise to partially interpreted models.

4. Conclusions

Johan van Benthem (1993, p. 116) contrasts games with those of proof and computation by paralleling three theses. According to Gentzen's Thesis, all rational inference admits of a Natural Deduction formalisation, while Church's Thesis asserts that any form of effective computation can be programmed on a Turing Machine. Hintikka's Thesis is that any rational human activity can be played via Logical Games.

This juxtaposing of Hintikka's Thesis does not, however, single out any particular role logical games have in human activities, as it could be understood to mean that any rational human activity can be modelled via playing Logical Games, or that any rational human activity can be explained via playing Logical Games. The former interpretation may paint a more charitable portrait of the situation, although it remains an open question where the universal import of the thesis comes from. It is also consistent with the evolutionary analogy, according to which games are meant to be ways of modelling what happens or might happen in nature, and not explanations of such things.

In the interest of placing the perspectives reached in this chapter into the wider context of my treatise, it could be said that the question of precisely what characteristics logical and semantic games of cognition will assume may be fruitfully addressed from the perspective of Peirce's system of logic, his semeiotics and the overall approach to scientific inquiry. After all, games serve various purposes in various circumstances. Hodges' examination paradigm is nothing more than a fossilised example of what might be going on in some proof-oriented and dialogue games. The activities of seeking and finding, or showing what one sees, are much more dynamic and symmetric, and derive their foundational value from the venerable systems of Kant, Peirce and others. Emerging from this multi-perspective is that games and logic involve cognate concepts that are relevant for the study of cognition. These include meaning, memory, learning, perception, decision-making, bounded versus unbounded rationality and identification.

On the question of what the purpose of the game is, one is merely reminded of Wittgenstein: the right way to study games is to vary and play them. This does not deprive them of their foundational value, but such value has to be weighed against the backdrop of the diversity they enjoy in formal inquiry.

Tables 10.1 and 10.2 summarise some of the main categories and characteristics of games in formal inquiries.

	Logic		*Mathematics*		*Computation*	
Game	Semantic	Dialogue	Comparison	Division	Automata	Ludics/Geom.Int.act.
Players	Verifier-Falsifier Myself-Nature	Utterer-Interpreter	Spoiler-Duplicator	Verifier-Falsifier	Automata	System-Environment
Operationalisation	Pick elements	Defender-Challenger	Pick elements	Pick elements Pick partitions Pick examples	Transitions	Defender-Challenger
Winning strategy	Material truth	Logical truth	Elementary equivalence	Learnability	Language acceptance	Proof
Related games	Comparison	Ludics Quantum	Semantic	Ultimatum Bargaining	?	Dialogue Quantum
Contributors	Peirce Hintikka	Peirce Lorenzen	Ehrenfeucht Fraïssé	Hirsch Hodkinson Valiant	Büchi Rabin Gurevich Harrington	Blass Abramsky Girard
Applications	Logical semantics NL semantics	Proof theory Argumentation Quantum theory	Expressivity Descriptive complexity Infinitary languages Finite model theory Distributive NFs Bisimulation	Algebraic logic Set theory Definability Machine learning	Computational complexity Formal language theory Model checking	Process algebra Linear logic Category theory Concurrency

Table 10.1. The variety of game theories in science.

	Philosophy of Science	*Physics*	*Linguistics*	*Biology*	
Game	Interrogative	Quantum	Optimality Theory	Evolution	Language evolution
Players	Myself-Nature	Proponent-Opponent	Utterer-Interpreter	Population-Environment	Hominids
Operationalisation	Questioning	Pick measures	Pick candidates	Pick commodity	?
Winning strategy	Interrogative derivability	Logical truth	Optimal interpretation	Non-invasion	Non-invasion
Related games	?	Dialogue	?	?	?
Contributors	Aristotle Frieden Hintikka	Mittelstaedt	McCarthy Prince Smolensky	Maynard Smith	Kirby
Applications	Philosophy of science	Quantum theory	Phonology Syntax-semantics-pragmatics Sociolinguistics Psychology	Evolutionary biology	Linguistics

Table 10.2. (cont.) The variety of game theories in science.

Notes

1 Aristotle, *Topica*, edited and translated by E.S. Forster, London: Harvard University Press, 1960.

2 See MS 623: 45 (6 June 1909, *Meaning: Pragmatism*) for Peirce's remarks on medieval *obligations*, cf. his *Reason's Rules*, MS 596–599.

3 Yrjönsuuri (2001) provides a substantial coverage of this area.

4 One might entertain that some dynamic concept of logic was anticipated here, because truth conditions are given up and the purpose of the argumentation is on consistency. This comparison is not entirely a happy one, however, since consistency will remain the purpose of interaction even if the falsity of the thesis was to be granted.

5 Nifo, Agostino, 1521. *Augustini Niphi de Medicis Philosophi Suessani Dialectica ludicra tyrunculis atque veteranis utillima peripatheticis consona: iunioribus sophisticantibus contraria: nunquam prius in lucem edita: Aureum opus viliori argento emite*, Venetiis 1521, (first ed. 1521, reprinted in 1525).

6 There was thus a serious gist in Brouwer's accusation that Hilbert was reducing mathematics to a game. For Hilbert was not reducing mathematics into formal symbol manipulation. Rather, he posed the question of what will ensue if the interpretation of non-logical symbols is distinguished from the interpretation of logical symbols.

7 Henkin was working at Princeton University at the time when research into game theory as launched by von Neumann & Morgenstern (1944) was intensively pursued. Henkin was occasionally visiting the game theory seminars.

8 In backward induction, one starts from the immediate predecessors of terminal nodes in a game tree, and selects a choice maximising the outcome over all alternative actions, deleting nodes that come immediately after this node, and assigning an outcome to it. Reasoning is then recursively applied to all histories, until the root of the tree is reached.

9 A cardinal number is a type of number defined in such a way that any method of counting sets using it gives the same result.

10 See Kiikeri (1997) for an interpretation of interrogative games of scientific inquiry in the context of formal learning theory.

11 One may also think of positions in Ehrenfeucht–Fraïssé games as Hintikka–Rantala constituents of mathematical structures. The notion of constituents was defined for certain infinitary languages in Hintikka & Rantala (1976) — in turn the offspring of distributive normal forms and other similar concepts that were revolving around the development of Ehrenfeucht–Fraïssé games in the 1950s.

12 Full completeness means that winning strategies in a game coincide with the notion of a proof within some sequential proof system.

13 In the game-theoretic parlance, the Falsifier and the Verifier are sometimes called Abélard and Heloïsé, respectively. The name R of the Referee then comes obviously from Abbott FulbeRt, Heloïsé's uncle.

14 A tensor product $\varphi_1 \otimes \varphi_2$ of two formulas φ_1 and φ_2 provides informational encapsulation: moves within a subformula φ_1 depend only on the information in φ_1, and moves within a subformula φ_2 depend only on the information in φ_2. In merging the two by the tensor product, the information need not become global. Furthermore, even if both φ_1 and φ_2 were non-determined, their tensor product may be determined, since the Falsifier has an additional move at the conjunctive part of the product, thereby signalling the information about the Verifier's earlier moves. In this sense these linear games are of imperfect information.

15 From Peirce's evolutionary perspective, justification is needed only for non-evident things. For such things, he drew an analogy between justifying a hypothesis and the game of whisk, in which cards should be played in a manner that if a certain assumption is true, there will be practical consequences.

16 See e.g., 4.530, 1905, *Prolegomena*.

17 For a system-theoretic view on games, see Klir (2001). The original goals of systems theory are an integral part of a general Peircean semeiotic approach to logic and cognition. Systems theory aims at investigating the arrangements and relations among the parts that connect subsystems into an organisation. As systems are by and large independent of the content of their constituents, the principles they involve are applicable across traditional disciplinary boundaries (Chapters 13 and 14).

18 Binmore's remark generalises to repeated games (Chapter 11).

19 For alternative rule-conditioned behaviour, see Weibull (1995, pp. 64–68).

20 Originally, evolutionary games were games against nature; the contemporary literature likes to view them as games between populations.

21 Natural selection may also be a candidate in restricting the set of hypotheses in inductive learning problems.

22 Wider impacts of the significance of an evolutionary game-theoretic approach to language can be derived from their proximity to Wittgenstein's later views on language (Chapter 7).

Chapter 11

THE EVOLUTION OF SEMANTICS AND LANGUAGE GAMES FOR MEANING

NOONE IS CAPABLE OF following a rule only once.
— Philip Mirowski, *Against Mechanism: Protecting Economics from Science*, 1988

ON THE OTHER HAND a language game does change with time.
— Ludwig Wittgenstein, *On Certainty*, 1969

To understand evolutionary aspects of communication is to understand the evolutionary development of the meaning relations between language and the world. In particular, such meaning relations are established by the application of the systems of games. I will explicate how semantic games may be subsumed under the evolutionary framework of repeated games, in which stable meanings survive populations of strategically interacting players. Consequently, I will assess the viability of the notions of compositionality, the common ground and salience in these evolutionary games. Foundationally, this discussion is rooted in Peirce's pragmatist philosophy.

1. Introduction

Peirce wrote in 1902:

> In linguistics, there is the question of the origin of language, which must be settled before linguistics takes its final form. The whole business of ~~drawing~~ [deriving] ancient history from documents that are always insufficient and, even when not conflicting, frequently pretty obviously false, must be carried on under the supervision of logic, or else be badly done.[1]

According to Peirce's famous classification of the sciences, linguistics belongs to "Classificatory Psychics", a discipline that divides into "Special Psychology", "Linguistics" and "Ethnology" (EP 2:261). Linguistics itself is "divided according to the families of speech, and cross-divided into (1) Word Linguistics; (2) Grammar; and there should be a comparative science of forms of composition" (EP 2:261).[2]

What Peirce said in the above quote is unashamedly skeptical. Since he advocated the well-known evolutionary perspective not only in explaining human activities and behaviour, including the institutional formation and development of scientific theories, but also in perceiving the sum and substance of natural laws themselves, it was only predictable that he held the question of the origin of language to precede other questions in linguistics.

Even today, language evolution is regarded as one of the hard scientific problems (Christiansen & Kirby, 2003; Wray, 2002). However, one could turn the tables and take evolution, for the most part, as a linguistic as well as logical (as opposed to, say, computational, primatological or archaeological) challenge. In this sense, evolution constitutes one piece of the puzzle comprising the secrets of the overall science of language — no more, no less.

I would nevertheless like to take a wider perspective on language evolution, and think of it as a problem that lies in the intersection of logic, the philosophy of language, the philosophy of mind, and theoretical linguistics. My reasoning is that I am primarily interested in the evolution of the meaning of linguistic expressions, not in the evolution of grammar. If we think of the origins and development of meaning as a puzzle concerning how words and compound expressions built up from a finite sample of words mean what they do, what made them represent the qualities of objects that they do, and how these meaning relations have evolved in the course of human development, we are dealing with the problem of the origin and evolution of semantics.

Semantics itself is a field that is best studied from the perspective of combining the aforementioned disciplines. Since it is not a self-standing discipline for the broad notion of meaning, it has to be fed by pragmatics. Hence the evolution of semantics and the evolution of pragmatics of language are intertwined. This is no hindrance to the purposes of this chapter, because the theoretical framework of games is synthetic and works independently of whether we think of linguistic meaning with semantic or pragmatic undertones. What is crucial is that, in the evolution of language, as in evolution in general, we need to work our way by way of some theories, and not solely by way of trying to derive a range of empirical evidence and generalising from that evidence. We need to invent hypotheses according to the theory at hand. I believe that language as an interactive, game-like system provides such a theory.[3]

The theory-internal findings and observations may be further filtered by means of empirical and historical studies, such as those pertaining to neuroscience, mammal studies, ontogenetic vs. phylogenetic arguments, and archaeological findings, in so far as they are, as Peirce noted, “carried on under the supervision of logic”. Subjective probabilities and preconceived notions that historians, historical and evolutionary linguists attach to hypotheses are unreliable. Thus, we must trust to the powers of the mind in creating and selecting the hypotheses that are to be tested (cf. Chapter 12, sect. 6).[4]

My thesis is that meaning emerged from interaction, or the intention to interact, between agents. For the time being, I wish to keep the terms general. By agents, we do not only mean early hominids, mammals, or any communicatively competent person or animal with a neural system serving goal-driven purposes. Rather, interactions take place between anyone or anything that, in the spirit of Peirce's semeiotic theory, is capable of putting forth a sign. This is important, especially in view of the recently mushrooming research on artificial, intelligent, cooperative and autonomous agent systems and their emergent features (Chapter 14; Fong et al. 2003).

I will not go on to recapitulate what may be meant by a sign, and how it hangs onto other elements in Peirce's semeiotics, but a succinct way of characterising the essentials is that a sign stands for or represents something of the idea that it produces (Chapter 1). It is a vehicle for conveying information into the mind. That for which it stands is its object, and that which it conveys is its meaning. The idea to which it gives rise in the mind of the interpreter is its interpretant. The concept of the mind, and accordingly the interpretant, are terms that may be attributed to any agent or interactor that has an aptitude for portraying signs.

Nor do I claim that the concept of meaning that is emerging from this pertains exclusively to symbolic expressions. In circles of language evolution, it has been customary to think of language, any language, as a symbolic means of expression (Deacon, 1997). This is unnecessarily restrictive. Humans are members of sign-theoretic species, among other agents accredited with signs.

Consequently, what is essential to the evolution of meaning is some minimal notion of intercommunication. However, a purely symbolic view of language does not yet tell us much about what communication is. To communicate is to use media, and this media may, alongside with actual languages and probably even preceding their existence, be equally well exhibited in the dance-like movements of bees, the mating calls of frogs, or in any non-linguistic system of communication including formal, heterogeneous and visual representations of thought and reasoning. Such methods of communication employed iconic rather than symbolic representations, and were likely to have antedated those of symbolic systems. Indexical signs were probably even earlier. The essence of being an icon is the representation of the object by some factual or conceptual resemblance or similarity of the sign to it.

Aside from actual interaction between agents, merely an intention to interact is sufficient for the initial formation of the general shape of semantic relations. Sign-theoretically, interaction need not take place between actual flesh-and-blood contestants. The medium is inbuilt in any sign carrier. What is essential are the opposing roles that mediate signs so as to give rise semantic variation. According to Peirce, this idea is explained via signs that suggest interpretants in the minds of those playing these different semeiotic roles. Among other things, the roles may represent different temporal phases of quasi-minds within the single mind of an agent. A fuller account of this line of thinking needs to address Peirce's communal and synechist account of the mind and its relation to thinking. I have presented aspects of it in other chapters.

What is also needed is differentiation into several kinds of interpretants. According to Peirce, those who put forward a sign have in mind intentional interpretants. Those who receive it create in their minds interpretants that are effectual. Their mutual merger is a communicational interpretant that may be used for communicational purposes to actually convey information. A mind may broadly be understood as any "*sign-creatory in connection with a reaction-machine*" (MS 318: 18). Actual conversations or dialogues between language users, the information provided in topic-comment and topic-focus structures or in syntactic schemes such as left and right dislocation and topic topicalisation, have all intentional content directing the focus needed in producing communicational interpretants.

My proposal to look at semeiotic activities in the evolution of meaning is that, whenever the communicational interpretant has a property of being final, in other words created at the point in which the process of semiosis terminates, it will represent semantic relations between objects and signs that are good candidates for the meaning of an assertion trafficed by the sign. I differ with Peirce in that I suggest these interpretants are not yet to be equated with the meaning of signs, despite the fact that they are products of systematic inquiry concerning them.

In other words, I take final interpretants to be either negative or positive in nature. They may represent either disagreement (a forbidding, denying, abrogate act) or agreement (a permitting, affirmative act) in communicational situations. This may be an idealisation, but without the assumption the truth-conditional aspect of semantics would be shallow. Whether these candidates persist throughout multiple cycles of interaction depends on the habits of the players of the semeiotic roles. If the final interpretants are invariably produced in different contexts in which the interaction takes place (in the evolutionary setting the context refers to different members of a population), the habits of agents playing these semeiotic roles are equated with the existence of stable strategies. The authenticity of language is found in the relative stableness of the significations of its parts. Logically, the signs that carry propositions will, in that case, represent true states of affairs that hold in some part of the world, model, or system. Stable strategies, I submit, produce the Wittgensteinian "certainties", propositions that "stand fast" (Wittgenstein, 1969, p. 144), propositions that no individual agent may be held responsible of coming to know or believe.[5]

This summarises the kinds of processes that institute the semantic relations that the community of partakers will come to endorse. Populations try to capture them in such a representational medium that seems fit. An expressive grammatically constrained instance of such a representational medium is the natural language. Grammaticalisation seems to be a handy mechanism to increase communicational fitness by constraining the combinatorial possibilities of representing features of the environment. The notion of intercommunication, in so far as it does not presuppose language, should be best viewed as 'proto-communication', which refers to the theoretical constructs of agents in the language game of meaning, not to actual communication that transmits information in linguistic structures using signals and other symbolic expressions.

Of essential importance is what happens in intercommunication and what is being produced both in the course of its progress and as its end product. I seek to explain these by game-theoretic terms.

2. Semantic games and linguistic meaning

The role of semantics in language evolution has received less attention than the evolution of phonology, syntax, grammar acquisition or learning. However, similar game-theoretic methods that have been used in these fields have recently played a visible role in semantics, logic, pragmatics, and in their interfaces.

In the game-theoretic approach to semantics (GTS) there are two players, Myself and Nature (the Utterer and the Interpreter), who play the roles of the verifier and the falsifier of the expressions presented to them (Chapter 7). These players undertake to play a non-cooperative game involving a shared commodity, which is the non-empty (usually infinite) domain of the model. The model is an interpreted structure in which the formulas of the underlying logic or natural language are true, the domains of which players are to pick elements from. Additional moves may refer to sentential connectives and non-logical constants, including a wide variety of lexical and morphological categories such as modals, intensional verbs, tense operators, pronouns, definite descriptions, possessives, genitives, prepositional phrases, eventualities, adverbs of quantification, aspectual particles and polarity items.

Legitimate moves are determined by the nature of the constituent expressions under evaluation. Since natural-language sentences and logical formulas are written linearly, the direction of the evaluation is from left-to-right and in heterogeneous systems such as graphs and diagrams, from the outside-in. In natural language, the customary game-theoretic approach assumes the set of ordering rules, which spell out the order in which the game rules are applied by giving a priority of interpretation to some of them. After the application of ordering rules, the surface structure of natural language is produced. Fuller details of are found in (Hintikka & Kulas, 1983).

For the purposes of my argument, I do not need to dwell on these aspects of semantic games here. The reason is that the use of ordering rules is a matter of taste rather than necessity, since the expressive structures of games in their extensive forms capture the interpretational order by writing out the histories of actions, thus producing a genuinely diachronic approach to meaning

(Chapter 7; Pietarinen 2004c; Pietarinen & Sandu 2004). The upshot is that ordering principles may be dispensed with.

More precisely, and reflecting the time-honoured division made in game theory (von Neumann & Morgenstern, 1944), semantic games may be thought of either as being in their normal form or in their extensive form. In the former case, the strategies correspond, logically speaking, to Skolem functions, mathematical entities expressing functional dependencies between logically active expressions. In the latter case, the games come in their extensive form, in which the root is the complete expression under evaluation and the terminal histories come with atomic formulas or their analogues in natural language. Extensive forms of a game are finite trees, but unlike the more common derivational phrase trees in linguistics, they are thoroughly semantic.[6]

Atomic expressions are interpreted (either totally or partially) prior to evaluation, so that they receive either the truth-value True or the truth-value False. The payoffs are assigned to strategies in normal-form games and to terminal histories in extensive-form games. An n-tuple of Skolem functions or strategies in the extensive game, which invariably leads the player, whose role at the beginning of the game is the verifying one, to the winning terminal position interpreted as True, will be her winning strategy. Likewise, an n-tuple of such strategies, invariably leading the falsifying player to the winning terminal position interpreted as False, will be his winning strategy. The existence of such winning strategies shows when a compound formula, a sentence of natural language or a segment of discourse is true in a given model.

The game-theoretic approach to semantics carries with it a couple of further assumptions. Just as in the traditional theory of games the players are, ceteris paribus, hyper-rational optimisers. This is problematic, but I will not dwell on this rather outdated notion here (Pietarinen & Sandu, 2004). Furthermore, the semantic games are static one-off games, having a finite horizon simply because the input strings are finite in length. The strategies are pure, and in terms of the conditions for truth of the expressions, Nature's function remains stationary. In addition, the game and its equilibrium are common knowledge.

In what follows, I seek to address these four points.

3. Evolutionary language-games

By suitably relaxing some of the aforementioned assumptions and characteristics gives us evolutionary analogues to semantic language-games. Such refinements underscore the role of repeated, gradually progressing interaction. Unlike static semantic games, we may now ask about the kinds of behaviours that survive in the population of strategically interacting members. Replacing the concept of a winning strategy are sequences of strategies uninvadable by any other strategies. There is no need to cling on to two players, one who acts as a verifier and the other who acts as a falsifier of the expression. In its place, think of the population as a team of players, assigned their verifying and falsifying roles by a continuous sampling, and of the 'mutant' strategies as not necessarily the opponent strategy of invading strategies of the adversaries, but also as one of invading strategies with the same purpose.

A generic rule that the players heed stipulates that whenever an assertion conveyed in the sign is encountered, certain activity is prompted in the interpreter. Based on that activity, a preliminary semantic relation is created between the utterance and the object of the sign. This is an irreducibly triadic relation in which the role of the interpretant of the sign created in the mind of the interpreter is essential. Whether the preliminary relation prevails depends on the properties of the interpretant and the habits of the participants.

Players' activities may be the genuine behavioural responses of seeking and finding actual individual instances of objects that are to function as the values of the predicate terms given in the utterance. Or they may be mental excitements in consciousness. Typically, these are related. The existence of an individual itself is an interaction between different phases of the mind. The phases are pairs of action-reaction that consciousness is constantly maintaining.

For instance, if the sign conveys the information that some property of some entity holds, then the utterer of that sign has to seek an individual who is predicated of the particular subject of the proposition given in the sign. If the sign articulates the universal statement that some property holds for every entity within the limits mutually understood, then the interpreter of that sign is given the responsibility of finding an individual applicable to the universally uttered subject term of the proposition in question. If, furthermore, it is conveyed that something exists, then in addition to the selection of a suitable individual by the utterer, and besides the fact that the object applies to the predicate, the answer to whether the chosen particularity exists is attained through a further inferential process by the interacting agents.

Further rules besides logical ones may then be realised. Semantic rules are by no means restricted to the search for individual values for nominal expressions on extensional domains, but may also refer to (i) discourse universes consisting of possible worlds linked by the accessibility relation denoting temporal points or time intervals and other modalities such as epistemic and other attitudinal and objectual locutions; (ii) events, states and processes (commonly termed eventualities); (iii) collections (higher-order typed domains); (iv) quantities relativised to some measure or distance, manner, motion.

The strategies identified with the habits are evolutionary. If the game is viewed in its extensive form, strategies map its non-terminal history onto some actions given by the types of domains listed above. Since the game is played repeatedly, a measure of fitness is either exhibited or inhibited after each period of the set of plays. The payoffs reflect the expected number of true or false interpretations as presented by the final interpretants.

More precisely, let $\mathcal{G}$ be an asymmetric game, and let the payoffs be functions $\mu: (\Phi \times S) \to \{1, -1\}$ from a set of signs Φ and the set of strategies S to positive and negative interpretants. Signs are taken to carry sentences, propositions, utterances, non-linguistic expressions, diagrammatic, graphical, visual, haptic or auditory representations, or any other stimuli imaginable. Let the set of players be the finite population i of diamorphic agents playing pure, deterministic choices in $\mathcal{G}$, adopting one of the two common strategies. Populations are thus not coalitions, but rather teams as defined in the branch of game theory known as team theory (Marshack & Radner, 1972). The members of the population i are playing either the role of the verifying player V or the role of the falsifying player F, determined by the type of sign under evaluation, and the strategies are dissected accordingly.

The sign is part of the sampling process of members from a large population. In each period t of positive integers in which the game $\mathcal{G}_t$ is played on $p \in \Phi$, a pair of members of the population is sampled to function as the verifiers V and the falsifiers F of the sign p. The members are informed about their own roles. The agents retain histories $h \in H$ transmitted from any earlier game $\mathcal{G}_{t-1}$ to $\mathcal{G}_t$. The role switches are denoted by the function $R: (\Phi \times H) \to \{\uparrow, \downarrow\}$ from signs and histories of the game $\mathcal{G}_t$ to the set of binary values. The purpose of R is to denote those histories $h \in H$ in $\mathcal{G}_t$ at any period t where the role is to flip from the verifying role to the falsifying one, or vice versa. This happens when some negative sign (other than contradictory negation, because it is not game-theoretic) occurring in $p \in \Phi$ is uttered.

Let ρ be a mutant strategy and let $\sigma^* = (\sigma^*_V, \sigma^*_F)$ be a strategy for population i from which two opposing players are sampled, σ^*_V being the strategy for any member of i called upon to play the role V, and σ^*_F being the strategy for any member of i called upon to play the role F. The purpose of the mutant ρ is to invade population i. Since $\mathcal{G}_t$ is asymmetric, some mutants may affect the behaviour of the players of only one role.

The purpose is to show which signs p are semantically stable among the population i. By following what is done in the traditional theory of evolutionary games, we may define the strategy σ^* as semantically stable if and only if (i) σ^* is a best reply to itself, and (ii) σ^* is a better reply to all other best replies ρ' than these are to themselves.[7]

Because $\mathcal{G}_t$ is asymmetric and may be embedded in the larger symmetric game $\mathcal{G}'_t$, σ^* is an evolutionarily stable strategy of the symmetric version $\mathcal{G}_t'$ of $\mathcal{G}_t$ if and only if σ^* is a strict Nash equilibrium of $\mathcal{G}_t$. A strict Nash equilibrium is a situation in which a player can only do worse by changing strategies. It provides a solution concept for $\mathcal{G}_t$ that is hard to destabilise if every player has a strict incentive to maintain his or her own behaviour.[8]

That the separation of one population into randomly-drawn pairs playing opposing roles is the right model for language interpretation is shown by the reasonableness of thinking one agent of the team or phases of quasi-minds as sometimes uttering and sometimes interpreting a sign. Whether such roles are actually found in nature among actual language users, or how the process of sampling invariably produces pairs of agents that agree to such polarised roles as if by miraculous coincidence, is irrelevant, since the theory does not assume extant agents that are entitled to deliberate on their roles.

In other words, even if the theory is applied to actual language users (or some computational simulation of them thereof), they would not make conscious choices concerning what sets of strategies to use. Peirce would have said that such strategies are parts of agents' habits, those innate rules, responses, guides, customs, dispositions, cognitive conceptions, generalisations and institutions that have influenced them through evolutionary time. In fact, habits are constituted through time. This is, of course, in perfect accordance with the much recent evolutionary and game-theoretic arguments employed in biology.[9]

I do not maintain that strategies analogous to evolutionarily stable ones are the sole true solution concepts for semantic games. They insist on strict Nash equilibria and try to narrow down the number of such points in repeated plays. In fact, interpretation is a process based on sequential interactive turns of actions and reactions over long periods of encounters. For this, extensive forms of $\mathcal{G}_\iota$ that bring out the subgame structure and successive responses to adversaries' choices tend to be more appropriate (Chapter 7). This was anticipated in my references to game histories. However, evolutionary analogues in extensive games are not straightforward (Cressman, 2003), and it is not obvious how to extend the concept of stableness in asymmetric games to extensive games: one problem is that there may be information sets at off-equilibrium paths that are reached with very small or zero probability (Samuelson, 1997). In addition, whether one can dispense with the untoward assumption of extensive evolutionary games that every player may occupy at most one history along each period t of plays of an extensive game is an open question, and so is the question of the right amount of derivational information that players should be able to garner from histories belonging to the periodical past.

Extensive-form representations contain essential information omitted in the normal forms. Such information refers, among others, to actions along a particular history, needed in accounting for many pragmatic phenomena, such as in anaphoric relations. The suitable or intended values of anaphoric pronouns are often found in the past discourse referring to derivational histories of that discourse. There is no way of recovering that information merely from the normal-form representation of the game (Pietarinen & Sandu, 2004).[10]

Apart from anaphora, similar backward references may occur in problems of resolving which particular states of affairs some modal expressions refer to that have occurred (that is, have been selected in the game) somewhere in the past. These two types of references (anaphoric and modal) may then be combined to create yet further semantic phenomena, the resolution of which revealing more complex structures of these games (Pietarinen, 2001a, 2003d).

In general, then, there are two levels of diachronism in language evolution: there is the semantic question of game-internal references to past actions, and the trans-structural question of the amount of actions transmitted from earlier periods to future ones. Pragmatic features per se are not subordinate solely to the latter kind of constraint, and may involve shorter intervals that take place within single games. However, their evolution is parasitic on aspects of evolutionary dynamics in sequences of periods comprising extensive games, because only then the games get to be played in several, changing and mutating environments and contexts that nurture the meaning.

To coin a slogan (and a pun), 'pragmatics is games minus equilibria'. The evolution of pragmatic aspects of language is then what happens in several such games that no longer disregard phenomena out of classical solution concepts.

Aside from these pragmatically-oriented considerations on language evolution, we could ask what it means to be a best reply to itself, in other words what it means for a member of the population to choose optimally in many games. It is characteristic in evolutionary games that players change over time. This is the fundamental reason behind the dynamics. My point is that attaining stable meanings is indicative of the fact that this meaning is approved by other post-entry factions of verifiers or falsifiers, with new possibilities of invasion by mutants in their strategies.

Other game-theoretic approaches to the evolution of language typically promote the communicability function as determining the payoffs of expressions or different languages (Kirby 1999; Komarova & Niyogi 2004; Nowak et al. 1999, 2000, 2001; Oliphant & Batali 1996). In contrast to these performatory or communicative paradigms, language evolution in the present context is based solely on its semantic attributes. The approach is also distinct from naming games (Steels, 1998) since following Wittgenstein's ingenious example, naming is not yet a genuine and complete move in a language game.[11]

I will proceed to comment on this evolutionary rendering of language games in the following sections, and put it into a wider conceptual perspective.

4. Truth, meaning and composition

The wedge between truth and meaning Traditional game-theoretic approaches to semantics assimilate the notions of truth and meaning, but enrich both by introducing strategic processes as the mediating notion. Assimilation of truth and meaning, routinely made in logics devoid of strategic concerns, is misleading in language evolution, in which the notion of utterance meaning is much more diverse. Words, sentences and segments of discourse appear rather as pieces in a larger puzzle of rule-governed social behaviour, and meanings appear rather as roles in the contexts of the conversational use of language.

By introducing evolutionary pressures to the processes of fixing meanings, we in fact make vital distinctions between truth and meaning. The separation of truth and meaning is, from the game-theoretic perspective, evident in the separation of the notions of equilibria and stable strategies from the actions and their diachronic structure that produce and assign values to signs. One cannot observe meaning from a series of actions alone. One also needs to see what it is to end up with interpretants at the point in which the interpretation process terminates. One needs to see the purpose of those actions. To describe a strategy is to put forward the general layout of the actions that it gives rise to. To implement it is to communicate what a sign means.

One should be wary of biological analogies here, however. The Peircean sign-theoretic outlook on communication suggests that the notion of populations in evolutionary language-games refers to the sequences of interpretations by 'quasi-minds' entering the process of semiosis. It may be useful to think of these processes as dialogues between the utterer and the interpreter, or the verifier and the falsifier, but it is not the case that it is invariably the role of the interpreter to try to invade the process and to disturb it. In so far as the utterer and the interpreter have agreed on the common purpose, they are peaceful co-habitants of the same evolutionary cycle of the dialogue. Once in a while, the habits of the utterer as well as the habits of the interpreter engaged in the dialogue are contested by a mutant who checks that they are stable in their fitness so as to persevere in future cycles.

The question that I turn to next concerns the extent of being social as regards to the embedded agents of our theoretical populations.

The evolution of semantics not social The aspiration to separate truth and meaning is endorsed by a number of elements in language, typically referred to as non-linguistic or pragmatic. The evolution of pragmatics is certainly complementary to the suggestion of putting the evolution of semantics higher up in the evolutionary agenda. But it needs to be recognised that these fields are inherently intertwined. At all events, many researchers have inferred from this methodological crisscross that the proper place for the truly pragmatic elements in the constitution of meaning is to be found in domains influenced by social factors appertaining to communities of language users (Burge, 1974).

A contrasting view is that there is no innate social character in evolutionary language-games for meaning, notwithstanding the elements of cooperation and coordination in them. Yet social dimensions come into play in the scientifically broad spectrum of language evolution, noticeably defective without cultural and biological feedback. It is nonetheless doubtful whether cultural, biological, ethnographic, anthropological, or neuropsychological data play any vital role in the theoretical question about the origins and development of the very semantic relations behind linguistic acts. The reason is simply that little is lost in dropping such excess assumptions from the theory. While progress will doubtless be made in understanding the evolution and emergence of social communication, it entails the much broader, semeiotic notion of communication, not vice versa.

In ruling social aspects out of the scope of language games for meaning, the purpose is not to suggest that language use is not relevant: together with the multiplicity of pragmatic ingredients, including the speaker's meaning as distinct from the literal meaning, the context of the utterances, the role played by the changing environment, the background information and beliefs, performatives, presumptive meanings and so on, language use is a central factor in the evolution of meaning. But it is a long conceptual step from pragmatic theories of language use to the convincing argument that the defining characteristics of language change, or the necessary condition for the emergence of language, would be in their inherently social character.[12] I do not see that gap to have been filled in recent evolutionary literature.

Admittedly, the concept of what is social may be stretched. This was, in effect, what Peirce himself did: "logic is rooted in the social principle" (2.654). It is vital in opening these sentiments out to recognise that for Peirce, logic takes in also all kinds of considerations of what one's rational action would be in situations that call for moral judgements. For Peirce, logic is a normative science, viz. the notion of truth has a normative component in it. The notion of 'social' refers to rationality of individual actions made in communities within which those individuals have been embedded.

The non-compositionality of language games More often than not, it has been argued in language evolution literature that compositionality gives an advantage to communicators in environments in which language is about to evolve (Nowak et al., 2001). It is by means of the compositional mapping of words or expressions on structures in the environment that describe complex events, so the story goes, that many limitations of communication attributable to linguistic errors can be surmounted.

Similar arguments for compositionality are put forward in Kirby (2000). Brighton & Kirby (2001) derive additional feedback on evolutionary learning and rewriting systems from the milieu of cultural evolution. According to Batali (1998), artificial learning and communicating agents are groups of neurons in neural-network architectures manipulated by connectionist learning algorithms. In general, these approaches resort to notions of evolutionary game theory in attempts to unearth thresholds to fitness parameters. When the thresholds are known, the preferences of payoffs can be determined, given the costs of the communication, by compositional means.

However, discrepancy exists between the arguments based on computer simulations and the overall game-theoretic formalism that I have endorsed in the evolutionary setting. The game-theoretic approach to meaning is, in essence, non-compositional. It does not automatically preserve the applicability of assigning the semantic attributes A that it assigns to the compound expression e (determined by the applicability of these attributes) to the proper subexpressions of e. It does not have to. It provides the meaning of e in terms of the existence of evolutionarily stable strategies for one of the dialogue parties. The subexpressions of e that are terminal are given by another attribute A' which, as I suggested, is defined in terms of the finality of the interpretant created, for their most part, in the minds of the players.

The key to resolving this discrepancy lies in the understanding of the term 'compositionality'. It is used in the evolutionary arguments of Kirby (2000) to refer to the opposite of holistic languages, typically proto-languages and sets of idiomatic expressions. The existence of idioms presents no counterexample to compositionality, however, as Westerståhl (2002) has shown. Despite the recurrent insistence that compositionality refers to the morphic construction of meaning of a compound expression from the meaning of its parts, those resorting to evolutionary arguments have not used 'meaning' in the sense of the assignment of semantic attributes to expressions. They have used it to refer to the construction of syntactically structured expressions from the space of algebraic meanings within genetically homogenous language users.

However, to have a compositional communication system evolving in the evolutionary game, in the sense that the determination of the applicability of semantic attributes A assigned to the compound expression e will also be determined by the applicability of some other semantic attribute A' to all constituents of e, one needs to settle on why the supplementary attributes A' are to be accepted as natural.[13] If we dispense with this naturality requirement, any partially (non-compositionally) interpreted grammar with finitely-generated syntax with co-finality properties (i.e., all uninterpreted expressions are subexpressions of interpreted ones) may be extended to a compositionally interpreted grammar. However, given the complexity of auxiliary attributes, such extensions are unlikely to be evolutionarily viable.[14]

Therefore, it is a non-starter to even try to make the line of approach I am advocating acceptable to the followers of the tradition centring on the syntactics/semantics interface. This tradition tries to study context-independent systematic effects of linguistic form on meaning, which is held to be non-negotiably and wholly compositional. Since their assumptions differ fundamentally from mine, their tradition cannot be used to criticise my approach to the evolutionary component found in these language games.

Cross-categorisation in semantic games The question of which sets of expressions constitute linguistically natural categories is related to the question of language universals. Finding natural categories is nonetheless muddled by the recognition that the game-theoretic analysis of meaning permeates through various such classes or categories. The same rule may account for a linguistic item that is a member of many different categories.

As prototypical examples of such rules, let us consider those that are intended to assign meaning to polarity items, in other words to expressions that have been delineated in terms of their sensitiveness to occurrences of negative words in their linguistic environment.[15] Let I be a choice set into which previous choices made in the game may be collected. Using the extensive-form representation of the semantic game, these elements may be drawn from the game histories in H, a suitable subset of them being preserved from previous periods. The parameter C denotes the category that can typically be derived from the main verb of the input sentence. The reference to time supposes a customary branching possible-worlds model of time from which time points or intervals are selected. The generic rule (G.lexical-NPI) for a subset of lexical negative polarity items, such as the idiomatic *budge (an inch)*, *lift a finger*, a superlative like *the slightest*, and so on, is the following:

(G.lexical-NPI) If the game $\mathcal{G}_\iota$ has reached sentence S of the form

X – lexical-NPI – Z,

then one of the players chooses some minimal quantity, amount, manner or degree from the choice set I (or one of the histories in H). Let this quantity be b. The game $\mathcal{G}$ is then continued with respect to the output sentence S'

X – have (has, does) b, and b is minimal Z of category C.

The capital letters X and Z are placeholders for an arbitrary grammatical environment.

Some other NPIs, such as *yet* and *anymore*, are also aspectual particles, and *already*, which is a positive polarity item is, together with *yet*, also aspectual and suppletive. If we move away from the suppletive function, their meaning is given by the same rules:

(G.yet) If the game $\mathcal{G}_\iota$ has reached sentence S of the form

X – Y – Z *yet*,

Myself chooses a time t_1, whereupon Nature chooses time t_2 from the reference interval T, $t_1 < t_2$ or $t_1 = t_2$, and the game continues with respect to the sentence S'

X – Y – Z at t_1, and X – was expected to *neg*(Z) at t_2.

Here $t_1 < t_2$ means that the time point t_1 occurs earlier than t_2. The operation *neg*(Z) is a verbal negation of Z.

(G.anymore) If the game $\mathcal{G}_\iota$ has reached sentence S of the form

X – Y – Z *anymore*,

Myself chooses a time point t_1, whereupon Nature chooses t_2 such that $t_2 < t_1$, and the game continues with respect to the sentence S'

At t_2, X – *neg*(Y – Z), and at t_1, X – Y – Z.

(G.already) If the game $\mathcal{G}_t$ has reached sentence S of the form

X – *already* Y – Z,

Myself chooses a time point t_1, whereupon Nature chooses time point t_2 from the reference interval T such that $t_1 < t_2$, and the game continues with respect to the sentence S'

X – Y – Z at t_1, and X – was expected to Y – Z at t_2.

The rules (G.yet) and (G.already) refer to the notion of expectation, which shows the flexibility of the game-theoretic approach in tackling phenomena that are pragmatically constrained. Such phenomena may be subsumed under the province of strategic meaning of utterances, which strives to incorporate what is contained in the strategies of the game that the players need to take into account in the process of attributing semantic values to such utterances. Hence, it goes beyond the question of the mere existence of equilibria.

A prototypical example of a linguistic phenomenon in which strategic meaning appears indispensable is anaphoric coreference, which I already noted to be one of those linguistic phenomena that may be identified through games in extensive forms. For instance, coreferences would be determined in the following sentences in terms of several overlapping phenomena, mixing generalised quantifiers, functional coreference, bridging, backwards anaphora and eventualities with one another (Pietarinen & Sandu, 2004):

Every man carried a gun. Most of them used it. (11.1)
(Generalised quantifier + plural + functional dependence)

Of course there is live music in our nightclub. Unfortunately, tonight they have a night off. (11.2)
(Bridging + plural)

Most students didn't get high grades. But everyone passed a math examination last week. (11.3)
(Functional dependency + plural + cataphora)

Everybody rode a rhino. Few liked it. (11.4)
(Generalised quantifier + functional dependence + eventuality)

Similar cross-categorial explanations are available in the game-theoretic framework for adverbs of quantification and eventualities, as well as for context-dependent quantifiers (Pietarinen, 2001b).

The upshot is that, while the problem of finding natural classes for sets of linguistic expressions may be insurmountable, the same interpretive processes are at work across different domains of discourse, including individuals, time and tense, propositional attitudes, manner and events. Games are domain-neutral, and in this respect constitute no argument against universal aspects of language. What universal aspects are preserved under evolutionary pressures may then be investigated by applying the semantic framework to the phenomena listed here.

5. Common knowledge in the evolution of semantics

What, then, is the true role played by common knowledge in game-theoretic arguments for the evolution of meaning? According to Lewis (1969), whose argument for the indispensability of the infinite hierarchy of mutual knowledge became popular, language is an example of a conventional system. He argued that coordinated communication would be stable in signalling games (kinds of proto-languages) unleashed to tackle the emergence of conventions, in other words that there will be equilibrium (or focal) points in the game between participants, provided that the communication is sufficiently accurate and hence successful. Since such points are not unique, conventions need to be established in order to choose the most appropriate ones. Signals are acts of coordination, but the question of which signals came to be associated with which meanings depends on considerations of salience (or prominence, preceding actions of the same history, vivid memories, regularity), together with convergence to a shared basis of common knowledge of these considerations.

This argument hinges on the assumption that there does not have to be any prior language for shared common knowledge regarding salience. Skyrms (1996) attempted to circumvent this assumption by taking the population to consist of inductive learners choosing signalling systems by chance. However, it should be noted that inductive learning also depends on some domain theory that indicates some ordering or measures of distance between predicates in the learning space.

Some common ground is inevitable in order for a language to evolve. Here semantic games come to the rescue. The players are mutually acquainted with the universes of discourse from which they are to choose elements or names as values of subject terms, predicates and other elementary expressions. A sign can be communicated only if suitable familiarity with it obtains, through acquaintance or instinctive and innate knowledge, or as Peirce held, through collateral observation, which may be either factual or conceptual, and experience and an understanding that the other party is also appropriately and similarly familiar with it. This does not mean that communicators have to be perfectly informed about the domains; they might be able to preview only fragments of them at a time. Of course, the domains are not required to be finite. Thus learning is essential. Various notions of learning may, indeed, be encoded into the strategies that the players use. For example, we may require them to be not only recursive, but also learnable in terms of formal learning theory.

Lewis was not the first to suggest the indispensability of common knowledge in communication. As early as in 1908, Peirce remarked in an unpublished paper: "No man can communicate the smallest item of information to his brother-man unless they have common familiar knowledge; where the word 'familiar' refers less to how well the object is known than to the manner of knowing. [...] Of course, two endless series of knowings are involved".[16] A suitable, mutually agreed platform needs to exist before any communication or dialogue can take place. Specifically, such common ground encompasses the fact that the universe of discourse is well known, and mutually known to be known, and agreed to exist between the utterer and the interpreter. In Peirce, we thus find an early but articulate account of the notion of mutual or common knowledge, a key to arguments for dialogical and interactive language evolution.

But does not this make the universe a closed, determined, and for these reasons a mutually and exhaustively known, totality? The answer is that the mere existence of some common ground and mutual acquaintance does not yet suggest that the proper and exhaustive identification of objects has taken place. While a sign may be communicated only if there is suitable familiarity with it, the object in question does not yet need to be identified. This is, I believe, what Peirce's addition of the qualification 'familiar' in common knowledge amounts to, referring to the degree of knowing rather than to a way of knowing. The act of naming an individual would already be a possible identification of it, albeit not yet a complete move in the semantic game. Even if identification fails, a sign can still be put forward, but in that case it will lack some of its qualities and hence be indeterminate.

This has a bearing on the concept of salience that Lewis and others have argued to be indispensable in order for a language or a conventional system to evolve. Salience has been quite elusive in game-theoretic analyses. More readily, in evolutionary games organisms do not need to recognise one type of behaviour as more successful than another, because they are genetically pre-programmed to act. Likewise, in semeiotic dialogues as in evolutionary semantic games, signs may be communicated and elements of domains selected without ideal identification of what the actions actually are, or of what their practical effects will be. This ought to be obvious given the generality of the notion of those who can communicate signs, and the point is inherently related to Peirce's understanding of the concept of a habit, that general organisational principle that all sign bearers, humans and non-humans alike, are prone to possess and educate. It is thus not merely embellishment to say that in evolutionary strategies, be they of focal or any other stable points of meaningful interaction, we find some contemporary correlates of Peirce's gone concept of a habit.

Another soft spot in discussions revolving around signalling games is that they tend to conflate signs with signals. The essential difference, nonetheless, is that signs are 'multiplexes' coming together with both their carriers, and interactive responses by their interpreters. They are a medium for communication, inseparable from their objects and interpretants. Signals, on the other hand, as used in Lewis' and his commentator's writings, are typically technical, non-contextual and non-situational devices by which one aims to predict what the future acts of a person, agent or a computational system are.

Even though focal points, drawing on considerations of salience (itself a species of Peircean category of firstness, of 'what strikes the mind') as well as 'common familiar knowledge', have not been fully incorporated into game-theoretic analyses (though there are attempts to that effect in Selten 1983; Sudgen 1995), it by no means follows that strategies, which are both formally and cognitively as concrete and as tangible entities as there can ever be in formal approaches to meaning, would lose their explanatory value in semantic theories.

Nor does it help to argue against my proposal by claiming that the cases in which evolutionary explanations do work are typically only those in which agents are entirely non-mental, automatic and blind actors with no real potential for deliberation and will. First, the subject matter here is not just the evolution of linguistic (i.e., symbolic) signs. Second, it is not needed for the purposes of my argument that such actors need to be actual language users operating in actual linguistic communities.

6. Comparison and outlook

That language evolved from interaction is not in itself a new idea. Similar observation has been, in one form or another, propounded in Batali 1998; Brighton & Kirby 2001; Deacon 1997; Gärdenfors 2002; Hurford 1989; Kirby 1999, 2000; Lewis 1969; Nowak et al. 1999, 2000, 2001; Oliphant 1994; Oliphant & Batali 1996; Skyrms 1996; Steels 1998 and Steels & McIntyre 1999. Most of these works are experimental, or give in to experimental results, and so provide important evidence for the communicational and game-theoretic aspects of language evolution. Nevertheless, they carry considerably different assumptions and presuppositions, and for the most part probe different problems with respect to one another or with respect to my approach.

One salient issue in current research on semantic aspects of the evolution of language is that it focuses on the emergence and development of lexical meaning. Most of the work done so far has not extended over individual words in a lexicon, let alone covered text beyond sentence boundaries. Far from resulting from any technical challenges, this appears to be a symptom of linguistic functionalism, which assumes that the smallest units associated with meaning should be as communicative as possible.

The position taken by Gärdenfors (2002) to the effect that cooperation and communication in humans, as facilitated by the ability to do goal-directed planning, imply that meaning is confined to symbolic expressions exhibited in human communication. I believe, and so did Peirce, that the ability to use symbols is only one aspect of human communication. The general strategy Gärdenfors adopts is nonetheless encouraging. Instead of starting with the phonological, syntactic and grammatical components of language, he takes the pragmatic and semantic sides as primary and bases the emergence of cooperative communication on them and not on notions of the stability of grammatical or morphosyntactic rules.

It has been repeatedly documented in recent literature on language evolution that Deacon (1997) is not only following Peirce's theory of signs in his exposition of the symbolic means of linguistic expressions, but he is also grounding the theory of human languages in Peirce's sign-theoretic system. I find this claim unfounded, not only for much the same reasons as there are limitations in Gärdenfors' cooperative account of symbolic communication, but also, and more importantly, because the missing component in Deacon's system of symbolic signs in human language use, despite its echoing of Peirce's theory, is the genuine (I almost said triadic) communicative dimension to their meaning. Such triadism is irreducible, as are the pragmatic notions of common ground and common knowledge. Accordingly, a broader view of communication, while of paramount importance to Peirce's semeiotics, is conspicuously lacking in recent computational theories that aim at simulating communication in population systems.

The experiments conducted by Nowak et al. (1999, 2000, 2001) in the setting of formal language theory aim at extending the learning of the grammatical structure of languages to the learning of mappings between linguistic form and meaning. If we consider meaning as to refer to the content of the utterance, this comes close to the recent optimality-theoretic approaches to language, some of them explicitly resorting to game-theoretic tools in describing Nash equilibrium points in the game of choosing form-content pairs (Dekker & van Rooy, 2000). However, the fundamental flaw in optimality theory is that its linguistic representation encodes dyadic form-content pairs discounting the role of the category of thirdness, namely how these structures are linked with their interpreting minds from which to emit their purpose and strategic aims.

The notion of interaction in the development of semantic relations for language has both theoretical and practical overtones. Although the analogy is by no means perfect, maybe not even viable, one may think of this interactive picture as a prima facie reflection of the distinction between two 'existences' of language (Chomsky, 1986), the I-language and the E-language. From a deliberately modified Peircean perspective, it could be said that the I-language ('internal' language) is something that is represented in the quasi-minds of quasi-utterers and quasi-interpreters, or in the minds of the verifiers and the falsifiers, all of which may be located within a single mind. Its constituents are signs, the meanings of which evolve through the dialogical process of semiosis. On the other hand, the E-language ('external' language) comprises the actual utterances in using language. The meaning of its constituents likewise requires a dynamic process, but the process takes place between the utterers and the interpreters of actual linguistic expressions. What is significant is that in both cases, the signs or expressions are put forward in games of the same overall semeiotic genre.

The methodological point I want to get across concerns the reasons why the science of the evolution of language, conceived of as an amalgamation of two elements — one with a neurophysiological base and the other with a mental base — is so important. They are both needed in order to understand language, and they are both realised in the synthesis of neurolinguistics and psycholinguistics on the one hand, and the philosophy of mind, language and logic on the other. The identification of elements needed in the combination of these for any feasible evolutionary argument to take place is one of the future's major challenges.

The relevance of such an enterprise is argued for in Pietarinen (2004a), in which findings in cognitive neuroscience, mainly those related to impairments in brain lesions, are seen to support a number of logical and linguistics phenomena, including the distinction between implicit and explicit knowledge, memory, belief and awareness. The semantic games of language evolution proposed here are in a like manner neuroscientifically faithful. For instance, there is the illustrious distinction between two functional extrastriate visual areas in the brain, the ventral pathway responsible for object perception and the dorsal pathway responsible for spatial perception between the relations of objects (Ungerleider & Mishkin 1982; Milner & Goodale 1995).

Game-theoretically, the processes of choosing actions correspond to the ventral 'what' systems (identity), while the processes of instantiating the elements so chosen as suitable attributes of predicates require coordination that corresponds to the dorsal 'where' system (localisation). Two main caveats are that (i) these two systems in the visual area are interconnected just are objects are to their spatial locations, and that (ii) they both subdivide into several 'subroutines', including the spatial vs. temporal and parallel vs. serial identifications.[17]

That semiotics is relevant to language evolution was largely established in Hurford (1989). Unfortunately, the semiotic background theory in that work was Saussure's psychologically and sociolinguistically influenced semiology, resulting in a fundamentally different theory from Peirce's semeiotics, which promotes a formal and logical outlook, and which needs to be kept apart from psychological and sociological concerns. Peirce's triadic theory of signs cannot be understood without the dialogical idea of communication between sign users, but as soon as that is done, it is seen how a game-theoretic system of meaning emerges. As argued here, its modern elaborations are amenable to some of the most vital aspects in the evolutionary genesis of logical and linguistic semantics.

* * *

According to the game approach to the evolution of semantics, there is a virtually indefinite number of meanings that may be assigned to an expression, for much the same reason as there is an indefinite number of different language games that may be played on the expression. This is, of course, a fundamentally Wittgensteinian perspective to language change. His ‘diachronic pragmatics’ was witnessed by constantly evolving lifeforms: “On the other hand a language game does change with time” (Wittgenstein, 1969, 256). In other words, as the context of the utterance or a fragment of dialogue changes, new language games emerge to reflect the change. Evolutionary games seen through their extensive form are tailor-made to encode these changes and the multiplicity of meaning.

One of the main points in game-theoretic arguments is that the rules of language define which choices are legitimate, while meaning evolves largely as a result of strategic considerations. An implication is that it is vital, for instance for the purposes of the learnability of language, to keep the defining rules of the game general, thereby maximising the number of language games that can be constructed on the basis of such rules.

Further research needs to establish evolutionary parameters deciding which of any two competing pairs of defining game rules or sets of rules have a greater chance of being maintained in the population of language learners. This problem has been addressed in computational simulations. Such evolution of constitutive rules was not my concern in this chapter. It was the evolution of strategies in repeated plays of semantic games, which has nothing to do with how the rules of semantic games are derived, how they evolve, how agents are involved in their genesis as well as in the rejections of adversary candidates, or how the rules may change in the course of linguistic change. Parameters such as simplicity and the ease of recalling a rule, resistance to ordering principles, and cross-categorial application may well be relevant factors in grammaticalisation, the evolution of constitutive rule systems for language games. Replicator dynamics with pure strategies is typically sufficient for such a purpose, but like other defining rules, that has to be kept separate from the primacy of the strategic component of language games.

For the time being, research addressing the strategic component is bound to remain mainly conceptual and philosophical, and its science is only just evolving. The relation of empirical research with the theoretical agenda of this paper is not to be asked, at least not yet.

Notes

1 MS 427: 77, 27 February 1902, *Logic II*, under the margin header "Psychical Science must rely on Philosophy"; 1.250, 1902, *Principles of Philosophy, Classification of the Sciences*. The reference to the business of deriving ancient history from documents is to his paper *On the Logic of Drawing History from Ancient Documents, especially from Testimonies*, MS 690, written in the previous year.

2 For details of the classification, see Anderson (1998); Kent (1987); Parker (1998); Pietarinen (2005i).

3 Pietarinen (2005d) has more on the role of semantic games in the evolution of pragmatics. In brief, no structural difference obtains between semantic and pragmatic change. The difference is epistemic: in order for a pragmatic change to take place, players must have some knowledge of the strategies in use in the game. In semantic change, on the other hand, the existence of stable strategies suffices.

4 However, we must also ponder on Peirce's sobering remark that "archeological researches have shown ancient testimony ought to be trusted in the main, with a small allowance for the changes in the meanings of words" (1.113, 1902, *The Scientific Attitude*). This may be understood as a claim that holds archaeology, due to its cumulative evidence, amenable to inductive methods, with the obvious caveat that any such admission is conditional on theories of semantic change, in other words on theories that strive to explain how meaning of the expressions has come to be varied during the course of producing the document and its interpretation.

5 Peirce explicated such certainties in terms of "propositions and inferences which Critical Commonsensism holds to be original, in the sense one cannot 'go behind' them (as the lawyers say)". They are "indubitable in the sense of being acritical" (5.440, 1905, *Issues of Pragmaticism*).

6 One could of course think of the nodes of an extensive game to be mapped onto syntactic parse trees.

7 We could furthermore assign probability measures Δ on the set of strategies S by defining vectors of frequencies with which the populations play each of the different strategies in S. Maynard Smith (1982) provides details on evolutionarily stable strategies. This concept was, according to him, first formulated in Maynard Smith & Price (1973). Concerning the history of this concept, the population genetist and information theorist R. A. Fisher proposed a measure of maximum likelihood frequency for learning probability distributions from finite samples in 1922. The information measure was independently co-invented by Leo Szilárd in the same year.

8 In the symmetric version $\mathcal{G}'_\iota$, σ^* specifies what to do in $\mathcal{G}_\iota$ no matter what the roles of the players are.

9 Meanwhile, the notion of a habit was expelled from the mainstream economics. Hodgson (1999) argues that the main reason was the emergence of positivist methodology, which relied on experimental methods and which affected the early 20th-century sociology, itself heavily influenced by behaviourist psychology in its excision of all purpose-driven biological explanations from human action.

10 See MS 318: 29: "Pronouns are words whose whole object is to indicate what kind of collateral observation must be made in order to determine the significance of some other part of the sentence". This is but an instance of the importance Peirce laid on the common ground, presupposed in any successful communication.

11 See especially Wittgenstein (2000–, 226, p. 35): "Within naming something we haven't yet made a move in the language game, — any more than you have made a move in chess by putting a piece on the board". Chapter 8 develops further the Wittgensteinian connection in relation to recent evolutional and computational perspectives to meaning.

12 For example, in their introduction to Knight et al. (2000), the authors state that, "Language — including its distinctive representational level — is intrinsically social, and can only have evolved under fundamentally social selection pressures".

13 For a number of newly-developed logics for which this naturality requirement of compositional semantics does not hold, see Sandu & Hintikka (2001).

14 In fact, in Jackendoff (2002) the use of the term 'compositionality' in evolutionary arguments is replaced by 'combinatorial', which refers to the generativity of syntax, but the motivations for this move are quite different from mine.

15 Pietarinen (2001e) argues for a 'negation-free' account of the semantics of these items.

16 MS 614: 1–2, 1908, *Common Ground*, see Chapters 12 and 13.

17 Let us recall that Peirce was acutely aware of the early findings in neuroscience, especially that certain brain lesions cause various handicaps in language (Chapter 2).

Chapter 12

COMMON GROUND, RELEVANCE AND OTHER NOTIONS OF PRAGMATICS: FROM PEIRCE TO GRICE AND BEYOND

According to the orthodox view of pragmatics, the esteemed notion of the common ground was introduced into the modern theory in the wake of the speech-act theories by David Lewis (1969), Stephen Schiffer (1972), and Robert Stalnaker (1974), who applied the concept of mutual or common knowledge and belief.[1] I will challenge this view. While these writers acknowledged that H. Paul Grice foresaw the central role of the common ground (his preferred terms was the "common-ground status" of discourse particles and assertions), it has not been noted that he was influenced by Peirce's pragmatic philosophy and his many writings on topics closely related to speech acts, assertions, conventional utterances, interpretants as implicatures, rationality, cooperation, dialogue strategies, and several other key pragmatic factors in communication. Peirce also acknowledged the importance of common knowledge in the very existence of the common ground.

In this chapter I will trace the development of pragmatic ideas on language, especially in terms of the emergence of the common ground and common knowledge. I will map out the development from Peirce's pragmatism of the late 19th century to Grice in mid-1900s, via the intermediaries of thinkers such as Philipp Wegener, William Dwight Whitney, Charles Kay Ogden, Ivor Amstrong Richards, Bronislaw Malinowski, Ludwig Wittgenstein and Alan H. Gardiner, and from Grice to some modern writers. I will also assess the extent to which these early ideas have been preserved and shared in present-day pragmatics, most of which is due to Grice's influence, and much of which has recently been centred on theories of relevance.

The other recent fields in which Peirce's presence may be felt in terms of these developments of pragmatic ideas on language — in addition to the question of the evolution of semantics discussed in Chapter 11 — are meaning in cognitive linguistics and language change in historical pragmatics.

1. Introduction

A good deal of present-day pragmatics is due to Grice: this much is plain. Nevertheless, in 1972, Schiffer wrote:

> Not only is Grice's account highly illuminating, it is also, as far as I know, the only published attempt ever made by a philosopher or anyone else to say precisely and completely what it is for someone to mean something. (Schiffer, 1972, p. 7).

Some might wish to argue that the crux of the matter is how to quantify 'precisely' or 'completely'. This is irrelevant: according to Sextus Empiricus, the Stoics already observed that "the barbarians do not understand, although they do hear the sound", and that "what is conveyed by the sign is the matter of discourse indicated thereby, which we apprehend over against and corresponding to our thought".[2] What the Stoics brought up, according to Sextus Empiricus, was the idea of truth and falsity subsisting in the "expression", which belongs to the speech, not only to the proposition (*ibid.*, II: 70). (Sextus himself rejected the view that the true and the false were attributes of speech.)

Nevertheless, what I would like to argue is that not only is Schiffer's view phoney in that there is an exceedingly rich history, as well as a proto-history, of closely studied phenomena of speaker intentions and interpretations in linguistic inquiries that were later institutionalised under the title of pragmatics, but also that his own take on what the notion of mutual knowledge means is subject to some sombre consequences.

I will limit myself to the end of the scale of pragmatics that borders on the theoretical or formal side. Admittedly, the study of 'formal pragmatics' — or the *ergon* (ἔργον) of it — may reveal a conceptual monster of the same genre as 'non-formal mathematics'. However, scholars have been able to say many useful things on both for centuries.

One way to approach the heterogeneous and messy field of pragmatics is to take it, simply, to be the study of assertions. A glance at its history reveals that it started as and has predominantly been about assertions as sentences, the independent parts of language aiming at expressing complete thoughts. This assertoric stance does not yet distinguish pragmatics from semantics, because semantics likewise studies the conditions in which expressions of complete thoughts hold. An element of artificiality exists in any attempt to demarcate between the two fields.

Aristotle studied kinds of assertions, or "statement-making sentences" in De interpretatione: "Not every sentence is a statement-making sentence, but only those in which there is truth or falsity".[3] The term "statement-making sentence" has originally been translated as "proposition", but, for reasons that leave much room for debate, not as "assertion".[4] He remitted the truth-valueless sentences to the realms of rhetoric, later to be glossed, quite prohibitively, as 'social acts' and 'social operations' by the Scottish philosopher Thomas Reid (1710–1790). He held an analysis of speeches outside Aristotelian propositions and associated operations of mind which they express of "real use" in logic and philosophy (Reid, 1967/1895, p. 692).

However, what Aristotle went on to investigate was the structure of the statement-making sentences, both when stated affirmatively and when stated so as to turn claims into denials. It is the affirmed quality of statement-making that is not altogether different from the qualities of assertion as distinct from the sentences that are merely propositional.

Because of this, it turns out to have been, to some extent at least, an illusory conviction of 20th-century philosophers of language that one would have to overcome Aristotle in order to capture the most sparkling pragmatic phenomena. Likewise, it has been fateful that the belief according to which all that is off-statement-making needs to be banished from exact scientific exploration was thought to have been thoroughly and single-handedly authorised by Aristotle.

Another pragmatic inclination in Aristotle's work is what is known in linguistics as the aspect of a verb, viz. the reference to the temporal layout of the action that is signified by verbs. In other words, a set of aspects of a verb phrase attempts to characterise all the different ways in which the internal temporal constituency represents itself in that expression, without external references to the constructions of tense. In Aristotelian terms, the temporal vs. aspectual distinction looms in the distinction between *energeiai* and *kineseis*. The former refers to states of affairs that represent that which obtains without being directed towards some endpoint or termination. Such affairs represent states rather than completed or perfected actions, and their internal temporal constituency is homogeneous throughout any given interval or duration. The latter refers to states of affairs that are realised by a given activity reaching its goal, and its internal temporal constituency does not remain homogeneous within some interval of time. It is customary in linguistics to refer to this Aristotelian distinction as one between atelic vs. telic verbs.

The development of aspectual characters of verbs may have started from Plato's Parmenides and Sophist. Accordingly, a rich and significant ancient history of pragmatics exists, by no means restricted to the aspectual classification of verbs. More generally, predicates, said or spoken phrases or things (the rhema, ῥῆμα, in Aristotle, deriving from Plato's idea of it performing an action given by the onoma, ὄνομα), are central elements in what the Stoics held to be forcefully conveyed, indicated, pointed out or made known by the linguistic sign as the matter of discourse, or as the matter of labour consumed in dealing with state-affairs (πράγματα).[5]

Then in the 19th century, what added to the confusion regarding the status of assertions was the unhappy disorder arising from Russell's translation of the German word 'Urtheil' in Frege's works as 'assertion' instead of the Kantian 'judgment'. Peirce was quick to note that there was a "serious error involved in making logic treat of 'judgments' in place of propositions. … A *proposition*

. . . is not to be understood as the lingual expression of a judgment. It is, on the contrary, that sign of which the judgment is one replica and the lingual expression another".[6] In one of the many places in which Peirce criticised the German tradition of logic, he added, "The German word Urtheil confounds the proposition itself with the psychological act of assenting to it" (MS L 75: 43). In another related place, he said, "Every assertion has a degree of energy" (MS L 75: 324). What Peirce is communicating here is that there is no assertion and no judgement unless there is someone or something for which it is to be a proposition. If this someone or something accepts a proposition, a judgement is made, and if he, she or it accepts the judgement, then an assertion of the proposition is made.

Peirce's prima facie casual statement that every assertion has a degree of energy nonetheless conveys a definite and significant idea. Acts of asserting should be considered analogous to action between objects in a concrete, physical sense. My assertion, by virtue of being a claim about its validity with reference to a situation, system or model of a suitable kind, exerts force upon you, the interpreter, largely because such validity is mutually testable or verifiable. Thus assertions become binding. Accordingly, I, the assertor, transmit a degree of energy to you via such assertions.

To push the analogy one step further, it could be said that assertions, like particles with gravitational mass in relativity theory, make the 'geometry' of the linguistic space of expressions to curve, so as to bring the communicating parties closer to each other in terms of mutual understanding of assertions. Such a model of a space of expressions, and the ensuing subspace of forceful assertions suitably manipulating the geometry of the space (together with speaker-oriented frames of reference), have not been much considered in pragmatics. This is an oversight, since the link is enforced by another analogue, namely one that obtains between the notion of linguistic information and the notion of entropy in physical systems, the latter also having been widely applied in communicative approaches to information. Energy was still the key outcome of the exertion of physical notion of force in the late-19th-century physics, and the notion of information and its probabilistic uses — for instance, in defining a priori probabilities in Bayesian reasoning — were impending. However, whereas it is easy to fall into the trap of analysing different notions of information, such as physical, computational, semantic or genetic, as if they were related to each other, the relation between assertoric force and physical force is nothing more than an analogue — but definitely a profitable one.

The emphasis placed on the importance of experience in tackling linguistic meaning is evident in the works of the early semanticists Dugald Stewart (1753–1828) and Michel Bréal (1832–1915). For instance, Bréal noted in his Essai de Sémantique (1897) the diachronic ingredient: "History alone can give to words that degree of precision which is needed for their right understanding"

(Bréal, 1964/1897, p. 111). According to Stewart, Thomas Reid's student in Edinburgh, "There is the *theoretical* or *conceptual* history of how human affairs may have been produced by natural causes, needed in showing how various parts of particular languages have gradually arisen" (Smith, 1980/1795, p. 293). He was giving a memorial lecture on Adam Smith, the notable analytical economist and philosopher and an early figure in the study of language and its evolution (sect. 3).

Peirce noted Stewart's sympathies towards common-sense philosophy, but considered his (and a fortiori Reid's) programme, which was mainly deployed to criticise the viability of metaphysical propositions, as quite different from his own "critical common sensism", which he took to be criticising the very methods by which any proposition is arrived at. Critical common sensism by no means accepted propositions at will, as it filtered out those that had not been logically contested.[7]

Bréal's work in Essai was, in contrast to common sensism, little more than an early tract in lexical semantics. It included considerations on analogy, metaphor, polysemy ("polysemia") and proper names. No philosophical probing into the notions of meaning or signification is to be found in his work. Illustrative of this is that any pragmatic involvedness that could have hampered his semantic analysis was unreservedly given the title "Subjective Element" of language (*ibid.*: 229–238) — a truly gigantic dustbin into which were consigned the speaker's or writer's intentions and mental components alike, including any assertoric force emanating in language use.

Aside from pragmatics, language "overflows the bound of logic on every side" (*ibid.*: 220), one example being grammar, which permits logically contradictory expressions while constituting its own non-logical system of rules for forming sentences. Welby wrote in a letter to Bréal, parts of which Bréal boldly communicated to the audience at University College, London in 1896, that there was nothing she found in his work "which contributed except indirectly, implicitly, or casually to the study of 'Sense, Meaning, or Interpretation"' (*ibid.*: 312). Bréal appears to have had a radically different conception of meaning in mind, and it is disheartening to note that, in this address, he continued to turn a blind eye to the kind of inquiry that Welby and, mutatis mutandis, also Peirce had in mind with these terms. This does not diminish the significance of Bréal in terms of the early phases of semantics and pragmatics. He anticipated the labelling of elements of language as speech acts, and introduced the concept of "latent ideas", namely presuppositions in the common ground concerning, among other things, the rudiments of grammar that speakers and hearers share in communication. But the breadth of this field of "semantics" as he came to define it was considerably limited in comparison to Peirce's semeiotics and Welby's significs. Welby's work came to define one influential trend in the pre-analytic philosophy of language, that of significs, only to be fallen into oblivion by the menacing ease of the Anglo-Saxon ordinary language philosophy. But that is another story told elsewhere (Pietarinen, 2003g; Schmitz, 1990).

2. Peirce's pragmatism vs. pragmatics

I noted in the preamble to this chapter that notions such as assertions, conventional utterances, interpretants, rationality, cooperation, habits and many others were Peirce's everyday wherewithal in his philosophy. These are also the key factors not only in pragmatic philosophy, but also in the main blocks upon which the later linguistic science of pragmatics was largely built during the 20th century.

Next I will present a couple of commonalities between Peirce's preferred concepts and the science of pragmatics. For instance, as far as assertions are concerned, Peirce noted several things that appeared later in speech-act theories. Before probing into these, I should note that, while assertions as such may be reinterpretable as linguistic acts, the consequences that Peirce assigned to the use of assertions are best interpreted along the lines of what the later game-theoretic or dialogical approaches to discourse and conversation suggest.

Accordingly, apart from physical connotations, what Peirce's "degree of energy" in assertions suggests is that assertions may be taken to possess utility, a parameter that could be used in measuring the successes and failures of assertions in conversational contexts. However, this idea is not to be implemented in the above-board sense that assertions cum assertions were to be assigned some utilities. Rather, utilities are assigned to the habits that guide agents' behaviour and their appliance and control of assertions. It was disclosed in Chapter 3 that later, due to the influence of game theory and the sociological movements of the early 20th century, the vernacular changed so that people began to refer to strategies rather than habits. Nevertheless, the concept of information, viz. what any force, such as that implicit in assertions was taken to illustrate, was codifiable as the payoffs assigned to the total strategies of the players. For sure, payoffs in game theory typically do not represent any degree of linguistic information. They are numerical values referring to virtual prices or commodity that exists in the market as a primitive notion, and used in evaluating the activities of trading. This was Adam Smith's time-honoured insight into the workings of economic systems and, as will be disclosed below, to the system of language.

What is implicit in the concept of assertion is what J. L. Austin identified not as some generic assertoric force, but as the dichotomy of illocutionary vs. perlocutionary force. This dichotomy identifies components of expressions exceeding the mere descriptive and immutable content of (declarative or non-declarative) propositions. Formally, they could be seen as functions from a proposition to some degree or quantity of force that the assertion produced in discourse is taken to possess. These refined notions of force, in that they appear to go beyond any semantic analysis of 'sense' of utterances, present us with one of the recurrent points that have been made in favour of pragmatics as a self-standing discipline separate from the goals of semantics.

This story is contestable on several counts. Peirce did not draw any dividing line between the meaning of the sentence and the differences the sentence may give rise to in terms of force. He simply could not have done that, because meaning was for him a multifarious concept that revealed several faces depending on whether the emphasis was on the interpretants created in the minds of the speaker or on the interpretants created in the minds of the hearer, or possibly on those shared by both the speaker and the interpreter. Examples of the first kind of interpretant include intentional interpretants, which embody the later-popularised idea of speaker-meaning. Examples of the second kind includes effectual interpretants created in the minds of the interpreters, and of the third kind immediate interpretants, which give the ordinary meaning of it, revealed in the right understanding of the sign.[8] The immediate interpretant is what the sign expresses, irrespective of its actual effects on the interpreter.

More to the concerns of contemporary pragmatics, the extra element that the first two interpretants in this triadic typology have, and which is used in differentiating between them, may be compared with the distinction between the illocutionary and perlocutionary force of utterances. Different illocutionary forces create different effects in the interpreters via utterers' deliberate ways of putting forth the utterance. Different perlocutionary forces create different interpretants in the interpreters via the utterer actually achieving something by the deliberate act of making the utterance. Since Peirce was merely interested in assertions, for him, utterances typically came with some illocutionary force, and the perlocutionary glut thus measured the coverage in which intentional interpretants were effectual.

Semantics, on the other hand, is not a theory of diversity of meanings, but merely one of translation. It shows what a sign means by translating it into another sign or a system of signs, such as a definition, paraphrase, translation, explication in another language, and the like. In this sense, semantics, if anyone wishes to anachronistically attribute this term to Peirce, does not reach beyond immediate interpretants.

Following scholastics, Peirce proposed a three-part division of semeiotic inquiry into speculative grammar, logic proper (critic), and methodeutic (speculative rhetoric). Charles W. Morris, accompanied by Carnap, advocated the distinction of linguistics into syntactics, semantics and pragmatics, by masking these classes on Peirce's trichotomy. True, the division between logic proper and speculative rhetoric drives a wedge between two perspectives on logic that have typically been addressed in different theories and using different conceptual resources in formal linguistics.

Morris and Carnap notwithstanding, however, that line does not quite fall where the so-called semantic/pragmatic interface is presumed to loom. The reason for this lies in the vastly more expansive breadth of Peirce's division. Methodeutic studies the general conditions of the reference of signs to their interpretants, and in this sense supersedes the notion of meaning given by the translational (semantic) account. Logic proper, in contrast, studies the general conditions of the reference of signs to their objects, and in this sense aims at providing a theory of truth. Speculative grammar concerns the general characters of the signs themselves, and in this sense does not refer beyond the signs that are its object matter.

Whether a theory of truth is purely a matter of semantics is an issue contested in many of the recent theories of logic, which criticise the overworked translational, or Tarski-type, semantic accounts on their quite unimaginative and rigid character (Chapter 8). Likewise, given the variety of different interpretants that Peirce offered, a positive answer to the question of whether the system of relating signs to their interpretants is exhausted by the received methods of pragmatics is at best arguable. For instance, the received methods in pragmatics are increasingly in need of methods employed in cognitive neurosciences and communicational studies related to the possibility of transmitting the utterer's intentions to the interpreter.

What, then, was the route by which Peirce's pragmatic ideas found their way into studies of language in the early 20th century? Linguistics, a relatively new field of investigation, was of course much more closely intertwined with philosophically and semiotically oriented cultural and anthropological studies than with any self-standing field of inquiry with its own theories and methodology.[9]

The official story goes something like the following. It was the trichotomy coined by Morris, the separation of studies on language into syntactics, semantics and pragmatics, that introduced the notion of pragmatics as a legitimate scientific field. Morris was heavily influenced by Peirce, and as noted, more often than not the underlying triad from which Morris derived his own version has been suggested to be one of Peirce's three grand divisions of inquiry, namely that of phenomenology (phaneroscopy), within which the trichotomy of speculative grammar, critic, and speculative rhetoric subsists. (The other two divisions are normative science and metaphysics, see Chapter 1). However, even though Morris would have had preserved Peirce's intentions (which he did not), the match could not have prevailed in the subsequent era of linguistics. For, in mapping the Peirce trichotomy onto the Morrisian one, not only would a vast residue result but the fields would also overlap considerably. To wit, different varieties of meaning that, from the contemporary perspective are semantic, would, in Peirce's distinction, pertain to critic and methodeutic. Moreover, what from the received perspective appear as terminological oddities such as Peirce's diagrammatic syntax (Chapter 4) is mostly semantics and even deals with pragmatic issues.

Questions also arise concerning where subjects such as cognitive science and psycholinguistics are to be put on Peirce's map of scientific classification. In so far as cognitive science is considered distinct from psychology, it mainly falls within speculative rhetoric, while psycholinguistics, for instance, seems entirely different in nature, and cannot dispense with psychology.[10] Are, and if so to what extent, psychological influences then permitted to infiltrate pragmatics? I do not intend to delve into these discrepancies here. The upshot nonetheless is that one would be well advised to take a cautious stance on the more narrow triad of syntax/semantics/pragmatics, and when needed, resume parts of the broader and intellectually more carefully considered trichotomy between speculative grammar, critic and methodeutic.

The view of pragmatics advocated by Morris primarily concerned the origin, use and effect of signs. His theory was fundamentally different from that of Peirce in that he confined these three functions of signs to contexts defined according to behavioural criteria. Accordingly, the contexts were radically narrower, because only living organisms were taken to function as sign interpreters, and because only dispositional parameters given by response sequences evoked by signs were approved within the notion of interpretants. His more stringent and deviant approach to Peirce's semeiotics is also apparent in his characterisation of pragmatics as "the science of the relation of signs to their interpreters" (Morris, 1946, p. 287). For Peirce, it was the science of the relation of signs to their interpretants, and as noted, provided for the latter a rich analytic classification.

Consequently, Morris had no real use for the concept of a habit, either, and thus contributed markedly to its insolvent demolition from the intellectual genres that followed. The only references to it were suffixed to his appendix *Some contemporary analyses of sign-processes* to Signs, Language and Behavior (Morris, 1946, pp. 285–310), the sole reason being its mockery in favour of behaviouristic stimulus-response models.

I am convinced that, as I hope will become evident, Peirce's influence was much wider than just a glimpse at Morris's misplaced promotion of it would have us believe. In fact, in tracing some of the most attractive and most revealing paths by which pragmatics found its way into late-20th-century linguistics, we do not need to rely on the informants who had access to Peirce's posthumously published writings, the first of which appeared in 1923. Peirce published an enormous amount of material during his lifetime. He was by no means a neglected figure in science or in philosophy, even though such erroneous claims have strenuously persisted, and even though proper acknowledgement of his influence has on more than a few occasions been ambivalent. People who had first-hand access to and acquaintance with Peirce's original manuscripts deposited at Harvard in 1915, were happy to publicise his findings as moulded by their own motivations, often without acknowledgement. One of the most awkward moments in the early history of the development of first-order logic took place when Russell, irritated by Peirce's dismissal of his Principles of Mathematics in his one-paragraph review published in the Nation in 1903, started to advertise the then-quite-exclusive Peirce–Peano logic of quantifiers as the Frege–Peano logic. Since three or more epithets are banned in philosophy, thereafter it came to be known as the 'Frege–Russell' logic.[11]

Peirce also recognised the existence of conventional utterances, subspecies of symbolic signs (mostly legisigns) in language, or icons the likeness of which is guided by conventional rules, which imply special habits or rules of interpretation affecting the conduct of the interpreter.[12] According to him, unlike other assertions, "upon [the falsehood of conventional utterances] no punishment at all is visited" (5.546). Such conventional utterances may pertain to person-to-person or intrapersonal communication and dialogue alike. This remark is thus only a solitary piece of evidence that adds to the cumulative weight of the totality of such evidence concerning his capacity of doing truly broad-spectrum pragmatics.

In fact, the conventionality of some signs and their pragmatic value was recognised betimes by young Peirce:

> [The major premiss of an arbitrary nature] is determined by the conventions of language, and expresses the occasion upon which a word is to be used; and in the formation of a sensation, it is determined by the constitution of our nature, and expresses the occasions upon which sensation, or a natural mental sign, arises. Thus, the sensation, so far as it represents something, is determined, according to a logical law, by previous cognitions; that is to say, these cognitions determine that there shall be a sensation. (5.291, 1868, *Some Consequences of Four Incapacities*).

But now, let us set Peirce in a broader context.

3. Economics, evolution and language change: some predecessors, contemporaries and followers

Let me observe in this section a couple of key insights from thinkers sharing similar views to Peirce's pragmatic and evolutionary concept of linguistic meaning. One topic to be emphasised here is on the question of to what extent economic considerations permeate language meaning and its change. This provides some historical background to the topic of language evolution discussed in the preceding chapter.

Adam Smith (1723–1790) Language and economics have been congenial, and to some extent complementary, issues of human behaviour since Adam Smith and the observation that that they both attempt to explain regularities in human interaction and the design of social systems. Several attempts have been made recently to revive this by studying Gricean pragmatics from the viewpoint of game theory and econometrics. Other, more recent contact points between language and economic considerations are to be found, for instance, in applications of optimality theory, according to which costs are visited on constraints given to multiple interpretations of an expression, together with a preference structure. Unfortunately, optimality theory will not be able to provide a systematic theory of interactive emergence and development of meaning, let alone pragmatics, since it lacks the third, referential or sense-giving dimension to the form-content pairs it attempts to constrain and evaluate.

In terms of just some of the aspects that are relevant to my topic, it was Smith's view that rules and protocols governing language arise by mutual consent, and that language as a rule-based system develops and is revised in accordance with usage over periods of time (Otteson, 2002). Like Wittgenstein almost two hundred years later, Smith made a major theoretical point in emphasising the priority of use over rigid rule-governing. He referred to the affinity between trade and language in The Wealth of Nations (1776) as two sides of the same underlying human process.[13] However, this reasoning is absent in his earlier Dissertation on the Origin of Languages (1767), in which he concentrates largely on what we recognise as lexical semantics: the focus is on the emergence of simple words such as proper names, adjectives, prepositions, comparatives, demonstrative pronouns and verbs (Smith, 1970/1767).

In Dissertation, Smith distinguishes between words that emerge from the desire to express qualities of objects, and those that emerge from the desire to express the relations in which objects stand to each other. This distinction, with its Kantian influence, covers Peirce's categories of firstness and secondness. Most of the discussion on thirdness, the appearance of the mediation of a mind that brings secondness and firstness into the relation, not found in this study, are nonetheless found elsewhere in Smith's studies on economics.

After all, the concept of the 'invisible hand' could in my view be seen as an instantiation of a law or a habit that organises the other two categories. Peirce was familiar with Smith's work, and considered his investigations on analytical economics to be a captivating attempt to develop a philosophy of common sense, of which pragmatism was to be the most elaborate and exact account, "especially by emphasizing the point that there is no intellectual value in mere feeling *per se*, but that the whole function of thinking consists in the regulation of conduct".[14]

William Dwight Whitney (1827–1894) Similarly Whitney, who wrote extensively on both theoretical and empirical aspects on language development and evolution, shared an uncluttered view of communication not merely as a system intended for pooling linguistic structures and the thoughts of different actors together, but as one signifying mediation: "Speech is not a personal possession, but a social; it belongs, not to the individual, but to the member of society. No item of existing language is the work of an individual; for what we may severally choose to say is not language until it be accepted and employed by our fellows. That is a word . . . as the sign of an idea; and their mutual understanding is the only tie which connects it with that idea".[15] Whitney displays here a linguistic counterpart to what Peirce made a philosophical doctrine, namely that no agent is absolutely an individual, but always a part of a community that has interconnected thought. Again, we come across the idea of speech as communal use, and the necessary existence of the common ground via mutual intelligibility and understanding as a test for the unity and coherence of language. These points clearly come out in comments such as, "Speech is strictly a social institution", and "The ideas of speech and community are inseparable" (*ibid.*: 105). Unlike Wittgenstein on these points (Chapter 8), Whitney was, like any linguist worth her salt at the time, eager to emphasise the necessity of the presence of social circumstances underlining the emergence, existence and development of language.

Furthermore, it was not only the social aspects of language, but also economic considerations of the Smithean kind that Whitney emphasised: "The history of the phonetic forms of words . . . has been recognized as a principal factor in that history, a tendency to economy, to the saving of effort, in the work of articulate utterance" (*ibid.*: 249). Similar arguments from economics have permeated contemporary linguistics in the form of optimality theory, and in computational simulations concerning how grammatical rules and simple vocabularies emerge in populations. These include the 'survival of the clearest' types of evolutionary arguments currently in vogue (Chapter 11).

Roman Jakobson, in *The World Response to Whitney's Principles of Linguistic Science*, which was his introduction to (Whitney, 1971), notes that parts of Whitney's writings are in close correspondence with the continual inquiry of Peirce, but that Peirce did not refer to his New England countryman (Whitney, 1971, p. xxvii). In fact, neither Jakobson nor, as far as I know, anyone else has noted that, as Whitney was the Editor-in-Chief of the first edition of the Century Dictionary and Cyclopedia (CD), he must have been familiar with Peirce's mindset, even though he probably would not have bothered to study his writings in depth, given his divergent interests and motivations.

Peirce contributed thousands of definitions to the CD during the 1880s. His subjects covered logic, metaphysics, mathematics, mechanics, astronomy, weights and measures, colour terms, and many common words of philosophical and scientific import, but not those to do with linguistics unless they overlapped with other fields, in which case multiple authorship is, of course, also possible. Strangely enough, Whitney was not among Peirce's correspondents, so apparently he was not very actively involved in editing Peirce's contributions. The two men must have been personal acquaintances, though. Whitney himself recognised Peirce's assistance not only by virtue of his contributions, but also from the numerous other suggestions that he made to the editors of the CD. Whitney was also appointed as a lecturer in comparative philology in 1879 at John Hopkins University: Peirce was appointed in June of the same year, and worked as a lecturer in logic there until 1884. This marked the beginning of their association. The design of the CD was started in early 1882.

Maurice Bloomfield (1855–1928) Yet another Hopkinsian was the orientalist Maurice Bloomfield, who contributed to the CD entries on Semasiology: "The science of the development and connections of the meanings of words; the department of significance in philology" (CD VIII: 5481, cf. Sematology). What is notable is his 'acoustic' account of the semantic expressivity of words, which he formulated as early as in 1895:

> Every word, in so far as it is semantically expressive, may establish, by haphazard favoritism, a union between its meaning and any of its sounds, and then send forth this sound (or sounds) upon predatory expeditions into domains where the sound is a first a stranger and parasite. . . . No word may consider itself permanently exempt from the call to pay tribute to some congeneric expression, no matter how distant the semasiological cousinship; . . .The signification of any word is arbitrarily attached to some sound element contained in it, and then cogeneric names are created by means of this infused, or we might say, irradiated, or inspired element. (Bloomfield, 1895).

Semantics warranted its own entry in the supplement volume of the Dictionary in 1909: "That branch of philology which is concerned with the meaning of words and the development of meanings; semasiology" (CDS XII: 1194). This was most likely also written by Bloomfield. The entry also refers to Bréal's inauguration of the word 'semantics' in the English translation (1900) of his Essais.

I believe this single reference has been exceptionally instrumental in getting the term 'semantics' ahead of 'semasiology', 'sematology', 'semology', 'significs', or any other of the variants that rivalled it. Although he published little on theoretical aspects of language meaning and philosophy, Bloomfield's small contribution to semantics appears to be of a magnitude of his much better-known namesake Leonard (1887–1949).

In the following couple of pages I will uncover some similarities between the pragmatic views that have surfaced in the writings of some early linguists and philosophers of language, such as Philipp Wegener, Bronislaw Malinowski, Ludwig Wittgenstein, and several others, in the period between Peirce and Grice, roughly from 1880s to 1950s.

Philipp Wegener (1848–1916) Many scholars besides Morris have played a part in linking pragmatism — mostly unintentionally and, loosely speaking, deliberately following either Peirce or James — with linguistic pragmatics. One of the original linguists of this ilk was the German linguist Philipp Wegener, whose 1885 monograph Untersuchungen über die Grundfragen des Sprachlebens maintained a modern view of language as a communicative system of interaction (Wegener, 1885). In the spirit of Kant, and like Peirce, he maintained that new information was given in the logical predicate of a sentence, which was to be communicated so that it became understood in the context, the situation of the appeared utterance ("die Situation der Anschauung", *ibid.*: 21). A similar idea comes out in the ancient logical subject vs. logical predicate and the theme vs. rhema distinctions, as well as in the the Kantian schema vs. content dichotomy. As an heir to functional linguistics, we nowadays tend to find the dichotomy in terms of topic (exposition, subject, focus) vs. comment (predicate) of assertions.

There were other kinds of situations revolving around the notion of speech and its comprehension that Wegener distinguished. Among them was one concerning speaker's and hearer's presuppositions, ("die Situation die Weltanschauung", *ibid.*: 26). Together with the cultural situation ("die Cultursituation" *ibid.*: 27), Wegener came to identify the chief ingredients that constitute the common ground of the interlocutors.

Given that cultural and historical factors shape linguistic understanding, Wegener made the observation that, in speech, what secured the understanding of the predicate in the embedded sentence was the recognition of the speaker's intentions and his or her meaning in communicating the information encoded in the utterance. The hearer or the interpreter plays a vital part in Wegener's outlook on communication in updating his or her beliefs and expectations according to the progress made in recognising the utterer's intentions.

As it will turn out in a more fleshed and boned fashion below, Wegener's account of communication was less one-sided than the much later pragmatic theories pressing on the precarious notion of speaker's meaning would have us believe. However, in doing so, it is unlikely that he knew anything of Peirce's work, especially his arguments for the 'collateral experience and observation' that are presupposed in order for any speech to survive, or, for that matter, of James or any of the other American pragmatists of his era.

Bronislaw Malinowski (1884–1942) Building on Wegener's findings, Bronislaw Malinowski, despite being mostly an ethnographer and a field linguist, leaned towards linguistic theories that were influenced by behaviourism and contextuality of speech. His empirical investigations were aimed at throwing light on the question of human language and semantics. He tried to develop an ethnolinguistic theory, which in The Language of Magic and Gardening (1935) was claimed to "show us what is essential in language and what, therefore, must remain the same throughout the whole range of linguistics varieties; how linguistic forms are influences by physiological, mental, social and cultural elements; what is the real nature of meaning and form, and how they correspond" (Malinowski, 1935, p. ix).

He was interested in linguistic universals, and in the possibility of a universal semantic theory of language. Like other empirically-minded linguists and behaviourist psychologists, he denied that language was a medium for the communication of conceptual elements of thought, or a medium for expressing propositional attitudes of the subject. Consequently, he became one of the researchers who contributed to the suppression of the view of language espoused by American pragmatism, namely the idea that the speaker and the hearer possess a growing idea potential in their habits. For him, the contextuality of language was proof that utterances were acts not to be considered independently of the situations in which they were uttered, and hence were bound to cultural and behavioural attributes. Furthermore, contextuality is vital for learning the meaning of words, which is best conducted in situations in which a given word for the speaker and the hearer is the most consequential. Therefore, even though Peirce would have approved some version of contextualism, Malinowski was certainly not among them in his dismissal of organisms' living influence upon each other in terms of their inner habit-growing potential.

Though thus leaving some conceptual gaps in the chain of pragmatic theories that started with Peirce and deteriorated with Morris, Malinowski made important remarks on pragmatism and the pragmatic meaning of utterances. He considered pragmatism to be the effective force behind all rules of conduct (*ibid.*: 49). As such, this is, in fact, closer to pragmatism than to pragmatics, the latter of which being suggested by Morris at around the same time.

Elements of pragmatism are also found in Malinowski's remark that it is in situations to which words belong that they "achieve an immediate, practical effect" (*ibid.*: 52), and in which they also acquire their meaning. Similarly, on page 214: "The meaning of any significant word, sentence or phrase is *the effective change* brought about by the utterance within the context of situation to which it is wedded". The use of the phrase "practical effect" may be thought to reflect Peirce's pragmatic maxim, and the idea of a meaning as a change in context is familiar both in dynamic theories of meaning in logic and in theories of relevance. However, the meaning of a single utterance is then defined as "the change produced by this sound in the behaviour of people" (*ibid.*: 59).

Here we may observe, once again, the fundamental differences and discrepancies between this contextual conception and the original intention of pragmatic philosophy. The mere reduction to the behaviour of language users overlooks the tenet of pragmatism, according to which it is the growth and change in agents' habits that will reflect the impact utterances have on a situation, and of which the change in the behaviour is merely a by-product. Nevertheless, these pragmatic tones were probably amplified by Odgen & Richards' The Meaning of Meaning (1923), in which snippets from the Peirce–Welby correspondence were reproduced, and in which one of Malinowski's earlier essays ('The problem of meaning in primitive languages') was reprinted as an appendix.

Some may hold that from the tenet of contextualism, irrespective of whether we dress it in linguistic or philosophical outfit, there is only a short step to be taken to arrive at relativism, the denial that there are universal truths concerning language and the world. Furthermore, the story goes, in virtue of being closely connected with pragmatic ideas, pragmatism may be seen, via contextualism, aligned with relativistic worldview. It is wrong to assume either of these reasonings. Pragmatism, even though co-existent with and to a degree congenial to contextualism, does not imply relativism, since it does not hold all analyses of knowledge as equally valid or invalid. Just as it is wrong to take pragmatism to be a value-free enterprise, possessing no ideals of successful inquiry, it is wrong to assume that its epistemology would not be grounded on some foundational principles (evolution is one such principle). Likewise, contextual views on language do not approve the denial of universal aspects of language, even though it may disapprove their absolute, eternal and all-encompassing character.

Thus the considerable intrinsic interest in the question of whether the philosophers commonly characterised as holding a version of contextual thought and meaning, may also be viewed as pragmatists.

Ludwig Wittgenstein (1889–1951) Wittgenstein dismissed the idea of he himself being a pragmatist by stating, "No. For I am not saying that a proposition is true if it is useful".[16] The term pragmatism was related here to the philosophical doctrine that Ramsey was communicating to Wittgenstein, and which was available both in Peirce's Monist series and in William James' writings. However, the kind of pragmatism that allies with 'truth as usefulness' is a far cry from Peirce's pragmatism. Therefore, the dismissal need not and ought not to interpret literally. It demonstrates Wittgenstein's alienation from James' version in his later years, while not necessarily alienating himself from Peirce. I contend that a little pragmatist lived inside Wittgenstein, even though it did not come out by way of canonical standards of philosophical doctrinisation.

What, then, were the elements of pragmatism that Wittgenstein endorsed? His thinking shifted during his middle period (roughly 1928–1929) towards what has slightly delusively been dubbed the 'meaning as use' doctrine.[17] How radical the change was is not for discussion here. I only wish to note that there were elements of continuity throughout, even though it may be more fashionable to emphasise the differences. Furthermore, as I noted in Chapter 2, if we take the 'meaning as use' doctrine at its face value, it is articulated in surprisingly similar terms in Peirce's article, published in the Nation in 1899: "The meaning of a word lies in the use that is to be made of it".

An example of such continuity is to be found in Wittgenstein's struggle in spelling out what 'non-literal' or 'non-linguistic' meaning was. In Tractatus, the distinction between saying and showing subsumed, in effect, a distinction concerning the problem of communicating 'literal' and 'non-literal' content by way of picture-like interiors of elementary propositions. When propositions grew deadly serious for him, the shift he had to make was towards the language-game idea. The reason for this was the blurred and imperfect quality of such pictures, which badly served the purposes of depicting general assertions. Games, on the other hand, were conveniently subservient to those human activities that create such pictures, and so the robust and static 'picture-world' link came to be replaced by dynamic language games as the primary medium of all communication.

Consequently, after the establishment of the language-game idea, the showing saying dualism was much less indispensable than it was before, because non-literal and non-linguistic content was now communicable by whatever properties there were in the overall family of such language-games. When Wittgenstein spoke of "one of the most important language-games", he essentially referred to the types of games that are simultaneously both able to show (point) and to say (or tell) what one sees to be the case (Chapter 8). From roughly 1930 onwards, the difference was no longer more than a matter of words.

These thoughts may be contrasted with Peirce's approach to logic. Aspects of the picture-theory of language, in capacity of mapping pictures to the object of their representation, have an appealing resemblance to Peirce's diagrammatic logic. I do not profess any strict connection between these quite differently motivated ideas, but it is unmistakable that some pictorial characters, for instance, the notion of isomorphism, were carried over to the language-game idea.

Diagrammatisation is nevertheless one of the most auspicious methods Wittgenstein could have employed in struggling to preserve the polarity idea of elementary propositions in the more complex cases of propositions. Diagrammatic images of general propositions need not to be indefinite or vague, even though their negations may be; it only leaves the determination of interpretation of the affirmative proposition to one who has a different polarity from the interpreter of its negation. The more indefinite the picture is, the more latitude of interpretation there is for its interpreter. Analogously, the more non-literal meaning exists in the assertion, the more there is for the interpreter to work out in terms of what interpretation was intended by the utterer.

Wittgenstein came exceedingly close to actually seeing what the conservation of aspects of the picture-like idea of propositions would have demanded. With reference to negation, he suggested: "Someone might show his understanding of the proposition 'The book is not red' by throwing away the red when preparing the model" (Wittgenstein, 1978, p. 10). This 'throwing away' is precisely what follows from the diagrammatisation of the proposition by means of cutting and severing the rhema (uninterpreted predicate term) ___ *is red* from the sheet of assertion on which the proposition is scribed. It even follows in the right order: first one interprets *the book* by connecting the subject with the predicate, but when encountering the cut he or she realises that what is enclosed within the cut is not asserted but denied. This occasional brainwave was never systematised, and it had to wait for later innovations to be put into use.

There is no immediate novelty in the basic idea, however. In Sophist, Plato wrote, "You see, Theaetetus, it is extremely difficult to understand how a man is to say or think that falsehood really exists and in saying this not be involved in contradiction".[18]

In our age, cognitive linguists have resurrected the idea that negation is really correlated with the idea of neuronal inhibition.

With respect to the overall genus of pictures, and especially during his transition phase, Wittgenstein was apt to call them 'portraits' instead of pictures. One is reminded here of Peirce's slogan describing his iconic graphs as "portraitures of thought" (Chapter 4). Chronologically, there appears to be a smooth transition from Peirce's diagrammatisation of logic to Wittgenstein's picture theory of language, but being independently invented for different purposes, they were bound to be substantially dissimilar. Peirce's aim was to examine and explore the relations represented by any graph instance, given its structural, iconic similarity with the objects it represents. This process was to account for necessary reasoning. In his early phase as laid out in Tractatus, Wittgenstein's motivation was to consider propositions as total pictures or images of states of affairs. The remnants of picture theory that were preserved throughout his writings were, in fact, quite close to Peirce's idea of iconic compositions of pictures as snapshots of the propositions presented in thoughts by conscious minds, although, unlike Peirce, he never went on to draw an analogy of pictures as dynamic, moving pictures of actions of thought.

Moreover, Wittgenstein's view of language marked a considerable step towards the science of pragmatics. To see the origins and growth of this view throughout his thinking we nevertheless need to go back to his formative years.

In 1914, he was of the opinion that all propositions were unasserted, and assertions were merely psychological.[19] His early conception of assertion revolved around this psychological view as distinct from strictly binary truth-valued propositions. The extraction of assertoric elements from statements was preserved until after his final revisions in the Philosophical Investigations, but at that time the subject had taken a linguistic turn towards speech acts.

In close agreement with such a turn, Wittgenstein considered and rejected the notion, suggested by Frege, that sentences might contain a supposed *Annahme* ('supposal') which was being asserted, and that there might be some special assertion signs such as question marks or signs pertaining to intonation, that delineated the part of the sentence that was assigned a truth-value from the rest of it. Wittgenstein contemplated that according to these proposals, an assertion could consist of the part of 'considering' and the part of 'asserting', and "that we perform these acts according to the signs of the sentence, almost as we sing from the notes".[20]

Wittgenstein nonetheless repudiated such acts. As the published version of the Investigations witnesses, what people say or assert is true or false, and they agree on what they say in that very language (Wittgenstein, 1953, 241). Such language is an example of the life-form of human beings, because any assertion turning out to be of contrary truth value than what was expected in intending it, shows an infringement of the mutual agreement made in the language, and for that reason by uttering false assertions one runs the risk of punishment.[21]

With the wonderful benefit of hindsight, we might say that not only false assertions, but any material breach of the conversational maxims formulated by Grice soon after Wittgenstein also counts as a case in which the party blameworthy of such an offence may incur penalties.[22] How such penalties are actually implemented is, of course, an entirely different matter, but they could well be considered in terms of quantities deducted from the payoffs assigned to the overall conversational strategies of the parties engaged in communication.

At this point I reach an intermediate conclusion. It is the idea of language as a form of life that provides us with what has been referred to as the common ground of language users. Forms of life are ways of experiencing, and the ways of enjoying common traits in experience is what everyone has and is mutually known and agreed to have by others. This also fits in with my previous comment on one of Wittgenstein's most important language games, namely the game of showing and saying what one sees considered in Chapter 8. What Wittgenstein took "using language" in "ordinary life" to presuppose was exactly these kinds of games (Wittgenstein, 2000–, 141: 1). Language games exist as roles in our ordinary life, and so we do not call using language a game at all if it is not hooked up with what experiences — all that can be given away, communicated or narrated by these most important activities — in human life provide.[23]

Nurturing the common ground requires a complex system of presuppositions in forms of life. According to Wittgenstein, "What we do in our language-game always rests on a tacit presupposition" (Wittgenstein, 1953, p. 179e). Such tacit presuppositions include the mutually agreed existence of presuppositions themselves: "Suppose we ever really express ourselves like this: "Naturally I am presupposing that...? — Or do we not do so only because the other person already knows that?" (Wittgenstein, 1953, p. 180e). Just how tacit these presuppositions and rules of language may be is shown by the fact that, under many circumstances, they are recognised and followed blindly, without intervention of conscious or aware interpretation. This is witnessed by such diverse issues as the implicit/explicit distinction uncovered in neuroscientific experiments and the habitual and non-consciously rational character of strategies ascribed to populations in evolutionary game theory. It has later turned out, however, that spelling out the structure of mutually agreed presuppositions and the system of propositional attitudes may be uncompromisingly hard.

It is worth considering whether Malinowski — presumably his *The problem of meaning in primitive languages* appended to Ogden & Richard's 1923 book — influenced Wittgenstein and his move towards the 'middle period' and further beyond, the period between Tractatus and the use-oriented view of language-games that was to emerge in the 1930s. Forms of life as public language-games suggests some initial sympathy between the two. Malinowski purported to spell out by his contextual theory of speech the essence of language by charting the invariants that are preserved through the range of linguistic variations, including linguistic forms influenced by physiological, mental, social and cultural elements. He thought that such an investigation would reveal the nature of the correspondence between meaning and form.

Despite appearances, this project did not impress Wittgenstein. Even though Wittgenstein played with the idea of 'virtual ethnography', that is, with thought experiments concerning what some particular tribe with a different culture might have meant by an expression recognised by us, his philosophical concept of language was not grounded on sociological, anthropological and ethnographic investigations as was Malinowski's, and showed little regard for such empirically established contexts of language use.

Another difference is that for Wittgenstein, talking about language or giving explanations of its use or function was something that could be done only in the very same language, not in some preparatory or provisional language devised for that purpose (Hintikka, 1996b). Whether a comparable view on the universality of language is found in Malinowski's work is suspicious, since he endorsed the pragmatic definition of meaning as an effective change in the context of utterance. Typically, such pragmatic leanings are allied with the view that the semantic relations between the language and the world can be varied from a metalinguistic perspective.[24]

Although there is more that could be said of Wittgenstein's role in pulling out numerous threads allied to pragmatism and in putting them to his personal use, and in anticipating many issues in the emerging field of pragmatics, I will leave the issue here, and after a few words on some other names, look into one of the most influential thinkers of the above-mentioned era, namely Grice and aspects of his pragmatic leanings in what has too often been mislabelled the 'psychological' programme of pragmatics.

Some other figures Other figures in the early history of semantics and pragmatics more or less related to Peirce's pragmatic philosophy include the English psychologist and philosopher George Frederick Stout (1860–1944), one of the first experimental psychologists alongside Peirce, with whom Peirce, together with Baldwin, wrote a definition "Whole (and parts)" in Baldwin's Dictionary, and the German Wilhelm Wundt (1832–1920), whose psychological philosophy Peirce harshly criticised, but whose physiological definition of thinking as a regulation of sensory organs he applauded (8.201, note 3, c.1905). How enthralled would both have been by the recent 'language organ' debate! Wundt's views on language were moulded by his numerous contacts with many like-minded colleagues, including Malinowski, and he shared, together with Peirce, the ultimate concern of linguistics to answer the question concerning the origins of language.

Following Odgen & Richards's book, another significant event contributing to the dissemination of Peirce's pragmatism was its transportation to the philosophical atmosphere of 1930s Vienna by Karl Menger. Whilst visiting Harvard in 1930 he was enthusiastically introduced to Peirce's work by one of the editors of the upcoming Collected Papers, Paul Weiss (Menger, 1994). Despite Menger's efforts, the influence did not bear immediate fruit. More significant in Europe was the Significs Movement in Amsterdam inspired by Welby, which was much closer and receptive to Peirce's way of thinking than either logical empiricism or the ill-fated analytic philosophy.

Furthermore, J. L. Austin, an archetype of the Oxonian 'run-of-the-mill' language philosophers, had an idea of what the prefiguring semioticians and semiologicians were after at the time he was embarking on his performative edition of the speech-act theory. As far as Peirce was concerned, Austin did not consider his statements to be particularly adequate, however: "With all his 66 divisions of signs, Peirce does not, I believe, distinguish between a sentence and a statement" (Austin, 1960, p. 87 n.).[25] By 'statement', Austin means assertion, "the utterance by a certain speaker or writer of certain words (a sentence) to an audience with reference to an historic situation, event or what not" (*ibid.*: 87–88). Needless to say, Peirce brought out such a division forcefully in his study of assertions decades earlier.

The English linguist Alan Henderson Gardiner's (1879–1963) statement in his 1932 book The Theory of Speech and Language that one of the generic situations within which utterances are put forward is the "Situation of Common Knowledge" (Gardiner, 1932, p. 51) is also relevant here. (Other situations are those of "Presence" and "Imagination".) All these resonate closely with Wegener's categorisation of situations in his 1885 monograph, which Gardiner held to have been written by the "pioneer of linguistic theory" (p. v). They both foresaw the distinction between meaning and speaker-meaning; for Gardiner this was the difference between "meaning" and "thing-meant" in and around words and sentences, and similarly as for Wegener, the pairing involved the listener's deduction of meaning partly from the words uttered and partly from the context of the situation. According to Gardiner, the thing-meant has to be "identified by the listener on the basis of the word-meanings submitted for him" (p. 11), and cannot be directly and unambiguously merely shown.

Since this aspect of Gardiner's theory of language and the German psychologist and linguist Karl Bühler's (1879–1963) *Appell* of his 'organon-model' of language were comparable, and since Wittgenstein was acquainted with Bühler's ideas by the late 1920s, it is possible that this burden of interpretation laid on the listener by such a distinction (or one of its several manifestations), marked the beginnings of Wittgenstein's alienation from any strict saying-showing dichotomy (Chapter 8).[26] Nevertheless, the remark on common knowledge is unmistakable, and demonstrates further the falsity of its postulated novelty in the post-Gricean pragmatics.

Before Peirce, the American philosopher and linguist Alexander Bryan Johnson (1786–1867) formulated a verification criteria for the meaningfulness of propositions used in asserting statements. Unlike many who succeeded him, he did not confine his explorations to lexical meaning, even though the distinction he made between a "verbal meaning" and a "sensible meaning" principally moved on the level of words (Johnson, 1959/1836, pp. 149, 264–266). According to him, the sensible signification is revealed in our senses, whereas the verbal signification is given in words, but this classification was also applicable to propositions composed of words, as indeed to definitions of physical and mathematical objects. Again, these considerations illustrate significant anticipation of the distinction between literal vs. non-literal meanings, viz. the meaning given in language vs. the meaning given in our senses. It should to be borne in mind that in the latter distinction, the object and the utterer may be one and the same, in which case the utterance originating from the object is that which is produced in the mind of the interpreter. According to this view, Johnson's considerations went beyond purely linguistic signs and speech acts and leaned towards Peircean phaneroscopy.

There is no reference to Johnson in Peirce's corpus, which is no surprise as Johnson's work went unnoticed until the republication of A Treatise on Language in 1947. Even when reissued in 1967, its editor David Rynin mourned: "Philosophers have still never heard of Johnson, almost nothing has been written about him, and to my knowledge his name does not yet appear in any history of philosophy . . . a most remarkable fact of nonhistory" (Todd & Blackwood, 1967, pp. 22–23). The same may be pronounced almost 40 years later.

According to Johnson, any answer to what meaning is needs to be derived, and often reconstructed, from the actions of language users situated and embodied in particular situations, such as cultural, historical, social, perspectival and recollective ones. This is a case in point in which Johnson comes close to Wittgenstein's pragmatically-influenced remarks on language games. In fact, Wittgenstein possessed a copy of Johnson's book, and thus it is quite possible that it had some impact on Wittgenstein's thinking during his last years.

4. Grice in the wake of Peirce

By the end of the 20th century, Herbert Paul Grice (1913–1988) was acknowledged to have virtually redefined the state of pragmatics. Influenced initially by the so-called English 'ordinary language philosophers', of late his impact has been felt in virtually all areas of pragmatics, ranging from computational dialogue systems to theoretical work on the logic of conversational maxims and the theory of games and decisions.

My aim in this section is to record some evidence of the claim that there is a reasonably direct and unswerving link between Grice's views on pragmatics and Peirce's pragmatism. Lest the extent of this resemblance is misunderstood, I will also note the main differences between the two.

To begin with, consider the following passage from Peirce:

> Honest people, when not joking, intend to make the meaning of their words determinate, so that there shall be no latitude of interpretation at all. That is to say, the character of their meaning consists in the implications and non-implications of their words; and they intend to fix what is implied and what is not implied. They believe that they succeed in doing so, and if their chat is about the theory of numbers, perhaps they may. But the further their topics are from such presciss, or "abstract," subjects, the less possibility is there of such precision of speech. In so far as the implication is not determinate, it is usually left vague; but there are cases where an unwillingness to dwell on disagreeable subjects causes the utterer to leave the determination of the implication to the interpreter; as if one says, "That creature is filthy, in every sense of the term." (5.447, 1905, *Issues of Pragmaticism*).

Now compare this to Grice:

> I wish to represent a certain subclass of nonconventional implicatures, which I shall call *conversational* implicatures, as being essentially connected with certain general features of discourse; . . . Our talk exchanges do not normally consist of a succession of disconnected remarks, and would not be rational if they did. They are characteristically, to some degree at least, cooperative efforts; and each participant recognizes in them, to some extent, a common purpose or set of purposes, or at least a mutually accepted direction. This purpose of direction may be fixed from the start (e.g., by an initial proposal of a question for discussion), or it may be so indefinite as to leave very considerable latitude to the participants (as in casual conversations). But at each stage, *some* possible conversational moves would be excluded as conversationally unsuitable. We might then formulate a rough general principle which participants will be expected (ceteris paribus) to observe, namely: Make your conversational contribution such as is required, at the stage at which it occurs, by the accepted purpose or direction of the talk exchange in which you are engaged. One might label this the Cooperative Principle. (Grice 1989, p. 26, *William James Lectures*, 1967).

I am not claiming that Peirce had in mind precisely the kind of Cooperative Principle coined by Grice sixty years later. Nor is it very plausible that Grice formulated his principle while being thoroughly cognisant of the above or similar unpublished passages in Peirce. In fact, the force of the cooperative principle seems to have been over-valued by commentators on Grice's work. For one thing, it does not single out competition. Many ordinary conversations are conducted under competitive conditions while preserving cooperation.

Nonetheless, the affinity of the above two samples is more than skin-deep. Both Peirce and Grice are seen to bring out as the main ingredient in successful communication and speech the common and shared purpose of the utterer and the interpreter. An interpreter to whom utterances are addressed is needed in order to be able to even begin a full-scale analysis of the meaning of a sign. It falls on the interpreter to recognise that the utterer is present both in the utterance and as a deliverer of it. This point reminds us of the pair of concepts that Grice famously emphasised, the distinction between the utterance's literal meaning and the speaker's meaning conveyed by it.

Furthermore, the notion of honesty that Peirce is alluding to is one of the properties needed to satisfy Grice's maxim of Quality, constituted by principles such as, “Try to make your contribution one that is true”, “Do not say what you believe to be false” and “Do not say that for which you lack adequate evidence” (Grice, 1989, p. 27).

What is equally remarkable here is that Peirce recognised the meaning of utterances in conversational settings as delivering both intended and non-intended content, and thus including in the overall meaning of such chains of utterances both implied and non-implied information. The more casual the topic of the conversation is, the more effort is required from the part of the utterer to balance between the two ends. As Peirce notes, sometimes, for the sake of expediency, it is the intention of the utterer to pronounce things that have deliberately non-determined implications.

Note, then, the following items that Grice requires the hearer to reply to when working out that a particular conversational implicature is present:

> (1) The conventional meaning of the words used, together with the identity of any references that may be involved; (2) the Cooperative Principle and its maxims; (3) the context, linguistic or otherwise, of the utterance; (4) other items of background knowledge; and (5) the fact (or supposed fact) that all relevant items falling under the previous headings are available to both participants and both participants know or assume this to be the case. (Grice, 1989, p. 31).

These constitute the common ground of the speaker and the hearer, referring to the knowledge, beliefs, expectations, discourse entities, propositions, presuppositions and attitudes common to the speaker and the addressee. Now let us compare this with the somewhat more verbose formulation of Peirce, addressing his reader in a characteristically dialogical and somewhat verbose style:

> There are some points concerning which you and I are thoroughly agreed, at the very outset. For instance, that you know the English language — at least tolerably. I am positively sure that you cannot deny that; — at any rate, not in English. There is much more that it will not be unreasonable to assume that you will assent to; such as that you know the rudiments of grammar, — meaning, of course, Aryan grammar, which is often called “universal grammar”; — that you have most of the leading attributes of the genus Homo, as set down in the books of physiology and of psychology. Nay, for more than that, you have had, I will wager, an experience of life quite similar in a general way, as regards the smaller and more elementary items of experience to mine. Among these I can instance this, that you, like me, have acquired considerable control not only of the movements of your limbs but also over your thoughts. If we were to meet in the flesh, we should both take it for granted. I should know that it was so, and know that you knew it, and knew that I knew that you knew it; and so on, *ad infinitum* and *vice versa*.[27]

Peirce took these items to constitute the most important characteristics of the common ground between the speaker and the hearer (or as is the case here, between the writer and the reader).

On the notion of common knowledge, Peirce wrote:

> No man can communicate the smallest item of information to his brother-man unless they have πoῦ στῶσι[28] of common familiar knowledge; where the word ‘familiar’ refers less to how well the object is known than to the manner of the knowing. This manner is such that when one knows anything familiarly, one familiarly knows that one knows it and can also distinguish it from other things. Common familiar knowledge is such that each knower knows that every other familiarly knows it, and familiarly knows that every other one of the knowers has a familiar knowledge of all this. Of course, two endless series of knowings are involved; but knowing is not an action but a habit, which may remain passive for an indefinite time. (MS 614: 1–2, 18 November 1908).

Despite its formally infinitary nature, the adoption of common knowledge in communicative situations is not essentially a diachronic process but represents a habitual state with a structure of attainable form, without having to span any actual period of time. This discredits Schiffer (1972), who doubted the feasibility of such ad infinitum reciprocity precisely because of its infinitary and practically infeasible character. Similar worries were raised in Johnson-Laird (1981). This matter did not worry Peirce, who had had frequent adoptions of infinite constructions in mathematics, especially in relation to continuity, infinitesimals and large collections. To be sure, beyond the question of how "common familiar knowledge" comes about in conversations or in the interpretation of utterances, an interpretation of a sign may take an indefinite time to be attained, depending on the interpretants that come into the play.

Peirce's "common familiar knowledge", being of a nature of a habit rather than related to sign-theoretic action, is not on a similar footing.[29] This has repercussions. For instance, the argument by Clark & Carlson (1982) is given some historical support in Peirce's writings. The argument is as follows. There is a mental primitive (assuming two agents, A and B) that 'A and B mutually believe that p' which, together with the recursive inference rule 'If A and B mutually believe that p, then: (a) A and B believe that p and believe that (a)' yields, if the need arises, to possibly infinite sequences of knowledge statements.

The mutually established common ground of familiar knowledge ought also to include primitives that establish not only knowledge but also common beliefs, common expectations and common presuppositions and possibly also other propositional attitudes, for instance concerning those propositions that the subject is capable of believing or knowing in addition to explicitly believing or knowing them. Peirce did not directly speak about beliefs, expectation and presuppositions, but in establishing a rigorous and broad enough common ground, he nonetheless made them implicit in his discussion both in the quoted and closely-related manuscripts.[30]

The qualification of "familiar" in the phrase "common familiar knowledge" only serves to reinforce these points, since it suggests that what constitutes the common ground for Peirce are, contra Stalnaker (1978), not presuppositions as propositions but the habitually grounded familiarities and attitudes with entities of different sorts.

An upshot is that the existence of the common ground based principally on manners of knowing rather that the depth of knowing does not jeopardise normativity of language, since common beliefs and presuppositions, unlike common knowledge, may well be correct as well as incorrect, in other words defeasible. The entities they refer to are not propositions that are believed to be either true or false, but manifestations (in Peircespeak: what are presented in the Phaneron) that constantly come into pass and disappear, often introduced by virtue of the exertion of force given in assertions and fading out as the force tends to an end.

Prompted by Schiffer's suggestion, Grice pondered the infinitely regressive character of analysing meaning in conversational utterances.[31] His conclusion was that, even if it were tempting to reason that no indefinite regress is involved in getting the requisite representations of the speaker's and the hearer's knowledge right, it is difficult to know what the proper cut-off point is when the iteration of intentions that the utterer wishes the addressee to recognise is no longer desirable. Notably, this conclusion was reached before the infinitary accounts of common knowledge by David Lewis and others came into market.

Grice (1989, p. 65) also remarked about expressions having "common ground status" that they "conventionally commit the speaker to the acceptance" of propositions. Grice is thus in agreement with Peirce's familiarity thesis concerning common knowledge not depending upon propositional presuppositions but upon conventional and habitual manners in which we become acquainted with objects.

The same passage in *Utterer's Meaning and Intentions* also refers to a three-way characterisation of "modes of correlation", for which Grice uses the terms "iconic", "associative" and "conventional". It is easy to see, quite apart from the terminological match of "iconical" with Peirce, that associative correlation comes close in meaning to the indexical relation between a sign and its objects, and that conventional correlation is in very close agreement, both substantially and etymologically, with the symbolic sign relation. Grice's choice of terms appears to have been influenced by the trends in psychology at that time.

Furthermore, in speaking of the "utterer's occasion meaning in the absence of an audience", Grice undertakes an elaboration of the issue that Peirce discussed in terms of situations in which there is no infinite collection in the series of utterers and interpreters, in which case a sign may fall short of having an interpreter. Situations in which utterances do not have interpreters do not necessarily lack interpretants, however, even though a "human interpreter is wanting" (MS 318; EP 2:404). Grice's rejoinder to what people have sometimes used as an objection to his surgery of speaker-meaning was that all utterances are performed as if there were an audience. This is a counterfactual reply to such objections voiced against Grice's theory. Peirce's reply would have been along similar lines: he held that even if an interpreter was not essential to a sign, its *quaesitum*, its essential ingredients, had to be present for the sign to fulfil its function. Since every sign gives rise to an interpretant of it, and since the interpreter and the interpretant will merge in cases in which there is no separate interpreter, the *quaesitum* of the interpreter is the determination of the object of the sign, that is, its interpretant.

The preceding couple of parallels may not yet represent conclusive proof that Peirce influenced Grice's work on issues of mutual interest. They may have just been investigating closely-related subjects on the varieties of meaning in conversational settings, their views converging only coincidentally. However, there is smoking-gun evidence that this was not the case on page 36 of Logic and Conversation, where Grice discusses an example intended to flout the supermaxim "Be perspicuous".[32] According to Grice, this happens in cases in which one interpretation is notably less straightforward than another. "Take the complex example of the British General who captured the province of Sind and sent back the message *Peccavi* [this was in 1843]. The ambiguity involved ('I have Sind'/'I have sinned') is phonemic, not morphemic; and the expression actually used is unambiguous, but since it is in a language foreign to speaker and hearer, translation is called for. . . .Whether or not the straightforward interpretant ('I have sinned') is being conveyed, it seems that the nonstraightforward interpretant must be. . ." Grice goes on to refer to the distinction between the straightforward and the nonstraightforward interpretant several times in the corresponding paragraph. The term interpretant is, of course, exclusively of Peircean origin. Unfortunately, Grice fails to cite Peirce, or for that matter whomever the term was taken from. Apart from this single paragraph, the term "interpretant" is not to be found anywhere else in Grice's published works.[33]

Even so, the sobering possibility remains that part of Grice's above-mentioned terms came from Morris, who was thoroughly inspired by Peirce's theories, but who came to give them some unfortunate and misleading behaviouristic and psychological twists. This is dubious, however, primarily for two reasons. First, there is no mention of 'straightforward' or 'nonstraightforward' interpretants in Morris. Second, unlike Morris, Grice did not link these interpretants with the psychological dispositions of speakers and hearers. Besides, even if he had primarily read Morris, the bulk of terminology Morris himself was using is traceable to Peirce, a fact that cannot be missed in his work.

Grice's term for an implicature that lacks the inferential demand for the interpreter to work out the implied content of the utterance was the conventional implicature. In this sense, what is equally fascinating and baffling is his argument concerning the central status of the principle of the economy of interpretation (*ibid.*: 69). It is about the amount of energy, time or space spent by rational speakers and interpreters engaged in a rational dialogue. It states that, for roughly equal outcomes, it is rational to employ strategies that consume less energy, time or space even if such strategies were less "ratiocinative" than those that consume more (*ibid.*: 83). Again, a game-theoretic flavour is discernible.

An obvious question that arises, then, is exactly how one is to understand the difference between ratiocinative and non-ratiocinative processes, so vital for Grice in terms of upholding the principle.

My response is that, as for Kant, for Grice, too, the concepts of rationality and reason were the assumptions from which all other principles, including the maxims emanated. However, what is ratiocinative needs to be distinguished from what is rational and explained in terms of the latter. If 'ratiocinative' is taken in its scholastic sense, namely as the inferential reasoning process that takes one from the known to the unknown, a mental passing or inner act of argumentation from the cognition of premisses to the cognition of consequences, Grice's meaning would be that it is sometimes rational to interpret assertions in a less inferential manner compared to the alternative interpretations that are not essentially inferior in terms of their consequences to the interpreter.

The question that remains is precisely what kind of reasoning process is admissible in this understanding of ratiocination. Did Grice, like John Stewart Mill (1806–1873) much earlier, intend it to be confined to necessary reasoning, or does it also take probabilistic and other non-deductive forms of inference into account? Grice does not answer these questions, but it is plausible that broadly conceived, ratiocination, unlike ideal rationality, bears a relation to presumptions, a version of non-necessary reasoning that is non-monotonic, namely retracts conclusions in the presence of new information. (Presumptive accounts of conversation have been studied by Levinson (2002) as an elaboration of the Gricean conversational implicature.)

Nor do I intend to address this issue here any further. I would merely like to point out what follows from the above reply, that it could be said that for rational agents who are guided by their reason, heeding the cooperative principle is primarily a moral, not empirical decision based on the realisation of utilities assigned to their linguistic actions. Hence, cooperation, and thus ultimately the whole of pragmatics becomes grounded on normative principles. Pragmatics falls within the same major branch of science as logic, ethics and aesthetics.

Related to these considerations is that the principle of economy of effort falls broadly within the class of similar principles also advocated by Peirce. For according to his account, "Knowledge that leads to other knowledge is more valuable in proportion to the trouble it saves in the way of expenditure to get that other knowledge" (1.122). Even though Peirce was speaking of knowledge in a general scientific and epistemological sense (Chapter 3), the principle applies to the interpretation of presuppositions in pragmatic contexts.

What also supports these conclusions is that Grice's programme turns out to be much less psychological than is typically thought. The first reference he actually made to psychology occurs only on page 137 of the Studies, in the concluding notes to the essay *Utterer's Meaning, Sentence-Meaning and Word-Meaning*, published in Foundations of Language in 1968. This is considerably later than his first outlines of a pragmatic theory of meaning for logic and language. The reason for this apparent mitigation of psychology is not difficult to point out. Reducing the speaker's meaning to propositional attitudes does not imply that the speaker's meaning is reduced to propositional attitude psychology. Speaker's meaning may refer to the mental, but does not commit its theories to any outright psychologism. Since Grice was mainly interested in the connection between logic on the one hand, and beliefs and intentions on the other, there was no need for him to embark on excess psychological theorising.

Thus the alignment of Peirce and Grice is no anachronism. This, despite the fact that, since Grice was sympathetic to the logical formalists' programme of the Frege-Russellian kind, there is little, sight unseen, in his thinking that associates with the grand pragmatists of the earlier era. Yet the methodological agenda that he laid is not alien to pragmatic concerns. For one thing, his anti-psychological and common sense stance in logic is unmistakable. Secondly, shreds of pragmatism show up in the wish to root acts of communication in human rationality and the shared goals of members of linguistic populations.

Leaving Grice aside for now, Peirce makes a significant remark, reminding us of views put forward later by Donald Davidson in support of a 'triangulated' view of interpersonal communication, a necessary condition for language and thought: "When two people are in heart to heart conversation, each is aware of what is passing in the other's mind ~~in exactly~~ [by substantially] the same ~~way in~~ [means by] which he is aware of what is passing in his own, though I do not say he is as completely cognizant of the one as of the other".[34] The emergence and history of the ideas related to Davidsonian triangulation deserve another study (but see Chapter 13 for a lite account). Similar triadic formulations surfaced already in Bühler's semantic studies, and were later absorbed by behaviouristically inclined linguists and psychologists working, for the most part, on decision-theoretic aspects of language use.

5. Post-Gricean pragmatics: towards relevance

After Grice, the science of pragmatics saw furcations in multiple directions. One influential theory, that of relevance (Sperber & Wilson, 1995) took up just one of Grice's particular maxims of cooperation, originally termed that of Relation, and moulded it into an attempt to actually establish what it means for an utterance to be relevant to the context and to previous chains of utterances. In Grice, this had remained an unanalysed primitive to be taken at face value.

However, far from what Sperber and Wilson, the authors of this theory of relevance, claimed in Grice (1986), it has not to date given us very rigorous insight into the logical workings of this admittedly indistinct and formidable concept. Some barriers were presented in Chapter 2. Let me add a couple of further points. First of all, such accounts are bound to be particularly subtle, and there will be many obstacles in creating a logical theory of relevance, in contrast to a psychological or cognitive theory. What is worth noting, however, is that typical law-like definitions using biconditional translation schemes are not much help. Instead, I believe it would be more fruitful to move along the lines of philosophical theories of actions and decisions that resort to the idea of agent-causality, the view that agents possess 'hyper-freewill', capable of overriding any ordinary causal law that may underlie their actions.

By this account, one may lose decidability as volitional acts and agents' best judgements are no longer in neat correspondence. So what; decisions are undecidable for countless other reasons, too. Not that Grice missed the ceteris paribus nature of relevance in company with the three other maxims of conversation: of course he did not. My suggestion is that agents have leeway of deciding, within reasonable limits mutually understood, which of the presented items of information are relevant for the purpose at hand. They may even choose ones that, according to some measure, would score less than other candidates. Given that decision, agents deliberate on accommodating the uttered and asserted information into their belief systems.

As far as the proponents of game-theoretic models of communication are concerned, this agent-causal account of relevance does not contradict the principle of utility maximisation. Unlike in the case of Sperber & Wilson, the fact that the agents are no longer forced to choose the maximally relevant actions in fact reinforces their strategic positions, because the most relevant information is by no means always the most consequential. Given that relevance is always relevance *for* something or *for* somebody, agent-causality appears a promising, albeit yet uncontested, candidate for the understanding and, I hope, formal and logical modelling of relevance.

This points in another inescapable direction that, regrettably, has remained fairly latent in pragmatic theories of conversation. It is the neglect of the role played by the interpreter. Grice's maxims refer exclusively to utterers, their intentions, their meaning, their implicatures. It is not that the role of the interpreters is given the irrevocable go-by within the theory — the cooperative principle does presuppose the existence of interpreters — but rather that such a role was largely de-emphasised. As a consequence, the subsequent studies perceived a too narrowly-focused agenda concerning relevance. As I have indicated, relevance is not solely what the maxim of relation literally means. Any half-sober conversation would be doomed from the start if only the utterer was required to be relevant. Relevance functions by way of mutual criteria, including decisions about what we take to be relevant in our utterances, and also what we, as interpreters, take to be relevant in the utterances addressed to us. No perfect match between the two needs to be presupposed in order for conversation to proceed in a mutually agreeable and understandable way.

It would be make-believe to claim that the core component of relevance in conversations was something novel with Grice, let alone Sperber & Wilson. Peirce, the great contextualiser of the 19th century, offered the following: "If the utterer says 'Fine day!' he does not dream of any possibility of the interpreter's thinking of any mere desire for a fine day that a Finn at the North Cape might have entertained on April 19, 1776. He means, of course, to refer to the actual weather, then and there, where he and the interpreter have it near the surface of their common consciousness" (MS 318: 32–33). The answer to what relevance theorists have been after is implicit in this example: the collaterality of what is given in observation for both the utterer and the interpreter of the utterance determines relevance.

Given Peirce's phenomenology, 'what is given' refers not only to real, dynamic, or physical objects, but to the ideas signs produce in consciousness. They consist of both factual and conceptual elements. There is no analytic/synthetic division in such collaterality. However, it needs to be borne in mind that ideas evoked by conscious minds depend on the situations or environments in which collateral observations can be made, and this includes the cases in which assertions are context-independent and could as well be made elsewhere, whereby the interpretants produced are, of course, likely to be different.

Since Peirce's theory of communication (to which I will return in the next chapter) is purpose-driven and full of accounts of meaningful intention, and since every utterance is made with some goal in sight, the notion of what is relevant is also to be assessed with that purpose in mind. What is relevant is relative to the circumstances at hand in the communicative situation, but what is really relevant is also, and most likely first and foremost, calculated for that purpose. By way of a slogan, thirdness ought to be ever-present in relevance.

Keeping the pragmatic maxim close to my heart, what I also wish to suggest here is that one of the key methods in assessing the scores and the overall scale according to which to weight items of information depends to a large extent on what the practical consequences of accommodating the chosen piece of information introduced in communication are, and on what will ensue in actually using that piece in further cycles of discourse. The most relevant information is that which provides the best toehold for an agent to continue the conversation. It is perfectly consistent with both cooperation and competition, which, contrary to what some have claimed, are themselves no rivals. Whatever is meant by practical consequences does not need to be actualised, even though it needs to be actualisable; it may just illustrate the Peircean would-be, a type of modality of which agents are able to say meaningful things in a similar way as they are able to say meaningful things about any modal proposition that may be presented to their consciousness and appear under their judged deliberation.

I strongly doubt that there is a better way of concisely capturing what relevance means. This pragmatic definition (not an explicit one, for it is a very broad-spectrum and I do not intend to flesh it out here, see Pietarinen 2005a) heeds Peirce's analogous pragmatic maxim. It denounces secondness, namely any attempt to provide necessary and sufficient conditions for relevance. Nevertheless, it is not one that requires the meaning of the concept under assessment to be relevant; as we know from Peirce's semeiotics, processes that aim at spelling out the meaning of an utterance may be very complex and of indefinite length. Relevance may be known even though the meaning, or significance, or habit-change, or logical interpretant, or nearly any of those Peircean concepts related to propositions with assertoric value, is not altogether recognised between the utterer and the interpreter. Since relevance is the hub and nub of the science of pragmatics, it is to be expected that something along the lines of the pragmatic account is indispensable.

Sperber & Wilson's idea that relevance is what infiltrates the context should not be read as opposed to the pragmatic account. What is missing in their proposal is nevertheless a way of grading relevance according to feasible criteria.

In view of these remarks, both the pragmatic definition and Sperber & Wilson's cognitive and contextual theory of relevance allow a tempting reinterpretation. In fact, in one of the early papers on relevance, a necessary condition was defined as one that, together with background knowledge and earlier assertions, yields new information not derivable from earlier assertions or background knowledge alone (Wilson & Sperber, 1979, p. 177). What is this notion called information here? Unfortunately, it is taken as a more or less primitive and logically unanalysed notion. Interestingly enough, though, if it is interpreted as what a proposition asserts and what it does not assert, in other words, what possible states of affairs it is capable of excluding and of not excluding, we get a fresh account of relevance along the lines of the time-honoured semantic definition of information. Allied to this definition then comes the notion of capability or force, which in turn may be explicated in terms of the complexity of including and excluding possible states of affairs by the proposition in question.

For instance, let us take one of Sperber & Wilson's own examples, the answer to the question "Where's my box of chocolates?", which is "The children were in your room this morning". This can be evaluated for its relevance to the question by computing the complexity involved for the interpreter of the answer to exclude the states of affairs that are not intended by it, as well as the complexity it involved in including the states that are intended by the answerer's utterance.

However, if the answerer himself or herself is one of the children, the situation is made more complicated by the answerer's intention to dupe the inquirer, in which case there are more alternative states of affairs to be included than in the normal case, and thus the answer is less informative and hence less relevant to the question.

Other notions of communicative force of relevance may be analysed along similar lines by using complexity and information as the key.

Relevance theory may have emerged in the wake of Grice, but it subsequently redefined its goals to the extent of now being somewhat of a rival. The emphasis on the search for the key principles of cognitive processing from which it is hoped that implicatures and other pragmatic notions ensue has had the effect of diminishing the force and depth of the all-powerful rationality postulate upon which Grice's programme was built. In so doing, relevance theorists have rubbed shoulders with computational sciences, sciences for the efficient accounting of information transmission and manipulation, while de-emphasising conceptual analyses of information. Accordingly, in fields such as economics and interactive epistemology, relevance theory has gained in status much in the manner of theories of less-than-hyperrational reasoning and action. These all share the methodological concern that the effort spent on any act of uttering and interpreting, or believing and decision-making, should be weighed against the practical consequences of such acts, and thus continue the venerable economy of research methodology that Peirce originated (Chapter 3). As I noted in the first section of this chapter, similar methodological attitude was also Grice's main preoccupation.

Unlike Peirce, Sperber & Wilson were drawn in opposing directions by their attempt to build the notion upon the psychological and cognitive theory of the competence of intelligent agents, while simultaneously trying to provide support to its inferential and logical dimensions. In contrast, Peirce's goal was not to spell out any theory of cognition of intelligent agents, let alone their psychology, but to dispense with these as much as possible. Given the scope of the problems that the subsequent research has unearthed, this was not perhaps a completely realistic undertaking, but at least he claims the priorities he felt were needed in the brands of rational inquiry concerning language and thought.

* * *

What, then, happened to the notion of the common ground after Grice? As I hope to have shown, it was well recognised in pre-Gricean pragmatics, but was never in full blossom in Grice's own writings. Stalnaker (1978) was singularly influential in suggesting that the common ground should be modelled as a set of possible worlds. He argued that it was a set of propositions, the presuppositions of agents, and could thus be represented by a possible-worlds framework, because presuppositions are claims about what the agent knows or assents to by means of suitable propositional attitudes. Stalnaker holds that any assertion nurtures the common ground by adding the content of the assertion to the set of presuppositions. This is the effect, the pragmatic bearing if you like, of assertions. The problem is that it is difficult to make sense of the idea that the common ground is constituted from assertions, since assertions are bound to the particular situations in which they are made. Stalnaker holds that it is the content of the assertions that is added to the common ground — their propositional core or essence — and that this addition allows for manipulation and modelling by possible-worlds semantics.

From the Peircean perspective this appears too rigid. It remains unclear what the non-assertoric content of utterances is that Stalnaker's proposal is bound to produce and how it should be analysed. How could sentences or expressions coinstantaneously be sufficiently similar to assertions and hence added to the common ground, while apparently retaining the notion of meaning that is non-truth-functional, be retractable and defeasible, perhaps merely manifested, implicit, familiar, potentially believable or residing in long-term memory, and yet function as a proper bridge between assertions and propositions?

In leaving this compartment it needs to be noted that there were others in the pre-Gricean and in some cases, pre-Peircean, era who would not have shared these difficulties. Notwithstanding Grice's original conservatism or the views of the soi-disant neo-Gricean henchmen, the question of the proper balance between logical and psychological aspects of meaning hardly arises as soon as it is observed through the glasses of economic and evolutionary methodology of research.

6. On historical and Peircean pragmatics

Towards methodeutic In this section, I wish to proceed to another related subject and to discuss the scientific methodology of historical pragmatics, especially in view of Peirce's notion of hypothesis selection in historical research, and in view of its links with his broad notion of logic as semeiotics.

The purpose of historical pragmatics is to understand the pragmatic aspects of language change. Many have observed that historical pragmatics constitutes both a subject matter and a methodology (Bertuccelli Papi, 2000). The former operates on an empirical, data-oriented domain, aiming at covering linguistic phenomena that are entangled with passages of time. The latter is a more disparate, on-going and loosely-grounded enterprise, which strives to find methods and tools by which to glean from documents some key principles concerning pragmatic change that has taken place in language.

The proposed characterisation of the divergence is apt, and I daresay little more than that the methodology of historical pragmatics is yet to be agreed upon. For instance, the definition given by Andreas H. Jucker is data-oriented, recommending the combination of the methods of historical linguistics with the methods of pragmatics.[35] For one, this excludes the development of pragmatic theories from its scope.

Four methodological remarks need to be made. The first is that historical pragmatics has chiefly been focussing on the subject of illocutionary development, or a little more generally, on the feasibility and nature of diachronic speech acts. This by no means covers the whole terrain. Even though such research questions have doubtless promised insights into the evolution of certain speech and language phenomena, in the interest of historical adequacy, it seems appropriate to trace the emergence and development of the theories of speech acts, their motivations, and their basic principles, rather than to take their soundness and applicability to pragmatic phenomena for granted.

The theoretical relevance of this position to historical pragmatics has been recognised. Bertuccelli Papi (2000) discusses some venerable questions arising in diachronic speech acts, related among others to the interplay of propositional vs. speaker meaning, grammatical structure vs. illocutions, social vs. cognitive factors, and so on. In this light, it is by no means obvious that these theories have been on the right track, at least if, philosophically and historically speaking, one is interested in the foundational import of pragmatics on what is typically classified as pertaining to the realm of semantics. The more so, given that the idea of tracking down the paths of the evolution of speech-act theories constitutes one aspect of the general task of tracking down the development of pragmatic theories at large.

The second, closely related point concerns the semantics/pragmatics distinction. I do not believe that any hard-and-fast separation between the two is feasible, even though attempts have been made to draw the boundaries in several alternative and non-trivial ways. Rather, the varieties appear to form a continuum of associations between conceptual representations, objects and their interpretations, leaving ample room for the updating and emergence of such associations. But if so, historical semantics and historical pragmatics must also be entangled, and so must their methods. Without delving into the details of the possibility of a semantics/pragmatics interface, defining pragmatics in the old-fashioned way as a non-truth-conditional meaning theory is simply void. For would semantics, contrariwise, be pragmatics minus non-truth-conditions?

In general, it seems that the distinctions that have so far been attempted tend to imply too narrow a view on both. My own belief is that we should revive the scholastic distinction, which Peirce so cogently underlined, between speculative grammar, logic proper and speculative rhetoric. All three possess elements of the current-day fields of syntax, semantics and pragmatics, but are not reducible to them.

Above and beyond, it was suggested a long time ago in Wegener's Unterschungen that, if we are interested in questions concerning the evolution of language, then it is undeniable that semantics follows pragmatics, or that pragmatics has priority over semantics, because any answer to what meaning is needs to be derived, and often reconstructed, from the actions of language users situated and embodied in particular cultural, historical, social, perspectival or recollective locations. The same principle was held by other like-minded early linguists, including Whitney and Johnson.

At all events, methods of historical semantics and historical pragmatics, while entangled, may be used to actually test the initial feasibility of any dividing line between semantics and pragmatics. This brings us to the third methodological point, the mutual interest between historical and evolutionary linguistics. Some cohesion between the two exists in coupling historical semantics with evolutionary approaches to meaning. Likewise, since the concept of meaning is not operational without some pragmatic factors, the ties that historical pragmatics have to evolutionary approaches to language deserve to be better known.

What are the best uses that can be made of evolutionary theories in historical pragmatics? To provide one topical suggestion, one perspective that I suggested in the previous chapter is to use evolutionary game theory in ascertaining which meanings come to live and persevere in finite populations of language agents. In brief, those strategies that are uninvadable by adversary mutants and thus score best against them (and against themselves) reveal meanings that are stable in the sense of semantic stableness. A possible reading of the notion of stableness is that the less context expressions required in order to be understood, the more stable the semantics is. One should thus note the considerable extend here of the sense in which semantic considerations are parasitic to pragmatic ones.

What follows from such an evolutionary and strategic perspective is the narrowing of the scientific gap between formal semantics on the one hand, and of historical semantics/pragmatics and the evolution of language on the other. It also appears to narrow the gap (or, alternatively, to identify parts of the overlap) between what is semantic and what is pragmatic in the evolution of language. The reason for this is that the strategies alluded to in evolutionary games do work for both realms: while the existence of certain stable strategies (typically those that are the winning verifying and the winning falsifying ones) spells out the truth-conditional meaning of sentences, the content of these strategies hinges on pragmatic factors concerning the lifecycles of the expressions used. [36]

Overall, the use of mathematical frameworks provides a toehold for linguistic phenomena that are not simply confined to the evolutionary emergence of primitive rule-based grammar systems.

The fourth methodological viewpoint that I wish to emphasise concerns the role of evidence in historical pragmatics, which has been noted to present a sustained challenge. In order to tackle this challenge, I wish to bring out Peirce's *On the Logic of Drawing History from Ancient Documents, Especially from Testimonies*. (To abridge my point, the title could perhaps be rewritten as *On the Logic of Drawing Pragmatics from Written Documents*). The point of Peirce's remarkable paper concerns the general use of the method of abduction in science together with the economic considerations of hypothesis generation and selection. Similar norms apply to the methodology of historical pragmatics, too, in which the quandary is to reconstruct the contexts of the documents within which something of interest has been uttered.

The primary conclusion of Peirce's paper is that one should not assign estimates of probability, or degrees of belief or credibility, to historical documents, because those estimates and degrees are bound to be unreliable. For the purposes of historical investigation, he recommends, instead, the method of experimental science. The method consists, in effect, of a list of factors that needs to be considered in selecting historical hypotheses, pertaining mostly to the economics of research. These factors are very basic and applicable across divergent historical disciplines, and may be gathered under three main qualities, which prefigured in Chapter 3. Let me illustrate some of their relevance to historical pragmatics.

First, there is the quality of caution, according to which a hypothesis should be broken into smaller logical components. 'Big' questions ought to be divided into a series of 'small' questions, and why-questions ought to be divided into a series of yes-no-questions.

For example, instead of asking, 'Is a diachronic speech act theory possible?', one is recommended to ask: 'What are the intervals of time that identify the lifecycles of certain classes of speech acts?', 'Are there some invariants in utterances that are preserved in these intervals?', or 'Is the propositional content of assertions subject to similar variation and change?' Instead of asking, 'Why do new meanings occur?', one may perhaps ask: 'Do populations of language users strive for increased effectiveness in coordination tasks?' In a similar vein, instead of asking, 'Why is there pragmatic change?', one is recommended to ask: 'Do performatives involve a change in the use of language over a given time?', 'Are the media of communication changing contemporaneously?' (e.g., from oral to written), or, 'Is grammaticalisation an epiphenomenon of pragmatic change, or vice versa?' And so on and so forth.

Second, there is the quality of breadth, according to which a scientific hypothesis is to be evaluated by its applicability to the same underlying phenomena occurring in other, related subjects, presumably across varying contexts, environments, circumstances, and linguistic classes and categories. Rival, manifold explanations of the same phenomena should be evaluated according to consequences that can be weighed and classified according to some feasible criteria.

By way of an example, if the hypothesis is that a pragmatic change occurred in discourse D because of a semantic change that took place in some particular set of propositional contents of the assertions in D, then according to the criteria, by this we are able to explain a change in another discourse D′ devoid of performatives. This hypothesis is thus more plausible than the one resorting to, say, a change in the performative structure of speech acts in the previous discourse D, because the former accounts also for the chance in D′.

Third, there is the quality of incomplexity, which means the absence of complexity, simplicity and perhaps something like artlessness. This is no forthright Ockham's razor. Rather, the principle is to be understood such that incomplete hypotheses (which they are bound to be) should "give a good leave" (EP 2:110), because they are likely to sooner or later be overridden by new hypotheses. They should point towards future investigation rather than past. Hypotheses are per se closer to the good and fruitful conducts to be followed than any static set of scientifically tested propositions.

In relation to diachronic pragmatics, this method may be understood to refer to the acceptance of both micro- and macro-level data in recreating contexts, in other words the taking into account evidence from both the cognitive and biological sides of a given set of theoretical assumptions. Moreover, the method suggests that cheaper (perhaps cognitively cheaper, that is, more effortless) hypotheses be tested before those that are more costly.

What Peirce is thus warning us against is the kind of philological fallacy of assigning implausible weights of credibility to the sources, which may appear to be of great importance, and which we may wish to use as a basis of a confirmation of the hypothesis upon which our theories are built, but which, being scarce, easily mislead.

Nevertheless, it is not obvious that the norms that Peirce puts forward would apply to historical pragmatics with the same force as they apply to the general science of history. The hypotheses in historical pragmatics need not hinge on the fact that what is stated is actually true (or at least credible), because the subject matter is, at least in part, intra-linguistic. To this it may be countered, however, that while the concepts of truth and falsity need not be taken into account in weighing the evidence (poets, plays or even legal documents are seldom plainly true), what are needed are the assurances that the samples are in some reasonable sense representative of the underlying pragmatic phenomena that are attempted to be explicated. Editing out, padding, and inserting narratives are concrete examples of actions that may have modified the reliability of the document. And of course, the problem of recreating contexts for the data depends on considerations of their fit with the reality. If so, philological fallacy lurks behind the corner after all.

To round off what we have discussed so far, how do the subtleties of hypothesis selection guide us in reconstructing the context for old documents? First of all, we should note the heterogeneousity of models of context. The two main ones are perhaps that (i) to be aware of the particulars of the conventions or normative grounds of language use may require a reconstruction of the socio-historical context within which the utterances must be understood, (ii) the utterer's and the interpreter's propositional attitudes, including beliefs and intentions, delineate a different, cognitive type of context, which due to the scarcity of supporting evidence is even more difficult to reconstruct from old documents than the former type of context.

The quality of caution in socio-historical contexts recommends that as many educated guesses as possible be made but that only one of them at a time is risked (HP: 756). Hence, one should prefer hypotheses that break such contexts into the smallest identifiable units (such as communities of different types of language users or saturated systems of conventions). The quality of breadth states that the best explanations account for the same phenomenon in other subjects (HP: 757) and so one should prefer, say, cross-categorial and cross-cultural explanations to singular, intra-cultural ones. Note that caution and breadth are by no means rivals. The quality of incomplexity recommends the creation of contexts which, while not entirely true or accurate, in comparison with the data will be suggestive of other hypotheses. There are better and worse false hypotheses, the former having more such pointers than the latter.

On the other hand, the quality of caution on cognitive contexts recommends isolating the smallest identifiable belief systems of linguistic communities. The quality of breadth recommends explaining several such systems by similar underlying conceptual patterns. The quality of incomplexity recommends the economic conduct of making the belief systems as simple as possible, for the sole fact that they are likely to be not entirely truthful.

The hermeneutics of historical pragmatics One final remark is that historical pragmatics appears to involve teleological explanations. But if so, are such explanations hermeneutic? In linguistics, teleological and functional explanations seek to address language change in terms of the purpose the language has, such as increasing the communicative effectiveness of a system of agents or the functionality of language within some specified context of use, be it oral, written, formal or informal (Short, 1999).

According to my analysis of historical pragmatics, the diachronic aspects of language change are indeed teleological, insofar as they involve reference to the purpose of human communicators to increase the mutually-shared, ever-growing supply of idea-potential, their *summum bonum*, that rational goal of human inquiry into which all agents with their varying ways of using and communicating with language are destined to tend, albeit perhaps unconsciously.

Whether these teleological explanations are also hermeneutic depends, however, on the scope and frequency in which references are made to the interpretation of human texts and documents, including historical artefacts, monuments and scriptures as well as culturally and narratively diverse oral traditions. The more weight one has to lay on the understanding of the meaning of these documents and artefacts in terms of the connection with their originators, the more hermeneutic the explication is. Since historical pragmatics operates on the domains of data that are not easily interpreted if parted with the acts of creation, performance or perhaps some particular juridical or political purpose to which they have been crafted, it is much more hermeneutic, say, than those areas of historical linguistics that attempt to explain grammaticalisation through the emergence of new linguistic forms. All the more so, since the mindsets behind creating a text, the different qualities of performing a play, or the social or institutional purpose a text has been envisaged to serve, must be recreated afresh from the limited data and background knowledge at our disposal.

We are hence facing a familiar dilemma: understanding the writer's, narrator's or law-maker's intentions rests solely on the written records concerning what he or she has said (together with some supporting collateral evidence), but interpreting these records involves a good insight into what his or her intentions have been. Various speech acts concerning politeness and performative phenomena are particularly disposed to fall prey to a hermeneutic circle.

To circumvent some of the core problems plaguing hermeneutically-inclined explanations in historical pragmatics and its demands for co-temporaneous empathy and *Verstehen*, an attempt can be made to identify some general, shared principles concerning the purpose served by pragmatic change. For one thing, language and economics have been congenial, and to some extent complementary, concerns of human behaviour since Adam Smith and the observation that they both attempt to explain regularities in human interaction and the design of social systems. Since I have addressed these principles in earlier sections and chapters, I need not return to them here.

Summarising, Peirce's value as the progenitor of pragmatic theories of language has not been fully acknowledged, partly because the key original textual sources to this effect are still unpublished, and partly because those who followed him did not pay enough attention to the history of semiotic thought relevant to pragmatics. The route by which Peirce's semeiotics found its way to modern linguistics, considerably mutated and deteriorated, is a curious story of its own. But I believe that understanding not only the early pragmatic theories and their development, but also the pragmatic historical data associated with documents, will greatly benefit from the understanding of his original ideas. After all, regarding the language evolution and, in particular, its emergence, there is a wider goal behind all this, expressed by Peirce in terms of the passage that I recycle here from the introductory section of Chapter 11, now in full swing: "In linguistics, there is the question of the origin of language, which must be settled before linguistics takes its final form. The whole business of deriving ancient history from documents that are always insufficient and, even when not conflicting, frequently pretty obviously false, *must be carried on under the supervision of logic*, or else be badly done".

Yes, this is precisely the concern of historical pragmatics, as well. We do not yet have a clear picture of what its logic is, but I hope that the points that I have raised provide preliminary indications of the places in which that logic may be sought.

7. Agenda cognitive linguistics

Another topic related to Peirce's pragmatics concerns the philosophy of cognitive linguistics. Unearthing the semeiotic roots of cognitive linguistics seems to me to be pressing, because it is commonly held among the practitioners of this area that there is no philosophical theory for the results of cognitive linguistics, which many have felt makes the field methodologically somewhat orphan and disparate. What I wish to argue is that, from the foundational perspective, cognitive linguistics has a venerable predecessor in Peirce, among whose systems such foundations is to be sought for.

First, some conceptual clarifications. I take the epithet ‘cognitive’ in cognitive linguistics to roughly mean ‘pertaining to, or being associated with, knowledge’. I thus take cognitive linguistics to refer to that recently-emerged discipline that is independent of both neurological, biological and psychological associations, although there may well be some factual overlap in the methodology actually professed by cognitive linguists, such as the important data coming from neuroscientific experiments pertaining to various dysfunctions of one’s language capacities (Deane, 1996), or related experiments in psycholinguistics.

Accordingly, then, in presenting what I take to be a historically significant contribution to the proto-history of cognitive linguistics, as well as to the early phases of pragmatics, namely Peirce’s semeiotic work, I subvert allusions to psychologism, the view according to which logical laws are subordinate to, or rest on, or at the very least are not entirely autonomous from, facts of human psychology. For certainly Peirce, and here he was echoing Kant, had a distaste for psychological concepts in exact sciences such as logic. He held it to be a normative science, and similar things may be voiced on language. What remains indispensable in his work, though, is mental vocabulary, involving as it did the notions of the interpreting mind and the actions of the mind exemplified in thoughts. That was essential to his broad conception of logic as the theory of signs, which was very useful in numerous formal, logical and linguistic tasks, including his multitudinous tackles on abduction.

Moreover, Peirce contemplated the possibility of the relevance of neurological facts to language skills. He surmised that his difficulties in understanding the logical structure of spoken or written language is due to his left-handedness, preferring thus diagrammatic means of reflection. As noted in Chapter 2, he knew Broca’s path-breaking investigations on the connections between lesions in parts of the frontal lobe and aphasia. Such observations witness considerable cross-disciplinary sensitivity. However, the first remark has to be taken cautiously, because Peirce was quite familiar with a dozen languages or so, including Inuktitut and Cuneiform.

He was also the first to propose a comprehensive diagrammatic approach to logic based on iconic system of graphs, which came to encompass not only the propositional and first-order logics, but also higher-order type-theoretic notions, abstractions, modalities, and so on. In a well-motivated sense, then, diagrammatic and graphical systems may be viewed as pre-linguistic structures well-suited for researching aspects of iconicity of language, which is of central interest in cognitive linguistics.

While being averse to psychologism, Peirce was sympathetic to both mentalism (that mental states are to actions as causes are to effects) and cognitivism (that such mental states and their representation may be studied by methods pertaining to knowledge and information processing). In particular, the mind is needed to generate signs that comprise thoughts but are not identical to them.

I nevertheless do not wish to suggest any pompous gloss to the effect that Peirce ought to be regarded as 'the first real cognitive linguist', even though the title would not, given his breathtaking scope of work on the relations between philosophy, logic, linguistics and psychological experiments, be entirely inappropriate. One such contact point is Peirce's evolutionary perspective that permeated his philosophy, including the development of scientific theories, discovery of natural laws, and the formation of intellectual habits of the human mind. Listing among these the emergence and development of human language and its meaning relations with the world, we come to view language as a cognitive tool among other tools that have catalysed our adaptation with the complex patterns of the world.

What I wish to focus on here are a couple of interrelated facets in Peirce's logic and semeiotics that seem to me to be of paramount significance to cognitive linguistics.[37] In the spirit of this work, I will mostly refer to theories of communication rather than theories of language, primarily for two reasons: firstly because Peirce's semeiotics was predominantly on any communication between the utterer and the interpreter mediated by signs, of which symbolic systems such as natural language — by which to express contents, ideas or deliberation on shared experiences — play only subordinate role. Secondly, his pragmatism was about the meaning of concepts, not only about the meaning of expressions, assertions or speech acts. What pragmatism mandates is that the meaning comes to be construed and is assessed according to the purpose of concepts and their practical effect.

This broad view of communication reached the mental realm. For instance, Peirce held thinking to be mental discourse or dialogue between two phases of the mind, or the 'theatres of consciousness' as he at one point came to describe these mental agents. Notable here is the introduction of 'the mind' and arguments for its indispensability, which has much later been entertained in cognitive linguistics, although quite independently of linking it with Peirce's philosophy (Langacker, 2000, p. 26).

Moreover, Peirce's pragmatism puts forward views akin to functionalist approaches in linguistics, according to which language, predominantly a goal-directed system of communication, is stratified into multiple layers or hierarchies according to the purpose of its separated units. This bears a relation not only with Peirce's pragmatism but also with his manifold innovations in logical representations, illustrated among others by his rhema (uninterpreted predicate) vs. content (dialogic actions) division, which has lived on in such divisions as the old vs. new information and topic vs. comment. His division was of fundamental importance in algebraic logic of relatives as well as in his topological logics based on a diagrammatic notation for iconic signs.

The first and foremost facet common to Peirce's semeiotics and cognitive linguistics falls within the broad genre of concepts that are known in knowledge representation, AI and cognitive science variously as frames, scripts, scenes or schemas, and in linguistic pragmatics as the common ground of the interlocutors (typically interspersed with notions such as 'accommodation' or 'grounding'). These notions have manifestly overlapping domains, and may be seen as converging attempts to provide memory storages for the agent's discretion, including episodic as well as conceptual information.

My quest here is twofold. First, I wish to ask when, how and by whom these notions came about during the early phases of the sciences of linguistics and primordial cognitive science. Second, I would like to know what the contribution to the field of cognitive linguistics may be that Peirce makes in his account of the notion of the common ground.

With respect to the former question, which deserves another study, I will simply recapitulat what I noted earlier in this chapter, namely that it is indeed in Peirce's writings that we find an advanced anticipation of the importance of the notion of the common ground. Abundant textual evidence from published and unpublished writings may be adduced to justify this.

The upshot is that communication and common ground are virtually synonymous, referring to both the factual material agents have collected during their existence and to the collateral observation (subsuming e.g. deixis, viewpoint, orientation, distribution of attention) derived from their situatedness in a particular location of the environment, be it spatio-temporal, epistemic or cultural.

What happened to these remarkable ideas in the subsequent literature, many of which bearing fundamental relevance to the agenda of cognitive linguistics, was that the bulk of them was put forward quite independently of Peirce's influence and with little thread of his semeiotics. The notion of the common ground is perhaps best-known from Stalnaker's analysis of presuppositions. He argued that the common ground is constituted by the system of (defeasible) beliefs, which in turn are created by the commonly held beliefs and expectations of the speaker and the hearer which, at least in principle, form an infinitely iterating hierarchy pertaining to the recognition of each other's beliefs and expectations. So the common ground is the set of propositions agreed by the conversants prior to conversation, nurtured by the addition of the content of the assertions to it. Lewis made this framework more dynamic, by adding the feature of accommodation, according to which presuppositions may come into existence in the course of conversation — if nothing else then for the sake of that conversation — if that is convenient for the overall purpose of communication, in other words prevents it from breaking down and from thus reducing the 'conversational scores' of the participants. Maybe the term 'cache beliefs' would be apposite in our information age.

From the Peircean vantage point, such a system of presuppositions is created by his notion of common familiar knowledge. It is reasonable to assent that other propositional attitudes, such as beliefs and expectations, were likewise admissible for him. What Peirce was nevertheless able to do was the addition of a rich logic by which to rigorously analyse natural language expressions.

What is problematic in Lewis' suggestion, and what we do not have in Peirce's account, is its shallow strategic repertoire. Which presuppositions come to be added to the common ground and which will be excluded? If accommodation is a dynamic process, what kinds of rules are there to prescribe these actions? Lewis only describes some salient classes of games (cooperative vs. competitive, permission, vagueness), noting how their scores evolve based on constitutive rules akin to explicit definitions. He deflects the issues to do with regulative rules governing these language games.

The common ground has to be actively constructed. The reason is not only to delineate legitimate from illegitimate, or permissive from non-permissive, actions and decisions in the conversational game, but to make the game in other respects playable. The conversants need to be acquainted with the universe of discourse, although not necessarily in any total fashion. The universe may be infinite, but it typically suffices that the players are familiar with just some part, nook or crank of it. This familiarity is counted among the tools that facilitate the construction of the common ground, and I believe it is a pre-eminent assumption in cognitive research on language, even though its status has been unclear in the 20th-century logic. Not only are parts of the universe readily interpreted before players will begin to draw from it elements and values intended by actual statements, but also the common ground will contain mutually known and mutually known to be known and observed aspects of language, including grammar, linguistic competence, self-awareness, and mutual experience.

It follows that it is not only the static network of presuppositions as propositions, but the measure of the relevance of such propositions, that needs to be taken into account in the construction of the common ground. Suppose my wife asks: "Where is the chocolate box that I saw on the table?" I say: "The children were here this morning". My answer is clearly relevant and conveys certain information, presupposing of course certain facts about the existence and character of out offspring. If I say "The refrigerator has been mended", it may be less clear how relevant (or wise) I am being, but there are common presuppositions all the same. To more so for "It is too hot to go for jogging today". There are several implications of utterances that may be suggested for the inclusion to the common ground, but such decisions are strategic and calculated according to the purposes of both the utterer and the addressee.

The question of exactly what is it that happened after Peirce that led us to where we stand today is relevant also in the context of cognitive linguistics. There are curious general strands by which semiotic thinking found its way to contemporary cognitive science, AI, non-symbolic and non-standard logics, or other knowledge representation techniques. These strands pertained predominantly not to the mainstream development of intellectual ideas, such as those that took us from logical atomism to logical empiricism, from the movement of significs to ordinary language philosophy, to speech acts, or to optimality theory. Typically they used the continental route via the phenomenological analysis of language (by Husserl) and the Gestalt conceptualisations of psyche and linguistic cognition (by other Germans), or else bypassed science altogether and took art and literature instead.

Nevertheless, as far as the cognitive content of linguistics is concerned, I believe that Peirce's potential is limitless, and displays a promising solution to the quest set out by Albertazzi (2000, p. 24), namely that there "is still no philosophical theory for the results of cognitive linguistics". Further, I should add that, since "missing is also an empirical geometry of cognitive and perceptual spaces" (*ibid.*: 24), it is all the more attractive to view Peirce's diagrammatic system of graphs, expounded in Part I, which he held to put before us "moving pictures of the action of the mind in thought", and to which a topological interpretation preserving its essential (not to even say Gestalt) structures under continuous deformations may be given, as such a candidate for the overall theory of cognitive mental spaces and their blends.

8. Conclusions

It is unfortunate that the true relevance of Peirce to pragmatics has been invariably missed or downplayed, even a hundred years after his most prolific period of such investigations. Just to expose a by no means uncommon sentiment, Clark (1996, p. 156) writes, "Peirce applied his theory to a wide range of philosophical issues, including logic, inference, belief, perception, and metaphysics, but oddly enough, not directly to communication or language use". As I hope to have shown, nothing could be further from the truth. Not only was the notion of communication a central concern in his theory of signs, it was also a strikingly articulate and versatile account of language use.

Clark continues: "Peirce also didn't distinguish between the type of thing a symbol (like 'give' or 'bird') could potentially signify and the type of thing a person actually uses it to signify on a particular occasion. Peirce was missing several distinctions that were made only fifty years later" (*ibid.*: 159–160). In so far as we can make sense of this opinion at all, it is misguided for similar reasons as the previous one was. Peirce studied speaker's meaning and distinguished it from other types of meaning in terms of a variety of interpretants.[38]

In the post-Gricean era, game-theoretic approaches have burgeoned especially due to the renewed interest in the evolutionary emergence of language. The impetus was given by Maynard Smith & Price's definition of evolutionarily stable strategies (1973), soon to be applied to linguistic considerations by Maynard Smith himself and others. The precursor of this was the early 20th-century information theorist and population genetist R. A. Fisher, who in 1922 suggested a measure of maximum likelihood frequency for learning probability distributions from finite samples. (Leo Szilárd coinvented the information measure independently in 1922.) What is notable here is the application of this measure to physics, within which it has also been suggested to evoke a game-theoretic process of deriving physical laws from the questioning dialogue between Nature and the Experimenter (Frieden, 1998). Thus, evidence recurs of the linguistic connotations of physical attempts to understand notions of energy, force and entropy. Besides, frequency measures based on maximum likelihood spring from Peirce's scientific methodology in weighing evidence for making inferences with uncertainties involved in hypotheses, and in avoiding biases in experiments involving human subjects, especially psychological experiments, by a proper randomisation of the data. When such randomisation is not possible given a scarcity of evidence, other, economically driven considerations of hypothesis generation and selection need to be introduced.

For many intents and purposes, economic and evolutionary notions are related. The whole industry of evolutionary economics is devoted to spelling out such connections (Hodgson, 1999) from communal and institutional point of view. The task is iffy, since not all connections are biological, even though they may employ self-organisation and complexity measures. Likewise, evolutionary considerations have been applied, for instance, to the simulation of the emergence of vocabularies, phonetics and simple grammar systems, but as the cumulative weight of this and the previous chapter suggests, little is known of such applications to genuinely strategic concerns arising in one particular institution of humans, the semantic/pragmatic continuum of language.

Since aspects of both economic and evolutionary notions may be tried to be spelled out in terms of game theory, the other relevant playground in which attempts to join such game-theoretic assets with pragmatic elements of language have been made is that of theories of conversation. However, even though, say, the Gricean cooperative principle would be preserved in conversation, it is not exactly right to simply equate it with the principle of cooperation in the sense of game theory. The reason is that Grice's technical definition of cooperation (according to which speaker's contribution ought to be such that is required by the accepted purpose of the exchange) is speaker-oriented and says little about the actual and quite complex process of interpretation. In game theory, on the other hand (according to which players' roles are, in normal cases, symmetric in the sense that no one cooperates less than the others), Grice's cooperation does not refer to interactive cooperation. Nor, clearly, does cooperation in Grice's sense refer to any folk-linguistic notion of carrying conversation on in tandem through something like the accustomed behaviour of the speakers and hearers.

Cooperation in the game-theoretic sense is quite different from cooperation in the sense in which Grice's defined it. Cooperative approaches in economics do not model how agents communicate with one other, because agents are assumed to endorse joint action and any communication is relegated to pre-play situations. Such games are coalitional, and likewise, Grice's cooperative principle could perhaps be renamed along the 'principle of required (or expected) contribution', in which by contribution it is referred to those parts and fragments of input an agent brings in to increase the 'idea-potential' of linguistic communities. When the presumptions are not met, the sole possibility of creating mutant strategies may alert the community to check if anything needs to be done to adjust the language-world associations, in other words the strategies by which such associations endure in these communities.

Moreover, theories of conversation differ fundamentally from theories of argumentation and debate, because in the latter, competing interests are pursued by the participants. For this reason, argumentation has found a more solid base in game theories, as witnessed by the work on dialogue games by Lorenzen, Lorenz and their posse (Chapters 7 and 9). However, there is no principled reason to expect that conversation could not be amenable to game-theoretic principles, by assuming a diminished amount of competition.

Many of these developments link with the pre-history of pragmatics that was briefly reviewed above. What is significant is the affinity between many of the ideas that have prevailed since Peirce's semeiotic studies. However, the influence of Peirce was greater on the early semanticists, significists and pragmaticists of the 20th century than on the overall analytic movement that took off from logical empiricism. Logical empiricism limped along just about to reach Grice, while analytic philosophy was enjoying its heyday. However, they both reached an impasse soon afterwards. The semeiotic foundation of pragmatics was almost entirely suppressed from mainstream analytical philosophy for the better part of its survival in the post-Gricean era.

Even if it were not particularly tantalising that there was more convergence and mutual interest between different disciplines during the early history of pragmatics (such as, say, between economics, evolution and linguistics), the conceptual arsenal that was already in use in the 19th century by Peirce and others vastly supersedes the image typically attributed to the status of pragmatics in that era. This shows the recurrent claims that speech-act theories and Grice's programme had virtually no predecessors to be unfounded. One particularly vital line of development that I did not enter here was accomplished by the Dutch significians, especially Gerrit Mannoury, who during the inter-War period was singularly successful in bringing the notion of speech acts to bear not only on psycholinguistic studies and experiments but also on the analysis of meaning in various communicative acts. Unlike the later speech-act theorists, his viewpoint was not confined to speaker's acts on the hearer, but recognised the need for accounting acts of the interpreter as well.[39]

The next two chapters sharpen the picture on Peirce's communicative signs, and seek applications in the contemporary era of electronic communication.

Notes

1 To these authors, common knowledge was anticipated by Thomas Schelling's game-theoretic analysis of cooperation and salience (Schelling, 1960).

2 Sextus Empiricus, *Adversus Mathematicos* (Against the Logicians) 8.11.

3 In Chapter 4 of Aristotle, *De Interpretatione*, Translated by J. L. Ackrill, in *The Complete Works of Aristotle, The Revised Oxford Translation*, Jonathan Barnes, ed., Vol. I, Guildford: Princeton, N. J., Princeton University Press, 1984.

4 Cf. the previous quotation translated as follows: "But while every sentence has meaning, though not as an instrument of nature but, as we observed, by convention, not all can be called propositions. We call propositions those only that have truth or falsity in them" (in the *Loeb Classical Library* translation by Harold P. Cookie, London: Harvard University Press, 1938).

5 Norman Kretzmann published a masterful article on the history of semantics in which these topics are investigated from the viewpoint of the philosophy of language (Kretzmann, 1967).

6 MS 517: 40, untitled; NEM 4:248.

7 5.523, c.1905, *Consequences of Critical Common-Sensism*.

8 Peirce makes the distinction between effectual and intended interpretants in his correspondence with Welby. Their merger is the communicational interpretant, shown to serve as the key sign-theoretic notion for later Peirce in Chapters 2, 13.

9 Bearing in mind that there is a history of linguistics going back to, among others, the Pāṇini grammar of Indian linguistics from c.400 BC.

10 This psycholinguistic trait was what the significians were largely after in their theories of language.

11 CN: 3.6; *Nation* 77 (15 October 1903) 308–309, *What is Meaning?* by V. Welby, *The Principles of Mathematics*, vol. I, by Bertrand Russell. According to this review, Russell's book, despite being "really important [work] on logic", "can hardly be called literature", is not "easy reading", and that "the matter of the second volume will probably consist, at least nine-tenths of it, of rows of symbols". Soon after Peirce's death, Russell was asked to edit his papers. Russell was not given a visa, which I believe was one of the scant strokes of fortune as far as Peirce's papers were concerned (cf. Houser 1992).

12 4.431, 1903, *On Existential Graphs, Euler's Diagrams, and Logical Algebra*.

13 According to Smith, "[The division of labour] is the necessary, though very slow and gradual, consequence of a certain propensity in human nature which has in view no such extensive utility; the propensity to truck, barter, and exchange one thing for another. . . . [it is] the necessary consequence of the faculties of reason and speech" (Smith, 1991/1776, p. 19).

14 8.199, c.1900, *Review of Wilhelm Wundt, Principles of Physiological Physiology*.

15 Whitney (1971, p. 100), written in 1867, *Language and the Study of Language*.

16 Nein. Denn ich sage nich, der Satz sei wahr, der nützlich ist (Wittgenstein, 2000–, 131: 70, 1946; 229: 932, 1947).

17 The phrase as such does not appear in Wittgenstein's writings, and when something like it is indeed meant, one wishes to know precisely what it is that Wittgenstein intends the verb 'use' to refers to.

18 Plato, *Theaetetus* and *Sophist*, translated by H. N. Fowler, London: Harvard University Press, 1921, p. 336.

19 Wittgenstein (2000–, 201a1: A5), *Notes on Logic*.

20 Wittgenstein (2000–, 226: 14), *Pre-war Investigations*.

21 See Wittgenstein's lines in the dialogue appended to Chapter 9.

22 In fact, mention of hindsight is not entirely necessary, given that notions resembling Grice's maxims of conversation appear to have been formulated throughout the history linguistic sciences, including Wegener, the Dutch significians, and many others.

23 For instance, when Wittgenstein became preoccupied with G. E. Moore's alleged proof of the external world, his remarks collected in *On Certainty* (1969) were not meant to be attempts to reject or uphold such a 'proof', but to bring out its truly astounding import: that the addressee recognises Moore's assertion, that he has a roughly similar perspective to him as the others, that he interprets his signs in ways sufficiently similar to the interpretations of the others, and so on. *On Certainty*, in as much as it illustrates a coherent whole of Wittgenstein's own accomplishment, is an attempt to articulate this shared capacity that users of language possess, not an exercise in exorcising skepticism.

24 This is not to say that Malinowski had no influence upon Wittgenstein. Some parallels are unearthed in Gellner (1995).

25 In 1945, Paul Weiss and Arthur Burks published a paper on Peirce's sixty-six divisions of signs (Weiss & Burks, 1945).

26 Other than this, Gardiner was a keen critic of Bühler, especially on his analysis of assertions and their relation to statements and propositions.

27 MS 612: 6–7, 2 November 1908, *Book I. Analysis of Thought. Chapter I. Common Ground*. Cf. MS 611: 1, 1908, *Chapter III. The Nature of Logical Inquiry*.

28 A "where to stand", referring to a remark by Archimedes c.235, see Appendix to Chapter 13, in which the whole manuscript is transcribed.

29 'Common familiar knowledge' was occasionally termed "common acknowledged information", affording us to act upon each others' opinions "to enable us to come forthwith to agreement upon all ordinary topics, or, at least, upon questions concerning our meaning in using familiar words such as knowledge, truth, and reality" (MS 612: 7).

30 For example, in MS 596, *Reason's Rules* [c.1902–03] (Chapter 1), Peirce lists several initial beliefs that ought to be assumed that any reasoner possesses. These beliefs are natural assets in the common ground of ordinary language users.

31 In 1967, Grice (1989, pp. 96–100), *Utterer's Meaning and Intentions*.

32 Flouts of this kind may be needed in order to breach the conversational maxim of Manner for the purpose of getting in a conversational implicature, thus revealing that the speaker, while breaking the rules of conversation on a superficial level, tries to ascertain that he or she is obeying them on a deeper, mutually agreed and understandable level of language.

33 Hans Reichenbach was an early user of Peirce's interpretant in his *Elements of Symbolic Logic* (New York: Macmillan, 1947). He started reading Pierce in 1934.

34 MS 612: 27–28, 11 November 1908.

35 Quoted in Watts (2000, p. 166): "Historical pragmatics tries to throw new light on historical language data and on the development of language by combining the traditional methodologies of historical linguistics with the methodologies developed in pragmatics".

36 One area of application that turns on both facets of strategies is natural language anaphora.

37 Some other interrelations are recorded in Shapiro & Haley (1999).

38 I also think that 'potential signification of a symbol' was no part of Peirce's nomenclature. Perhaps it could be charitably reconstructed as referring to rhemas, unsaturated predicate terms. However, if this were the case, it would give us yet another reason to suspect Clark's assertion, namely that a person actually uses a symbol to produce its intended signification by filling out blanks of rhemas and selecting suitable objects from the universe of discourse in question, which of course would be a process dependent on a particular context of symbols.

39 Schmitz (1984) rediscovered the importance of Mannoury's speech and hear-act account.

Chapter 13

PEIRCE'S THEORY OF COMMUNICATION AND ITS CONTEMPORARY RELEVANCE

O MONSTROUS, dead, unprofitable world,
That thou canst hear, and hearing, hold thy way!
— Matthew Arnold, *Sonnets: Written in Emerson's Essays*, 1849

THIS PROBLEM OF THE CLASSIFICATION OF SIGNS is at present occupying the attention of one of the greatest minds of the day, Dr C. S. Peirce, and it is likely that anything attempted prematurely will be rendered obsolete by his long delayed work on Semeiotic.
— C. K. Ogden, *The Progress of Significs*, 1911

1. Introduction

Our mobile era of electronic communication has created a huge dynamic and semeiotic system of information flow, constructed out of the triadic components envisaged by Peirce, such as icons, indices and symbols, and signs, objects and interpretants. Iconic signs bear some semblance or likeness, whatever that is, to what they represent. Indices point at something and say "there!", and symbols signify objects by conventions of a community. In fact, according to Peirce: "The only way of directly communicating an idea is by means of an icon; and every indirect method of communicating an idea must depend for its establishment upon the use of an icon".[1] And signs give rise to interpretants in the minds of the interpreters.

Regrettably, this somewhat simplistic triadic exposé of Peirce's theory of signs has persisted in semeiotics or in one of its neighbouring disciplines as the somehow exhaustive and final description of what he intended. I believe that the more fascinating and richer structure of the signs comes out of their character of "intercommunication" and "interaction" (Peirce's terms, EP 2:389), which has been acknowledged much less frequently.

Despite this shortcoming, the full Peircean road to inquiry — as travelled by the dynamic community of inquisitive learners, or the "community of quasi-minds" consisting of "liquid in a number of bottles which are ~~connected~~ [in intricate connexion] by tubes filled with liquid",[2] or the scientific communities of users of the data that is being provided by Nature, or the vastly increasing electronic sources — reflects the contemporary weight put on all kinds of multi-agent systems in computation. I will discuss multi-agent systems in the next chapter. However, this weight ought still to be strengthened by incorporating Peirce's view of communication into the semeiotic picture emerging from a transdisciplinary multi-agent research. The agents are not only abstract communicators, they are also signs, and thus also minds and in a bona fide relationship with objects. As some signs are phenomenal, they are apt for framing the electronic communication of machine-like quasi-minds. The correlates of Peirce's concepts of representamen (a sign put forward by the utterer), interpretant (what the sign determines within the mind of the interpreter), and various subspecies of the interpretant (e.g., the intentional, the effectual and the communicational) in the context of contemporary media-driven communication and learning need to be determined in the general amalgamation of his sign-theoretic triadism and communication as sign transmission. This is yet to be accomplished. Its importance is evident, for instance, from Peirce's unexpected late idea of the *commens* as the locus at which the thoughts of all minds that participated in the creation of the common ground congregate.

Appendix provides a diplomatic transcription of one of Peirce's central previously unpublished manuscript on the common ground, written in November 1908. Peirce intended it to be the opening chapter for his book entitled simply Logic, which was never completed.

The creation of the common ground by continuous intercommunication and interaction reflects the computational desire to furnish multi-agent systems with properties that would enable them to entertain appropriate interoperation. Thus the initiatives of semantic and pragmatic webs are also given increased semeiotic motivation as soon as adjoined with an understanding of Peirce's open-systems theory of communication (sect. 5). Above all, the phrase "medium of communication" was taken by Peirce to illustrate a broader notion than just the noun *sign*, namely a species of thirdness, a category of mediation, the synthetic consciousness, a prediction of future courses of events, continuity, learning, and growth (MS 283: 106).

A rewarding possibility to evaluate the interplay between technological growth and philosophy is to draw focal parallels and to make comparisons between the notions used in both fields, rather than to seek some overarching foundation for some particular set of technological innovations. Technological advancements have often been made, and sometimes rightfully so, quite regardless of philosophical problems. This has entailed the invention as well as the reinvention of some philosophical concepts. In some cases, terminology has just been hijacked by hackers. This happened with the all-pervading use of ontology in computer science, which has hardly anything to do with its metaphysical homograph. There is no single ontology in web technology, only a library of possible modes of being. It is up to the users to make queries and pick relevant ontologies to be the shared formal specifications of the conceptualisations of what there is. Ontologies tend to reflect interpretations of terms of logical or representational languages, and thus become dependent on the universes of discourse that are common and shared between the agents who operate in them. No self-sustaining substances prevail in user-independent reality.

In the long run, we may witness some convergence of these vocabularies. This is likely to happen as a result of the recently-emerged ideas that aim at new approaches to the organisation, acquisition and evolution of the data contained in the web, namely the 'semantic' and 'pragmatic' renderings of the web concept (the scare quotes will become evident as I proceed). The aim that has been announced quite openly is to ensure that these systems are, or will be, built on the sign-theoretic principles of pragmatic philosophy, most notably on the principles that Peirce is claimed to have envisioned.

I want to know why. My purpose is to concentrate on two interrelated issues. First, my aim is not to unravel Peirce's overall and certainly very complex pragmatist and sign-theoretic philosophy, aspects of which were exposed in earlier chapters, but rather to understand his theory of communication. Of course, this theory cannot be severed from other parts of his thinking, such as his categories, pragmaticism, semeiotics, or the logic of EGs, but as I hope will become clear, the essentials can be understood without overkill from phaneroscopy, evolutionary metaphysics, or his mature theory of signs.

My other aim in this and the next chapter is to assess the relevance of Peirce's theory of communication to some of the emerging contemporary issues in computer science, web technology, and the overall modern era of communicating systems. I have no interest in presenting details of these innovations; I hope that many of them will be familiar. As it turns out, a number of technological and computational innovations have roots in Peirce's scientific method. Being semeiotic, his philosophical and logical concepts are widely applicable and not limited to human users or inquirers. For that reason they are not limited to the linguistic notion of communication either, but reach over virtually all that communication is or would be, now or in the future, including what AI, neuroscience, quantum theory or bioinformatics are able to provide.

My second task is easier. Even if there is as yet nothing like a full picture of Peirce's theory of communication, I believe that we understand it well enough to perceive its relevance to a host of issues, from general systems theory to the applied sciences of computation, communication and information.

2. Triangulate them all

What, then, is Peirce's theory of communication? There is no simple answer to this, and the question has become ever more widespread (Bergman 2000;

Habermas 1995; Johansen 1993; Ransdell 1977). I made some remarks concerning it in Chapter 2. The topic is indispensable in attempts to understand his philosophy from the perspective that aims at strengthening the coherence of his writings, and avoids fragmenting and isolating his boundless fields of interests.

A curious aspect of Peirce's communicative approach to signs is its seemingly dyadic, two-place nature. Prima facie, one may think that the approach is related to the transmissional idea of signs between two (possibly interpersonal) agents, the utterer and the interpreter, in a suitable medium of communication — not unlike Shannon's and others' later syntactic theories that focus on the question of how and via what media information should be propagated. This does, I submit, hardly any justice to Peirce's own intentions.

Second, Peirce's theory of communication is logical. This is the reason why some researchers, including Richard J. Parmentier, have dismissed it as unsuitable for inquiries involving social and cultural issues (Parmentier, 1994). The truth, I believe, is rather that the concepts of what is social and what is cultural are more liable to be stretched and given Peircean twists. For instance, a broad understanding of the social transpires in the currently popular research on multi-agent systems in computation. I will argue in the next chapter that one rarely noted virtue of multi-agent systems is that they provide a much more precise sociological analysis of social codes and practices than the semi-formal notions of social inquiry resorting to 'games' or the 'games people play' (Combs, 2000). Whether such attribution is justified in the end I do not seek to address here (for a masterful study of related questions, see Tuomela 2000). Likewise, the 20th-century concept of logic, set apart from the semeiotics, is limited and not representative of Peirce's overall aims. For, according to Peirce, "Logic is rooted in the social principle" (2.654).[3] In opening up this sentiment, it is essential to recognise that, for Peirce, logical research should also take in the considerations of what one's rational action would be in situations that called for moral judgements. This is of course connected with the fact that his logic is a normative science in the sense that the notion of truth in logic conceals a normative component.

Peirce laid out practically all his divisions in a triadic, three-place format. He did this for many reasons, the most notable being mathematical: Given his assumptions concerning mathematical relations (he was one of the founders of the algebraic logic of relations), no non-degenerate three-place relation can be constructed out of just one- and two-place ones. Therefore, it seems that his overall method of communicating via signs is in some way discrepant or in disagreement with the triadic nature of the other parts of his theory.

The question is: how does the idea of communication between two agents fit into this triadic picture? The answer is, in fact, to be found in his MS 318 (partially reprinted in EP 2), in which he explains his sign theory from the communicational perspective. First of all, there are signs that have no utterers. These are the signs found in nature. Then there are signs that have no interpreters, such as encrypted messages, or the golden plate on the side of the Pioneer 11 probe at the moment of uttering this sentence. The utterers and the interpreters associated with these kinds of signs will receive the prefix "quasi", and they could be thought of as positions, phases of the thinking mind, or semeiotic roles in the process of semiosis. In other words, they are theoretical entities devoid of actual minds connected to brains.[4] In the special case of signs that are symbolic natural-language assertions, the utterers and the interpreters are characteristically human beings. In that interpersonal situation, the utterers and the interpreters are, to a degree, distinct from those of the object and the interpretant.

What, then, are we to say about the residual cases? According to Peirce, the object-interpretant axis represents a continuum that is not meant to demarcate objects and interpretants in any non-fuzzy, clear-cut manner. Some utterers may be assimilated or equated with objects, and some interpreters may likewise be assimilated with interpretants. There are two dynamic scales within the triadic division of signs, one representing the object-interpretant continuum and the other representing the utterer-interpreter continuum. Depending on the nature of the signs, these two scales may coincide, as is the case with non-linguistic signs that have utterers and interpreters, for instance.

The general picture that emerges is schematised in Figure 13.1. There are two main trichotomies, the sign (representamen)-object-interpretant and the sign (representamen)-utterer-interpreter. By moving along the base of the latter triangle towards interpretants and the interpreter, the utterer's state of information increases. Conversely, by moving from the interpreter towards the object and the utterer, the state of the information of the interpreter increases. The dashed arrows show the increase and decrease in the states of information of the utterers and the interpreters. The overlapping area is the common ground in which the communicational interpretants are determined. The α-angle measures the degree to which the objects and utterers converge, and the β-angle measures the degree to which the interpretants and their interpreters converge. Both angles measure the degree of interpersonality in communicational sign-theoretic situations. It may also be concluded from this figure that it is the breadth of the base of the sign-object-interpretant triangle that measures the distance between objects and their interpretants.[5]

Peirce calls what there is in the gravitation between the utterer and its object "the essential ingredient of the utterer", or the utterer's "*quaesitum*" (MS 318: 22), its right to enquire further the components of its object. The *quaesitum* involves the open-ended, dynamic and diachronic model of logical semiosis moving through both historical and real time. Notions of general equilibria may well constitute its parts, but is not exhausted by them. It is found in the utterer's delineation of the class of the universe of discourse that is understood

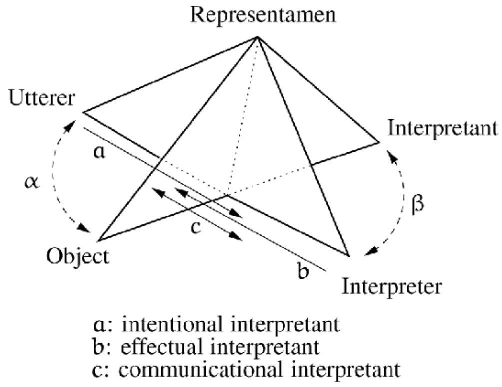


Figure 13.1. The semeiotic pyramid

to be at issue in the dialogue.[6] The utterer chooses the object, sample or its instance from the domain as intended by the utterance and as understood to be predicable by the the rhemas of the proposition in question. I will not delve into the issue of how such choices are made (see Chapter 4): I can override comprehensive explanation by noting that in the case the act of uttering and the object intended by the utterance amount to one and the same thing, there is no decision to be made. If not, the purpose and strategic considerations of the utterer and the interpreter are of prime importance. These are, in turn, related to Peirce's notion of a habit and its cultivation in possibly infinitely repeated runs of semeiotic plays, discussed in Part I of this book.

An open question that has not been posed before is whether Peirce intended this interpretation to supply an objectual interpretation of the quantifiers Σ and Π in the sense of choosing objects from the domain, and intending the names of these objects to function as values of the rhemas and the quantifiers. Or alternatively, did he espouse substitutional interpretation, namely the participants in the dialogue picking out names that are instances of some given substitution-class of non-logical constants?[7] His typical choice of the term was a selection of "instances" as proper values for logical and non-logical constants, and they typically referred to objects, but often he just left unspecified what he the selections can be. For instance, as I noted in Chapter 1, his improvement on Kantian logical analysis ordained that sometimes, some instances need to interpret indefinite expressions.[8] A question that arises is, to what extent are we justified in taking them as corresponding to model-theoretical entities? Or, is the contemporary conception of what model-theoretical entities may be too narrow and inadequate?

Quite another aspect of Peirce's theory of communication is that, if the interpretant is what he termed the (ultimate) logical interpretant, then, in representing a state of perfect knowledge, the ontological and epistemological distinctions make no difference in scientific inquiry. The object merges with its interpretant, disintegrating the triangle into a dyadic relation between the sign and its ultimation. In other words, the maximal state of information leaves no latitude for interpretation because there is no longer any difference between objects and interpretants.

Many interpretant triads exist in Peirce's writings, which I will not go on to review here (see Chapter 1). They are all intended to make negligible distinctions in order to broaden the base of the relevant triad in the semeiotic pyramid.

Because Peirce was keen to repudiate all psychological influence on the province of logic and semeiotics, he might have wished to eliminate the concept of utterers and interpreters from the dominion of sign action. This is evidenced in his frequent tendency, I submit, to assimilate utterers and objects on the one hand, and interpreters and interpretants on the other. Upon closer inspection, this assimilation does not mean reduction at all. The concepts of the utterers and the interpreters are, as Peirce puts it, "welded" into one sign (4.551), but they move along the distinct scale from that of the objects and the interpretants, as the bases of the triangles have independently variable breadths. It is the 'gravitational' force between the utterer and the object, and between the interpreter and the interpretant, provided by the utterer's or interpreter's *quaesitum* of meeting the obligations concerning the object or the interpretant.

Since every thought is a sign, no thought can evolve unless conceived of as dialogic, either between multiple, interpersonal parties or as a quasi-dialogue within one mind. This quasi-dialogical perspective offers a method of assigning semantic values to logical propositions, whereas person-to-person dialogues work for pragmatic theories of communication and discourse.[9]

According to Peirce, however, there is little difference between multi-party interaction in, say, a social setting and the intrapersonal reasoning and action in logic, because "a person is not absolutely an individual" (5.421).[10]

Peirce's theory of communication comes extraordinarily close to the dialogical and game-theoretic interpretations of logic. These interpretations could be considered formal (logical) and strategic regimentations of relevant parts of Peircean semiosis. Although invented quite independently, they endorse elements of communal or social approaches to meaning in that the idea is to check the truth-values of the propositions of a logical language. For the most part, they lurk behind his diagrammatic and iconic systems of EGs, and point towards ways of extending these systems. The difference between dialogue games and GTS is that dialogues aim at validity of propositions, whereas semantic games seek to establishing when the propositions are true in a model and when they are false in a model. These theories both distinguish players' roles in terms of the polarity of the logical constant encountered in the formula, including logical connectives, and switch the roles when negation is encountered. Dialogue games and GTS are also both strategic in that the notion of a winning strategy represents the concrete concept that agrees with the notions of validity (dialogues) and truth (semantic games) of propositions. Peirce had most of the features of both in his system of logic, although he did not come to endorse any unequivocal game-theoretic terminology.

Even so, his theory is somewhat richer still. Its all-important concepts of the common ground and the universe of discourse are not limited to logical or model-theoretic domains. His theory is applicable beyond proof theories and game semantics, placing pragmatic and discourse-related phenomena in linguistics under logical and semeiotic scrutiny. As was seen in the previous chapter, it also contained the origins of speech-acts and theories of relevance.

The collaterally-acquired common ground secures the very success of communication in semeiotic dialogues. The ground refers to what is mutually understood and shared between dialogue participants, determined in their common mind as the common communicational interpretant, which allows them to understand each other's utterances. As noted, Peirce once gave this common mind the special name of *commens*.

The title of this section is not coincidental. The emerging idea which was never explained in full by Peirce, is not unlike what ensues from Donald Davidson's triangulation scheme (Davidson, 2001). According to it, whereas individually and communally the speakers and hearers of language may go wrong in their interpretations, in a broader context, for any communication to be successful, the beliefs of others are not to be taken to be radically different from our own. Anyone having a belief must be interpreted as having a true belief, even if the belief in question would turn out to be false. Peirce's guarantee of a similar outcome was the inevitability of collateral observation and mutual experience plus the normative maxim of *summum bonum* that the communities of inquirers share in communication. The main idea is thus also similar to Davidson's principle of charity in interpretation.

However, there are points in which Davidson's triangulation may be insufficient for Peircean purposes. Apart from the communicational link between two or more subjects, which Davidson considers to be essentially richer than any monological incident, the 'third' in this theory is simply the object, of which the conversations are intended to be about within the subjects in question. To make this picture richer, and to account for its multi-culturalised versions that have been argued as being consistent with the original scheme (van Braakel, 2003), one would be well advised to replace the notion of the object and the associated concept of objectivity with the Peircean understanding of the sign, with the more phenomenological undertones of the latter.

This, mutatis mutandis, gives rise to a 'double triangulation' a version of which is depicted in Figure 13.1. What accompanies this understanding is the extent to which the two corners of the bases of both triangulations are related, in other words the affinity of the object with its utterer and the interpretant with its interpreter.

Furthermore, the presence of two similarity responses, which Davidson assumed to be persons, may be replaced by the pragmatic concepts of the ego and non-ego, the interaction of which gives rise to existence. In Davidson's view, similarity responses evoked by the two parties also give rise to objectivity of what exists.

Such a semeiotic enrichment of triangulation ipso facto accounts for all the cultural differences that language users may exhibit, and the ensuing 'multiple-worlds views' that examples of peculiar ways of conceptualising the world may imply.[11]

The universe, or multiple universes of discourse, is thus a key element in Peirce's theory of communication. In dialogues, they are not just total domains in that overworned logistic sense, but encompass also presuppositions shared in the conversation and established by the same principles as the existence of mutually-gained collaterality of the common ground.

What, then, are the practical outcomes of Peirce's concept of communication?

3. Applications and complications

Interoperability Interoperability is a boast recently made in all corners of computing. Dictionaries tend to define it as 'compatible software or hardware', but it does not merely represent the technical challenge of some coding or manufacturing problem. On the contrary, it has been described as "the ongoing process of ensuring that the systems, procedures and culture of an organisation are managed in such a way as to maximise opportunities for exchange and re-use of information, whether internally or externally" (Miller, 2001).

This pragmatic definition focuses on what goes on in communities of engineers, researchers, managers, and other users of knowledge. It is also reminiscent of utilitarism, implying the maximising of something (here: opportunities to exchange information). Intentions to do this clearly depend on the scope of common interest in having interoperative systems and products in the first place. Indeed, humans can be stunningly interoperable at will.

Interoperability also has linguistic, social/communal, legal and normative aspects, and so it is a good example of Peircean inquiry as an indefinitely extendible and inexhaustible activity. The goal is to create communicational interpretants in a variety of cases, across the boundaries of what is artificial and what is human, whatever entities the subjects engaged in communication are taken to be.[12]

Weave this: semantic and pragmatic webs The goal of next-generation web technology is to define meaning in web documents. The increasingly popular albeit not yet very widely implemented approach known as the semantic web uses mark-up methods instead of plain keywords, which define the class and subclass hierarchies and the relations between the concepts that appear on the page (Bernes-Lee, 1999; Fensel & Musen, 2001; Bernes-Lee & Miller, 2002; The Semantic Web Agreement Group, 2001). This metadata information provides the 'semantics' (or in this restricted sense the 'meaning') of the document. According to this project, it is hoped that the increased production of metadata for ontology languages will create a network of documents, the content of which could be automatically processed in a much more elastic and adaptable manner than in standard syntax-driven string-matching search methods.

One might think that this approach really has nothing to do with semantics. However, as a practical present-day version of the Peircean notion, it provides the meaning of the data or of a code by translation. Peircean semantics is, after all, a theory of translation, a rendition of a given symbolic statement into some other statement, diction, or paraphrase, or into some other language, or perhaps a dictionary-like definition of it.

However, this understanding of semantics lacks the semeiotic components of the utterers and the interpreters of the data. We still need to understand how the metadata, such as that provided by the Resource Description Framework (RDFS), is connected to the interpreters and objects of data. This connection defines the pragmatic meaning of data. However, as such it does not fulfil the vision of a pragmatic web, as announced, for instance, in de Moor et al. (2002). The pragmatic web draws the community of inquirers, most notably web users, into issues to do with the purpose of information. While such intentions and contexts of users surely play a significant role in pragmatic accounts of meaning, and while these researchers are certainly right in criticising the semantic-web approach for its limitation to the metadata idea and neglect of the communities of human users and engineers, the approach sidesteps the perhaps more profitable possibility of incorporating truly semeiotic pragmatics into the automatised and computational level of the web. Rightly, it points out that a semantic web devoid of human users is insufficient. It asserts that new meanings or concepts do not simply emerge by adding more and more structural features to the web pages or by linking them more and more efficiently. Even so, there is still a need for a methodeutic here, making contact with the third main class of normative logic beyond grammar and logical semantics, to fosted the methods of communication between human users, computerised agents as well as humans and computers.

Multi-agent systems We can only hope that from the ashes of the vast amount of research done on multi-agent systems will rise precisely this pragmatic web challenge. The challenge involves an attempt to build agents, or pieces of advanced software, that are designed to play the different semeiotic roles of quasi-utterers and quasi-interpreters. In other words, they would play the different positions in the cycles of dialogic semiosis as prescribed in Peirce's theory. This is the way in which they are intended to contribute to the generation of new objects and the evolution of new meanings in the web.

Agent systems still lack the truly goal-directed specifications of processes. Only when that is accomplished could they be considered to create habits and produce the wherewithal for revision. This is a long way off. Autonomous and proactive agents need to build second-order evaluations of their own strategies, noting when a habit-change occurs, namely when the logical interpretants considered in Chapter 1 are produced in the quasi-minds of agents as the end products of the process of semiosis that terminates or is about to terminate. They need to learn whenever they 'feel' pain, whenever something meaningful happens to one of the individual agents. Knowing when that is to happen depends on the correct evaluation of the habits that are already in the agent's possession. The next chapter takes up this theme of Peirce's thought.

Questioning the web Elements of goal-directed agent systems are emerging in the vision of the knowledge web. The aim is to overcome these shortcomings and supersede both semantic and pragmatic web enterprises by taking agents as constructors building a huge question-answering system on the web data, and responding to queries on an information-need basis. This is certainly also a long way off, because one needs to accomplish two things: (i) a comprehensive logic of questions and answers, and (ii) a definition of a workable possible-worlds structure of the web. Neither has been accomplished as yet.

On the first point, the quest for a logical relation between questions and answers stands upon the edge of theories of presuppositions, because requests for information can be viewed as epistemic statements. (This task is also related to the separate question of how to extend Peirce's theory of abduction.) The command 'Bring it about so that S' has root in the non-imperative epistemic sentence 'I know that S'. As to the second point, web nodes should be viewed as knowledge providers, and via that emerging structure epistemic statements are translated to mean, 'The user knows S in the information state w if and only if S holds in all the web nodes accessible from w'.

Initial states for the users of information then need to be agreed. As they are software agents querying other software agents, w codifies the knowledge in the position they have reached within a "model-checking" game on the web.

An alternative, and I believe complementary, way of building a knowledge web is to use conceptual graphs (Mineau, 2002), which closely resemble Peirce's EGs. Their purpose in the web domain is to provide good representational formalisms to describe the workings of software agents. Many of Peirce's ideas, however, especially those related to the gamma part and their extensions are still to be incorporated into the conceptual-graph framework.

The semantic + pragmatic web = the semeiotic web? Both semantic and pragmatic web visions share a mutual concern about the inadequacy of the current architectonics of the web. The semantic web aims at providing a 'logical analysis' of data, while the pragmatic component adds the human perspective. Both of these approaches are somewhat inadequate, but in their combination, and from the perspective of the semeiotic and logical approach to inquiry, they have rich emergent features and promising applications to the web context, in which semantic and pragmatic initiatives are inadequate alone. The outcome of these considerations the truly semeiotic web. This combines both semantic-web and pragmatic-web initiatives, but exceeds them in that it takes both in a way that is faithful to Peirce's pragmatic approach to inquiry, its methodeutic.

Unlike the related concepts of semantic and pragmatic webs, such plans need also to be operationalised in an effective and thorough fashion. Hence a new approach to multi-agent systems is needed that addresses the weaknesses of both semantic and pragmatic webs by the new logic of questioning and answering and by taking agents as roles in the dialogical, semeiotic inquiry of signs in the universe of the web. This task is taken up in the next chapter. The upshot is that semantic and pragmatic conceptions cannot and should not be separated (Pietarinen, 2003f).

4. Pragmatism from a communicational perspective

In the light of previous remarks, the common ground of pragmatically-inclined software agents is bound to be quite different from that of humans. Software agents do not have similar self-awareness, such as what it is to be a member of the common genus of *homo*. It is also quite clear that knowledge of the language and knowledge of the universal aspects of grammar or specifications of a code lies in the hands of the programmer. It is equally obvious that experience of the world differs.

Nevertheless, agents do not need to be taken as fundamentally different from humans in all their aspects. For instance, common knowledge of rationality and common knowledge, essential in the creation of communicational interpretants, are definable on the logical level. Linking languages to the world presupposes that there is a great deal of common experience shared by their utterers and interpreters. In Peirce's system, it is typically the copula that ties predicate terms to the elements of the domains of discourse. Nowadays, the interpretation of languages is given by the static valuation function that assigns values to their non-logical constants (functions, predicates and symbols). This provides the boundary conditions upon which the semantic clauses are defined. However, from the semeiotic perspective such a valuation is itself subservient to dynamic and dialogical reinterpretations. A way to spell out the difference is in terms of closed vs. open systems (Chapter 3; sect. 5 below), in which for the latter, no function or law exists that associates the system with its environment.

The characteristics of the latter system are emblematic of non-deterministic functions, relations with non-unique outputs.

The upshot is that two main components of being pragmatic should equally be taken into account in semantic/pragmatic web enterprises.

The first is the contextual/situational/environmental dependency of signs. There are logical ways of tackling this, witness the conceptual-graph research anchored in the diagrammatisation of assertions. Logics based on diagrammatic reasoning plus other heterogeneous representation formalisms are typically context-dependent by their very nature.

The second component is the utterer's meaning as distinct from the literal meaning of the utterance. Recalling the divisions between triadic interpretants, we say that the utterer's meaning is in the intentional interpretant, mediated in as meaning-preserving a way as possible to the receiving effectual interpretant created in the mind of the interpreter. In contrast, the literal meaning is in the immediate interpretant of the sign. The immediate interpretant is then that which is created even if there is no interpreter.

These points relate to the Peircean concept of the universe of discourse, which can be conceived in the following two ways.

First, there is the contextualisation task, which is made easier by there being collateral observation and mutual experience shared by agents. This is the task of 'model-building'. It is described in the presuppositions of EGs by way of collaboration between the Graphist and the Grapheus, namely between the agent who proposes modifications to the graphs and the agent who creates the universe and decides the truth of atomic expressions. The Grapheus does this by either authorising or refuting the actions proposed by the Graphist. Peirce held that there is no opponency or competition in this description. The aim is to capture and agree on what the relevant aspects of the system are that are to be modelled and the properties of which are to be studied.

Tableaux methods are examples of model building in which the aim is to search for a counterexample to the proposed assertion. In a similar vein, to check consistency of the assertion is to perform a satisfiability check, which means that one tries is engaged in building a model for the assertion. These modern approaches introduce competition in the sense that a set of assertions having a model is tantamount to the existence of a winning strategy for the Graphist (the Builder, the Proponent). Likewise, the existence of the winning strategy for the Grapheus (the Critic, the Opponent) is tantamount to the demonstration that the negation of a given assertion holds. What the Grapheus is doing is to search for counterexamples that would show the invalidity of initial assertions.

In the Scottish Book from 1930s Ulam and others proposed that the game-like idea of forcing (Banach-Mazur games) applies to building of a model. The game is one of teamwork and cooperation. Peirce's anticipation of this was the idea that the Graphist and the Grapheus "collaborate" in building a "Pheme" (a model) for assertions (4.538; 4.552). In that game, the Graphist "proposes modifications to the graphs", while the Grapheus "creates the universe" and decides upon the "determinations", namely the interpretations of atomic formulas by "authorising" or "refuting" the "actions" of the Graphist (4.538; 4.552).

After building, the semantic game on assertions commences. Second, therefore, the sign-theoretic communicative phase takes prcedence. In that phase, signs represent objects, and their instances are chosen from the mutually observed domain of discourse by the dialogue participants. This is the task of model-interpreting or model-checking. Peirce described it in the constitutive rules of interpretation and considerations pertaining to the education of partakers' habits towards stableness. The participants will now have conflicting purposes. This description antedated the modern methods of semantic games.

Both tasks are of great concern for those working on formal methods: logicians in their model-theoretic activities, econometricians in their attempt to identify relevant parts of economic systems, and natural language semanticists in their judgements concerning the amount of non-truth-conditional material that infiltrates linguistic theories of meaning.

In a closer relation to the topics of the present chapter, the aforementioned points give rise to the following general observations and suggestions:

(i) The easier it has become to transmit data through computerised networks, the more difficult it has become to share data for mutual processing and understanding. This is not as much a shortcoming due to technological challenges as the failure to admit non-individualistic Peircean persons. This concern was recognised by the early significians, who wished to facilitate improved understanding through post-Peircean analysis of communication. In our day and age, it appears almost as if multi-agent systems were the proxy forces that have been set out to do what humans have failed to accomplish.

(ii) What is more important than the complex attempt to make incompatible vocabularies of databases and web documents understand each other is to refurbish methodologies for sharing meaningful information. Peirce laid emphasis on the importance of methodeutic for the community of inquirers in their study of "the methods that ought to be pursued in the investigation, in the exposition, and in the application of truth" (1.191). This is essential in communication, for "it is the doctrine of the general conditions of the reference of Symbols and other Signs to the Interpretants which they aim to determine".[13] Ultimately, there is then a need for finding a "method of discovering methods" (2.108, 1902), a logical task that would enable inquiries to manage the ever-increasing streams of computerised information.

5. Towards open-systems philosophy

To bend the perspective somewhat, we may think of communication as the abstract activity in any open system that receives persistent streams of through-put and is not yet in the state of equilibrium or in the state of maximum negative entropy. Accordingly, information may be held to be whatever there is that is needed in such activities to ascertain that the system is not in such a state. The meaning of signs would, according to this view, when arrived at in their final, ultimate state of interpretation, have reached an equilibrium in which much of the information has been consumed. In a well-defined logical sense, tautologies (analytic logical truths) are uninformative in not giving away any new information that agents could make genuine use by means other than those of logical information processing, including theorem proving. All in all, languages may be viewed as open systems that operate with specific kinds of signs. Maybe logic, too, has taken tendencies to reflect such systems increasingly more.[14]

Information plays a similar role in biological and physical systems. They start with some random state or a distribution of values and weights such as a priori probabilities for different alternative states of the world. As these primordial states gradually evolve, together with the law-like features that govern them, they tend to states that reduce the need for the consumption and exchange of information in order for the agent or the experimenter to measure the approximate distance from that goal state. There are various parameters that the notion of information may denote in such processes, including symmetry constraints, various conservation principles and degrees of freedom of physical systems.

Logically and mathematically, information may mark those boundary conditions that are fixed in any particular local system under investigation, such as in a particular model in the mathematical sense of model theory. As the tendency of model theory is to study what is similar and what is dissimilar between classes of structures that are parts of some large homogeneous structure, the role of information is even more crucial. Such propensity suggests one to regard models as systems that possess the kinds of characteristics that define openness in the sense of general systems theory, the large homogeneous structures playing the role of the environment.

Systems or models of that kind, like organisations, institutions, economies, social groups and sets of conventions and laws, however man-made or natural, may be seen as maintaining systemic habits with adaptability, self-organisation and change as their core features. There are countless examples of the usefulness of this way of looking at interactive systems. Chapter 14 provides one parallel between those of games and multi-agent systems, recently employed in computation and AI. This parallel has several points of contact between Peirce. From the general systems perspective, the concept of a habit appears as particularly appealing. Among other things, it may be seen as a reflection of aspects of the logica utens of open systems (Chapter 1), whereas its companion of logica docens refers to the compartment of logic by which we, as engineers or system modellers, come to understand these things.[15]

That open systems are closer to game-theoretic ways of thinking than static transition systems (such as automata) is shown by the closed and inactive nature of transitions in their process-like notion of producing outputs, in which the behaviour of the system is unable to cope with the variability of input, such as its timing, pace, quality and the methods of feeding. This observation serves to support my thesis that large portions of Peirce's philosophy may be productively conceived through a game-theoretic lens.

A slogan says that 'evolution is chaos with feedback'. This is Peirce's evolutionary metaphysics in a nutshell. Chaos is his firstness, tychism that prevails in the universe of unorganised substance. Feedback provides a law, secondness that flows from input to produce output that is of certain value. Finally, evolution is the habit-taking tendency of the universe, which brings the law of feedback and the payoff values it provides in continuous relation with chaotic ripples of chance. Peirce expressed this view in his *A Guess at the Riddle*, written in 1887–88 in his attempt to summarise the answer to the puzzle of the cause and origins of the universe (reprinted in W6). That article outlines a grand philosophical agenda for 'general informatics', ranging from information-processing sciences to general systems theory, still awaiting full-bodied investigation.

6. Conclusions

Having identified some main issues involved in a general amalgamation of communicative and triadic viewpoints on signs, and assessed the contribution it has made to the emerging contour of a full Peircean notion of communication as sculpted by the recent era of intercommunicating computational systems, what are the repercussions? Peirce's philosophy represents a drastic departure from the Cartesian view, the one-time programme of those who were trying to understand the discourse of the distinction between mind and matter on the one hand, and the discourse of the interaction between them on the other. Peirce presents all interaction as triadic between signs, objects and interpretants. His thinking may be reproduced as open-systems philosophy in which systems, be they artefacts or human beings, react to environments in a non-programmed, habitual manner.

For that reason, some may regard it as a never-never philosophy, a Peircean would-be, a hypostatically abstracted metaphilosophical *Erewhon*. Nevertheless, it is strictly rational, adhering to principles of logic, while keeping a critical eye on other socio-logical principles of inquiry, including the much less logical post-Marxist utopias of global communication communities, or dystopias of all-pervading power relations. It denounces skepticism. It emphasises the positive role of the community of inquirers, be they quasi-minds of software agents or human interpreters, in creating new objects and events, developing new meanings and concepts, and ultimately achieving the main goal of scientific inquiry, namely the attainment of the truth.

It is remarkable how well Peirce's never-never philosophy has kept its promises in the light of current technological advances — I see this as a self-returning pragmatic maxim. I predict prosperity for Peirce's philosophy as the 21st century kicks off, not only because of its pragmatic solutions to ever-increasing pragmatic questions, but also because we are only beginning to see the grave limitations of the last century's conceptions of logic and the impasse of analytic philosophy.

Notes

1 2.278, 1895, *Speculative Grammar: The Icon, Index and Symbol*. Peirce's *chef d'œuvre* came into being shortly after these remarks were made, in the form of his diagrammatic system of EGs. As I noted in Part I, this was a thoroughly iconic representation of and reasoning about 'moving pictures of thought', which encompassed not only propositional and predicate logic, but also modalities, higher-order notions, abstraction and category-theoretic notions. The importance of iconic representation in scientific and everyday communication has frequently been noted, starting with the works of Russell, Wittgenstein and Neurath, although as logics they had to await the heterogeneous systems of the late 20th century.

2 MS 318: 133. Peirce's attempt was to explain what it means for a sign to be "a determination of a quasi-mind" (MS 318: 131) by the synechist metaphor of the chemical continuity of fluids, for "a pure idea without metaphor or other significant clothing is an onion without a peel" (MS 318: 132).

3 There is a somewhat converse declaration elsewhere, "The social principle is rooted intrinsically in logic", 5.354.

4 This may be related to the concept of 'natural intelligence' that has been deployed in AI, especially in automated reasoning research (Pietarinen, 2004a).

5 Johansen (1993) has studied interactions between sign-theoretic triads from the communicational point of view. My interpretation of Peirce's theory differs from those suggestions, however.

6 MS 318: 29: "Pronouns are words whose whole object is to indicate what kind of collateral observation must be made in order to determine the significance of some other part of the sentence. 'Which' directs us to ~~turn our attention in what has been said,~~ [seek the quaesitum in the previous context;] the personal pronouns to observe who is the speaker, who the hearer, etc. The demonstrative pronouns usually direct ~~attention~~ [this sort of] observation to the circumstances of the utterance (perhaps to the way a finger points) rather than to words."

7 This is the interpretation that was so named mainly after Kripke (1976).

8 Some passages suggest that Peirce came close to the 'discourse referent' idea in DRT (see Chapters 4 and 6).

9 Strategic versions of such dialogues give rise to optimality-based theories for phonological, syntactic, semantic and pragmatic inquiries in linguistics (Dekker & van Rooy, 2000), and various conversational and dialogue games for actual language users (Carlson, 1983), for instance.

10 The multi-agent nature of communities has multiple contact points with Peirce's theological, cosmological, evolutionary and agapistic metaphysics. That he was caught between the two fires of exact sciences and religious thought prevented him from presenting his systems in a sustained, unitary form.

11 Such cross-cultural issues are the key ingredients in historical research on pragmatic change (Chapter 12).

12 Shared ontologies are good examples of communicational interpretants in artificial systems.

13 2.93, 1902, *Partial Synopsis of a Proposed Work in Logic*.

14 Such tendencies would, among others, (i) endorse the return to pictorial and graphical methods of expressing logical assertions instead of symbolic predicates, (ii) refer to semeiotic processes instead of interpretations of non-logical constants, (iii) manipulate logical relations in a Gestalt, context-dependent manner rather than compositionally, and (iv) refer to indefinite and indeterminate objects and events in addition to well-defined individuals (Pietarinen, 2004f, 2005a,c).

15 It is also curious to note the Peircean premonition that in many difference equations for logistical examples, if the growth factor increases beyond three, the systems do not tend to stable equilibria.

Appendix 13.A: Manuscript 614 on Common Ground

Logic. Book I. Analysis of Thought.

Chapter I. Common Ground.

18 November 1908

Your purpose in reading these pages ~~was~~ [has been] mine in writing them, namely that you should be enabled to reach the truth the more surely and expeditiously for having studied them. If we succeed it will be a great achievement for us both. For my part I shall feel ~~as if I could~~ [that having so succeeded with one, I may hope to succeed with so many that all of us together shall] move the world. But as Archimedes said he could do that with his lever only if he had a ποῦ στῶ, a where to stand, so I am obliged to remember that no man can communicate the smallest item of information to his brother-man unless they have a ποῦ στῶσι of common familiar knowledge; where the word 'familiar' refers less to how well the object is known than to the manner of the knowing.[1] ~~that is, directly in the object~~ [p. 2] This manner is such that when one knows anything familiarly, one familiarly knows that one knows it and can also distinguish it from other things. Common familiar knowledge is such that each knower knows that every other familiarly knows it, and familiarly knows that every other one of the knowers has a familiar knowledge of all this. Of course, two endless series of knowings are involved; but knowing is not an action but a habit, which may remain passive for an indefinite time.

You have an advantage over me in this matter, since you know something about me, while I do not even know that you exist. Nevertheless I know that if you exist you have some acquaintance with the English language, and that you have some notion of the grammar of our Aryan languages; and it will be safe to assume that we have a common familiar knowledge of the ordinary truths of human life. I shall risk the assumption that you [p. 3] are neither a child nor a dullard, but are a normal adult of sufficient intelligence to be interested in methods and their adaption to ends and not to confine your admirations to successful results, ~~which are~~ (these being actually more or less fortuitous), a character which places your intelligence, in my estimation, in a class decidedly above that of the average of mankind.

Such being the case, I risk nothing in assuming that you are well aware that the exercize ~~over our~~ of control over our habits, if it is not the most important business of life, is at least very near to being so, and I dare say you have taken some pains to discover just how that control is effected. The word 'habit', as it is ordinarily used, ~~is not~~ does not convey ~~the precise~~ [quite the] idea that I seek to convey. It is, I think, usually taken to denote any character of a [p. 4] person which conforms to these two conditions: firstly, that it ~~has~~ [shall have] resulted from that person's having many times behaved in one general way under circumstances of one general kind; and secondly, it shall consist in a tendency, on the part of that person after the fulfilment of the first condition, to behave in the same general way under circumstances of the same general kind. This makes the habit to consist in an impulse or psychic cause of a resemblance between a person's actions after repetitions and his actions during those repetitions. The sense in which I use the word but slightly differs from that, except that I disregard both the manner in which the habit has been established and the difference between the behaviour of a person and that of a thing. I mean by a habit ~~the accord of the behaviour of a person~~ a power acting through ~~the~~ [a person's] soul and tending to make his behaviour accord with, or conform to, a general idea.

[p. 5, 19 November 1908] Everybody knows that people are able to exercise some considerable control over their habits; and some day I ought to endeavour to give you some aid in tracing out the *modus operandi* of that control somewhat minutely. At present, it comes in my way to direct your attention to certain features of the process only. In doing so, I shall not meddle with the science of psychology; and this is a remark that I shall often have occasion to repeat.

Many persons, perhaps most persons have the idea that every observation about the human mind is a psychological observation. They might as well regard the sight or sound of an apple dropping from a tree as an astronomical observation in view of what is said to have befallen Isaac Newton. The notion is due to the utter want of comprehension, on the part of the public, [p. 6] of what science, in its modern sense, really consists in; and in here speaking of the public, I not only include ninety-nine per cent of the members of the most enlightened of unscientific circles, many of whom call themselves 'scientists', — a word, by the way, that very rarely drops from the lips of a genuine man of science; — but I include not a few true scientific men, besides. No doubt, all well-educated people, — in this country, at least, — understand that 'science', in its modern sense, is neither what the ancients meant by *scientia*, or ἐπιστήμη, which was nearly what we should call 'comprehension', which a mental acquirement, nor is it what Coleridge called 'science', and defined as systematized, or organized knowledge, — ~~knowledge~~ [thus] being, in truth, the fossilized remains of science. They know, or the wisest of them do, [p. 7] that when they ~~hear~~ [overhear] mathematicians, or physicists, or chemists, or physiologists, or comparative anatomists, or astronomers, or geologists, or the like speak of 'science', what will be meant in every nine cases out of ten will be the collective activity, — the 'business', — that is being carried on in social groups each consisting of men who possess special facilities, external and internal, for the solution of problems of a certain kind, and are devoting their ~~whole power furthering such~~ [total energies to] discovering those solutions. So much, I say, the best-instructed of the public do understand; but not even they nor all scientific men of great eminence, either, seem to be aware that no man has any special facility for solving any hitherto unsolved problem he chooses of however narrow a class that is marked out by any rational characters. *He only finds himself in* [p. 8] *condition to attack this and that individual problem*; and ~~if there be a~~ if these belong to a class of problems out of which single problems come within the range of facilities of members of a social group, so that they will with competence examine his work upon the problems he attacks, and pronounce their approval of it, then, and not otherwise he is what scientific men usually mean when they speak of a person as a scientific man. The pioneer of an entirely new line of inquiry cannot be pronounced a scientific man, except by those who afterward follow in his foot-steps; and they will commonly detect grave errors in his procedure, or at any rate, will think they do. But ~~for the~~ [in] most cases, a quite new road through the dark ~~thick~~ jungle ~~virgin forest~~ of ignorance does not get broken at all until science reaches a stage in its development at which several men make the epochal discovery at once. That the best-instructed of [p. 9] the public do not understand the condition ~~whose~~ [the] statement of which I have italicized above is shown by their often demanding that the 'scientists' should investigate this or that phenomenon not to ascertain whether it accords with established principles (which can always be done), but to discover what new secret it involves; and now and then we hear scientific men themselves acknowledge the reasonableness of such demand. Yet ~~that principle~~ [that the italicized statement] is true, several ~~eminent~~ [great] discoverers have virtually declared; and a much ~~better~~ stronger argument ~~of~~ for it, as not depending upon any fallible opinions is the frequency with which, in the history of science, epochal discoveries of the most astounding ~~character~~ [novelty] have been discovered simultaneously by ~~different~~ more men than one up to half a dozen. For this is an inevitable consequence of the italicized condition, which otherwise seems inexplicable. If the substance [p. 10] of my italicized sentence had merely been an ingenious hypothesis framed to account for the strange historical series of coincidences, the latter (as I shall show you later) would have furnished no support for belief in the former: but in point of fact it was quite the other way. The former was an induction by which my life-long intimacy with many scientific men had led me to believe that what I had remarked of my own quite hap-hazard competence to attack problems was true of men of science generally, — a belief that had some additional support. Still, though that induction was quite legitimate, I remarked two elements of weakness in it, first, that it was of that crude kind to which one ought not to trust exclusively when one can avoid doing so, and secondly that the observations had not been regularly recorded but had preserved only by a treacherous memory that [p. 11] in some cases had made but vague reports. I thought, therefore, that it might be that what my inductive theory required in some exceptional case might be clearly negatived by memory of the facts, which would at once refute the theory; and therefore, as my object was to ascertain the very truth of the matter, I tried to ~~find~~ think of a case which, according to the induction, ought to present some exceptional feature. With this view ~~considered how it would be with~~ [I determined to study out] what the theory would require in the case of a problem whose solution should involve some great novelty; for I had a vague inkling that the requirement of the theory in that case would be exceptional. What I ~~had in~~ looked forward to as possible was that the facts might ~~be~~ have no such exceptional character as the theory would require, and would thus refute the latter. I began by considering the case of a scientific man [p. 12, 20 November 1908] engaged upon a problem of a familiar kind. To fix my ideas I imagined the problem to be that of determining the atomic weight of tellurium

[a.p. 3, 18 November 1908] are neither a child nor a dullard, but are a normal adult of more than average intelligence.

As such, you are well aware that the exercise of control over one's own habits is perhaps the most important business of life, meaning by a habit an object whose being lies in an enduring state of a person which consists in ~~his acting~~ ~~the~~ [a] tendency to act, mentally or bodily, in certain ~~sort of~~ [general] way, whenever he has been acted on in a certain general ~~sort of~~ [general] way, regardless of how this tendency may have been established.

[a.p. 5, 18 November 1908] Everybody knows that persons are able to exercize some considerable control over their habits. Later we shall have occasion to consider in some measuer the *modus operandi* of this control. At present, we need only note that a review, or reminiscential repetition, of one's conduct upon a given occasion often excites a feeling of repulsion, and that this leads him to imagine behaviour governed by other general ideas, and finally he will imagine a line of conduct, or controlled behaviour, on the given occasion which excites a more intense feeling of attractions than any other that occurs to him; and ~~this behav he will repe~~ behaviour goverened by the same general idea as the behaviour that is most attractive to him will be copied in imagination in more or less varied forms all governed by the same general idea. Now, it is an important law of the soul that repeated performances ~~of~~ governed by one [a.p. 6] general idea ~~to give~~ impart a power to that idea ~~be~~ alike over imaginary and over real behaviour, and that whether the performances ~~are~~ repeated ~~in~~ [be] actual or ~~in~~ merely imaginary, although the influence of actual performance is generally more powerful, owing to a secondary cause. It follows, therefore, that in a mind ~~much given~~ which has a habit of reviewing ~~conduct~~ [behaviour], there will be a constant tendency toward the formation of habits which give rise to conduct that is 'approved' on reflexion, that is which excites a feeling of attraction. ~~This is~~ [I have thus] submitted to your criticism a very slight and poverty stricken sketch of a complex phenomenon that lies at the corner-stone of morality. There is one class of habits which, "from the nature of things", — as we say of what is true by logical necessity, — [end of the manuscript]

Notes

1 ["Give me where to stand and I will move the earth", a remark by Archimedes c.235 BC quoted by Pappus of Alexandria in *Synagoge* VIII, c.340 AD, Berlin: Hultsch, 1878, p. 1060. Cf. "Give me somewhere to stand and I will move the earth", *Great Mathematical Works* II, Ivor Thomas, Loeb Classical Library, Cambridge: Harvard University Press, 1941, p. 35.]

Chapter 14

GAMES *VIS-À-VIS* MULTI-AGENT SYSTEMS: A PEIRCEAN MANIFESTO

An older and more mature field than multi-agent systems, game theory demonstrates its importance in a number of new connections. Being goal-oriented processes, games introduce strategic dimensions into systems that otherwise may rest on unstable methodology, and economise agent interaction by tackling agent rationality as well as its bounded manifestations. Since games are open systems, heeding this methodological stance turns agents into such open systems too. These connections have remained in the shadow of the mainstream agent theories.

However, games have some intrinsic limitations in representing players' information and knowledge in terms of extensive forms and the partitional information structure. Multi-agent systems may be better suited to some representational tasks.

In this chapter, I relate some foundational aspects of the interplay between the two fields to Peirce's pragmatist philosophy. I will also investigate some applications that the amalgamation of games and agents give rise to, including bargaining and negotiation, and assess the use of semantic and interrogative games in areas such as electronic commerce and in the development of semantic/pragmatic interfaces in web technology anticipated in the previous chapter.

1. A semeiotic perspective

Game theory, as it was originally conceived, is aimed at investigating interactive processes governed by a coherent set of rules that would spell out the possible actions for a participant, given the actions of other participants. At the heart of this idea is the notion of a strategy, a non-constitutive rule intended to provide a mechanism that chooses from the set of possible actions the actual ones that the players should come up with in the course of the game, given the principles of rationality and related epistemic postulates.

The much more recent theories of multi-agent systems aim at providing theoretical and practical computational models of intercommunicating and co-operating agents. These recently mushrooming systems operate on similar principles as games, but without comparable emphasis on strategic concerns. The mutual interplay of these two systems defines my agenda here.

The main difference may be spelled out in Peircean terms. Games are open-ended entities pertaining to the category of thirdness, all that is capable of bringing firstness (how one conceives being) and secondness (how one conceives reaction with something) into a relation (Chapter 1). It is not just intercommunication, dialogue or cooperative action between agents, but also the mediation that takes place in systems. Thirdness is purposeful, self-conscious and meaningful action, with signs as the media between the communicating actors.

Strategies are instructions that evaluate actions, and hence are species of thirdness. They indicate what the actions of a player or an agent ought to be in an inventive manner. In their capacity of providing functions that evaluate individual choices, they also provide a route by which one might hope to be able to understand how intelligence emerges, namely through the constant evaluation of individual actions, and with the aid of the associated notions of learning and recognition of new concepts as implied by these actions.

From the standpoint of multi-agent systems, which are computational entities based on process-theoretic concepts of interaction and communication, such systems fall short of taking the crucial step from the category of secondness to the category of thirdness. This is because agent systems lack a repertoire from which to accommodate representations of strategic reasoning. This failure is inevitable, among other things because of the ill-fated emphasis laid on the concept of a 'protocol' (Sandholm, 1999). As it stands, a protocol is a defining, typically algorithmically designed rule that, certainly not unlike a strategy, states what the agents' plans of actions are. It governs the legitimate interactions between the participants. It does not exhaust the strategic dimensions of an action, however, nor does it dictate what a player ought to do at any conceivable position of the state-transition diagram or game tree, however unlikely the positions may be. What is needed is an evaluative higher-order rule that indicates how the protocols are achieving their goals. Without such an evaluative functional, in the end one falls sort of bridging the gap between rule-governed behaviour and truly inventive, strategically accomplished actions.

This point is reinforced by the recognition of the lack of actual game-theoretic concepts in logics for multi-agent systems. I do not mean that logics to tackle significant game-theoretic questions do not exist. Nor do I claim that there would be a shortage of sufficiently advanced ways of addressing a host of problems arising in multi-agent architectures and interaction. Nevertheless, typical logics devised for multi-agent purposes, including the family of BDI logics (belief-desire-intention) and many other modal and epistemic logics thereof, employ the concept of an agent that has little to do with the strategic outlook. As their titles are apt to imply, logics for 'intelligent agents' define some inactive constituent parts, not the active contributors such as goal-oriented players.

At least a partial remedy to this is to take the formalisms intended for multi-agent systems themselves to be amenable to semantic treatment in terms of the resources provided by game theory, for instance by game-theoretic semantics (GTS). Likewise, from the general systems perspective games are goal-seeking and goal-oriented processes that try to solve some decision problem, such as optimisation, satisfaction, control or verification (Mesarovic, 1975; de Alfaro, 2003).

Such processes are typically generated by open systems. The behaviour of an open system is determined by the interaction of its structure with the environment. Games between the System and the Environment in computation (especially in reactive and embedded systems), or between the Scientist (the Experimenter) and Nature in philosophy of science may be viewed as open systems (Chapters 10 and 13). In contrast, transition systems such as automata or process algebra are closed. Their behaviour is not affected by the context. The entire state-transition system or automaton is a single agent or a player, and its internal structure determines its behaviour.[1] Accordingly, state-transition systems do not represent a purposeful behaviour comparable with open and reactive systems, let alone games, and so the distinction between what is closed and what is open may be drawn analogously with drawing the distinction between Peirce's semeiotic categories of secondness and thirdness.

Admittedly, aspects of the convergence of game and multi-agent theories have been recognised relatively recently (Rosenschein & Zlotkin 1994; Kraus 2001). One of the original motivations came through the rediscovery of the para-Nash results concerning cooperation between agents without assuming communication. Advocates of this line of research have been keen to emphasise the viability of the ensuing methodology in terms of the success that it has had on two principal fronts: situations involving negotiation (bargaining), and coordination scenarios via the assumption of mutual rationality.

It is easy to appreciate that these are important areas of application, but to say that game and decision theoretic methodology is crucial to multi-agent systems is, according to current knowledge, an overstatement. On the whole, the hitherto substantiated aspects of this convergence have not shed much light on the idea that strategic reasoning as originally conceived provided the main motivation to pursue games in the first place, even though Kraus (2001) takes a step in the right direction in considering a range of negotiation situations in a strategic setting. The reason is precisely that the idea of interaction, as important as it may be, remains on the level of secondness, and is not automatically, not even typically, amenable to an analysis that brings it into proper relation with actions

— for instance, because of the lack of clearly definable preferences or because of the lack of salient situational elements.

From the perspective of pragmatist philosophy, Peirce would have thought of what we currently call strategies as instances of his far-reaching notion of a habit. In habits, one might hope to get a hold of precisely the kind of evaluative functional that accounts for the facts that make the behaviour of agents and living beings alike constantly improving and ultimately optimal in achieving their goals. Among many other things, a habit is also a term from Peirce's logical art that comes to the fore in his sign-theoretic writings, in which he discusses the class of interpretants that are in a definite sense final, that is, are the end products of the process of semiosis.

If I am right, a habit involved in such a process may be characterised as being the winning for the interlocutor. In other words, the habit in question embodies attributes that enable one of the players of the semeiotic process to invariably succeed in its aims.[2] Besides, as Peirce argued, habits are not signs, but are innate, organic characters of all beings. For instance, the essentials of the methodology in the approach taken in AI in terms of reactive and embedded systems found in robotics (as opposed to representational or deliberative systems) may be traced back to Peirce's evolutionary and strategic view of intelligence. No internal model of the environment is constructed by the modeller in reactive architectures, while in representational ones it is the symbolic manipulation and reasoning that is assumed to depict the behaviour of artificial agents.

Another major aspect of Peirce's logic was his diagrammatic theory of EGs. As I observed in Chapter 5, diagrammatic logic was influential in contemporary research on conceptual graphs in AI and web-technology, and instrumental to the later development of possible-worlds semantics. Here again, the idea of confrontation and collaboration between two parties, the Graphist and the Grapheus, is vital. The Graphist proposes modifications to the graph, while the Grapheus creates the universe (the universe of discourse) and monitors the utterances produced by the Graphist. It is quite characteristic of Peirce that he fell short of bringing the idea of a habit to bear on the model-theoretic aspects of EGs.

In what follows, my purpose is to put forward and assess a couple of tactics in which games enter the realm of multi-agent systems. It turns out, for example, that the role and methodology of multi-agent schemes in computing can be 'economised' by taking note of related aspects in game theory pertaining to the way in which the concept of information is understood and handled. Summarising, the following points ensue.

(i) Like multi-agent systems, games assume rationality, but they embed that notion into strategic considerations by and large lacking in the methodology of agent-based systems.

(ii) Taking games as general systems (Klir, 2001) reveals that there are some fundamental impediments that the theory of extensive-form games currently has to face. It cannot capture everything that agents are supposed to be informed about in any given position in which they are to make decisions. I will discuss some of these limitations and suggest that it is possible to revise the notion of information sets, and to generalise the ordinary partitional representation of information within the framework of extensive-form games to the dynamic and non-partitional representation of information. This paves the way for a more general theory of information flow in multi-agent systems.

(iii) There are applications of games and systems in the field of web technology. In particular, the concept of bargaining games turns out to be useful in understanding electronic commerce, but it usually requires incomplete information. I will also argue that games known as semantic games in logic and formal linguistics contribute to the development of the semantic web or one of its pragmatisations.

My disclaimer is that I have no intention of providing any in-depth survey of the large amount of recent work done at the interface at which games and multi-agent systems congregate.

2. On the foundations of agent methodology

There is no agreed definition of a multi-agent system. For that reason, I will in this section discuss informally some of the properties that are typically attributed to such systems.

What is an agent? An agent in computational systems is typically just a piece of software. On a slightly more general level, any entity that acts upon the environment (or system) it inhabits could also be considered to incorporate the concept of the agent. Nevertheless, it is routinely assumed to have certain further properties. It is a widespread assumption that these properties should include at least rationality, autonomy, proactiveness and reactivity, and often even something resembling social ability (Wooldridge, 2000). Most of these qualities pertain to all life forms from congeries to costermongers, but they may also refer to inanimate matter, reflecting a kind of 'computational hylozoism' that stretches the concept of the semeiotic mind so as to apply life-like qualities to man-made contraptions.

Other recurring buzzwords in the literature include 'selfishness' and 'virtual', which may be largely ignored for my purposes. The former is an instance of one's social capacities, but is also reflected in game theory as the foundationally essential principle of expected utility maximisation. The term 'virtual' refers to the concept of an agent that is strictly confined to computational theories.[3]

Far from being a defect, such generosity of definition may be, and sometimes has been, turned into a virtue in research about rational agents. Things that have blurry or fuzzy edges may indeed be useful in so far as we have at least a rough idea of what we are talking about. This is because the concept in question would be less theory-laden, to borrow somewhat old-fashioned empiricist terminology. Observations about agents are observations concerning the intentions of the designer rather than any real-world phenomenon.

Even so, typical books about agents contain little or no theoretical presuppositions or methodological assumptions concerning these basic properties (Wooldridge, 2000). Given our conception of an agent as something or someone that is rational, autonomous, proactive, reactive and operating in socially constrained situations, we need a theory that turns these vague notions into concepts that can be captured in some formal or rigorous way, and then explore the consequences of these formalisations. For this reason, I would like to suggest that game theory provides a methodology on which multi-agent properties can be grounded. This is not to say that such a connection has not been discovered before; numerous studies have been devoted to the issue. However, the contribution and interest of the kind of connection suggested lies in the fact that the effects of the interplay are not merely in describing what agent properties are and then making assumptions on that basis. We also need to know how to make predictions about agent behaviour, and perhaps with a bit of luck then to prescribe their emergent properties.

Let us consider each of these properties of intelligent agenthood, and some of their underlying assumptions, in turn.

Rationality Wooldridge (2000, p. 1) describes agent rationality as an assumption that agents prefer to perform actions that are in their own best interests, given the background beliefs they have about the world. This is the basic route to rational decisions in economic theory, too: we accept only actions that are based on certain objectives, such as the satisfaction of a certain representation of preferences or, if subjective probabilities are taken to exist, on the suitable calculation of maximal expected utilities. As it happens, any reasonable attempt to guarantee self-interest may actually thwart it.

Since rational choice is necessarily an idealisation, economists have turned to a broad-ranging investigation of accounts of it that are less-than-perfect, by having in mind theories that could take into consideration the cognitive limitations of agents. In reality, such limitations on cognitive repertoires may take many forms, such as incomplete knowledge of and incomplete information on certain parts and aspects of the structure of the game in question, and likewise imperfect information on and perhaps imperfect recall of other players' knowledge and information or their preferences. Moreover, issues such as computational and algorithmic complexity, various manifestations of uncertainty, the refusal or incapacity to perform deductive reasoning, the refusal to or having difficulty in performing statistical analyses of equilibria, and a dynamically evolving set of alternative actions, are all instances of bounded forms of rationality.

Many of these manifestations define a natural agenda for multi-agent systems, too. Replacing players with Turing machines, finite automata or neural-network algorithms, for example, produces agents whose strategic resources are limited by constraints on the computational power of these devices. Other than computability and recursivity, we may also require some version of learnability of strategies and restrictions on how players are able to acquire concepts that are at least potentially learnable.

Given that it is such a heterogeneous concept, not everything falling under the broad notion of bounded rationality is easily recognised. Such cognitive limitations also need to be investigated in order to mitigate assumptions of common knowledge of rationality concerning other players in the game. As it happens, common knowledge of rationality is a multi-modal concept that concerns all participants, not unlike solution concepts such as sequential equilibria, which concern players' beliefs and expectations in terms of how the other players will behave on future occasions.

A further point that will arise not very far in the future is that, as quantum computers spring into existence, the notion of rationality will have to abide by the new boundaries drawn according to the transgressed limits of theories of computability.[4]

Autonomy The property of autonomy states that agents make independent decisions. This means that a decision made by one agent, a, does not involve anything about the decision made by another agent, b.[5] In terms of the autonomy, it is also sensible to assume that the information and the epistemic structure of an agent is private, and can be shared with others only by its own pronouncement.

The assumption of independence means that if we were to regard multi-agent systems as games, the decisions are made on the basis of non-cooperative principles. In a basic setting this means that each individual i has a utility function $u_i\colon X \to R$ from a set of outcomes X to the set R of reals. The outcome is a function $f\colon A_1 \times \ldots \times A_n \to X$ of the actions A_i agent can take independently of the others. No notion of intercommunication between agents need to arise, as the maximatisation of this utility function guarantees that optimal outcomes are reached.

The latter part of the assumption of autonomy may be analysed by resorting to intensional concepts and their iteration. Here, epistemic logic may be resorted to in representing the agent's knowledge, information, belief and other propositional attitudes. According to this paradigm, inaugurated in Hintikka (1962) and advanced in computational circles in Fagin et al. (1995), an agent i knows a proposition p in a given world or state w, if and only if p is true in all of i's epistemic alternatives to w. As such, there is no game theory in the modal logic of agents. However, the proposition may be a statement of what actions other players have carried out earlier in the game, for example. Alternatively, one could interpret modal notions by means of semantic games deriving evaluational superstructures for modal models (Pietarinen, 2003d, 2004c). A proliferation of different notions of knowledge took place since the 1980s in terms of distributed, common, shared and focussed knowledge abound in computer science and AI (Pietarinen, 2002b).

Proactiveness Proactiveness means that agent behaviour is goal-directed, or that agents have a purpose through which their activities and dealings could be understood and assessed. In order to clarify this multifaceted concept, we may again fall back on intensionality. This is because goal-directed behaviour presupposes that agents have intentions toward goals, while intentions in turn presuppose the possibility to choose from several states or alternative courses of action.

Here, the alternative courses of actions, or possible worlds, are perhaps best understood in the sense of being 'small worlds', figments of total universes (Savage, 1972). This means that a sequence of events that contains sets of states is temporally and spatially limited and cannot encompass all the information about the actual world. As far as an actual decision maker is concerned, this means that his or her attention needs to be restricted to "relatively simple situations" in any circumstance (Savage, 1972, p. 82), or to be able to isolate them from the larger context.

Aside from these modal aspects, it has been customary in game theory to assume that agents have a purpose, and are capable of acting upon it. This is by no means an unproblematic assumption, and as a form of scientific explanation has long been in the bad books of many scientific fields, perhaps most notably in biology. In sharp contrast, Peirce's pragmatic philosophy is full of meaningful intentions and purposes in action.

Reactiveness The concept of reactiveness has been taken to mean the ability to respond to changes in an agent's surroundings. In game-theoretic terms, reactivity is a means of acting strategically upon environmental transformations. There are different opinions on precisely how dynamic such a competence is. In strategic (normal) forms of a game, each player chooses his or her strategy once and for all, the relevant information being coded into the strategy functions. In extensive-form games, on the other hand, strategies are developed across the board, as calculated responses to adversaries' actions.

The traditional theory of games, in the sense first formulated in von Neumann & Morgenstern (1944), is a static theory of strategic interaction. The games defined therein remain static even if they were conceived in their extensive forms depicting the total structure of actions and responses to them. Although its importance was recognised by von Neumann and Morgenstern, the truly dynamic theory of games is still under intense development (Basar, 1994).

One of the major problems in the literature on agents appears to be finding the right balance between proactive and reactive behaviour. There are natural bounds to deliberation and computation given by rationality considerations, and intentions alone do not always carry enough weight to bring proper plans about. In this sense, one faces precisely the same problem that game theory has wrestled with, too.

Social ability It may seem somewhat anthropomorphic to refer to computational, virtual agents in terms of their social ability, even more so when the literature attributes notions and tasks such as auctions, negotiation, bargaining and cooperation to them. In real situations, such as the game-theoretic modelling of social constructions, they nonetheless represent commonplace, albeit very complex phenomena.

Social action is orientation towards others. While this much may be true, what multi-agent systems need to focus on is the question of what kind of orientation is best suited for their purposes. Given that social processes emerge as soon as there are autonomous agents, perhaps acting in their own interests and taking into account the actions of others, we have not yet gone beyond a prototypical game-theoretic situation, not even beyond that of a non-cooperative game. We still need to know how orientation and accounts of other agents are formed. A comprehensive understanding of the social habitat of agents is still being sought (Johnson, 1998; Tuomela, 2000).

One may also bring sociological analysis to bear on agent theory via the notion of a game. Addressing the question of how to establish that the semi-formal idea of a game in sociology (Giddens, 1971) applies to multi-agent theories would improve on the extant sociological analyses, which so far have not been of much interest among computational agent theorists. Social concepts enrich current theories. The problem is that social codes are not typically expressed explicitly as command lines in a software package or in definitions of computational theories, but are implicit or tacit, yet mutually shared, reciprocal features constituting actions and reactions to them.

What else do game-theoretic analyses have in the offing here? In the 1950s, one of the pioneers, Jessie Bernard, urged sociologists to translate their problems into the language of game theory, especially notions such as coalitions, institutions, coordination, conflict resolution, and similar cultural influences. Worries were voiced on the feasibility of such an approach, which was often thought to clash with qualitative methodology. More recently, Wooldridge (2000, p. 11) wrote, "Like decision theory, game theory is *quantitative* in nature". This is at best a half-truth, however. Just to mention one thing, qualitative decision theory has become increasingly influential (Lehmann, 1997), and there are many areas in which it overlaps with game theory and its aims. Here again, many of the common characteristics between the two fields remain to be charted.

3. Games, agents and information

Strategisation Extensive games described in Chapter 7 provide a rich framework in which to represent agent information, strategic actions and so forth. There are more recent versions, such as evolutionary game theory, that extend strategic concepts to the realm of non-rational objects (yet intelligence in some form is still required). And if so, strategic concepts are something that make sense in the inanimate realm of computing, too. Peirce endorsed similar views on signs requiring quasi-minds that are at the same time "distinct" and "welded" (4.551). For one thing, a system and its environment are such quasi-minds welded into one sign, which is an open system.

There are numerous areas in multi-agent research in which strategic conceptualisations turn out to be important. For instance, heterogeneous agent societies (Subrahmanian et al., 2000), despite concerning groups of agents, use strategies that are individualistic rather than collective. Such societies would thus fall quite naturally within team theories (see below) rather than coalition games, which use collective strategies instead. Preference-based non-monotonic reasoning is also a field in which strategic issues should be given increasing attention (Lin & Shoham, 1988). It is vital that such concerns are extended to the whole field of non-monotonic reasoning and non-monotonic logic, which have to date been based on rather ad hoc definitions of non-monotonic entailment relations.

Viewing games as multi-agent systems, the constraints given in the previous subsection may appear too restrictive. If such systems lack strategic reasoning powers, it is natural to assume that this weakness shows up in the relaxation of these constraints. For instance, the assumption of observed choices (aka perfect foresight, see Chapter 7) no longer holds. Consequently, strategies may not be defined only on total information sets, but perhaps only on some subset of histories contained in them. Logics developed for various strategic multi-agent tasks, such as communication, coordination, negotiation and verification of computer program correctness may be enriched accordingly.

Games as open systems In order to be truly operational, agent systems need to be open. Open systems receive constant input from their environment. They cannot be characterised merely functionally, and so they are closer to concurrent processes in computation than to any automata-theoretic description.

Since games may be viewed as models of interaction between agents and their environment, the scope of the theories of multi-agent systems is broadened as they are not determined solely by their internal properties.

Alternatively, games may also be open. In that sense, games are not just interactive processes providing a mechanism of choosing from the set of possible actions the actual ones calculated to be the best replies to the actions of the adversaries. For strategic decisions would also have to be made concerning the input, delineating appropriate contexts or states of affairs within which best plays are to proceed.

Such a relation of games to their environment may be characterised by chance moves by Nature that decides the contexts within which further plays are conducted. These games are thus ones of incomplete information (Chapter 7; Harsanyi 1967).

Summarising, there are two perspectives to openness: as two-player game models of a system reacting with its environment, or as games in which decisions concerning input are being made alongside with strategic planning concerning the output and future actions, given the outcome of the played defined by the payoff function.

Let us next turn to one philosophical cornerstone in the interface between systems and games: Wittgenstein's systemic concept of a language game.

Wittgenstein's language games One of the major problems with game theory is that, in some circumstances, it either imposes too much structure, or then the structure is too rigid. Originally, the structure of extensive games was meant to depict the actions, their sequential order, and the outcomes of decisions. Since then, it has gradually been realised that the information (knowledge) the players have concerning the game structure itself, the rules of the game, as well as other players' knowledge, beliefs and actions, also need to be rigorously modelled and understood. It is unclear whether it is the actions alone that give rise to genuine and coherent notions of a game and of players' strategic interaction, or whether one should rather start with a suitable, orderly 'proto-game' with some slots or 'pre-positions' waiting to be filled with actions. As surprising as it may seem though, the latter concept of games is actually what game-theoretic literature usually assumes. This is particularly evident in the assumption of perfect foresight, namely that players are assumed to be able always and everywhere to observe their immediate available options.

The former approach is a more liberal, one is almost tempted to say Wittgenstein's approach, explained with reference to something that actually lies outside of the structural notion of a game. Wittgenstein's game was to get words of a text, or a complete primitive language, to derive their meaning from the role they have in certain non-linguistic activities. For later Wittgenstein, this meant that games were not really found in logic or in language, but conceptually prior to them; they are activities from which logic and language acquire their meaning: " For what we call the meaning of the word lies in the game we play with it" (Wittgenstein, 2000–, 149: 18).

What this idea of a 'proto-game' actually means is illustrated by the familiar scenario in which you and I agree to meet in some big capital city (Schelling, 1960). If this is all we have agreed, we are both most definitely thinking about meeting at as unique a place in that city as possible. The upshot is that our language game already constitutes a variety of hidden assumptions, a desire to establish a common ground, and numerous other parameters intertwined with a vast array of habits of language use, such as salience, prominence, earlier choices, vividness of memory, and so on. This may also be compared with the by now familiar like-minded comment from Peirce: "No man can communicate the smallest item of information to his brother-man unless they have common familiar knowledge; where the word 'familiar' refers less to how well the object is known than to the manner of knowing. [. . .] Of course, two endless series of knowings are involved" (MS 614: 1–2; see Chapter 13). In this comment we also find an anticipation of the notion of common knowledge as it was crafted in Lewis (1969) to make systems of communication function and to give rise to simple systems of language through attempts of coordination.[6]

The indispensability of mutually-acquired common ground is something to be kept in mind whenever we argue that applications of multi-agent systems tend to become more and more realistic.

Limitations of extensive games Quite apart from the preconceptions we have about games, the founding fathers of game theory designed an extensive-form representation that was not as rigorous and realistic model of player knowledge and information as one might wish for. According to Shubik (1992, p. 160), however, "The development of the notation for the extensive form had a considerable impact on decision theory and psychology. The ability to represent and analyze different information structures was a breakthrough of the first magnitude". True, the invention was remarkable. Nevertheless, there are some fundamental impediments in the traditional way of representing players' information by extensive games. This is evident in extensive games that partition histories into information sets thus assimilating the essentially different notions of players making moves within an information set and players possessing an information set (Pietarinen, 2003b). Actions within an information set means that the range of uncertainty is modelled with reference to the sets of histories in which the players are supposed to plan and execute their decisions. Having an information set, on the other hand, is less restrictive, since it only means that we may choose any set of histories and take that selection to refer to players' information sets. Once these two concepts are set apart, extensive games can capture notions that cannot possibly be represented without the distinction.

In order to properly distinguish these notions from each other and to analyse them, we need to have the option of drawing information sets in a more flexible manner, and not only around the histories within which the respective player moves. We need to include these histories in information sets that correspond to moves referring to the past events hidden from a player. A player acting under imperfect information has to be annotated with this relocated information set.

Consequently, players' range of uncertainty becomes modelled in a more realistic way, since it will be located where their information sets really should occur. Naturally, the players do not have to be the same ones who are making decisions on the histories included within the information sets that have been repositioned.

This relatively simple quirk provides a more general solution to the problem of representing players' information than is accomplished by Bonanno & Battigalli (1997), for example. They attempt to free the conventional static informational structure of extensive games from some of their limitations by referring to players' information about their payoffs, which nonetheless is not plausible from a strategic point of view, especially if such an assumption is made whenever any of the multiplicity of plays is not yet finished. A more general solution is obtained as soon as information sets are taken to be truly dynamic, so that they can disappear and come into existence while the game is in progress.

Imperfect recall for internal agenthood It is not necessary to think of agents as independent players. A type of multi-agent interaction exists that takes place within a single player, or within some other entity or agent, that we require to have certain specific properties. This is the case whenever we allow information sets to be marked either (i) around the histories of any one play of the game, or (ii) around the histories of different plays, but in locations that hide some of the player's own previous moves. In both cases, the information sets give rise to the classes of games that do not have perfect recall. All these give rise to open multi-agent architectures with specific kinds of uncertainties.

Games with the latter type of information sets give rise to games of imperfect recall. What happens in (i) is that there are plays that visit an information set more than once. Games with such information sets have the additional property of absentmindedness (Piccione & Rubinstein, 1997), for when faced with one, whether any decision was made is forgotten. These games give rise to time inconsistency, because the agent's strategy has to produce the same action in all the histories within an information set that are taken to represent the agent's absentmindedness. If the player is better off by changing strategies, say, from mixed to behavioural ones within such an information set, he or she would be able to resume perfect recall just by observing the identity of the strategy that is being used, the blocking of which is not plausible in self-conscious systems.[7]

An additional but in reality quite important further point here is that, by allowing the more dynamic concept of information sets in the sense argued in the previous section, the property of absentmindedness may also mean that

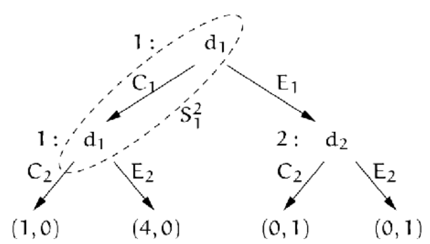


Figure 14.1. Player 2 absentminded about absentmindedness.

players forget whether or not some other player is absentminded and thus cannot recall his or her location in a game (Figure 3). This vindicates the overall view of games as systems that can be extended to cover further modelling tasks, as such games of iterated, higher-order absentmindedness ('absentminded about absentmindedness') do not seem to have been considered before.

Team theory: the forerunner of agent systems A related tradition exists in the game-theoretic literature known as team theory (Kim & Roush 1987, Marshack & Radner 1972). This flourished long before multi-agent systems and web applications emerged. Even so, this theory is directly applicable to a variety of multi-agent problems. Team theory studied some of the central aspects of multi-agent interaction in depth way ahead of agent theories.

According to team theory, teams are groups of individuals with a common goal but individual information, knowledge and actions. Essentially, the theory, which underscores the central place games and strategic interactions have in it, states that all solutions of two-person zero-sum games still hold for games played by teams (Ho & Sun, 1974). Admittedly, team theory is a heterogeneous enterprise that aims to bring together various theoretical constructions, such as decision and systems theory, operations research, dynamic games, search and coordination, plus parallel processing. But this also applies to multi-agent systems, and indeed they overlap a great deal with team theory.

Within strictly noncooperative game theory, there is never a need for teams instead of players, because all conflict situations can be modelled by all parties involved forming a single player and the notion of payoff that each of its members would have. In order to obtain nonidentical payoffs within teams, any excess cooperation and coordination have to be explicitly represented as identifiable elements in the (extensive) game framework. Accordingly, cooperative games may be viewed as coordinate systems with teams.

Moreover, an understanding of coalitions and coalition formation is topical in multi-agent systems that typically desist from being strictly noncooperative.

Social-choice theory Social-choice theory investigates methods that aim at obtaining certain group rankings of alternatives from individual rankings (Johnson, 1998). This is an iffy task, since an agent may misrepresent its own preferences in order to gain advantages, or then there are certain 'dictatorial' methods allowing all possible individual rankings to be generated. In particular, one of the main results states that a social-choice function cannot be both Pareto optimal (no outcome is better for all players) and independent of irrelevant alternatives (if A is a subset of B and $x \in A$ is optimal in B, then x is optimal in A).[8]

This is typical in bargaining games, for example, a class of games representative of various cooperative situations between agents.

Bargaining games: an e-commerce application Electronic commerce (e-commerce) has given rise to a mixture of approaches to fault tolerance, since the various communication channels have much more uncertainty and unreliability between the parties than the conventional market does. Models of games applied to bargaining and negotiating situations need imperfect information. Moreover, the likelihood that negotiations break down is much greater than in the conventional market (companies cease to exist, or just do not respond or react properly). Clearly, the participants suffer from such communication failures, but it is not uncommon that they also pretend to be affected by them and hence increase the probabilities of certain preferred actions. Since in negotiation games players' preferences are typically taken to satisfy the axiom that no agreement is worse than a disagreement, sudden and unexpected changes in the bargaining process based on the simple accept-reject schema are intolerable.

Two comments are in order concerning typical bargaining models based on alternating-offers framework, in which a negotiating game is modelled by an alternating sequence of actions consisting of players' acceptances and rejections (Rubinstein, 1998). First, such a framework is fairly general and there remains considerable latitude as to what its best interpretation is. It may be criticised for lacking a realistic interpretation faithful to real bargaining situations. For example, in real-life bargaining and negotiation, offers and refutations are typically accompanied by some persuasion or argumentation crucial to decisions. This is why the model is actually more suitable for e-commerce, in which the number of arguments siding with the negotiation process is, in fact, greatly reduced because of the increased means of controlling the process.

Second, the Nash solution and other solution concepts are known only if games have perfect information. It is an open question as to how such a solution concept that attempts to account for the players' beliefs under uncertainty (sequential equilibria) could be incorporated into bargaining games.

The reason for the latter impediment is that certain incomplete-information games in which chance moves are hidden from players (Chapter 7) do not make much sense in bargaining situations, since the players ought to know if the negotiations break down. If a chance move delivers players' types, that would give rise to choices that can be hidden. Indeed, a vast literature on such kinds of incomplete-information bargaining games exists. For example, the Vendor may bargain with a team of Customers so that the Customers are told only their own shares. Baliga & Serrano (1995) define a bargaining game for a situation in which each responder (a coalition) knows only the share offered to him or her by the Proposer, not those offered to any other responder. This is a plausible and realistic scenario, albeit not the most general one. A generalised bargaining game with imperfect information would need to be fault-tolerant in the sense that players are not always perfectly informed of whether their response to an offer becomes known to the Proposer, or whether the Proposer's offer becomes known to the respondent. This kind of information flow typical in electronic communication is related the old Byzantine problem of coordinated attack, which states that neither party will attack a common enemy unless it is sure that the other party will attack the enemy too. If there is a delay in the coordination between the parties, it never becomes common knowledge that the other one is attacking. Likewise with the alternating offerings.

The upshot is that we need models of extensive bargaining games that allow information sets around nodes that are able to hide offers from respondents, as well as around nodes that accordingly are able to hide responses from the Proposers. If the period of negotiation is protracted enough, it becomes reasonable even to allow information sets that are marked around nodes along the same histories, in other words players occasionally forget their own offers and responses as well. Since electronic negotiations, bargaining, and commerce have to be tailored to be operational in environments in which there is an increased risk of all kinds of faults and uncertainty, game-theoretic models have to be prepared to reflect that change.

Logic, games and the emergence of a semantic web Knowledge representation is undergoing rapid development in the race to express structured information suitable for efficient agent manipulation. The underlying logics should be powerful enough to describe complex objects, but not too powerful to be intractable or to give rise to outright inconsistencies.

These aims correspond with the desiderata of the project of developing what is known as a vision of a semantic web, a kind of web that is able to use structured information so that (i) suitable knowledge-representation languages would be available and directly applicable to the tasks at hand, and (ii) the system's question-answering on the web-based data would be facilitated in a warranted manner (cf. the previous chapter).

My point is that GTS could be made to realise these desiderata. Semantic games capture interactive situations between the utterer and the interpreter, or the User and the Client (the Web, the Oracle), providing a powerful alternative to classical semantics, and replacing the denotational secondness of the latter with strategic thirdness.[9]

The second prospect is that we may make use of another type of games, namely interrogative games (Chapter 10). These fill a natural lacuna for question-answering systems. Interrogative games are played between the participants of the Inquirer and Nature (or the User and the Agent, or the Agent and a semantic web). If the Inquirer is the Agent, its task is also to draw autonomous logical inferences from the answers that a semantic web is able to provide. In addition, the User may request inferences or 'proofs' of the semantic web's answers via these Agents. Thus these interrogative games represent many kinds of dialogues, as well as scientific inquiry (Hintikka, 1995).

Furthermore, in developing reliable questioning-answering agents to work on semantic web building, we need to take many linguistic issues into consideration. These include presuppositions (a question must not be asked before its presupposition is established), conditions for conclusive answers, conversational implicature (inferred non-literal meaning of utterances or data), the common ground between the interlocutors, the topicalisation of a dialogue, plus numerous other communicative goals. These items gradually build up the future agenda, the goal of which is to seriously address the field of pragmatics and the semantics/pragmatics interface using computational and agent-based methods, in addition to the ready-made structured information that current languages such as XML, RDF and OIL are able to provide. Beyond the vision of a semantic web, a truly pragmatic web is thus our next challenge in the realm of web technology.

As noted in the preceding chapter, a pragmatic web should distinguish between the two major roles that pragmatics as a linguistic theory has played over the years. On the one hand, it needs to know when the data on the web should take the context of use of expressions into account. To do this, knowledge-representation languages need to be made to somehow understand and resolve how to deal with indexical expressions and demonstratives. On the other hand, the second major role is also to take the Speaker's (the Utterer's, the User's) intentions; that is, his, her or its meaning plus implicatures and presumptive meaning (as distinguished from the literal meaning) into account.

4. Conclusions

What contributed to the suppression of game-theoretic arguments in agent research may well have been a consequence of their presumed applicability to restricted situations that meet stringent assumptions of rational acts, unlimited computational resources, and rigid rule-following. Such cautiousness is warranted in some cases, but in general, these parameters may be relaxed. It is not even clear what the notion of rationality in games in the end amounts to, even though it was laid out as the bedrock of the science of economics and its coinage of *homo œconomicus*, presumably to differentiate it from other notions of *homo* prevailing in the social sciences. Rather than attempting to salvage such ideals, the modern creature of *homo ludens* strives to save agents from all-encompassing irrationality, and to dress them in the outfit of a localised or theory-internal notion of rationality.

It is far from clear that game theory has fulfilled the promises it set out to accomplish. Even so, the current research on agent systems now has the opportunity to make do without such strategies that invariably have to maximise expected utility, or without the concept of players that have to be ideally rational. Computational theories provide more rigorous ways to model agent information than what is accomplished in the traditional layout of game theory, summoning *homo faber* to join *ludens* in the battle against *œconomicus*.

Game theory, in turn, is better suited to the task of making some sense out of the agent's rationality and its bounded manifestations. It is an implement of one aspect of Peircean thirdness, incorporating meaning into state transitions, on its way to growing into a theory that allows the spontaneous evaluation of its own strategies, reflected in the semeiotic interpretation that stabilise into equilibria after multiple runs of games. Moreover, notions of limited access to information, imperfect recall, team theories, logically non-omniscient modal and epistemic logics, social constructs, and realistic negotiation scenarios are just isolated examples of the kind of steps that need to be taken in order to attain an overall theory of intelligent agency.

The relevance to the wider agenda of epistemological issues in science is that the interactive and strategic perspective into which pragmatic philosophy may be recast is in close connection with the venerable tradition of science as 'putting questions to Nature' (or perhaps 'putting questions to a semantic web'). One illustrative aspect of the connection not explored here in depth is the strategic dimension needed in applying the concept of the abductive method. As computational theories of multi-agent systems progress, this interactive setting will become more and more automated. For instance, the production of a sufficient range of scientific hypotheses based on a large amount of data contested by induction hinge increasingly on the amount of semeiotically interpreted concept of 'the mind' derivable from intelligent databases.

These hypotheses, and *eo ipso* the multi-agent systems that produce them, would then need to satisfy the qualitatively given criteria for hypothesis selection, such as the one given by Peirce in his economy of research programme (Chapter 3). After all, I have provided merely some preliminary indications of the places in which some building blocks for the design of that sort of intelligence could be found. Peirce's pragmatist philosophy and semeiotics is a fertile ground upon which to develop computational systems of intelligent, active, social, strategically interacting and cooperating agents.

Notes

1 This was Michael O. Rabin's original insight to automata in the 1950s. Games were explicitly connected with automata much later via monadic second-order logic.

2 This may be compared with the idea of general equilibrium systems, in which general systems strive for stationary, stable states (Hansen, 1970). A stable state is a property of a dynamic model of processes in which nothing changes over time. In relation to final interpretants, the meaning of a concept is agreed upon by the Utterer and the Interpreter through semiosis that strives at stable states as regards the conventionalised meaning of concepts.

3 Peirce defined virtual as follows: "a virtual X (where X is a common noun) is something, not an X, which has the efficiency (*virtus*) of an X" (6.372, 1901). A virtual agent would thus be an 'agentless agent', not an actual one but a displacement of it by something that preserves the actual efficiency of the original agent.

4 Game theory may clarify foundations of quantum theory and quantum computing. No wonder that John von Neumann, the chief engineer of both game theory and quantum logic, had a parallel interest in game- and quantum-theoretic interaction (Pietarinen, 2002a).

5 Peirce defined dependence as a variational principle: "Anything, X, is said to be *Dependent* upon anything else, Y, in so far as an alteration in Y involves an alteration in X" (MS 644, 1909, *On Definition of the Analysis of Meaning*).

6 Note also: "When one person informs another of any fact, and the ~~second~~ [hearer] rightly understands what the utterer of the information means, that information refers to something which they do not know and could not know by any mere description of what it was like, but which they well know and know each other to know by ~~common~~ [an] experience both have had, something which has happened to them which has forced them both to acknowledge that the object referred to has ~~noted~~ resisted their wills" (MS 49: 16–17 a.p., *Numeration*).

7 The received way to deal with games of imperfect recall and those of absentmindedness is to split any one player into a set of agents, or multiple-selves, each of which is taken to make private decisions and to obtain private information (Chapter 7).

8 This refers to the classical work by Arrow (1951). However, if we were to allow membership functions to be local in the sense of topos theory (McLarty, 1995) rather than being based on classical set theory, we would get a different independence axiom where Arrow's analysis would no longer be valid. To my knowledge, such local membership functions have not been considered in the area of social-choice theory.

9 As I noted in Part I, Peirce termed such game-theoretic evaluation 'endoporeutic', since it proceeds outside-in, from context to the atomic expressions. This runs counter to the nowadays much more customary inside-out Tarski semantics. I believe endoporeutics is needed in 'explicit semantics' for agent communication in open systems. Peirce would have called the shared meanings concerning the messages that mediate between agents 'communicational interpretants'.

Chapter 15

FINAL WORDS

Allow me to conclude by selecting a couple of excerpts from one of the leading game theorists, Robert Aumann, and point out some lines of thought that both parallel and contrast with the theses that emerge from the present treatise. These remarks also put Peirce's philosophy in a novel contemporary perspective.

> Game theory is a sort of umbrella or "unified field" theory for the rational side of social science, where "social" is interpreted broadly, to include human as well as non-human players (computers, animals, plants). Unlike other approaches to disciplines like economics and political science, game theory does not use different, ad-hoc constructs to deal with various specific issues, such as perfect competition, monopoly, oligopoly, international trade, taxation, voting, deterrence, and so on. Rather, it develops methodologies that apply in principle to all interactive situations, then sees where these methodologies lead in each specific application. (Aumann, 1987a, p. 460).

In a similar vein, Peirce's semeiotics provides a methodology applicable across a variety of scientific inquiry. Humans, bees and crystals alike give birth to signs. Groups of players, or agents in a community of inquirers, are paramount to the success of the scientific method.

Scientific methodology must endure irrespective of the individual situations in which signs are put forward and interpreted. Its value lies precisely in the pragmatic outcomes of applying the methodology. In the theory of signs, the methods are inherently and thoroughly interactive and intercommunicative between the sign carriers.

> Indeed, the influence of an individual participant on the economy cannot be mathematically negligible, as long as there are only finitely many participants. Thus a mathematical model appropriate to the intuitive notion of perfect competition must contain infinitely many participants. We submit that the most natural model for this purpose contains a continuum of participants, similar to the continuum of points on a line or the continuum of particles in a fluid. (Aumann, 1964, p. 39).

Aumann studies continuums of participants by alluding to infinitesimal notions. One might well recall that Peirce's scientific method requires a potentially endless multitude of the community of researchers. Moreover, his synechistic semeiosis not only prefigured Aumann's view, but also looked further ahead, for instance in terms of his reluctance to associate the true continuum with the real line consisting of points. One might also recall his prurient remark on the liquid in a number of interconnected bottles as an allegory to the common mind.

Peirce's approach to scientific inquiry maintains that no single individual actor can attain much alone. Only a myriad of ever-increasing and repeating cycles of inquiry coordinated between a continuum of investigators will converge to the truth, or in the very least, to the positive increase of the *summum bonum*, the common idea-potential of humankind.

What's more, Aumann's recourse to perfect competition need not side with the kind of Gospel of Greed that Peirce passionately opposed. Perfect competition is as improbable as perfect communication and complete information. An ideal limit indefinitely deferred to the future, it does not and need not represent any actualised state of affairs in any domain of investigation.

> [One wishes] to point out a fundamental difference between the game-theoretic and other approaches to social sciences. The more conventional approaches take institutions as given, and ask where they lead. The game theoretic approach asks how the institutions came about, what led to them? Thus general equilibrium theory takes the idea of market prices for granted; it concerns itself with their existence and properties, calculating them, and so on. Game Theory asks, why are there market prices? How did they come about? (Aumann, 1987a, p. 467).

In view of the theory of semantic games one may ask an analogous question: Why is there truth or falsity to assertions in the first place? This happens irrespective of whether these assertions are purely logical, heterogeneous, or pertain to our natural language of self-communion. Unlike the dominant Tarski semantics of the 20$^{\text{th}}$-century logical tradition, let alone neoclassical general equilibria analyses, Peirce's diagrammatic and dialogic theory, with architecture similar to open systems, construes interpretations through the mutual, habitual and coordinated interaction between players sharing a common ground and common familiar knowledge. Accordingly, his approach by no means takes these interpretations — and consequently the payoffs assigned to atomic expressions — for granted.

> The theory of repeated games of complete information is concerned with the evolution of fundamental patterns of interaction between people (or for that matter, animals; the problems it attacks are similar to those of social biology). Its aim is to account for phenomena such as cooperation, altruism, revenge, threats (self-destructive or otherwise), etc. — phenomena which may at first seem irrational — in terms of the usual "selfish" utility-maximizing paradigm of game theory and neoclassical economics. (Aumann, 1981, p. 11).

Perhaps we could add that the meaning of assertions changes because (i) the language games are being played in different contexts, environments and situations, (ii) potentially infinite populations of language users may employ them, and (iii) they are constantly tested against the threats of mutant meanings. Peirce's philosophy is likely to reach even further than the game theories of neoclassical descent. He emphasised the habitual nature of human action, which is not any reflected, self-centred reasoning about the optimal outcome (even though individual actions may involve self-purposive and cogitated reasoning), but rather supports finitely rational and purposeful common behaviour. His philosophy is not undermined by 'value depends upon utility' or 'truth-values depend upon atomic payoffs' types of arguments. Sets of strategies are not pre-established; there are open-ended collections of 'micro-institutions' from which semantics, pragmatics and methodeutic proper gradually emerge. Much more is to be expected from applying and eventually injecting Peirce's ideas into the modern theories of games and rational behaviour than is currently realised.

> [Games put forth] a subtle interplay of concealing and revealing information: concealing, to prevent the other players from using the information to your disadvantage; revealing, to use the information yourself, and to permit the other players to use it to your advantage. (Aumann, 1985, pp. 46–47).

In Wittgenstein's later philosophy, the micro-institutions inherent in human endeavours in mediating the meaning between assertions and the world were collectively dubbed as language games. The varying streams of information are the essentia of such institutions. Through time, they tend to stabilise, sustain and counterbalance the semantic relations that these institutions are destined to construct. To grasp the relation between acts and samples is not to vaunt different speech acts, but to go about the use of the expression in relation to that which one has found to be the case in its context.

> The common knowledge assumption underlies all of game theory and much of economic theory. Whatever be the model under discussion, . . .the model itself must be common knowledge; otherwise the model is insufficiently specified, and the analysis incoherent. (Aumann, 1987a, p. 473).

Peirce's construal of the common ground of strategically intercommunicating sign-carriers introduced the infinite reciprocity of common familiar knowledge, but avoided associating it with propositional content. Only much later, the significance of common knowledge caught on in theories of economics, communication, dialogue, signalling and coordination, as well as modal and epistemic logics.

All in all, it has been a privilege to dip into an erupting volcano.

PEIRCE'S LOGIC
Exercises 1

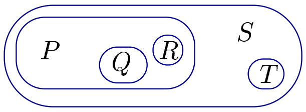
1. Draw alpha graphs for the following expressions of propositional logic:

   (a) $P \wedge \neg\neg Q$

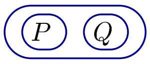
   (b) $(P \vee Q) \vee (R \vee S)$

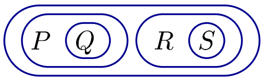
   (c) $(P \rightarrow (Q \vee R)) \rightarrow (S \rightarrow T)$

2. In the alpha graph below, find its (i) areas, (ii) enclosures, (iii) evenly and oddly enclosed subgraphs and (iv) maximal nests.



3. Find all the ways of reading the following alpha graphs in propositional logic:

(a) 

(b) 

4. Using the rules of transformations for alpha graphs show that the following inferences hold:

   (a) $\{P \vee P\} \vdash P$

   (b) $\{P \vee Q, P \rightarrow R, Q \rightarrow S\} \vdash R \vee S$

   (c) $\{\neg Q \rightarrow (P \wedge \neg P)\} \vdash Q$

   (d) $\{P \rightarrow ((Q \wedge R \wedge \neg S) \vee (T \wedge U))\} \vdash (P \wedge V) \rightarrow ((Q \wedge \neg(S \wedge W)) \vee T)$

5. (a) Using alpha graphs and their rules of transformation show the following logical equivalences:

      i. $P \rightarrow Q \Leftrightarrow \neg(P \wedge \neg Q) \Leftrightarrow \neg P \vee Q \Leftrightarrow (\neg\neg\neg P \vee \neg\neg Q) \wedge (P \vee \neg P)$.

      ii. $P \rightarrow (Q \rightarrow R) \Leftrightarrow (P \wedge Q) \rightarrow R$.

      iii. $P \oplus Q \Leftrightarrow \neg(P \leftrightarrow Q)$, where $\oplus$ is the exclusive disjunction.

   (b) Construct an algorithm which maps all sentences of propositional logic into alphagraphs. Use that algorithm to verify (5.a.ii).

PEIRCE'S LOGIC
Exercises 2

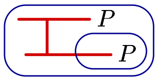
1. Show using the transformation rules for alpha graphs:

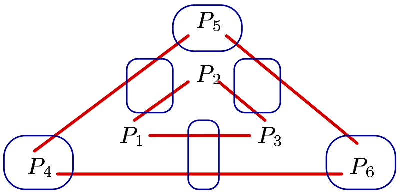
   (a) $\vdash\ P \rightarrow (Q \rightarrow P)$ (b) $\vdash\ (P \rightarrow (Q \rightarrow R)) \rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R))$

2. (a) Draw beta graphs for the following sentences of predicate logic:
   i. $\exists x(\neg P_1(x) \wedge \neg P_2(x))$
   ii. $\exists x \neg(P_1(x) \wedge P_2(x))$
   iii. $\forall x \exists y \forall z \exists u(P_1(x, y) \rightarrow P_2(z, u))$.

   (b) Construe the smallest beta graph (a graph with the fewest number of spots) in which a bridge must be used. Translate it into a sentence of predicate logic.

3. Translate the following beta graphs into predicate logic sentences:

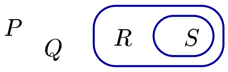
(a) 

(b) 

4. Show using the transformation rules for beta graphs:

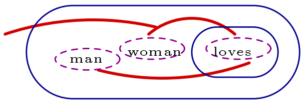
   (a) 3.a is a tautology
   (b) $\neg\neg\exists x P(x) \quad \Leftrightarrow \quad \exists x \neg\neg P(x)$
   (c) $\{\forall x\,(P_1(x) \vee P_2(x))\}\ \vdash\ \exists x P_1(x) \vee \exists x P_2(x)$.

5. (a) Let the truth values be $v(P) = t, v(Q) = f, v(R) = t, v(S) = t$. Construe a semantic game tree for the alpha graph

   Who has a winning strategy?

   (b) Let the model be $M = \langle D, I\rangle$, $D = \{\text{John}, \text{Mary}\}$, $I(a) = \text{John}$, $I(b) = \text{Mary}$, $I(\text{man}) = \{\text{John}\}$, $I(\text{woman}) = \{\text{Mary}\}$, $I(\text{loves}) = \{(\text{John}, \text{Mary}), (\text{Mary}, \text{John})\}$. Construe a semantic game tree for the beta graph

   Who has a winning strategy?

PEIRCE'S LOGIC
Exercises 3

1. Using transformation rules for beta graphs show that
   (a) $\{\forall x(P_1(x) \vee P_2(x)), \neg\exists y P_1(y), \forall z \neg P_2(z)\}$ is contradictory.
   (b) $\{\exists x \forall y\, P_1(x,y)\} \vdash \forall y \exists x\, P_1(x,y)$.
2. Draw beta graphs that represent the sentences
   (a) *Something is different than everything.*
   (b) *All parents love all of their children.*
   (c) *A man walks in the park. He is happy. He whistles.*
3. (a) Construct the semantic game tree for the alpha graph that represents exclusive disjunction $P \oplus Q$, given the valuations (i) $v(P) = t, v(Q) = t$; (ii) $v(P) = t, v(Q) = f$; (iii) $v(P) = f, v(Q) = f$. Who has a winning strategy? What is it?
   (b) Construct the semantic game tree for the beta graph that represents the predicate-logic sentence $\forall x \neg(P_1(x) \rightarrow P_2(x,x))$. Let $M_1 = \langle D, I\rangle, D = \{1,2\}, I(a) = 1, I(b) = 2, I(P_1) = 1, I(P_2) = \{\langle 1,1\rangle, \langle 1,2\rangle\}$ and $M_2 = \langle D, I\rangle$, $D = \{1,2\}$, $I(a) = 1$, $I(b) = 2$, $I(P_1) = 1$, $I(P_2) = \{\langle 2,2\rangle\}$. Who has a winning strategy in M_1? What is it? What about M_2?
4. (a) Draw gamma graphs that represent the sentences
      i. *It is possible that if necessarily P, then necessarily not P.*
      ii. *It could be that none of the problems has a solution.*
      iii. *It could be that every problem has a solution.*
   (b) Using transformation rules for gamma graphs show
      i. $\vdash \Box(P \wedge Q) \rightarrow \Box P$
      ii. $\vdash \Box(P \rightarrow Q) \rightarrow (\Box P \rightarrow \Box Q)$
      iii. $\Box(P \wedge Q) \Leftrightarrow \Box P \wedge \Box Q$.
   (c) Using tinctures for objective possibility draw gamma graphs that represent the sentences
      i. *There is a person who will commit a suicide if she fails in business.*
      ii. *There is a person who will commit a suicide if everybody fails in business.*

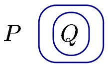
   Try to write these in the language of first-order logic. What happens?
5. (a) Using potentials draw a gamma graph that represents the property of a two-place relation 'being later than'.
   (b) Using metagraphs draw a gamma graph that represents the following property of existential graphs:

   If a line of identity l_1 exists, then there are two lines of identities, l_2, l_3, such that every point of the original line of identity l_1 is a point of either l_2 or l_2.

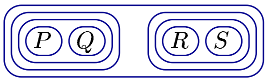
PEIRCE'S LOGIC
Solutions 1

1. Draw the corresponding alpha graphs for the following expressions of propositional logic:

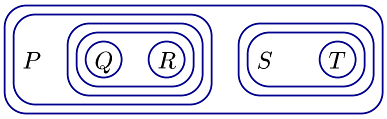
   (a) $P \wedge \neg\neg Q$

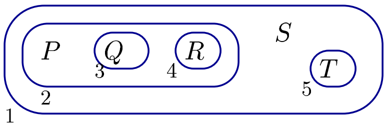
   (b) $(P \vee Q) \vee (R \vee S)$

   (c) $(P \to (Q \vee R)) \to (S \to T)$

2. In the alpha graph below, find its (i) areas, (ii) enclosures, (iii) evenly and oddly enclosed subgraphs and (iv) maximal nests (P, Q, R, S, T are atomic).

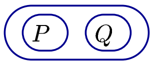
   

   Let the alpha graph be G_1. Number the cuts $1 \ldots 5$.

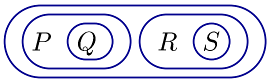
Areas:	Enclosures:	Evenly encl.:	Oddly encl.:	max. nests.:
Interior of 1	G_1	P	S	SA-1
2	2+its area	$\boxed{Q}$	$\boxed{T}$	SA-1-2
3	3+its area	$\boxed{R}$	$S\ \boxed{T}$	SA-1-5
4	4+its area	$P\ \boxed{Q}\ \boxed{R}$	$\boxed{P\ \boxed{Q}\ \boxed{R}}$	SA-1-2-3
5	5+its area	$P\ \boxed{Q}$	$\boxed{P\ \boxed{Q}\ \boxed{R}}\ S\ \boxed{T}$	SA-1-2-4
		$\boxed{Q}\ \boxed{R}$	$\boxed{P\ \boxed{Q}\ \boxed{R}}\ S$	
		$P\ \boxed{R}$	$\boxed{P\ \boxed{Q}\ \boxed{R}}\ \boxed{T}$	
		T	Q	
		G_1	R	

   18 subgraphs in total.

3. Find all the ways of reading the following alpha graphs in propositional logic:

(a) 

$P \vee Q, Q \vee P,$
$\neg(\neg P \wedge \neg Q), \neg(\neg Q \wedge \neg P),$
$\neg P \rightarrow Q, \neg Q \rightarrow P.$

(b) 

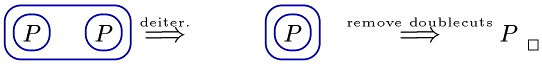
$(P \wedge \neg Q) \vee (R \wedge \neg S),$
$(P \wedge \neg Q) \vee (\neg S \wedge R),$
$(\neg Q \wedge P) \vee (R \wedge \neg S),$
$(\neg Q \wedge P) \vee (\neg S \wedge R),$
$(R \wedge \neg S) \vee (P \wedge \neg Q),$
$(\neg S \wedge R) \vee (P \wedge \neg Q),$
$(R \wedge \neg S) \vee (\neg Q \wedge P),$
$(\neg S \wedge R) \vee (\neg Q \wedge P),$

$\neg((\neg(P \wedge \neg Q) \wedge \neg(R \wedge \neg S))),$
$\neg((\neg(R \wedge \neg S) \wedge \neg(P \wedge \neg Q))),$
$\neg((\neg(\neg Q \wedge P) \wedge \neg(\neg S \wedge R))),$
$\neg((\neg(R \wedge \neg S) \wedge \neg(\neg Q \wedge P))),$
$\neg((\neg(P \wedge \neg Q) \wedge \neg(\neg S \wedge R))),$
$\neg((\neg(\neg S \wedge R) \wedge \neg(P \wedge \neg Q))),$
$\neg((\neg(\neg Q \wedge P) \wedge \neg(R \wedge \neg S))),$
$\neg((\neg(\neg S \wedge R) \wedge \neg(\neg Q \wedge P))),$
$\neg((P \rightarrow Q) \wedge (R \rightarrow S)),$
$\neg((R \rightarrow S) \wedge (P \rightarrow Q)),$

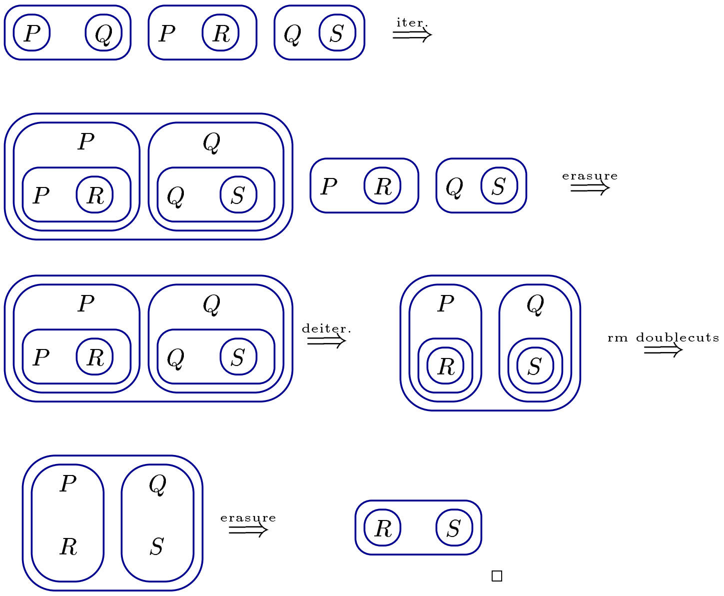
$(P \rightarrow Q) \rightarrow (R \wedge \neg S),$
$(P \rightarrow Q) \rightarrow (\neg S \wedge R),$
$(R \rightarrow S) \rightarrow (P \wedge \neg Q),$
$(R \rightarrow S) \rightarrow (\neg Q \wedge P),$
$\neg(R \wedge \neg S) \rightarrow (P \wedge \neg Q),$
$\neg(R \wedge \neg S) \rightarrow (\neg Q \wedge P),$
$\neg(\neg S \wedge R) \rightarrow (P \wedge \neg Q),$
$\neg(\neg S \wedge R) \rightarrow (\neg Q \wedge P),$
$\neg(P \wedge \neg Q) \rightarrow (R \wedge \neg S),$
$\neg(\neg Q \wedge P) \rightarrow (R \wedge \neg S),$
$\neg(P \wedge \neg Q) \rightarrow (\neg S \wedge R),$
$\neg(\neg Q \wedge P) \rightarrow (\neg S \wedge R).$

4. Using the rules of transformations for alpha graphs show that the following inferences hold:

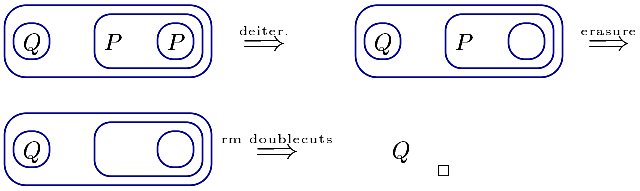
(a) $\{P \vee P\} \vdash P$



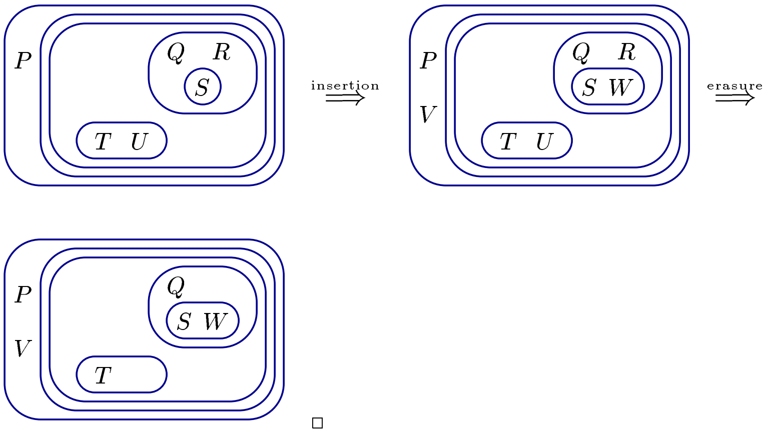
(b) $\{P \vee Q, P \to R, Q \to S\} \vdash R \vee S$



(c) $\{\neg Q \to (P \wedge \neg P)\} \vdash Q$



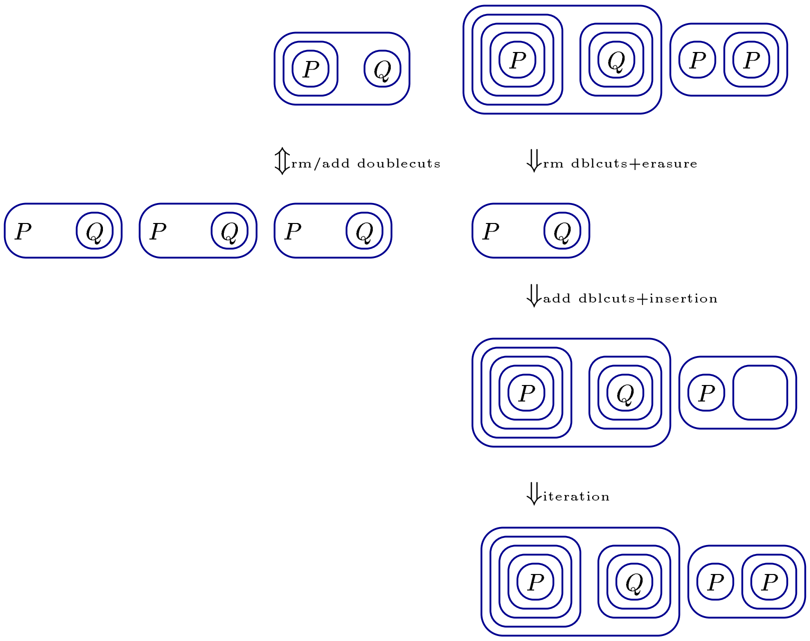
(d) $\{P \to ((Q \wedge R \wedge \neg S) \vee (T \wedge U))\} \vdash (P \wedge V) \to ((Q \wedge \neg(S \wedge W)) \vee T)$



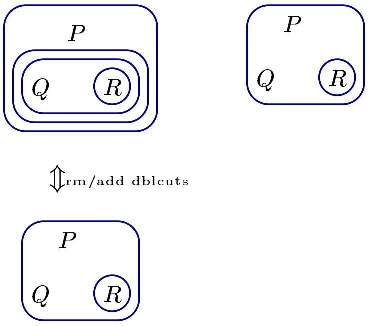
□

5. (a) Using alpha graphs and their rules of transformation show the following logical equivalences:

   i. $P \to Q \Leftrightarrow \neg(P \wedge \neg Q) \Leftrightarrow \neg P \vee Q \Leftrightarrow (\neg\neg\neg P \vee \neg\neg Q) \wedge (P \vee \neg P)$.

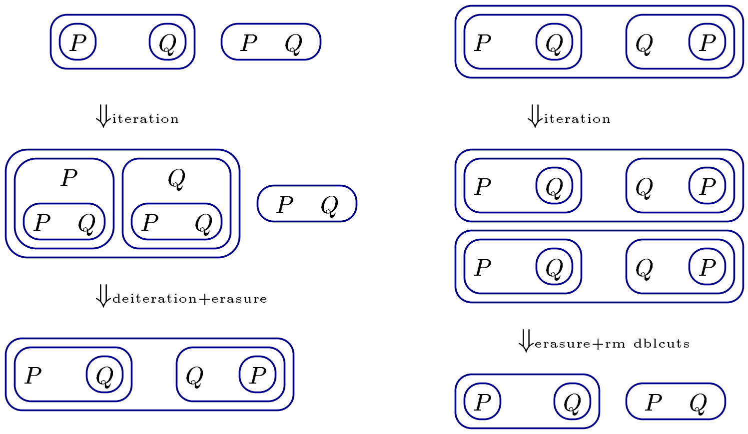


ii. $P \to (Q \to R) \Leftrightarrow\ (P \wedge Q) \to R$.



iii. $P \oplus Q \Leftrightarrow \neg(P \leftrightarrow Q)$, where $\oplus$ is exclusive disjunction.

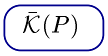
$P \oplus Q :$ $\neg(P \leftrightarrow Q) :$



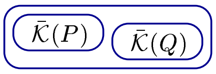
(b) Construct an algorithm that maps all sentences of propositional logic into alpha graphs. Use that algorithm to verify (5.a.ii).

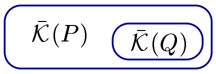
Let the algorithm be $\mathcal{A}$. Define a mapping $\mathcal{K}\colon \Phi \to \mathcal{G}_\alpha$ from atomic formulas to alpha graphs, where $\mathcal{K}(P_i) = P_i$, $P_i \in \Phi, i = 1 \ldots n$, and $\mathcal{K}(\top) = \mathbf{SA}$, $\mathcal{K}(\bot) =$ empty cut. Extend $\mathcal{K}$ to $\bar{\mathcal{K}}\colon \mathrm{WFF} \to \mathcal{G}_\alpha$ from well-founded formulas to alpha graphs such that

i. If $P \in \Phi$, then $\bar{\mathcal{K}}(P) = \mathcal{K}(P)$.

ii. If $\bar{\mathcal{K}}(\neg P)$, then 

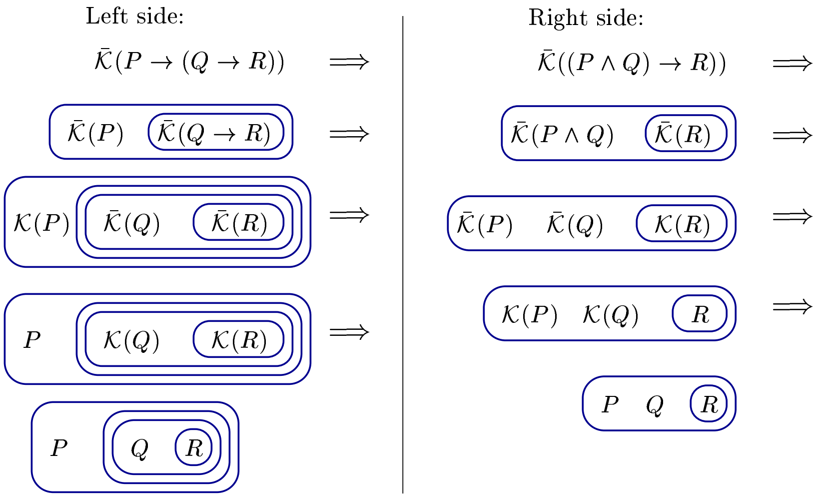
iii. If $\bar{\mathcal{K}}(P \wedge Q)$, then $\bar{\mathcal{K}}(P)$ $\bar{\mathcal{K}}(Q)$

iv. If $\bar{\mathcal{K}}(P \vee Q)$, then 

v. If $\bar{\mathcal{K}}(P \to Q)$, then 

Then check (5.a.ii). Assume that $P, Q, R \in \Phi$. We show that

$$\bar{\mathcal{K}}(P \to (Q \to R)) \Leftrightarrow \bar{\mathcal{K}}((P \wedge Q) \to R).$$

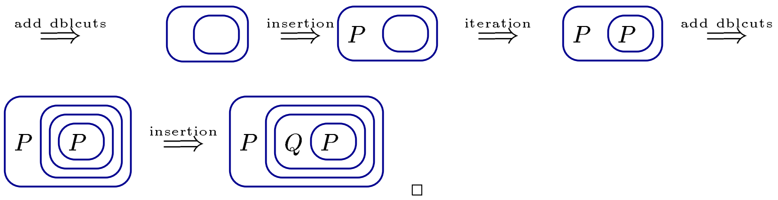


So $\mathcal{A}$ rewrites these wffs into alpha graphs that are the same modulo double cuts.

PEIRCE'S LOGIC
Solutions 2

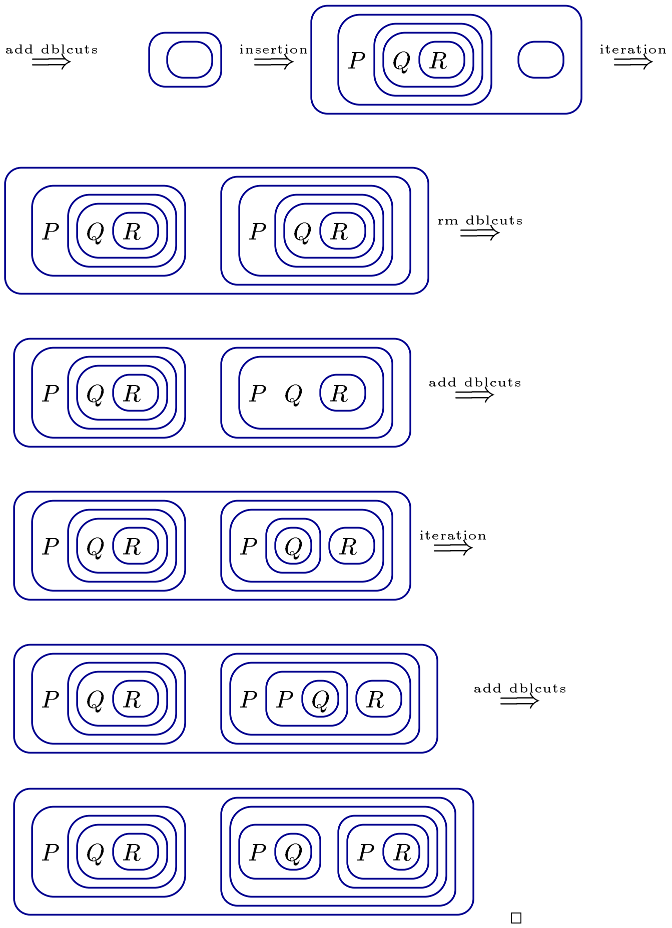
1. Show using the transformation rules for alpha graphs:

   (a) $\vdash P \to (Q \to P)$



□

   (b) $\vdash (P \to (Q \to R)) \to ((P \to Q) \to (P \to R))$



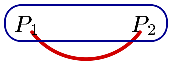
□

2. (a) Draw beta graphs for the following sentences of predicate logic:

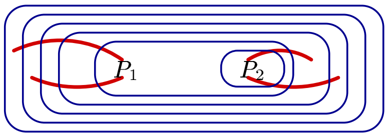
   i. $\exists x(\neg P_1(x) \wedge \neg P_2(x))$

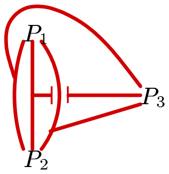
   ii. $\exists x \neg (P_1(x) \wedge P_2(x))$

   iii. $\forall x \exists y \forall z \exists u (P_1(x, y) \rightarrow P_2(z, u))$.

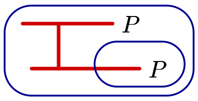
   (b) Construe the smallest beta graph (a graph with the fewest number of spots), in which a bridge must be used. Translate it to predicate logic.

   $\exists x_1 \exists x_2 \exists x_3 \, (P_1(x_1, x_2, x_3) \wedge P_2(x_1, x_2, x_3) \wedge P_3(x_1, x_2, x_3))$.

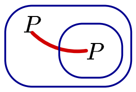
   Change any position of any two lines attached to P_1, P_2 or P_3 and at least one bridge must be used. (It is assumed that all relations are symmetric.)

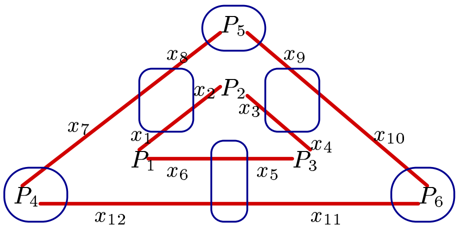
3. Translate the following beta graphs to predicate logic:

   (a) 

   $\forall x \forall y \, ((P(x) \wedge (x = y)) \rightarrow P(y))$.

   This is logically equivalent to $\forall x \, (P(x) \rightarrow P(x))$. By deiteration, we can rewrite it into (and iterate back to the original):

   (a*) 

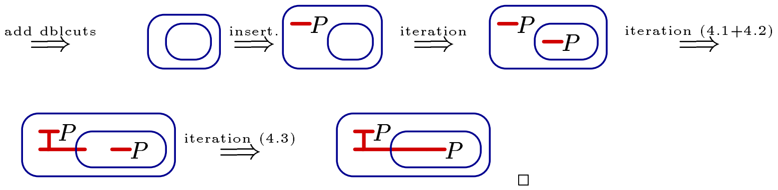
(b) 

Let us enumerate the outermost portions of ligatures and assign variables to them:

$\exists x_1 \ldots \exists x_{12} \, (P_1(x_1, x_6) \wedge P_2(x_2, x_3) \wedge P_3(x_4, x_5) \wedge (x_1 \neq x_2) \wedge (x_3 \neq x_4) \wedge (x_5 \neq x_6) \wedge \neg P_4(x_7, x_{12}) \wedge \neg P_5(x_8, x_9) \wedge \neg P_6(x_{10}, x_{11}) \wedge (x_7 \neq x_8) \wedge (x_9 \neq x_{10}) \wedge (x_{11} \neq x_{12}))$.

4. Show using the transformation rules for beta graphs:

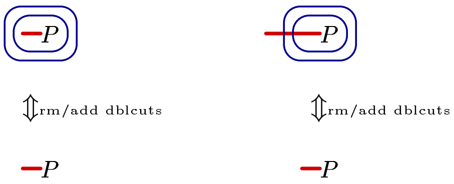
   (a) 3.a. is a tautology



□

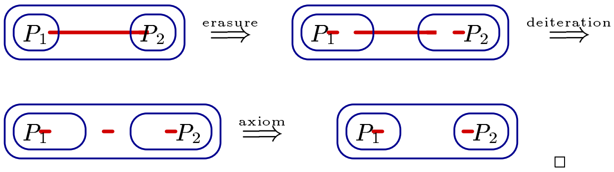
(Alternatively, we can cut the graph and derive an empty cut by applying the converses of the rules above: deiteration (5.3) ⇒ deiteration (5.2)+(5.1) ⇒ deiteration ⇒ rm dblcuts ⇒ empty cut.)

   (b) $\neg\neg\exists x P(x) \quad \Leftrightarrow \quad \exists x \neg\neg P(x)$



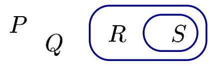
You may not show the equivalence without first removing the dblcut and then adding a dblcut.

   (c) $\{\forall x \, (P_1(x) \vee P_2(x))\} \ \vdash \ \exists x P_1(x) \vee \exists x P_2(x)$.

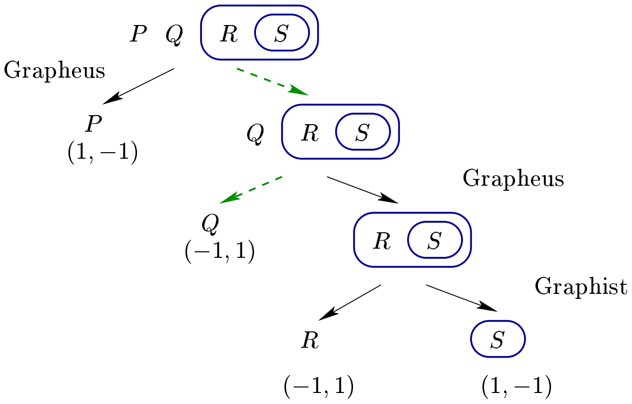


□

5. (a) Let the truth values be $v(P) = t, v(Q) = f, v(R) = t, v(S) = t$. Construe a semantic game tree for the alpha graph

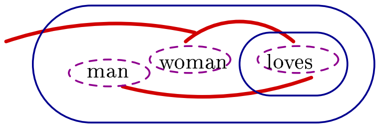
(*) 

Who has a winning strategy? What is it?

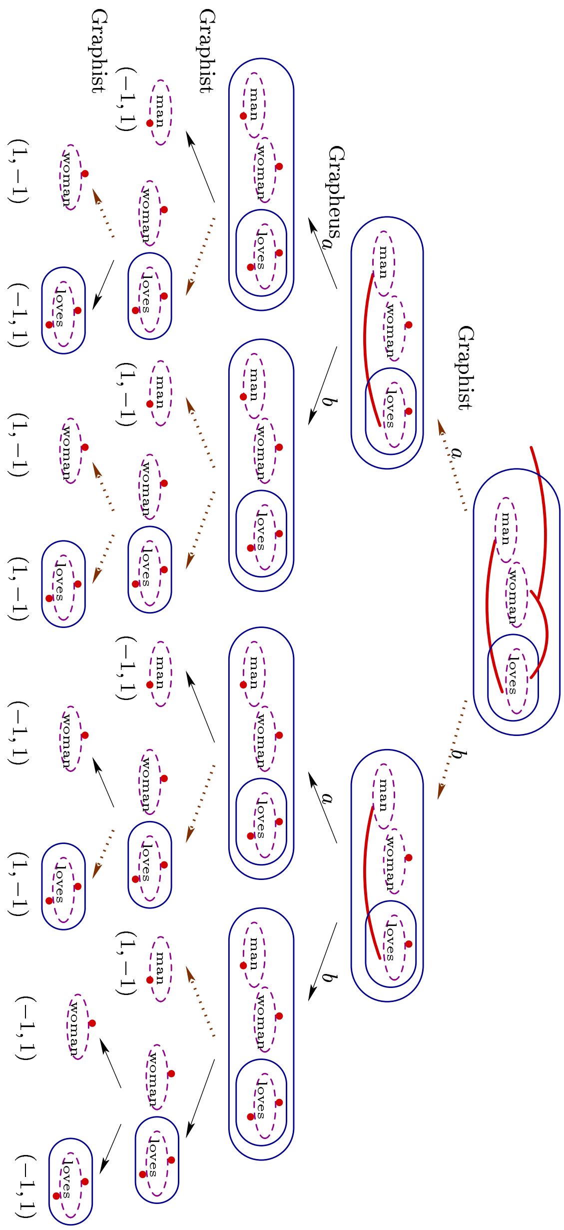


Grapheus has a winning strategy (dashed arrows) and so the graph (*) is false.

(b) Let $M = \langle D, I \rangle$, $D = \{\text{John}, \text{Mary}\}$, $I(a) = \{\text{John}\}$, $I(b) = \{\text{Mary}\}$, $I(\text{man}) = \{\text{John}\}$, $I(\text{woman}) = \{\text{Mary}\}$, $I(\text{loves}) = \{\langle \text{John}, \text{Mary} \rangle, \langle \text{Mary}, \text{John} \rangle\}$. Construe a semantic game tree for the beta graph *There exists a woman who loves and is loved by every man*:

(**) 

Who has a winning strategy? What is it?

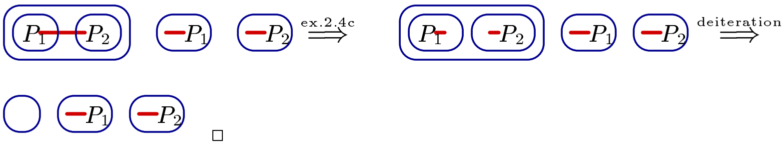


Graphist has a winning strategy (dotted arrows), and so the graph (**) is true in M.

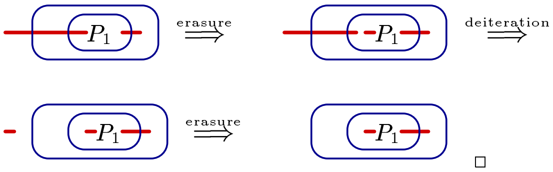
PEIRCE'S LOGIC
Solutions 3

1. Using transformation rules for beta graphs show that

   (a) $\{\forall x(P_1(x) \vee P_2(x)), \neg\exists y P_1(y), \forall z \neg P_2(z)\}$ is contradictory.

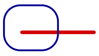


   (b) $\{\exists x \forall y\, P_1(x,y)\} \vdash \forall y \exists x\, P_1(x,y)$.

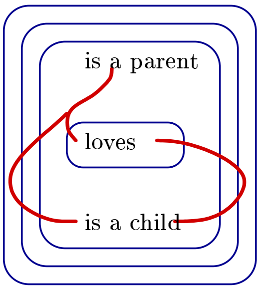


2. Draw beta graphs that represent the sentences

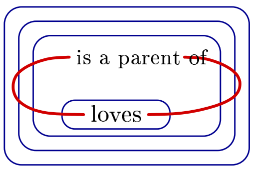
   (a) *Something is different than everything.*



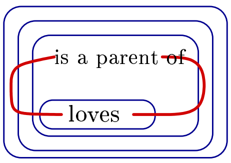
   (b) *All parents love (and are loved by) all of their children.*



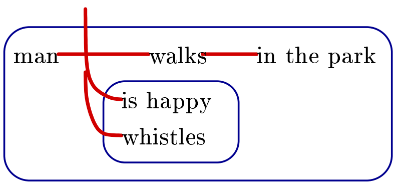
or



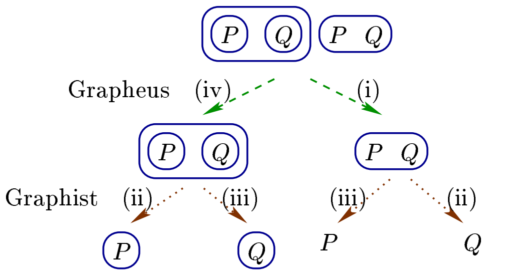
In the sentence *All parents love some of their children*, the line of identity for *some* needs to be nested within that for *every*:



   (c) *A man walks in the park. He is happy. He whistles.*



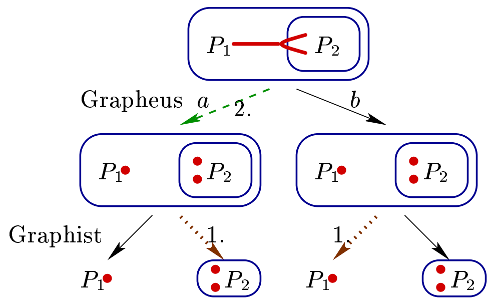
3. (a) Construe a semantic game tree for the alpha graph that represents exclusive disjunction $P \oplus Q$. Let the valuations be (i) $v(P) = t, v(Q) = t$; (ii) $v(P) = t, v(Q) = f$; (iii) $v(P) = f, v(Q) = t$; (iv) $v(P) = f, v(Q) = f$. Who has a winning strategy? What is it?



(i)	$(1,-1)$	$(1,-1)$	$(-1,1)$	$(-1,1)$
(ii)	$(1,-1)$	$(-1,1)$	$(-1,1)$	$(1,-1)$
(iii)	$(-1,1)$	$(1,-1)$	$(1,-1)$	$(-1,1)$
(iv)	$(-1,1)$	$(-1,1)$	$(1,-1)$	$(1,-1)$

Grapheus has a winning strategy (dashed arrows, (i) and (iv)). Graphist has a winning strategy (dotted arrows, (ii) and (iii)).

(b) Construe a semantic game tree for the beta graph that represents the predicate-logic sentence $\forall x(P_1(x) \to P_2(x,x))$. Let $M_1 = \langle D, I \rangle$, $D = \{1,2\}$, $I(a) = 1$, $I(b) = 2$, $I(P_1) = 1$, $I(P_2) = \{\langle 1,1 \rangle, \langle 1,2 \rangle\}$ and $M_2 = \langle D, I \rangle$, $D = \{1,2\}$, $I(a) = 1$, $I(b) = 2$, $I(P_1) = 1$, $I(P_2) = \{\langle 2,2 \rangle\}$. Who has a winning strategy in M_1? What is it? What about M_2?



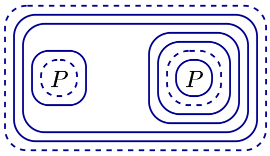
1.	$(-1,1)$	$(1,-1)$	$(1,-1)$	$(-1,1)$
2.	$(-1,1)$	$(-1,1)$	$(1,-1)$	$(1,-1)$

Graphist has a winning strategy in the model M_1 (dotted arrows). Grapheus has a winning strategy in the model M_2 (dashed arrows).

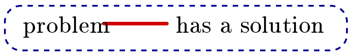
PEIRCE'S LOGIC
Solutions 3

4. (a) Draw gamma graphs that represent the sentences

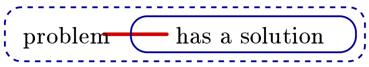
i. *It is possible that if necessarily P, then necessarily not P.*



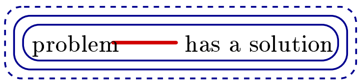
ii. *It could be that none of the problems has a solution.*

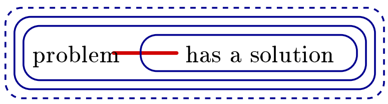


iii. *It could be that every problem has a solution.*



In (ii) and (iii), add double cuts within the broken cut:

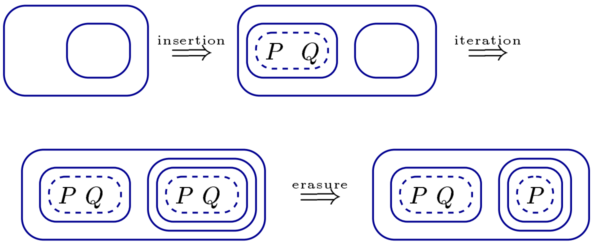
(ii*) 

(iii*) 

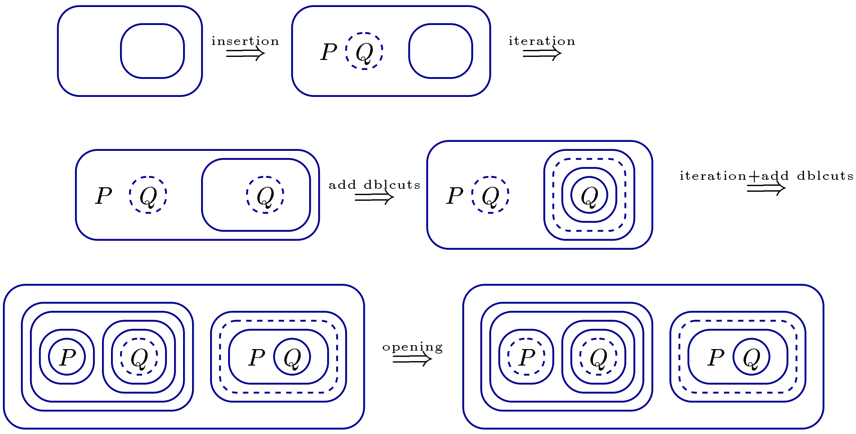
Now (ii*) is: *It may be that there is not any problem that will have a solution*, that is, *It may be that for all problems, they do not have a solution*. Likewise (iii*) reads: *It may be that there is not any problem that will not have a solution*, in other words *It may be, that for all problems, they have a solution*.

(b) Using the transformation rules for gamma graphs show

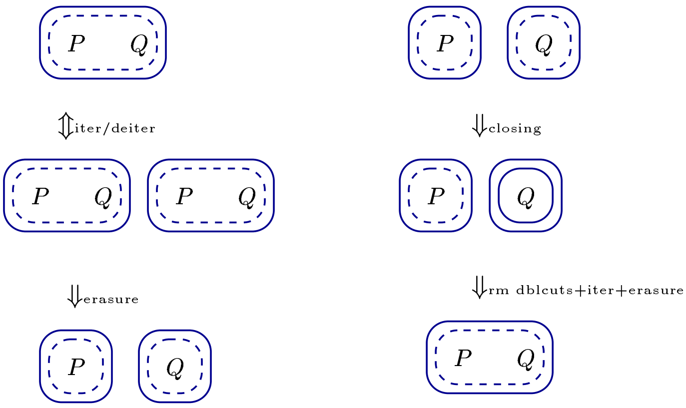
i. $\vdash \Box(P \wedge Q) \rightarrow \Box P$



ii. $\vdash \Box(P \rightarrow Q) \rightarrow (\Box P \rightarrow \Box Q)$

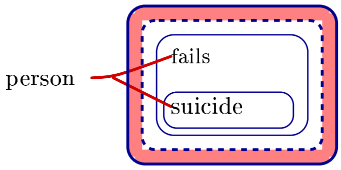


iii. $\Box(P \wedge Q) \quad \Leftrightarrow \quad \Box P \wedge \Box Q.$

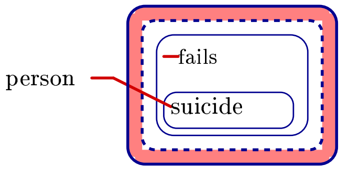


(c) Using tinctures for objective possibility draw gamma graphs that represent the sentences

i. *There is a person who will commit a suicide if she fails in business.*



ii. *There is a person who will commit a suicide if everybody fails in business.*

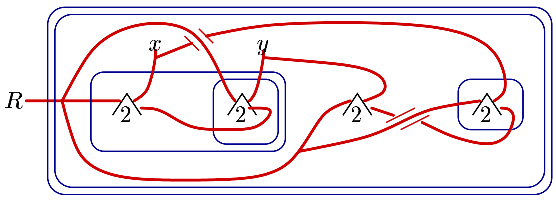


Try to write these in predicate logic. What happens?

(i) $= \exists x(E(x) \to I(x)) \Leftrightarrow \exists x E(x) \to \exists x\, I(x) \Leftrightarrow \forall y E(x) \to \exists x\, I(x) \Leftrightarrow \exists x(\forall y E(y) \to I(x)) =$ (ii).

The predicate-logic sentences are logically equivalent, although (i) and (ii) clearly do not mean the same thing!

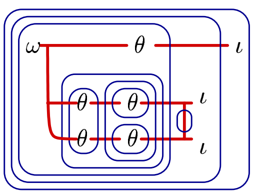
5. (a) Using potentials draw a gamma graph that represents the property of a two-place relation 'being later than' ('posteriority').



Here x is anterior to y since there is a two-place relation R between y and anything that stands in an R-relation with x, and something to which y but not x stands in the R-relation.

(b) Using metagraphs draw a gamma that represents the following property of beta graphs:

If a line of identity l_1 exists, then there are two lines of identities, l_2, l_3, such that every point of the original line of identity l_1 is a point of either l_2 or l_3.



___ι: 'is a line of identity'
___θ___: 'is a place of'
___ω: 'is a point'.